US012403177B2

(12) **United States Patent**
Delisle et al.

(10) **Patent No.: US 12,403,177 B2**
(45) **Date of Patent: *Sep. 2, 2025**

(54) **SILK-BASED PRODUCT FORMULATIONS AND METHODS OF USE**

(71) Applicant: **Cocoon Biotech Inc.**, Mansfield, MA (US)

(72) Inventors: **Scott Delisle**, Mansfield, MA (US); **Michael Santos**, Mansfield, MA (US)

(73) Assignee: **Cocoon Biotech Inc.**, Mansfield, MA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 125 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/176,903**

(22) Filed: **Mar. 1, 2023**

(65) **Prior Publication Data**

US 2023/0263859 A1 Aug. 24, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/972,021, filed as application No. PCT/US2019/035306 on Jun. 4, 2019, now Pat. No. 11,633,455.

(60) Provisional application No. 62/757,995, filed on Nov. 9, 2018, provisional application No. 62/757,984, filed on Nov. 9, 2018, provisional application No. 62/717,025, filed on Aug. 10, 2018, provisional application No. 62/680,376, filed on Jun. 4, 2018.

(51) **Int. Cl.**

A61K 38/17	(2006.01)
A61K 9/00	(2006.01)
A61K 9/08	(2006.01)
A61K 45/06	(2006.01)
A61K 47/10	(2017.01)
A61K 47/26	(2006.01)
A61P 27/04	(2006.01)

(52) **U.S. Cl.**

CPC ***A61K 38/1767*** (2013.01); ***A61K 9/0048*** (2013.01); ***A61K 9/08*** (2013.01); ***A61K 45/06*** (2013.01); ***A61K 47/10*** (2013.01); ***A61K 47/26*** (2013.01); ***A61P 27/04*** (2018.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,233,211	A	11/1980	Ohtomo	
9,187,538	B2	11/2015	Altman	
9,511,012	B2	12/2016	Altman	
9,517,191	B2	12/2016	Altman	
9,522,107	B2	12/2016	Altman	
9,522,108	B2	12/2016	Altman	
9,545,369	B2	1/2017	Altman	
10,588,843	B2	3/2020	Altman	
10,610,478	B2	4/2020	Altman	
10,987,294	B2	4/2021	Altman	
11,298,310	B2	4/2022	Altman	
11,298,311	B2	4/2022	Altman	
11,376,329	B2	7/2022	Kluge	
11,857,663	B2	1/2024	Altman	
11,857,664	B2	1/2024	Altman	
12,024,538	B2 *	7/2024	Delisle	C08L 89/00
2004/0005363	A1	1/2004	Tsukada	
2004/0219630	A1	11/2004	Tsubouchi	
2007/0297981	A1	12/2007	Ousler, III et al.	
2009/0286959	A1 *	11/2009	Lee	A61P 17/18 530/353
2014/0235554	A1	8/2014	Lawrence et al.	
2015/0094269	A1 *	4/2015	Altman	A61K 8/27 206/524.1
2016/0046679	A1 *	2/2016	Kluge	C07K 14/43518 428/221
2016/0095695	A1	4/2016	Altman et al.	
2016/0215103	A1	7/2016	Omenetto et al.	
2016/0331815	A1	11/2016	Lawrence et al.	
2021/0228684	A1	7/2021	Delisle et al.	

FOREIGN PATENT DOCUMENTS

CN	1431023	A	7/2003
KR	20170062028	A	6/2017
WO	WO 03022909	A1	3/2003
WO	WO 2010126640	A2	11/2010
WO	WO 2011022771	A1	3/2011
WO	WO 2014012099	A1	1/2014
WO	WO 2017123383	A2	7/2017
WO	2017139684	A1	8/2017
WO	2019094700	A1	5/2019
WO	2019094702	A1	5/2019
WO	2020247594	A1	12/2020

OTHER PUBLICATIONS

National Center for Biotechnology Information (2024). PubChem Compound Summary for CID 10340, Sodium Carbonate. Retrieved Aug. 30, 2024 from https://pubchem.ncbi.nlm.nih.gov/compound/Sodium-Carbonate. (Year: 2024).*

Cho et al., "Molecular weight distribution and solution properties of silk fibroins with different dissolution conditions", International Journal of Biological Macromolecules, 51 (2012) 336-341.

Pritchard and Kaplan, "Silk fibroin biomaterials for controlled release drug delivery", Expert Opin. Drug Deliv. 8(6): 797-811, 2011.

(Continued)

Primary Examiner — Christina Bradley

(74) *Attorney, Agent, or Firm* — Karen A. LeCuyer; DeWitt LLP

(57) **ABSTRACT**

Embodiments of the present disclosure include formulations of silk-based product (SBP) formulations and related methods of preparation and use in a variety of applications in the fields of human therapeutics, veterinary medicine, agriculture, and material science.

6 Claims, No Drawings

Specification includes a Sequence Listing.

(56) **References Cited**

OTHER PUBLICATIONS

Pritchard et al., "Effect of Silk Protein Processing on Drug Delivery from Silk Films", Macromol Biosci., 3(3), 311-320 (2013).
Pritchard et al., "Incorporation of Proteinase Inhibitors into Silk-Based Delivery Devices for Enhanced Control of Degradation and Drug Release", Biomaterials 32(3): 909-918 (2011).
Rockwood et al., "Materials fabrication from Bombyx mori silk fibroin", Nature Protocols, 6(10), 2011, 1612-1631.
Sah et al., 'Regenerated Silk Fibroin from B. mori Silk Cocoon for Tissue Engineering Applications, Int. J. Sci. Dev. (2010) 1(5), pp. 404-408.
Fan et al. "Silk materials for medical, electronic and optical applications", Sci China Tech Sci, 62 (2019) 903-918.
Wenk et al., "Silk fibroin as a vehicle for drug delivery applications", Journal of Controlled Release 150 (2011) 128-141.
Wray et al., "Effect of Processing on silk-based biomaterials: Reproducibility and Biocompatibility", J Biomed Mater Res Part B 2011:99B:89-101.
Cole, J.; "Artificial Tears: What Matters and Why"; Review of Optometry; 2020; 9 pages.
European Search Report for Application #19816014.5 [PCT/US2019/035306] dated Mar. 29, 2022; 8 pages.
International Search Report and Written Opinion for International Application PCT/US2019/035306; International Filing Date: Jun. 4, 2019; Date of Mailing: Sep. 4, 2019; 8 pages.
Tran, S. et al.; "A Review of the Emerging Role of Silk for the Treatment of the Eye"; Pharmaceutical Research, vol. 35, Issue No. 12; 2018; 16 pages; doi: 10.1007/s11095-018-2534-y.
Volkov, V. et al.; On the Routines of Wild-Type Silk Fibroin Processing Toward Silk-Inspired Materials: A Review:I Macromolecular Materials and Engineering, vol. 12; 2015; pp. 1199-1216.
Koza, Stephan M. et al. Successful Transfer of Size-Exclusion Separations Between HPLC and UPLC. https://www.waters.com/content/dam/waters/en/app-notes/2015/720005214/720005214-en.pdf.
U.S. Appl. No. 16/972,021, filed Dec. 4, 2020, Scott Delisle.

* cited by examiner

SILK-BASED PRODUCT FORMULATIONS AND METHODS OF USE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/972,021, filed on Dec. 4, 2020, which is a 371 U.S. National Stage filing of PCT/US2019/035306, filed on Jun. 4, 2019, which claims priority to 62/680,376 filed on Jun. 4, 2018 entitled Silk-Based Products for Ocular Lubrication; 62/717,025 filed Aug. 10, 2018, entitled Silk-Based Product Formulations and Methods of Use; 62/757,984 filed Nov. 9, 2018, entitled Silk-Based Products for Ocular Lubrication; and 62/757,995 filed Nov. 9, 2018, entitled Silk-Based Product Formulations and Methods of Use, the contents of each of which are herein incorporated by reference in their entirety.

SEQUENCE LISTING

The Instant Application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on May 12, 2023 is named "CBH0009USC" and is 2,185 bytes in size. The Sequence Listing does not go beyond the disclosure in the application as filed.

FIELD OF DISCLOSURE

The present disclosure relates to formulations and methods. Specifically provided are silk-based product formulations.

BACKGROUND OF THE DISCLOSURE

Silk is a naturally occurring polymer. Most silk fibers are derived from silkworm moth (*Bombyx mori*) cocoons and include silk fibroin and sericin proteins. Silk fibroin is a fibrous material that forms a polymeric matrix bonded together with sericin. In nature, silk is formed from a concentrated solution of these proteins that are extruded through silkworm spinnerets to produce a highly insoluble fiber. These fibers have been used for centuries to form threads used in garments and other textiles.

Many properties of silk make it an attractive candidate for products serving a variety of industries. Polymer strength and flexibility has supported classical uses of silk in textiles and materials, while silk biocompatibility has gained attention more recently for applications in the fields of medicine and agriculture. Additional uses for silk in applications related to material science are being explored as technologies for producing and processing silk advance.

Although a variety of products and uses related to silk are being developed, there remains a need for methods of producing and processing silk and silk-based products that can meet the demands of modern medicine. Additionally, there remains a need for silk-based products that can leverage silk polymer strength, flexibility, and biocompatibility to meet needs in the fields of medicine, agriculture, and material sciences. The present disclosure addresses these needs by providing methods for producing and processing silk as well as formulations of silk-based products useful in a variety of industries.

SUMMARY OF THE DISCLOSURE

In some embodiments, the present disclosure provides silk-based product (SBP) formulations that comprise processed silk and at least one excipient, wherein the processed silk comprises or is derived from one or more articles, said one or more articles is selected from the group consisting of raw silk, silk fiber, silk fibroin, and a silk fibroin fragment. The SBP formulation may comprises or may be combined with one or more members selected from the group consisting of: (a) a therapeutic agent; (b) a cargo; (c) a microorganism; and (d) a biological system.

The processed silk and/or other SBP component (excipient, therapeutic agent, microbe, cargo, and/or biological system) may be present in SBP formulations at a concentration (by weight, volume, or concentration) of from about 0.0001% to about 0.001%, from about 0.001% to about 0.01%, from about 0.01% to about 1%, from about 0.05% to about 2%, from about 1% to about 5%, from about 2% to about 10%, from about 4% to about 16%, from about 5% to about 20%, from about 8% to about 24%, from about 10% to about 30%, from about 12% to about 32%, from about 14% to about 34%, from about 16% to about 36%, from about 18% to about 38%, from about 20% to about 40%, from about 22% to about 42%, from about 24% to about 44%, from about 26% to about 46%, from about 28% to about 48%, from about 30% to about 50%, from about 35% to about 55%, from about 40% to about 60%, from about 45% to about 65%, from about 50% to about 70%, from about 55% to about 75%, from about 60% to about 80%, from about 65% to about 85%, from about 70% to about 90%, from about 75% to about 95%, from about 80% to about 96%, from about 85% to about 97%, from about 90% to about 98%, from about 95% to about 99%, from about 96% to about 99.2%, from about 97% to about 99.5%, from about 98% to about 99.8%, from about 99% to about 99.9%, or greater than 99.9%.

The SBP formulation may have processed silk and/or other SBP components (excipient, therapeutic agent, microbe, cargo, and/or biological system) present at a concentration of from about 0.01 pg/mL to about 1 pg/mL, from about 0.05 pg/mL to about 2 pg/mL, from about 1 pg/mL to about 5 pg/mL, from about 2 pg/mL to about 10 pg/mL, from about 4 pg/mL to about 16 pg/mL, from about 5 pg/mL to about 20 pg/mL, from about 8 pg/mL to about 24 pg/mL, from about 10 pg/mL to about 30 pg/mL, from about 12 pg/mL to about 32 pg/mL, from about 14 pg/mL to about 34 pg/mL, from about 16 pg/mL to about 36 pg/mL, from about 18 pg/mL to about 38 pg/mL, from about 20 pg/mL to about 40 pg/mL, from about 22 pg/mL to about 42 pg/mL, from about 24 pg/mL to about 44 pg/mL, from about 26 pg/mL to about 46 pg/mL, from about 28 pg/mL to about 48 pg/mL, from about 30 pg/mL to about 50 pg/mL, from about 35 pg/mL to about 55 pg/mL, from about 40 pg/mL to about 60 pg/mL, from about 45 pg/mL to about 65 pg/mL, from about 50 pg/mL to about 75 pg/mL, from about 60 pg/mL to about 240 pg/mL, from about 70 pg/mL to about 350 pg/mL, from about 80 pg/mL to about 400 pg/mL, from about 90 pg/mL to about 450 pg/mL, from about 100 pg/mL to about 500 pg/mL, from about 0.01 ng/mL to about 1 ng/mL, from about 0.05 ng/mL to about 2 ng/mL, from about 1 ng/mL to about 5 ng/mL, from about 2 ng/mL to about 10 ng/mL, from about 4 ng/mL to about 16 ng/mL, from about 5 ng/mL to about 20 ng/mL, from about 8 ng/mL to about 24 ng/mL, from about 10 ng/mL to about 30 ng/mL, from about 12 ng/mL to about 32 ng/mL, from about 14 ng/mL to about 34 ng/mL, from about 16 ng/mL to about 36 ng/mL, from about 18 ng/mL to about 38 ng/mL, from about 20 ng/mL to about 40 ng/mL, from about 22 ng/mL to about 42 ng/mL, from about 24 ng/mL to about 44 ng/mL, from about 26 ng/mL to about 46 ng/mL, from about 28 ng/mL to about 48 ng/mL, from about 30 ng/mL to about 50 ng/mL, from about 35 ng/mL to about 55 ng/mL, from about 40 ng/mL to about 60 ng/mL, from about 45 ng/mL to about 65 ng/mL, from about 50 ng/mL to about 75 ng/mL, from about 60 ng/mL to about 240 ng/mL, from about 70 ng/mL to about 350 ng/mL, from about 80 ng/mL to about 400 ng/mL, from about 90 ng/mL to about 450 ng/mL, from about 100 ng/mL to about 500 ng/mL, from about 0.01 μg/mL to about 1 μg/mL, from about 0.05 μg/mL to about 2 μg/mL, from about 1 μg/mL to about 5 μg/mL, from about 2 μg/mL to about 10 μg/mL, from about 4 μg/mL to about 16 μg/mL, from about 5 μg/mL to about 20 μg/mL, from about 8 μg/mL to about 24 μg/mL, from about 10 μg/mL to about 30 μg/mL, from about 12 μg/mL to about 32 μg/mL, from about 14 μg/mL to about 34 μg/mL, from about 16 μg/mL to about 36 μg/mL, from about 18 μg/mL to about 38 μg/mL, from about 20 μg/mL to about 40 μg/mL, from about 22 μg/mL to about 42 μg/mL, from about 24 μg/mL to about 44 μg/mL, from about 26 μg/mL to about 46 μg/mL, from about 28 μg/mL to about 48 μg/mL, from about 30 μg/mL to about 50 μg/mL, from about 35 μg/mL to about 55 μg/mL, from about 40 μg/mL to about 60 μg/mL, from about 45 μg/mL to about 65 μg/mL, from about 50 μg/mL to about 75 μg/mL, from about 60 μg/mL to about 240 μg/mL, from about 70 μg/mL to about 350 μg/mL, from about 80 μg/mL to about 400 μg/mL, from about 90 μg/mL to about 450 μg/mL, from about 100 μg/mL to about 500 μg/mL, from about 0.01 mg/mL to about 1 mg/mL, from about 0.05 mg/mL to about 2 mg/mL, from about 1 mg/mL to about 5 mg/mL, from about 2 mg/mL to about 10 mg/mL, from about 4 mg/mL to about 16 mg/mL, from about 5 mg/mL to about 20 mg/mL, from about 8 mg/mL to about 24 mg/mL, from about 10 mg/mL to about 30 mg/mL, from about 12 mg/mL to about 32 mg/mL, from about 14 mg/mL to about 34 mg/mL, from about 16 mg/mL to about 36 mg/mL, from about 18 mg/mL to about 38 mg/mL, from about 20 mg/mL to about 40 mg/mL, from about 22 mg/mL to about 42 mg/mL, from about 24 mg/mL to about 44 mg/mL, from about 26 mg/mL to about 46 mg/mL, from about 28 mg/mL to about 48 mg/mL, from about 30 mg/mL to about 50 mg/mL, from about 35 mg/mL to about 55 mg/mL, from about 40 mg/mL to about 60 mg/mL, from about 45 mg/mL to about 65 mg/mL, from about 50 mg/mL to about 75 mg/mL, from about 60 mg/mL to about 240 mg/mL, from about 70 mg/mL to about 350 mg/mL, from about 80 mg/mL to about 400 mg/mL, from about 90 mg/mL to about 450 mg/mL, from about 100 mg/mL to about 500 mg/mL, from about 0.01 g/mL to about 1 g/mL, from about 0.05 g/mL to about 2 g/mL, from about 1 g/mL to about 5 g/mL, from about 2 g/mL to about 10 g/mL, from about 4 g/mL to about 16 g/mL, or from about 5 g/mL to about 20 g/mL.

The SBP formulation may have processed silk and/or other SBP components (excipient, therapeutic agent, microbe, cargo, and/or biological system) present in SBPs at a concentration of from about 0.01 pg/kg to about 1 pg/kg, from about 0.05 pg/kg to about 2 pg/kg, from about 1 pg/kg to about 5 pg/kg, from about 2 pg/kg to about 10 pg/kg, from about 4 pg/kg to about 16 pg/kg, from about 5 pg/kg to about 20 pg/kg, from about 8 pg/kg to about 24 pg/kg, from about 10 pg/kg to about 30 pg/kg, from about 12 pg/kg to about 32 pg/kg, from about 14 pg/kg to about 34 pg/kg, from about 16 pg/kg to about 36 pg/kg, from about 18 pg/kg to about 38 pg/kg, from about 20 pg/kg to about 40 pg/kg, from about 22 pg/kg to about 42 pg/kg, from about 24 pg/kg to about 44 pg/kg, from about 26 pg/kg to about 46 pg/kg, from about 28 pg/kg to about 48 pg/kg, from about 30 pg/kg to about 50 pg/kg, from about 35 pg/kg to about 55 pg/kg, from about 40 pg/kg to about 60 pg/kg, from about 45 pg/kg to about 65 pg/kg, from about 50 pg/kg to about 75 pg/kg, from about 60 pg/kg to about 240 pg/kg, from about 70 pg/kg to about 350 pg/kg, from about 80 pg/kg to about 400 pg/kg, from about 90 pg/kg to about 450 pg/kg, from about 100 μg/kg to about 500 pg/kg, from about 0.01 ng/kg to about 1 ng/kg, from about 0.05 ng/kg to about 2 ng/kg, from about 1 ng/kg to about 5 ng/kg, from about 2 ng/kg to about 10 ng/kg, from about 4 ng/kg to about 16 ng/kg, from about 5 ng/kg to about 20 ng/kg, from about 8 ng/kg to about 24 ng/kg, from about 10 ng/kg to about 30 ng/kg, from about 12 ng/kg to about 32 ng/kg, from about 14 ng/kg to about 34 ng/kg, from about 16 ng/kg to about 36 ng/kg, from about 18 ng/kg to about 38 ng/kg, from about 20 ng/kg to about 40 ng/kg, from about 22 ng/kg to about 42 ng/kg, from about 24 ng/kg to about 44 ng/kg, from about 26 ng/kg to about 46 ng/kg, from about 28 ng/kg to about 48 ng/kg, from about 30 ng/kg to about 50 ng/kg, from about 35 ng/kg to about 55 ng/kg, from about 40 ng/kg to about 60 ng/kg, from about 45 ng/kg to about 65 ng/kg, from about 50 ng/kg to about 75 ng/kg, from about 60 ng/kg to about 240 ng/kg, from about 70 ng/kg to about 350 ng/kg, from about 80 ng/kg to about 400 ng/kg, from about 90 ng/kg to about 450 ng/kg, from about 100 ng/kg to about 500 ng/kg, from about 0.01 μg/kg to about 1 μg/kg, from about 0.05 μg/kg to about 2 μg/kg, from about 1 μg/kg to about 5 μg/kg, from about 2 μg/kg to about 10 μg/kg, from about 4 μg/kg to about 16 μg/kg, from about 5 μg/kg to about 20 μg/kg, from about 8 μg/kg to about 24 μg/kg, from about 10 μg/kg to about 30 μg/kg, from about 12 μg/kg to about 32 μg/kg, from about 14 μg/kg to about 34 μg/kg, from about 16 μg/kg to about 36 μg/kg, from about 18 μg/kg to about 38 μg/kg, from about 20 μg/kg to about 40 μg/kg, from about 22 μg/kg to about 42 μg/kg, from about 24 μg/kg to about 44 μg/kg, from about 26 μg/kg to about 46 μg/kg, from about 28 μg/kg to about 48 μg/kg, from about 30 μg/kg to about 50 μg/kg, from about 35 μg/kg to about 55 μg/kg, from about 40 μg/kg to about 60 μg/kg, from about 45 μg/kg to about 65 μg/kg, from about 50 μg/kg to about 75 μg/kg, from about 60 μg/kg to about 240 μg/kg, from about 70 μg/kg to about 350 μg/kg, from about 80 μg/kg to about 400 μg/kg, from about 90 μg/kg to about 450 μg/kg, from about 100 μg/kg to about 500 μg/kg, from about 0.01 mg/kg to about 1 mg/kg, from about 0.05 mg/kg to about 2 mg/kg, from about 1 mg/kg to about 5 mg/kg, from about 2 mg/kg to about 10 mg/kg, from about 4 mg/kg to about 16 mg/kg, from about 5 mg/kg to about 20 mg/kg, from about 8 mg/kg to about 24 mg/kg, from about 10 mg/kg to about 30 mg/kg, from about 12 mg/kg to about 32 mg/kg, from about 14 mg/kg to about 34 mg/kg, from about 16 mg/kg to about 36 mg/kg, from about 18 mg/kg to about 38 mg/kg, from about 20 mg/kg to about 40 mg/kg, from about 22 mg/kg to about 42 mg/kg, from about 24 mg/kg to about 44 mg/kg, from about 26 mg/kg to about 46 mg/kg, from about 28 mg/kg to about 48 mg/kg, from about 30 mg/kg to about 50 mg/kg, from about 35 mg/kg to about 55 mg/kg, from about 40 mg/kg to about 60 mg/kg, from about 45 mg/kg to about 65 mg/kg, from about 50 mg/kg to about 75 mg/kg, from about 60 mg/kg to about 240 mg/kg, from about 70 mg/kg to about 350 mg/kg, from about 80 mg/kg to about 400 mg/kg, from about 90 mg/kg to about 450 mg/kg, from about 100 mg/kg to about 500 mg/kg, from about 0.01 g/kg to about 1 g/kg, from about 0.05 g/kg to about 2 g/kg, from about 1 g/kg to about 5 g/kg, from about 2 g/kg to about 10 g/kg, from about 4 g/kg to about 16 g/kg, or from about 5 g/kg to about 20 g/kg, from about 10 g/kg to about 50 g/kg, from about 15 g/kg to about 100 g/kg, from about 20 g/kg to about 150 g/kg, from about 25 g/kg to about 200 g/kg, from about 30 g/kg to about 250 g/kg, from about 35 g/kg to about 300 g/kg, from about 40 g/kg to about 350 g/kg, from about 45 g/kg to about 400 g/kg, from about 50 g/kg to about 450 g/kg, from about 55 g/kg to about 500 g/kg, from about 60 g/kg to about 550 g/kg, from about 65 g/kg to about 600 g/kg, from about 70 g/kg to about 650 g/kg, from about 75 g/kg to about 700 g/kg, from about 80 g/kg to about 750 g/kg, from about 85 g/kg to about 800 g/kg, from about 90 g/kg to about 850 g/kg, from about 95 g/kg to about 900 g/kg, from about 100 g/kg to about 950 g/kg, or from about 200 g/kg to about 1000 g/kg.

The SBP formulation may comprise processed silk and/or other SBP components (excipient, therapeutic agent, microbe, cargo, and/or biological system) present in SBPs at a concentration of from about 0.1 pM to about 1 pM, from about 1 pM to about 10 pM, from about 2 pM to about 20 pM, from about 3 pM to about 30 pM, from about 4 pM to about 40 pM, from about 5 pM to about 50 pM, from about 6 pM to about 60 pM, from about 7 pM to about 70 pM, from about 8 pM to about 80 pM, from about 9 pM to about 90 pM, from about 10 pM to about 100 pM, from about 11 pM to about 110 pM, from about 12 pM to about 120 pM, from about 13 pM to about 130 pM, from about 14 pM to about 140 pM, from about 15 pM to about 150 pM, from about 16 pM to about 160 pM, from about 17 pM to about 170 pM, from about 18 pM to about 180 pM, from about 19 pM to about 190 pM, from about 20 pM to about 200 pM, from about 21 pM to about 210 pM, from about 22 pM to about 220 pM, from about 23 pM to about 230 pM, from about 24 pM to about 240 pM, from about 25 pM to about 250 pM, from about 26 pM to about 260 pM, from about 27 pM to about 270 pM, from about 28 pM to about 280 pM, from about 29 pM to about 290 pM, from about 30 pM to about 300 pM, from about 31 pM to about 310 pM, from about 32 pM to about 320 pM, from about 33 pM to about 330 pM, from about 34 pM to about 340 pM, from about 35 pM to about 350 pM, from about 36 pM to about 360 pM, from about 37 pM to about 370 pM, from about 38 pM to about 380 pM, from about 39 pM to about 390 pM, from about 40 pM to about 400 pM, from about 41 pM to about 410 pM, from about 42 pM to about 420 pM, from about 43 pM to about 430 pM, from about 44 pM to about 440 pM, from about 45 pM to about 450 pM, from about 46 pM to about 460 pM, from about 47 pM to about 470 pM, from about 48 pM to about 480 pM, from about 49 pM to about 490 pM, from about 50 pM to about 500 pM, from about 51 pM to about 510 pM, from about 52 pM to about 520 pM, from about 53 pM to about 530 pM, from about 54 pM to about 540 pM, from about 55 pM to about 550 pM, from about 56 pM to about 560 pM, from about 57 pM to about 570 pM, from about 58 pM to about 580 pM, from about 59 pM to about 590 pM, from about 60 pM to about 600 pM, from about 61 pM to about 610 pM, from about 62 pM to about 620 pM, from about 63 pM to about 630 pM, from about 64 pM to about 640 pM, from about 65 pM to about 650 pM, from about 66 pM to about 660 pM, from about 67 pM to about 670 pM, from about 68 pM to about 680 pM, from about 69 pM to about 690 pM, from about 70 pM to about 700 pM, from about 71 pM to about 710 pM, from about 72 pM to about 720 pM, from about 73 pM to about 730 pM, from about 74 pM to about 740 pM, from about 75 pM to about 750 pM, from about 76 pM to about 760 pM, from about 77 pM to about 770 pM, from about 78 pM to about 780 pM, from about 79 pM to about 790 pM, from about 80 pM to about 800 pM, from about 81 pM to about 810 pM, from about 82 pM to about 820 pM, from about 83 pM to about 830 pM, from about 84 pM to about 840 pM, from about 85 pM to about 850 pM, from about 86 pM to about 860 pM, from about 87 pM to about 870 pM, from about 88 pM to about 880 pM, from about 89 pM to about 890 pM, from about 90 pM to about 900 pM, from about 91 pM to about 910 pM, from about 92 pM to about 920 pM, from about 93 pM to about 930 pM, from about 94 pM to about 940 pM, from about 95 pM to about 950 pM, from about 96 pM to about 960 pM, from about 97 pM to about 970 pM, from about 98 pM to about 980 pM, from about 99 pM to about 990 pM, from about 100 pM to about 1 nM, from about 0.1 nM to about 1 nM, from about 1 nM to about 10 nM, from about 2 nM to about 20 nM, from about 3 nM to about 30 nM, from about 4 nM to about 40 nM, from about 5 nM to about 50 nM, from about 6 nM to about 60 nM, from about 7 nM to about 70 nM, from about 8 nM to about 80 nM, from about 9 nM to about 90 nM, from about 10 nM to about 100 nM, from about 11 nM to about 110 nM, from about 12 nM to about 120 nM, from about 13 nM to about 130 nM, from about 14 nM to about 140 nM, from about 15 nM to about 150 nM, from about 16 nM to about 160 nM, from about 17 nM to about 170 nM, from about 18 nM to about 180 nM, from about 19 nM to about 190 nM, from about 20 nM to about 200 nM, from about 21 nM to about 210 nM, from about 22 nM to about 220 nM, from about 23 nM to about 230 nM, from about 24 nM to about 240 nM, from about 25 nM to about 250 nM, from about 26 nM to about 260 nM, from about 27 nM to about 270 nM, from about 28 nM to about 280 nM, from about 29 nM to about 290 nM, from about 30 nM to about 300 nM, from about 31 nM to about 310 nM, from about 32 nM to about 320 nM, from about 33 nM to about 330 nM, from about 34 nM to about 340 nM, from about 35 nM to about 350 nM, from about 36 nM to about 360 nM, from about 37 nM to about 370 nM, from about 38 nM to about 380 nM, from about 39 nM to about 390 nM, from about 40 nM to about 400 nM, from about 41 nM to about 410 nM, from about 42 nM to about 420 nM, from about 43 nM to about 430 nM, from about 44 nM to about 440 nM, from about 45 nM to about 450 nM, from about 46 nM to about 460 nM, from about 47 nM to about 470 nM, from about 48 nM to about 480 nM, from about 49 nM to about 490 nM, from about 50 nM to about 500 nM, from about 51 nM to about 510 nM, from about 52 nM to about 520 nM, from about 53 nM to about 530 nM, from about 54 nM to about 540 nM, from about 55 nM to about 550 nM, from about 56 nM to about 560 nM, from about 57 nM to about 570 nM, from about 58 nM to about 580 nM, from about 59 nM to about 590 nM, from about 60 nM to about 600 nM, from about 61 nM to about 610 nM, from about 62 nM to about 620 nM, from about 63 nM to about 630 nM, from about 64 nM to about 640 nM, from about 65 nM to about 650 nM, from about 66 nM to about 660 nM, from about 67 nM to about 670 nM, from about 68 nM to about 680 nM, from about 69 nM to about 690 nM, from about 70 nM to about 700 nM, from about 71 nM to about 710 nM, from about 72 nM to about 720 nM, from about 73 nM to about 730 nM, from about 74 nM to about 740 nM, from about 75 nM to about 750 nM, from about 76 nM to about 760 nM, from about 77 nM to about 770 nM, from about 78 nM to about 780 nM, from about 79 nM to about 790 nM, from about 80 nM to about 800 nM, from about 81 nM to about 810 nM, from about 82 nM to about 820 nM, from about 83 nM to about 830 nM, from about 84 nM to about 840 nM, from about 85 nM to about 850 nM, from about 86 nM to about 860 nM, from about 87 nM to about 870 nM, from about 88 nM to about 880 nM, from about 89 nM to about 890 nM, from about 90 nM to about 900 nM, from about 91 nM to about 910 nM, from about 92 nM to about 920 nM, from about 93 nM to about 930 nM, from about 94 nM to about 940 nM, from about 95 nM to about 950 nM, from about 96 nM to about 960 nM, from about 97 nM to about 970 nM, from about 98 nM to about 980 nM, from about 99 nM to about 990 nM, from about 100 nM to about 1 μM, from about 0.1 μM to about 1 μM, from about 1 μM to about 10 μM, from about 2 μM to about 20 μM, from about 3 μM to about 30 μM, from about 4 μM to about 40 μM, from about 5 μM to about 50 μM, from about 6 μM to about 60 μM, from about 7 μM to about 70 μM, from about 8 μM to about 80 μM, from about 9 μM to about 90 μM, from about 10 μM to about 100 μM, from about 11 μM to about 110 μM, from about 12 μM to about 120 μM, from about 13 μM to about 130 μM, from about 14 μM to about 140 μM, from about 15 μM to about 150 μM, from about 16 μM to about 160 μM, from about 17 μM to about 170 μM, from about 18 μM to about 180 μM, from about 19 μM to about 190 μM, from about 20 μM to about 200 μM, from about 21 μM to about 210 μM, from about 22 μM to about 220 μM, from about 23 μM to about 230 μM, from about 24 μM to about 240 μM, from about 25 μM to about 250 μM, from about 26 μM to about 260 μM, from about 27 μM to about 270 μM, from about 28 μM to about 280 μM, from about 29 μM to about 290 μM, from about 30 μM to about 300 μM, from about 31 μM to about 310 μM, from about 32 μM to about 320 μM, from about 33 μM to about 330 μM, from about 34 μM to about 340 μM, from about 35 μM to about 350 μM, from about 36 μM to about 360 μM, from about 37 μM to about 370 μM, from about 38 μM to about 380 μM, from about 39 μM to about 390 μM, from about 40 μM to about 400 μM, from about 41 μM to about 410 μM, from about 42 μM to about 420 μM, from about 43 μM to about 430 μM, from about 44 μM to about 440 μM, from about 45 μM to about 450 μM, from about 46 μM to about 460 μM, from about 47 μM to about 470 μM, from about 48 μM to about 480 μM, from about 49 μM to about 490 μM, from about 50 μM to about 500 μM, from about 51 μM to about 510 μM, from about 52 μM to about 520 μM, from about 53 μM to about 530 μM, from about 54 μM to about 540 μM, from about 55 μM to about 550 μM, from about 56 μM to about 560 μM, from about 57 μM to about 570 μM, from about 58 μM to about 580 μM, from about 59 μM to about 590 μM, from about 60 μM to about 600 μM, from about 61 μM to about 610 μM, from about 62 μM to about 620 μM, from about 63 μM to about 630 μM, from about 64 μM to about 640 μM, from about 65 μM to about 650 μM, from about 66 μM to about 660 μM, from about 67 μM to about 670 μM, from about 68 μM to about 680 μM, from about 69 μM to about 690 μM, from about 70 μM to about 700 μM, from about 71 μM to about 710 μM, from about 72 μM to about 720 pM, from about 73 μM to about 730 μM, from about 74 μM to about 740 μM, from about 75 μM to about 750 μM, from about 76 μM to about 760 μM, from about 77 μM to about 770 μM, from about 78 μM to about 780 μM, from about 79 μM to about 790 μM, from about 80 μM to about 800 μM, from about 81 μM to about 810 μM, from about 82 μM to about 820 μM, from about 83 μM to about 830 μM, from about 84 μM to about 840 μM, from about 85 μM to about 850 pM, from about 86 μM to about 860 μM, from about 87 μM to about 870 μM, from about 88 μM to about 880 μM, from about 89 μM to about 890 μM, from about 90 μM to about 900 μM, from about 91 μM to about 910 μM, from about 92 μM to about 920 μM, from about 93 μM to about 930 μM, from about 94 μM to about 940 μM, from about 95 μM to about 950 μM, from about 96 μM to about 960 μM, from about 97 μM to about 970 μM, from about 98 μM to about 980 pM,

from about 99 μM to about 990 μM, from about 100 μM to about 1 mM, from about 0.1 mM to about 1 mM, from about 1 mM to about 10 mM, from about 2 mM to about 20 mM, from about 3 mM to about 30 mM, from about 4 mM to about 40 mM, from about 5 mM to about 50 mM, from about 6 mM to about 60 mM, from about 7 mM to about 70 mM, from about 8 mM to about 80 mM, from about 9 mM to about 90 mM, from about 10 mM to about 100 mM, from about 11 mM to about 110 mM, from about 12 mM to about 120 mM, from about 13 mM to about 130 mM, from about 14 mM to about 140 mM, from about 15 mM to about 150 mM, from about 16 mM to about 160 mM, from about 17 mM to about 170 mM, from about 18 mM to about 180 mM, from about 19 mM to about 190 mM, from about 20 mM to about 200 mM, from about 21 mM to about 210 mM, from about 22 mM to about 220 mM, from about 23 mM to about 230 mM, from about 24 mM to about 240 mM, from about 25 mM to about 250 mM, from about 26 mM to about 260 mM, from about 27 mM to about 270 mM, from about 28 mM to about 280 mM, from about 29 mM to about 290 mM, from about 30 mM to about 300 mM, from about 31 mM to about 310 mM, from about 32 mM to about 320 mM, from about 33 mM to about 330 mM, from about 34 mM to about 340 mM, from about 35 mM to about 350 mM, from about 36 mM to about 360 mM, from about 37 mM to about 370 mM, from about 38 mM to about 380 mM, from about 39 mM to about 390 mM, from about 40 mM to about 400 mM, from about 41 mM to about 410 mM, from about 42 mM to about 420 mM, from about 43 mM to about 430 mM, from about 44 mM to about 440 mM, from about 45 mM to about 450 mM, from about 46 mM to about 460 mM, from about 47 mM to about 470 mM, from about 48 mM to about 480 mM, from about 49 mM to about 490 mM, from about 50 mM to about 500 mM, from about 51 mM to about 510 mM, from about 52 mM to about 520 mM, from about 53 mM to about 530 mM, from about 54 mM to about 540 mM, from about 55 mM to about 550 mM, from about 56 mM to about 560 mM, from about 57 mM to about 570 mM, from about 58 mM to about 580 mM, from about 59 mM to about 590 mM, from about 60 mM to about 600 mM, from about 61 mM to about 610 mM, from about 62 mM to about 620 mM, from about 63 mM to about 630 mM, from about 64 mM to about 640 mM, from about 65 mM to about 650 mM, from about 66 mM to about 660 mM, from about 67 mM to about 670 mM, from about 68 mM to about 680 mM, from about 69 mM to about 690 mM, from about 70 mM to about 700 mM, from about 71 mM to about 710 mM, from about 72 mM to about 720 mM, from about 73 mM to about 730 mM, from about 74 mM to about 740 mM, from about 75 mM to about 750 mM, from about 76 mM to about 760 mM, from about 77 mM to about 770 mM, from about 78 mM to about 780 mM, from about 79 mM to about 790 mM, from about 80 mM to about 800 mM, from about 81 mM to about 810 mM, from about 82 mM to about 820 mM, from about 83 mM to about 830 mM, from about 84 mM to about 840 mM, from about 85 mM to about 850 mM, from about 86 mM to about 860 mM, from about 87 mM to about 870 mM, from about 88 mM to about 880 mM, from about 89 mM to about 890 mM, from about 90 mM to about 900 mM, from about 91 mM to about 910 mM, from about 92 mM to about 920 mM, from about 93 mM to about 930 mM, from about 94 mM to about 940 mM, from about 95 mM to about 950 mM, from about 96 mM to about 960 mM, from about 97 mM to about 970 mM, from about 98 mM to about 980 mM, from about 99 mM to about 990 mM, from about 100 mM to about 1 M, from about 1 M to about 10 M, from about 2 M to about 20 M, from about 3 M to about 30 M, from about

4 M to about 40 M, from about 5 M to about 50 M, from about 6 M to about 60 M, from about 7 M to about 70 M, from about 8 M to about 80 M, from about 9 M to about 90 M, from about 10 M to about 100 M, from about 11 M to about 110 M, from about 12 M to about 120 M, from about 13 M to about 130 M, from about 14 M to about 140 M, from about 15 M to about 150 M, from about 16 M to about 160 M, from about 17 M to about 170 M, from about 18 M to about 180 M, from about 19 M to about 190 M, from about 20 M to about 200 M, from about 21 M to about 210 M, from about 22 M to about 220 M, from about 23 M to about 230 M, from about 24 M to about 240 M, from about 25 M to about 250 M, from about 26 M to about 260 M, from about 27 M to about 270 M, from about 28 M to about 280 M, from about 29 M to about 290 M, from about 30 M to about 300 M, from about 31 M to about 310 M, from about 32 M to about 320 M, from about 33 M to about 330 M, from about 34 M to about 340 M, from about 35 M to about 350 M, from about 36 M to about 360 M, from about 37 M to about 370 M, from about 38 M to about 380 M, from about 39 M to about 390 M, from about 40 M to about 400 M, from about 41 M to about 410 M, from about 42 M to about 420 M, from about 43 M to about 430 M, from about 44 M to about 440 M, from about 45 M to about 450 M, from about 46 M to about 460 M, from about 47 M to about 470 M, from about 48 M to about 480 M, from about 49 M to about 490 M, or from about 50 M to about 500 M.

The processed silk of the SBP formulation may comprise silk fibroin, wherein the silk fibroin comprises a beta-sheet, an alpha-helix, a coiled coil, and/or a random coil. Silk fibroin may comprise a silk fibroin polymer, a silk fibroin monomer, and/or a silk fibroin fragment. The processed silk may comprise a silk fibroin fragment, wherein the silk fibroin fragment comprises a silk fibroin heavy chain fragment and/or a silk fibroin light chain fragment. The processed silk may comprise silk fibroin, wherein the silk fibroin comprises a plurality of silk fibroin fragments. The plurality of silk fibroin fragments may comprise a molecular weight of from about 1 kDa to about 350 kDa.

The SBP may comprise one or more formats selected from the group consisting of adhesives, capsules, cakes, coatings, cocoons, combs, cones, cylinders, discs, emulsions, fibers, films, foams, gels, grafts, hydrogels, implants, mats, membranes, microspheres, nanofibers, nanoparticles, nanospheres, nets, organogels, particles, patches, powders, rods, scaffolds, sheets, solids, solutions, sponges, sprays, spuns, suspensions, tablets, threads, tubes, vapors, and yarns. The format may be a solution. The format may be a hydrogel. The format may be a cake. The format may be a powder. The format may be a film.

The processed silk of the SBP formulation may comprise silk fibroin at a concentration between 0.5% and 5%. In one aspect, the silk fibroin is present at a concentration of 0.5%. In one aspect, the silk fibroin is present at a concentration of 1%. In one aspect, the silk fibroin is present at a concentration of 2.5%. In one aspect, the silk fibroin is present at a concentration of 3%. In one aspect, the silk fibroin is present at a concentration of 5%.

The SBP formulation is in a solution which may be, but is not limited to, phosphate buffer, borate buffer, and phosphate buffered saline. The solution may further comprise propylene glycol, sucrose and/or trehalose. Propylene glycol may be present in a concentration of 1%. Sucrose may be present in a concentration such as, but not limited to, 10 mM, 50 mM, 100 mM and 150 mM. Trehalose may be present in a concentration such as, but not limited to, 10 mM, 50 mM, 100 mM and 150 mM.

In some embodiments, the present disclosure provides a silk-based product (SBP) for ocular lubrication that includes processed silk and an ocular therapeutic agent. The processed silk may be silk fibroin. The SBP may include from about 0.0001% to about 35% (w/v) of silk fibroin. The silk fibroin may be prepared by degumming for a time of a 30-minute boil, a 60-minute boil, a 90-minute boil, a 120-minute boil, and a 480-minute boil. The SBP may be stressed. The SBP may be stressed by one or more methods which includes heating the SBP to 60° C. and autoclaving the SBP. The SBP may include one or more excipients. The one or more excipients may include one or more of sucrose, lactose, phosphate salts, sodium chloride, potassium phosphate monobasic, potassium phosphate dibasic, sodium phosphate dibasic, sodium phosphate monobasic, polysorbate 80, phosphate buffer, phosphate buffered saline, sodium hydroxide, sorbitol, mannitol, lactose USP, Starch 1500, microcrystalline cellulose, potassium chloride, sodium borate, boric acid, sodium borate decahydrate, magnesium chloride hexahydrate, calcium chloride dihydrate, sodium hydroxide, Avicel, dibasic calcium phosphate dehydrate, tartaric acid, citric acid, fumaric acid, succinic acid, malic acid, hydrochloric acid, polyvinylpyrrolidone, copolymers of vinylpyrrolidone and vinylacetate, hydroxypropylcellulose, hydroxyethylcellulose, hydroxypropylmethylcellulose, polyvinyl alcohol, polyethylene glycol, acacia, trehalose, and sodium carboxymethylcellulose. One or more of the excipients may include phosphate buffer. One or more of the excipients may include phosphate buffered saline. One or more of the excipients may include sucrose. The excipients may include boric acid, sodium borate decahydrate, sodium chloride, potassium chloride, magnesium chloride hexahydrate, calcium chloride dihydrate, sodium hydroxide, and hydrochloric acid. The SBP may include at least one excipient selected from one or more members of the group consisting of sorbitol, triethylamine, 2-pyrrolidone, alpha-cyclodextrin, benzyl alcohol, beta-cyclodextrin, dimethyl sulfoxide, dimethylacetamide (DMA), dimethylformamide, ethanol, gamma-cyclodextrin, glycerol, glycerol formal, hydroxypropyl beta-cyclodextrin, kolliphor 124, kolliphor 181, kolliphor 188, kolliphor 407, kolliphor EL (cremophor EL), cremophor RH 40, cremophor RH 60, dalpha-tocopherol, PEG 1000 succinate, polysorbate 20, polysorbate 80, solutol HS 15, sorbitan monooleate, poloxamer-407, poloxamer-188, Labrafil M-1944CS, Labrafil M-2125CS, Labrasol, Gellucire 44/14, Softigen 767, mono- and di-fatty acid esters of PEG 300, PEG 400, or PEG 1750, kolliphor RH60, N-methyl-2-pyrrolidone, castor oil, corn oil, cottonseed oil, olive oil, peanut oil, peppermint oil, safflower oil, sesame oil, soybean oil, hydrogenated vegetable oils, hydrogenated soybean oil, medium chain triglycerides of coconut oil, medium chain triglycerides of palm seed oil, beeswax, d-alpha-tocopherol, oleic acid, medium-chain mono-glycerides, medium-chain di-glycerides, alpha-cyclodextrin, beta-cyclodextrin, hydroxypropyl-beta-cyclodextrin, sulfo-butylether-beta-cyclodextrin, hydrogenated soy phosphatidylcholine, distearoylphosphatidylglycerol, L-alphadimyristoylphosphatidylcholine, L-alpha-dimyristoylphosphatidylglycerol, PEG 300, PEG 300 caprylic/capric glycerides (Softigen 767), PEG 300 linoleic glycerides (Labrafil M-2125CS), PEG 300 oleic glycerides (Labrafil M-1944CS), PEG 400, PEG 400 caprylic/capric glycerides (Labrasol), polyoxyl 40 stearate (PEG 1750 monosterate), polyoxyl 8 stearate (PEG 400 monosterate), polysorbate 20, polysorbate 80, polyvinyl pyrrolidone, propylene carbonate, propylene glycol, solutol HS15, sorbitan monooleate (Span 20), sulfobutylether-beta-cyclodextrin, transcutol, triacetin, 1-dodecylazacyclo-heptan-2-one, caprolactam, castor oil, cottonseed oil, ethyl acetate, medium chain triglycerides, methyl acetate, oleic acid, safflower oil, sesame oil, soybean oil, tetrahydrofuran, glycerin, and PEG 4 kDa. The SBP may be formulated, and the formulation may be as hydrogels and solutions. The formulation may be a hydrogel. The formulation may be a solution. The silk fibroin concentration in the solution may be below 1% (w/v). The SBP may be a solution, and the SBP may be stressed. The SBP may be a hydrogel, and the SBP may be stressed. The SBP may be a solution, and the solution may shear thin. The solutions may have the viscosity of a gel at a lower shear rate. The solutions may have the viscosity of a fluid at higher shear rates. The ocular therapeutic agent may be a nonsteroidal anti-inflammatory drug (NSAID) or protein. The SBP may be formulated for topical administration. The SBP may be formulated for ocular administration. The SBP may be biocompatible. The SBP may include any of the samples listed in any of the Tables 1-4.

In some embodiments, the present disclosure provides a method of preparing the SBP formulations comprising: (a) preparing the processed silk, wherein the processed silk comprises or is derived from one or more articles selected from the group consisting of raw silk, silk fiber, silk fibroin, and a silk fibroin fragment; and (b) preparing the SBP formulation using the processed silk. In some embodiments, the present disclosure provides a method of treating an ocular indication of a subject that includes administering to the subject any of the SBPs described herein. The ocular indication may be dry eye disease. The SBP may be administered to the eye. The SBP may be administered via topical administration. The topical administration of SBP may be as drops or sprays. The SBP may shear thin. The shear thinning of the SBPs may tune the residence time in the eye. The residence time of the SBP may be increased.

DETAILED DESCRIPTION OF THE DISCLOSURE

Embodiments of the present disclosure relate to silk-based products (SBPs), formulations and their methods of use. The term "silk" generally refers to a fibrous material formed by insects and some other species that includes tightly bonded protein filaments. Herein, the term "silk" is used in the broadest sense and may embrace any forms, variants, or derivatives of silk discussed.

Silk fibers from silkworm moth (*Bombyx mori*) cocoons include two main components, sericin (usually present in a range of 20-30%) and silk fibroin (usually present in a range of 70-80%). Structurally, silk fibroin forms the center of the silk fibers and sericin acts as the gum coating the fibers. Sericin is a gelatinous protein that holds silk fibers together with many of the characteristic properties of silk (see Qi et al. (2017) Int J Mol Sci 18:237 and Deptuch et al. (2017) Materials 10:1417, the contents of each of which are herein incorporated by reference in their entireties). Silk fibroin is an insoluble fibrous protein consisting of layers of antiparallel beta sheets. Its primary structure mainly consists of recurrent serine, alanine, and glycine repeating units. The isoelectric point of silk fibroin has been determined to be around 4.2. Silk fibroin monomers include a complex of heavy chain (around 350 kDa) and light chain (around 25 kDa) protein components. Typically, the chains are joined by a disulfide bond. With some forms, heavy chain and light chain segments are non-covalently bound to a glycoprotein, p25. Polymers of silk fibroin monomers may form through hydrogen bonding between monomers, typically increasing mechanical strength (see Qi et al. (2017) Int J Mol Sci 18:237). During silk processing, fragments of silk fibroin monomers may be produced, including, but not limited to, fragments of heavy and/or light chains. These fragments may retain the ability to form hydrogen bonds with silk fibroin monomers and fragments thereof. Herein, the term "silk fibroin" is used in its broadest sense and embraces silk fibroin polymers, silk fibroin monomers, silk fibroin heavy and light chains, silk fibroin fragments, and variants, derivatives, or mixtures thereof from any of the wild type, genetically modified, or synthetic sources of silk described herein.

The present disclosure includes methods of preparing processed silk and SBPs, different forms of SBP formulations, and a variety of applications for utilizing processed silk, SBPs, and SBP formulations alone or in combination with various compounds, compositions, and devices.

Silk-Based Products and Formulations

As used herein, the term "silk-based product" or "SBP" refers to any compound, mixture, or other entity that is made up of or that is combined with processed silk. "Processed silk," as used herein, refers to any forms of silk harvested, obtained, synthesized, formatted, manipulated, or altered through at least one human intervention. SBPs may include a variety of different formats suited for a variety of different applications. Examples of SBP formats include, but are not limited to, fibers, nanofibers, mats, films, foams, membranes, rods, tubes, gels, hydrogels, microspheres, nanospheres, solutions, patches, grafts, adhesives, capsules, cones, cylinders, cakes, discs, emulsions, nanoparticles, nets, organogels, particles, scaffolds, sheets, solids, sponges, sprays, spuns, suspensions, tablets, threads, vapors, yarns, and powders. Additional formats are described herein.

SBPs may find utility in variety of fields and for a variety of applications. Such utility may be due to the unique physical and chemical properties of silk. These physical and chemical properties include, but are not limited to, biocompatibility, biodegradability, bioresorbability, solubility, crystallinity, porosity, mechanical strength, thermal stability, hydrophobicity, and transparency. In some embodiments, SBPs may be used for one or more therapeutic applications, agricultural applications, and/or material science applications. Such SBPs may include processed silk, wherein the processed silk is or is derived from one or more of raw silk, silk fibers, silk fibroin, and silk fibroin fragments. Processed silk present in some SBPs may include one or more silk fibroin polymers, silk fibroin monomers, and/or silk fibroin fragments. In some embodiments, silk fibroin fragments include silk fibroin heavy chain fragments and/or silk fibroin light chain fragments. Some silk fibroin present in SBPs include a plurality of silk fibroin fragments. Each of the plurality of silk fibroin fragments may have a molecular weight of from about 1 kDa to about 400 kDa.

In some embodiments, SBPs may be formulations (e.g., SBP formulations). As used herein, the term "formulation" refers to a mixture of two or more components or the process of preparing such mixtures. In some embodiments, the formulations are low cost and eco-friendly. In some embodiments, the preparation or manufacturing of formulations is low cost and eco-friendly. In some embodiments, the preparation or manufacturing of formulations is scalable. In some embodiments, SBPs are prepared by extracting silk fibroin via degumming silk yarn. In some embodiments, the yarn is medical grade. In some embodiments the yarn may be silk sutures. The extracted silk fibroin may then be dissolved in a solvent (e.g. water, aqueous solution, organic solvent). The dissolved silk fibroin may then be dried (e.g., oven dried, air dried, or freeze-dried). In some embodiments, dried silk fibroin is formed into formats described herein. In some embodiments, that format is a solution. In some embodiments, that format is a powder. In some embodiments, that format is a hydrogel. In some embodiments, formulations include one or more excipients, carriers, additional components, and/or therapeutic agents to generate SBPs. In some embodiments, formulations of processed silks are prepared during the manufacture of SBPs.

Formulation components and/or component ratios may be modulated to affect one or more SBP properties, effects, and/or applications. Variations in the concentration of silk fibroin, choice of excipient, the concentration of excipient, the osmolarity of the formulation, and the method of formulation represent non-limiting examples of differences in formulation that may alter properties, effects, and applications of SBPs. In some embodiments, the formulation of SBPs may modulate their physical properties. Examples of physical properties include solubility, density, and thickness. In some embodiments, the formulation of SBPs may modulate their mechanical properties. Examples of mechanical properties that may be modulated include, but are not limited to, mechanical strength, tensile strength, elongation capabilities, elasticity, compressive strength, stiffness, shear strength, toughness, torsional stability, temperature stability, moisture stability, viscosity, and reeling rate.

In some embodiments, the formulations are prepared to be sterile. As used herein, the term "sterile" refers to something that is aseptic. In some embodiments, SBPs are prepared from sterile materials. In some embodiments, SBPs are prepared and then sterilized. In some embodiments, processed silk is degummed and then sterilized. In some embodiments, processed silk is sterilized and then degummed. Processed silk and/or SBPs may be sterilized via gamma radiation, autoclave (e.g., autoclave sterilization), filtration, electron beam, and any other method known to those skilled in the art.

A pharmaceutical composition (e.g., SBP formulation) in accordance with the present disclosure may be prepared, packaged, and/or sold in bulk, as a single unit dose, and/or as a plurality of single unit doses. As used herein, a "unit dose" refers to a discrete amount of the pharmaceutical composition comprising a predetermined amount of therapeutic agent or other compounds. The amount of therapeutic agent may generally be equal to the dosage of therapeutic agent administered to a subject and/or a convenient fraction of such dosage including, but not limited to, one-half or one-third of such a dosage.

Sources of Silk

SBP formulations may include processed silk obtained from one or more of a variety of sources. Processed silk may include raw silk. "Raw silk," as used herein, refers to silk that has been harvested, purified, isolated, or otherwise collected from silk producers. The term "silk producer," as used herein, refers to any organism capable of producing silk. Raw silk has been processed in large quantities for thousands of years, primarily from silkworms (*Bombyx mori*), which use silk to form their cocoon. Raw silk from silkworm cocoons includes silk fibroin and sericin that is secreted onto silk fibroin during cocoon formation. Raw silk may be harvested as a silk fiber. As used herein, the term "silk fiber" refers to any silk that is in the form of a filament or thread. Silk fibers may vary in length and width and may include, but are not limited to, yarns, strings, threads, and nanofibers. In some embodiments, raw silk may be obtained in the form of a yarn.

SBPs may include processed silk obtained from any one of a variety of sources. Processed silk may include raw silk. "Raw silk," as used herein, refers to silk that has been harvested, purified, isolated, or otherwise collected from silk producers. The term "silk producer," as used herein, refers to any organism capable of producing silk. Raw silk has been processed in large quantities for thousands of years, primarily from silkworms (*Bombyx mori*), which use silk to form their cocoon. Raw silk from silkworm cocoons includes silk fibroin and sericin that is secreted onto silk fibroin during cocoon formation. Raw silk may be harvested as a silk fiber. As used herein, the term "silk fiber" refers to any silk that is in the form of a filament or thread. Silk fibers may vary in length and width and may include, but are not limited to, yarns, strings, threads, and nanofibers. In some embodiments, raw silk may be obtained in the form of a yarn.

In some embodiments, processed silk includes silk obtained from a silk producer. Silk producers may be organisms found in nature (referred to herein as "wild type organisms") or they may be genetically modified organisms. There are many species of silk producers in nature capable of producing silk. Silk producers may be insect species, such as silkworms. Some silk producers include arachnid species. In some embodiments, silk producers include species of mollusk. Silk produced by different silk producing species may vary in physical and/or chemical properties. Such properties may include amino acid content, secondary structure (e.g. beta-sheet content), mechanical properties (e.g. elasticity), and others. In some embodiments, the present disclosure provides blends of processed silk from multiple silk producers or other sources (e.g., recombinant or synthetic silk). Such blends may have synergistic properties that are absent from processed silk obtained from single sources or from alternative blends. For example, Janani G et al. describe a silk scaffold fabricated by blending *Bombyx mori* silk fibroin with cell adhesion motif (RGD) rich *Antheraea assamensis* silk fibroin which displays enhanced liver-specific functions of cultured hepatocytes (Acta Biomater. 2018 February; 67:167-182, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, processed silk may be obtained from the silkworm species *Bombyx mori*. Other examples of silk producer species include, but are not limited to, *Bombyx mandarina, Bombyx sinesis, Anaphe moloneyi, Anaphe panda, Anaphe reticulate, Anaphe ambrizia, Anaphe carteri, Anaphe venata, Anapha infracta, Antheraea assamensis, Antheraea assama, Antheraea mylitta, Antheraea pernyi, Antheraea yamamai, Antheraea polyphemus, Antheraea oculea, Anisota senatoria, Apis mellifera, Araneus diadematus, Araneus cavaticus, Automeris io, Atticus atlas, Copaxa multifenestrata, Coscinocera hercules, Callosamia promethea, Eupackardia calleta, Eurprosthenops australis, Gonometa postica, Gonometa rufobrunnea, Hyalophora cecropia, Hyalophora euryalus, Hyalophora gloveri, Miranda auretia, Nephila madagascarensis, Nephila clavipes, Pachypasa otus, Pachypasa atus, Philosamia ricini, Pinna squamosa, Rothschildia hesperis, Rothschildia lebeau, Sarnia Cynthia*, and *Sarnia ricini.*

Genetically Modified Organisms

In some embodiments, silk producers are genetically modified organisms. As used herein, the term "genetically modified organism" or "GMO" refers to any living entity that includes or is derived from some form of genetic manipulation. The genetic manipulation may include any human intervention that alters the genetic material of an organism. In some embodiments, the genetic manipulation is limited to selecting organisms for reproduction based on genotype or phenotype. In some embodiments, genetic manipulation includes adding, deleting, and/or substituting one or more nucleotides of a wild type DNA sequence. The genetic manipulation may include the use of recombinant DNA technology. Recombinant DNA technology involves the exchange of DNA sections between DNA molecules. Some genetic manipulation involves the transfer of genetic material from another organism to the GMO. GMOs including such transferred genetic material are referred to as "transgenic organisms." Some genetic materials may be synthetically produced (see e.g., Price et al. (2014) J Control Release 190:304-313; and Deptuch et al. (2017) Materials 10:1417, the contents of each of which are herein incorporated by reference in their entirety). The genetic material may be transferred by way of a vector. The vector may be a plasmid. In some embodiments the vector is a virus. Some genetic manipulations involve the use of inhibitory RNA. In some embodiments, genetic manipulations are carried out using clustered regularly interspaced short palindromic repeats (CRISPR) technology.

GMO silk producers may be species generally known to produce silk (e.g., any of those described above). Some GMO silk producers are species not generally known to produce silk, but that are genetically manipulated to produce silk. Such organisms may be genetically modified to include at least one nucleic acid encoding at least one silk protein (e.g., silk fibroin, silk fibroin heavy chains, silk fibroin light chains, sericin, or fragments or derivates thereof). Some GMO silk producers are genetically manipulated to produce silk with one or more altered silk properties (e.g., strength, stability, texture, etc.). Some genetic manipulations affect characteristics of the GMO that are not directly related to silk production or silk properties (e.g., disease resistance, reproduction, etc.).

In some embodiments, GMO silk producers include genetically modified silkworms (e.g., *Bombyx mori*). Genetically modified silkworms may include genetic manipulations that result in silkworm production of silk fibroin strands that include degradable linkers. In some embodiments, GMOs are arachnids (e.g., spiders).

In some embodiments, GMO silk producers are cells. Such cells may be grown in culture and may include any type of cell capable of expressing protein. The cells may be prokaryotic or eukaryotic cells. In some embodiments, silk producer cells include bacterial cells, insect cells, yeast cells, mammalian cells, or plant cells. Cells may be transformed or transduced with nucleic acids encoding one or more silk proteins (e.g., silk fibroin, sericin, or fragments or derivates thereof).

In some embodiments, GMO silk producers may include, but are not limited to, *Bombyx mori*, soybeans, *Arabidopsis, Escherichia coli, Pichia pastoris*, potato, tobacco, baby hamster kidney cells, mice, and goats (e.g., see Tokareva et al. (2013) Microb Biotechnol 6(6):651-63 and Deptuch et al. (2017) Materials 10:1417). In some embodiments, silk may be produced in green plants (e.g., see International Publication Number WO2001090389, the contents of which are herein incorporated by reference in their entirety).

Recombinant Silk

As used herein, the term "recombinant silk" refers to any form of silk produced using recombinant DNA technology. Recombinant silk proteins may include amino acid sequences corresponding to silk proteins produced by wild type organisms; amino acid sequences not found in nature; and/or amino acid sequences found in nature, but not associated with silk. Some recombinant silk includes amino acid sequences with repetitive sequences that contribute to polymer formation and/or silk properties (e.g., see Deptuch et al. (2017) Materials 10:1417). Nucleic acid segments encoding repetitive sequences may be incorporated into plasmids after self-ligation into multimers (e.g., see Price et al. (2014) J Control Release 190:304-313).

In some embodiments, recombinant silk may be encoded by expression plasmids.

In some embodiments, recombinant silk may be expressed as a monomer. The monomers may be combined with other monomers or other silk proteins to obtain multimers (e.g., see Deptuch et al. (2017) Materials 10:1417). Some monomers may be combined according to methods known in the art. Such methods may include, but are not limited to, ligation, step-by-step ligation, recursive directional ligation, native chemical ligation, and concatemerization.

In some embodiments, recombinant silk may be expressed using the "PiggyBac" vector. The PiggyBac vector includes a spider transposon that is compatible with expression in silkworms.

In some embodiments, recombinant silk may be produced in a silk producing species. Examples of silk producing species include, but are not limited to, *Bombyx mori, Bombyx mandarina, Bombyx sinesis, Anaphe moloneyi, Anaphe panda, Anaphe reticulate, Anaphe ambrizia, Anaphe carteri, Anaphe venata, Anapha infracta, Antheraea assamensis, Antheraea paphis, Antheraea assama, Antheraea mylitta, Antheraea pernyi, Antheraea yamamai, Antheraea polyphemus, Antheraea oculea, Anisota senatoria, Apis mellifera, Araneus diadematus, Araneus cavaticus, Automeris io, Atticus atlas, Coscinocera hercules, Callosamia promethea, Copaxa multifenestrata, Eupackardia calleta, Eurprosthenops australis, Gonometa postica, Gonometa rufobrunnea, Hyalophora cecropia, Hyalophora euryalus, Hyalophora gloveri, Miranda auretia, Nephila madagascarensis, Nephila clavipes, Pachypasa otus, Pachypasa atus, Philosamia ricini, Pinna squamosa, Rothschildia hesperis, Rothschildia lebeau, Sarnia Cynthia*, and *Sarnia ricini.*

Synthetic Silk

In some embodiments, SBP formulations include synthetic silk. As used herein, the term "synthetic silk" refers to silk prepared without the aid of a silk producer. Synthetic silk may be prepared using standard methods of peptide synthesis. Such methods typically include the formation of amino acid polymers through successive rounds of polymerization. Amino acids used may be obtained through commercial sources and may include natural or non-natural amino acids. In some embodiments, synthetic silk polypeptides are prepared using solid-phase synthesis methods. The polypeptides may be linked to resin during synthesis. In some embodiments, polypeptide synthesis may be conducted using automated methods.

Synthetic silk may include polypeptides that are identical to wild type silk proteins (e.g., silk fibroin heavy chain, silk fibroin light chain, or sericin) or fragments thereof. In some embodiments, synthetic silk includes polypeptides that are variants of silk proteins or silk protein fragments. Some synthetic silk includes polypeptides with repeating units that correspond with or are variations of those found in silk fibroin heavy chain proteins.

Processed Silk

In some embodiments, SBP formulations include processed silk. Various processing methods may be used to obtain specific forms or formats of processed silk. Such processing methods may include, but are not limited to, acidifying, air drying, alkalinizing, annealing, autoclaving, chemical crosslinking, chemical modification, concentration, cross-linking, degumming, diluting, dissolving, dry spinning, drying, electrifying, electrospinning, electrospraying, emulsifying, encapsulating, extraction, extrusion, gelation, harvesting, heating, lyophilization, molding, oven drying, pH alteration, precipitation, purification, shearing, sonication, spinning, spray drying, spray freezing, spraying, vapor annealing, vortexing, and water annealing. The processing steps may be used to prepare final SBPs or they may be used to generate processed silk preparations. As used herein, the term "processed silk preparation" is generally used to refer to processed silk or compositions that include processed silk that are prepared for or obtained during or after one or more processing steps. Processed silk preparations may be SBPs, may be components of SBPs, SBP formulations or may be used as a starting or intermediate composition in the preparation of SBPs. Processed silk preparations may include other components related to processing (e.g., solvents, solutes, impurities, catalysts, enzymes, intermediates, etc.). Processed silk preparations that include silk fibroin may be referred to as silk fibroin preparations. In some embodiments, processed silk manufacturing is simple, scalable, and/or cost effective.

In some embodiments, processed silk may be prepared as, provided as, or included in a yarn, thread, string, a nanofiber, a textile, a cloth, a fabric, a particle, a nanoparticle, a microsphere, a nanosphere, a powder, a solution, a gel, a hydrogel, an organogel, a mat, a film, a foam, a membrane, a rod, a tube, a patch, a sponge, a scaffold, a capsule, an excipient, an implant, a solid, a coating, and/or a graft.

In some embodiments, processed silk may be stored frozen or dried to a stable soluble form. Processed silk may be frozen with cryoprotectants. Cryoprotectants may include, but are not limited to, phosphate buffer, sucrose, trehalose, histidine, and any other cryoprotectant known to one of skill in the art. In some embodiments, SBPs may be stored frozen or dried to a stable soluble form. In some embodiments, the SBPs may be solutions.

In some embodiments, preparation of processed silk and/or SBP formulations may be scaled up for manufacturing at a large scale. In some embodiments, production of processed silk and/or SBP formulations may be accomplished with automated machinery.

Any of the methods known in the art and/or described herein may be used to extract silk fibroin. The yield of silk fibroin from extraction may be, but is not limited to, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or greater than 99%.

Silk Properties

In some embodiments, processed silk may be selected based on or prepared to include features affecting one or more properties of the processed silk. Such properties may include, but are not limited to, stability, complex stability, composition stability, payload retention or release, payload release rate, wettability, mechanical strength, tensile strength, elongation capabilities, elasticity, compressive strength, stiffness, shear strength, toughness, hydrophobicity, torsional stability, temperature stability, moisture stability, strength, flexibility, solubility, crystallinity, viscosity, and porosity. Features affecting one or more processed silk properties may include silk secondary structure. Secondary structure refers to three-dimensional arrangements of polypeptide chains based on local interactions between neighboring residues. Common secondary structures include β-pleated sheets and α-helices. Silk secondary structure may enhance or attenuate solubility. In some embodiments, β-sheet secondary structure content may enhance processed silk crystallinity. "Crystallinity" refers to the degree of structure and arrangement between atoms or molecules in a compound, with increased structure yielding greater crystallinity. β-sheet structures may be antiparallel β-sheets. In some embodiments, processed silk includes polypeptides with random coil secondary structure. Some processed silk includes polypeptides with coiled coil secondary structure. In some embodiments, processed silk includes a combination of two or more forms of secondary structure. In some embodiments, processed silk may include polypeptides with multiple repeats. As used herein when referring to polypeptides, the term "multiple repeat" refers to an amino acid sequence that is duplicated two or more times in succession within a polypeptide. Silk fibroin heavy chains include multiple repeats that enable static interactions between parallel silk fibroin heavy chains. Multiple repeats may include repeats of the sequences GAGAGS (SEQ ID NO: 1) and/or GA. In some embodiments, the A of GA dipeptides may be replaced with S or Y. In some embodiments, multiple repeats may include any of those presented in Qi et al. (2017) Int J Mol Sci 18:237, the contents of which are herein incorporated by reference in their entirety. Multiple repeats may enable formation of stable, crystalline regions of antiparallel β-sheets.

Processed silk may include silk fibroin forms described by Qi et al. (2017) Int J Mol Sci 18:237 and Cao et al. (2009) Int J Mol Sci 10:1514-1524, the contents of each of which are herein incorporated by reference in their entirety. These silk fibroin forms are referred to as silk I, silk II, and silk III. Silk I and silk II forms are commonly found in nature. Silk I predominantly includes random coil secondary structures. Silk II predominantly includes β-sheet secondary structure. Silk III predominantly includes an unstable structure.

Processed silk may be treated to modulate β-sheet content and/or crystallinity. In some embodiments these treatments are used to reduce the solubility and/or hydrophobicity of the silk fibroin or silk fibroin composition. Treatments may include, but are not limited to, alteration of the pH, sonication of the silk fibroin, incorporation of an excipient, increasing or decreasing the temperature, treatment with acid, treatment with formic acid, treatment with glycerol, treatment with an alcohol, treatment with methanol, treatment with ethanol, treatment with isopropanol, and/or treatment with a mixture of alcohol and water. In some embodiments, treatments result in transition between forms of silk I, II, or III. Such methods may include any of those described in Cao et al. (2009) Int J Mol Sci 10:1514-1524).

Strength and Stability

Processed silk strength and stability are important factors for many applications. In some embodiments, processed silk may be selected based on or prepared to maximize mechanical strength, tensile strength, elongation capabilities, elasticity, flexibility, compressive strength, stiffness, shear strength, toughness, torsional stability, biological stability, resistance to degradation, and/or moisture stability. In some embodiments, processed silk had a non-acidic microenvironment. In some embodiments, the non-acidic microenvironment enhances the stability of processed silk and or SBPs. In some embodiments, the non-acidic microenvironment enhances the stability of therapeutic agents formulated with the processed silk and/or SBP.

Biocompatibility

In some embodiments, processed silk may be selected based on or prepared to maximize biocompatibility. As used herein, the term "biocompatibility" refers to the degree with which a substance avoids provoking a negative biological response in an organism exposed to the substance. The negative biological response may include an inflammatory response and/or local sensitization, hemorrhage, and/or other complications known to those skilled in the art. In some embodiments, administration of processed silk or an SBP does not induce an inflammatory response, local sensitization, hemorrhage, and/or other complications known to those skilled in the art. In some embodiments, contact with processed silk or an SBP does not induce an inflammatory response, local sensitization, hemorrhage, and/or other complications known to those skilled in the art. In some embodiments, no inflammatory response, local sensitization, hemorrhage, and/or other complications occur after up to 7 months of contact with processed silk or an SBP. In some embodiments, processed silk biocompatibility is enhanced through preparations that produce only non-toxic byproducts during degradation. In some embodiments, exposure to an SBP generates a tolerable biological response, within an acceptable threshold known to those skilled in the art. In some embodiments, processed silk is biocompatible in humans and human whole blood. In some embodiments, processed silk is biocompatible in animals. In some embodiments, processed silk produces no adverse reactions, no acute inflammation, and no immunogenicity in vivo. In some embodiments, the processed silk or SBP is safe to use in vivo. In some embodiments, processed silk or SBPs are biocompatible and/or tolerable in vitro. In some embodiments, processed silk or SBPs are biocompatible and/or tolerable in vivo. In some embodiments, no inflammatory response, local sensitization, hemorrhage, and/or other complications occur after up to 1 day, up to 3 days, up to 1 week, up to 1 month, up to 2 months, up to 3 months, up to 4 months, up to 5 months, up to 6 months, up to 7 months, up to 8 months, up to 9 months, up to 10 months, up to 11 months, or up to 1 year of contact with processed silk or an SBP.

Biodegradability

In some embodiments, processed silk may be selected based on or prepared to maximize biodegradability. As used herein, the term "biodegradability" refers to the degree with which a substance avoids provoking a negative response to an environment exposed to the substance as it deteriorates. The negative environmental response may include a response to toxic byproducts generated as a substance deteriorates. In some embodiments, processed silk biodegradability is enhanced through preparations that produce only non-toxic byproducts during degradation. In some embodiments, processed silk biodegradability is enhanced through preparations that produce only inert amino acid byproducts. In some embodiments, the SBP and/or SBP by products are considered naturally derived and environmentally and/or eco-friendly.

Anti-Evaporative Properties

In some embodiments, processed silk may be selected based on or prepared to reduce the evaporation of a solution. In some embodiments, processed silk may reduce the evaporation of a solution. In some embodiments, an SBP may demonstrate anti-evaporative properties by creating a water and/or water vapor barrier, as taught in Marelli et al. (2008) Sci Rep 6:25263, the contents of which are herein incorporated by reference in their entirety. In some embodiments, processed silk may extend the lifetime or residence time of an SBP product due to its ability to prevent evaporation. In some embodiments, processed silk may increase the amount of time required for a solution to evaporate. In some embodiments, processed silk may be selected based on or prepared to reduce the evaporation of a solution. In some embodiments, processed silk may reduce the evaporation of a solution. In some embodiments, processed silk may extend the lifetime or residence time of an SBP product due to its ability to prevent evaporation. In some embodiments, processed silk may increase the amount of time required for a solution to evaporate.

Demulcent

In some embodiments, processed silk and/or SBPs may act as demulcents. As used herein, the term "demulcent" refers to a substance that relieves irritation or inflammation of the mucous membranes by forming a protective film. This film may mimic a mucous membrane. Demulcents may also provide lubrication. Demulcents may include non-polymeric demulcents and polymer demulcents. Added demulcents may modulate the viscosity of an SBP or product containing an SBP.

Surfactant

In some embodiments, processed silk and/or SBPs may act as a surfactant. As used herein, the term "surfactant" refers to a substance that reduces the surface tension between two materials. In some embodiments, the SBP is a solution. In some embodiments, the SBP is a hydrogel. In some embodiments, the SBP has a surface tension similar to that of water. In some embodiments, the SBP has a surface tension similar to that of human tears. Human tears have been reported to have a surface tension of 43.6 mN/m, as described in Sweeney et al. (2013) Experimental Eye Research 117: 28-38, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the surface tension of the SBP may be controlled by the concentration of processed silk. In some embodiments, the surface tension is about 30-60 mN/m. In some embodiments, the surface tension of an SBP is about 35-55 mN/m. In some embodiments, the surface tension of an SBP is about 40-50 mN/m.

Antimicrobial and Bacteriostatic Properties

In some embodiments, processed silk may be based on or prepared to maximize antimicrobial properties. As used herein, the term "antimicrobial" properties refer to the ability of processed silk or SBPs to inhibit, deter the growth of microorganisms and/or kill the microorganisms. Microorganisms may include bacteria, fungi, protozoans, and viruses. In some embodiments, the antimicrobial properties may include but are not limited to antibacterial, antifungal, antiseptic, and/or disinfectant properties. In some embodiments, antimicrobial properties of silk may be modulated during one or more processing steps or during fabrication of a SBP. In some embodiments, antimicrobial properties may be modulated by the varying the source of silk utilized for the preparation of SBPs (Mirghani, M et al. 2012, Investigation of the spider web of antibacterial activity, (MICOTriBE) 2012; the contents of which are incorporated by reference in their entirety). In some embodiments, processed silk and SBPs described herein may possess antimicrobial properties against gram positive bacteria. In some embodiments, processed silk and SBPs described herein may possess antimicrobial properties against gram negative bacteria.

In some embodiments, processed silk may be based on or prepared to maximize bacteriostatic properties. As used herein, the term "bacteriostatic" refers to a substance that prevents bacterial reproduction and may or may not kill said bacteria. Bacteriostatic agents prevent the growth of bacteria. In some embodiments, bacteriostatic properties of silk may be modulated during one or more processing steps or during fabrication of a SBP. In some embodiments, bacteriostatic properties may be modulated by the varying the source of silk utilized for the preparation of SBPs. In some embodiments, processed silk and SBPs described herein may possess bacteriostatic properties against gram positive bacteria. In some embodiments, processed silk and SBPs described herein may possess bacteriostatic properties against gram negative bacteria.

Anti-Inflammatory Properties

In some embodiments, processed silk or SBPs may have or be prepared to maximize anti-inflammatory properties. It has been reported that silk fibroin peptide derived from silkworm *Bombyx mori* exhibited anti-inflammatory activity in a mice model of inflammation (Kim et al., (2011) BMB Rep 44(12):787-92; the contents of which are incorporated by reference in their entirety). In some embodiments, processed silk or SBPs may be administered to a subject alone or in combination with other therapeutic agents to elicit anti-inflammatory effects. It is contemplated that processed silk or SBPs alone or combination with other therapeutic agents may be used to treat various inflammatory diseases. For example, processed silk or SBPs may reduce signs and symptoms of inflammation, such as but not limited to, swelling, redness, tenderness, rashes, fever, and pain.

Harvesting Silk

In some embodiments, processed silk is harvested from silk producer cocoons. Cocoons may be prepared by cultivating silkworm moths and allowing them to pupate. Once fully formed, cocoons may be treated to soften sericin and allow for unwinding of the cocoon to form raw silk fiber. The treatment may include treatment with hot air, steam, and/or boiling water. Raw silk fibers may be produced by unwinding multiple cocoons simultaneously. The resulting raw silk fibers include both silk fibroin and sericin. Subsequent processing may be carried out to remove sericin from the raw silk fibers or from later forms of processed silk or SBPs. In some embodiments, raw silk may be harvested directly from the silk glands of silk producers. Raw silk may be harvested from wild type or GMO silk producers.

Extraction of Sericin/Degumming

In some embodiments, sericin may be removed from processed silk, a process referred to herein as "degumming." The processed silk may include raw silk, which includes sericin secreted during cocoon formation. Methods of degumming may include heating (e.g., boiling) in a degumming solution. As used herein, the term "degumming solution" refers to a composition used for sericin removal that includes at least one degumming agent. As used herein, a "degumming agent" refers to any substance that may be used for sericin removal. Heating in degumming solution may reduce or eliminate sericin from processed silk. In some embodiments, heating in degumming solution includes boiling. Heating in degumming solution may be followed by rinsing to enhance removal of sericin that remains after heating. In some embodiments, raw silk is degummed before further processing or utilization in SBPs. In other embodiments, raw silk is further processed or otherwise incorporated into a SBP prior to degumming. Such methods may include any of those presented in European Patent No. EP2904134 or United States Patent Publication No. US2017031287, the contents of each of which are herein incorporated by reference in their entirety.

Degumming agents and/or degumming solutions may include, but are not limited to water, alcohols, soaps, acids, alkaline solutions, and enzyme solutions. In some embodiments, degumming solutions may include salt-containing alkaline solutions. Such solutions may include sodium carbonate. Sodium carbonate concentration may be from about 0.01 M to about 0.3 M. In some embodiments, sodium carbonate concentration may be from about 0.01 M to about 0.05 M, about 0.05 M to about 0.1 M, from about 0.1 M to about 0.2 M, or from about 0.2 M to about 0.3 M. In some embodiments, sodium carbonate concentration may be 0.02 M. In some embodiments, degumming solutions may include from about 0.01% to about 1% (w/v) sodium carbonate. In some embodiments, degumming solutions may include from about 0.01% to about 10% (w/v) sodium carbonate. In some embodiments, degumming solutions may include from about 0.01% (w/v) to about 1% (w/v), from about 1% (w/v) to about 2% (w/v), from about 2% (w/v) to about 3% (w/v), from about 3% (w/v) to about 4% (w/v), from about 4% (w/v) to about 5% (w/v), or from about 5% (w/v) to about 10% (w/v) sodium carbonate. In some embodiments, degumming solutions may include sodium dodecyl sulfate (SDS). Such degumming solutions may include any those described in Zhang et al. (2012) J Translational Med 10:117, the contents of which are herein incorporated by reference in their entirety. In some embodiments, degumming solutions include boric acid. Such solutions may include any of those taught in European Patent No. EP2904134, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the degumming solution may have a pH of from about 0 to about 5, from about 2 to about 7, from about 4 to about 9, from about 5 to about 11, from about 6 to about 12, from about 6.5 to about 8.5, from about 7 to about 10, from about 8 to about 12, and from about 10 to about 14. In some embodiments, processed silk is present in degumming solutions at concentrations of from about 0.1% to about 2%, from about 0.5% to about 3%, from about 1% to about 4%, or from about 2% to about 5% (w/v). In some embodiments, processed silk is present in degumming solutions at concentrations of greater than 5% (w/v).

Degumming may be carried out by "boiling" in degumming solutions at or near atmospheric boiling temperatures. As used herein, "boiling" does not necessarily mean at or above 100° C. Boiling may be properly used to describe heating the solution at a temperature that is less than or greater than 100° C. Some boiling temperatures may be from about 60° C. to about 115° C. In some embodiments, boiling is carried out at 100° C. In some embodiments, boiling is carried out at about 60° C., about 65° C., about 70° C., about 75° C., about 80° C., about 85° C., about 86° C., about 87° C., about 88° C., about 89° C., about 90° C., about 91° C., about 92° C., about 93° C., about 94° C., about 95° C., about 96° C., about 97° C., about 98° C., about 99° C., about 100° C., about 101° C., about 102° C., about 103° C., about 104° C., about 105° C., about 106° C., about 107° C., about 108° C., about 109° C., about 110° C., about 111° C., about 112° C., about 113° C., about 114° C., about 115° C. or greater than 115° C.

In some embodiments, degumming includes heating in degumming solution for a period of from about 10 seconds to about 45 seconds, from about 30 seconds to about 90 seconds, from about 1 min to about 5 min, from about 2 min to about 10 min, from about 5 min to about 15 min, from about 10 min to about 25 min, from about 20 min to about 35 min, from about 30 min to about 50 min, from about 45 min to about 75 min, from about 60 min to about 95 min, from about 90 min to about 125 min, from about 120 min to about 175 min, from about 150 min to about 200 min, from about 180 min to about 250 min, from about 210 min to about 350 min, from about 240 min to about 400 min, from about 270 min to about 450 min, from about 300 min to about 500 min, from about 330 min to about 550 min, from about 360 min to about 600 min, from about 390 min to about 700 min, from about 420 min to about 800 min, from about 450 min to about 900 min, from about 480 min to about 1000 min, from about 510 min to about 1100 min, from about 540 min to about 1200 min, from about 570 min to about 1300 min, from about 600 min to about 1400 min, from about 630 min to about 1500 min, from about 660 min to about 1600 min, from about 690 min to about 1700 min, from about 720 min to about 1800 min, from about 1440 min to about 1900 min, from about 1480 min to about 2000 min, or longer than 2000 min.

In some embodiments, processed silk preparations are characterized by the number of minutes boiling was carried out for preparation, a value referred to herein as "minute boil" or "mb." The minute boil value of a preparation may be associated with known or presumed characteristics of similar preparations with the same minute boil value. Such characteristics may include concentration and/or molecular weight of preparation compounds, proteins, or protein fragments altered during boiling. In some embodiments, processed silk preparations (e.g., silk fibroin preparations) have an mb value of from about 1 mb to about 5 mb, from about 2 mb to about 10 mb, from about 5 mb to about 15 mb, from about 10 mb to about 25 mb, from about 20 mb to about 35 mb, from about 30 mb to about 50 mb, from about 45 mb to about 75 mb, from about 60 mb to about 95 mb, from about 90 mb to about 125 mb, from about 120 mb to about 175 mb, from about 150 mb to about 200 mb, from about 180 mb to about 250 mb, from about 210 mb to about 350 mb, from about 240 mb to about 400 mb, from about 270 mb to about 450 mb, from about 300 mb to about 480 mb, or greater than 480 mb.

In some embodiments, degumming may be carried out by treatment with high temperatures and/or pressures. Such methods may include any of those presented International Patent Application Publication No. WO2017200659, the contents of which are herein incorporated by reference in their entirety.

Silk Fibroin Boiling Time

SBP formulations may comprise processed silk with varying molecular weights. SBP formulations may include low molecular weight silk fibroin. As used herein, the term "low molecular weight silk fibroin" refers to silk fibroin with a molecular weight below 200 kDa. Some SBP formulations may include high molecular weight silk fibroin. As used herein, the term "high molecular weight silk fibroin" refers to silk fibroin with a molecular weight equal to or greater than 200 kDa. In some embodiments, the silk fibroin molecular weight is defined by the degumming boiling time. In some embodiments, silk fibroin with a 480-minute boil, or "mb" may produce be a low molecular weight silk fibroin when compared to a silk fibroin produced with a 120-minute boil, or "mb". In some aspects, the 120 mb silk fibroin is considered to be high molecular weight silk fibroin in comparison to the 480 mb silk fibroin. In some embodiments, a longer boiling time is considered to be lower molecular weight silk fibroin. In some embodiments, a shorter boiling time is considered to be a higher molecular weight silk fibroin. In some embodiments, the boiling time is about 15 minutes, about 30 minutes, about 60 minutes, about 90 minutes, about 120 minutes, or about 480 minutes. In some embodiments, an SBP is prepared with processed silk with a single boiling time. In some embodiments, an SBP contains a blend of processed silk with different boiling times.

In one embodiment, the SBP formulation includes 30 mb silk fibroin.

In one embodiment, the SBP formulation includes 60 mb silk fibroin.

In one embodiment, the SBP formulation includes 90 mb silk fibroin.

In one embodiment, the SBP formulation includes 120 mb silk fibroin.

In one embodiment, the SBP formulation includes 480 mb silk fibroin.

Processed Silk Preparation Characterization

Preparations of processed silk sometimes include mixtures of silk fibroin polymers, silk fibroin monomers, silk fibroin heavy chains, silk fibroin light chains, sericin, and/or fragments of any of the foregoing. Where the exact contents and ratios of components in such processed silk preparations are unknown, the preparations may be characterized by one or more properties of the preparation or by conditions or methods used to obtain the preparations.

Solubility and Concentration

Processed silk preparations may include solutions that include processed silk (also referred to herein as "processed silk solutions"). Processed silk solutions may be characterized by processed silk concentration. For example, processed silk may be dissolved in a solvent after degumming to generate a processed silk solution of silk fibroin for subsequent use. Solvent used to dissolve processed silk may be a buffer. In some embodiments, solvent used is an organic solvent. Organic solvents may include, but are not limited to hexafluoroisopropanol (HFIP), methanol, isopropanol, ethanol, or combinations thereof. In some embodiments, solvents include a mixture of an organic solvent and water or an aqueous solution. Solvents may include water or aqueous solutions. Aqueous solutions may include aqueous salt solutions that include one or more salts. Such salts may include but are not limited to lithium bromide (LiBr), lithium thiocyanate, Ajisawa's reagent, a chaotropic agent, calcium nitrate, or other salts capable of solubilizing silk, including any of those disclosed in U.S. Pat. No. 9,623,147 (the content of which is herein incorporated by reference in its entirety). In some embodiments, solvents used in processed silk solutions include high salt solutions. In some embodiments, the solution comprises 5 to 13 M LiBr. The concentration of LiBr may be 9.3 M. In some embodiments, solvents used in processed silk solutions may include Ajisawa's reagent, as described in Zheng et al. (2016) Journal of Biomaterials Applications 31:450-463, the content of which is herein incorporated by reference in its entirety. Ajisawa's reagent comprises a mixture of calcium chloride, ethanol, and water in a molar ratio of 1:2:8 respectively.

In some embodiments, processed silk may be present in processed silk solutions at a concentration of from about 0.01% (w/v) to about 1% (w/v), from about 0.05% (w/v) to about 2% (w/v), from about 1% (w/v) to about 5% (w/v), from about 2% (w/v) to about 10% (w/v), from about 4% (w/v) to about 16% (w/v), from about 5% (w/v) to about 20% (w/v), from about 8% (w/v) to about 24% (w/v), from about 10% (w/v) to about 30% (w/v), from about 12% (w/v) to about 32% (w/v), from about 14% (w/v) to about 34% (w/v), from about 16% (w/v) to about 36% (w/v), from about 18% (w/v) to about 38% (w/v), from about 20% (w/v) to about 40% (w/v), from about 22% (w/v) to about 42% (w/v), from about 24% (w/v) to about 44% (w/v), from about 26% (w/v) to about 46% (w/v), from about 28% (w/v) to about 48% (w/v), from about 30% (w/v) to about 50% (w/v), from about 35% (w/v) to about 55% (w/v), from about 40% (w/v) to about 60% (w/v), from about 45% (w/v) to about 65% (w/v), from about 50% (w/v) to about 70% (w/v), from about 55% (w/v) to about 75% (w/v), from about 60% (w/v) to about 80% (w/v), from about 65% (w/v) to about 85% (w/v), from about 70% (w/v) to about 90% (w/v), from about 75%

(w/v) to about 95% (w/v), from about 80% (w/v) to about 96% (w/v), from about 85% (w/v) to about 97% (w/v), from about 90% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 96% (w/v) to about 99.2% (w/v), from about 97% (w/v) to about 99.5% (w/v), from about 98% (w/v) to about 99.8% (w/v), from about 99% (w/v) to about 99.9% (w/v), or greater than 99.9% (w/v). In some embodiments, the processed silk is silk fibroin.

Processed silk solutions may be characterized by the length of time and/or temperature needed for processed silk to dissolve. The length of time used to dissolve processed silk in solvent is referred to herein as "dissolution time." Dissolution times for dissolution of processed silk in various solvents may be from about 1 min to about 5 min, from about 2 min to about 10 min, from about 5 min to about 15 min, from about 10 min to about 25 min, from about 20 min to about 35 min, from about 30 min to about 50 min, from about 45 min to about 75 min, from about 60 min to about 95 min, from about 90 min to about 125 min, from about 120 min to about 175 min, from about 150 min to about 200 min, from about 180 min to about 250 min, from about 210 min to about 350 min, from about 240 min to about 360 min, from about 270 min to about 420 min, from about 300 min to about 480 min, or longer than 480 minutes.

The temperature used to dissolve processed silk in solvent is referred to herein as "dissolution temperature." Dissolution temperatures used for dissolution of processed silk in solvent may include room temperature. In some embodiments, dissolution temperature may be from about 0° C. to about 10° C., from about 4° C. to about 25° C., from about 20° C. to about 35° C., from about 30° C. to about 45° C., from about 40° C. to about 55° C., from about 50° C. to about 65° C., from about 60° C. to about 75° C., from about 70° C. to about 85° C., from about 80° C. to about 95° C., from about 90° C. to about 105° C., from about 100° C. to about 115° C., from about 110° C. to about 125° C., from about 120° C. to about 135° C., from about 130° C. to about 145° C., from about 140° C. to about 155° C., from about 150° C. to about 165° C., from about 160° C. to about 175° C., from about 170° C. to about 185° C., from about 180° C. to about 200° C., or greater than 200° C. In some embodiments, the processed silk is silk fibroin. Dissolution of some processed silk solutions may use a dissolution temperature of 60° C. Dissolution of some processed silk solutions may use a dissolution temperature of 80° C., as described in Zheng et al. (2016) Journal of Biomaterials Applications 31:450-463. In some embodiments, dissolution includes boiling. In some embodiments, dissolution may be carried out by autoclaving. In some embodiments, silk fibroin solutions may be prepared according to any of the methods described in International Patent Application Publication No. WO2017200659 or Abdel-Naby (2017) PLoS One 12(11):e0188154), the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, one or more of sucrose, phosphate buffer, tris buffer, trehalose, mannitol, citrate buffer, ascorbate, histidine, and/or a cryoprotective agent is added to processed silk solutions.

Chaotropic Agents

In some embodiments, processed silk may be dissolved with the aid of a chaotropic agent. As used herein, a "chaotropic agent" refers to a substance that disrupts hydrogen bonding networks in aqueous solutions to facilitate dissolution of a solute. Chaotropic agents typically modify the impact of hydrophobicity on dissolution. Chaotropic agents may be organic compounds. Such compounds may include, but are not limited to, sodium dodecyl sulfate, ethanol, methanol, phenol, 2-propanol, thiourea, urea, n-butanol, and any other chemicals capable of solubilizing silk. In some embodiments, the chaotropic agent is a salt, including, but not limited to, zinc chloride, calcium nitrate, lithium perchlorate, lithium acetate, sodium thiocyanate, calcium thiocyanate, magnesium thiocyanate, calcium chloride, magnesium chloride, guanidinium chloride, lithium bromide, lithium thiocyanate, copper salts, and other salts capable of solubilizing silk. Such salts typically create high ionic strength in the aqueous solutions which destabilizes the beta-sheet interactions in silk fibroin. In some embodiments, a combination of chaotropic agents is used to facilitate the dissolution of silk fibroin. In some embodiments, a chaotropic agent is used to dissolve raw silk during processing.

Molecular Weight

In some embodiments, processed silk preparations may be characterized by the molecular weight of proteins present in the preparations. Different molecular weights may be present as a result of different levels of silk fibroin dissociation and/or fragmentation during degumming or other processing. When referring to silk fibroin molecular weight herein, it should be understood that the molecular weight may be associated with silk fibroin polymers, silk fibroin monomers, silk fibroin heavy and/or light chains, silk fibroin fragments, or variants, derivates, or mixtures thereof. Accordingly, silk fibroin molecular weight values may vary depending on the nature of the silk fibroin or silk fibroin preparation. In some embodiments, processed silk preparations are characterized by average molecular weight of silk fibroin fragments present in the preparation; by a range of silk fibroin fragment molecular weights; by a threshold of silk fibroin fragment molecular weights; or by combinations of averages, ranges, and thresholds.

In some embodiments, processed silk preparation may include silk fibroin, fibroin fragments, or a plurality of fibroin fragments with a molecular weight of, average molecular weight of, upper molecular weight threshold of, lower molecular weight threshold of, or range of molecular weights with an upper or lower range value of from about 1 kDa to about 4 kDa, from about 2 kDa to about 5 kDa, from about 3.5 kDa to about 10 kDa, from about 5 kDa to about 20 kDa, from about 10 kDa to about 35 kDa, from about 15 kDa to about 40 kDa, from about 20 kDa to about 45 kDa, from about 25 kDa to about 50 kDa, from about 30 kDa to about 55 kDa, from about 35 kDa to about 60 kDa, from about 40 kDa to about 65 kDa, from about 45 kDa to about 70 kDa, from about 50 kDa to about 75 kDa, from about 55 kDa to about 80 kDa, from about 60 kDa to about 85 kDa, from about 65 kDa to about 90 kDa, from about 70 kDa to about 95 kDa, from about 75 kDa to about 100 kDa, from about 80 kDa to about 105 kDa, from about 85 kDa to about 110 kDa, from about 90 kDa to about 115 kDa, from about 95 kDa to about 120 kDa, from about 100 kDa to about 125 kDa, from about 105 kDa to about 130 kDa, from about 110 kDa to about 135 kDa, from about 115 kDa to about 140 kDa, from about 120 kDa to about 145 kDa, from about 125 kDa to about 150 kDa, from about 130 kDa to about 155 kDa, from about 135 kDa to about 160 kDa, from about 140 kDa to about 165 kDa, from about 145 kDa to about 170 kDa, from about 150 kDa to about 175 kDa, from about 160 kDa to about 200 kDa, from about 170 kDa to about 210 kDa, from about 180 kDa to about 220 kDa, from about 190 kDa to about 230 kDa, from about 200 kDa to about 240 kDa, from about 210 kDa to about 250 kDa, from about 220 kDa to about 260 kDa, from about 230 kDa to about 270 kDa, from about 240 kDa to about 280 kDa, from about 250 kDa to about 290 kDa, from about 260 kDa to about 300 kDa, from about 270 kDa to about 310 kDa, from about 280 kDa to about 320 kDa, from about 290 kDa to about 330 kDa, from about 300 kDa to about 340 kDa, from about 310 kDa to about 350 kDa, from about 320 kDa to about 360 kDa, from about 330 kDa to about 370 kDa, from about 340 kDa to about 380 kDa, from about 350 kDa to about 390 kDa, from about 360 kDa to about 400 kDa, from about 370 kDa to about 410 kDa, from about 380 kDa to about 420 kDa, from about 390 kDa to about 430 kDa, from about 400 kDa to about 440 kDa, from about 410 kDa to about 450 kDa, from about 420 kDa to about 460 kDa, from about 430 kDa to about 470 kDa, from about 440 kDa to about 480 kDa, from about 450 kDa to about 490 kDa, from about 460 kDa to about 500 kDa, or greater than 500 kDa.

In one embodiment, the silk preparation may include silk fibroin with a molecular weight of or an average molecular weight of 5-60 kDa.

In one embodiment, the silk preparation may include silk fibroin with a molecular weight of or an average molecular weight of 30-60 kDa. In one aspect, silk fibroin in this range maybe referred to as low molecular weight.

In one embodiment, the silk preparation may include silk fibroin with a molecular weight of or an average molecular weight of 100-300 kDa. In one aspect, silk fibroin in this range maybe referred to as high molecular weight.

In one embodiment, the silk preparation may include silk fibroin with a molecular weight of or an average molecular weight of 361 kDa.

Processed silk preparations may be analyzed, for example, by polyacrylamide gel electrophoresis (PAGE) alongside molecular weight standards to determine predominate molecular weights of proteins and/or polymers present. Additional methods for determining the molecular weight range or average molecular weight for a processed silk preparation may include, but are not limited to, sodium dodecyl sulfate (SDS)-PAGE, size-exclusion chromatography (SEC), high pressure liquid chromatography (HPLC), non-denaturing PAGE, and mass spectrometry (MS).

In some embodiments, silk fibroin molecular weight is modulated by the method of degumming used during processing. In some embodiments, longer heating times during degumming are used (e.g., see International Patent Application Publication No. WO2014145002, the contents of which are herein incorporated by reference in their entirety). Longer heating (e.g., boiling) time may be used during the degumming process to prepare silk fibroin with lower average molecular weights. In some embodiments, heating times may be from about 1 min to about 5 min, from about 2 min to about 10 min, from about 5 min to about 15 min, from about 10 min to about 25 min, from about 20 min to about 35 min, from about 30 min to about 50 min, from about 45 min to about 75 min, from about 60 min to about 95 min, from about 90 min to about 125 min, from about 120 min to about 175 min, from about 150 min to about 200 min, from about 180 min to about 250 min, from about 210 min to about 350 min, from about 240 min to about 400 min, from about 270 min to about 450 min, from about 300 min to about 480 min, or more than 480 min. Additionally, the sodium carbonate concentration used in the degumming process, as well as the heating temperature, may also be altered to modulate the molecular weight of silk fibroin.

In some embodiments, silk fibroin molecular weight is presumed, without actual analysis, based on methods used to prepare the silk fibroin. For example, silk fibroin may be presumed to be low molecular weight silk fibroin or high molecular weight silk fibroin based on the length of time that heating is carried out (e.g., by minute boil value). In some embodiments, the molecular weight range for silk fibroin with a 480 mb is between 5-20 kDa. In some embodiments, the molecular weight as defined by the minute boil is as described in International Patent Application Publication No. WO2017139684.

In some embodiments, SBPs include a plurality of silk fibroin fragments generated using a dissociation procedure. The dissociation procedure may include one or more of heating, acid treatment, chaotropic agent treatment, sonication, and electrolysis. Some SBPs include a plurality of silk fibroin fragments dissociated from raw silk, silk fiber, and/or silk fibroin by heating. The heating may be carried out at a temperature of from about 30° C. to about 1,000° C. In some embodiments, heating is carried out by boiling. The raw silk, silk fiber, and/or silk fibroin may be boiled for from about 1 second to about 24 hours.

Bond and Amino Acid Content

In some embodiments, processed silk preparations may be characterized by the content of various amino acids present in the preparations. Different ratios and/or percentages of one or more amino acids may be present as a result of degumming or other processing. Such amino acids may include serine, glycine, and alanine. Amino acid content of processed silk preparations may be measured by any method known to one of skill in the art, including, but not limited to amino acid analysis and mass spectrometry. In some embodiments, the amino acid content of a processed silk preparation is measured for one amino acid (e.g. serine). In some embodiments, the amino acid content of a processed silk preparation may be measured for a combination of two or more amino acids (e.g. serine, glycine, and alanine). In some embodiments, processed silk preparations of the present disclosure may contain from about 0% to about 1%, from about 1% to about 5%, from about 5% to about 10%, from about 10% to about 15%, from about 15% to about 20%, from about 20% to about 25%, from about 25% to about 30%, from about 30% to about 35%, from about 35% to about 40%, or from about 40% to about 45% of any of the one or more amino acids described herein.

In some embodiments, the amino acid content of silk fibroin may be altered after processing (e.g. degumming). In some embodiments, the serine content of silk fibroin may decrease after processing (e.g. degumming). The serine content of silk fibroin in processed silk preparations may decrease by about 0%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, or about 25%.

In some embodiments, processed silk preparations may be characterized by the content of disulfide bonds present in the preparations. Different ratios and/or percentages of disulfide bonds may be present as a result of degumming or other processing. Disulfide bond content of processed silk preparations may be measured by any method known to one of skill in the art. In some embodiments, the disulfide bond content of silk fibroin may be altered after processing (e.g. degumming and/or boiling). In some embodiments, the disulfide bond content of silk fibroin may decrease after processing (e.g. degumming and/or boiling). The disulfide bond content of silk fibroin in processed silk preparations may decrease by about 0%, 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, 20%, 21%, 22%, 23%, 24%, 25%, 26%, 27%, 28%, 29%, 30%, 31%, 32%, 33%, 34%, 35%, 36%, 37%, 38%, 39%, 40%, 41%, 42%, 43%, 44%, 45%, 46%, 47%, 48%, 49%, 50%, 51%, 52%, 53%, 54%, 55%, 56%, 57%, 58%, 59%, 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or about 100%.

Purification and Concentration

In some embodiments, processed silk preparations may be purified. Purification, as used herein, refers to any process used to segregate or extract one entity from another. In some embodiments, purification is manual or automated. Purification may include the removal of salts, impurities, or contaminants from processed silk preparations.

In some embodiments, processed silk may be purified by concentration from a processed silk solution. Methods of concentrating silk fibroin from processed silk solutions may include any of those described in the International Patent Application Publication No. WO2017139684, the contents of which are incorporated herein by reference in their entirety. In some embodiments, purification and/or concentration may be carried out by one or more of dialysis, centrifugation, air drying, vacuum drying, filtration, and/or Tangential Flow Filtration (TFF).

In some embodiments, processed silk solutions are purified by dialysis. Dialysis may be carried out to remove undesired salts and/or contaminants. In some embodiments, processed silk solutions are concentrated via dialysis. Purification and/or concentration of processed silk by dialysis may be carried out as described in International Patent Application Publication No. WO2005012606, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the dialysis is performed against a hygroscopic polymer to concentrate the silk fibroin solution. In some embodiments the dialysis is manual, with the use of a membrane and manual solvent changes. In some embodiments, the solvent is changed between 1 and 10 times over the course of the procedure. In some embodiments, the membrane is a dialysis cassette. The dialysis cassette may be a slide-a-lyzer dialysis cassette. In some embodiments, the membrane is dialysis tubing. The dialysis tubing may be regenerated cellulose dialysis tubing and/or snake skin. The dialysis tubing or cassette may be rinsed in distilled water for 30 minutes to prepare the membrane for use. In some embodiments, the dialysis tubing has a molecular weight cutoff of 3.5 kDa. In some embodiments, the dialysis is performed at a temperature of from about 1° C. to about 30° C. In some embodiments, dialysis is performed at room temperature. In other embodiments, the dialysis is performed at 4° C. Dialysis may be performed until desired concentrations of silk fibroin and salt are obtained from processed silk solutions. Dialysis may be performed for periods of time from about 30 minutes to about 24 hours or beyond. For example, dialysis may be carried out for from about 30 minutes to about 2 hours, from about 1 hour to about 6 hours, from about 3 hours to about 10 hours, from about 5 hours, to about 12 hours, from about 7 hours to about 15 hours, from about 11 hours to about 20 hours, or from about 16 hours to about 24 hours.

In some embodiments, dialysis may be automated. The dialysis may use an automated water change system. Such systems may include tanks of up to 10 L and may be able to hold multiple dialysis cassettes (e.g., see International Patent Application Publication No. WO2017106631, the contents of which are herein incorporated by reference in their entirety). Automated equipment may enable purification of larger volumes of solution with greater efficiency. Automated controllers, programmed with the proper times and volumes, may be used to facilitate changes of solvent or buffer over the course of dialysis. The solvent may be replaced from about 1 to about 20 times or more during dialysis. In some embodiments, automated dialysis may be completed in about 48 hours.

Dialysis may be performed with various solvents depending on the nature of the preparation being processed. In some embodiments the solvent is water. In some embodiments, the solvent is an aqueous solution. In some embodiments the solvent includes a hygroscopic polymer. Hygroscopic polymers may include, but are not limited to, polyethylene glycol (PEG), polyethylene oxide (PEO), collagen, fibronectin, keratin, polyaspartic acid, polylysine, alginate, chitosan, chitin, hyaluronic acid, pectin, polycaprolactone, polylactic acid, polyglycolic acid, polyhydroxyalkanoates, dextrans, and polyanhydrides. Additional examples of polymers, hygroscopic polymers, and related dialysis methods that may be employed include any of those found in International Patent Application Publication Nos. WO2005012606, WO2005012606, and WO2017106631, and U.S. Pat. Nos. 6,302,848; 6,395,734; 6,127,143; 5,263,992; 6,379,690; 5,015,476; 4,806,355; 6,372,244; 6,310,188; 5,093,489; 6,325,810; 6,337,198; 6,267,776; 5,576,881; 6,245,537; 5,902,800; and 5,270,419; the contents of each of which are herein incorporated by reference in their entirety. Hygroscopic polymer concentrations may be from about 20% (w/v) to about 50% (w/v). In some embodiments, dialysis may be performed in a stepwise manner in a urea solution, and the urea solution may be subsequently be replaced with urea solutions of a lower concentration during buffer changes, until it is ultimately replaced with water, as described in Zheng et al. (2016) Journal of Biomaterials Applications 31:450-463.

In some embodiments, processed silk preparations may be purified by filtration. Such filtration may include trans flow filtration (TFF), also known as tangential flow filtration. During TFF, solutions may be passed across a filter membrane. Anything larger than the membrane pores is retained, and anything smaller passes through the membrane (e.g., see International Patent Application Publication No. WO2017106631, the contents of which are herein incorporated by reference in their entirety). With the positive pressure and flow along the membrane, instead of through it, particles trapped in the membrane may be washed away. TFF may be carried out using an instrument. The instrument may be automated. The membranes may be housed in TFF tubes with vertical inlets and outlets. The flow of solvent may be controlled by peristaltic pumps. Some TFF tubes may include a dual chamber element. The dual chamber element may enable TFF filtration of processed silk solutions at higher concentrations, while reducing aggregation via the reduction of shear forces.

In some embodiments, processed silk solutions are purified and/or concentrated by centrifugation. Centrifugation may be performed before or after other forms of purification, which include, but are not limited to dialysis and tangential flow filtration. Centrifugation times and speeds may be varied to optimize purification and/or concentration according to optimal time frames. Purification and/or concentration by centrifugation may include pelleting of the processed silk and removal of supernatant. In some cases, centrifugation is used to push solvent through a filter, while retaining processed silk. Centrifugation may be repeated as many times as needed. In some embodiments, silk fibroin solutions are centrifuged two or more times during concentration and/or purification.

In some embodiments, processed silk may be purified by any method known to one of skill in the art. In some embodiments, processed silk is purified to remove salts (e.g.

lithium bromide). In some embodiments, processed silk is purified to isolate processed silk of a desired molecular weight. In some embodiments, processed silk is purified by chromatography. Chromatography may include preparatory-scale, gravity, size exclusion chromatography (SEC). In some embodiments, processed silk is purified by gel permeation chromatography. Processed silk may be purified at any scale. In some embodiments, processed silk is purified on a milligram scale. In some embodiments, processed silk is purified on a gram scale. In some embodiments, processed silk is purified on a kilogram scale.

In some embodiments, SBP formulations may be directly prepared from dialyzed silk fibroin. In some embodiments, SBP formulations may be directly prepared from dialyzed and filtered silk fibroin.

Drying Methods

In some embodiments, processed silk preparations are dried to remove solvent. In some embodiments, SBP formulations may be rinsed prior to drying. Methods of drying may include, but are not limited to, air drying, oven drying, lyophilization, spray drying, spray freezing, and vacuum drying. Drying may be carried out to alter the consistency and/or other properties of processed silk preparations. One or more compounds or excipients may be combined with processed silk preparations to improve processed silk recovery and/or reconstitution after the drying process. For example, sucrose may be added to improve silk fibroin recovery and reconstitution from dried solutions. In some embodiments, drying may be carried out in the fabrication of a processed silk format or a SBP. Examples include, but are not limited to fabrication of fibers, nanofibers, mats, films, foams, membranes, rods, tubes, gels, hydrogels, microspheres, nanospheres, solutions, patches, grafts and powders. In some embodiments, drying processed silk is carried out by oven drying, lyophilizing, and/or air drying.

Oven drying refers to any drying method that uses an oven. According to some methods, ovens are maintained at temperatures of from about 30° C. to about 90° C. or more. In some embodiment, oven drying is carried out at a temperature of 60° C. Processed silk preparations may be placed in ovens for a period of from about 1 hour to about 24 hours or more. In one embodiment, SBP formulations are oven dried at 60° C. for 2 hours. Oven drying may be used to dry silk fibroin preparations. In some embodiments, silk fibroin preparations are oven dried for 16 hours at 60° C. to obtain a desired format. In some cases, silk fibroin solutions are oven dried overnight. Examples of formats obtained by oven drying may include, but are not limited to, fibers, nanofibers, mats, films, foams, membranes, rods, tubes, gels, hydrogels, microspheres, nanospheres, solutions, patches, grafts, and powders.

In some embodiments, processed silk preparations are freeze dried. Freeze drying may be carried out by lyophilization. Freeze drying may require processed silk preparations to be frozen prior to freeze drying. Freezing may be carried out at temperatures of from about 5° C. and about −85° C. In some embodiments, freeze drying is carried out by lyophilization for up to 75 hours. In some embodiments, lyophilization is used to prepare processed silk formats or SBPs. Such formats may include, but are not limited to, fibers, nanofibers, mats, films, foams, membranes, rods, tubes, gels, hydrogels, microspheres, nanospheres, solutions, patches, grafts and powders. The use of lyophilization to fabricate SBPs may be carried out according to any of the methods described in Zhou et al. (2017) Acta Biomater S1742-7061(17)30569; Yang et al. (2017) Int J Nanomedicine 12:6721-6733; Seo et al. (2017) J Biomater Appl 32(4):484-491; Ruan et al. (2017) Biomed Pharmacother 97:600-606; Wu et al. (2017) J Mech Behav Biomed Mater 77:671-682; Zhao et al. (2017) Materials Letters 211:110-113; Chen et al. (2017) PLoS One 12(11):e0187880; Min et al. (2017) Int J Biol Macromol 17: 32855-8; Sun et al. Journal of Materials Chemistry B 5:8770; and Thai et al. J Biomed Mater (2017) 13(1):015009, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, processed silk preparations may be dried by air drying. "Air drying," as used herein refers to the removal of moisture by exposure to ambient or circulated gasses. Air drying may include exposing a preparation to air at room temperature (from about 18° C. to about 29° C.). Air drying may be carried out for from about 30 minutes to about 24 hours or more. In some embodiments, silk fibroin preparations are air dried to prepare SBPs. SBP formats that may be prepared may include, but are not limited to, fibers, nanofibers, mats, films, foams, membranes, rods, tubes, gels, hydrogels, microspheres, nanospheres, solutions, patches, grafts and powders. Some examples of the use of air drying for fabrication of SBPs are presented in Susanin et al. (2017) Fibre Chemistry 49(2):88-96; Lo et al. J Tissue Eng Regen Med (2017) doi.10.1002/term.2616; and Mane et al. Scientific Reports 7:15531, the contents of each of which are herein incorporated by reference in their entirety.

Spinning

In some embodiments, processed silk may be prepared by spinning. As used herein, the term "spinning" refers to a process of twisting materials together. Spinning may include the process of preparing a silk fiber by twisting silk proteins as they are secreted from silk producers. Other forms of spinning include spinning one or more forms of processed silk together to form a thread, filament, fiber, or yarn. The processed silk may already consist of a filamentous format prior to spinning. In some embodiments, processed silk is processed by spinning from a non-filamentous format (e.g., from a film, mat, or solution).

In some embodiments, spinning includes the technique of electrospinning. Electrospinning may be used to prepare silk fibers from silk fibroin. The silk fibroin may be dissolved in water or an aqueous solution before electrospinning. In other embodiments, silk fibroin is dissolved in an organic solvent before electrospinning. The organic solvent may be hexafluoroisopropanol (HFIP). In some embodiments, electrospinning may be carried out as described in Yu et al. (2017) Biomed Mater Res A doi. 10.1002/jbm.a.36297 or Chantawong et al. (2017) Mater Sci Mater Med 28(12):191, the contents of each of which are herein incorporated by reference in their entirety.

Electrospinning typically includes the use of an electrospinning apparatus. Processed silk may be added to the apparatus to produce silk fiber. The processed silk may be silk fibroin in solution. Electrospinning apparatus components may include one or more of a spinneret (also referred to spinnerette), needle, mandrel, power source, pump, and grounded collector. The apparatus may apply voltage to the dissolved silk fibroin, causing electrostatic repulsion that generates a charged liquid that is extruded from the end. Electrostatic repulsion also enables fiber elongation as it forms, and charged liquid cohesion prevents it from breaking apart. Resulting fiber may be deposited on the collector. In some embodiments, electrospinning methods may be carried out according to those described in European Patent No. EP3206725; Manchineella et al. (2017) European Journal of Organic Chemistry 30:4363-4369; Park et al. (2017) Int J Biomacromol 50141-8130(17):32645-4; Wang et al. (2017) J Biomed Mater Res A doi.10.1002/jbm.a.36225;

Chendang et al. (2017) J Biomaterials and Tissue Engineering 7:858-862; Kambe et al. (2017) Materials (Basel) 10(10):E1153; Chouhan et al. (2017) J Tissue Eng Reneg Med doi.10.1002/term.2581; Genovese et al. (2017) ACS Appl Mater Interfaces doi.10.1021acsami.7b13372; Yu et al. (2017) Biomed Mater Res A doi. 10.1002/jbm.a.36297; Chantawong et al. (2017) Mater Sci Mater Med 28(12):191, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, spinning may be carried out as dry spinning. Dry spinning may be carried out using a dry spinning apparatus. Dry spinning may be used to prepare silk fibers from SBP formulations. The preparations may include silk fibroin solutions. The preparations may be aqueous solutions. Dry spinning apparatuses typically use hot air to dry processed silk as it is extruded. In some embodiments, dry spinning may be carried out according to any of the methods presented in Zhang et al. (2017) Int J Biol Macromol pii:S0141-8130(17):32857, the contents of which are herein incorporated by reference in their entirety.

Processing Methods: Spraying

In some embodiments, processing methods include spraying. As used herein, the term "spraying" refers to the sprinkling or showering of a compound or composition in the form of small drops or particles. Spraying may be used to prepare SBPs by spraying processed silk. Spraying may be carried out using electrospraying. Processed silk used for spraying may include processed silk in solution. The solution may be a silk fibroin solution. Solutions may be aqueous solutions. Some solutions may include organic solvents. Electrospraying may be carried out in a manner similar to that of electro spinning, except that the charged liquid lacks cohesive force necessary to prevent extruding material from breaking apart. In some embodiments, spraying methods may include any of those presented in United States Publication No. US2017/333351 or Cao et al. (2017) Scientific Reports 7:11913, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, electrospray methods include a coaxial system for coaxial spraying.

In some embodiments, spraying is carried out as spray drying. Spray drying is a method of producing a dry powder from a liquid or slurry by rapidly drying with a hot gas. For example, the silk fibroin solution may be introduced as a fine spray or mist into a tower or chamber with heated air. The large surface area of the spray droplets causes evaporation of the water to occur rapidly, converting the droplets into dry powder particles. The heat and drying process may induce beta-sheet formation in the silk fibroin. Additional advantages of spray drying may include low heat, speed, reproducibility, and scalability.

In one embodiment, the spraying is carried out as spray drying using the electrostatic spray drying methods known in the art.

In some embodiments, spraying is carried out as spray coating. For example, SBP formulations may be sprayed onto the surface of a substance to form a coating. The spray coating processing may be a thermal spray coating process where SBP formulations are heated or melted by a heat source, for example, by electrical means (plasma or arc) or chemical means (combustion flame). Thermal spraying techniques that may be used herein include, but are not limited to, plasma spraying, detonation spraying, wire arc spraying, flame spraying, high velocity oxy-fuel coating spraying (HVOF), high velocity air fuel (HVAF), warm spraying, and cold spraying.

In one embodiment, the spray coating may be used for enteric capsules.

Processing Methods: Precipitation

In some embodiments, processing methods include precipitation. As used herein, the term "precipitation" refers to the deposition of a substance in solid form from a solution. Precipitation may be used to obtain solid processed silk from processed silk solutions. The processed silk may be silk fibroin. Processed silk may be precipitate from a solution. The solvent may be aqueous. In some embodiments, the solvent is organic. Examples of organic solvents include, but are not limited to, HFIP, methanol, ethanol, and other alcohols. In some embodiments, the solvent is water. In some embodiments the solvent is a mixture of an organic solvent and water. Aqueous solvents may contain one or more salts. Processed silk may be precipitated from processed silk solutions by modulating one or more components of the solution to alter the solubility of the processed silk and promote precipitation. Additional processing steps may be employed to initiate or speed precipitation. Such methods may include, but are not limited to sonication, centrifugation, increasing the concentration of processed silk, altering the concentration of salt, adding additional salt or salts, altering the pH, applying shear stress, adding excipients, or applying chemical modifications.

Processing Methods: Milling

In some embodiments, processing methods include milling. As used herein, "milling" generally refers to the process of breaking down a solid substance into smaller pieces using physical forces such as grinding, crushing, pressing and/or cutting. As a non-limiting example, SBP formulations may be milled to create powders. The density of powder formulations may be controlled during the milling process. As another non-limiting example, solid encapsulation of a therapeutic agent or cargo with another substance (e.g., SBPs) may be prepared by milling. The therapeutic agent or cargo may include any one of those described herein. In some embodiments, the therapeutic agent or cargo to be encapsulated by another substance may include SBPs.

Altering Mechanical Properties

In some embodiments, the mechanical properties of processed silk may be altered by modulating physical and/or chemical properties of the processed silk. The mechanical properties include, but are not limited to, mechanical strength, tensile strength, elongation capabilities, elasticity, compressive strength, stiffness, shear strength, toughness, torsional stability, temperature stability, moisture stability, viscosity and reeling rate. In some embodiments, the tensile strength of processed silk is stronger than steel. In some embodiments, the tensile strength of SBPs is stronger than steel. Examples of the physical and chemical properties used to tune the mechanical properties of processed silk include, but are not limited to, the temperature, formulations, silk concentration, β-sheet content, crosslinking, the molecular weight of the silk, the storage of the silk, storage, methods of preparation, dryness, methods of drying, purity, and degumming. Methods of tuning the mechanical strength of processed silk are taught in International Patent Application Publication No. WO2017123383, European Patent No. EP2904134, European Patent No. EP3212246, Fang et al., Wu et al., Susanin et al., Zhang et al., Jiang et al., Yu et al., Chantawong et al., and Zhang et al. (Fang et al. (2017) Journal of Materials Chemistry B 5(30):6042-6048; Wu et al. (2017) J Mech Behav Biomed Mater 77:671-682; Susanin et al. (2017) Fibre Chemistry 49(2):88-96; Zhang et al. (2017) Fibers and Polymers 203:9-16; Jiang et al. (2017) J Biomater Sci Polym Ed 15:1-36; Yu et al. (2017) Biomed

Mater Res A doi. 10.1002/jbm.a.36297; Chantawong et al. (2017) Mater Sci Mater Med 28(12):191; Zhang et al. (2017) Int J Biomacromol 50141-8310(17):32857), the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, the excipients which may be incorporated in a formulation may be used to control the modulus of SBP formulations. In some embodiments, these SBP formulations are hydrogels.

In some embodiments, processed silk hydrogels are prepared with different excipients and tested for their mechanical properties, including the modulus. SBP formulations may be assessed for modulus, shear storage modulus, shear loss modulus, phase angle, and viscosity using a rheometer, and/or any other method known to one skilled in the art. Rheometer geometry may be selected based on sample viscosity, shear rates, and shear stresses desired, as well as sample volumes. Geometries that are suitable for measuring the rheological properties of SBP formulations include, not are not limited to, cone and plate, parallel plates, concentric cylinders (or Bob and Cup), and double gap cylinders. In one embodiment, a cone and plate geometry is used. In another embodiment, a concentric cylinder geometry is used. SBP formulations may be tested both before and after gelation. In some embodiments, SBP formulations are prepared, optionally with different excipients, and tested for their mechanical properties, including the shear storage modulus, the shear loss modulus, phase angle, and viscosity. As used herein, the term "shear storage modulus" refers to the measure of a material's elasticity or reversible deformation as determined by the material's stored energy. As used herein, the term "shear loss modulus" refer to the measure of a material's ability to dissipate energy, usually in the form of heat. As used herein, the term "phase angle" refers to the difference in the stress and strain applied to a material during the application of oscillating shear stress. As used herein, the term "viscosity" refers to a material's ability to resist deformation due to shear forces, and the ability of a fluid to resist flow. In some embodiments, processed silk hydrogels may possess similar viscosities, but vary in the modulus.

In some embodiments, the concentration of processed silk may enable silk preparations to shear thin. In some embodiments the silk preparation is an SBP. In some embodiments, the SBP is a hydrogel. In some embodiments, the molecular weight of processed silk hydrogels may enable hydrogels to shear thin. In some embodiments, hydrogels prepared with low molecular weight silk fibroin may be injected with much less force than hydrogels of similar viscosity that are prepared with higher molecular weight silk fibroin. In some embodiments, hydrogels with low molecular weight silk fibroin display higher viscosity than hydrogels with high molecular weight silk fibroin.

In some embodiments, the concentration of silk fibroin may be used to control the shear storage modulus and/or the shear loss modulus of processed silk preparations. In some embodiments, a preparation with stressed silk may be used to control the shear storage modulus and the shear loss modulus. In some embodiments, the excipients incorporated in a formulation may be used to control the shear storage modulus and/or the shear loss modulus of processed silk preparations. In some embodiments, these processed silk preparations are hydrogels. In some embodiments, these processed silk preparations are solutions. In some embodiments, processed silk preparations are prepared, optionally with different excipients, and tested for their mechanical and physical properties, including the shear storage modulus, the shear loss modulus, phase angle, and viscosity. As used herein, the term "shear storage modulus" refers to the measure of a material's elasticity or reversible deformation as determined by the material's stored energy. As used herein, the term "shear loss modulus" refers to the measure of a material's ability to dissipate energy, usually in the form of heat. As used herein, the term "phase angle" refers to the difference in the stress and strain applied to a material during the application of oscillating shear stress. As used herein, the term "viscosity" refers to a material's ability to resist deformation due to shear forces, and the ability of a material to resist flow. Processed silk preparations may be assessed for shear storage modulus, shear loss modulus, phase angle, and viscosity using a rheometer, and/or any other method known to one skilled in the art. Rheometer geometry may be selected based on sample viscosity, shear rates, and shear stresses desired, as well as sample volumes. Geometries that are suitable for measuring the rheological properties of SBP formulations include, not are not limited to, cone and plate, parallel plates, concentric cylinders (or Bob and Cup), and double gap cylinders. In one embodiment, a cone and plate geometry is used. In another embodiment, a concentric cylinder geometry is used. Processed silk preparations may be tested both before and after gelation. In some embodiments, processed silk preparations may possess similar viscosities, but vary in the modulus. In some embodiments, the processed silk preparations may have the viscosity of a liquid. In some embodiments, the processed silk preparations may have the viscosity of a gel.

In some embodiments, the processed silk preparations may shear thin or display shear thinning properties. As used herein, the term "shear thinning" refers to a decrease in viscosity at increasing shear rates. As used herein, the term "shear rate" refers to the rate of change in the ratio of displacement of material upon the application of a shear force to the height of the material. This ratio is also known as strain. In some embodiments, the boiling time during degumming of processed silk may enable processed silk preparations to shear thin. In some embodiments, the concentration of processed silk may enable silk preparations to shear thin. In some embodiments, the processed silk preparations may have the viscosity of a liquid at higher shear rates. In some embodiments, the processed silk preparations may have the viscosity of a gel at lower shear rates.

In some embodiments, the mechanical properties of processed silk preparations may be tuned by a preparation with stressed silk. As used herein, the term "stress" or "stressed" refers to a treatment that may alter the shelf life and/or stability of processed silk and/or an SBP. In some embodiments, processed silk is stressed by treatment with heat. In some embodiments, processed silk is stressed by heating to 60° C. In some embodiments, processed silk is stressed by heating overnight. In some embodiments, processed silk is stressed by autoclave. In some embodiments, processed silk is stressed by overnight heating to 60° C. followed by autoclave. In some embodiments, silk is stressed during the preparation of processed silk. In some embodiments, processed silk is stressed during the preparation of SBPs. In some embodiments, SBPs are stressed. Stressed silk or SBPs may be used in any of the embodiments described in the present disclosure.

In some embodiments, boiling silk fibroin in 0.02M sodium carbonate for 480 minutes may result in a polydisperse mixture of peptides ranging in molecular weight from about 200,000 Da to about 7000 Da, with an average molecular weight of about 35,000 Da. In some embodiments, the molecular weight of polymers (e.g. processed silk) may have a dramatic effect on properties such as stability, viscosity, surface tension, gelation and bioactivity. In some embodiments, polydisperse processed silk (e.g. silk fibroin degummed with a 480 minute boil) may be separated into narrow molecular weight fractions. In some embodiments, the separation of polydisperse processed silk may optimize one or more properties of an SBP (e.g. stability, viscosity, surface tension, gelation and bioactivity). Polydisperse mixtures of processed silk may be separated into fractions by any method known to one of skill in the art. In some embodiments, fractionation of processed silk may be used to isolate processed silk with narrower polydispersity. In some embodiments, processed silk is fractionated by preparatory-scale, gravity, size exclusion chromatography (SEC). In some embodiments, processed silk is fractionated by gel permeation chromatography. Processed silk may be fractionated at any scale. In some embodiments, processed silk is fractionated on a milligram scale. In some embodiments, processed silk is fractionated on a gram scale. In some embodiments, processed silk is fractionated on a kilogram scale.

Modulating Degradation and Resorption

In some embodiments, processed silks are, or are processed to be, biocompatible. As used herein, a "biocompatible" substance is any substance that is not harmful to most living organisms or tissues. With some processed silk, degradation may result in products that are biocompatible, making such processed silk attractive for a variety of applications. Some processed silk may degrade into smaller proteins or amino acids. Some processed silk may be resorbable under physiological conditions. In some embodiments, products of silk degradation may be resorbable in vivo. In some embodiments, the rate of degradation of processed silk may be tuned by altering processed silk properties. Examples of these properties include, but are not limited to, type and concentration of certain proteins, β-sheet content, crosslinking, silk fibroin molecular weight, and purity. In some embodiments, rate of processed silk degradation may be modulated by method of storage, methods of preparation, dryness, methods of drying, reeling rate, and degumming process.

In some embodiments, the bioresorbability and degradation of processed silk is modulated by the addition of sucrose, as taught in Li et al. (Li et al. (2017) Biomacromolecules 18(9):2900-2905), the contents of which are herein incorporated by reference in their entirety. Processed silk may be formulated with sucrose to enhance thermal stability. Furthermore, processed silk with sucrose may also be formulated with antiplasticizing agents to further enhance thermal stability of processed silk, SBPs, and/or therapeutic agents included in SBPs. Methods of increasing thermal stability using antiplasticizing agents may include any of those described in Li et al. (Li et al. (2017) Biomacromolecules 18(9):2900-2905), the contents of which are herein incorporated by reference in their entirety. In some embodiments, the addition of sucrose to processed silk preparations prior to lyophilization leads to an increased reconstitution efficiency. In some embodiments, the addition of sucrose may be used to create higher molecular weight processed silk preparations as well as to maintain long term storage stability. In some embodiments, the incorporation of sucrose into processed silk preparations described herein enables slower freezing during lyophilization cycle.

In some embodiments, an SBP maintains and or improves stability by at least 1 day, at least 2 days, at least 3 days, at least 4 days, at least 5 days, at least 6 days, at least 7 days, at least 8 days, at least 9 days, at least 10 days, at least 11 days, at least 12 days, at least 13 days, at least 2 weeks, at least 3 weeks, at least 1 month, at least 6 weeks, at least 2 months, at least 10 weeks, at least 3 months, at least 14 weeks, at least 4 months, at least 18 weeks, at least 5 months, at least 22 weeks, at least 6 months, at least 7 months, at least 8 months, at least 9 months, at least 10 months, at least 11 months, at least a year, at least 2 years, at least 3 years, at least 4 years, at least 5 years, or more than 5 years. In some embodiments, an SBP preparation reduces stability by In some embodiments, an SBP maintains and or improves stability by at least 1 day, at least 2 days, at least 3 days, at least 4 days, at least 5 days, at least 6 days, at least 7 days, at least 8 days, at least 9 days, at least 10 days, at least 11 days, at least 12 days, at least 13 days, at least 2 weeks, at least 3 weeks, at least 1 month, at least 6 weeks, at least 2 months, at least 10 weeks, at least 3 months, at least 14 weeks, at least 4 months, at least 18 weeks, at least 5 months, at least 22 weeks, at least 6 months, at least 7 months, at least 8 months, at least 9 months, at least 10 months, at least 11 months, at least a year, at least 2 years, at least 3 years, at least 4 years, at least 5 years, or more than 5 years. In some embodiments, an SBP may have a shelf life of least 1 day, at least 2 days, at least 3 days, at least 4 days, at least 5 days, at least 6 days, at least 7 days, at least 8 days, at least 9 days, at least 10 days, at least 11 days, at least 12 days, at least 13 days, at least 2 weeks, at least 3 weeks, at least 1 month, at least 6 weeks, at least 2 months, at least 10 weeks, at least 3 months, at least 14 weeks, at least 4 months, at least 18 weeks, at least 5 months, at least 22 weeks, at least 6 months, at least 7 months, at least 8 months, at least 9 months, at least 10 months, at least 11 months, at least a year, at least 2 years, at least 3 years, at least 4 years, at least 5 years, or more than 5 years.

In some embodiments, the bioresorbability and degradation of processed silk may be tuned through formulation with additional bioresorbable polymer matrices, as taught in International Patent Application Publication Numbers WO2017177281 and WO2017179069, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, the polymer matrix is polyurethane. In some embodiments, these polymer matrices may be polycaprolactone and a ceramic filler. The ceramic filler may include MgO.

In some embodiments, the bioresorbability and degradation of processed silk is tuned through the fabrication of a composite scaffold. Composite scaffolds, combinations of scaffolds or scaffolds formed from more than one material, may be formed from two or more SBP formulations. In some embodiments, processed silk scaffolds comprising a combination of silk fibroin microspheres within a larger processed silk preparation may demonstrate slower degradation in comparison with other scaffolds, as taught in European Patent No. EP3242967, the contents of which are herein incorporated by reference in their entirety.

Analytics

In some embodiments, processed silk products may be analyzed for properties such as molecular weight, aggregation, amino acid content, lithium content, heavy metal content, bromide content and endotoxin level. Such properties may be evaluated via any analytical methods known in the art. As a non-limiting example, the Ultra-Performance Liquid Chromatography (UPLC)-Size Exclusion Chromatography (SEC) method may be used to assess the molecular weight and/or aggregation of the silk fibroin proteins in the processed silk products.

In some embodiments, processed silk products may be analyzed for silk fibroin concentration. Such properties may be evaluated via any analytical methods known in the art. As a non-limiting example, gravimetry and/or ultraviolet-visible spectroscopy (UV-Vis) may be used.

In some embodiments, silk fibroin molecular weight is modulated by the method of degumming used during processing. In some embodiments, longer heating times during degumming are used (e.g., see International Publication No. WO2014145002, the contents of which are herein incorporated by reference in their entirety). Longer heating (e.g., boiling) time may be used during the degumming process to prepare silk fibroin with lower average molecular weights. In some embodiments, heating times may be from about 1 min to about 5 min, from about 2 min to about 10 min, from about 5 min to about 15 min, from about 10 min to about 25 min, from about 20 min to about 35 min, from about 30 min to about 50 min, from about 45 min to about 75 min, from about 60 min to about 95 min, from about 90 min to about 125 min, from about 120 min to about 175 min, from about 150 min to about 200 min, from about 180 min to about 250 min, from about 210 min to about 350 min, from about 240 min to about 400 min, from about 270 min to about 450 min, from about 300 min to about 480 min, or more than 480 min. Additionally, the sodium carbonate concentration used in the degumming process, as well as the heating temperature, may also be altered to modulate the molecular weight of silk fibroin. In one embodiment, the alteration may cause an increase in the molecular weight of silk fibroin. As compared to silk fibroin where the sodium carbonate concentration and/or the heating temperature was not altered, the increase of the molecular weight may be 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or greater than 99% higher. In one embodiment, the alteration may cause a decrease in the molecular weight of silk fibroin. As compared to silk fibroin where the sodium carbonate concentration and/or the heating temperature was not altered, the decrease of the molecular weight may be 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or greater than 99% lower.

In some embodiments, silk fibroin molecular weight may be presumed, without actual analysis, based on methods used to prepare the silk fibroin. For example, silk fibroin may be presumed to be low molecular weight silk fibroin or high molecular weight silk fibroin based on the length of time that heating is carried out (e.g., by minute boil value).

In some embodiments, SBP formulations include a plurality of silk fibroin fragments generated using a dissociation procedure. The dissociation procedure may include one or more of heating, acid treatment, base treatment, chaotropic agent treatment, sonication, and electrolysis. Some SBPs include a plurality of silk fibroin fragments dissociated from raw silk, silk fiber, and/or silk fibroin by heating. The heating may be carried out at a temperature of from about 30° C. to about 1,000° C. In some embodiments, heating is carried out by boiling. The raw silk, silk fiber, and/or silk fibroin may be boiled for from about 1 second to about 24 hours.

Porosity

In some embodiments, processed silk may include variations in porosity. As used herein, the term "porosity" refers to the frequency with which holes, pockets, channels, or other spaces occur in a material, in some cases influencing the movement of elements to and/or from the material. Processed silk porosity may influence one or more other silk properties or properties of an SBP that includes the processed silk. These properties may include, but are not limited to, stability, payload retention or release, payload release rate, wettability, mechanical strength, tensile strength, elongation capabilities, density, thickness, elasticity, compressive strength, stiffness, shear strength, toughness, torsional stability, temperature stability, and moisture stability. In some embodiments, processed silk porosity may control the diffusion or transport of agents from, within, or into the processed silk or SBP. Such agents may include, but are not limited to, therapeutics, biologics, chemicals, small molecules, oxidants, antioxidants, macromolecules, microspheres, nanospheres, cells, or any payloads described herein.

Processed silk porosity may be modulated during one or more processing steps or during fabrication of a SBP (e.g., see International Publication No. WO2014125505 and U.S. Pat. No. 8,361,617, the contents of each of which are herein incorporated by reference in their entirety). In some embodiments, processed silk porosity may be modulated by one or more of sonication, centrifugation, modulating silk fibroin concentration, modulating salt concentration, modulating pH, modulating secondary structural formats, applying shear stress, modulating excipient concentration, chemical modification, crosslinking, or combining with cells, bacteria, and/or viral particles.

Strength and Stability

Processed silk strength and stability are important factors for many applications. In some embodiments, processed silk may be selected based on or prepared to maximize mechanical strength, tensile strength, elongation capabilities, elasticity, flexibility, compressive strength, stiffness, shear strength, toughness, torsional stability, biological stability, resistance to degradation, and/or moisture stability. In some embodiments, processed silk has a non-acidic microenvironment. In some embodiments, the non-acidic microenvironment enhances the stability of processed silk and or SBPs. In some embodiments, the non-acidic microenvironment enhances the stability of therapeutic agents formulated with processed silk and/or SBP. In some embodiments, the tensile strength of processed silk is stronger than steel. In some embodiments, the tensile strength of an SBP is stronger than steel.

In some embodiments, processed silk may demonstrate stability and/or is determined to be stable under various conditions. As used herein, "stability" and "stable" refers to the capacity of a substance (e.g. an SBP) to remain unchanged over time under the described conditions. Those conditions may be in vitro, in vivo, or ex vivo. In some embodiments, an SBP may be stable for up to 1 hour, up to 3 days, up to 1 week, up to 1 month, up to 3 months, up to 4 months, up to 6 months, up to 7 months, up to 1 year, up to 2 years, or up to 5 years.

Injectability

In some embodiments, processed silk may be selected based on or prepared to modulate the injectability of an SBP formulation. As used herein, the term "injectability" refers to the force required to push a composition through a syringe or syringe and needle. Injections may be used to administer SBP formulations. The SBP formulations may be administered via syringe to a subject. Injectability may be measured by the force required to push the composition through the desired syringe. The force may be, but is not limited to 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150, 160, 170, 180, 190, or 200 N, or a range of 10-50, 10-60, 10-90, 10-100, 10-110, 10-150, 10-200, 20-50, 20-70, 20-100, 20-120, 20-150, 20-200, 30-50, 30-80, 30-100, 30-110, 30-130, 30-150, 30-200, 40-50, 40-90, 40-100, 40-120, 40-140, 40-150, 40-200, 50-100, 50-130, 50-150, 50-200, 60-100, 60-110, 60-140, 60-150, 60-160, 60-200, 70-100, 70-120, 70-150, 70-170, 70-200, 80-100, 80-130, 80-150, 80-160, 80-180, 80-200, 90-100, 90-140, 90-150, 90-170, 90-190, 90-200, 100-150, 100-180, 100-200, 110-150, 110-160, 110-190, 110-200, 120-150, 120-170, 120-200, 130-150, 130-180, 130-200, 140-150, 140-190, 140-200, 150-200, 160-200, 170-200, 180-200, or 190-200 N.

In some embodiments, the SBP formulations described herein may be injected with a force of 200 N or less.

In some embodiments, the SBP formulations described herein may be injected with a force of 150 N or less.

In some embodiments, the SBP formulations described herein may be injected with a force of 100 N or less.

In some embodiments, the SBP formulations described herein may be injected with a force of 50 N or less.

In some embodiments, the SBP formulations described herein may be injected with a force of 20 N or less.

In some embodiments, the SBP formulations described herein may be injected with a force of 10 N or less.

In some embodiments, the SBP formulations described herein may be injected with a force of 5 N or less.

In some embodiments, injectability may also be analyzed by maximum force. As used herein, the term "maximum force" refers to the highest force achieved during injection. The maximum force may occur at the beginning of an injection. The maximum force may be, but is not limited to, 5, 10, 20, 25, 30, 40, 50, 60, 70, 75, 80, 90, 100, 110, 120, 125, 130, 140, 150, 160, 170, 175, 180, 190, or 200 N, or 5-10, 5-25, 10-50, 10-60, 10-90, 10-100, 10-110, 10-150, 10-200, 20-50, 20-70, 20-100, 20-120, 20-150, 20-200, 25-50, 30-50, 30-80, 30-100, 30-110, 30-130, 30-150, 30-200, 40-50, 40-90, 40-100, 40-120, 40-140, 40-150, 40-200, 50-100, 50-130, 50-150, 50-200, 60-100, 60-110, 60-140, 60-150, 60-160, 60-200, 70-100, 70-120, 70-150, 70-170, 70-200, 75-100, 80-100, 80-130, 80-150, 80-160, 80-180, 80-200, 90-100, 90-140, 90-150, 90-170, 90-190, 90-200, 100-150, 100-180, 100-200, 110-150, 110-160, 110-190, 110-200, 120-150, 120-170, 120-200, 125-150, 130-150, 130-180, 130-200, 140-150, 140-190, 140-200, 150-175, 150-200, 160-200, 170-200, 175-200, 180-200, or 190-200 N.

In some embodiments, the maximum force is from about 5 N to about 200 N. In some embodiments, the maximum force may be from about 0.001 N to about 5 N, from about 5 N to about 25 N, from about 25 N to about 50 N, from about 50 N to about 75 N, from about 75 N to about 100 N, from about 100 N to about 125 N, from about 125 N to about 150 N, from about 150 N to about 175 N, or from about 175 N to about 200 N.

In some embodiments, the SBP formulation may be delivered using a syringe with a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100 mL syringe which has an applicator which is 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, or more than 10 mm.

In one embodiment, the SBP formulation may be delivered using a 3 mL syringe with a 1.5 mm applicator.

pH

In some embodiments, the SBP formulation may be optimized for a specific pH. The pH of the SBP formulation may be, but is not limited, to 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, 5, 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.8, 5.9, 6, 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7, 6.8, 6.9, 7, 7.1, 7.2, 7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9, 8, 8.1, 8.2, 8.3, 8.4, 8.5, 8.6, 8.7, 8.8, 8.9, 9, 9.1, 9.2, 9.3, 9.4, 9.5, 9.6, 9.7, 9.8, 9.9, 10, 10.1, 10.2, 10.3, 10.4, 10.5, 10.6, 10.7, 10.8, 10.9, 11, 11.1, 11.2, 11.3, 11.4, 11.5, 11.6, 11.7, 11.8, 11.9, 12, 12.1, 12.2, 12.3, 12.4, 12.5, 12.6, 12.7, 12.8, 12.9, 13, 13.1, 13.2, 13.3, 13.4, 13.5, 13.6, 13.7, 13.8, 13.9, and 14.

In one embodiment, the SBP formulation may be optimized for a specific pH range. The pH range may be, but is not limited to, 0-4, 1-5, 2-6, 3-7, 4-8, 5-9, 6-10, 7-11, 8-12, 9-13, 10-14, 0-4, 1-5, 2-6, 3-7, 4-8, 5-9, 6-10, 7-11, 8-12, 9-13, 10-14, 0-4.5, 1-5.5, 2-6.5, 3-7.5, 4-8.5, 5-9.5, 6-10.5, 7-11.5, 8-12.5, 9-13.5, 0-1.5, 1-2.5, 2-3.5, 3-4.5, 4-5.5, 5-6.5, 6-7.5, 7-8.5, 8-9.5, 9-10.5, 10-11.5, 11-12.5, 12-13.5, 0-1, 1-2, 2-3, 3-4, 4-5, 5-6, 6-7, 6.5-7.5, 7-8, 8-9, 9-10, 10-11, 11-12, 12-13, 13-14, 0-0.5, 0.5-1, 1-1.5, 1.5-2, 2-2.5, 2.5-3, 3-3.5, 3.5-4, 4-4.5, 4.5-5, 5-5.5, 5.5-6, 6-6.5, 6.5-7, 7-7.5, 7.5-8, 8-8.5, 8.5-9, 9-9.5, 9.5-10, 10-10.5, 10.5-11, 11-11.5, 11.5-12, 12-12.5, 12.5-13, 13-13.5, or 13.5-14.

In one embodiment, the pH of the SBP formulation is between 4-8.5.

In one embodiment, the pH of the SBP formulation is between 6.5-7.5

In one embodiment, the pH of the SBP formulation is between 7-7.5.

Specific Gravity

In some embodiments, the SBP formulation may be optimized for a specific gravity. The specific gravity of the SBP formulation may be, but is not limited, to 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.1, 3.2, 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8, 4.9, and 5.

In some embodiments, the specific gravity of the SBP formulation may be, but is not limited to, 0.1-5 g/ml, 0.2-5 g/ml, 0.3-5 g/ml, 0.4-5 g/ml, 0.5-5 g/ml, 0.6-5 g/ml, 0.7-5 g/ml, 0.8-5 g/ml, 0.9-5 g/ml, 1-5 g/ml, 1.1-5 g/ml, 1.2-5 g/ml, 1.3-5 g/ml, 1.4-5 g/ml, 1.5-5 g/ml, 1.6-5 g/ml, 1.7-5 g/ml, 1.8-5 g/ml, 1.9-5 g/ml, 2-5 g/ml, 2.1-5 g/ml, 2.2-5 g/ml, 2.3-5 g/ml, 2.4-5 g/ml, 2.5-5 g/ml, 2.6-5 g/ml, 2.7-5 g/ml, 2.8-5 g/ml, 2.9-5 g/ml, 3-5 g/ml, 3.1-5 g/ml, 3.2-5 g/ml, 3.3-5 g/ml, 3.4-5 g/ml, 3.5-5 g/ml, 3.6-5 g/ml, 3.7-5 g/ml, 3.8-5 g/ml, 3.9-5 g/ml, 4-5 g/ml, 4.1-5 g/ml, 4.2-5 g/ml, 4.3-5 g/ml, 4.4-5 g/ml, 4.5-5 g/ml, 4.6-5 g/ml, 4.7-5 g/ml, 4.8-5 g/ml, 4.9-5 g/ml, 0.1-0.3 g/ml, 0.2-0.4 g/ml, 0.3-0.5 g/ml, 0.4-0.6 g/ml, 0.5-0.7 g/ml, 0.6-0.8 g/ml, 0.7-0.9 g/ml, 0.8-1 g/ml, 0.9-1.1 g/ml, 1-1.2 g/ml, 1.1-1.3 g/ml, 1.2-1.4 g/ml, 1.3-1.5 g/ml, 1.4-1.6 g/ml, 1.5-1.7 g/ml, 1.6-1.8 g/ml, 1.7-1.9 g/ml, 1.8-2 g/ml, 1.9-2.1 g/ml, 2-2.2 g/ml, 2.1-2.3 g/ml, 2.2-2.4 g/ml, 2.3-2.5 g/ml, 2.4-2.6 g/ml, 2.5-2.7 g/ml, 2.6-2.8 g/ml, 2.7-2.9 g/ml, 2.8-3 g/ml, 2.9-3.1 g/ml, 3-3.2 g/ml, 3.1-3.3 g/ml, 3.2-3.4 g/ml, 3.3-3.5 g/ml, 3.4-3.6 g/ml, 3.5-3.7 g/ml, 3.6-3.8 g/ml, 3.7-3.9 g/ml, 3.8-4 g/ml, 3.9-4.1 g/ml, 4-4.2 g/ml, 4.1-4.3 g/ml, 4.2-4.4 g/ml, 4.3-4.5 g/ml, 4.4-4.6 g/ml, 4.5-4.7 g/ml, 4.6-4.8 g/ml, 4.7-4.9 g/ml, 4.8-5 g/ml, 0.1-0.4 g/ml, 0.2-0.5 g/ml, 0.3-0.6 g/ml, 0.4-0.7 g/ml, 0.5-0.8 g/ml, 0.6-0.9 g/ml, 0.7-1 g/ml, 0.8-1.1 g/ml, 0.9-1.2 g/ml, 1-1.3 g/ml, 1.1-1.4 g/ml, 1.2-1.5 g/ml, 1.3-1.6 g/ml, 1.4-1.7 g/ml, 1.5-1.8 g/ml, 1.6-1.9 g/ml, 1.7-2 g/ml, 1.8-2.1 g/ml, 1.9-2.2 g/ml, 2-2.3 g/ml, 2.1-2.4 g/ml, 2.2-2.5 g/ml, 2.3-2.6 g/ml, 2.4-2.7 g/ml, 2.5-2.8 g/ml, 2.6-2.9 g/ml, 2.7-3 g/ml, 2.8-3.1 g/ml, 2.9-3.2 g/ml, 3-3.3 g/ml, 3.1-3.4 g/ml, 3.2-3.5 g/ml, 3.3-3.6 g/ml, 3.4-3.7 g/ml, 3.5-3.8 g/ml, 3.6-3.9 g/ml, 3.7-4 g/ml, 3.8-4.1 g/ml, 3.9-4.2 g/ml, 4-4.3 g/ml, 4.1-4.4 g/ml, 4.2-4.5 g/ml, 4.3-4.6 g/ml, 4.4-4.7 g/ml, 4.5-4.8 g/ml, 4.6-4.9 g/ml, 4.7-5 g/ml, 0.1-0.5 g/ml, 0.2-0.6 g/ml, 0.3-0.7 g/ml, 0.4-0.8 g/ml, 0.5-0.9 g/ml, 0.6-1 g/ml, 0.7-1.1 g/ml, 0.8-1.2 g/ml, 0.9-1.3 g/ml, 1-1.4 g/ml, 1.1-1.5 g/ml, 1.2-1.6 g/ml, 1.3-1.7 g/ml, 1.4-1.8 g/ml, 1.5-1.9 g/ml, 1.6-2 g/ml, 1.7-2.1 g/ml, 1.8-2.2 g/ml, 1.9-2.3 g/ml, 2-2.4 g/ml, 2.1-2.5 g/ml, 2.2-2.6 g/ml, 2.3-2.7 g/ml, 2.4-2.8 g/ml, 2.5-2.9 g/ml, 2.6-3 g/ml, 2.7-3.1 g/ml, 2.8-3.2 g/ml, 2.9-3.3 g/ml, 3-3.4 g/ml, 3.1-3.5 g/ml, 3.2-3.6 g/ml, 3.3-3.7 g/ml, 3.4-3.8 g/ml, 3.5-3.9 g/ml, 3.6-4 g/ml, 3.7-4.1 g/ml, 3.8-4.2 g/ml, 3.9-4.3 g/ml, 4-4.4 g/ml, 4.1-4.5 g/ml, 4.2-4.6 g/ml, 4.3-4.7 g/ml, 4.4-4.8 g/ml, 4.5-4.9 g/ml, 4.6-5 g/ml, 0.1-0.6 g/ml, 0.2-0.7 g/ml, 0.3-0.8 g/ml, 0.4-0.9 g/ml, 0.5-1 g/ml, 0.6-1.1 g/ml, 0.7-1.2 g/ml, 0.8-1.3 g/ml, 0.9-1.4 g/ml, 1-1.5 g/ml, 1.1-1.6 g/ml, 1.2-1.7 g/ml, 1.3-1.8 g/ml, 1.4-1.9 g/ml, 1.5-2 g/ml, 1.6-2.1 g/ml, 1.7-2.2 g/ml, 1.8-2.3 g/ml, 1.9-2.4 g/ml, 2-2.5 g/ml, 2.1-2.6 g/ml, 2.2-2.7 g/ml, 2.3-2.8 g/ml, 2.4-2.9 g/ml, 2.5-3 g/ml, 2.6-3.1 g/ml, 2.7-3.2 g/ml, 2.8-3.3 g/ml, 2.9-3.4 g/ml, 3-3.5 g/ml, 3.1-3.6 g/ml, 3.2-3.7 g/ml, 3.3-3.8 g/ml, 3.4-3.9 g/ml, 3.5-4 g/ml, 3.6-4.1 g/ml, 3.7-4.2 g/ml, 3.8-4.3 g/ml, 3.9-4.4 g/ml, 4-4.5 g/ml, 4.1-4.6 g/ml, 4.2-4.7 g/ml, 4.3-4.8 g/ml, 4.4-4.9 g/ml, 4.5-5 g/ml, 0.1-0.7 g/ml, 0.2-0.8 g/ml, 0.3-0.9 g/ml, 0.4-1 g/ml, 0.5-1.1 g/ml, 0.6-1.2 g/ml, 0.7-1.3 g/ml, 0.8-1.4 g/ml, 0.9-1.5 g/ml, 1-1.6 g/ml, 1.1-1.7 g/ml, 1.2-1.8 g/ml, 1.3-1.9 g/ml, 1.4-2 g/ml, 1.5-2.1 g/ml, 1.6-2.2 g/ml, 1.7-2.3 g/ml, 1.8-2.4 g/ml, 1.9-2.5 g/ml, 2-2.6 g/ml, 2.1-2.7 g/ml, 2.2-2.8 g/ml, 2.3-2.9 g/ml, 2.4-3 g/ml, 2.5-3.1 g/ml, 2.6-3.2 g/ml, 2.7-3.3 g/ml, 2.8-3.4 g/ml, 2.9-3.5 g/ml, 3-3.6 g/ml, 3.1-3.7 g/ml, 3.2-3.8 g/ml, 3.3-3.9 g/ml, 3.4-4 g/ml, 3.5-4.1 g/ml, 3.6-4.2 g/ml, 3.7-4.3 g/ml, 3.8-4.4 g/ml, 3.9-4.5 g/ml, 4-4.6 g/ml, 4.1-4.7 g/ml, 4.2-4.8 g/ml, 4.3-4.9 g/ml, 4.4-5 g/ml, 0.1-0.8 g/ml, 0.2-0.9 g/ml, 0.3-1 g/ml, 0.4-1.1 g/ml, 0.5-1.2 g/ml, 0.6-1.3 g/ml, 0.7-1.4 g/ml, 0.8-1.5 g/ml, 0.9-1.6 g/ml, 1-1.7 g/ml, 1.1-1.8 g/ml, 1.2-1.9 g/ml, 1.3-2 g/ml, 1.4-2.1 g/ml, 1.5-2.2 g/ml, 1.6-2.3 g/ml, 1.7-2.4 g/ml, 1.8-2.5 g/ml, 1.9-2.6 g/ml, 2-2.7 g/ml, 2.1-2.8 g/ml, 2.2-2.9 g/ml, 2.3-3 g/ml, 2.4-3.1 g/ml, 2.5-3.2 g/ml, 2.6-3.3 g/ml, 2.7-3.4 g/ml, 2.8-3.5 g/ml, 2.9-3.6 g/ml, 3-3.7 g/ml, 3.1-3.8 g/ml, 3.2-3.9 g/ml, 3.3-4 g/ml, 3.4-4.1 g/ml, 3.5-4.2 g/ml, 3.6-4.3 g/ml, 3.7-4.4 g/ml, 3.8-4.5 g/ml, 3.9-4.6 g/ml, 4-4.7 g/ml, 4.1-4.8 g/ml, 4.2-4.9 g/ml, 4.3-5 g/ml, 0.1-0.9 g/ml, 0.2-1 g/ml, 0.3-1.1 g/ml, 0.4-1.2 g/ml, 0.5-1.3 g/ml, 0.6-1.4 g/ml, 0.7-1.5 g/ml, 0.8-1.6 g/ml, 0.9-1.7 g/ml, 1-1.8 g/ml, 1.1-1.9 g/ml, 1.2-2 g/ml, 1.3-2.1 g/ml, 1.4-2.2 g/ml, 1.5-2.3 g/ml, 1.6-2.4 g/ml, 1.7-2.5 g/ml, 1.8-2.6 g/ml, 1.9-2.7 g/ml, 2-2.8 g/ml, 2.1-2.9 g/ml, 2.2-3 g/ml, 2.3-3.1 g/ml, 2.4-3.2 g/ml, 2.5-3.3 g/ml, 2.6-3.4 g/ml, 2.7-3.5 g/ml, 2.8-3.6 g/ml, 2.9-3.7 g/ml, 3-3.8 g/ml, 3.1-3.9 g/ml, 3.2-4 g/ml, 3.3-4.1 g/ml, 3.4-4.2 g/ml, 3.5-4.3 g/ml, 3.6-4.4 g/ml, 3.7-4.5 g/ml, 3.8-4.6 g/ml, 3.9-4.7 g/ml, 4-4.8 g/ml, 4.1-4.9 g/ml, 4.2-5 g/ml, 0.1-1 g/ml, 0.2-1.1 g/ml, 0.3-1.2 g/ml, 0.4-1.3 g/ml, 0.5-1.4 g/ml, 0.6-1.5 g/ml, 0.7-1.6 g/ml, 0.8-1.7 g/ml, 0.9-1.8 g/ml, 1-1.9 g/ml, 1.1-2 g/ml, 1.2-2.1 g/ml, 1.3-2.2 g/ml, 1.4-2.3 g/ml, 1.5-2.4 g/ml, 1.6-2.5 g/ml, 1.7-2.6 g/ml, 1.8-2.7 g/ml, 1.9-2.8 g/ml, 2-2.9 g/ml, 2.1-3 g/ml, 2.2-3.1 g/ml, 2.3-3.2 g/ml, 2.4-3.3 g/ml, 2.5-3.4 g/ml, 2.6-3.5 g/ml, 2.7-3.6 g/ml, 2.8-3.7 g/ml, 2.9-3.8 g/ml, 3-3.9 g/ml, 3.1-4 g/ml, 3.2-4.1 g/ml, 3.3-4.2 g/ml, 3.4-4.3 g/ml, 3.5-4.4 g/ml, 3.6-4.5 g/ml, 3.7-4.6 g/ml, 3.8-4.7 g/ml, 3.9-4.8 g/ml, 4-4.9 g/ml, 4.1-5 g/ml, 0.1-1.1 g/ml, 0.2-1.2 g/ml, 0.3-1.3 g/ml, 0.4-1.4 g/ml, 0.5-1.5 g/ml, 0.6-1.6 g/ml, 0.7-1.7 g/ml, 0.8-1.8 g/ml, 0.9-1.9 g/ml, 1-2 g/ml, 1.1-2.1 g/ml, 1.2-2.2 g/ml, 1.3-2.3 g/ml, 1.4-2.4 g/ml, 1.5-2.5 g/ml, 1.6-2.6 g/ml, 1.7-2.7 g/ml, 1.8-2.8 g/ml, 1.9-2.9 g/ml, 2-3 g/ml, 2.1-3.1 g/ml, 2.2-3.2 g/ml, 2.3-3.3 g/ml, 2.4-3.4 g/ml, 2.5-3.5 g/ml, 2.6-3.6 g/ml, 2.7-3.7 g/ml, 2.8-3.8 g/ml, 2.9-3.9 g/ml, 3-4 g/ml, 3.1-4.1 g/ml, 3.2-4.2 g/ml, 3.3-4.3 g/ml, 3.4-4.4 g/ml, 3.5-4.5 g/ml, 3.6-4.6 g/ml, 3.7-4.7 g/ml, 3.8-4.8 g/ml, 3.9-4.9 g/ml, 4-5 g/ml, 0.1-1.6 g/ml, 0.2-1.7 g/ml, 0.3-1.8 g/ml, 0.4-1.9 g/ml, 0.5-2 g/ml, 0.6-2.1 g/ml, 0.7-2.2 g/ml, 0.8-2.3 g/ml, 0.9-2.4 g/ml, 1-2.5 g/ml, 1.1-2.6 g/ml, 1.2-2.7 g/ml, 1.3-2.8 g/ml, 1.4-2.9 g/ml, 1.5-3 g/ml, 1.6-3.1 g/ml, 1.7-3.2 g/ml, 1.8-3.3 g/ml, 1.9-3.4 g/ml, 2-3.5 g/ml, 2.1-3.6 g/ml, 2.2-3.7 g/ml, 2.3-3.8 g/ml, 2.4-3.9 g/ml, 2.5-4 g/ml, 2.6-4.1 g/ml, 2.7-4.2 g/ml, 2.8-4.3 g/ml, 2.9-4.4 g/ml, 3-4.5 g/ml, 3.1-4.6 g/ml, 3.2-4.7 g/ml, 3.3-4.8 g/ml, 3.4-4.9 g/ml, 3.5-5 g/ml, 0.1-2.1 g/ml, 0.2-2.2 g/ml, 0.3-2.3 g/ml, 0.4-2.4 g/ml, 0.5-2.5 g/ml, 0.6-2.6 g/ml, 0.7-2.7 g/ml, 0.8-2.8 g/ml, 0.9-2.9 g/ml, 1-3 g/ml, 1.1-3.1 g/ml, 1.2-3.2 g/ml, 1.3-3.3 g/ml, 1.4-3.4 g/ml, 1.5-3.5 g/ml, 1.6-3.6 g/ml, 1.7-3.7 g/ml, 1.8-3.8 g/ml, 1.9-3.9 g/ml, 2-4 g/ml, 2.1-4.1 g/ml, 2.2-4.2 g/ml, 2.3-4.3 g/ml, 2.4-4.4 g/ml, 2.5-4.5 g/ml, 2.6-4.6 g/ml, 2.7-4.7 g/ml, 2.8-4.8 g/ml, 2.9-4.9 g/ml, 3-5 g/ml, 0.1-2.6 g/ml, 0.2-2.7 g/ml, 0.3-2.8 g/ml, 0.4-2.9 g/ml, 0.5-3 g/ml, 0.6-3.1 g/ml, 0.7-3.2 g/ml, 0.8-3.3 g/ml, 0.9-3.4 g/ml, 1-3.5 g/ml, 1.1-3.6 g/ml, 1.2-3.7 g/ml, 1.3-3.8 g/ml, 1.4-3.9 g/ml, 1.5-4 g/ml, 1.6-4.1 g/ml, 1.7-4.2 g/ml, 1.8-4.3 g/ml, 1.9-4.4 g/ml, 2-4.5 g/ml, 2.1-4.6 g/ml, 2.2-4.7 g/ml, 2.3-4.8 g/ml, 2.4-4.9 g/ml, 2.5-5 g/ml, 0.1-3.1 g/ml, 0.2-3.2 g/ml, 0.3-3.3 g/ml, 0.4-3.4 g/ml, 0.5-3.5 g/ml, 0.6-3.6 g/ml, 0.7-3.7 g/ml, 0.8-3.8 g/ml, 0.9-3.9 g/ml, 1-4 g/ml, 1.1-4.1 g/ml, 1.2-4.2 g/ml, 1.3-4.3 g/ml, 1.4-4.4 g/ml, 1.5-4.5 g/ml, 1.6-4.6 g/ml, 1.7-4.7 g/ml, 1.8-4.8 g/ml, 1.9-4.9 g/ml, and 2-5 g/ml.

In one embodiment, the specific gravity of the SBP formulation may be between 1.2-2 g/ml.

In one embodiment, the specific gravity of the SBP formulation may be 1.8-2 g/ml.

Shear Recovery

In some embodiments, the SBP formulation may optimized for shear recovery. As described herein, "shear recovery" describes the ability of a physical property of an SBP formulation to recover to a specific percent of its original measure within a specified time post-shear application. Properties that can be measured by methods known in the art may include, but are not limited to, G', G", phase angle, and/or viscosity.

In one embodiment, the shear recovery of the SBP formulation is greater than 75% at 1 second, 2 seconds, 3 seconds, 4 seconds, 5 seconds, 6 seconds, 7 seconds, 8 seconds, 9 seconds, 10 seconds, 11, seconds, 12 seconds, 13 seconds, 14 seconds, 15 seconds, 16 seconds, 17 seconds, 18 seconds, 19 seconds, 20 seconds, 25 seconds, 30 seconds, 35 seconds, 40 seconds, 45 seconds, 50 seconds, 55 seconds, 1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 7 minutes, 8 minutes, 9 minutes, 10 minutes, 11 minutes, 12 minutes, 13 minutes, 14 minutes, 15 minutes, 16 minutes 17 minutes, 18 minutes, 19 minutes, 20 minutes, 25 minutes, 30 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, 60 minutes or more than 60 minutes. As a non-limiting example, the shear recovery of the SBP formulation is greater than 75% at 1 minute. As a non-limiting example, the shear recovery of the SBP formulation is greater than 75% at 10 seconds.

In one embodiment, the shear recovery of the SBP formulation is greater than 80% at 1 second, 2 seconds, 3 seconds, 4 seconds, 5 seconds, 6 seconds, 7 seconds, 8 seconds, 9 seconds, 10 seconds, 11, seconds, 12 seconds, 13 seconds, 14 seconds, 15 seconds, 16 seconds, 17 seconds, 18 seconds, 19 seconds, 20 seconds, 25 seconds, 30 seconds, 35 seconds, 40 seconds, 45 seconds, 50 seconds, 55 seconds, 1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 7 minutes, 8 minutes, 9 minutes, 10 minutes, 11 minutes, 12 minutes, 13 minutes, 14 minutes, 15 minutes, 16 minutes 17 minutes, 18 minutes, 19 minutes, 20 minutes, 25 minutes, 30 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, 60 minutes or more than 60 minutes. As a non-limiting example, the shear recovery of the SBP formulation is greater than 80% at 1 minute. As a non-limiting example, the shear recovery of the SBP formulation is greater than 80% at 10 seconds.

In one embodiment, the shear recovery of the SBP formulation is greater than 85% at 1 second, 2 seconds, 3 seconds, 4 seconds, 5 seconds, 6 seconds, 7 seconds, 8 seconds, 9 seconds, 10 seconds, 11, seconds, 12 seconds, 13 seconds, 14 seconds, 15 seconds, 16 seconds, 17 seconds, 18 seconds, 19 seconds, 20 seconds, 25 seconds, 30 seconds, 35 seconds, 40 seconds, 45 seconds, 50 seconds, 55 seconds, 1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 7 minutes, 8 minutes, 9 minutes, 10 minutes, 11 minutes, 12 minutes, 13 minutes, 14 minutes, 15 minutes, 16 minutes 17 minutes, 18 minutes, 19 minutes, 20 minutes, 25 minutes, 30 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, 60 minutes or more than 60 minutes. As a non-limiting example, the shear recovery of the SBP formulation is greater than 85% at 1 minute. As a non-limiting example, the shear recovery of the SBP formulation is greater than 85% at 10 seconds.

In one embodiment, the shear recovery of the SBP formulation is greater than 90% at 1 second, 2 seconds, 3 seconds, 4 seconds, 5 seconds, 6 seconds, 7 seconds, 8 seconds, 9 seconds, 10 seconds, 11, seconds, 12 seconds, 13 seconds, 14 seconds, 15 seconds, 16 seconds, 17 seconds, 18 seconds, 19 seconds, 20 seconds, 25 seconds, 30 seconds, 35 seconds, 40 seconds, 45 seconds, 50 seconds, 55 seconds, 1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 7 minutes, 8 minutes, 9 minutes, 10 minutes, 11 minutes, 12 minutes, 13 minutes, 14 minutes, 15 minutes, 16 minutes 17 minutes, 18 minutes, 19 minutes, 20 minutes, 25 minutes, 30 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, 60 minutes or more than 60 minutes. As a non-limiting example, the shear recovery of the SBP formulation is greater than 90% at 1 minute. As a non-limiting example, the shear recovery of the SBP formulation is greater than 90% at 10 seconds.

In one embodiment, the shear recovery of the SBP formulation is greater than 95% at 1 second, 2 seconds, 3 seconds, 4 seconds, 5 seconds, 6 seconds, 7 seconds, 8 seconds, 9 seconds, 10 seconds, 11, seconds, 12 seconds, 13 seconds, 14 seconds, 15 seconds, 16 seconds, 17 seconds, 18 seconds, 19 seconds, 20 seconds, 25 seconds, 30 seconds, 35 seconds, 40 seconds, 45 seconds, 50 seconds, 55 seconds, 1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 7 minutes, 8 minutes, 9 minutes, 10 minutes, 11 minutes, 12 minutes, 13 minutes, 14 minutes, 15 minutes, 16 minutes 17 minutes, 18 minutes, 19 minutes, 20 minutes, 25 minutes, 30 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, 60 minutes or more than 60 minutes. As a non-limiting example, the shear recovery of the SBP formulation is greater than 95% at 1 minute. As a non-limiting example, the shear recovery of the SBP formulation is greater than 95% at 10 seconds.

In one embodiment, the shear recovery of the SBP formulation is greater than 99% at 1 second, 2 seconds, 3 seconds, 4 seconds, 5 seconds, 6 seconds, 7 seconds, 8 seconds, 9 seconds, 10 seconds, 11, seconds, 12 seconds, 13 seconds, 14 seconds, 15 seconds, 16 seconds, 17 seconds, 18 seconds, 19 seconds, 20 seconds, 25 seconds, 30 seconds, 35 seconds, 40 seconds, 45 seconds, 50 seconds, 55 seconds, 1 minute, 2 minutes, 3 minutes, 4 minutes, 5 minutes, 6 minutes, 7 minutes, 8 minutes, 9 minutes, 10 minutes, 11 minutes, 12 minutes, 13 minutes, 14 minutes, 15 minutes, 16 minutes 17 minutes, 18 minutes, 19 minutes, 20 minutes, 25 minutes, 30 minutes, 35 minutes, 40 minutes, 45 minutes, 50 minutes, 55 minutes, 60 minutes or more than 60 minutes. As a non-limiting example, the shear recovery of the SBP formulation is greater than 99% at 1 minute. As a non-limiting example, the shear recovery of the SBP formulation is greater than 99% at 10 seconds.

Stability and Degradation

In some embodiments, the SBP formulation may optimized for stability.

In one embodiment, the SBP formulation may have an in vivo degradation rate of greater than 10 days, 20 days, 30 days, 40 days, 50 days, 60 days, 70 days, 80 days, 90 days, 100 days, 120 days, 140 days, 160 days, 180 days, 200 days, 250 days, 300 days, 350 days, or 400 days. As a non-limiting example, the in vivo degradation rate is greater than 60 days. As another non-limiting example, the in vivo degradation rate is greater than 120 days.

In one embodiment, the SBP formulation may have an in vivo degradation rate of greater than 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 hours, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 days, 1, 2, 3, 4 weeks, or 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 months. In some instances, there is no change in weight of the sample over the period. As a non-limiting example, the SBP formulation may have an in vivo degradation rate of greater than 7 days and there is no change in weight in the sample. As a non-limiting example, the SBP formulation may have an in vivo degradation rate of greater than 14 days and there is no change in weight in the sample.

In one embodiment, the SBP formulation may be stable at room temperature for at least 1 month, 2 months, 3 months, 4 months, 5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 12 months, 18 months, 1 year, 2 years, 3 years, 4 years, 5 years, 6 years, 7 years, 8 years, 9 years or more than 9 years. As a non-limiting example, the SBP formulation is stable at room temperature for 2 years. As another non-limiting example, the SBP formulation is stable at room temperature for 3 years.

Endotoxin

In some embodiments, the SBP formulation may optimized for a lower endotoxin level.

In one embodiment, the endotoxin level in the SBP formulation is less than 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 EU/g. As a non-limiting example, the endotoxin level is less than 100 EU/g.

In one embodiment, the endotoxin level in the SBP formulation is less than 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, or 100 EU/mL. As a non-limiting example, the endotoxin level is less than 1 EU/mL by the Limulus amebocyte lysate (LAL) method.

In one embodiment, the endotoxin level in the SBP formulation is between 0.5-5, 1-10, 5-10, 5-15, 10-20, 10-25, 10-30, 10-40, 10-50, 10-60, 10-70, 10-80, 10-90, 10-100, 25-50, 25-75, 25-100, 50-75, 50-100, or 75-100 EU/g. As a non-limiting example, the endotoxin level of the SBP formulation is between 0.5-5 EU/g.

Rheological Properties

In some embodiments, SBP formulations may be optimized to modulate SBP rheological properties, including, but not limited to, viscosity, storage modulus (G'), loss modulus, and phase angle. As used herein, the term "viscosity" refers to a measure of a material's resistance to flow and may include shear viscosity or interfacial viscosity. As used herein, the term "shear storage modulus" refers to the measure of a material's elasticity or reversible deformation as determined by the material's stored energy. As used herein, the term "shear loss modulus" refers to the measure of a material's ability to dissipate energy, usually in the form of heat. As used herein, the term "phase angle" refers to the difference in the stress and strain applied to a material during the application of oscillating shear stress. The viscosity and other rheological properties of a composition (e.g., a gel, e.g., hydrogel or organogel) provided herein can be determined using a rotational viscometer or rheometer. Additional methods for determining the rheological properties of a composition (e.g., gel, e.g., hydrogel or organogel) and other properties of the composition are known in the art. In some embodiments, SBP rheological properties may be altered by the incorporation of an excipient that is a gelling agent. In some embodiments, the identity of the excipient (e.g. PEG or poloxamer) may be altered to tune the rheological properties of SBPs. In some embodiments, the rheological properties of SBPs may be tuned for the desired application (e.g. tissue engineering scaffold, drug delivery system, surgical implant, etc.).

In some embodiments, the viscosity of SBPs is tunable between 1-1000 centipoise (cP). In some embodiments, the viscosity of an SBP is tunable from about 0.0001 to about 1000 Pascal seconds (Pa*s). In some embodiments, the viscosity of an SBP is from about 1 cP to about 10 cP, from about 2 cP to about 20 cP, from about 3 cP to about 30 cP, from about 4 cP to about 40 cP, from about 5 cP to about 50 cP, from about 6 cP to about 60 cP, from about 7 cP to about 70 cP, from about 8 cP to about 80 cP, from about 9 cP to about 90 cP, from about 10 cP to about 100 cP, from about 100 cP to about 150 cP, from about 150 cP to about 200 cP, from about 200 cP to about 250 cP, from about 250 cP to about 300 cP, from about 300 cP to about 350 cP, from about 350 cP to about 400 cP, from about 400 cP to about 450 cP, from about 450 cP to about 500 cP, from about 500 cP to about 600 cP, from about 550 cP to about 700 cP, from about 600 cP to about 800 cP, from about 650 cP to about 900 cP, or from about 700 cP to about 1000 cP. In some embodiments, the viscosity of an SBP is from about 0.0001 Pa*s to about 0.001 Pa*s, from about 0.001 Pa*s to about 0.01 Pa*s, from about 0.01 Pa*s to about 0.1 Pa*s, from about 0.1 Pa*s to about 1 Pa*s, from about 1 Pa*s to about 10 Pa*s, from about 2 Pa*s to about 20 Pa*s, from about 3 Pa*s to about 30 Pa*s, from about 4 Pa*s to about 40 Pa*s, from about 5 Pa*s to about 50 Pa*s, from about 6 Pa*s to about 60 Pa*s, from about 7 Pa*s to about 70 Pa*s, from about 8 Pa*s to about 80 Pa*s, from about 9 Pa*s to about 90 Pa*s, from about 10 Pa*s to about 100 Pa*s, from about 100 Pa*s to about 150 Pa*s, from about 150 Pa*s to about 200 Pa*s, from about 200 Pa*s to about 250 Pa*s, from about 250 Pa*s to about 300 Pa*s, from about 300 Pa*s to about 350 Pa*s, from about 350 Pa*s to about 400 Pa*s, from about 400 Pa*s to about 450 Pa*s, from about 450 Pa*s to about 500 Pa*s, from about 500 Pa*s to about 600 Pa*s, from about 550 Pa*s to about 700 Pa*s, from about 600 Pa*s to about 800 Pa*s, from about 650 Pa*s to about 900 Pa*s, from about 700 Pa*s to about 1000 Pa*s or from about 10 Pa*s to about 2500 Pa*s.

In some embodiments, the SBP formulations may shear thin or display shear thinning properties. As used herein, the term "shear thinning" refers to a decrease in viscosity at increasing shear rates. As used herein, the term "shear rate" refers to the rate of change in the ratio of displacement of material upon the application of a shear force to the height of the material. This ratio is also known as strain.

In some embodiments, the storage modulus and/or the loss modulus (G' and G" respectively) of SBPs is tunable between 0.0001-20000 Pascals (Pa). In some embodiments, the storage modulus and/or the loss modulus of SBPs is from about 0.0001 Pa to about 0.001 Pa, from about 0.001 Pa to about 0.01 Pa, from about 0.01 Pa to about 0.1 Pa, from about 0.1 Pa to about 1 Pa, from about 1 Pa to about 10 Pa, from about 2 Pa to about 20 Pa, from about 3 Pa to about 30 Pa, from about 4 Pa to about 40 Pa, from about 5 Pa to about 50 Pa, from about 6 Pa to about 60 Pa, from about 7 Pa to about 70 Pa, from about 8 Pa to about 80 Pa, from about 9 Pa to about 90 Pa, from about 10 Pa to about 100 Pa, from about 100 Pa to about 150 Pa, from about 150 Pa to about 200 Pa, from about 200 Pa to about 250 Pa, from about 250 Pa to about 300 Pa, from about 300 Pa to about 350 Pa, from about 350 Pa to about 400 Pa, from about 400 Pa to about 450 Pa, from about 450 Pa to about 500 Pa, from about 500 Pa to about 600 Pa, from about 550 Pa to about 700 Pa, from about 600 Pa to about 800 Pa, from about 650 Pa to about 900 Pa, from about 700 Pa to about 1000 Pa, from about 1000 Pa to about 1500 Pa, from about 1500 Pa to about 2000 Pa, from about 2000 Pa to about 2500 Pa, from about 2500 Pa to about 3000 Pa, from about 3000 Pa to about 3500 Pa, from about 3500 Pa to about 4000 Pa, from about 4000 Pa to about 4500 Pa, from about 4500 Pa to about 5000 Pa, from about 5000 Pa to about 5500 Pa, from about 5500 Pa to about 6000 Pa, from about 6000 Pa to about 6500 Pa, from about 6500 Pa to about 7000 Pa, from about 7000 Pa to about 7500 Pa, from about 7500 Pa to about 8000 Pa, from about 8000 Pa to about 8500 Pa, from about 8500 Pa to about 9000 Pa, from about 9000 Pa to about 9500 Pa, from about 9500 Pa to about 10000 Pa, from about 10000 Pa to about 11000 Pa, from about 11000 Pa to about 12000 Pa, from about 12000 Pa to about 13000 Pa, from about 13000 Pa to about 14000 Pa, from about 14000 Pa to about 15000 Pa, from about 15000 Pa to about 16000 Pa, from about 16000 Pa to about 17000 Pa, from about 17000 Pa to about 18000 Pa, from about 18000 Pa to about 19000 Pa, or from about 19000 Pa to about 20000 Pa.

In some embodiments, the phase angle of SBPs is tunable between 1°-90°). In some embodiments, the phase angle of SBPs is from about 1° to about 2°, from about 2° to about 3°, from about 3° to about 4°, from about 4° to about 5°, from about 5° to about 6°, from about 6° to about 7°, from about 7° to about 8°, from about 8° to about 9°, from about 9° to about 10°, from about 10° to about 15°, from about 15° to about 20°, from about 20° to about 25°, from about 25° to about 30°, from about 30° to about 35°, from about 35° to about 40°, from about 40° to about 45°, from about 45° to about 50°, from about 50° to about 55°, from about 55° to about 60°, from about 60° to about 65°, from about 65° to about 70°, from about 70° to about 75°, from about 75° to about 80°, from about 80° to about 85°, or from about 85° to about 90°.

Stress Resistance

In some embodiments, SBPs may be formulated to modulate SBP resistance to stress. Resistance to stress may be measured using one or more rheological measurements. Such measurements may include, but are not limited to tensile elasticity, shear or rigidity, volumetric elasticity, and compression. Additional rheological measurements and properties may include any of those taught in Zhang et al. (2017) Fiber and Polymers 18(10):1831-1840; McGill et al. (2017) Acta Biomaterialia 63:76-84; and Choi et al. (2015) In-Situ Gelling Polymers, Series in BioEngineering doi. 10.1007/978-981-287-152-7_2, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, stress resistance may be modulated through incorporation of excipients (e.g., PEG or poloxamer). In some embodiments, SBP stress-resistance properties may be modulated to suit a specific application (e.g., tissue engineering scaffold, drug delivery system, surgical implant, etc.).

In some embodiments, stress resistance may be measured by shear recovery experiments. In some embodiments, the SBPs recover 100% of their viscosity from before the application of shear forces. In some embodiments, the SBPs recover from 0.1%-5%, from 5%-10%, from 10% to 25%, from 25% to 50%, from 50% to 75%, or from 75% to 100% of their viscosity from before the application of shear forces. Shear recovery may be measured via any method known to one skilled in the art. In some embodiments, shear recovery occurs over the course of 1 second, 10 seconds, 30 seconds, or one minute.

Osmolarity

In some embodiments, SBP formulations may include processed silk with or without other components (e.g., excipients and cargo). The SBP formulations may contain an from about 1 mOsm to about 10 mOsm, from about 2 mOsm to about 20 mOsm, from about 3 mOsm to about 30 mOsm, from about 4 mOsm to about 40 mOsm, from about 5 mOsm to about 50 mOsm, from about 6 mOsm to about 60 mOsm, from about 7 mOsm to about 70 mOsm, from about 8 mOsm to about 80 mOsm, from about 9 mOsm to about 90 mOsm, from about 10 mOsm to about 100 mOsm, from about 15 mOsm to about 150 mOsm, from about 25 mOsm to about 200 mOsm, from about 35 mOsm to about 250 mOsm, from about 45 mOsm to about 300 mOsm, from about 55 mOsm to about 350 mOsm, from about 65 mOsm to about 400 mOsm, from about 75 mOsm to about 450 mOsm, from about 85 mOsm to about 500 mOsm, from about 125 mOsm to about 600 mOsm, from about 175 mOsm to about 700 mOsm, from about 225 mOsm to about 800 mOsm, from about 275 mOsm to about 285 mOsm, from about 280 mOsm to about 900 mOsm, or from about 325 mOsm to about 1000 mOsm. The SBPs may have an osmolarity of from about 1 mOsm/L to about 10 mOsm/L, from about 2 mOsm/L to about 20 mOsm/L, from about 3 mOsm/L to about 30 mOsm/L, from about 4 mOsm/L to about 40 mOsm/L, from about 5 mOsm/L to about 50 mOsm/L, from about 6 mOsm/L to about 60 mOsm/L, from about 7 mOsm/L to about 70 mOsm/L, from about 8 mOsm/L to about 80 mOsm/L, from about 9 mOsm/L to about 90 mOsm/L, from about 10 mOsm/L to about 100 mOsm/L, from about 15 mOsm/L to about 150 mOsm/L, from about 25 mOsm/L to about 200 mOsm/L, from about 35 mOsm/L to about 250 mOsm/L, from about 45 mOsm/L to about 300 mOsm/L, from about 55 mOsm/L to about 350 mOsm/L, from about 65 mOsm/L to about 400 mOsm/L, from about 75 mOsm/L to about 450 mOsm/L, from about 85 mOsm/L to about 500 mOsm/L, from about 125 mOsm/L to about 600 mOsm/L, from about 175 mOsm/L to about 700 mOsm/L, from about 225 mOsm/L to about 800 mOsm/L, from about 275 mOsm/L to about 285 mOsm/L, from about 280 mOsm/L to about 900 mOsm/L, or from about 325 mOsm/L to about 1000 mOsm/L

In some embodiments, the SBP formulation has an osmolarity from about 290-320 mOsm/L.

In some embodiment, the SBP formulation has an osmolarity of 280 mOsm/L.

In some embodiment, the SBP formulation has an osmolarity of 290 mOsm/L.

Concentrations and Ratios of SBP Components

SBP formulations may include formulations of processed silk with other components (e.g., excipient, therapeutic agent, microbe, cargo, and/or biological system), wherein each component is present at a specific concentration, ratio, or range of concentrations or ratios, depending on application. In some embodiments, the concentration of processed silk or other SBP component (e.g., excipient, therapeutic agent, microbe, cargo, and/or biological system) is present in SBP formulations at a concentration (by weight, volume, or concentration) of from about 0.0001% to about 0.001%, from about 0.001% to about 0.01%, from about 0.01% to about 1%, from about 0.05% to about 2%, from about 1% to about 5%, from about 2% to about 10%, from about 4% to about 16%, from about 5% to about 20%, from about 8% to about 24%, from about 10% to about 30%, from about 12% to about 32%, from about 14% to about 34%, from about 16% to about 36%, from about 18% to about 38%, from about 20% to about 40%, from about 22% to about 42%, from about 24% to about 44%, from about 26% to about 46%, from about 28% to about 48%, from about 30% to about 50%, from about 35% to about 55%, from about 40% to about 60%, from about 45% to about 65%, from about 50% to about 70%, from about 55% to about 75%, from about 60% to about 80%, from about 65% to about 85%, from about 70% to about 90%, from about 75% to about 95%, from about 80% to about 96%, from about 85% to about 97%, from about 90% to about 98%, from about 95% to about 99%, from about 96% to about 99.2%, from about 97% to about 99.5%, from about 98% to about 99.8%, from about 99% to about 99.9%, or greater than 99.9%.

In some embodiments, the concentration of processed silk or other SBP component (e.g., excipient, therapeutic agent, microbe, cargo, and/or biological system) is present in SBP formulations at a concentration of from about 0.0001% (w/v) to about 0.001% (w/v), from about 0.001% (w/v) to about 0.01% (w/v), from about 0.01% (w/v) to about 1% (w/v), from about 0.05% (w/v) to about 2% (w/v), from about 1% (w/v) to about 5% (w/v), from about 2% (w/v) to about 10% (w/v), from about 4% (w/v) to about 16% (w/v), from about 5% (w/v) to about 20% (w/v), from about 8% (w/v) to about 24% (w/v), from about 10% (w/v) to about 30% (w/v), from about 12% (w/v) to about 32% (w/v), from about 14% (w/v) to about 34% (w/v), from about 16% (w/v) to about 36% (w/v), from about 18% (w/v) to about 38% (w/v), from about 20% (w/v) to about 40% (w/v), from about 22% (w/v) to about 42% (w/v), from about 24% (w/v) to about 44% (w/v), from about 26% (w/v) to about 46% (w/v), from about 28% (w/v) to about 48% (w/v), from about 30% (w/v) to about 50% (w/v), from about 35% (w/v) to about 55% (w/v), from about 40% (w/v) to about 60% (w/v), from about 45% (w/v) to about 65% (w/v), from about 50% (w/v) to about 70% (w/v), from about 55% (w/v) to about 75% (w/v), from about 60% (w/v) to about 80% (w/v), from about 65% (w/v) to about 85% (w/v), from about 70% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 80% (w/v) to about 96% (w/v), from about 85% (w/v) to about 97% (w/v), from about 90% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 96% (w/v) to about 99.2% (w/v), from about 97% (w/v) to about 99.5% (w/v), from about 98% (w/v) to about 99.8% (w/v), from about 99% (w/v) to about 99.9% (w/v), or greater than 99.9% (w/v).

In some embodiments, the concentration of processed silk or other SBP component (e.g., excipient, therapeutic agent, microbe, cargo, and/or biological system) is present in SBP formulations at a concentration of from about 0.0001% (v/v) to about 0.001% (v/v), from about 0.001% (v/v) to about 0.01% (v/v), from about 0.01% (v/v) to about 1% (v/v), from about 0.05% (v/v) to about 2% (v/v), from about 1% (v/v) to about 5% (v/v), from about 2% (v/v) to about 10% (v/v), from about 4% (v/v) to about 16% (v/v), from about 5%

(v/v) to about 20% (v/v), from about 8% (v/v) to about 24% (v/v), from about 10% (v/v) to about 30% (v/v), from about 12% (v/v) to about 32% (v/v), from about 14% (v/v) to about 34% (v/v), from about 16% (v/v) to about 36% (v/v), from about 18% (v/v) to about 38% (v/v), from about 20% (v/v) to about 40% (v/v), from about 22% (v/v) to about 42% (v/v), from about 24% (v/v) to about 44% (v/v), from about 26% (v/v) to about 46% (v/v), from about 28% (v/v) to about 48% (v/v), from about 30% (v/v) to about 50% (v/v), from about 35% (v/v) to about 55% (v/v), from about 40% (v/v) to about 60% (v/v), from about 45% (v/v) to about 65% (v/v), from about 50% (v/v) to about 70% (v/v), from about 55% (v/v) to about 75% (v/v), from about 60% (v/v) to about 80% (v/v), from about 65% (v/v) to about 85% (v/v), from about 70% (v/v) to about 90% (v/v), from about 75% (v/v) to about 95% (v/v), from about 80% (v/v) to about 96% (v/v), from about 85% (v/v) to about 97% (v/v), from about 90% (v/v) to about 98% (v/v), from about 95% (v/v) to about 99% (v/v), from about 96% (v/v) to about 99.2% (v/v), from about 97% (v/v) to about 99.5% (v/v), from about 98% (v/v) to about 99.8% (v/v), from about 99% (v/v) to about 99.9% (v/v), or greater than 99.9% (v/v).

In some embodiments, the concentration of processed silk or other SBP component (e.g., excipient, therapeutic agent, microbe, cargo, and/or biological system) is present in SBP formulations at a concentration of from about 0.0001% (w/w) to about 0.001% (w/w), from about 0.001% (w/w) to about 0.01% (w/w), from about 0.01% (w/w) to about 1% (w/w), from about 0.05% (w/w) to about 2% (w/w), from about 1% (w/w) to about 5% (w/w), from about 2% (w/w) to about 10% (w/w), from about 4% (w/w) to about 16% (w/w), from about 5% (w/w) to about 20% (w/w), from about 8% (w/w) to about 24% (w/w), from about 10% (w/w) to about 30% (w/w), from about 12% (w/w) to about 32% (w/w), from about 14% (w/w) to about 34% (w/w), from about 16% (w/w) to about 36% (w/w), from about 18% (w/w) to about 38% (w/w), from about 20% (w/w) to about 40% (w/w), from about 22% (w/w) to about 42% (w/w), from about 24% (w/w) to about 44% (w/w), from about 26% (w/w) to about 46% (w/w), from about 28% (w/w) to about 48% (w/w), from about 30% (w/w) to about 50% (w/w), from about 35% (w/w) to about 55% (w/w), from about 40% (w/w) to about 60% (w/w), from about 45% (w/w) to about 65% (w/w), from about 50% (w/w) to about 70% (w/w), from about 55% (w/w) to about 75% (w/w), from about 60% (w/w) to about 80% (w/w), from about 65% (w/w) to about 85% (w/w), from about 70% (w/w) to about 90% (w/w), from about 75% (w/w) to about 95% (w/w), from about 80% (w/w) to about 96% (w/w), from about 85% (w/w) to about 97% (w/w), from about 90% (w/w) to about 98% (w/w), from about 95% (w/w) to about 99% (w/w), from about 96% (w/w) to about 99.2% (w/w), from about 97% (w/w) to about 99.5% (w/w), from about 98% (w/w) to about 99.8% (w/w), from about 99% (w/w) to about 99.9% (w/w), or greater than 99.9% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 1% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 2% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 3% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 4% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 5% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 6% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 10% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 20% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 30% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 16.7% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 20% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 23% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 25% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 27.3% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 28.6% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 33.3% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 40% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 50% (w/w).

In some embodiments, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient, therapeutic agent, microbe, cargo, and/or biological system) is present in SBP formulations at a concentration of from about 0.01 pg/mL to about 1 pg/mL, from about 0.05 pg/mL to about 2 pg/mL, from about 1 pg/mL to about 5 pg/mL, from about 2 pg/mL to about 10 pg/mL, from about 4 pg/mL to about 16 pg/mL, from about 5 pg/mL to about 20 pg/mL, from about 8 pg/mL to about 24 pg/mL, from about 10 pg/mL to about 30 pg/mL, from about 12 pg/mL to about 32 pg/mL, from about 14 pg/mL to about 34 pg/mL, from about 16 pg/mL to about 36 pg/mL, from about 18 pg/mL to about 38 pg/mL, from about 20 pg/mL to about 40 pg/mL, from about 22 pg/mL to about 42 pg/mL, from about 24 pg/mL to about 44 pg/mL, from about 26 pg/mL to about 46 pg/mL, from about 28 pg/mL to about 48 pg/mL, from about 30 pg/mL to about 50 pg/mL, from about 35 pg/mL to about 55 pg/mL, from about 40 pg/mL to about 60 pg/mL, from about 45 pg/mL to about 65 pg/mL, from about 50 pg/mL to about 75 pg/mL, from about 60 pg/mL to about 240 pg/mL, from about 70 pg/mL to about 350 pg/mL, from about 80 pg/mL to about 400 pg/mL, from about 90 pg/mL to about 450 pg/mL, from about 100 pg/mL to about 500 pg/mL, from about 0.01 ng/mL to about 1 ng/mL, from about 0.05 ng/mL to about 2 ng/mL, from about 1 ng/mL to about 5 ng/mL, from about 2 ng/mL to about 10 ng/mL, from about 4 ng/mL to about 16 ng/mL, from about 5 ng/mL to about 20 ng/mL, from about 8 ng/mL to about 24 ng/mL, from about 10 ng/mL to about 30 ng/mL, from about 12 ng/mL to about 32 ng/mL, from about 14 ng/mL to about 34 ng/mL, from about 16 ng/mL to about 36 ng/mL, from about 18 ng/mL to about 38 ng/mL, from about 20 ng/mL to about 40 ng/mL, from about 22 ng/mL to about 42 ng/mL, from about 24 ng/mL to about 44 ng/mL, from about 26 ng/mL to about 46 ng/mL, from about 28 ng/mL to about 48 ng/mL, from about 30 ng/mL to about 50 ng/mL, from about 35 ng/mL to about 55 ng/mL, from about 40 ng/mL to about 60 ng/mL, from about 45 ng/mL to about 65 ng/mL, from about 50 ng/mL to about 75 ng/mL, from about 60 ng/mL to about 240 ng/mL, from about 70 ng/mL to about 350 ng/mL, from about 80 ng/mL to about 400 ng/mL, from about 90 ng/mL to about 450 ng/mL, from about 100 ng/mL to about 500 ng/mL, from about 0.01 μg/mL to about 1 μg/mL, from about 0.05 μg/mL to about 2 μg/mL, from about 1 μg/mL to about 5 μg/mL, from about 2 μg/mL to about 10 μg/mL, from about 4 μg/mL to about 16 μg/mL, from about 5 μg/mL to about 20 μg/mL, from about 8 μg/mL to about 24 μg/mL, from about 10 μg/mL to about 30 μg/mL, from about 12 μg/mL to about 32 μg/mL, from about 14 μg/mL to about 34 μg/mL, from about 16 μg/mL to about 36 μg/mL, from about 18 μg/mL to about 38 μg/mL, from about 20 μg/mL to about 40 μg/mL, from about 22 μg/mL to about 42 μg/mL, from about 24 μg/mL to about 44 μg/mL, from about 26 μg/mL to about 46 μg/mL, from about 28 μg/mL to about 48 μg/mL, from about 30 μg/mL to about 50 μg/mL, from about 35 μg/mL to about 55 μg/mL, from about 40 μg/mL to about 60 μg/mL, from about 45 μg/mL to about 65 μg/mL, from about 50 μg/mL to about 75 μg/mL, from about 60 μg/mL to about 240 μg/mL, from about 70 μg/mL to about 350 μg/mL, from about 80 μg/mL to about 400 μg/mL, from about 90 μg/mL to about 450 μg/mL, from about 100 μg/mL to about 500 μg/mL, from about 0.01 mg/mL to about 1 mg/mL, from about 0.05 mg/mL to about 2 mg/mL, from about 1 mg/mL to about 5 mg/mL, from about 2 mg/mL to about 10 mg/mL, from about 4 mg/mL to about 16 mg/mL, from about 5 mg/mL to about 20 mg/mL, from about 8 mg/mL to about 24 mg/mL, from about 10 mg/mL to about 30 mg/mL, from about 12 mg/mL to about 32 mg/mL, from about 14 mg/mL to about 34 mg/mL, from about 16 mg/mL to about 36 mg/mL, from about 18 mg/mL to about 38 mg/mL, from about 20 mg/mL to about 40 mg/mL, from about 22 mg/mL to about 42 mg/mL, from about 24 mg/mL to about 44 mg/mL, from about 26 mg/mL to about 46 mg/mL, from about 28 mg/mL to about 48 mg/mL, from about 30 mg/mL to about 50 mg/mL, from about 35 mg/mL to about 55 mg/mL, from about 40 mg/mL to about 60 mg/mL, from about 45 mg/mL to about 65 mg/mL, from about 50 mg/mL to about 75 mg/mL, from about 60 mg/mL to about 240 mg/mL, from about 70 mg/mL to about 350 mg/mL, from about 80 mg/mL to about 400 mg/mL, from about 90 mg/mL to about 450 mg/mL, from about 100 mg/mL to about 500 mg/mL, from about 0.01 g/mL to about 1 g/mL, from about 0.05 g/mL to about 2 g/mL, from about 1 g/mL to about 5 g/mL, from about 2 g/mL to about 10 g/mL, from about 4 g/mL to about 16 g/mL, or from about 5 g/mL to about 20 g/mL.

In some embodiments, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient, therapeutic agent, microbe, cargo, and/or biological system) is present in SBP formulations at a concentration of from about 0.01 pg/kg to about 1 pg/kg, from about 0.05 pg/kg to about 2 pg/kg, from about 1 μg/kg to about 5 pg/kg, from about 2 pg/kg to about 10 pg/kg, from about 4 pg/kg to about 16 pg/kg, from about 5 pg/kg to about 20 pg/kg, from about 8 pg/kg to about 24 pg/kg, from about 10 pg/kg to about 30 pg/kg, from about 12 pg/kg to about 32 pg/kg, from about 14 pg/kg to about 34 pg/kg, from about 16 pg/kg to about 36 pg/kg, from about 18 pg/kg to about 38 pg/kg, from about 20 pg/kg to about 40 pg/kg, from about 22 pg/kg to about 42 pg/kg, from about 24 pg/kg to about 44 pg/kg, from about 26 pg/kg to about 46 pg/kg, from about 28 pg/kg to about 48 pg/kg, from about 30 pg/kg to about 50 pg/kg, from about 35 pg/kg to about 55 pg/kg, from about 40 pg/kg to about 60 pg/kg, from about 45 pg/kg to about 65 pg/kg, from about 50 pg/kg to about 75 pg/kg, from about 60 pg/kg to about 240 pg/kg, from about 70 pg/kg to about 350 pg/kg, from about 80 pg/kg to about 400 pg/kg, from about 90 pg/kg to about 450 pg/kg, from about 100 pg/kg to about 500 pg/kg, from about 0.01 ng/kg to about 1 ng/kg, from about 0.05 ng/kg to about 2 ng/kg, from about 1 ng/kg to about 5 ng/kg, from about 2 ng/kg to about 10 ng/kg, from about 4 ng/kg to about 16 ng/kg, from about 5 ng/kg to about 20 ng/kg, from about 8 ng/kg to about 24 ng/kg, from about 10 ng/kg to about 30 ng/kg, from about 12 ng/kg to about 32 ng/kg, from about 14 ng/kg to about 34 ng/kg, from about 16 ng/kg to about 36 ng/kg, from about 18 ng/kg to about 38 ng/kg, from about 20 ng/kg to about 40 ng/kg, from about 22 ng/kg to about 42 ng/kg, from about 24 ng/kg to about 44 ng/kg, from about 26 ng/kg to about 46 ng/kg, from about 28 ng/kg to about 48 ng/kg, from about 30 ng/kg to about 50 ng/kg, from about 35 ng/kg to about 55 ng/kg, from about 40 ng/kg to about 60 ng/kg, from about 45 ng/kg to about 65 ng/kg, from about 50 ng/kg to about 75 ng/kg, from about 60 ng/kg to about 240 ng/kg, from about 70 ng/kg to about 350 ng/kg, from about 80 ng/kg to about 400 ng/kg, from about 90 ng/kg to about 450 ng/kg, from about 100 ng/kg to about 500 ng/kg, from about 0.01 μg/kg to about 1 μg/kg, from about 0.05 μg/kg to about 2 μg/kg, from about 1 μg/kg to about 5 μg/kg, from about 2 μg/kg to about 10 μg/kg, from about 4 μg/kg to about 16 μg/kg, from about 5 μg/kg to about 20 μg/kg, from about 8 μg/kg to about 24 μg/kg, from about 10 μg/kg to about 30 μg/kg, from about 12 μg/kg to about 32 μg/kg, from about 14 μg/kg to about 34 μg/kg, from about 16 μg/kg to about 36 μg/kg, from about 18 μg/kg to about 38 μg/kg, from about 20 μg/kg to about 40 μg/kg, from about 22 μg/kg to about 42 μg/kg, from about 24 μg/kg to about 44 μg/kg, from about 26 μg/kg to about 46 μg/kg, from about 28 μg/kg to about 48 μg/kg, from about 30 μg/kg to about 50 μg/kg, from about 35 μg/kg to about 55 μg/kg, from about 40 μg/kg to about 60 μg/kg, from about 45 μg/kg to about 65 μg/kg, from about 50 μg/kg to about 75 μg/kg, from about 60 μg/kg to about 240 μg/kg, from about 70 μg/kg to about 350 μg/kg, from about 80 μg/kg to about 400 μg/kg, from about 90 μg/kg to about 450 μg/kg, from about 100 μg/kg to about 500 μg/kg, from about 0.01 mg/kg to about 1 mg/kg, from about 0.05 mg/kg to about 2 mg/kg, from about 1 mg/kg to about 5 mg/kg, from about 2 mg/kg to about 10 mg/kg, from about 4 mg/kg to about 16 mg/kg, from about 5 mg/kg to about 20 mg/kg, from about 8 mg/kg to about 24 mg/kg, from about 10 mg/kg to about 30 mg/kg, from about 12 mg/kg to about 32 mg/kg, from about 14 mg/kg to about 34 mg/kg, from about 16 mg/kg to about 36 mg/kg, from about 18 mg/kg to about 38 mg/kg, from about 20 mg/kg to about 40 mg/kg, from about 22 mg/kg to about 42 mg/kg, from about 24 mg/kg to about 44 mg/kg, from about 26 mg/kg to about 46 mg/kg, from about 28 mg/kg to about 48 mg/kg, from about 30 mg/kg to about 50 mg/kg, from about 35 mg/kg to about 55 mg/kg, from about 40 mg/kg to about 60 mg/kg, from about 45 mg/kg to about 65 mg/kg, from about 50 mg/kg to about 75 mg/kg, from about 60 mg/kg to about 240 mg/kg, from about

70 mg/kg to about 350 mg/kg, from about 80 mg/kg to about 400 mg/kg, from about 90 mg/kg to about 450 mg/kg, from about 100 mg/kg to about 500 mg/kg, from about 0.01 g/kg to about 1 g/kg, from about 0.05 g/kg to about 2 g/kg, from about 1 g/kg to about 5 g/kg, from about 2 g/kg to about 10 g/kg, from about 4 g/kg to about 16 g/kg, or from about 5 g/kg to about 20 g/kg, from about 10 g/kg to about 50 g/kg, from about 15 g/kg to about 100 g/kg, from about 20 g/kg to about 150 g/kg, from about 25 g/kg to about 200 g/kg, from about 30 g/kg to about 250 g/kg, from about 35 g/kg to about 300 g/kg, from about 40 g/kg to about 350 g/kg, from about 45 g/kg to about 400 g/kg, from about 50 g/kg to about 450 g/kg, from about 55 g/kg to about 500 g/kg, from about 60 g/kg to about 550 g/kg, from about 65 g/kg to about 600 g/kg, from about 70 g/kg to about 650 g/kg, from about 75 g/kg to about 700 g/kg, from about 80 g/kg to about 750 g/kg, from about 85 g/kg to about 800 g/kg, from about 90 g/kg to about 850 g/kg, from about 95 g/kg to about 900 g/kg, from about 100 g/kg to about 950 g/kg, or from about 200 g/kg to about 1000 g/kg.

In some embodiments, the concentration of processed silk or other SBP component (e.g., excipient, therapeutic agent, microbe, cargo, and/or biological system) is present in SBP formulations at a concentration of from about 0.1 pM to about 1 pM, from about 1 pM to about 10 pM, from about 2 pM to about 20 pM, from about 3 pM to about 30 pM, from about 4 pM to about 40 pM, from about 5 pM to about 50 pM, from about 6 pM to about 60 pM, from about 7 pM to about 70 pM, from about 8 pM to about 80 pM, from about 9 pM to about 90 pM, from about 10 pM to about 100 pM, from about 11 pM to about 110 pM, from about 12 pM to about 120 pM, from about 13 pM to about 130 pM, from about 14 pM to about 140 pM, from about 15 pM to about 150 pM, from about 16 pM to about 160 pM, from about 17 pM to about 170 pM, from about 18 pM to about 180 pM, from about 19 pM to about 190 pM, from about 20 pM to about 200 pM, from about 21 pM to about 210 pM, from about 22 pM to about 220 pM, from about 23 pM to about 230 pM, from about 24 pM to about 240 pM, from about 25 pM to about 250 pM, from about 26 pM to about 260 pM, from about 27 pM to about 270 pM, from about 28 pM to about 280 pM, from about 29 pM to about 290 pM, from about 30 pM to about 300 pM, from about 31 pM to about 310 pM, from about 32 pM to about 320 pM, from about 33 pM to about 330 pM, from about 34 pM to about 340 pM, from about 35 pM to about 350 pM, from about 36 pM to about 360 pM, from about 37 pM to about 370 pM, from about 38 pM to about 380 pM, from about 39 pM to about 390 pM, from about 40 pM to about 400 pM, from about 41 pM to about 410 pM, from about 42 pM to about 420 pM, from about 43 pM to about 430 pM, from about 44 pM to about 440 pM, from about 45 pM to about 450 pM, from about 46 pM to about 460 pM, from about 47 pM to about 470 pM, from about 48 pM to about 480 pM, from about 49 pM to about 490 pM, from about 50 pM to about 500 pM, from about 51 pM to about 510 pM, from about 52 pM to about 520 pM, from about 53 pM to about 530 pM, from about 54 pM to about 540 pM, from about 55 pM to about 550 pM, from about 56 pM to about 560 pM, from about 57 pM to about 570 pM, from about 58 pM to about 580 pM, from about 59 pM to about 590 pM, from about 60 pM to about 600 pM, from about 61 pM to about 610 pM, from about 62 pM to about 620 pM, from about 63 pM to about 630 pM, from about 64 pM to about 640 pM, from about 65 pM to about 650 pM, from about 66 pM to about 660 pM, from about 67 pM to about 670 pM, from about 68 pM to about 680 pM, from about 69 pM to about 690 pM, from about 70 pM to about 700 pM, from about 71 pM to about 710 pM, from about 72 pM to about 720 pM, from about 73 pM to about 730 pM, from about 74 pM to about 740 pM, from about 75 pM to about 750 pM, from about 76 pM to about 760 pM, from about 77 pM to about 770 pM, from about 78 pM to about 780 pM, from about 79 pM to about 790 pM, from about 80 pM to about 800 pM, from about 81 pM to about 810 pM, from about 82 pM to about 820 pM, from about 83 pM to about 830 pM, from about 84 pM to about 840 pM, from about 85 pM to about 850 pM, from about 86 pM to about 860 pM, from about 87 pM to about 870 pM, from about 88 pM to about 880 pM, from about 89 pM to about 890 pM, from about 90 pM to about 900 pM, from about 91 pM to about 910 pM, from about 92 pM to about 920 pM, from about 93 pM to about 930 pM, from about 94 pM to about 940 pM, from about 95 pM to about 950 pM, from about 96 pM to about 960 pM, from about 97 pM to about 970 pM, from about 98 pM to about 980 pM, from about 99 pM to about 990 pM, from about 100 pM to about 1 nM, from about 0.1 nM to about 1 nM, from about 1 nM to about 10 nM, from about 2 nM to about 20 nM, from about 3 nM to about 30 nM, from about 4 nM to about 40 nM, from about 5 nM to about 50 nM, from about 6 nM to about 60 nM, from about 7 nM to about 70 nM, from about 8 nM to about 80 nM, from about 9 nM to about 90 nM, from about 10 nM to about 100 nM, from about 11 nM to about 110 nM, from about 12 nM to about 120 nM, from about 13 nM to about 130 nM, from about 14 nM to about 140 nM, from about 15 nM to about 150 nM, from about 16 nM to about 160 nM, from about 17 nM to about 170 nM, from about 18 nM to about 180 nM, from about 19 nM to about 190 nM, from about 20 nM to about 200 nM, from about 21 nM to about 210 nM, from about 22 nM to about 220 nM, from about 23 nM to about 230 nM, from about 24 nM to about 240 nM, from about 25 nM to about 250 nM, from about 26 nM to about 260 nM, from about 27 nM to about 270 nM, from about 28 nM to about 280 nM, from about 29 nM to about 290 nM, from about 30 nM to about 300 nM, from about 31 nM to about 310 nM, from about 32 nM to about 320 nM, from about 33 nM to about 330 nM, from about 34 nM to about 340 nM, from about 35 nM to about 350 nM, from about 36 nM to about 360 nM, from about 37 nM to about 370 nM, from about 38 nM to about 380 nM, from about 39 nM to about 390 nM, from about 40 nM to about 400 nM, from about 41 nM to about 410 nM, from about 42 nM to about 420 nM, from about 43 nM to about 430 nM, from about 44 nM to about 440 nM, from about 45 nM to about 450 nM, from about 46 nM to about 460 nM, from about 47 nM to about 470 nM, from about 48 nM to about 480 nM, from about 49 nM to about 490 nM, from about 50 nM to about 500 nM, from about 51 nM to about 510 nM, from about 52 nM to about 520 nM, from about 53 nM to about 530 nM, from about 54 nM to about 540 nM, from about 55 nM to about 550 nM, from about 56 nM to about 560 nM, from about 57 nM to about 570 nM, from about 58 nM to about 580 nM, from about 59 nM to about 590 nM, from about 60 nM to about 600 nM, from about 61 nM to about 610 nM, from about 62 nM to about 620 nM, from about 63 nM to about 630 nM, from about 64 nM to about 640 nM, from about 65 nM to about 650 nM, from about 66 nM to about 660 nM, from about 67 nM to about 670 nM, from about 68 nM to about 680 nM, from about 69 nM to about 690 nM, from about 70 nM to about 700 nM, from about 71 nM to about 710 nM, from about 72 nM to about 720 nM, from about 73 nM to about 730 nM, from about 74 nM to about 740 nM, from about 75 nM to about 750 nM, from about 76 nM to about 760 nM, from about 77 nM to about 770 nM, from about 78 nM to about 780 nM, from about 79 nM to about 790 nM, from about 80 nM to about 800 nM, from about 81 nM to about 810 nM, from about 82 nM to about 820 nM, from about 83 nM to about 830 nM, from about 84 nM to about 840 nM, from about 85 nM to about 850 nM, from about 86 nM to about 860 nM, from about 87 nM to about 870 nM, from about 88 nM to about 880 nM, from about 89 nM to about 890 nM, from about 90 nM to about 900 nM, from about 91 nM to about 910 nM, from about 92 nM to about 920 nM, from about 93 nM to about 930 nM, from about 94 nM to about 940 nM, from about 95 nM to about 950 nM, from about 96 nM to about 960 nM, from about 97 nM to about 970 nM, from about 98 nM to about 980 nM, from about 99 nM to about 990 nM, from about 100 nM to about 1 μM, from about 0.1 μM to about 1 μM, from about 1 μM to about 10 μM, from about 2 μM to about 20 μM, from about 3 μM to about 30 μM, from about 4 μM to about 40 μM, from about 5 μM to about 50 μM, from about 6 μM to about 60 μM, from about 7 μM to about 70 μM, from about 8 μM to about 80 μM, from about 9 μM to about 90 μM, from about 10 μM to about 100 μM, from about 11 μM to about 110 μM from about 12 μM to about 120 μM, from about 13 μM to about 130 μM, from about 14 μM to about 140 μM from about 15 μM to about 150 μM, from about 16 μM to about 160 μM, from about 17 μM to about 170 μM, from about 18 μM to about 180 μM, from about 19 μM to about 190 μM, from about 20 μM to about 200 μM, from about 21 μM to about 210 μM, from about 22 μM to about 220 μM, from about 23 μM to about 230 μM from about 24 μM to about 240 μM, from about 25 μM to about 250 μM, from about 26 μM to about 260 μM, from about 27 μM to about 270 μM from about 28 μM to about 280 μM from about 29 μM to about 290 μM, from about 30 μM to about 300 μM, from about 31 μM to about 310 μM, from about 32 μM to about 320 μM, from about 33 μM to about 330 μM, from about 34 μM to about 340 μM, from about 35 μM to about 350 μM, from about 36 μM to about 360 μM, from about 37 μM to about 370 μM, from about 38 μM to about 380 μM, from about 39 μM to about 390 μM, from about 40 μM to about 400 μM from about 41 μM to about 410 μM from about 42 μM to about 420 μM, from about 43 μM to about 430 μM, from about 44 μM to about 440 μM, from about 45 μM to about 450 μM, from about 46 μM to about 460 μM, from about 47 μM to about 470 μM, from about 48 μM to about 480 μM, from about 49 μM to about 490 μM, from about 50 μM to about 500 μM, from about 51 μM to about 510 μM, from about 52 μM to about 520 μM, from about 53 μM to about 530 μM, from about 54 μM to about 540 μM from about 55 μM to about 550 μM, from about 56 μM to about 560 μM, from about 57 μM to about 570 μM, from about 58 μM to about 580 μM, from about 59 μM to about 590 μM, from about 60 μM to about 600 μM, from about 61 μM to about 610 μM, from about 62 μM to about 620 μM, from about 63 μM to about 630 μM, from about 64 μM to about 640 μM, from about 65 μM to about 650 μM, from about 66 μM to about 660 μM from about 67 μM to about 670 μM from about 68 μM to about 680 μM from about 69 μM to about 690 μM, from about 70 μM to about 700 μM, from about 71 μM to about 710 μM, from about 72 μM to about 720 μM, from about 73 μM to about 730 μM, from about 74 μM to about 740 μM, from about 75 μM to about 750 μM, from about 76 μM to about 760 μM from about 77 μM to about 770 μM, from about 78 μM to about 780 μM, from about 79 μM to about 790 μM from about 80 μM to about 800 μM from about 81 μM to about 810 μM, from about 82 μM to about 820 μM, from about 83 μM to about 830 μM, from about 84 μM to

about 840 μM, from about 85 μM to about 850 μM, from about 86 μM to about 860 μM, from about 87 μM to about 870 μM, from about 88 μM to about 880 μM, from about 89 μM to about 890 μM, from about 90 μM to about 900 μM, from about 91 μM to about 910 μM, from about 92 μM to about 920 μM, from about 93 μM to about 930 μM, from about 94 μM to about 940 μM, from about 95 μM to about 950 μM, from about 96 μM to about 960 μM, from about 97 μM to about 970 μM, from about 98 μM to about 980 μM, from about 99 μM to about 990 μM, from about 100 μM to about 1 mM, from about 0.1 mM to about 1 mM, from about 1 mM to about 10 mM, from about 2 mM to about 20 mM, from about 3 mM to about 30 mM, from about 4 mM to about 40 mM, from about 5 mM to about 50 mM, from about 6 mM to about 60 mM, from about 7 mM to about 70 mM, from about 8 mM to about 80 mM, from about 9 mM to about 90 mM, from about 10 mM to about 100 mM, from about 11 mM to about 110 mM, from about 12 mM to about 120 mM, from about 13 mM to about 130 mM, from about 14 mM to about 140 mM, from about 15 mM to about 150 mM, from about 16 mM to about 160 mM, from about 17 mM to about 170 mM, from about 18 mM to about 180 mM, from about 19 mM to about 190 mM, from about 20 mM to about 200 mM, from about 21 mM to about 210 mM, from about 22 mM to about 220 mM, from about 23 mM to about 230 mM, from about 24 mM to about 240 mM, from about 25 mM to about 250 mM, from about 26 mM to about 260 mM, from about 27 mM to about 270 mM, from about 28 mM to about 280 mM, from about 29 mM to about 290 mM, from about 30 mM to about 300 mM, from about 31 mM to about 310 mM, from about 32 mM to about 320 mM, from about 33 mM to about 330 mM, from about 34 mM to about 340 mM, from about 35 mM to about 350 mM, from about 36 mM to about 360 mM, from about 37 mM to about 370 mM, from about 38 mM to about 380 mM, from about 39 mM to about 390 mM, from about 40 mM to about 400 mM, from about 41 mM to about 410 mM, from about 42 mM to about 420 mM, from about 43 mM to about 430 mM, from about 44 mM to about 440 mM, from about 45 mM to about 450 mM, from about 46 mM to about 460 mM, from about 47 mM to about 470 mM, from about 48 mM to about 480 mM, from about 49 mM to about 490 mM, from about 50 mM to about 500 mM, from about 51 mM to about 510 mM, from about 52 mM to about 520 mM, from about 53 mM to about 530 mM, from about 54 mM to about 540 mM, from about 55 mM to about 550 mM, from about 56 mM to about 560 mM, from about 57 mM to about 570 mM, from about 58 mM to about 580 mM, from about 59 mM to about 590 mM, from about 60 mM to about 600 mM, from about 61 mM to about 610 mM, from about 62 mM to about 620 mM, from about 63 mM to about 630 mM, from about 64 mM to about 640 mM, from about 65 mM to about 650 mM, from about 66 mM to about 660 mM, from about 67 mM to about 670 mM, from about 68 mM to about 680 mM, from about 69 mM to about 690 mM, from about 70 mM to about 700 mM, from about 71 mM to about 710 mM, from about 72 mM to about 720 mM, from about 73 mM to about 730 mM, from about 74 mM to about 740 mM, from about 75 mM to about 750 mM, from about 76 mM to about 760 mM, from about 77 mM to about 770 mM, from about 78 mM to about 780 mM, from about 79 mM to about 790 mM, from about 80 mM to about 800 mM, from about 81 mM to about 810 mM, from about 82 mM to about 820 mM, from about 83 mM to about 830 mM, from about 84 mM to about 840 mM, from about 85 mM to about 850 mM, from about 86 mM to about 860 mM, from about 87 mM to about 870 mM, from about 88 mM to about 880 mM, from about 89 mM to about

890 mM, from about 90 mM to about 900 mM, from about 91 mM to about 910 mM, from about 92 mM to about 920 mM, from about 93 mM to about 930 mM, from about 94 mM to about 940 mM, from about 95 mM to about 950 mM, from about 96 mM to about 960 mM, from about 97 mM to about 970 mM, from about 98 mM to about 980 mM, from about 99 mM to about 990 mM, from about 100 mM to about 1 M, from about 1 M to about 10 M, from about 2 M to about 20 M, from about 3 M to about 30 M, from about 4 M to about 40 M, from about 5 M to about 50 M, from about 6 M to about 60 M, from about 7 M to about 70 M, from about 8 M to about 80 M, from about 9 M to about 90 M, from about 10 M to about 100 M, from about 11 M to about 110 M, from about 12 M to about 120 M, from about 13 M to about 130 M, from about 14 M to about 140 M, from about 15 M to about 150 M, from about 16 M to about 160 M, from about 17 M to about 170 M, from about 18 M to about 180 M, from about 19 M to about 190 M, from about 20 M to about 200 M, from about 21 M to about 210 M, from about 22 M to about 220 M, from about 23 M to about 230 M, from about 24 M to about 240 M, from about 25 M to about 250 M, from about 26 M to about 260 M, from about 27 M to about 270 M, from about 28 M to about 280 M, from about 29 M to about 290 M, from about 30 M to about 300 M, from about 31 M to about 310 M, from about 32 M to about 320 M, from about 33 M to about 330 M, from about 34 M to about 340 M, from about 35 M to about 350 M, from about 36 M to about 360 M, from about 37 M to about 370 M, from about 38 M to about 380 M, from about 39 M to about 390 M, from about 40 M to about 400 M, from about 41 M to about 410 M, from about 42 M to about 420 M, from about 43 M to about 430 M, from about 44 M to about 440 M, from about 45 M to about 450 M, from about 46 M to about 460 M, from about 47 M to about 470 M, from about 48 M to about 480 M, from about 49 M to about 490 M, or from about 50 M to about 500 M.

SBPs may include a ratio of silk fibroin (by weight, volume, or concentration) to at least one excipient and/or therapeutic agent (by weight, volume, or concentration) of from about 0.001:1 to about 1:1, from about 0.005:1 to about 5:1, from about 0.01:1 to about 0.5:1, from about 0.01:1 to about 10:1, from about 0.02:1 to about 20:1, from about 0.03:1 to about 30:1, from about 0.04:1 to about 40:1, from about 0.05:1 to about 50:1, from about 0.06:1 to about 60:1, from about 0.07:1 to about 70:1, from about 0.08:1 to about 80:1, from about 0.09:1 to about 90:1, from about 0.1:1 to about 100:1, from about 0.2:1 to about 150:1, from about 0.3:1 to about 200:1, from about 0.4:1 to about 250:1, from about 0.5:1 to about 300:1, from about 0.6:1 to about 350:1, from about 0.7:1 to about 400:1, from about 0.8:1 to about 450:1, from about 0.9:1 to about 500:1, from about 1:1 to about 550:1, from about 2:1 to about 600:1, from about 3:1 to about 650:1, from about 4:1 to about 700:1, from about 5:1 to about 750:1, from about 6:1 to about 800:1, from about 7:1 to about 850:1, from about 8:1 to about 900:1, from about 9:1 to about 950:1, from about 10:1 to about 960:1, from about 50:1 to about 970:1, from about 100:1 to about 980:1, from about 200:1 to about 990:1, or from about 500:1 to about 1000:1. In some embodiments, SBP formulations contain trace amounts of excipient.

In some embodiments, the concentration processed silk and/or other components may be determined by absorbance. In some embodiments, the concentration of processed silk and/or other components may be determined by their absorbance at 280 nm.

Appearance: Transparent, Opaque, Translucent

In some embodiments, the appearance of SBP formulations described in the present disclosure may be tuned for the application for which they were designed. In some embodiments, SBP formulations may be transparent. In some embodiments, SBP formulations may be translucent. In some embodiments, SBP formulations may be opaque. In some embodiments, SBP preparation methods may be used to modulate clarity, as taught in International Patent Application Publication No. WO2012170655, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the incorporation of excipients may be used to tune the clarity of processed silk preparations. In some embodiments, the excipient is sucrose. In some embodiments, the sucrose may also increase protein reconstitution during lyophilization. In some embodiments, sucrose may improve processed silk hydrogel clarity (optically transparent). The transparency of SBP formulations, as well as other properties, may render resulting labels edible, biodegradable, and/or holographic.

Solubility

In some embodiments, SBP formulations or components thereof are water soluble. The water solubility, along with the rate of degradation, of SBPs may modulate payload (e.g., therapeutic agent) release rate and/or release period. An increasing amount of payload may be released into surrounding medium as surrounding matrix dissolves (e.g., see International Publication Numbers WO2013126799 and WO2017165922; and U.S. Pat. No. 8,530,625, the contents of each of which are herein incorporated by reference in their entirety). Longer time periods required to dissolve SBPs or components thereof may result in longer release periods. In some embodiments, SBP solubility may be modulated in order to control the rate of payload release in the surrounding medium. The solubility of SBPs may be modulated via any method known to those skilled in the art. In some embodiments, SBP solubility may be modulated by altering included silk fibroin secondary structure (e.g., increasing beta-sheet content and/or crystallinity). In some embodiments, SBP solubility may be modulated by altering SBP format. In some embodiments, SBP solubility and/or rate of degradation may be modulated to facilitate extended release of therapeutic agent payloads in vitro and/or in vivo.

Residence Time

In some embodiments, SBP formulations may be prepared to have desired residence time according to the application for which they are designed. As used herein, the term "residence time" refers to the average length of time during which a substance (e.g., SBP formulations) is in a given location or condition. In some embodiments, residence time of SBP formulations described herein may vary from a few hours to several months. For example, residence time of SBP formulations may be about 1 hour, about 2 hours, about 3 hours, about 4 hours, about 5 hours, about 6 hours, about 7 hours, about 8 hours, about 9 hours, about 10 hours, about 11 hours, about 12 hours, about 13 hours, about 14 hours, about 15 hours, about 16 hours, about 17 hours, about 18 hours, about 19 hours, about 20 hours, about 21 hours, about 22 hours, about 23 hours, about 1 day, about 2 days, about 3 days, about 4 days, about 5 days, about 6 days, about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, about 1 month, about 2 months, about 3 months, about 4 months, about 5 months, about 6 months, about 7 months, about 8 months, about 9 months, about 10 months, about 11 months, or longer than 1 year.

Excipients

In some embodiments, SBP formulations include one or more excipients. In some embodiments, SBP formulation may not include an excipient. As used herein, the term "excipient" refers to any substance included in a composition with an active agent or primary component, often serving as a carrier, diluent, or vehicle for the active agent or primary component. In some embodiments, excipients may be compounds or compositions approved for use by the US Food and Drug Administration (FDA). In some embodiments, SBPs may include excipients that increase SBP stability or stability of one or more other SBP components. Some SBPs may include an excipient that modulates payload release. Excipients may include, but are not limited to, solvents, diluents, liquid vehicles, dispersion or suspension media or aids, surfactants, thickening agents, emulsifying agents, lipids, liposomes, isotonic agents, buffers, and preservatives. In some embodiments, excipients include lipidoids, lipid nanoparticles, polymers, lipoplexes, particles, core-shell nanoparticles, peptides, proteins, cells, hyaluronidase, and/or nanoparticle mimics. In some embodiments, processed silk and/or SBPs may be used as an excipient. In some embodiments, excipients included in SBPs are selected from one or more of sucrose, lactose, phosphate salts, sodium chloride, potassium phosphate monobasic, potassium phosphate dibasic, sodium phosphate dibasic, sodium phosphate monobasic, polysorbate 80, phosphate buffer, phosphate buffered saline, sodium hydroxide, sorbitol, mannitol, lactose USP, Starch 1500, microcrystalline cellulose, potassium chloride, sodium borate, boric acid, sodium borate decahydrate, magnesium chloride hexahydrate, calcium chloride dihydrate, sodium hydroxide, Avicel, dibasic calcium phosphate dehydrate, tartaric acid, citric acid, fumaric acid, succinic acid, malic acid, hydrochloric acid, polyvinylpyrrolidone, copolymers of vinylpyrrolidone and vinylacetate, hydroxypropylcellulose, hydroxyethylcellulose, hydroxypropylmethylcellulose, polyvinyl alcohol, polyethylene glycol, acacia, and sodium carboxymethylcellulose. Excipients may include phosphate buffered saline. Excipients may be present in SBPs at any concentration. In some embodiments, excipients are present at a concentration of from about 0.0001% weight per weight (w/w) of excipient to total SBP weight to about 20% (w/w). In some embodiments, excipients are present at a concentration of from about 0.0001% weight per weight (w/w) of excipient to total SBP weight to about 50% (w/w).

In some embodiments, excipients included in SBPs may be selected from one or more of sorbitol, triethylamine, 2-pyrrolidone, alpha-cyclodextrin, benzyl alcohol, beta-cyclodextrin, dimethyl sulfoxide, dimethylacetamide (DMA), dimethylformamide, ethanol, gamma-cyclodextrin, glycerol, glycerol formal, hydroxypropyl beta-cyclodextrin, kolliphor 124, kolliphor 181, kolliphor 188, kolliphor 407, kolliphor EL (cremophor EL), cremophor RH 40, cremophor RH 60, dalpha-tocopherol, PEG 1000 succinate, polysorbate 20, polysorbate 80, solutol HS 15, sorbitan monooleate, poloxamer-407, poloxamer-188, Labrafil M-1944CS, Labrafil M-2125CS, Labrasol, Gellucire 44/14, Softigen 767, mono- and di-fatty acid esters of PEG 300, PEG 400, or PEG 1750, kolliphor RH60, N-methyl-2-pyrrolidone, castor oil, corn oil, cottonseed oil, olive oil, peanut oil, peppermint oil, safflower oil, sesame oil, soybean oil, hydrogenated vegetable oils, hydrogenated soybean oil, medium chain triglycerides of coconut oil, medium chain triglycerides of palm seed oil, beeswax, d-alpha-tocopherol, oleic acid, medium-chain mono-glycerides, medium-chain di-glycerides, alpha-cyclodextrin, betacyclodextrin, hydroxypropyl-beta-cyclodextrin, sulfo-butylether-beta-cyclodextrin, hydrogenated soy phosphatidylcholine, distearoylphosphatidylglycerol, L-alphadimyristoylphosphatidylcholine, L-alpha-dimyristoylphosphatidylglycerol, PEG 300, PEG 300 caprylic/capric glycerides (Softigen 767), PEG 300 linoleic glycerides (Labrafil M-2125CS), PEG 300 oleic glycerides (Labrafil M-1944CS), PEG 400, PEG 400 caprylic/capric glycerides (Labrasol), polyoxyl 40 stearate (PEG 1750 monosterate), polyoxyl 8 stearate (PEG 400 monosterate), polysorbate 20, polysorbate 80, polyvinyl pyrrolidone, propylene carbonate, propylene glycol, solutol HS15, sorbitan monooleate (Span 20), sulfobutylether-beta-cyclodextrin, transcutol, triacetin, 1-dodecylazacyclo-heptan-2-one, caprolactam, castor oil, cottonseed oil, ethyl acetate, medium chain triglycerides, methyl acetate, oleic acid, safflower oil, sesame oil, soybean oil, tetrahydrofuran, glycerin, and PEG 4 kDa. Such SBPs may include hydrogels. In some embodiments, SBP hydrogels include one or more of polysorbate 80, poloxamer-188, PEG 4 kDa, and glycerol.

In some embodiments, excipients included in SBPs are selected from one or more of those listed in Table 1. In the Table, example categories are indicated for each excipient. These categories are not limiting and each excipient may fall under multiple categories (e.g., any of the categories of excipients described herein).

TABLE 1

Excipients

Excipient	Example Category
Avicel	bulking agent
bulking agent	bulking agent
copolymers of vinylpyrrolidone and vinylacetate	bulking agent
dibasic calcium phosphate dehydrate	bulking agent
fumaric acid	bulking agent
hydroxypropylmethylcellulose	bulking agent
lactose USP	bulking agent
malic acid	bulking agent
microcrystalline cellulose	bulking agent
polyvinylpyrrolidone	bulking agent
tartaric acid	bulking agent
(12Z,15Z)-N,N-dimethyl-2-nonylhenicosa-12,15-dien-1-amine	cationic lipid
(12Z,15Z)-N,N-dimethylhenicosa-12,15-dien-4-amine	cationic lipid
(13Z,16Z)-N,N-dimethyl-3-nonyldocosa-13,16-dien-1-amine	cationic lipid
(13Z,16Z)-N,N-dimethyldocosa-13,16-dien-5-amine	cationic lipid
(14Z)-N,N-dimethylnonacos-14-en-10-amine	cationic lipid
(14Z,17Z)-N,N-dimethyltricosa-14,17-dien-4-amine	cationic lipid
(14Z,17Z)-N,N-dimethyltricosa-14,17-dien-6-amine	cationic lipid
(15Z)-N,N-dimethyl eptacos-15-en-10-amine	cationic lipid
(15Z,18Z)-N N-dimethyltetracosa-15,18-dien-7-amine	cationic lipid
(15Z,18Z)-N,N-dimethyltetracosa-15,18-dien-5-amine	cationic lipid
(162)-N,N-dimethylpentacos-16-en-8-amine	cationic lipid
(16Z,19Z)-N,N-dimethylpentacosa-16,19-dien-6-amine	cationic lipid
(17Z)-N,N-dimethylhexacos-17-en-9-amine	cationic lipid
(17Z)-N,N-dimethylnonacos-17-en-10-amine	cationic lipid

TABLE 1-continued

Excipients

Excipient	Example Category
(17Z,20Z)-N,N-dimemylhexacosa-17,20-dien-9-amine	cationic lipid
(17Z,20Z)-N,N-dimethylhexacosa-17,20-dien-7-amine	cationic lipid
(18Z)-N,N-dimetylheptacos-18-en-10-amine	cationic lipid
(18Z,21 Z)-N,N-dimethylheptacosa-18,21-dien-8-amine	cationic lipid
(18Z,21Z)-N,N-dimethylheptacosa-18,21-dien-10-amine	cationic lipid
(19Z,22Z)-N,N-dimethyloctacosa-19,22-dien-9-amine	cationic lipid
(19Z,22Z)-N,N-dimethyloctacosa-19,22-dien-7-amine	cationic lipid
(11E,20Z,23Z)-N,N-dimethylnonacosa-11,20,2-trien-10-amine	cationic lipid
(1Z,19Z)-N5N-dimethylpentacosa-16,19-dien-8-amine	cationic lipid
(20Z)-N,N-dimethylheptacos-20-en-10-amine	cationic lipid
(202)-N,N-dimethylnonacos-20-en-10-amine	cationic lipid
(20Z,23Z)-N,N-dimethylnonacosa-20,23-dien-10-amine	cationic lipid
(20Z,23Z)-N-ethyl-N-methylnonacosa-20,23-dien-10-amine	cationic lipid
(21Z,24Z)-N,N-dimethyltriaconta-21,24-dien-9-amine	cationic lipid
(22Z)-N,N-dimethylhentriacont-22-en-10-amine	cationic lipid
(22Z,25Z)-N,N-dimethylhentriaconta-22,25-dien-10-amine	cationic lipid
(24Z)-N,N-dimethyltritriacont-24-en-10-amine	cationic lipid
(2R)-1-[(3,7-dimethyloctyl)oxy]-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine	cationic lipid
(2R)-N,N-dimethyl-H(1-metoylo ctyl) oxy]-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine	cationic lipid
(2S)-1-(heplyloxy)-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy] propan-2-amine	cationic lipid
(2S)-1-(hexyloxy)-3-[(11Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethylpropan-2-amine	cationic lipid
(2S)-1-(hexyloxy)-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxylpropan-2-amine	cationic lipid
(2S)-1-(1Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethyl-3-(pentyloxy) propan-2-amine	cationic lipid
(2S)-1-[(13Z)-docos-13-en-1-yloxy]-3-(hexyloxy)-N,N-dimethylpropan-2-amine	cationic lipid
(2S)-1-[(13Z,16Z)-docosa-13,16-dien-1-yloxy]-3-(hexyloxy)-N,N-dimethylpropan-2-amine	cationic lipid
(2S)-N,N-dimethyl-1-[(6Z,9Z,12Z)-octadeca-6,9,12-trien-1-yloxy]-3-(octyloxy)propan-2-amine	cationic lipid
(2S)-N,N-dimethyl-1-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-[(5Z)-oct-5-en-1-yloxy]propan-2-amine	cationic lipid
1,2-dilinolenyloxy-3-dimethylaminopropane (DLenDMA)	cationic lipid
1,2-distearloxy-N,N-dimethylaminopropane (DSDMA)	cationic lipid
1-[(11Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine	cationic lipid
1-[(11Z,14Z)-1-nonylicosa-11,14-dien-1-yl] pyrrolidine	cationic lipid
1-[(13Z)-docos-13-en-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine	cationic lipid
1-[(13Z,16Z)-docosa-13,16-dien-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine	cationic lipid
1-[(1R,2S)-2-hepty lcyclopropyl]-N,N-dimethyloctadecan-9-amine	cationic lipid
1-[(1S,2R)-2-decylcyclopropyl]-N,N-dimethylpentadecan-6-amine	cationic lipid
1-[(1S,2R)-2-hexylcyclopropyl]-N,N-dimethylnonadecan-10-amine	cationic lipid
1-[(9Z)-hexadec-9-en-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine	cationic lipid
1-{2-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-1-[(octyloxy)methyl]ethyl}azetidine	cationic lipid
1-{2-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-1-[(octyloxy)methyl]ethyl}pyrrolidine	cationic lipid
cationic lipid	cationic lipid
CLI-CLXXIX of International Publication No. WO2008103276	cationic lipid
DLin-DMA	cationic lipid
DODMA	cationic lipid
formula CLI-CLXXIX of U.S. Pat. No. US7893302	cationic lipid
formula CLI-CLXXXXII of U.S. Pat. No. US7404969	cationic lipid
formula I-VI of U.S. Pat. Publication No. US20100036115	cationic lipid
N,N-dimethyl-1-(octyloxy)-3-({8-[(1S,2S)-2-{[(1R,2R)-2-pentylcyclopropyl]methyl} cyclopropyl]octyl}oxy)propan-2-amine	cationic lipid
N,N-dimethyl-1-{[8-(2-oclylcyclopropyl) octyl]oxy}-3-(octyloxy)propan-2-amine	cationic lipid
N,N-dimethyl-21-[(1S,2R)-2-octylcyclopropyl] henicosan-10-amine	cationic lipid
N,N-dimethyl-3-{7-[(1S.2R)-2-octylcyclopropyl]heptyl} dodecan-1-amine	cationic lipid
N,N-dimethylheptacosan-10-amine	cationic lipid
N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl] eptadecan-8-amine	cationic lipid
N,N-dimethyl-1-((1S,2R)-2-octylcyclopropyl] pentadecan-8-amine	cationic lipid
R-N,N-dimethyl-1-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-(octyloxy)propan-2-amine	cationic lipid
S-N,N-dimethyl-1-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-(octyloxy)propan-2-amine	cationic lipid
N,N-dimethyl-[(1R,2S)-2-undecylcyclopropyl] tetradecan-5-amine	cationic lipid
N,N-dimethyl-1-(nonyloxy)-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine	cationic lipid
N,N-dimethyl-1-(1S,2R)-2-octylcyclopropyl] nonadecan-10-amine	cationic lipid
N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl] hexadecan-8-amine	cationic lipid
N,N-dimethyl-1-[(1S,2S)-2-{[(1R,2R)-2-pentylcydopropyl]methyl}cyclopropyl]nonadecan-10-amine	cationic lipid
N,N-dimethyl-1-[(9Z)-oetadec-9-en-1-yloxy]-3-(octyloxy)propan-2-amine	cationic lipid
coating agents	coating agent
poly(alkyl)(meth)acrylate	coating agent
poly(ethylene-co-vinyl acetate)	coating agent
zein	coating agent
apocarotenal	colorant
apocatotenal derivative	colorant
astaxanthin	colorant
astaxanthin derivative	colorant
bixin	colorant
canthaxanthin	colorant
canthaxanthin derivative	colorant
capsanthin	colorant
capsanthin derivative	colorant
capsorubin derivative	colorant
capso rubin occurring in paprika ole-oresin	colorant
caretinoids	colorant
colorant	colorant
crocin	colorant
crocin derivative	colorant

TABLE 1-continued

Excipients	
Excipient	Example Category
dyes	colorant
FD&C Blue No. 2 (indigotine)	colorant
FD&C colorant	colorant
FD&C Red No. 3 (ervthrosine)	colorant
FD&C Red No. 40 (allura red AC)	colorant
food coloring	colorant
inks	colorant
lutein	colorant
lutein derivative	colorant
lycopene	colorant
pigments	colorant
rhodoxanthin	colorant
rubixanthin	colorant
saffron	colorant
saffron derivative	colorant
turmeric	colorant
violaxanthin	colorant
β-carotene	colorant
β-carotene derivative	colorant
flowability agents	flowability agent
1-dodecylazacyclo-heptan-2-one	gelling agent
2-pyrrolidone	gelling agent
acacia	gelling agent
alginic acid	gelling agent
alpha-cyclodextrin	gelling agent
beeswax	gelling agent
bentonite	gelling agent
benzyl alcohol	gelling agent
beta-cyclodextrin	gelling agent
caprolactam	gelling agent
CARBOPOL ® (also known as carbomer)	gelling agent
carboxymethyl cellulose	gelling agent
castor oil	gelling agent
com oil	gelling agent
cottonseed oil	gelling agent
cremaphor RH 40	gelling agent
cremaphor RH 60	gelling agent
d-alpha-tocopherol	gelling agent
di-fatty acid ester of PEG 1750	gelling agent
di-fatty acid ester of PEG 300	gelling agent
di-fatty acid ester of PEG 400	gelling agent
dimethyl sulfoxide	gelling agent
dimethylacetamide (DMA)	gelling agent
dimethylformamide	gelling agent
distearoylphosphatidylglycerol	gelling agent
ethanol	gelling agent
ethyl acetate	gelling agent
ethylcellulose	gelling agent
gamma-cyclodextrin	gelling agent
gelatin	gelling agent
Gellucire 44/14	gelling agent
glycerin	gelling agent
glycerol	gelling agent
glycerol formal	gelling agent
glycerophosphate	gelling agent
hydrogenated soy phosphatidylcholine	gelling agent
hydrogenated soybean oil	gelling agent
hydrogenated vegetable oils	gelling agent
hydroxy ethyl cellulose	gelling agent
hydroxyethyl cellulose	gelling agent
hydroxypropyl beta-cyclodextrin	gelling agent
hydroxypropyl cellulose	gelling agent
hydroxypropyl-beta-cyclodextrin	gelling agent
kolliphor 124	gelling agent
kolliphor 181	gelling agent
kolliphor 188	gelling agent
kolliphor 407	gelling agent
kolliphor EL (cremaphor EL)	gelling agent
kolliphor RH60	gelling agent
Labrafil M-1944CS	gelling agent
Labrafil M-2125CS	gelling agent
Labrasol	gelling agent
L-alpha-dimyristoylphosphatidylcholine	gelling agent
L-alphadimyristoylphosphatidylglycerol	gelling agent

TABLE 1-continued

Excipients	
Excipient	Example Category
magnesium aluminum silicate	gelling agent
medium chain triglyceride	gelling agent
medium-chain diglyceride	gelling agent
medium-chain mono-glyceride	gelling agent
medium-chain triglyceride of coconut oil	gelling agent
medium-chain triglyceride of palm seed oil	gelling agent
methyl acetate	gelling agent
methylcellulose	gelling agent
mono-fatty acid ester of PEG 1750	gelling agent
mono-fatty acid ester of PEG 300	gelling agent
mono-fatty acid ester of PEG 400	gelling agent
N-methyl-2-pyrrolidone	gelling agent
oleic acid	gelling agent
olive oil	gelling agent
peanut oil	gelling agent
PEG 1000 succinate	gelling agent
PEG 1750	gelling agent
PEG 300	gelling agent
PEG 300 caprylic/capric glyceride (Softigen 767)	gelling agent
PEG 300 linoleic glyceride (Labrafil M-2125CS)	gelling agent
PEG 300 oleic glyceride (Labrafil M-1944CS)	gelling agent
PEG 400	gelling agent
PEG 400 caprylic/capric glyceride (Labrasol)	gelling agent
PEG 4000 (PEG 4 kDa)	gelling agent
peppermint oil	gelling agent
polaxamer	gelling agent
poloxamer-188	gelling agent
poloxamer-407	gelling agent
polyoxyl 40 stearate (PEG 1750 monosterate)	gelling agent
polyoxyl 8 stearate (PEG 400 monosterate)	gelling agent
polysorbate 20	gelling agent
polysorbate-80 (tween-80)	gelling agent
polysorbate-SO	gelling agent
polyvinyl alcohol	gelling agent
polyvinyl pyrrolidone	gelling agent
polyvinyl pyrrolidone-12	gelling agent
polyvinyl pyrrolidone-17	gelling agent
propylene carbonate	gelling agent
propylene glycol	gelling agent
safflower oil	gelling agent
sesame oil	gelling agent
sodium alginate	gelling agent
Softigen 767	gelling agent
solutol HS 15	gelling agent
sorbitan monooleate	gelling agent
sorbitan monooleate (Span 20)	gelling agent
sorbitol	gelling agent
soybean oil	gelling agent
sulfobutylether-beta-cydodextrin	gelling agent
sulfo-butylether-beta-cyclodextrin	gelling agent
tetrahydrofuran	gelling agent
tragacanth	gelling agent
transcutol	gelling agent
triacetin	gelling agent
triethanolamine	gelling agent
triethylamine	gelling agent
xanthan gum	gelling agent
(50:50, Poly(D1-Lactic-Co-Glycolic Acid)	general
(50:50, Polyacrylic Acid (250000 Mw)	general
1,2,6-Hexanetriol	general
1.2-Dimyristoyl-Sn-Glycero-3-(Phospho-S-(1-Glycerol))	general
1,2-Dimyristoyl-Sn-Glycero-3-Phosphocholine	general
1,2-Dioleoyl-Sn-Glycero-3-Phosphocholine	general
1,2-Dipalimtoyl-Sn-Glycero-3-(Phospho-Rac-(1-Glycerol))	general
1,2-Distearoyl-Sn-Glycero-3-(Phospho-Rac-(1-Glycerol))	general
1,2-Distearoyl-Sn-Glycero-3-Phosphocholine	general
1-O-Tolylbiguanide	general
2-Ethyl-1,6-Hexanediol	general
Acetic Acid	general
Acetic Anhydride	general
Acetone	general
Acetone Sodium Bisulfite	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Acetylated Lanolin Alcohol	general
Acetylated Monoglyceride	general
Acetylcysteine	general
Acetyltryptophan (DL-)	general
Acrylates Copolymer	general
Acrylic Acid-Isooctyl Acrylate Copolymer	general
Acrylic Adhesive 788	general
Activated Charcoal	general
Adcote 72A103	general
Adhesive Tape	general
Adipic Acid	general
Aerotex Resin 3730	general
Alanine	general
albumin	general
Albumin Aggregated	general
Albumin Colloidal	general
Albumin Human	general
Alcohol	general
Alfadex	general
Alkyl Ammonium Sulfonic Acid Betaine	general
Alkyl Aryl Sodium Sulfonate	general
Allantoin	general
Allyl Alpha-Ionone	general
Almond Oil	general
Alpha Terpineol	general
Alpha-Tocopherol (DL-)	general
Alpha-Tocopherol Acetate (DL-)	general
Aluminum Acetate	general
Aluminum Chiorhydroxy Allantoinate	general
Aluminum Hydroxide	general
Aluminum Hydroxide - Sucrose	general
Aluminum Hydroxide Gel	general
Aluminum Hydroxide Gel F 500	general
Aluminum Hydroxide Gel F 5000	general
Aluminum Monostearate	general
Aluminum Oxide	general
Aluminum Polyester	general
Aluminum Silicate	general
Aluminum Starch Octenyl succinate	general
Aluminum Stearate	general
Aluminum Subacetate	general
Aluminum Sulfate Anhydrous	general
Amerchol C	general
Amerchol-Cab	general
Aminomethylpropanol	general
Ammonia	general
Ammonia Solution	general
Ammonium Acetate	general
Ammonium Hydroxide	general
Ammonium Lauryl Sulfate	general
Ammonium Nonoxynol-4 Sulfate	general
Ammonium Salt Of C-12-C-15 Linear Primary Alcohol Ethoxylate	general
Ammonium Sulfate	general
Ammonyx	general
Amphoteric-2	general
Amphoteric-9	general
Anethole	general
Anhydrous Citric Acid	general
Anhydrous Dextrose	general
Anhydrous Lactose	general
Anhydrous Trisodium Citrate	general
Aniseed Oil	general
Artoxid Sbn	general
Antifoam	general
Antipyrine	general
Apaflurane	general
Apricot Kernel Oil Peg-6 Esters	general
Aquaphor	general
Arginine	general
Arlacel	general
Ascorbic Acid	general
Ascorbyl Palmitate	general
Aspartic Acid	general
Bacteriostatic	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Balsam Peru	general
Barium Sulfate	general
Beheneth-10	general
Benzalkonium Chloride	general
Benzenesulfonic Acid	general
Benzethonium Chloride	general
Benzododecinium Bromide	general
Benzoic Acid	general
Benzyl Benzoate	general
Benzyl Chloride	general
Betadex	general
Bibapcitide	general
Bismuth Subgallate	general
Boric Acid	general
Brocrinat	general
Butane	general
Butyl Alcohol	general
Butyl Ester Of Vinyl Methyl Ether/Maleic Anhydride Copolymer (125000 Mw)	general
Butyl Stearate	general
Butylated Hydroxyanisole	general
Butylated Hydroxytoluene	general
Butylene Glycol	general
Butylparaben	general
Butyric Acid	general
C20-40 Pareth-24	general
Caffeine	general
Calcium	general
Calcium Carbonate	general
Calcium Chloride	general
Calcium Gluceptate	general
Calcium Hydroxide	general
Calcium Lactate	general
Calcobutrol	general
Caldiamide Sodium	general
Caloxetate Trisodium	general
Calteridol Calcium	general
Canada Balsam	general
Caprylic/Capric Triglyceride	general
Caprylic/Capric/Stearic Triglyceride	general
Caplan	general
Captisol	general
Caramel	general
Carbomer 1342	general
Carbomer 1382	general
Carbomer 934	general
Carbomer 934p	general
Carbomer 940	general
Carbomer 941	general
Carbomer 980	general
Carbomer 981	general
Carbomer Homopolymer Type B (Allyl Pentaerythritol Crosslinked)	general
Carbomer Homopolymer Type C (Allyl Pentaerythritol Crosslinked)	general
Carbon Dioxide	general
Carboxy Vinyl Copolymer	general
Carboxymethylcellulose (CMC)	general
Carboxymethylcellulose Sodium	general
Carboxypoly methylene	general
Carrageenan	general
Carrageenan Salt	general
Cedar Leaf Oil	general
Cellobiose	general
Cellulose	general
Cerasynt-Se	general
Ceresin	general
Ceteareth-12	general
Ceteareth-15	general
Ceteareth-30	general
Cetearyl Alcohol/Ceteareth-20	general
Cetearyl Ethylhexanoate	general
Ceteth-10	general
Ceteth-2	general
Ceteth-20	general

TABLE 1-continued

Excipients

Excipient	Example Category
Ceteth-23	general
Cetostearyl Alcohol	general
Cetrimonium Chloride	general
Cetyl Alcohol	general
Cetyl Esters Wax	general
Cetyl Palmitate	general
Cetylpyridinium Chloride	general
Chlorobutanol	general
Chlorobutanol Hemihydrate	general
Chlorocresol	general
Chloroutanol anhydrous	general
Chloroxylenol	general
Cholesterol	general
Choleth	general
Choleth-24	general
Citrate	general
Citric Acid	general
citric acid (hydrous)	general
Citric Acid Monohydrate	general
Cocamide Ether Sulfate	general
Cocamine Oxide	general
Coco Betaine	general
Coco Diethanolamide	general
Coco Monoethanolamide	general
Cocoa Butter	general
Coco-Glycerides	general
Coconut Oil	general
Coconut Oil glycerides	general
Cocoyl Caprylocaprate	general
Cola Nitida Seed Extract	general
Collagen	general
Colloidal	general
Coloring Suspension	general
Corn	general
Cream Base	general
Creatine	general
Creatinine	general
Cresol	general
Croscarmellose Sodium	general
Crospovidone	general
Cupric Sulfate	general
Cupric Sulfate Anhydrous	general
Cyclomethicone	general
Cyclo methicone/Dimethicone Copolyol	general
Cysteine	general
Cysteine (DL-)	general
Cysteine Hydrochloride	general
Cysteine Hydrochloride Anhydrous	general
D&CRed No. 28	general
D&C Red No. 33	general
D&C Red No. 36	general
D&C Red No. 39	general
D&C Yellow No. 10	general
Dalfampridine	general
Dauber 1-5 Pestr (Matte) 164z	general
Decyl Methyl Sulfoxide	general
Dehydag Wax Sx	general
Dehydrated	general
Dehydroacetic Acid	general
Dehymuls E	general
Denatonium Benzoate	general
Denatured	general
Dental	general
Deoxycholic Acid	general
Dextran	general
Dextran 40	general
Dextrin	general
Dextrose	general
Dextrose Monohydrate	general
Dextrose Solution	general
Diatrizoic Acid	general
Diazolidinyl Urea	general
Dichlorobenzyl Alcohol	general
Dichlorodifluoromethane	general
Dichlorotetrafluoroethane	general

TABLE 1-continued

Excipients

Excipient	Example Category
Diethanolamine	general
Diethyl Pyrocarbonate	general
Diethyl Sebacate	general
Diethylene Glycol Monoethyl Ether	general
Diethylhexyl Phthalate	general
Dihydroxyaluminum Aminoacetate	general
Diisopropanolamine	general
Diisopropyl Adipate	general
Diisopropyl Dilinoleate	general
Dimethicone 350	general
Dimethicone Copolyol	general
Dimethicone Mdx4-4210	general
Dimethicone Medical Fluid .360	general
Dimethyl Isosorbide	general
Dimethylaminoethyl Methacrylate - Butyl Methacrylate - Methyl Methacrylate Copolymer	general
Dimethyldioctadecylammonium Bentonite	general
Dimethylsiloxane/Methylvinylsiloxane Copolymer	general
Dinoseb Ammonium Salt	general
Dipalmitoylphosphatidylglycerol (DL-)	general
Dipropylene Glycol	general
Disodium Cocoamphodiacetate	general
Disodium Laureth Sulfosuccinate	general
Disodium Lauryl Sulfosuccinate	general
Disodium Sulfosalicylate	general
Disofenin	general
Divinylbenzene Styrene Copolymer	general
Dmdm Hydantoin	general
Docosanol	general
Docusate Sodium	general
Duro-Tak 280-2516	general
Duro-Tak 387-2516	general
Duro-Tak 80-1196	general
Duro-Tak 87-2070	general
Duro-Tak 87-2194	general
Duro-Tak 87-2287	general
Duro-Tak 87-2296	general
Duro-Tak 87-2888	general
Duro-Tak 87-2979	general
Edetate Calcium Disodium	general
Edetate Disodium	general
Edetate Disodium Anhydrous	general
Edetate Sodium	general
Edetic Acid	general
Egg	general
Egg Phospholipid	general
Entsufon	general
Entsufon Sodium	general
Epilactose	general
Epitetracycline Hydrochloride	general
Essence Bouquet 9200	general
Ethanolamine Hydrochloride	general
Ethoxylated	general
Ethyl Ester Terminated	general
Ethyl Oleate	general
Ethylene Glycol	general
Ethylene Vinyl Acetate Copolymer	general
Ethylenediamine	general
Ethylenediamine Dihydrochloride	general
Ethylenediaminetetracetic acid (EDTA)	general
Ethylene-Propylene Copolymer	general
Ethylene-Vinyl Acetate Copolymer (28% Vinyl Acetate)	general
Ethylene-Vinyl Acetate Copolymer (9% Vinylacetate)	general
Ethylhexyl Hydroxystearate	general
Ethylparaben	general
Eucalyptol	general
Exametazime	general
F&C Red No. 40	general
Fat (Edible)	general
Fat (Hard)	general
Fatty Acid	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Fatty Acid Ester	general
Fatty Acid Pentaerythriol Ester	general
Fatty Alcohol	general
Fatty Alcohol Citrate	general
FD&C Blue No. 1 (brilliant blue FCF)	general
FD&C Green No. 3 (fast green FCF)	general
FD&C Red No. 4	general
FD&C Yellow No. 10 (Delisted)	general
FD&C Yellow No. 5 (tartrazine)	general
FD&C Yellow No. 6 (sunset yellow)	general
Ferric Chloride	general
Ferric Oxide	general
Flavor 89-186	general
Flavor 89-259	general
Flavor Df-119	general
Flavor Df-1530	general
Flavor Enhancer	general
Flavor Fig 827118	general
Flavor Raspberry Pfc-8407	general
Flavor Rhodia Pharmaceutical No. Rf 451	general
Fluorochlorohydrocarbon	general
Formaldehyde	general
Formaldehyde Solution	general
Fractionated Coconut Oil	general
Fragrance 3949-5	general
Fragrance 520a	general
Fragrance 6.007	general
Fragrance 91-122	general
Fragrance 9128-Y	general
bragrance 93498g	general
Fragrance Balsam Pine No. 5124	general
Fragrance Bouquet 10328	general
Fragrance Chemoder 6401-B	general
Fragrance Chemoderm 6411	general
Fragrance Cream No. 73457	general
Fragrance Cs-28197	general
Fragrance Felton 066m	general
Fragrance Firmenich 47373	general
Fragrance Givaudan Ess 9090/1c	general
Fragrance H-6540	general
Fragrance Herbal 10396	general
Fragrance Nj-1085	general
Fragrance P O Fl-147	general
Fragrance Pa 52805	general
Fragrance Pera Derm D	general
Fragrance Rbd-9819	general
Fragrance Shaw Mudge U-7776	general
Fragrance Tf 044078	general
Fragrance Ungerer Honeysuckle K 2771	general
Fragrance Ungerer N5195	general
Fructose	general
Gadolinium Oxide	general
Galactose	general
Gamma Cyclodextrin	general
Gelatin (Crosslinked)	general
Gelfoam Sponge	general
Gellan Gum (Low Acyl)	general
Gelva 737	general
Gentisic Acid	general
Gentisic Acid Ethanolamide	general
Glacial acetic acid	general
Gluceptate Sodium	general
Gluceptate Sodium Dihydrate	general
Gluconolactone	general
Glucuronic Acid	general
Glutamic Acid (DL-)	general
Glutathione	general
Glycerol Ester Of Hydrogenated Rosin	general
Glyceryl Citrate	general
Glyceryl Isostearate	general
Glyceryl Laurate	general
Glyceryl Monostearate	general
Glyceryl Oleate	general
Glyceryl Oleate/Propylene Glycol	general
Glyceryl Palmitate	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Glyceryl Ricinoleate	general
Glyceryl Stearate	general
Glyceryl Stearate - Laureth-23	general
Glyceryl Stearate/Peg, Stearate	general
Glyceryl Stearate/Peg-100 Stearate	general
Glycervi Stearate/Peg-40 Stearate	general
Glyceryl Stearate-Stearamidoethyl Diethylamine	general
Glyceryl Trioleate	general
Glycine	general
Glycine Hydrochloride	general
Glycol Distearate	general
Glycol Stearate	general
Guanidine Hydrochloride	general
Guar Gum	general
Hair Conditioner (18n195-1m)	general
Heptane	general
Hetastarch	general
Hexylene Glycol	general
High Density Polyethylene	general
Histidine	general
Human Albumin Microspheres	general
Hyaluronate Sodium	general
Hydrocarbon	general
Hydrocarbon Gel	general
Hydrochloric Acid	general
Hydrocortisone	general
Hydrogel Polymer	general
Hydrogen Peroxide	general
Hydrogenated Castor Oil	general
Hydrogenated coconut oil	general
Hydrogenated Coconut Oil Glyceride	general
Hydrogenated palm kernel oil	general
Hydrogenated Palm Kernel Oil glyceride	general
Hydrogenated Palm Oil	general
Hydrogenated Palm/Palm Kernel Oil Peg-6 Ester	general
Hydrogenated Polvbutene 635-690	general
Hydrogenated Soy	general
Hydrogenated soy phosphotidylcholine	general
Hydroxide Ion	general
Hydroxyethylpiperazine Ethane Sulfonic Acid	general
Hydroxymethyl Cellulose	general
Hydroxyoctacosanyl Hydroxystearate	general
Hydroxypropyl Methylcellulose 2906	general
Hydroxypropyl-B-cyclodextrin	general
Hypromellose	general
Hypromellose 2208 (15000 Mpa · S)	general
Hypromellose 2910 (15000 Mpa · S)	general
Imidurea	general
Iodine	general
Iodoxamic Acid	general
Iofetamine Hydrochloride	general
Irish Moss Extract	general
Isobutane	general
Isoceteth-20	general
Isoleucine	general
Isooctyl Acrylate	general
Isopropyl Alcohol	general
Isopropyl Isostearate	general
Isopropyl Myristate	general
Isopropyl Palmitate	general
Isopropyl Stearate	general
Isostearic Acid	general
Isostearyl Alcohol	general
Isotonic Sodium Chloride Solution	general
Jelene	general
Kaolin	general
Kathon Cg	general
Kathon Cg II	general
Lactate	general
Lactic Acid	general
Lactic Acid (DL-)	general
Lactobionic Acid	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Lactose	general
Lactose hydrous	general
Lactose Monohydrate	general
Laneth	general
Lanolin	general
Lanolin (ethoxylated)	general
Lanolin (hydrogenated)	general
Lanolin Alcohol	general
Lanolin Anhydrous	general
Lanolin Cholesterol	general
Lanolin Nonionic Derivatives	general
Lauralkonium Chloride	general
Lauramine Oxide	general
Laurdimonium Hydrolyzed Animal Collagen	general
Laureth Sulfete	general
Laureth-2	general
Laureth-23	general
Laureth-4	general
Laurie Diethanolamide	general
Lauric Myristic Diethanolamide	general
Lauroyl Sarcosine	general
Lauryl Lactate	general
Lauryl Sulfate	general
Lavandula Angustifolia Flowering Top	general
Lecithin	general
Lecithin (hydrogenated)	general
Lecithin Unbleached	general
Lemon Oil	general
Leucine	general
Levulinic Acid	general
Lidofenin	general
Light Mineral Oil	general
Light Mineral Oil (85 Ssu)	general
Limonene (+/−)	general
Lipocol Sc-15	general
Lysine	general
Lysine Acetate	general
Lysine Monohydrate	general
Magnesium Aluminum Silicate Hydrate	general
Magnesium Chloride	general
Magnesium Nitrate	general
Magnesium Stearate	general
Maleic Acid	general
Maltitol	general
Maltodextrin	general
Mannitol	general
Mannose	general
Maprofix	general
Mebrofenin	general
Medical Adhesive Modified 5-15	general
Medical Antiform A-F Emulsion	general
Medium Chain	general
Medronate Disodium	general
Medronic Acid	general
Meglumine	general
Melezitose	general
Menthol	general
Metacresol	general
Metaphosphoric Acid	general
Methanesulfonic Acid	general
Methionine	general
Methyl Alcohol	general
Methyl Gluceth-10	general
Methyl Gluceth-20	general
Methyl Gluceth-20 Sesquistearate	general
Methyl Glucose Sesquistearate	general
Methyl Laurate	general
Methyl Pyrrolidone	general
Methyl Salicylate	general
Methyl Stearate	general
Methylboronic Acid	general
Methylcellulose (4000 Mpa · S)	general
Methylchloroisothiazolinone	general
Methylene Blue	general
Methylisothiazolinone	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Methylparaben	general
Microcry stalline	general
Microcrystalline Wax	general
Mineral Oil	general
Monostearyl Citrate	general
Monothioglycerol	general
Multisterol Extract	general
Myristyl Alcohol	general
Myristyl Lactate	general
Myristyl-.Gamma.-Picolinium Chloride	general
N-(Carbamoyl-Methoxy Peg-40)-1,2-Distearoyl-Cephalin Sodium	general
N,N-Dimethylacetamide	general
Niacinamide	general
Nioxime	general
Nitric Acid	general
Nitrogen	general
Nonoxynol Iodine	general
Nonoxynol-15	general
Nonoxynol-9	general
Norflurane	general
Oatmeal	general
Octadecene-1/Maleic Acid Copolymer	general
Octanoic Acid	general
Octisalate	general
Octoxynol-1	general
Octoxynol-40	general
Octoxynol-9	general
Octyldodecanol	general
Octylphenol Polymethylene	general
Oleth-10/Oleth-5	general
Oleth-2	general
Oleth-20	general
Oleyl Alcohol	general
Oleyl Oleate	general
Oxidronate Disodium	general
Oxyquinoline	general
Palm Kernel Oil	general
Palm Kernel Oil Glyceride	general
Palmitamine Oxide	general
Parabens	general
Paraffin	general
Parfum Creme 45/3	general
Peanut Oil (Refined)	general
Pectin	general
Peg 6-32 Stearate/Glycol Stearate	general
Peg Vegetable Oil	general
Peg-100 Stearate	general
Peg-12 Glyceryl Laurate	general
Peg-120 Glyceryl Stearate	general
Peg-120 Methyl Glucose Dioleate	general
Peg-15 Cocamine	general
Peg-150 Distearate	general
Peg-2 Stearate	general
Peg-20 Sorbitan Isostearate	general
Peg-22 Methyl Ether/Dodecyl Glycol Copolymer	general
Peg-25 Propylene Glycol Stearate	general
Peg-4 Dilaurate	general
Peg-4 Laurate	general
Peg-40 Castor Oil	general
Peg-40 Sorbitan Diisostearate	general
Peg-45/Dodecyl Glycol Copolymer	general
Peg-5 Oleate	general
Peg-50 Stearate	general
Peg-54 Hydrogenated Castor Oil	general
Peg-6 Isostearate	general
Peg-60 Castor Oil	general
Peg-60 Hydrogenated Castor Oil	general
Peg-7 Methyl Ether	general
Peg-75 Lanolin	general
Peg-8 Laurate	general
Peg-8 Stearate	general
Pegoxol 7 Stearate	general
Pentadecalactone	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Pentaerythritol Cocoate	general
Pentasodium Pentetate	general
Pentetate Calcium Trisodium	general
Pentetic Acid	general
Perflutren	general
Perfume 25677	general
Perfume Bouquet	general
Perfume E-1991	general
Perfume Gd 5604	general
Perfume Tana 90/42 Scba	general
Perfume W-1952-1	general
Petrolatum	general
Petroleum Distillate	general
Phenol	general
Phenol (Liquefied)	general
Phenonip	general
Phenoxyethanol	general
Phenylalanine	general
Phenylethyl Alcohol	general
Phenylmercuric Acetate	general
Phenylmercuric Nitrate	general
Phosphate buffer	general
Phosphate buffered saline	general
Phosphate salts	general
Phosphatidyl Glycerol	general
Phospholipid	general
Phospholipid (Egg)	general
Phospholipon 90 g	general
Phosphoric Acid	general
Pine Needle Oil (Pinus Sylvestris)	general
Piperazine Hexahydrate	general
Plastibase-50w	general
Polacrilin	general
Polidromum Chloride	general
Poloxamer 124	general
Poloxamer 181	general
Poloxamer 182	general
Poloxamer-188	general
Poloxamer 237	general
Poloxamer-407	general
Poly(Bis(P-Catboxyphenoxy)Propane Anhydride):Sebacic Acid	general
Poly(Dimethylsiloxane/Methylvinylsiloxane/ Methylhydrogensiloxane) Dimethylvinyl Or Dimethylhydroxy Or Trimethyl Endblocked	general
Poly(Dl-Lactic-Co-Glycolic Acid)	general
Polybutene (1400 Mw)	general
Polycarbophil	general
Polyester Polyamine Copolymer	general
Polyester Rayon	general
Polyethylene Glycol 1000	general
Polyethylene Glycol 1450	general
Polyethylene Glycol 1500	general
Polyethylene Glycol 1540	general
Polyethylene Glycol 200	general
Polyethylene Glycol 300	general
Polyethylene Glycol 300-1600	general
Polyethylene Glycol 3350	general
Polyethylene Glycol 400 (PEG 400)	general
Polyethylene Glycol 4000 (PEG 4000, PEG 4 kDa)	general
Polyethylene Glycol 540	general
Polyethylene Glycol 600	general
Polyethylene Glycol 6000	general
Polyethylene Glycol 8000	general
Polyethylene Glycol 900	general
Polyethylene High Density Containing Ferric Oxide Black (<1%)	general
Polyethylene Low Density Containing Barium Sulfate (20-24%)	general
Polyethylene T	general
Polyethylene Terephthalate	general
Polyglactin	general
Polyglyceryl-3 Oleate	general
Polyglyceryl-4 Oleate	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Polyhydroxymethyl Methacrylate	general
Polyisobutylene	general
Polyisobutylene (1100000 Mw)	general
Polyisobutylene (35000 Mw)	general
Polyisobutylene 178-236	general
Polyisobutylene 241-294	general
Polyisobutylene 35-39	general
Polyisobutylene Low Molecular Weight	general
Polyisobutylene Medium Molecular Weight	general
Polyisobutylene/Polybutene Adhesive	general
Polylactide	general
Polyol	general
Polyoxyethylene	general
Polyoxyethylene Alcohol	general
Polyoxyethylene Fatly Acid Ester	general
Polyoxyethylene Propylene	general
Polyoxyl 20 Cetostearyl Ether	general
Polyoxyl 32 Palmitostearate	general
Polyoxyl 35 Castor Oil	general
Polyoxyl 40 Hydrogenated Castor Oil	general
Polyoxyl 40 Stearate	general
Polyoxyl 400 Stearate	general
Polyoxyl 6	general
Polyoxyl Distearate	general
Polyoxyl Glyceryl Stearate	general
Polyoxyl Lanolin	general
Polyoxyl Palmitate	general
Polyoxyl Stearate	general
Polyoxylpropylene 1800	general
Polypropylene	general
Polypropylene Glycol	general
Polyquaternium-10	general
Polyquaternium-7 (70/30) Acrylamide/ Dadmac	general
Polysiloxane	general
Polysorbate 40	general
Polysorbate 60	general
Polysorbate 65	general
Polyurethane	general
Polyvinyl Acetate	general
Polyvinyl Chloride	general
Polyvinyl Chloride-Polyvinyl Acetate Copolymer	general
Polyvinylpyridine	general
Poppy Seed Oil	general
Potash	general
Potassium Acetate	general
Potassium Alum	general
Potassium Bicarbonate	general
Potassium Bisulfite	general
Potassium Chloride	general
Potassium Citrate	general
Potassium Hydroxide	general
Potassium Metabisulfite	general
Potassium Phosphate (Dibasic)	general
Potassium Phosphate (Monobasic)	general
Potassium Soap	general
Povidone	general
Povidone Acrylate Copolymer	general
Povidone Hydrogel	general
Povidone K17	general
Povidone K25	general
Povidone K29/32	general
Povidone K30	general
Povidone K90	general
Povidone K90f	general
Povidone/Eicosene Copolymer	general
Ppg-12/Smdi Copolymer	general
Ppg-15 Stearyl Ether	general
Ppg-20 Methyl Glucose Ether Distearate	general
Ppg-26 Oleate	general
Pregelatinized	general
Product Wat	general
Proline	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Promulgen D	general
Promulgen G	general
Propane	general
Propellant A-46	general
Propyl Gallate	general
Propylene Glycol Diacetate	general
Propylene Glycol Dicaprylate	general
Propylene Glycol Monolaurate	general
Propylene Glycol Monopalmitostearate	general
Propylene Glycol Palmito stearate	general
Propylene Glycol Ricinoleate	general
Propylene Glycol/Diazolidinyl Urea/ Methylparaben/Propylparben	general
Propylparaben	general
Protamine Sulfate	general
Protein Hydrolysate	general
Pvm/Ma Copolymer	general
Quaternium-15	general
Quaternium-15 Cis-Form	general
Quaternium-52	general
Ra-2397	general
Ra-3011	general
Raffinose	general
Saccharin	general
Saccharin Sodium	general
Saccharin Sodium Anhydrous	general
Sd Alcohol 3a	general
Sd Alcohol 40	general
Sd Alcohol 40-2	general
Sd Alcohol 40b	general
Sepineo P 600	general
Serine	general
Shea Butter	general
Silastic Brand Medical Grade Tubing	general
Silastic Medical Adhesive	general
Silica	general
Silicon	general
Silicon Dioxide	general
Silicone	general
Silicone Adhesive 4102	general
Silicone Adhesive 4502	general
Silicone Adhesive Bio-Psa Q7-4201	general
Silicone Adhesive Bio-Psa Q7-4301	general
Silicone Emulsion	general
Silicone Type A	general
Silicone/Polyester Film Strip	general
Simethicone	general
Simethicone Emulsion	general
Sipon Ls 20np	general
Soda Ash	general
Sodium Acetate	general
Sodium Acetate Anhydrous	general
Sodium Alkyl Sulfate	general
Sodium Ascorbate	general
Sodium Benzoate	general
Sodium Bicarbonate	general
Sodium Bisulfate	general
Sodium Borate	general
Sodium Borate Decahydrate	general
Sodium Carbonate	general
Sodium Carbonate Decahydrate	general
Sodium Carbonate Monohydrate	general
Sodium Cetostearyl Sulfate	general
Sodium Chlorate	general
Sodium Chloride	general
Sodium Chloride Injection	general
Sodium Cholesteryl Sulfate	general
Sodium Citrate	general
Sodium Citrate Dihydrate	general
Sodium Cocoyl Sarcosinate	general
Sodium Desoxycholate	general
Sodium Dithionite	general
Sodium Dodecylbenzenesulfonate	general
Sodium Formaldehyde Sulfoxylate	general
Sodium Gluconate	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Sodium Hydroxide	general
Sodium Hypochlorite	general
Sodium Iodide	general
Sodium Lactate	general
Sodium Lactate (L-)	general
Sodium Laureth-2 Sulfate	general
Sodium Laureth-3 Sulfate	general
Sodium Laureth-5 Sulfate	general
Sodium Lauroyl Sarcosinate	general
Sodium Lauryl Sulfate	general
Sodium Laurel Sulfoacetate	general
Sodium Metabisulfite	general
Sodium Phosphate	general
Sodium Phosphate (Dibasic)	general
Sodium Phosphate (Dibasic, Anhydrous)	general
Sodium Phosphate (Dibasic, Dihydrate)	general
Sodium Phosphate (Dibasic, Dodecahydrate)	general
Sodium Phosphate (Dibasic, Heptahydrate)	general
Sodium Phosphate (Monobasic)	general
Sodium Phosphate (Monobasic, Anhydrous)	general
Sodium Phosphate (Monobasic, Dihydrate)	general
Sodium Phosphate (Monobasic, Monohydrate)	general
Sodium Phosphate Dihydrate	general
Sodium Polyacrylate (2500000 Mw)	general
Sodium Pyrophosphate	general
Sodium Pyrrolidone Carboxylate	general
Sodium Starch Glycolate	general
Sodium Succinate Hexahydrate	general
Sodium Sulfate	general
Sodium Sulfate Anhydrous	general
Sodium Sulfate Decahydrate	general
Sodium Sulfite	general
Sodium Sulfosuccinated Undecyclenic Monoalkylolamide	general
Sodium Tartrate	general
Sodium Thioglycolate	general
Sodium Thiomalate	general
Sodium Thiosulfate	general
Sodium Thiosulfate Anhydrous	general
Sodium Trimetaphosphate	general
Sodium Xylenesulfonate	general
Somay 44	general
Sorbic Acid	general
Sorbitan	general
Sorbitan Isostearate	general
Sorbitan Monolaurate	general
Sorbitan Monopalmitate	general
Sorbitan Monostearate	general
Sorbitan Sesquioleate	general
Sorbitan Trioleate	general
Sorbitan Tristearale	general
Sorbitol Solution	general
Sorbose	general
Soybean	general
Soybean Flour	general
Spearmint Oil	general
Spermaceti	general
Squalane	general
Stabilized Oxychloro Complex	general
Stannous 2-Ethylhexanoate	general
Stannous Chloride	general
Stannous Chloride Anhydrous	general
Stannous Fluoride	general
Stannous Tartrate	general
Starch	general
Starch 1500	general
Stearalkonium Chloride	general
Stearalkonium Hectorite/Propylene Carbonate	general
Stearamidoethyl Diethylamine	general
Steareth-10	general
Steareth-100	general
Steareth-2	general
Steareth-20	general
Steareth-21	general
Steareth-40	general

TABLE 1-continued

Excipients	
Excipient	Example Category
Stearic Acid	general
Stearic Diethanolamide	general
Stearoxytrimethylsilane	general
Steartrimonum Hydrolyzed Animal Collagen	general
Stearyl Alcohol	general
Styrene/Isoprene/Styrene Block Copolymer	general
Succimer	general
Succinic Acid	general
Sucralose	general
Sucrose	general
Sucrose Distearate	general
Sucrose Polyester	general
Sugar	general
Sulfacetamide Sodium	general
Sulfobutylether.Beta.-Cyclodextrin	general
Sulfur Dioxide	general
Sulfuric Acid	general
Sulfurous Acid	general
Surfactol Qs	general
Lagatose (D-)	general
Talc	general
Tall Oil	general
Tallow Glycerides	general
Tartaric Acid (DL-)	general
Tenox	general
Tenox-2	general
Tert-Butyl Alcohol	general
Tert-Butyl Hydroperoxide	general
Tert-Butylhydroquinone	general
Tetrakis(2-Methoxylsobutylisocyanide) Copper(I) Tetrafluoroborate	general
Tetrapropyl Orthosilicate	general
Tetrofosmin	general
Theophylline	general
Thimerosal	general
Threonine	general
Thymol	general
Tin	general
Titanium Dioxide	general
Tocopherol	general
Tocophersolan	general
Trehalose	general
Tricaprylin	general
Trichloromonofluoromethane	general
Trideceth-10	general
Triethanolamine Lauryl Sulfate	general
Trifluoroacetic Acid	general
Triglycerides	general
Trihalose	general
Trihydroxystearin	general
Trilaneth-4 Phosphate	general
Trilaureth-4 Phosphate	general
Trisodium Citrate Dihydrate	general
Trisodium Hedta	general
Triton 720	general
Triton X-200	general
Trolamine	general
Tromantadine	general
Tromethamine	general
Tryptophan	general
Tyloxapol	general
Tyrosine	general
Undecylenic Acid	general
Union 76 Amsco-Res 6038	general
Urea	general
Valine	general
Vegetable Oil	general
Vegetable Oil Glyceride	general
Versetamide	general
Viscarin	general
Viscose/Cotton	general
Vitamin E	general
Water	general
Wax	general
Wecobee P	general

TABLE 1-continued

Excipients	
Excipient	Example Category
White	general
White Ceresin Wax	general
White Soft	general
White Wax	general
Zinc	general
Zinc Acetate	general
Zinc Carbonate	general
Zinc Chloride	general
Zinc Oxide	general
DSPC	lipid nanoparticle
lipid nanoparticle	lipid nanoparticle
PEG-DMG 2000 (1,2-dimyristoyl-sn-glycero-3-phophoethanolamine-N-[methoxy(polyethylene glycol)-2000)	lipid nanoparticle
1,2-dilauroyl-sn-glycero-3-phosphocholine (DLPC)	lipids
1,2-dimyristoyl-sn-glycero-3-phosphocholine (DMPC)phosphatidylinositol	lipids
1,2-dioleoyl-sn-glycero-3-phophoethanolamine (DOPE)	lipids
1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC)	lipids
diglyceride	lipids
dilinoleoylphosphatidylcholine	lipids
dioleoylphosphatidylcholine	lipids
dipalmitoylphosphatidylcholine	lipids
distearoylphosphatidylcholine	lipids
fats	lipids
lysolipids	lipids
lysophosphatidylethanolamine	lipids
lysophospholipid	lipids
monoglyceride	lipids
mono-myristoyl-phosphatidylethanolamine (MMPE)	lipids
mono-oleoyl-phosphatidic acid (MOPA)	lipids
mono-oleoyl-phosphatidylethanolamine (MOPE)	lipids
mono-oleoyl-phosphatidylglycerol (MOPG)	lipids
mono-oleoyl-phosphatidylserine (MOPS)	lipids
palmitoyloleoyl	lipids
palmitoyloleoyl phosphatidylcholine	lipids
palmitoyl-oleoyl-phosphatidylethanolamine (POPE)	lipids
phosphatidic acid	lipids
phosphatidylcholines	lipids
phosphatidylethanolamine	lipids
phosphatidylserine	lipids
phosphotidylglycerol	lipids
sterol	lipids
1,2-dilinoleyloxy-3-dimethylaminopropane (DLin-DMA)	liposomes
1,2-dioleyloxy-N,N-dimethylaminopropane (DODMA) liposomes	liposomes
2,2-dilinoleyl-4-(2-dimethylaminoethyl)-[1,3]-dioxolane (DLin-KC2-DMA)	liposomes
DiLa2 liposomes from Marina Biotech (Bothell, WA)	liposomes
hyaluronan-coated liposomes	liposomes
liposome	liposomes
MC3	liposomes
neutral DOPC (1,2-dioleoyl-sn-glycero-3-phosphocholine) based liposome	liposomes
SMARTICLES ® (Marina Biotech, Bothell, WA)	liposomes
stabilized nucleic acid lipid particle (SNALP)	liposomes
stabilized plasmid-lipid particles (SPLP)	liposomes
alkali salt	lubricant
alkaline earth salt	lubricant
aqueous solution	lubricant
calcium stearate	lubricant
fumed silica	lubricant
high molecular weight polyalkylene ghxol	lubricant
high molecular weight polyethylene glycol	lubricant
hyaluronic acid	lubricant

TABLE 1-continued

Excipients	
Excipient	Example Category
hydrogenated vegetable oil	lubricant
hydrous magnesium silicate	lubricant
lipids	lubricant
lubricants	lubricant
lubricin	lubricant
micelle	lubricant
microsphere	lubricant
monoester of propylene glycol	lubricant
oils	lubricant
polymer	lubricant
saturated fatty acid containing about 16-20 carbon atoms	lubricant
saturated fatty acid containing about 8-22 carbon atoms	lubricant
solvents	lubricant
stearate salts	lubricant
transition metal salt	lubricant
vegetable oil derivative	lubricant
acrylic acid	nanoparticles
acrylic polymer	nanoparticles
amino alkyl methacrylate copolymer	nanoparticles
anhydride-modified material	nanoparticles
anhydride-modified phytoglycogen beta-dextrin	nanoparticles
carbon nanoparticles	nanoparticles
ceramic silicon carbide nanoparticle	nanoparticles
cerium oxide nanoparticle	nanoparticles
curcumin nanoparticle	nanoparticles
cyanoethyl methacrylate	nanoparticles
DLin-KC2-DMA	nanoparticles
DLin-M C3 -DMA	nanoparticles
ethoxyethyl methacrylate	nanoparticles
glycogen-type material	nanoparticles
gold nanoparticle	nanoparticles
iron nanoparticles	nanoparticles
iron oxide nanoparticle	nanoparticles
magnetic nano particle	nanoparticles
methacrylic acid	nanoparticles
methacrylic acid copolymer	nanoparticles
methyl methacrylate copolymer	nanoparticles
nanodiamond	nanoparticles
nickel nanoparticle	nanoparticles
phytoglvcogen beta-dextrin	nanoparticles
phytoglycogen octenyl succinate	nanoparticles
platinum nanoparticles	nanoparticles
poly(4-hydroxy-L-proline ester)	nanoparticles
poly(acrylic acid)	nanoparticles
poly(ethylene imine)	nanoparticles
poly(L-lactide-co-L-lysine)	nanoparticles
poly(methacrylic acid)	nanoparticles
poly(orthoesters)	nanoparticles
poly(serine ester)	nanoparticles
polyacetal	nanoparticles
polyacrylate	nanoparticles
polycyanoacrylate	nanoparticles
polyester	nanoparticles
polyether	nanoparticles
polyethylene	nanoparticles
polyhydroxyacid	nanoparticles
polylysine	nanoparticles
polymer coated iron oxide nanoparticle	nanoparticles
polymeric mycelle	nanoparticles
polymethacrylate	nanoparticles
polyphosphazene	nanoparticles
polypropylfumerate	nanoparticles
polyureas	nanoparticles
protein filled nanoparticle	nanoparticles
silica nanoparticle	nanoparticles
silicon dioxide crystalline nanoparticle	nanoparticles
silver nanoparticles	nanoparticles
silver oxide nanoparticle	nanoparticles
titanium dioxide nanoparticle	nanoparticles
natural polymers	natural polymers
natural rubbers	natural polymers

TABLE 1-continued

Excipients	
Excipient	Example Category
ceramic	other
cobalt-chromum-molydenum composite	other
duck's feet collagen	other
ionic liquids	other
magnesium oxide	other
melanin	other
metal scaffold	other
nano-hydroxyapatite	other
poly(α-estef)	other
SBA15	other
alginate	polymers
alkyl cellulose	polymers
amber	polymers
bacterial cellulose	polymers
bioplastic	polymers
bioresorbable polymer matrix	polymers
carbohydrate polymers	polymers
cellulose acetate	polymers
cellulose ester	polymers
cellulose ether	polymers
chitin	polymers
chitosan	polymers
copolymers of acrylic and methacrylic acid esters	polymers
derivatized cellulose	polymers
elastin	polymers
ethylene vinyl acetate polymer (EVA)	polymers
EUDRAGIT ® RL	polymers
EUDRAGIT ® RS	polymers
fibrin	polymers
genetically modified bioplastics	polymers
glycogen	polymers
high-density polyethylene (HOPE)	polymers
hydroxypropyl methylcellulose (HPMC)	polymers
hydroxyalkyl celluloses	polymers
hydroxypropyl ethylcellulose (HEC)	polymers
hydroxypropyl methacrylate (HPMA)	polymers
hy droxypropylcellulose	polymers
keratins	polymers
lignin	polymers
lipid-derived polymer	polymers
low-density polyethylene (LDPE)	polymers
methacrylates	polymers
natural rubber	polymers
neoprene	polymers
nitro cellulose	polymers
nucleic acid	polymers
nylon	polymers
nylon 6	polymers
nylon 6.6	polymers
nylone	polymers
phenol formaldehyde resin	polymers
poloxamer	polymers
poly(butyl(meth)acrylate)	polymers
poly(butyric acid)	polymers
poly(caprolactone) (PCL)	polymers
poly(D,L-lactide) (PDLA)	polymers
poly(D,L-lactide-co-caprolactone)	polymers
poly(D,L-lactide-co-caprolactone-co-glycolide)	polymers
poly(D,L-lactide-co-PPO-co-D,L-lactide)	polymers
poly(ester amides)	polymers
poly(ester ethers)	polymers
poly(ethyl(meth)acrylate)	polymers
poly(ethylene terephthalate)	polymers
poly(glycolic acid) (PGA)	polymers
poly(hexyl(meth)acrylate)	polymers
poly(hydroxy acids)	polymers
poly(isobutyl acrylate)	polymers
poly(isobutyl(meth)acrylate)	polymers
poly(isodecyl(meth)acrylate)	polymers
poly(isopropyl acrylate)	polymers
poly(lactic acid) (PLA)	polymers
poly(lactic acid-co-glycolic acid) (PLGA)	polymers
poly(lactide-co-caprolactone)	polymers
poly(lactide-co-glycolide)	polymers
poly(lauryl(meth)acrylate)	polymers

TABLE 1-continued

Excipients	
Excipient	Example Category
poly(L-lactic acid) (PLLA)	polymers
poly(L-lactic acid-co-glycolic acid) (PLLGA)	polymers
poly(L-lactide) (PLLA)	polymers
poly(methyl acrylate)	polymers
poly(methyl(meth)acrylate) (PMMA)	polymers
poly(octadecyl acrylate)	polymers
poly(ortho)esters	polymers
poly(phenyl(meth)acrylate)	polymers
poly(valeric acid)	polymers
poly(vinyl acetate)	polymers
polyacrylonitrile	polymers
polyalkyl cyanoacralate	polymers
polyalkylene	polymers
polyalkylene glycol	polymers
polyalkylene oxide	polymers
polyalkylene terephthalate	polymers
polyamide	polymers
polyamino acid	polymers
polyanhydride	polymers
polyaniline	polymers
polycaprolactone	polymers
polycarbonate	polymers
polydioxanone	polymers
polydioxanone copolymer	polymers
polyethtylene glycol diglycidl ester	polymers
polyethylene glycol	polymers
polyethylene oxide	polymers
polyethyleneglycol	polymers
polyglycolide	polymers
polyhydroxyalkanoate	polymers
polyhydroxybutyrate (also known as polyhydroxyalkanoate)	polymers
polyhydroxyurethane	polymers
polyisoprene	polymers
polyketal	polymers
polylactic acid	polymers
poly-L-glutamic acid	polymers
poly-L-lysine (PLL)	polymers
polymer of acrylic acid	polymers
polyorthoester	polymers
polyoxymethylene	polymers
polypeptides	polymers
polyphosphoester	polymers
polypropylene fumarate	polymers
polysaccharide	polymers
polystyrene	polymers
polytetrafluoroethylene	polymers
polyvinyl butyral	polymers
polyvinyl ester	polymers
polyvinyl ether	polymers
polyvinyl halides such as poly(vinyl chloride) (PVC)	polymers
poyphosphazene	polymers
quarternary ammonium chitosan	polymers
shellac	polymers
sodium carboxymethylcellulose	polymers
synthetic polyether	polymers
synthetic rubber	polymers
thermoplastic polyurethane	polymers
trimethylene carbonate	polymers
ultra-high-molecular-weight-polyethylene (UHMWPE)	polymers
wool	polymers
β-keratin	polymers
alkylparaben	preservative
amino acids	preservative
Antioxidant	preservative
BHA	preservative
BHT	preservative
calcium propionate	preservative
disodium EDTA	preservative
glutaraldehyde	preservative
magnesium chloride hexahydrate	preservative
m-cresol	preservative
methyl paraben	preservative

TABLE 1-continued

Excipients	
Excipient	Example Category
o-cresol	preservative
p-cresol	preservative
phenylmercuric nitrite	preservative
potassium hydrogen sulfite	preservative
potassium sorbate	preservative
preservative	preservative
propyl paraben	preservative
selenium	preservative
sodium dehydroacetate	preservative
sodium nitrate	preservative
sodium nitrite	preservative
sulfites	preservative
vitamin A	preservative
vitamin C	preservative
acesulfame potassium	sweetener
advantame	sweetener
artificial sweetener	sweetener
aspartame	sweetener
brazzein	sweetener
curculin	sweetener
cyclamates	sweetener
erythritol	sweetener
glucose	sweetener
glycyrrhizin	sweetener
hydrogenated starch hydrolysate	sweetener
inulin	sweetener
ismalt	sweetener
isomaltooligosaccharide	sweetener
isomaltulose	sweetener
lactitol	sweetener
lead acetate	sweetener
mabinlin	sweetener
miraculin	sweetener
mogroside	sweetener
monantin	sweetener
neotame	sweetener
osladin	sweetener
pentadin	sweetener
polydextrose	sweetener
psicose	sweetener
stevia	sweetener
sweetener	sweetener
tagatose	sweetener
thaumatin	sweetener
xylitol	sweetener
xylose	sweetener
elastomer	synthetic polymer
synthetic fiber	synthetic polymer
synthetic polymer	synthetic polymer
thermoplastic	synthetic polymer
thermoset	synthetic polymer
Non-polymeric diol	Demulcent
Non-polymeric glycol	Demulcent
Cellulose derivative	Demulcent
Dextran 70	Demulcent
Cationic cellulose derivative	Demulcent

Some excipients may include pharmaceutically acceptable excipients. The phrase "pharmaceutically acceptable" as used herein, refers to suitability within the scope of sound medical judgment for contacting subject (e.g., human or animal) tissues and/or bodily fluids with toxicity, irritation, allergic response, or other complication levels yielding reasonable benefit/risk ratios. As used herein, the term "pharmaceutically acceptable excipient" refers to any ingredient, other than active agents, that is substantially nontoxic and non-inflammatory in a subject. Pharmaceutically acceptable excipients may include, but are not limited to, solvents, dispersion media, diluents, inert diluents, buffering agents, lubricating agents, oils, liquid vehicles, dispersion or suspension aids, surface active agents, isotonic agents, thickening or emulsifying agents, preservatives, and the like, as suited to the particular dosage form desired. Various excipients for formulating pharmaceutical compositions and techniques for preparing the composition are known in the art (see Remington: The Science and Practice of Pharmacy, 21st Edition, A. R. Gennaro, Lippincott, Williams & Wilkins, Baltimore, MD, 2006; incorporated herein by reference in its entirety). The use of a conventional excipient medium may be contemplated within the scope of the present disclosure, except insofar as any conventional excipient medium may be incompatible with a substance or its derivatives, such as by producing any undesirable biological effect or otherwise interacting in a deleterious manner with any other component(s) of pharmaceutical compositions.

In one embodiment, the excipient is sorbitol.

In one embodiment, the excipient is mannitol.

Polymers

In some embodiments, excipients may include polymers. As used herein, the term "polymer" refers to any substance formed through linkages between similar modules or units. Individual units are referred to herein as "monomers." Common polymers found in nature include, but are not limited to, carbon chains (e.g., lipids), polysaccharides, nucleic acids, and proteins. In some embodiments, polymers may be synthetic (e.g., thermoplastics, thermosets, elastomers, and synthetic fibers), natural (e.g., chitosan, cellulose, polysaccharides, glycogen, chitin, polypeptides, β-keratins, nucleic acids, natural rubber, etc.), or a combination thereof. In some embodiments, polymers may be irradiated. Non limiting examples of polymers include ethylcellulose and co-polymers of acrylic and methacrylic acid esters (EUDRAGIT® RS or RL), alginates, sodium carboxymethylcellulose, carboxypolymethylene, hydroxpropyl methylcellulose, hydroxypropyl cellulose, collagen, hydroxypropyl ethylcellulose, hydroxyethylcellulose, methylcellulose, xanthum gum, polyethylene oxide, polyethylene glycol, polysiloxane, poyphosphazene, low-density polyethylene (LDPE), high-density polyethylene (HDPE), polyvinyl chloride, polystyrene, nylon, nylon 6, nylon 6.6, polytetrafluoroethylene, thermoplastic polyurethanes, polycaprolactone, polyamide, polycarbonate, chitosan, cellulose, polysaccharides, glycogen, starch, chitin, polypeptides, keratins, β-keratins, nucleic acids, natural rubber, hyaluronan, polylactic acid, methacrylates, polyisoprene, shellac, amber, wool, synthetic rubber, silk, phenol formaldehyde resin, neoprene, nylon, polyacrylonitrile, silicone, polyvinyl butyral, polyhydroxybutyrate (also known as polyhydroxyalkanoate), polyhydroxyurethanes, bioplastics, genetically modified bioplastics, lipid-derived polymers, lignin, carbohydrate polymers, ultra-high-molecular-weight-polyethylene (UHMWPE), gelatin, dextrans, and polyamino acids.

Specific non-limiting examples of specific polymers include, but are not limited to poly(caprolactone) (PCL), ethylene vinyl acetate polymer (EVA), poly(lactic acid) (PLA), poly(L-lactic acid) (PLLA), poly(glycolic acid) (PGA), poly(lactic acid-co-glycolic acid) (PLGA), poly(L-lactic acid-co-glycolic acid) (PLLGA), poly(D,L-lactide) (PDLA), poly(L-lactide) (PLLA), poly(D,L-lactide-co-caprolactone), poly(D,L-lactide-co-caprolactone-co-glycolide), poly(D,L-lactide-co-PEO-co-D,L-lactide), poly(D,L-lactide-co-PPO-co-D,L-lactide), polyalkyl cyanoacralate, polyurethane, poly-L-lysine (PLL), hydroxypropyl methacrylate (HPMA), polyethyleneglycol, poly-L-glutamic acid, poly(hydroxy acids), polyanhydrides, polyorthoesters, poly(ester amides), polyamides, poly(ester ethers), polycarbonates, polyalkylenes such as polyethylene and polypropylene, polyalkylene glycols such as poly(ethylene glycol) (PEG), polyalkylene oxides (PEO), polyalkylene terephthalates such as poly(ethylene terephthalate), polyvinyl alcohols (PVA), polyvinyl ethers, polyvinyl esters such as poly (vinyl acetate), polyvinyl halides such as poly(vinyl chloride) (PVC), polyvinylpyrrolidone, polysiloxanes, polystyrene (PS), polyurethanes, derivatized celluloses such as alkyl celluloses, hydroxyalkyl celluloses, cellulose ethers, cellulose esters, nitro celluloses, hydroxypropylcellulose, carboxymethylcellulose, polymers of acrylic acids, such as poly(methyl(meth)acrylate) (PMMA), poly(ethyl(meth) acrylate), poly(butyl(meth)acrylate), poly(isobutyl(meth) acrylate), poly(hexyl(meth)acrylate), poly(isodecyl(meth) acrylate), poly(lauryl(meth)acrylate), poly(phenyl(meth) acrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), poly(octadecyl acrylate) and copolymers and mixtures thereof, polydioxanone and its copolymers, polyhydroxyalkanoates, polypropylene fumarate, polyoxymethylene, poloxamers, poly(ortho)esters, poly(butyric acid), poly(valeric acid), poly(lactide-co-caprolactone), and trimethylene carbonate, polyvinylpyrrolidone. In some embodiments, polymer excipients may include any of those presented in Table 1, above.

Particles

In some embodiments, excipients may include particles. Such particles may be of any size and shape, depending on the nature of associated SBPs. In some embodiments, excipient particles are nanoparticles. Non-limiting examples of nanoparticles include gold nanoparticles, silver nanoparticles, silver oxide nanoparticles, iron nanoparticles, iron oxide nanoparticles, platinum nanoparticles, silica nanoparticles, titanium dioxide nanoparticles, magnetic nanoparticles, cerium oxide nanoparticles, protein filled nanoparticles, carbon nanoparticles, nanodiamonds, curcumin nanoparticles, polymeric micelles, polymer coated iron oxide nanoparticles, ceramic silicon carbide nanoparticles, nickel nanoparticles, and silicon dioxide crystalline nanoparticles.

In some embodiments, nanoparticles may include carbohydrate nanoparticles. Carbohydrate nanoparticles may include carbohydrate carriers. As a non-limiting example, carbohydrate carriers may include, but are not limited to, anhydride-modified or glycogen-type materials, phytoglycogen octenyl succinate, phytoglycogen beta-dextrin, or anhydride-modified phytoglycogen beta-dextrin. (See e.g., International Publication Number WO2012109121, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, excipient nanoparticles may include lipid nanoparticles. Lipid nanoparticle excipients may be carriers in some embodiments. In some embodiments, lipid nanoparticles may be formulated with cationic lipids. In some embodiments, cationic lipids may be biodegradable cationic lipids. Such cationic lipids may be used to form rapidly eliminated lipid nanoparticles. Cationic lipids may include, but are not limited, DLinDMA, DLin-KC2-DMA, and DLin-MC3-DMA. Biodegradable lipid nanoparticles may be used to avoid toxicity associated with accumulation of more stable lipid nanoparticles in plasma and tissues over time.

In some embodiments, nanoparticles include polymeric matrices. As used herein, the term "polymeric matrix" refers to a network of polymer fibers that are bound together to form a material. The polymer fibers may be uniform or may include different lengths or monomer subunits. In some embodiments, polymer matrices may include one or more of polyethylenes, polycarbonates, polyanhydrides, polyhydroxyacids, polypropylfumerates, polycaprolactones, polyamides, polyacetals, polyethers, polyesters, poly(orthoesters), polycyanoacrylates, polyvinyl alcohols, polyurethanes, polyphosphazenes, polyacrylates, polymethacrylates, polycyanoacrylates, polyureas, polystyrenes, polyamines, polylysine, poly(ethylene imine), poly(serine ester), poly(L-lactide-co-L-lysine), poly(4-hydroxy-L-proline ester), or combinations thereof.

In some embodiments, polymers include diblock copolymers. As used herein, the term "diblock copolymer" refers to polymers with two different monomer chains grafted to form a single chain. Diblock polymers may be designed to aggregate in different ways, including aggregation as a particle. In some embodiments, diblock copolymers include polyethylene glycol (PEG) in combination with polyethylenes, polycarbonates, polyanhydrides, polyhydroxyacids, polypropylfumerates, polycaprolactones, polyamides, polyacetals, polyethers, polyesters, poly(orthoesters), polycyanoacrylates, polyvinyl alcohols, polyurethanes, polyphosphazenes, polyacrylates, polymethacrylates, polycyanoacrylates, polyureas, polystyrenes, polyamines, polylysine, poly(ethylene imine), poly(serine ester), poly(L-lactide-co-L-lysine), or poly(4-hydroxy-L-proline ester).

In some embodiments, nanoparticles include acrylic polymers. As used herein, the term "acrylic polymer" refers to a polymer made up of acrylic acid monomers or derivatives or variants of acrylic acid. Monomers included in acrylic polymers may include, but are not limited to, acrylic acid, methacrylic acid, acrylic acid and methacrylic acid copolymers, methyl methacrylate copolymers, ethoxyethyl methacrylates, cyanoethyl methacrylate, amino alkyl methacrylate copolymer, poly(acrylic acid), poly(methacrylic acid), and polycyanoacrylates.

In some embodiments, particle excipients may include any of those presented in Table 1, above.

Lipids

In some embodiments, excipients include lipids. As used herein, the term "lipid" refers to members of a class of organic compounds that include fatty acids and various derivatives of fatty acids that are soluble in organic solvents, but not in water. Examples of lipids include, but are not limited to, fats, triglycerides, oils, waxes, sterols (e.g. cholesterol, ergosterol, hopanoids, hydroxysteroids, phytosterol, and steroids), stearin, palmitin, triolein, fat-soluble vitamins (e.g., vitamins A, D, E, and K), monoglycerides (e.g. monolaurin, glycerol monostearate, and glyceryl hydroxystearate), diglycerides (e.g. diacylglycerol), phospholipids, glycerophospholipids (e.g., phosphatidic acid, phosphatidylethanolamine, phosphatidylcholine, phosphatidylserine, phosphoinositides), sphingolipids (e.g., sphingomyelin), and phosphosphingolipids. In some embodiments, lipids may include, but are not limited to, any of those listed (e.g., fats and fatty acids) in Table 1, above.

In some embodiments, lipid excipients include amphiphilic lipids (e.g., phospholipids). As used herein, the term "amphiphilic lipid" refers to a class of lipids with both hydrophilic and hydrophobic domains. Amphiphilic lipids may be used to prepare vesicles as these molecules typically form layers along water:lipid interfaces. Non-limiting examples of amphiphilic lipids include, but are not limited to, phospholipids, phosphatidylcholines, phosphatidylethanolamines, palmitoyl-oleoyl-phosphatidylethanolamine (POPE), phosphatidylserines, phosphotidylglycerols, lysophospholipids such as lysophosphatidylethanolamines, mono-oleoyl-phosphatidylethanolamine (MOPE), mono-myristoyl-phosphatidylethanolamine (MMPE), lysolipids, mono-oleoyl-phosphatidic acid (MOPA), mono-oleoyl-phosphatidylserine (MOPS), mono-oleoyl-phosphatidylglycerol (MOPG), palmitoyloleoyl phosphatidylcholine, lysophosphatidylethanolamine, dipalmitoylphosphatidylcholine, dioleoylphosphatidylcholine; distearoylphosphatidylcholine, dilinoleoylphosphatidylcholine, 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC), 1,2-dioleoyl-sn-glycero-3-phophoethanolamine (DOPE), 1,2-dilauroyl-sn-glycero-3-phosphocholine (DLPC), 1,2-dimyristoyl-sn-glycero-3-phosphocholine (DMPC)phosphatidylinositol, phosphatidic acid, palmitoyloleoyl phosphatidylcholine, lysophosphatidylethanolamines, monoglycerides, diglycerides, triglycerides.

Lipid Vesicles

In some embodiments, excipients may include lipid vesicles or components of lipid vesicles. As used herein, the term "lipid vesicle" refers to a particle enveloped by an amphiphilic lipid membrane. Examples of lipid vesicles include, but are not limited to, liposomes, lipoplexes, and lipid nanoparticles. SBPs may include lipid vesicles as cargo or payloads. In some embodiments, SBPs are or encompassed by lipid vesicles. Such lipid vesicles may be used to deliver SBPs as a payload. Such SBPs may themselves include cargo or payload. As used herein, the term "liposome" refers generally to any vesicle that includes a phospholipid bilayer and aqueous core. Liposomes may be artificially prepared and may be used as delivery vehicles. Liposomes can be of different sizes. Multilamellar vesicles (MLVs) may be hundreds of nanometers in diameter and contain two or more concentric bilayers separated by narrow aqueous compartments. Small unicellular vesicles (SUVs) may be smaller than 50 nm in diameter. Large unilamellar vesicles (LUVs) may be between 50 and 500 nm in diameter. Liposomes may include opsonins or ligands to improve liposome attachment to unhealthy tissue or to activate events (e.g., endocytosis). Liposome core pH may be modulated to improve payload delivery. In some embodiments, lipid vesicle excipients may include, but are not limited to, any of those listed in Table 1, above.

In some embodiments, liposomes may include 1,2-dioleyloxy-N,N-dimethylaminopropane (DODMA) liposomes, DiLa2 liposomes (Marina Biotech, Bothell, WA), 1,2-dilinoleyloxy-3-dimethylaminopropane (DLin-DMA) liposomes, 2,2-dilinoleyl-4-(2-dimethylaminoethyl)-[1,3]-dioxolane (DLin-KC2-DMA) liposomes, and MC3 liposomes (e.g., see US Publication Number US20100324120, the contents of which are herein incorporated by reference in their entirety). In some embodiments, liposomes may include small molecule drugs (e.g., DOXIL® from Janssen Biotech, Inc., Horsham, PA).

Liposomes may be formed from the synthesis of stabilized plasmid-lipid particles (SPLP) or stabilized nucleic acid lipid particle (SNALP) that have been previously described and shown to be suitable for delivery of oligonucleotides in vitro and in vivo (see Wheeler et al. Gene Therapy. 1999 6:271-281; Zhang et al. Gene Therapy. 1999 6:1438-1447; Jeffs et al. Pharm Res. 2005 22:362-372; Morrissey et al., Nat Biotechnol. 2005 2:1002-1007; Zimmermann et al., Nature. 2006 441:111-114; Heyes et al. J Contr Rel. 2005 107:276-287; Semple et al. Nature Biotech. 2010 28:172-176; Judge et al. J Clin Invest. 2009 119:661-673; deFougerolles *Hum Gene Ther.* 2008 19:125-132). These liposomes are designed for the delivery of DNA, RNA, and other oligonucleotide constructs, and they may be adapted for the delivery of SBPs with oligonucleotides. These liposome formulations may be composed of 3 to 4 lipid components in addition to SBPs. As an example, a liposome may contain 55% cholesterol, 20% disteroylphosphatidyl choline (DSPC), 10% PEG-S-DSG, and 15% 1,2-dioleyloxy-N,N-dimethylaminopropane (DODMA), as described by Jeffs et al. As another example, certain liposome formulations may contain, but are not limited to, 48% cholesterol, 20% DSPC, 2% PEG-c-DMA, and 30% cationic lipid, where the cationic lipid can be 1,2-distearloxy-N,N-dimethylaminopropane (DSDMA), DODMA, DLin-DMA, or 1,2-dilinolenyloxy-3-dimethylaminopropane (DLenDMA), as described by Heyes et al.

In some embodiments, SBPs may be encapsulated within liposomes and/or contained in an encapsulated aqueous liposome core. In another embodiment, SBPs may be formulated in an oil-in-water emulsion where the emulsion particle comprises an oil core and a cationic lipid which can interact with SBPs, anchoring them to emulsion particles (e.g., see International Publication. Number WO2012006380, the contents of which are herein incorporated by reference in their entirety. In another embodiment, SBPs may be formulated in lipid vesicles which may have crosslinks between functionalized lipid bilayers (e.g., see United States Publication Number US20120177724, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, lipid vesicles may include cationic lipids selected from one or more of formula CLI-CLXXIX of International Publication Number WO2008103276; formula CLI-CLXXIX of U.S. Pat. No. 7,893,302; formula CLI-CLXXXXII of U.S. Pat. No. 7,404,969; and formula I-VI of United States Publication Number US20100036115, the contents of each of which are herein incorporated by reference in their entirety. As non-limiting examples, cationic lipids may be selected from (20Z,23Z)—N,N-dimethylnonacosa-20,23-dien-10-amine, (17Z,20Z)—N,N-dimemylhexacosa-17,20-dien-9-amine, (1Z,19Z)—N5N-dimethylpentacosa-16, 19-dien-8-amine, (13Z,16Z)—N,N-dimethyldocosa-13,16-dien-5-amine, (12Z,15Z)—N,N-dimethylhenicosa-12,15-dien-4-amine, (14Z,17Z)—N,N-dimethyltricosa-14,17-dien-6-amine, (15Z,18Z)—N,N-dimethyltetracosa-15,18-dien-7-amine, (18Z,21Z)—N,N-dimethylheptacosa-18,21-dien-10-amine, (15Z,18Z)—N,N-dimethyltetracosa-15,18-dien-5-amine, (14Z,17Z)—N,N-dimethyltricosa-14,17-dien-4-amine, (19Z,22Z)—N,N-dimeihyloctacosa-19,22-dien-9-amine, (18Z,21 Z)—N,N-dimethylheptacosa-18,21-dien-8-amine, (17Z,20Z)—N,N-dimethylhexacosa-17,20-dien-7-amine, (16Z,19Z)—N,N-dimethylpentacosa-16,19-dien-6-amine, (22Z,25Z)—N,N-dimethylhentriaconta-22,25-dien-10-amine, (21Z,24Z)—N,N-dimethyltriaconta-21,24-dien-9-amine, (18Z)—N,N-dimetylheptacos-18-en-10-amine, (17Z)—N,N-dimethylhexacos-17-en-9-amine, (19Z,22Z)—N,N-dimethyloctacosa-19,22-dien-7-amine, N,N-dimethylheptacosan-10-amine, (20Z,23Z)—N-ethyl-N-methylnonacosa-20,23-dien-10-amine, 1-[(11Z,14Z)-1-nonylicosa-11,14-dien-1-yl]pyrrolidine, (20Z)—N,N-dimethylheptacos-20-en-10-amine, (15Z)—N,N-dimethyl eptacos-15-en-10-amine, (14Z)—N,N-dimethylnonacos-14-en-10-amine, (17Z)—N,N-dimethylnonacos-17-en-10-amine, (24Z)—N,N-dimethyltritriacont-24-en-10-amine, (20Z)—N,N-dimethylnonacos-20-en-10-amine, (22Z)—N,N-dimethylhentriacont-22-en-10-amine, (16Z)—N,N-dimethylpentacos-16-en-8-amine, (12Z,15Z)—N,N-dimethyl-2-nonylhenicosa-12,15-dien-1-amine, (13Z,16Z)—N,N-dimethyl-3-nonyldocosa-13,16-dien-1-amine, N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]eptadecan-8-amine, 1-[(1S,2R)-2-hexylcyclopropyl]-N,N-dimethylnonadecan-10-amine, N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]nonadecan-10-amine, N,N-dimethyl-21-[(1S,2R)-2-octylcyclopropyl]henicosan-10-amine,N,N-dimethyl-1-[(1S,2S)-2-{[(1R,2R)-2-pentylcyclopropyl]methyl}cyclopropyl]nonadecan-10-amine,N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]hexadecan-8-amine, N,N-dimethyl-[(1R,2S)-2-undecylcyclopropyl]tetradecan-5-amine, N,N-dimethyl-3-{7-[(1S,2R)-2-octylcyclopropyl]heptyl}dodecan-1-amine, 1-[(1R,2S)-2-heptylcyclopropyl]-N,N-dimethyloctadecan-9-amine, 1-[(1S,2R)-2-decylcyclopropyl]-N,N-dimethylpentadecan-6-amine, N,N-dimethyl-1-[(1S,2R)-2-octylcyclopropyl]pentadecan-8-amine, R—N,N-dimethyl-1-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-(octyloxy)propan-2-amine, S—N,N-dimethyl-1-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-(octyloxy)propan-2-amine, 1-{2-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-1-[(octyloxy)methyl]ethyl}pyrrolidine, (2S)—N,N-dimethyl-1-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-3-[(5Z)-oct-5-en-1-yloxy]propan-2-amine, 1-{2-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]-1-[(octyloxy)methyl]ethyl}azetidine, (2S)-1-(hexyloxy)-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, (2S)-1-(heptyloxy)-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, N,N-dimethyl-1-(nonyloxy)-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, N,N-dimethyl-1-[(9Z)-octadec-9-en-1-yloxy]-3-(octyloxy)propan-2-amine; (2S)—N,N-dimethyl-1-[(6Z,9Z,12Z)-octadeca-6,9,12-trien-1-yloxy]-3-(octyloxy)propan-2-amine, (2S)-1-[(11Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethyl-3-(pentyloxy)propan-2-amine, (2S)-1-(hexyloxy)-3-[(11Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethylpropan-2-amine, 1-[(11Z,14Z)-icosa-11,14-dien-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine, 1-[(13Z,16Z)-docosa-13,16-dien-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine, (2S)-1-[(13Z,16Z)-docosa-13,16-dien-1-yloxy]-3-(hexyloxy)-N,N-dimethylpropan-2-amine, (2S)-1-[(13Z)-docos-13-en-1-yloxy]-3-(hexyloxy)-N,N-dimethylpropan-2-amine, 1-[(13Z)-docos-13-en-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine, 1-[(9Z)-hexadec-9-en-1-yloxy]-N,N-dimethyl-3-(octyloxy)propan-2-amine, (2R)—N,N-dimethyl-H(1-metoyloctyl)oxy]-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, (2R)-1-[(3,7-dimethyloctyl)oxy]-N,N-dimethyl-3-[(9Z,12Z)-octadeca-9,12-dien-1-yloxy]propan-2-amine, N,N-dimethyl-1-(octyloxy)-3-({8-[(1S,2S)-2-{[(1R,2R)-2-pentylcyclopropyl]methyl}cyclopropyl]octyl}oxy)propan-2-amine, N,N-dimethyl-1-{[8-(2-octylcyclopropyl)octyl]oxy}-3-(octyloxy)propan-2-amine, (11E,20Z,23Z)—N,N-dimethylnonacosa-11,20,2-trien-10-amine, or pharmaceutically acceptable salts or stereoisomers thereof.

In some embodiments, lipids may be cleavable lipids. Such lipids may include any of those described in International Publication Number WO2012170889, the contents of which are herein incorporated by reference in their entirety. In some embodiments, SBPs may be formulated with at least one of the PEGylated lipids described in International Publication Number WO2012099755, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, excipients include lipid nanoparticles. As used herein, the term "lipid nanoparticle" or "LNP" refers to a tiny colloidal particle of solid lipid and surfactant, typically ranging in size of from about 10 nm in diameter to about 1000 nm in diameter. LNPs may contain PEG-DMG 2000 (1,2-dimyristoyl-sn-glycero-3-phophoethanolamine-N-[methoxy(polyethylene glycol)-2000). In some embodiments, LNPs may contain PEG-DMG 2000, a cationic lipid known in the art and at least one other component. LNPs may contain PEG-DMG 2000, a cationic lipid known in the art, DSPC and cholesterol. As a non-limiting example, LNPs may contain PEG-DMG 2000, DLin-DMA, DSPC, and cholesterol.

In some embodiments, excipients may include DiLa2 liposomes (Marina Biotech, Bothell, WA), SMARTICLES® (Marina Biotech, Bothell, WA), neutral DOPC (1,2-dioleoyl-sn-glycero-3-phosphocholine) based liposomes, and hyaluronan-coated liposomes (Quiet Therapeutics, Israel).

In some embodiments, excipients may include lipidoids. As used herein, the term "lipidoid" refers to any non-lipid material that mimics lipid properties. The synthesis of lipidoids may be carried out as described by others (e.g., see Mahon et al., Bioconjug Chem. 2010 21:1448-1454; Schroeder et al., J Intern Med. 2010 267:9-21; Akinc et al., Nat Biotechnol. 2008 26:561-569; Love et al., Proc Natl Acad Sci USA. 2010 107:1864-1869; and Siegwart et al., Proc Natl Acad Sci USA. 2011 108:12996-3001, the contents of each of which are herein incorporated by reference in their entireties). Lipidoids may be included in complexes, micelles, liposomes, or particles. In some embodiments, SBPs may include lipidoids.

In some embodiments, lipidoids may be combined with lipids to form particles. Such lipids may include cholesterol. Some lipidoids may be combined with PEG (e.g., C14 alkyl chain length). As another example, formulations with certain lipidoids, include, but are not limited to, C12-200 and may contain a combination of lipidoid, disteroylphosphatidyl choline, cholesterol, and PEG-DMG.

Coating Agents

In some embodiments, excipients may include coating agents. Polymers are commonly used as coating agents and may be layered over SBPs. Non-limiting examples of polymers for use as coating agents include polyethylene glycol, methylcellulose, hypromellose, ethylcellulose, gelatin, hydroxypropyl cellulose, titanium dioxide, zein, poly(alkyl)(meth)acrylate, poly(ethylene-co-vinyl acetate), and combinations thereof. In some embodiments, coating agents may include one or more compounds listed in Table 1, above.

Bulking Agents

In some embodiments, excipients include bulking agents. As used herein, the term "bulking agent" refers to a substance that adds weight and volume to a composition. Examples of hulking agents include, but are not limited to, lactose, sorbitol, sucrose, mannitol, lactose USP, Starch 1500, microcrystalline cellulose, Avicel, dibasic calcium phosphate dehydrate, sucrose, tartaric acid, citric acid, fumaric acid, succinic acid, malic acid, polyvinylpyrrolidone, copolymers of vinylpyrrolidone and vinylacetate, hydroxypropylcellulose, hydroxyethylcellulose, hydroxypropylmethylcellulose, polyvinyl alcohol, polyethylene glycol, acacia, sodium carboxymethylcellulose, and combinations thereof. In some embodiments, bulking agents may include any of those presented in Table 1, above.

Lubricants

In some embodiments, excipients may include lubricants. As used herein, the term "lubricant" refers to any substance used to reduce friction between two contacting materials. Lubricants may be natural or synthetic. Lubricants may comprise oils, lipids, microspheres, polymers, water, aqueous solutions, liposomes, solvents, alcohols, micelles, stearate salts, alkali, alkaline earth, and transition metal salts thereof (e.g., calcium, magnesium, or zinc), stearic acid, polyethylene oxide, talc, hydrogenated vegetable oil, and vegetable oil derivatives, fumed silica, silicones, high molecular weight polyalkylene glycol (e.g. high molecular weight polyethylene glycol), monoesters of propylene glycol, saturated fatty acids containing about 8-22 carbon atoms and/or 16-20 carbon atoms, and any other component known to one skilled in the art. Other examples of lubricants include, but are not limited to, hyaluronic acid, magnesium stearate, calcium stearate, and lubricin. In some embodiments, lubricant excipients may include any of those presented in Table 1, above.

Sweeteners and Colorants

In some embodiments, excipients may include sweeteners and/or colorants. As used herein, a "sweetener" refers to a substance that adds a sweet taste to or improves the sweetness of a composition. Sweeteners may be natural or artificial. Non-limiting examples of sweeteners include glucose, aspartame, sucralose, neotame, acesulfame potassium, saccharin, advantame, cyclamates, sorbitol, xylitol, lactitol, xylose, stevia, lead acetate, mogrosides, brazzein, curculin, erythritol, glycyrrhizin, glycerol, hydrogenated starte hydrolysates, inulin, ismalt, isomaltooligosaccharide, isomaltulose, mabinlin, maltodextrin, miraculin, monantin, osladin, pentadin, polydextrose, psicose, tagatose, thaumatin, mannitol, lactose, and sucrose. In some embodiments, sweetener excipients may include any of those presented in Table 1, above.

As used herein, the term "colorant" refers to any substance that adds color to a composition (e.g., a dye). Non-limiting examples of colorants include dyes, inks, pigments, food coloring, turmeric, titanium dioxide, caretinoids (e.g., bixin, β-carotene, apocarotenals, canthaxanthin, saffron, crocin, capsanthin and capsorubin occurring in paprika ole-oresin, lutein, astaxanthin, rubixanthin, violaxanthin, rhodoxanthin, lycopene, and derivatives thereof), and FD&C colorants [e.g., FD&C Blue No. 1 (brilliant blue FCF); FD&C Blue No. 2 (indigotine); FD&C Green No. 3 (fast green FCF); FD&C Red No. 40 (allura red AC); FD&C Red No. 3 (erythrosine); FD&C Yellow No. 5 (tartrazine); and FD&C Yellow No. 6 (sunset yellow)]. In some embodiments, colorant excipients may include any of those presented in Table 1, above.

Preservatives

In some embodiments, excipients may include preservatives. As used herein a "preservative" is any substance that protects against decay, decomposition, or spoilage. Preservatives may be natural or synthetic. They may be antimicrobial preservatives, which inhibit the growth of bacteria or fungi, including mold, or antioxidants such as oxygen absorbers, which inhibit the oxidation of food constituents. Common antimicrobial preservatives include calcium propionate, sodium nitrate, sodium nitrite, sulfites (sulfur dioxide, sodium bisulfite, potassium hydrogen sulfite, etc.) and disodium EDTA. Antioxidants include BHA and BHT. Other preservatives include formaldehyde (usually in solution), glutaraldehyde (kills insects), vitamin A, vitamin C, vitamin E, selenium, amino acids, methyl paraben, propyl paraben, potassium sorbate, sodium chloride, ethanol, phenol, m-cresol, p-cresol, o-cresol, chlorocresol, benzyl alcohol, phenylmercuric nitrite, phenoxyethanol, methylchloroisothiazolinone, chlorobutanol, magnesium chloride (e.g., hexahydrate), alkylparaben (methyl, ethyl, propyl, butyl and the like), benzalkonium chloride, benzethonium chloride, sodium dehydroacetate, thimerosal, and combinations thereof. In some embodiments, preservative excipients may include any of those presented in Table 1, above.

Flowability Agents

In some embodiments, excipients may include flowability agents. As used herein, the term "flowability agent" refers to a substance used to reduce viscosity and/or aggregation in a composition. Flowability agents are particularly useful for the formulation of powders, particles, solutions, gels, polymers, and any other form of matter capable of flow from one area to another. Flowability agents have been used to improve powder flowability for the manufacture of therapeutics, as taught in Morin et al. (2013) AAPS PharmSciTech 14(3):1158-1168, the contents of which are herein incorporated by reference in their entirety. In some embodiments, flowability agents are used to modulate SBP viscosity. In some embodiments, flowability agents may be lubricants. Non-limiting examples of flowability agents include magnesium stearate, stearic acid, hydrous magnesium silicate, and any other lubricant used to promote flowability known to one skilled in the art. In some embodiments, flowability agent excipients may include any of those presented in Table 1, above.

Gelling Agents

In some embodiments, excipients may include gelling agents. As used herein, the term "gelling agent" refers to any substance that promotes viscosity and/or polymer cross-linking in compositions. Non-limiting examples of gelling agents include glycerol, glycerophosphate, sorbitol, hydroxyethyl cellulose, carboxymethyl cellulose, triethylamine, triethanolamine, 2-pyrrolidone, alpha-cyclodextrin, benzyl alcohol, beta-cyclodextrin, dimethyl sulfoxide, dimethylacetamide (DMA), dimethylformamide, ethanol, gamma-cyclodextrin, glycerol formal, hydroxypropyl beta-cyclodextrin, kolliphor 124, kolliphor 181, kolliphor 188, kolliphor 407, kolliphor EL (cremaphor EL), cremaphor RH 40, cremaphor RH 60, d-alpha-tocopherol, PEG 1000 succinate, polysorbate 20, polysorbate-80, solutol HS 15, sorbitan monooleate, poloxamer-407, poloxamer-188, Labrafil M-1944CS, Labrafil M-2125CS, Labrasol, Gellucire 44/14, Softigen 767, mono- and di-fatty acid esters of PEG 300, PEG 400, or PEG 1750, kolliphor RH60, N-methyl-2-pyrrolidone, castor oil, corn oil, cottonseed oil, olive oil, peanut oil, peppermint oil, safflower oil, sesame oil, soybean oil, hydrogenated vegetable oils, hydrogenated soybean oil, and medium-chain triglycerides of coconut oil and palm seed oil, beeswax, d-alpha-tocopherol, oleic acid, medium-chain mono- and diglycerides, alpha-cyclodextrin, beta-cyclodextrin, hydroxypropyl-beta-cyclodextrin, sulfo-butylether-beta-cyclodextrin, hydrogenated soy phosphatidylcholine, distearoylphosphatidylglycerol, L-alpha-dimyristoylphosphatidylcholine, L-alphadimyristoylphosphatidylglycerol, PEG 300, PEG 300 caprylic/capric glycerides (Softigen 767), PEG 300 linoleic glycerides (Labrafil M-2125CS), PEG 300 oleic glycerides (Labrafil M-1944CS), PEG 400, PEG 400 caprylic/capric glycerides (Labrasol), polyoxyl 40 stearate (PEG 1750 monosterate), PEG 4000 (PEG 4 kDa), polyoxyl 8 stearate (PEG 400 monosterate), polysorbate 20, polysorbate-SO, polyvinyl pyrrolidone, polyvinyl pyrrolidone-12, polyvinyl pyrrolidone-17, propylene carbonate, propylene glycol, solutol HS 15, sorbitan monooleate (Span 20), sulfobutylether-beta-cyclodextrin, transcutol, triacetin, 1-dodecylazacyclo-heptan-2-one, caprolactam, castor oil, cottonseed oil, ethyl acetate, medium chain triglycerides, methyl acetate, oleic acid, safflower oil, sesame oil, soybean oil, tetrahydrofuran, and glycerin. Additional examples of gelling agents include acacia, alginic acid, bentonite, CARBOPOLS® (also known as carbomers), carboxymethyl cellulose, ethylcellulose, gelatin, hydroxy ethyl cellulose, hydroxypropyl cellulose, magnesium aluminum silicate, methylcellulose, poloxamers, polyvinyl alcohol, sodium alginate, tragacanth, and xanthan gum. In some embodiments, gelling agent excipients may include any of those presented in Table 1, above.

PEGs which may be used as gelling agents and/or excipients may be selected from a variety of chain lengths and molecular weights. These compounds are typically prepared through ethylene oxide polymerization. In some embodiments, PEGs may have a molecular weight of from about 300 g/mol to about 100,000 g/mol. In some embodiments, PEGs may have a molecular weight of from about 3600 g/mol to about 4400 g/mol. In some embodiments, PEGs with a molecular weight of from about 300 g/mol to about 3000 g/mol, from about 350 g/mol to about 3500 g/mol, from about 400 g/mol to about 4000 g/mol, from about 450 g/mol to about 4500 g/mol, from about 500 g/mol to about 5000 g/mol, from about 550 g/mol to about 5500 g/mol, from about 600 g/mol to about 6000 g/mol, from about 650 g/mol to about 6500 g/mol, from about 700 g/mol to about 7000 g/mol, from about 750 g/mol to about 7500 g/mol, from about 800 g/mol to about 8000 g/mol, from about 850 g/mol to about 8500 g/mol, from about 900 g/mol to about 9000 g/mol, from about 950 g/mol to about 9500 g/mol, from about 1000 g/mol to about 10000 g/mol, from about 1100 g/mol to about 12000 g/mol, from about 1200 g/mol to about 14000 g/mol, from about 1300 g/mol to about 16000 g/mol, from about 1400 g/mol to about 18000 g/mol, from about 1500 g/mol to about 20000 g/mol, from about 1600 g/mol to about 22000 g/mol, from about 1700 g/mol to about 24000 g/mol, from about 1800 g/mol to about 26000 g/mol, from about 1900 g/mol to about 28000 g/mol, from about 2000 g/mol to about 30000 g/mol, from about 2200 g/mol to about 35000 g/mol, from about 2400 g/mol to about 40000 g/mol, from about 2600 g/mol to about 45000 g/mol, from about 2800 g/mol to about 50000 g/mol, from about 3000 g/mol to about 55000 g/mol, from about 10000 g/mol to about 60000 g/mol, from about 13000 g/mol to about 65000 g/mol, from about 16000 g/mol to about 70000 g/mol, from about 19000 g/mol to about 75000 g/mol, from about 22000 g/mol to about 80000 g/mol, from about 25000 g/mol to about 85000 g/mol, from about 28000 g/mol to about 90000 g/mol, from about 31000 g/mol to about 95000 g/mol, or from about 34000 g/mol to about 100000 g/mol are utilized.

Demulcents

In some embodiments, excipients may include demulcents. As used herein, the term "demulcent" refers to a substance that relieves irritation or inflammation of the mucous membranes by forming a protective film. Demulcents may include non-polymeric demulcents and polymer demulcents. Non-limiting examples of non-polymeric demulcents include glycerin, gelatin, propylene glycol, and other non-polymeric diols and glycols. Non-limiting examples of polymer demulcents include polyvinyl alcohol (PVA), povidone or polyvinyl pyrrolidone (PVP), cellulose derivatives, polyethylene glycol (e.g., PEG 300, PEG 400), polysorbate 80, and dextran (e.g., dextran 70). Specific cellulose derivatives may include hydroxypropyl methyl cellulose, carboxymethyl cellulose, carboxymethylcellulose sodium, methyl cellulose, hydroxyethyl cellulose, hypromellose, and cationic cellulose derivatives. In some embodiments, demulcent excipients may include any of those presented in Table 1, above.

Formats

SBPs may include or be prepared to conform to a variety of formats. In some embodiments, such formats include formulations of processed silk with various excipients and/or cargo. In some embodiments, SBP formats include, but are not limited to, gels, hydrogels, drops, creams, microspheres, implants, and solutions. In some embodiments, the formats are formulated with a therapeutic agent.

Formulations

In some embodiments, SBPs may be formulations. As used herein, the term "formulation" refers to a mixture of two or more components or the process of preparing such mixtures. In some embodiments, the formulations are low cost and eco-friendly. In some embodiments, the preparation or manufacturing of formulations is low cost and eco-friendly. In some embodiments, the preparation or manufacturing of formulations is scalable. In some embodiments, SBPs are prepared by extracting silk fibroin via degumming silk yarn. In some embodiments, the yarn is medical grade. In some embodiments the yarn may be silk sutures. The extracted silk fibroin may then be dissolved in a solvent (e.g. water, aqueous solution, organic solvent). The dissolved silk fibroin may then be dried (e.g., oven dried, air dried, or freeze-dried). In some embodiments, dried silk fibroin is formed into formats described herein. In some embodiments, that format is a solution. In some embodiments, that format is a hydrogel. In some embodiments, formulations include one or more excipients, carriers, additional components, and/or therapeutic agents to generate SBPs. In some embodiments, formulations of processed silk are prepared during the manufacture of SBPs. In some embodiments, the silk is graded from 3-6, wherein the higher graded silk denotes higher quality silk. Grades of silk may vary in several properties, including, but not limited to, color, number of knots, lustrousness, and cleanliness. In some embodiments, the silk is grade 3 (grade AAA). In some embodiments, the silk is grade 4 (grade AAAA). In some embodiments, the silk is grade 5 (grade AAAAA). In some embodiments, the silk is grade 6 (grade AAAAAA). Formulations, preparations, and SBPs of the present disclosure may use silk of any grade. In some embodiments, properties of SBPs and ocular SBPs may not be affected or altered by the grade of silk (e.g. clarity, solubility, rheology, viscosity, hydrogel formation, and SEC results). In some embodiments, properties of SBPs and ocular SBPs may be affected or altered by the grade of silk (e.g. clarity, solubility, rheology, viscosity, hydrogel formation, and SEC results).

Formulation components and/or component ratios may be modulated to affect one or more SBP properties, effects, and/or applications. Variations in the concentration of silk fibroin, choice of excipient, the concentration of excipient, the osmolarity of the formulation, and the method of formulation represent non-limiting examples of differences in formulation that may alter properties, effects, and applications of SBPs. In some embodiments, the formulation of SBPs may modulate their mechanical properties. Examples of mechanical properties that may be modulated include, but are not limited to, mechanical strength, tensile strength, elongation capabilities, elasticity, compressive strength, stiffness, shear strength, toughness, torsional stability, temperature stability, moisture stability, viscosity, and reeling rate.

Cargo

In some embodiments, SBPs and SBP formulations are or include cargo. As used herein, the term "cargo" refers to any substance that is embedded in, enclosed within, attached to, or otherwise associated with a carrier. SBP formulations may be carriers for a large variety of cargo. Such cargo may include, but are not limited to, compounds, compositions, therapeutic agents, biological agents, materials, cosmetics, devices, agricultural compositions, particles, lipids, liposomes, sweeteners, colorants, preservatives, carbohydrates, small molecules, supplements, tranquilizers, ions, metals, minerals, nutrients, pesticides, herbicides, fungicides, and cosmetics.

In some embodiments, the cargo is or includes a payload. As used herein, the term "payload" refers to cargo that is delivered from a source or carrier to a target. Payloads may be released from SBP formulations, where SBP formulations serve as a carrier. Where SBPs are the payload, the SBPs may be released from a source or carrier. In some embodiments, payloads remain associated with carriers upon delivery. Payloads may be released in bulk or may be released over a period of time, also referred to herein as the "delivery period." In some embodiments, payload release is by way of controlled release. As used herein, the term "controlled release" refers to distribution of a substance from a source or carrier to a surrounding area, wherein the distribution occurs in a manner that includes or is affected by some manipulation, some property of the carrier, or some carrier activity.

In some embodiments, controlled release may include a steady rate of release of payload from carrier. In some embodiments, payload release may include an initial burst, wherein a substantial amount of payload is released during an initial release period followed by a period where less payload is released. As used herein, the term "initial burst" refers to a rate of payload release from a source or depot over an initial release period (e.g., after administration or other placement, for example in solution during experimental analysis) that is higher than rates during one or more subsequent release periods. In some embodiments, release rate slows over time. Payload release may be measured by assessing payload concentration in a surrounding area and comparing to initial payload concentration or remaining payload concentration in a carrier or source area. Payload release rate may be expressed as a quantity or mass of payload released over time (e.g., mg/min). Payload release rate may be expressed as a percentage of payload released from a source or carrier over a period of time (e.g., 5%/hour). Controlled release of a payload that extends the delivery period is referred to herein as "sustained release." Sustained release may include delivery periods that are extended over a period of hours, days, months, or years.

Some controlled release may be mediated by interactions between payload and carrier. Some controlled release is mediated by interactions between payload or carrier with surrounding areas where payload is released. With sustained payload release, payload release may be slowed or prolonged due to interactions between payload and carrier or payload and surrounding areas where payload is released. Payload release from SBPs may be controlled by SBP viscosity. Where the SBP includes processed silk gel, gel viscosity may be adjusted to modulate payload release.

In some embodiments, payload delivery periods may be from about 1 second to about 20 seconds, from about 10 seconds to about 1 minute, from about 30 seconds to about 10 minutes, from about 2 minutes to about 20 minutes, from about 5 minutes to about 30 minutes, from about 15 minutes to about 1 hour, from about 45 minutes to about 2 hours, from about 90 minutes to about 5 hours, from about 3 hours to about 20 hours, from about 10 hours to about 50 hours, from about 24 hours to about 100 hours, from about 48 hours to about 2 weeks, from about 72 hours to about 4 weeks, from about 1 week to about 3 months, from about 1 month to about 6 months, from about 3 months to about 1 year, from about 9 months to about 2 years, or more than 2 years.

In some embodiments, payload release may be consistent with near zero-order kinetics. In some embodiments, payload release may be consistent with first-order kinetics. In some embodiments, payload release may be modulated based on the density, loading, molecular weight, and/or concentration of the payload. Where the carrier is an SBP, payload release may be modulated by one or more of SBP drying method, silk fibroin molecular weight, and silk fibroin concentration.

In some embodiments, SBP formulations maintain and/or improve cargo stability, purity, and/or integrity. For example, SBP formulations may be used to protect therapeutic agents or macromolecules during lyophilization. The maintenance and/or improvement of stability during lyophilization may be determined by comparing SBP cargo stability to formulations lacking processed silk or to standard formulations in the art.

Viscosity

In some embodiments, SBPs may be formulated to modulate SBP viscosity. The viscosity of a composition (e.g., a solution, a gel or hydrogel) provided herein can be determined using a rotational viscometer or rheometer. Additional methods for determining the viscosity of a composition and other rheological properties may include any of those known in the art. In some embodiments, the SBP viscosity may be controlled via the concentration of processed silk. In some embodiments, the SBP viscosity may be controlled via the molecular weight of processed silk. In some embodiments, the SBP viscosity may be controlled via the boiling time or mb of the processed silk. In some embodiments, the SBP viscosity may be altered by the incorporation of stressed silk. In some embodiments, SBP viscosity is altered by the incorporation of an excipient. In some embodiments, SBP viscosity may be altered by the incorporation of an excipient that is a gelling agent. In some embodiments, the identity of the excipient (e.g., PEG or poloxamer) may be altered to modulate SBP viscosity. In some embodiments, the viscosity of SBPs may be tuned for the desired application (e.g., drug delivery system, surgical implant, lubricant, etc.). In some embodiments, the viscosity of SBPs is tunable between 1-1000 centipoise (cP). In some embodiments, the viscosity of an SBP is tunable from about 0.0001 to about 1000 Pascal seconds (Pa*s). In some embodiments, the viscosity of an SBP is from about 1 cP to about 10 cP, from about 2 cP to about 20 cP, from about 3 cP to about 30 cP, from about 4 cP to about 40 cP, from about 5 cP to about 50 cP, from about 6 cP to about 60 cP, from about 7 cP to about 70 cP, from about 8 cP to about 80 cP, from about 9 cP to about 90 cP, from about 10 cP to about 100 cP, from about 100 cP to about 150 cP, from about 150 cP to about 200 cP, from about 200 cP to about 250 cP, from about 250 cP to about 300 cP, from about 300 cP to about 350 cP, from about 350 cP to about 400 cP, from about 400 cP to about 450 cP, from about 450 cP to about 500 cP, from about 500 cP to about 600 cP, from about 550 cP to about 700 cP, from about 600 cP to about 800 cP, from about 650 cP to about 900 cP, or from about 700 cP to about 1000 cP. In some embodiments, the viscosity of an SBP is from about 0.0001 Pa*s to about 0.001 Pa*s, from about 0.001 Pa*s to about 0.01 Pa*s, from about 0.01 Pa*s to about 0.1 Pa*s, from about 0.1 Pa*s to about 1 Pa*s, from about 1 Pa*s to about 10 Pa*s, from about 2 Pa*s to about 20 Pa*s, from about 3 Pa*s to about 30 Pa*s, from about 4 Pa*s to about 40 Pa*s, from about 5 Pa*s to about 50 Pa*s, from about 6 Pa*s to about 60 Pa*s, from about 7 Pa*s to about 70 Pa*s, from about 8 Pa*s to about 80 Pa*s, from about 9 Pa*s to about 90 Pa*s, from about 10 Pa*s to about 100 Pa*s, from about 100 Pa*s to about 150 Pa*s, from about 150 Pa*s to about 200 Pa*s, from about 200 Pa*s to about 250 Pa*s, from about 250 Pa*s to about 300 Pa*s, from about 300 Pa*s to about 350 Pa*s, from about 350 Pa*s to about 400 Pa*s, from about 400 Pa*s to about 450 Pa*s, from about 450 Pa*s to about 500 Pa*s, from about 500 Pa*s to about 600 Pa*s, from about 550 Pa*s to about 700 Pa*s, from about 600 Pa*s to about 800 Pa*s, from about 650 Pa*s to about 900 Pa*s, or from about 700 Pa*s to about 1000 Pa*s.

Interfacial Viscosity

In some embodiments, silk fibroin may preferentially partition to the air-water interface when in solution. Similar to a surfactant, hydrophobic regions of hydrophobic proteins (e.g. silk fibroin) may exclude themselves from solution. The protein may migrate to the air-water boundary, resulting in an increase in the local concentration at this interface. This increase in the local concentration of a high molecular weight, hydrophobic protein may lead to an increase in local viscosity, as described in Sharma et al. (2011) Soft Matter 7(11): 5150-5160 and in Jaishankar et al. (2011) Soft Matter 7: 7623-7634, the contents of each of which are herein incorporated by reference in their entirety. This effect is termed "interfacial viscosity". In some embodiments, processed silk may demonstrate the effects of interfacial viscosity. In some embodiments, the interfacial viscosity is independent of the concentration of processed silk. As long as there is sufficient protein in solution to create a film at the air-water interface, the viscosity of this film will increase to a similar effect.

This phenomenon may be observed with the use of a rotational rheometer fit with a cone and plate configuration. This setup allows for an air-water interface located at the edges of the cone and plate. If protein migrates and leads to local concentration build-up at the edges of the plate, increasing shear viscosity will be observed due the increased local viscosity on the torque of the rotating cone spindle. Viscosity may increase with decreasing shear rate, as the decreasing shear allows for more efficient stacking of proteins, increased protein concentration, and more stable film of protein at the air-water interface.

Incorporation of a surfactant, a molecule that can better and more efficiently associate to the air-water interface, may block the hydrophobic protein association and may negate increase in viscosity due to the protein buildup at the boundary. When a competing surfactant is added to the formulations, preventing the silk fibroin from partitioning this interface, the viscosity may be reduced; formulations may have shear viscosity properties similar to controls (without silk fibroin).

In some embodiments, the viscosity of an SBP as measured by a rotational rheometer fit with a cone and plate configuration may be the interfacial viscosity. Unlike the rotational rheometer, a capillary rheometer has a very low air-water interface. This may greatly reduce if not eliminate interfacial viscosity effects that are observed using other rheologic methods (cone and plate rotational rheology). In some embodiments, the viscosity of an SBP may be measured by a capillary rheometer to measure the viscosity without measuring the interfacial viscosity. In some embodiments, the viscosity of an SBP may be measured with a surfactant incorporated to negate the effects of interfacial viscosity.

Stress Resistance

In some embodiments, SBPs may be formulated to modulate SBP resistance to stress. Resistance to stress may be measured using one or more rheological measurements. Such measurements may include, but are not limited to tensile elasticity, shear or rigidity, volumetric elasticity, and compression. Additional rheological measurements and properties may include any of those taught in Zhang et al. (2017) Fiber and Polymers 18(10):1831-1840; McGill et al. (2017) Acta Biomaterialia 63:76-84; and Choi et al. (2015)

In-Situ Gelling Polymers, Series in BioEngineering doi. 10.1007/978-981-287-152-7_2, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, stress resistance may be modulated through incorporation of excipients (e.g., PEG or poloxamer). In some embodiments, SBP stress-resistance properties may be modulated to suit a specific application (e.g., lubricant, etc.).

Concentrations and Ratios of SBP Components

SBPs may include formulations of processed silk with other components (e.g., excipients and cargo), wherein each SBP component is present at a specific concentration, ratio, or range of concentrations or ratios, depending on SBP format and/or application. In some embodiments, the concentration of processed silk (e.g. silk fibroin) or other SBP component (e.g., excipient or cargo) is present in SBPs and SBP formulations at a concentration of from about 0.0001% (w/v) to about 0.001% (w/v), from about 0.001% (w/v) to about 0.01% (w/v), from about 0.01% (w/v) to about 1% (w/v), from about 0.05% (w/v) to about 2% (w/v), from about 1% (w/v) to about 5% (w/v), from about 2% (w/v) to about 10% (w/v), from about 4% (w/v) to about 16% (w/v), from about 5% (w/v) to about 20% (w/v), from about 8% (w/v) to about 24% (w/v), from about 10% (w/v) to about 30% (w/v), from about 12% (w/v) to about 32% (w/v), from about 14% (w/v) to about 34% (w/v), from about 16% (w/v) to about 36% (w/v), from about 18% (w/v) to about 38% (w/v), from about 20% (w/v) to about 40% (w/v), from about 22% (w/v) to about 42% (w/v), from about 24% (w/v) to about 44% (w/v), from about 26% (w/v) to about 46% (w/v), from about 28% (w/v) to about 48% (w/v), from about 30% (w/v) to about 50% (w/v), from about 35% (w/v) to about 55% (w/v), from about 40% (w/v) to about 60% (w/v), from about 45% (w/v) to about 65% (w/v), from about 50% (w/v) to about 70% (w/v), from about 55% (w/v) to about 75% (w/v), from about 60% (w/v) to about 80% (w/v), from about 65% (w/v) to about 85% (w/v), from about 70% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 80% (w/v) to about 96% (w/v), from about 85% (w/v) to about 97% (w/v), from about 90% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 96% (w/v) to about 99.2% (w/v), from about 97% (w/v) to about 99.5% (w/v), from about 98% (w/v) to about 99.8% (w/v), from about 99% (w/v) to about 99.9% (w/v), or greater than 99.9% (w/v).

In some embodiments, the concentration of processed silk (e.g. silk fibroin) or other SBP component (e.g., excipient or cargo) may be present in SBPs or SBP formulations at a concentration of from about 0.0001% (v/v) to about 0.001% (v/v), from about 0.001% (v/v) to about 0.01% (v/v), from about 0.01% (v/v) to about 1% (v/v), from about 0.05% (v/v) to about 2% (v/v), from about 1% (v/v) to about 5% (v/v), from about 2% (v/v) to about 10% (v/v), from about 4% (v/v) to about 16% (v/v), from about 5% (v/v) to about 20% (v/v), from about 8% (v/v) to about 24% (v/v), from about 10% (v/v) to about 30% (v/v), from about 12% (v/v) to about 32% (v/v), from about 14% (v/v) to about 34% (v/v), from about 16% (v/v) to about 36% (v/v), from about 18% (v/v) to about 38% (v/v), from about 20% (v/v) to about 40% (v/v), from about 22% (v/v) to about 42% (v/v), from about 24% (v/v) to about 44% (v/v), from about 26% (v/v) to about 46% (v/v), from about 28% (v/v) to about 48% (v/v), from about 30% (v/v) to about 50% (v/v), from about 35% (v/v) to about 55% (v/v), from about 40% (v/v) to about 60% (v/v), from about 45% (v/v) to about 65% (v/v), from about 50% (v/v) to about 70% (v/v), from about 55% (v/v) to about 75% (v/v), from about 60% (v/v) to about 80% (v/v), from about 65% (v/v) to about 85% (v/v), from about 70% (v/v) to about 90% (v/v), from about 75% (v/v) to about 95% (v/v), from about 80% (v/v) to about 96% (v/v), from about 85% (v/v) to about 97% (v/v), from about 90% (v/v) to about 98% (v/v), from about 95% (v/v) to about 99% (v/v), from about 96% (v/v) to about 99.2% (v/v), from about 97% (v/v) to about 99.5% (v/v), from about 98% (v/v) to about 99.8% (v/v), from about 99% (v/v) to about 99.9% (v/v), or greater than 99.9% (v/v).

In some embodiments, the concentration of processed silk (e.g. silk fibroin) or other SBP component (e.g., excipient or cargo) may be present in SBPs and SBP formulations at a concentration of from about 0.0001% (w/w) to about 0.001% (w/w), from about 0.001% (w/w) to about 0.01% (w/w), from about 0.01% (w/w) to about 1% (w/w), from about 0.05% (w/w) to about 2% (w/w), from about 1% (w/w) to about 5% (w/w), from about 2% (w/w) to about 10% (w/w), from about 4% (w/w) to about 16% (w/w), from about 5% (w/w) to about 20% (w/w), from about 8% (w/w) to about 24% (w/w), from about 10% (w/w) to about 30% (w/w), from about 12% (w/w) to about 32% (w/w), from about 14% (w/w) to about 34% (w/w), from about 16% (w/w) to about 36% (w/w), from about 18% (w/w) to about 38% (w/w), from about 20% (w/w) to about 40% (w/w), from about 22% (w/w) to about 42% (w/w), from about 24% (w/w) to about 44% (w/w), from about 26% (w/w) to about 46% (w/w), from about 28% (w/w) to about 48% (w/w), from about 30% (w/w) to about 50% (w/w), from about 35% (w/w) to about 55% (w/w), from about 40% (w/w) to about 60% (w/w), from about 45% (w/w) to about 65% (w/w), from about 50% (w/w) to about 70% (w/w), from about 55% (w/w) to about 75% (w/w), from about 60% (w/w) to about 80% (w/w), from about 65% (w/w) to about 85% (w/w), from about 70% (w/w) to about 90% (w/w), from about 75% (w/w) to about 95% (w/w), from about 80% (w/w) to about 96% (w/w), from about 85% (w/w) to about 97% (w/w), from about 90% (w/w) to about 98% (w/w), from about 95% (w/w) to about 99% (w/w), from about 96% (w/w) to about 99.2% (w/w), from about 97% (w/w) to about 99.5% (w/w), from about 98% (w/w) to about 99.8% (w/w), from about 99% (w/w) to about 99.9% (w/w), or greater than 99.9% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs and SBP formulations at a concentration of 1% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 2% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 3% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 4% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 5% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 6% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 10% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 20% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 30% (w/v).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 16.7% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 20% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 23% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 25% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 27.3% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 28.6% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 33.3% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 40% (w/w).

In one embodiment, the concentration of processed silk or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 50% (w/w).

In some embodiments, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient or cargo) may be present in SBPs and SBP formulations at a concentration of from about 0.01 pg/mL to about 1 pg/mL, from about 0.05 pg/mL to about 2 pg/mL, from about 1 pg/mL to about 5 pg/mL, from about 2 pg/mL to about 10 pg/mL, from about 4 pg/mL to about 16 pg/mL, from about 5 pg/mL to about 20 pg/mL, from about 8 pg/mL to about 24 pg/mL, from about 10 pg/mL to about 30 pg/mL, from about 12 pg/mL to about 32 pg/mL, from about 14 pg/mL to about 34 pg/mL, from about 16 pg/mL to about 36 pg/mL, from about 18 pg/mL to about 38 pg/mL, from about 20 pg/mL to about 40 pg/mL, from about 22 pg/mL to about 42 pg/mL, from about 24 pg/mL to about 44 pg/mL, from about 26 pg/mL to about 46 pg/mL, from about 28 pg/mL to about 48 pg/mL, from about 30 pg/mL to about 50 pg/mL, from about 35 pg/mL to about 55 pg/mL, from about 40 pg/mL to about 60 pg/mL, from about 45 pg/mL to about 65 pg/mL, from about 50 pg/mL to about 75 pg/mL, from about 60 pg/mL to about 240 pg/mL, from about 70 pg/mL to about 350 pg/mL, from about 80 pg/mL to about 400 pg/mL, from about 90 pg/mL to about 450 pg/mL, from about 100 pg/mL to about 500 pg/mL, from about 0.01 ng/mL to about 1 ng/mL, from about 0.05 ng/mL to about 2 ng/mL, from about 1 ng/mL to about 5 ng/mL, from about 2 ng/mL to about 10 ng/mL, from about 4 ng/mL to about 16 ng/mL, from about 5 ng/mL to about 20 ng/mL, from about 8 ng/mL to about 24 ng/mL, from about 10 ng/mL to about 30 ng/mL, from about 12 ng/mL to about 32 ng/mL, from about 14 ng/mL to about 34 ng/mL, from about 16 ng/mL to about 36 ng/mL, from about 18 ng/mL to about 38 ng/mL, from about 20 ng/mL to about 40 ng/mL, from about 22 ng/mL to about 42 ng/mL, from about 24 ng/mL to about 44 ng/mL, from about 26 ng/mL to about 46 ng/mL, from about 28 ng/mL to about 48 ng/mL, from about 30 ng/mL to about 50 ng/mL, from about 35 ng/mL to about 55 ng/mL, from about 40 ng/mL to about 60 ng/mL, from about 45 ng/mL to about 65 ng/mL, from about 50 ng/mL to about 75 ng/mL, from about 60 ng/mL to about 240 ng/mL, from about 70 ng/mL to about 350 ng/mL, from about 80 ng/mL to about 400 ng/mL, from about 90 ng/mL to about 450 ng/mL, from about 100 ng/mL to about 500 ng/mL, from about 0.01 μg/mL to about 1 μg/mL, from about 0.05 μg/mL to about 2 μg/mL, from about 1 μg/mL to about 5 μg/mL, from about 2 μg/mL to about 10 μg/mL, from about 4 μg/mL to about 16 μg/mL, from about 5 μg/mL to about 20 μg/mL, from about 8 μg/mL to about 24 μg/mL, from about 10 μg/mL to about 30 μg/mL, from about 12 μg/mL to about 32 μg/mL, from about 14 μg/mL to about 34 μg/mL, from about 16 μg/mL to about 36 μg/mL, from about 18 μg/mL to about 38 μg/mL, from about 20 μg/mL to about 40 μg/mL, from about 22 μg/mL to about 42 μg/mL, from about 24 μg/mL to about 44 μg/mL, from about 26 μg/mL to about 46 μg/mL, from about 28 μg/mL to about 48 μg/mL, from about 30 μg/mL to about 50 μg/mL, from about 35 μg/mL to about 55 μg/mL, from about 40 μg/mL to about 60 μg/mL, from about 45 μg/mL to about 65 μg/mL, from about 50 μg/mL to about 75 μg/mL, from about 60 μg/mL to about 240 μg/mL, from about 70 μg/mL to about 350 μg/mL, from about 80 μg/mL to about 400 μg/mL, from about 90 μg/mL to about 450 μg/mL, from about 100 μg/mL to about 500 μg/mL, from about 0.01 mg/mL to about 1 mg/mL, from about 0.05 mg/mL to about 2 mg/mL, from about 1 mg/mL to about 5 mg/mL, from about 2 mg/mL to about 10 mg/mL, from about 4 mg/mL to about 16 mg/mL, from about 5 mg/mL to about 20 mg/mL, from about 8 mg/mL to about 24 mg/mL, from about 10 mg/mL to about 30 mg/mL, from about 12 mg/mL to about 32 mg/mL, from about 14 mg/mL to about 34 mg/mL, from about 16 mg/mL to about 36 mg/mL, from about 18 mg/mL to about 38 mg/mL, from about 20 mg/mL to about 40 mg/mL, from about 22 mg/mL to about 42 mg/mL, from about 24 mg/mL to about 44 mg/mL, from about 26 mg/mL to about 46 mg/mL, from about 28 mg/mL to about 48 mg/mL, from about 30 mg/mL to about 50 mg/mL, from about 35 mg/mL to about 55 mg/mL, from about 40 mg/mL to about 60 mg/mL, from about 45 mg/mL to about 65 mg/mL, from about 50 mg/mL to about 75 mg/mL, from about 60 mg/mL to about 240 mg/mL, from about 70 mg/mL to about 350 mg/mL, from about 80 mg/mL to about 400 mg/mL, from about 90 mg/mL to about 450 mg/mL, from about 100 mg/mL to about 500 mg/mL, from about 0.01 g/mL to about 1 g/mL, from about 0.05 g/mL to about 2 g/mL, from about 1 g/mL to about 5 g/mL, from about 2 g/mL to about 10 g/mL, from about 4 g/mL to about 16 g/mL, or from about 5 g/mL to about 20 g/mL.

In one embodiment, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 5 mg/mL.

In one embodiment, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 2.5 mg/mL.

In one embodiment, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 1.25 mg/mL.

In one embodiment, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 0.625 mg/mL.

In one embodiment, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient or cargo) is present in SBPs at a concentration of 0.3125 mg/mL.

In some embodiments, the concentration of processed silk (e.g., silk fibroin) or other SBP component (e.g., excipient or cargo) is present in SBPs and SBP formulations at a concentration of from about 0.01 pg/kg to about 1 pg/kg, from about 0.05 pg/kg to about 2 pg/kg, from about 1 pg/kg to about 5 pg/kg, from about 2 pg/kg to about 10 pg/kg, from about 4 pg/kg to about 16 pg/kg, from about 5 pg/kg to about 20 pg/kg, from about 8 pg/kg to about 24 pg/kg, from about 10 pg/kg to about 30 pg/kg, from about 12 pg/kg to about 32 pg/kg, from about 14 pg/kg to about 34 pg/kg, from about 16 pg/kg to about 36 pg/kg, from about 18 pg/kg to about 38 pg/kg, from about 20 pg/kg to about 40 pg/kg, from about 22 pg/kg to about 42 pg/kg, from about 24 pg/kg to about 44 pg/kg, from about 26 pg/kg to about 46 pg/kg, from about 28 pg/kg to about 48 pg/kg, from about 30 pg/kg to about 50 pg/kg, from about 35 pg/kg to about 55 pg/kg, from about 40 pg/kg to about 60 pg/kg, from about 45 pg/kg to about 65 pg/kg, from about 50 pg/kg to about 75 pg/kg, from about 60 pg/kg to about 240 pg/kg, from about 70 pg/kg to about 350 pg/kg, from about 80 pg/kg to about 400 pg/kg, from about 90 pg/kg to about 450 pg/kg, from about 100 pg/kg to about 500 pg/kg, from about 0.01 ng/kg to about 1 ng/kg, from about 0.05 ng/kg to about 2 ng/kg, from about 1 ng/kg to about 5 ng/kg, from about 2 ng/kg to about 10 ng/kg, from about 4 ng/kg to about 16 ng/kg, from about 5 ng/kg to about 20 ng/kg, from about 8 ng/kg to about 24 ng/kg, from about 10 ng/kg to about 30 ng/kg, from about 12 ng/kg to about 32 ng/kg, from about 14 ng/kg to about 34 ng/kg, from about 16 ng/kg to about 36 ng/kg, from about 18 ng/kg to about 38 ng/kg, from about 20 ng/kg to about 40 ng/kg, from about 22 ng/kg to about 42 ng/kg, from about 24 ng/kg to about 44 ng/kg, from about 26 ng/kg to about 46 ng/kg, from about 28 ng/kg to about 48 ng/kg, from about 30 ng/kg to about 50 ng/kg, from about 35 ng/kg to about 55 ng/kg, from about 40 ng/kg to about 60 ng/kg, from about 45 ng/kg to about 65 ng/kg, from about 50 ng/kg to about 75 ng/kg, from about 60 ng/kg to about 240 ng/kg, from about 70 ng/kg to about 350 ng/kg, from about 80 ng/kg to about 400 ng/kg, from about 90 ng/kg to about 450 ng/kg, from about 100 ng/kg to about 500 ng/kg, from about 0.01 μg/kg to about 1 μg/kg, from about 0.05 μg/kg to about 2 μg/kg, from about 1 μg/kg to about 5 μg/kg, from about 2 μg/kg to about 10 μg/kg, from about 4 μg/kg to about 16 μg/kg, from about 5 μg/kg to about 20 μg/kg, from about 8 μg/kg to about 24 μg/kg, from about 10 μg/kg to about 30 μg/kg, from about 12 μg/kg to about 32 μg/kg, from about 14 μg/kg to about 34 μg/kg, from about 16 μg/kg to about 36 μg/kg, from about 18 μg/kg to about 38 μg/kg, from about 20 μg/kg to about 40 μg/kg, from about 22 μg/kg to about 42 μg/kg, from about 24 μg/kg to about 44 μg/kg, from about 26 μg/kg to about 46 μg/kg, from about 28 μg/kg to about 48 μg/kg, from about 30 μg/kg to about 50 μg/kg, from about 35 μg/kg to about 55 μg/kg, from about 40 μg/kg to about 60 μg/kg, from about 45 μg/kg to about 65 μg/kg, from about 50 μg/kg to about 75 μg/kg, from about 60 μg/kg to about 240 μg/kg, from about 70 μg/kg to about 350 μg/kg, from about 80 μg/kg to about 400 μg/kg, from about 90 μg/kg to about 450 μg/kg, from about 100 μg/kg to about 500 μg/kg, from about 0.01 mg/kg to about 1 mg/kg, from about 0.05 mg/kg to about 2 mg/kg, from about 1 mg/kg to about 5 mg/kg, from about 2 mg/kg to about 10 mg/kg, from about 4 mg/kg to about 16 mg/kg, from about 5 mg/kg to about 20 mg/kg, from about 8 mg/kg to about 24 mg/kg, from about 10 mg/kg to about 30 mg/kg, from about 12 mg/kg to about 32 mg/kg, from about 14 mg/kg to about 34 mg/kg, from about 16 mg/kg to about 36 mg/kg, from about 18 mg/kg to about 38 mg/kg, from about 20 mg/kg to about 40 mg/kg, from about 22 mg/kg to about 42 mg/kg, from about 24 mg/kg to about 44 mg/kg, from about 26 mg/kg to about 46 mg/kg, from about 28 mg/kg to about 48 mg/kg, from about 30 mg/kg to about 50 mg/kg, from about 35 mg/kg to about 55 mg/kg, from about 40 mg/kg to about 60 mg/kg, from about 45 mg/kg to about 65 mg/kg, from about 50 mg/kg to about 75 mg/kg, from about 60 mg/kg to about 240 mg/kg, from about 70 mg/kg to about 350 mg/kg, from about 80 mg/kg to about 400 mg/kg, from about 90 mg/kg to about 450 mg/kg, from about 100 mg/kg to about 500 mg/kg, from about 0.01 g/kg to about 1 g/kg, from about 0.05 g/kg to about 2 g/kg, from about 1 g/kg to about 5 g/kg, from about 2 g/kg to about 10 g/kg, from about 4 g/kg to about 16 g/kg, or from about 5 g/kg to about 20 g/kg, from about 10 g/kg to about 50 g/kg, from about 15 g/kg to about 100 g/kg, from about 20 g/kg to about 150 g/kg, from about 25 g/kg to about 200 g/kg, from about 30 g/kg to about 250 g/kg, from about 35 g/kg to about 300 g/kg, from about 40 g/kg to about 350 g/kg, from about 45 g/kg to about 400 g/kg, from about 50 g/kg to about 450 g/kg, from about 55 g/kg to about 500 g/kg, from about 60 g/kg to about 550 g/kg, from about 65 g/kg to about 600 g/kg, from about 70 g/kg to about 650 g/kg, from about 75 g/kg to about 700 g/kg, from about 80 g/kg to about 750 g/kg, from about 85 g/kg to about 800 g/kg, from about 90 g/kg to about 850 g/kg, from about 95 g/kg to about 900 g/kg, from about 100 g/kg to about 950 g/kg, or from about 200 g/kg to about 1000 g/kg.

Appearance: Transparent, Opaque, Translucent

In some embodiments, the appearance of SBPs described in the present disclosure may be tuned for the application for which they were designed. In some embodiments, SBPs may be transparent. In some embodiments, SBPs may be translucent. In some embodiments, SBPs may be opaque. In some embodiments, SBP preparation methods may be used to modulate clarity, as taught in International Patent Application Publication No. WO2012170655, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the incorporation of excipients may be used to tune the clarity of processed silk preparations. In some embodiments, the excipient is sucrose. In some embodiments, the sucrose may also increase protein reconstitution during lyophilization. In some embodiments, sucrose may improve processed silk hydrogel clarity (optical transparency).

The optical clarity of an SBP may be measured by any method known to one of skill in the art. SBPs may be determined to be optically clear because silk materials may be naturally optically clear, as described in Lawrence et al. (2009) Biomaterials 30(7): 1299-1308, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the optical clarity of an SBP is measured by absorbance. As used herein, the term "absorbance" refers to a measurement of a substance's ability to take in light. Absorbance may be measured at any wavelength. In some embodiments, the absorbance of an SBP is measured from 200 nm-850 nm. The absorbance of an SBP may optionally be converted to transmittance. As used herein, the term "transmittance" refers to a measurement of the amount of light that has passed through a substance unchanged, without absorbance, reflection, or scattering. Transmittance may be determined by the ratio of the intensity of transmitted light to the ratio of the intensity of the incident light. In some embodiments, an SBP absorbs light at around 280 nm, which is the range of light absorbed for a protein. Said SBP may be otherwise determined to be optically clear. In some embodiments, the optical clarity of an SBP is measured with any of the methods described in Toytziaridis et al. (2016) International Journal of Molecular Sciences 17: 1897, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, optically transparent SBPs may be used for ocular applications, e.g., treatment of ocular conditions, diseases, and/or indications.

Combinations

In some embodiments, SBP formulations are presented in a combinatorial format. A combinatorial format may consist of two or more different materials that have been combined to form a single composition. In some embodiments, two or more SBPs of different formats (e.g. rod, hydrogel etc.) are combined to form a single composition (e.g., see European Publication Number EP3212246, the contents of which are herein incorporated by reference in their entirety). In some embodiments, one or more SBP is combined with a different material (e.g. a polymer, a mat, a particle, a microsphere, a nanosphere, a metal, a scaffold, etc.) to form a single composition (e.g., see International Publication Number WO2017179069, the contents of which are herein incorporated by reference in their entirety. In some embodiments, combinatorial formats are prepared by formulating two or more SBPs of different formats as a single composition (e.g., see Kambe et al. (2017) Materials (Basel) 10(10):1153, the contents of which are herein incorporated by reference in their entirety). In some embodiments, combinatorial formats are prepared by formulating two or more SBPs of different formats, along with another material, as a single composition (e.g., see International Publication Number WO2017177281, the contents of which are herein incorporated by reference in their entirety). In some embodiments, combinatorial formats include adding one or more SBPs to a first SBP of a different format (e.g., see European Patent Number EP3212246, the contents of which are herein incorporated by reference in their entirety). In some embodiments, combinatorial formats include adding one or more SBPs to a first composition comprising a different material (e.g., see Jiang et al. (2017) J Biomater Sci Polym Ed 15:1-36, the contents of which are herein incorporated by reference in their entirety). In some embodiments, the combinatorial formats are prepared by adding one or more materials to one or more first formed SBPs (e.g., see Babu et al. (2017) J Colloid Interface Sci 513:62-72, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, SBP formulations may be administered in combination with other therapeutic agent and/or methods of treatment, e.g., with known pharmaceuticals and/or known therapeutic methods, such as, for example, those which are currently employed for treating these disorders. For example, SBP formulations used to treat cancer may be administered in combination with other anti-cancer treatments (e.g., biological, chemotherapy, or radiotherapy treatments).

Distribution

SBP components may be distributed equally or unequally, depending on format and application. Non-limiting examples of unequal distribution include component localization in SBP regions or compartments, on SBP surfaces, etc. In some embodiments, components include cargo. Such cargo may include payloads, for example, therapeutic agents. In some embodiments, therapeutic agents are present on the surface of an SBP (e.g., see Han et al. (2017) Biomacromolecules 18(11):3776-3787.; Ran et al. (2017) Biomacromolecules 18(11):3788-3801, the contents of each of which are herein incorporated by reference in their entirety). In some embodiments, components (e.g., therapeutic agents) are homogenously mixed with processed silk to generate a desired distribution (e.g., see United States Publication No. US20170333351; Sun et al. (2017) Journal of Materials Chemistry B 5:8770-8779; and Du et al. (2017) Nanocale Res Lett 12(1):573, the contents of each of which are herein incorporated by reference in their entirety). In some embodiments, components (e.g., therapeutic agents) are encapsulated in SBPs (e.g., see Shi et al. (2017) Nanoscale 9:14520, the contents of which are herein incorporated by reference in their entirety).

Rods

In some embodiments, SBP formulation includes rods. As used herein when referring to SBPs, the term "rod" refers to an elongated format, typically cylindrical, that may have blunted or tapered ends. Rods may be suitable for implantation or similar administration methods as it may be possible to deliver rods by injection. Rods may also be obtained simply by passing suitably viscous SBP formulations through a needle, cannula, tube, or opening. In some embodiments, rods are prepared by one or more of injection molding, heated or cooled extrusion, extrusion through a coating agent, milling with a therapeutic agent, and combining with a polymer followed by extrusion.

In some embodiments, SBP rods include processed silk (e.g., silk fibroin) rods. Some rods may include coterminous luminal cavities in whole or in part running through the rod. Rods may be of any cross-sectional shape, including, but not limited to, circular, square, oval, triangular, irregular, or combinations thereof.

In some embodiments, rods are prepared from silk fibroin preparations. The silk fibroin preparations may include lyophilized silk fibroin. The lyophilized silk fibroin may be dissolved in water to form silk fibroin solutions used in rod preparation. Silk fibroin solutions may be prepared as stock solutions to be combined with additional components prior to rod preparation. In some embodiments silk fibroin stock solutions have a silk fibroin concentration of between 10% (w/v) and 40% (w/v). In some embodiments, the silk fibroin stock solution for the preparation of silk fibroin rods has a concentration of at least 10% (w/v), at least 20% (w/v), at least 30% (w/v), at least 40% (w/v), or at least 50% (w/v).

In some embodiments, silk fibroin stock solution prepared for rod formation are mixed with one or more other components intended to be include in the final processed silk rods. Examples of such other components include, but are not limited to, excipients, salts, therapeutic agents, biological agents, proteins, small molecules, and polymers. In some embodiments, processed silk rods may include between 20 to 55% (w/w) silk fibroin. In some embodiments, processed silk rods may include between 40 to 80% (w/w) therapeutic agent. In some embodiments, processed silk rods may include 35% (w/w) silk fibroin and 65% (w/w) therapeutic agent. In some embodiments, processed silk rods may include 30% (w/w) silk fibroin and 70% (w/w) therapeutic agent. In some embodiments, processed silk rods may include 40% (w/w) silk fibroin and 60% (w/w) therapeutic agent. In some embodiments, processed silk rods may include 26% (w/w) silk fibroin and 74% (w/w) therapeutic agent. In some embodiments, processed silk rods may include 37% (w/w) silk fibroin and 63% (w/w) therapeutic agent. In some embodiments, processed silk rods may include 33% (w/w) silk fibroin and 66% (w/w) therapeutic agent. In some embodiments, processed silk rods may include 51% (w/w) silk fibroin and 49% (w/w) therapeutic agent. In some embodiments, silk fibroin may be included at a concentration (w/w) of 0.01% to about 1%, from about 0.05% to about 2%, from about 0.1% to about 30%, from about 1% to about 5%, from about 2% to about 10%, from about 3% to about 15%, from about 4% to about 20%, from about 5% to about 25%, from about 6% to about 30%, from about 7% to about 35%, from about 8% to about 40%, from about 9% to about 45%, from about 10% to about 50%, from about 12% to about 55%, from about 14% to about 60%, from about 16% to about 65%, from about 18% to about 70%, from about 20% to about 75%, from about 22% to about 80%, from about 24% to about 85%, from about 26% to about 90%, from about 28% to about 95%, from about 30% to about 96%, from about 32% to about 97%, from about 34% to about 98%, from about 36% to about 98.5%, from about 38% to about 99%, from about 40% to about 99.5%, from about 42% to about 99.6%, from about 44% to about 99.7%, from about 46% to about 99.8%, or from about 50% to about 99.9%.

In some embodiments, processed silk rods are prepared by extrusion. As used herein, the term "extrusion" refers to a process by which a substance is forced through an opening, tube, or passage. In some embodiments, processed silk rods are formed by extruding SBP formulations through a needle or cannula. SBP formulations used for rod formation may have varying levels of viscosity. Preparation viscosity may depend on the presence and/or identity of excipients present. In some embodiments, SBP formulations may include compounds or compositions intended to be embedded in rods prepared by extrusion. Excipients, compounds, or compositions included in SBP formulations used for extrusion may include, but are not limited to, salts, therapeutic agents, biological agents, proteins, small molecules, and polymers. Extrusion may be carried out manually or by an automated process.

In some embodiments, extrusion may be carried out using a syringe. The syringe may be fitted with a needle, tube, or cannula. The needle, tube, or cannula may have a sharpened end or a blunt end. The needle may have a diameter of from about 0.1 mm to about 0.3 mm, from about 0.2 mm to about 0.7 mm, from about 0.4 mm to about 1.1 mm, from about 0.6 mm to about 1.5 mm, from about 0.8 mm to about 1.9 mm, from about 1 mm to about 2.3 mm, from about 1.2 mm to about 2.7 mm, from about 1.6 mm to about 3.1 mm, or from about 2 mm to about 3.5 mm. SBP formulations may be used to fill tubes, wherein the SBP formulations are incubated in the tubes for various periods of time under various conditions (e.g., various temperatures). In some embodiments, tubing filled with processed silk preparation may be incubated at 37° C. for from about 2 hours to about 36 hours or more. In some embodiments, processed silk filled tubing is incubated for 24 hours. In some embodiments, SBP formulations remain in tubing after the 37° C. incubation. In some embodiments, SBP formulations are removed from the tubing after the incubation at 37° C. SBP formulations removed from tubing may maintain a rod-shaped format. Such preparations may be dried after removal from tubing. In some embodiments, SBP formulations may be encased in tubing while drying. Rods may be dried by one or more of freeze-drying, oven drying, and air drying. Some SBP formulations may be removed tubing after drying.

Tubing used for extrusion may be composed of various materials. In some embodiments, tubing is made from one or more of silicone, polyetheretherketone (PEEK), polytetrafluoroethylene (PTFE), amorphous fluoroplastics, fluorinated ethylene propylene, perfluoroalkoxy copolymers, ethylene-tetrafluoroethylene, polyolefins, and nylon.

In some embodiments, rods may have a diameter of from about 0.05 μm to about 10 μm, from about 1 μm to about 20 μm, from about 2 μm to about 30 μm, from about 5 μm to about 40 μm, from about 10 μm to about 50 μm, from about 20 μm to about 60 μm, from about 30 μm to about 70 μm, from about 40 μm to about 80 μm, from about 50 μm to about 90 μm, from about 0.05 mm to about 2 mm, from about 0.1 mm to about 3 mm, from about 0.2 mm to about 4 mm, from about 0.5 mm to about 5 mm, from about 1 mm to about 6 mm, from about 2 mm to about 7 mm, from about 5 mm to about 10 mm, from about 8 mm to about 16 mm, from about 10 mm to about 50 mm, from about 20 mm to about 100 mm, from about 40 mm to about 200 mm, from about 60 mm to about 300 mm, from about 80 mm to about 400 mm, from about 250 mm to about 750 mm, or from about 500 mm to about 1000 mm. In some embodiments, rods include a diameter of at least 0.5 μm, at least 1 μm, at least 10 μm, at least 100 μm, at least 500 μm, at least 1 mm, at least 10 mm, or at least 100 mm. In one embodiment, the rods have a diameter of 1 mm. In another embodiment, the rods have a diameter of 0.5 mm. In another embodiment, the rods have a diameter of 400 μm. In another embodiment, the rods have a diameter of 430 μm.

In some embodiments, the rods described herein may have a density of from about 0.01 μg/mL to about 1 μg/mL, from about 0.05 μg/mL to about 2 μg/mL, from about 1 μg/mL to about 5 μg/mL, from about 2 μg/mL to about 10 μg/mL, from about 4 μg/mL to about 16 μg/mL, from about 5 μg/mL to about 20 μg/mL, from about 8 μg/mL to about 24 μg/mL, from about 10 μg/mL to about 30 μg/mL, from about 12 μg/mL to about 32 μg/mL, from about 14 μg/mL to about 34 μg/mL, from about 16 μg/mL to about 36 μg/mL, from about 18 μg/mL to about 38 μg/mL, from about 20 μg/mL to about 40 μg/mL, from about 22 μg/mL to about 42 μg/mL, from about 24 μg/mL to about 44 μg/mL, from about 26 μg/mL to about 46 μg/mL, from about 28 μg/mL to about 48 μg/mL, from about 30 μg/mL to about 50 μg/mL, from about 35 μg/mL to about 55 μg/mL, from about 40 μg/mL to about 60 μg/mL, from about 45 μg/mL to about 65 μg/mL, from about 50 μg/mL to about 75 μg/mL, from about 60 μg/mL to about 240 μg/mL, from about 70 μg/mL to about 350 μg/mL, from about 80 μg/mL to about 400 μg/mL, from about 90 μg/mL to about 450 μg/mL, from about 100 μg/mL to about 500 μg/mL, from about 0.01 mg/mL to about 1 mg/mL, from about 0.05 mg/mL to about 2 mg/mL, from about 1 mg/mL to about 5 mg/mL, from about 2 mg/mL to about 10 mg/mL, from about 4 mg/mL to about 16 mg/mL, from about 5 mg/mL to about 20 mg/mL, from about 8 mg/mL to about 24 mg/mL, from about 10 mg/mL to about 30 mg/mL, from about 12 mg/mL to about 32 mg/mL, from about 14 mg/mL to about 34 mg/mL, from about 16 mg/mL to about 36 mg/mL, from about 18 mg/mL to about 38 mg/mL, from about 20 mg/mL to about 40 mg/mL, from about 22 mg/mL to about 42 mg/mL, from about 24 mg/mL to about 44 mg/mL, from about 26 mg/mL to about 46 mg/mL, from about 28 mg/mL to about 48 mg/mL, from about 30 mg/mL to about 50 mg/mL, from about 35 mg/mL to about 55 mg/mL, from about 40 mg/mL to about 60 mg/mL, from about 45 mg/mL to about 65 mg/mL, from about 50 mg/mL to about 75 mg/mL, from about 60 mg/mL to about 240 mg/mL, from about 70 mg/mL to about 350 mg/mL, from about 80 mg/mL to about 400 mg/mL, from about 90 mg/mL to about 450 mg/mL, from about 100 mg/mL to about 500 mg/mL, from about 0.01 g/mL to about 1 g/mL, from about 0.05 g/mL to about 2 g/mL, from about 1 g/mL to about 5 g/mL, from about 2 g/mL to about 10 g/mL, from about 4 g/mL to about 16 g/mL, or from about 5 g/mL to about 20 g/mL.

Gels and Hydrogels

In some embodiments, SBP formulations include gels or hydrogels. As used herein, the term "gel" refers to a dispersion of liquid molecules in a solid medium. Gels in which the dispersed liquid molecules include water are referred to herein as "hydrogels." Gels in which the dispersed liquid molecules include an organic phase are referred to herein as "organogels." The solid medium may include polymer networks. Hydrogels may be formed with silk of any grade (e.g. grade 3, grade 4, grade 5, and/or grade 6).

In some embodiments, SBP gels or hydrogels are prepared with processed silk. In processed silk gels, polymer networks may include silk fibroin. In some embodiments, gels are prepared with one or more therapeutic agents. In some embodiments, gels include one or more excipients. The excipients may be selected from any of those described herein. In some embodiments, excipients may include salts. In some embodiments, the excipients may include gelling agents. In some embodiments, gels are prepared with one or more therapeutic agents, biological agents, proteins, small molecules, and/or polymers. In some embodiments, gels may be prepared by mixing a solution comprising processed silk with a gelling agent. The gelling agent may be in a second solution. In some embodiments, the therapeutic agent may be in solution with processed silk. In some embodiments, the therapeutic agent may be in solution with the gelling agent. In some embodiments, a stock solution of therapeutic agent may be used to dissolve processed silk for the preparation of a hydrogel. The ratio of the solution comprising processed silk to the gelling agent or solution comprising the gelling agent may be from about 5:1 to about 4.5:1, from about 4.5:1 to about 4:1, from about 4:1 to about 3.5:1, from about 3.5:1 to about 3:1, from about 3:1 to about 2.5:1, from about 2.5:1 to about 2:1, from about 2:1 to about 1.5:1, from about 1.5:1 to about 1:1, from about 1:1 to about 1:1.5, from about 1:1.5 to about 1:2, from about 1:2 to about 1:2.5, from about 1:2.5 to about 1:3, from about 1:3 to about 1:3.5, from about 1:3.5 to about 1:4, from about 1:4 to about 1:4.5, or from about 1:4.5 to about 1:5.

Gel preparation may require varying temperatures and incubation times for gel polymer networks to form. In some embodiments, SBP formulations are heated to 37° C. to prepare gels. In some embodiments, SBP formulations are incubated at 4° C. to prepare gels. In some embodiments, SBP formulations are incubated for from about 2 hours to about 36 hours or more to promote gel formation. In some embodiments, gel formation requires mixing with one or more gelling agents or excipients. Mixing may be carried out under various temperatures and lengths of time to allow gel polymer networks to form. Gel formation may require homogenous dispersion of gelling agents or excipients. In some embodiments, SBP formulations used to prepare gels include silk fibroin. Gel formation for processed silk gels may require incubation at 37° C. for up to 24 hours. Gel formation for processed silk gels may require incubation at 4° C. for up to 24 hours. Some gels may be stored for later use or processing. In some embodiments, gels are stored at 4° C.

In some embodiments, processed silk gels include one or more excipients and/or gelling agents at a concentration of from about 0.01% (w/v) to about 0.1% (w/v), from about 0.1% (w/v) to about 1% (w/v), from about 0.5% (w/v) to about 5% (w/v), from about 1% (w/v) to about 10% (w/v), from about 5% (w/v) to about 15% (w/v), from about 10% (w/v) to about 30% (w/v), from about 15% (w/v) to about 45% (w/v), from about 20% (w/v) to about 55% (w/v), from about 25% (w/v) to about 65% (w/v), from about 30% (w/v) to about 70% (w/v), from about 35% (w/v) to about 75% (w/v), from about 40% (w/v) to about 80% (w/v), from about 50% (w/v) to about 85% (w/v), from about 60% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 90% (w/v) to about 96% (w/v), from about 92% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 98% (w/v) to about 99.5% (w/v), or from about 99% (w/v) to about 99.9% (w/v).

In some embodiments, processed silk gels (e.g., hydrogels or organogels) include silk fibroin at a concentration of from about 0.01% (w/v) to about 0.1% (w/v), from about 0.1% (w/v) to about 1% (w/v), from about 0.5% (w/v) to about 5% (w/v), from about 1% (w/v) to about 10% (w/v), from about 5% (w/v) to about 15% (w/v), from about 10% (w/v) to about 30% (w/v), from about 15% (w/v) to about 45% (w/v), from about 20% (w/v) to about 55% (w/v), from about 25% (w/v) to about 65% (w/v), from about 30% (w/v) to about 70% (w/v), from about 35% (w/v) to about 75% (w/v), from about 40% (w/v) to about 80% (w/v), from about 50% (w/v) to about 85% (w/v), from about 60% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 90% (w/v) to about 96% (w/v), from about 92% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 98% (w/v) to about 99.5% (w/v), or from about 99% (w/v) to about 99.9% (w/v).

Silk fibroin included may be from a silk fibroin preparation with an average silk fibroin molecular weight or range of molecular weights of from about 3.5 kDa to about 10 kDa, from about 5 kDa to about 20 kDa, from about 10 kDa to about 30 kDa, from about 15 kDa to about 40 kDa, from about 20 kDa to about 50 kDa, from about 25 kDa to about 60 kDa, from about 30 kDa to about 70 kDa, from about 35 kDa to about 80 kDa, from about 40 kDa to about 90 kDa, from about 45 kDa to about 100 kDa, from about 50 kDa to about 110 kDa, from about 55 kDa to about 120 kDa, from about 60 kDa to about 130 kDa, from about 65 kDa to about 140 kDa, from about 70 kDa to about 150 kDa, from about 75 kDa to about 160 kDa, from about 80 kDa to about 170 kDa, from about 85 kDa to about 180 kDa, from about 90 kDa to about 190 kDa, from about 95 kDa to about 200 kDa, from about 100 kDa to about 210 kDa, from about 115 kDa to about 220 kDa, from about 125 kDa to about 240 kDa, from about 135 kDa to about 260 kDa, from about 145 kDa to about 280 kDa, from about 155 kDa to about 300 kDa, from about 165 kDa to about 320 kDa, from about 175 kDa to about 340 kDa, from about 185 kDa to about 360 kDa, from about 195 kDa to about 380 kDa, from about 205 kDa to about 400 kDa, from about 215 kDa to about 420 kDa, from about 225 kDa to about 440 kDa, from about 235 kDa to about 460 kDa, or from about 245 kDa to about 500 kDa.

In some embodiments, hydrogels include one or more therapeutic agents at a concentration of from about 0.0001% (w/v) to about 0.001% (w/v), from about 0.001% (w/v) to about 0.01% (w/v), from about 0.01% (w/v) to about 0.1% (w/v), from about 0.1% (w/v) to about 1% (w/v), from about 0.5% (w/v) to about 5% (w/v), from about 1% (w/v) to about 10% (w/v), from about 5% (w/v) to about 15% (w/v), from about 10% (w/v) to about 30% (w/v), from about 15% (w/v) to about 45% (w/v), from about 20% (w/v) to about 55% (w/v), from about 25% (w/v) to about 65% (w/v), from about 30% (w/v) to about 70% (w/v), from about 35% (w/v) to about 75% (w/v), from about 40% (w/v) to about 80% (w/v), from about 50% (w/v) to about 85% (w/v), from about 60% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 90% (w/v) to about 96% (w/v), from about 92% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 98% (w/v) to about 99.5% (w/v), or from about 99% (w/v) to about 99.9% (w/v).

Gelling agents may be used to facilitate sol-gel transition. As used herein, the term "sol-gel transition" refers to the shift of a formulation from a solution to a gel. In some embodiments, the use of gelling agents may be carried out according to any of such methods described in International Publication No. WO2017139684, the contents of which are herein incorporated by reference in their entirety. Gelling agents may be water-soluble, waxy solids. In some embodiments, gelling agents may be water-soluble and hygroscopic in nature. In some embodiments, gelling agents may include polar molecules. Gelling agents may have net positive, net negative, or net neutral charges at a physiological pH (e.g., pH of about 7.4). Some gelling agents may be amphipathic. Additional examples of gelling agents include oils (e.g., castor, corn oil, cottonseed oil, olive oil, peanut oil, peppermint oil, safflower oil, sesame oil, soybean oil, hydrogenated vegetable oil, hydrogenated soybean oil, and medium-chain triglycerides of coconut oil and/or palm seed oil), emulsifiers [e.g., polyoxyl 40 stearate (PEG 1750 monosterate), polyoxyl 8 stearate (PEG 400 monosterate), polysorbate 20, polysorbate 80, polysorbate-SO, or poloxamer], surfactants (e.g., polysorbate, poloxamer, sodium dodecyl sulfate, Triton X100, or tyloxapol), and suspending agents (e.g., polyvinyl pyrrolidone, polyvinyl pyrrolidone-12, polyvinyl pyrrolidone-17, hydroxyethyl cellulose, or carboxymethyl cellulose). Any gelling agent listed in Table 1 may be used.

In some embodiments, gel formation is induced by applying one or more of the following to processed silk preparations: ultrasound, sonication, shear forces, temperature change (e.g., heating), addition of precipitants, modulation of pH, changes in salt concentration, chemical cross-linking, chemical modification, seeding with preformed hydrogels, increasing silk fibroin concentration, modulating osmolarity, use of electric fields, or exposure to electric currents. In some embodiments, methods of inducing gel formation may include, but are not limited to any of those described in International Patent Application Publication No. WO2005012606 or United States Patent Publication No. US2011/0171239, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, processed silk gel preparation may be carried with the aid of sonication. As used herein, the term "sonication" refers to a process of agitation using sound energy. Sonication conducted at frequencies greater than 20 kHz is referred to as ultrasonication. Sonication may aid in gel formation by dispersing and/or agitating polymer components within a solution to foster an arrangement that favors polymer network formation. The polymer network may include silk fibroin. In some embodiments, the use of sonication for gel preparation may be carried out according to any of the methods described in Zhao et al. (2017) Materials Letters 211:110-113 or Mao et al. (2017) Colloids Surf B Biointerfaces 160:704-714), the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, processed silk gel formation may be carried out using shear forces. As used herein, the term "shear forces" refers to unaligned forces that apply pressure to two or more different parts of an object or medium from different and/or opposing directions. Shear forces are distinct from compression forces, which are directed toward each other. Shear forces may be applied during processed silk gel preparation using a syringe, tubing, needle, or other apparatus capable of increasing shear forces. Processed silk preparation may be pushed through a syringe, tubing, needle, or other apparatus to generate shear forces. The use of shear forces in gel formation may include any of those described in United States Patent Publication No. US2011/0171239, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, changes in temperature may be used to aid in processed silk gel formation. Changes in temperature may be used to disperse or align polymer components in an arrangement that promotes gel polymer network formation. The polymer components may include silk fibroin. In some embodiments, gel formation may be carried out by raising or lowering the temperature of a processed silk preparation to from about 0° C. to about 5° C., from about 2° C. to about 6° C., from about 4° C. to about 12° C., from about 8° C. to about 16° C., from about 10° C. to about 26° C., from about 15° C. to about 28° C., from about 20° C. to about 32° C., from about 25° C. to about 34° C., from about 30° C. to about 45° C., from about 35° C. to about 55° C., from about 37° C. to about 65° C., from about 40° C. to about 75° C., from about 50° C. to about 100° C., from about 60° C. to about 120° C., from about 70° C. to about 140° C., from about 80° C. to about 160° C., or from about 100° C. to about 300° C. In some embodiments, one or more excipients or gelling agents may be included to lower the temperature necessary for gel formation to occur. Such embodiments may be employed to protect temperature-sensitive components embedded within gels. In some embodiments, gel formation is carried out at 4° C. Glycerol, polyethylene glycol (PEG), and/or polymers of PEG (e.g., PEG400) may be included in SBP formulations as excipients to lower the temperature necessary to form a gel. The gel may be a silk fibroin gel. Excipient concentration may be about 30% (w/v). Silk fibroin concentration may be from about 2% to about 30%.

In some embodiments, gel formation is carried out by applying an electric current, also referred to as "electrogelation." Electrogelation may be carried out according to any of the methods presented in International Publication No. WO2010036992, the contents of which are herein incorporated by reference in their entirety. In some embodiments, a reverse voltage may be applied to reverse gel formation and regenerate a processed silk solution.

In some embodiments, gel formation is carried out by modulating the pH of processed silk preparations. Gel formation through pH modulation may be carried out according to the methods described in International Patent Application Publication No. WO2005012606, United States Patent Publication No. US2011/0171239, and Dubey et al. (2017) Materials Chemistry and Physics 203:9-16, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, gel formation is carried out in association with modulating the osmolarity of a processed silk preparation. As used herein, the term "osmolarity" or "osmotic concentration" refers to the number of osmoles of solute in solution on a per liter basis (Osm/L). Unlike molarity, which is a measure of the number of moles solute per liter of solvent (M), osmolarity factors in the effect of ions on osmotic pressure. For example, a 1 M solution of NaCl would have an osmolarity of 2 Osm/L while a 1 M solution of $MgCl_2$ would have an osmolarity of 3 Osm/L. Hypo- or hyper-osmotic formulations can lead to local tissue damage and reduced biocompatibility. In some embodiments, the osmolarity of processed silk gels is modulated by controlling the type, molecular weight, and/or concentration of excipients included. Osmolarity may be modulated by varying the concentration and/or molecular weight of salts used in processed silk preparations. In some embodiments, osmolarity is reduced by using lower molecular weight gelling agents. For example, 4 kDa PEG may be used in place of PEG400. The use of Poloxamer-188 at 10% (w/v) may reduce osmolarity in comparison to lower molecular weight species such as glycerol. In some embodiments, sodium chloride may be added to increase osmolarity. In some embodiments, osmolarity is adjusted to fall between 280 and 320 mOsm/L.

In some embodiments, gel formation is carried out through seeding. As used herein when referring to gel formation, "seeding" refers to a process of inducing gel formation using a small amount of pre-formed gel. Seeding may promote gel formation by encouraging polymer network formation to build off of the pre-formed gel introduced. In some embodiments the gel includes silk fibroin. Seeding with a pre-formed silk fibroin hydrogel may be used to promote transition of a silk fibroin solution into a silk fibroin gel. In some embodiments, seeding reduces the need for gelling agents and/or excipients to form gels.

In some embodiments, gel formation is carried out using chemical cross-linking. As used herein, the term "chemical cross-linking" refers to a process of forming covalent bonds between chemical groups from different molecules or between chemical groups present on different parts of the same molecule. In some embodiments, chemical cross-linking may be carried out by contacting SBP formulations with ethanol. Such methods may be carried out according to those described in Shi et al. (2017) Advanced Material 29(29):1701089, the contents of which are herein incorporated by reference in their entirety. In some embodiments, cross-linking may be carried out using enzymes. Methods of enzyme cross-linking using horse radish peroxidase may include any of those described in McGill et al. (2017) Acta Biomaterialia 63:76-84 or Guo et al. (2017) Biomaterials 145:44-55, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, chemical cross-linking may be photo-initiated, as disclosed in International Publication No. WO2017123383 and in Zhang et al. (2017) Fibers and Polymers 18(10):1831-1840, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, other chemical modifications may be used during processed silk gel preparation. Some chemical modifications may be used to induce silk fibroin β-sheet conformations. In some embodiments, this process involves contact with a chemical. Chemicals may include, but are not limited to, ethanol. In some embodiments, silk fibroin may be chemically crosslinked with other materials during gel preparation. Such materials may include other peptides (e.g., see Guo et al. (2017) Biomaterials 145:44-55, the contents of which are herein incorporated by reference in their entirety). In some embodiments, processed silk gels are prepared by formation of internal chemical cross-links. These crosslinks may be dityrosine crosslinks (e.g., see International Patent Application Publication No. WO2017123383, the contents of which are herein incorporated by reference in their entirety). In some embodiments, photosensitive materials may be used to promote chemical modifications. Such materials may include riboflavin (e.g., see International Publication No. WO2017123383). In some embodiments, processed silk gels may be functionalized with particles. These particles may be microspheres and/or nanospheres (e.g., see Ciocci et al. (2017) Int J Biol Macromol S0141-8130(17):32839-8, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, the SBPs are prepared as hydrogels. In some embodiments, the hydrogels have a concentration between about 3% (w/v) to about 15% (w/v) silk fibroin. In some embodiments the silk fibroin has a boiling time of 90 mb, 120 mb, or 480 mb. In some embodiments, the hydrogels are prepared from silk fibroin lyophilized in phosphate buffer. In some embodiments, the hydrogels have trace amounts of phosphate salts (e.g. potassium phosphate dibasic and potassium phosphate monobasic). In some embodiments, the hydrogels comprise between about 10% (w/v) to about 50% (w/v) excipient. In some embodiments, the excipient is poloxamer-188 (P188), in some embodiments, the excipient is glycerol. In some embodiments, the excipient is PEG 4000 (PEG 4 kDa) and the formulation may optionally include hydrochloric acid. In some embodiments, the excipient is PEG400 and the formulation may optionally include hydrochloric acid. In some embodiments, the hydrogels comprise 15 mM hydrochloric acid. In some embodiments, the formulations are as described in Table 2. In the sample named 90mb; hyd; 5% SFf; 10% P188f, "90mb" refers to silk degummed with a 90-minute boil, "hyd" refers to the formulation of the sample as a hydrogel, "5% SFf" refers to a formulation with 5% (w/v) silk fibroin, and "40% Glycf" refers to a formulation with 40% (w/v) glycerol.

TABLE 2

HYDROGEL FORMULATIONS

Sample	Boil Time (mb)	[Silk] (%)	Excipient	[Excipient] (%)	Sample Name
1	90	5	P188	10	90 mb; hyd; 10% st; 5% SFf; 10% P188f
2	90	5	PEG 4kDa	40	90 mb; hyd; 10% st; 5% SFf; 40% PEG4kf
3	90	5	Glycerol	40	90 mb: hyd; 10% st; 5% SFf; 40% Glycf
4	90	10	P188	10	90 mb; hyd; 20% st; 10% SFf; 10% P188f
5	90	10	PEG 4kDa	40	90 mb; hyd; 20% st; 10% SFf; 40% PEG4kf
6	90	10	Glycerol	40	90 mb; hyd; 20% st; 10% SFf; 40% Glycf
7	480	5	P188	10	480 mb; hyd; 10% st; 5% SFf; 10% P188f
8	480	5	PEG 4kDa	40	480 mb; hyd; 10% st; 5% SFf; 40% PEG4kf
9	480	5	Glycerol	40	480 mb; hyd; 10% st; 5% SFf; 40% Glycf
10	480	10	P188	10	480 mb; hyd; 120% st; 0% SFf; 10% P188f
11	480	10	PEG 4kDa	40	480 mb; hyd; 20% st; 10% SFf; 40% PEG4kf
12	480	10	Glycerol	40	480 mb; hyd; 20% st; 10% SFf; 40% Glycf
13	480	15	P188	10	480 mb; hyd; 30% st; 15% SFf; 10% P188f
14	480	15	PEG 4kDa	40	480 mb; hyd; 30% st; 15% SFf; 40% PEG4kf
15	480	15	Glycerol	40	480 mb; hyd; 30% st; 15% SFf; 40% Glycf

In some embodiments, SBP gels or hydrogels have sufficient internal strength to maintain themselves as a cohesive matrix without the need for mechanical reinforcement. The cohesive property of SBP gels or hydrogels may be tuned according to their intended applications. As a non-limiting example, SBP gels or hydrogels may be capable of tolerating physiological concentrations of ionized salt without breakdown of the gel.

Solutions

In some embodiments, SBPs, SBP formulations are or include solutions. As used herein, the term "solution" refers to a dispersion of molecules or solute in a liquid medium or solvent. In some embodiments, the solute is a processed silk or an SBP. In some embodiments, the solvent is water. In some embodiments, the solvent is an aqueous buffer. Non-limiting examples of buffer include citrate buffer, phosphate buffer, phosphate buffer saline, borate buffer, sodium borate, glycine-HCl, sodium acetate, citrate buffered saline, Tris buffer, HEPES buffer, MOPS buffer, and cacodylate buffer. In some embodiments, SBP formulations may be prepared as a solution, as taught in International Patent Application Publication No. WO2005012606 and Cheng et al. (2015) J. App. Polym. Sci. 132(22): 41959. In some embodiments, an SBP is prepared as a solution using any of the methods described herein. In some embodiments, the solution is prepared by dissolving processed silk in water or buffer. In some embodiments the solution is mixed to facilitate dissolution. In some embodiments, the solution is heated to facilitate dissolution. In some embodiments, solutions are prepared with one or more therapeutic agents. In some embodiments, solutions include one or more excipients. The excipients may be selected from any of those described herein. In some embodiments, excipients may include salts. In some embodiments, the excipients may be any of those described in Table 1. In some embodiments, solutions are prepared with one or more therapeutic agents, biological agents, proteins, small molecules, and/or polymers. In some embodiments, a stock solution of the therapeutic agent may be used to dissolve processed silk in the preparation of a solution. In some embodiments, a stock solution of the therapeutic agent may be mixed with a stock solution of processed silk to prepare the SBP. In some embodiments, a solution may be diluted to obtain additional solutions with processed silk at varying concentrations.

In some embodiments, processed silk solutions include one or more excipients at a concentration of from about 0.0001% (w/v) to about 0.001% (w/v), from about 0.001% (w/v) to about 0.01% (w/v), from about 0.01% (w/v) to about 0.1% (w/v), from about 0.1% (w/v) to about 1% (w/v), from about 0.5% (w/v) to about 5% (w/v), from about 1% (w/v) to about 10% (w/v), from about 5% (w/v) to about 15% (w/v), from about 10% (w/v) to about 30% (w/v), from about 15% (w/v) to about 45% (w/v), from about 20% (w/v) to about 55% (w/v), from about 25% (w/v) to about 65% (w/v), from about 30% (w/v) to about 70% (w/v), from about 35% (w/v) to about 75% (w/v), from about 40% (w/v) to about 80% (w/v), from about 50% (w/v) to about 85% (w/v), from about 60% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 90% (w/v) to about 96% (w/v), from about 92% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 98% (w/v) to about 99.5% (w/v), or from about 99% (w/v) to about 99.9% (w/v).

In some embodiments, processed silk solutions include silk fibroin at a concentration of 0.0001% (w/v) to about 0.001% (w/v), from about 0.001% (w/v) to about 0.01% (w/v), from about 0.01% (w/v) to about 0.1% (w/v), from about 0.1% (w/v) to about 1% (w/v), from about 0.5% (w/v) to about 5% (w/v), from about 1% (w/v) to about 10% (w/v), from about 5% (w/v) to about 15% (w/v), from about 10% (w/v) to about 30% (w/v), from about 15% (w/v) to about 45% (w/v), from about 20% (w/v) to about 55% (w/v), from about 25% (w/v) to about 65% (w/v), from about 30% (w/v) to about 70% (w/v), from about 35% (w/v) to about 75% (w/v), from about 40% (w/v) to about 80% (w/v), from about 50% (w/v) to about 85% (w/v), from about 60% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 90% (w/v) to about 96% (w/v), from about 92% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 98% (w/v) to about 99.5% (w/v), or from about 99% (w/v) to about 99.9% (w/v). Silk fibroin included may be from a silk fibroin preparation with an average silk fibroin molecular weight or range of molecular weights of from about 3.5 kDa to about 10 kDa, from about 5 kDa to about 20 kDa, from about 10 kDa to about 30 kDa, from about 15 kDa to about 40 kDa, from about 20 kDa to about 50 kDa, from about 25 kDa to about 60 kDa, from about 30 kDa to about 70 kDa, from about 35 kDa to about 80 kDa, from about 40 kDa to about 90 kDa, from about 45 kDa to about 100 kDa, from about 50 kDa to about 110 kDa, from about 55 kDa to about 120 kDa, from about 60 kDa to about 130 kDa, from about 65 kDa to about 140 kDa, from about 70 kDa to about 150 kDa, from about 75 kDa to about 160 kDa, from about 80 kDa to about 170 kDa, from about 85 kDa to about 180 kDa, from about 90 kDa to about 190 kDa, from about 95 kDa to about 200 kDa, from about 100 kDa to about 210 kDa, from about 115 kDa to about 220 kDa, from about 125 kDa to about 240 kDa, from about 135 kDa to about 260 kDa, from about 145 kDa to about 280 kDa, from about 155 kDa to about 300 kDa, from about 165 kDa to about 320 kDa, from about 175 kDa to about 340 kDa, from about 185 kDa to about 360 kDa, from about 195 kDa to about 380 kDa, from about 205 kDa to about 400 kDa, from about 215 kDa to about 420 kDa, from about 225 kDa to about 440 kDa, from about 235 kDa to about 460 kDa, or from about 245 kDa to about 500 kDa.

In some embodiments, processed silk solutions include one or more therapeutic agents at a concentration of from about 0.0001% (w/v) to about 0.001% (w/v), from about 0.001% (w/v) to about 0.01% (w/v), from about 0.01% (w/v) to about 0.1% (w/v), from about 0.1% (w/v) to about 1% (w/v), from about 0.5% (w/v) to about 5% (w/v), from about 1% (w/v) to about 10% (w/v), from about 5% (w/v) to about 15% (w/v), from about 10% (w/v) to about 30% (w/v), from about 15% (w/v) to about 45% (w/v), from about 20% (w/v) to about 55% (w/v), from about 25% (w/v) to about 65% (w/v), from about 30% (w/v) to about 70% (w/v), from about 35% (w/v) to about 75% (w/v), from about 40% (w/v) to about 80% (w/v), from about 50% (w/v) to about 85% (w/v), from about 60% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 90% (w/v) to about 96% (w/v), from about 92% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 98% (w/v) to about 99.5% (w/v), or from about 99% (w/v) to about 99.9% (w/v). In some embodiments, the solutions have a concentration between about 0.25% (w/v) to about 2% (w/v) silk fibroin. In some embodiments the silk fibroin has a boiling time of 90 mb, 120 mb, or 480 mb. In some embodiments, the solutions are prepared from silk fibroin lyophilized in phosphate buffer. In some embodiments, the solutions have trace amounts of phosphate salts (e.g. potassium phosphate dibasic and potassium phosphate monobasic). In some embodiments, the solutions are formulated with one or more therapeutic agents. In some embodiments, one or more therapeutic agents may be a biological agent. In some embodiments, the biological agent is a protein.

In some embodiments, the solutions have a concentration between about 0.25% (w/v) to about 2% (w/v) silk fibroin. In some embodiments the silk fibroin has a boiling time of 90 mb, 120 mb, or 480 mb. In some embodiments, the solutions are prepared from silk fibroin lyophilized in phosphate buffer. In some embodiments, the solutions have trace amounts of phosphate salts (e.g. potassium phosphate dibasic and potassium phosphate monobasic). In some embodiments, the solutions are formulated with one or more therapeutic agents. In some embodiments, one or more therapeutic agents may be a biological agent. In some embodiments, the biological agent is a protein.

In some embodiments, silk fibroin may be prepared as a stock solution.

In one embodiment, the silk fibroin stock solution has a concentration of 10% (w/v).

In one embodiment, the silk fibroin stock solution has a concentration of 20% (w/v).

In one embodiment, the silk fibroin stock solution has a concentration of 30% (w/v).

In one embodiment, the silk fibroin stock solution has a concentration of 40% (w/v).

In one embodiment, the silk fibroin stock solution has a concentration of 50% (w/v).

In some embodiments, processed silk formulations may be prepared as solutions. In these solutions, the processed silk may be silk fibroin. Silk fibroin may be degummed with any minute boil (mb) described herein, including, but not limited to, 15, 30, 60, 90, 120 and 480 mb. These solutions may be prepared in water. These solutions may also be prepared with any buffer or excipient described herein, including, but not limited to, phosphate buffer (PB), borate buffer (DED), and propylene glycol (PG). Buffers and excipients may be present at any concentration described herein (e.g. 10 mM PB). Phosphate buffer may be prepared as 10 mM PB. Borate buffer (DED) may be prepared as 6 mg/mL boric acid, 0.45 mg/mL sodium borate, 3.4 mg/mL sodium chloride, 1.4 mg/mL potassium chloride, 0.06 mg/mL magnesium chloride, and 0.06 mg/mL calcium chloride, pH 7.3. In some embodiments, the solutions may comprise 1% PG. In some embodiments, the solution has a pH between 7.0 and 8.0. In some embodiments, the solution has a pH of 7.3. In some embodiments, the solution has a pH of 7.4. In some embodiments, these solutions may comprise silk fibroin at a concentration of from about 0.01% to about 30% (w/v). In some embodiments, these solutions may comprise silk fibroin at a concentration of from about 0.01% to about 0.1%, about 0.1% to about 1%, about 1% to about 2%, about 2% to about 3%, about 3% to about 4%, about 4% to about 5%, about 5% to about 10%, about 10% to about 15%, about 15% to about 20%, or about 20% to about 30% (w/v) silk fibroin.

In some embodiments, the solution comprises 1% (w/v) silk fibroin.

In some embodiments, the solution comprises 2% (w/v) silk fibroin.

In some embodiments, the solution comprises 3% (w/v) silk fibroin.

In some embodiments, the solution comprises 1% (w/v) silk fibroin and 1% (w/v) PG.

In some embodiments, the solutions may be any of those described in Table 3.

TABLE 3

SOLUTION FORMULATIONS			
Sample No.	Description	Silk %	Buffer
82-1	3% Water, Baseline	3	Water
82-2	3% PB, Baseline	3	10 mMPB
82-3	3% DED Baseline	3	DED
82-4	1% PBS, Baseline	1	PBS
82-5	1% DED, Baseline	1	DED
82-6	1% PBS + PG, Baseline	1	PBS + 1% PG
82-7	1% DED + PG, Baseline	1	DED + 1% PG
83-1	0.5% SF in 10 mM phosphate, pH 7.4	5	10 mMPB
83-2	1% SF in 10 mM phosphate, pH 7.4	10	10 mMPB
83-3	2.5% SF in 10 mM phosphate, pH 7.4	25	10 mM PB
83-4	5% SF in 10 mM phosphate, pH 7.4	50	10 mMPB

TABLE 3-continued

SOLUTION FORMULATIONS			
Sample No.	Description	Silk %	Buffer
83-5	5% SF, 10 mM sucrose in 10 mM phosphate, pH 7.4	50	10 mM PB + 10 mM sucrose
83-6	5% SF, 50 mM sucrose in 10 mM phosphate, pH 7.4	50	10 mMPB + 50 mM sucrose
83-7	5% SF, 100 mM sucrose in 10 mM phosphate, pH 7.4	50	10 mMPB + 100 mM sucrose
83-8	5% SF, 150 mM sucrose in 10 mM phosphate, pH 7.4	50	10 mMPB + 150 mM sucrose
83-9	5% SF, 10 mM trehalose in 10 mM phosphate, pH 7.4	50	10 mMPB + 10 mM trehalose
83-10	5% SF, 50 mM trehalose in 10 mM phosphate, pH 7.4	50	10 mM PB + 50 mM trehalose
83-11	5% SF, 100 mM trehalose in 10 mM phosphate, pH 7.4	50	10 mMPB + 100 mM trehalose
83-12	5% SF, 150 mM trehalose in 10 mM phosphate, pH 7.4	50	10 mMPB + 150 mM trehalose

Particles

In some embodiments, SBP formulation include SBP particles. As used herein, the term "particle" refers to a minute portion of a substance. SBP particles may include particles of processed silk. Processed silk particles may include silk fibroin particles. Silk fibroin particles may be tiny clusters of silk fibroin or they may be arranged as more ordered structures. Particles may vary in size. Processed silk particles may be visible or may be too tiny to view easily with the naked eye. Particles with a width of from about 0.1 μm to about 100 μm are referred to herein as "microparticles." Particles with a width of about 100 nm or less are referred to herein as "nanoparticles." Microparticles and nanoparticles that are spherical in shape are termed microspheres and nanospheres, respectively. Processed silk particle preparations may include particles with uniform width or with ranges of widths. In some embodiments, processed silk particle preparations include average particle widths of or ranges of particle widths of from about 10 nm to about 25 nm, from about 20 nm to about 50 nm, from about 30 nm to about 75 nm, from about 40 nm to about 80 nm, from about 50 nm to about 100 nm, from about 0.05 μm to about 10 μm, from about 1 μm to about 20 μm, from about 2 μm to about 30 μm, from about 5 μm to about 40 μm, from about 10 μm to about 50 μm, from about 20 μm to about 60 μm, from about 30 μm to about 70 μm, from about 40 μm to about 80 μm, from about 50 μm to about 90 μm, from about 0.05 mm to about 2 mm, from about 0.1 mm to about 3 mm, from about 0.2 mm to about 4 mm, from about 0.5 mm to about 5 mm, from about 1 mm to about 6 mm, from about 2 mm to about 7 mm, from about 5 mm to about 10 mm, from about 10 nm to about 100 μm, from about 10 μm to about 10 mm, from about 50 nm to about 500 μm, from about 50 μm to about 5 mm, from about 100 nm to about 10 mm, or from about 1 μm to about 10 mm. In some embodiments, processed silk particle preparations include average particle widths of at least 10 nm, at least 100 nm, at least 0.5 μm, at least 1 μm, at least 10 μm, at least 100 μm, at least 500 μm, at least 1 mm, or at least 10 mm.

Processed silk particles may be formed through spraying of a processed silk preparation. In some embodiments, electrospraying is used. Electrospraying may be carried out using a coaxial electrospray apparatus (e.g., see Cao et al. (2017) Scientific Reports 7:11913, the contents of which are herein incorporated by reference in their entirety). In some embodiments, silk fibroin microspheres or nanospheres may be obtained by electrospraying a silk fibroin preparation into a collector and flash freezing the sprayed particles (e.g., see United States Publication No. US2017/0333351, the contents of which are herein incorporated by reference in their entirety). The flash frozen silk fibroin particles may then be lyophilized. In some embodiments, processed silk particles may be prepared using centrifugal washing, followed by lyophilization, as taught in United States Publication No. US2017/0340575, the contents of which are herein incorporated by reference in their entirety. In some embodiments, processed silk microspheres may be formed through the use of a microfluidic device (e.g., see Sun et al. (2017) Journal of Materials Chemistry B 5:8770-8779, the contents of which are herein incorporated by reference in their entirety). In some embodiments, microspheres are formed via coagulation in a methanol bath, as taught in European Patent No. EP3242967, the contents of which are herein incorporated by reference in their entirety.

Devices

In some embodiments, SBP formulations may be included as or in device components. As used herein, the term "device" refers to any article constructed or modified to suit a particular purpose. Devices may be designed for a variety of purposes, including, but not limited to, therapeutic applications, material science applications, and agricultural applications. In some embodiments, SBPs are embedded or incorporated into devices. Some devices include SBPs as coatings or lubricants. In some embodiments, devices include implants, patches, mesh, sponges, grafts, insulators, pipes, prosthetics, resistors, bedding, blankets, liners, ropes, plugs, fillers, electronic devices, mechanical devices, medical devices, surgical devices, veterinary devices, and agricultural devices. Additional devices are described herein.

In some embodiments, SBPs may be or may be included in therapeutic devices. In some embodiments, therapeutic devices may be coated with SBPs described herein. Some therapeutic devices may include therapeutic agents. In some embodiments, the use of SBPs within therapeutic devices may enable the delivery of therapeutic agents via such therapeutic devices. Some therapeutic devices may include synthetic materials. In some embodiments, therapeutic devices include, but are not limited to, artificial blood vessels, artificial organ, bandage, cartilage replacement, filler, hemostatic sponge, implant, silk contact lens, contact lens solution, stem cell, surgical mesh, surgical suture, tissue replacement, vascular patch, wound dressing, antenna, applier, assembly, balloon, barrier, biosensor, biotransducer, cable assembly, caliper, capacitor, carrier, clamp, connector, corneal implant, coronary stent, cryotome, degradable device, delivery device, dermatome, detector, diagnostic device, dilator, diode, discharge device, display technology, distractor, drill bit, electronic device, graft, grasper, harmonic scalpel, hemostatic device, imaging apparatus, implant, implant for continuous drug delivery, integrated circuit, intraocular lens, lancet, LIGASURE™, liner, magnetic or inductive device, magnetic resonance imaging apparatus, mechanical assembly, medical device, memristor, module, needle, nerve stimulator, network, neurostimulator, occluder, optoelectronic device, pacemaker, patch, pen, piezoelectric device, pin, pipe, plate, positioner, power source, probe, prosthesis, prosthetic, protection device, removable device, resistor, retractor, rod, rongeur, rope, ruler, scalpel, scope, screw, semiconductor, sensor, solution, specula, stent, stent, sterotactic device, suction tip, suction tube, surgical device, surgical mesh, surgical scissor, surgical staple, suture, switch, temperature sensor, terminal, tie, tip, transducer, transistor, tube, ultrasound tissue disruptor, vacuum tube, vacuum valve, ventilation system, water balloon, wire, bleb, gel, gel that hardens after implantation, implant, lacrimal plug, lens, plug, punctal plug, rod, slurry, slurry that hardens after implantation, and solids.

Solid Implants

In some embodiments, SBP formulations may be included as or in solid implants. SBP solid implants may be prepared by gelation of processed silk followed by drying and/or injection molding. Methods of drying may include any of those described herein, such as heat, air drying or lyophilization. The drying process may induce beta-sheet formation. Density of the SBP formulation is properly controlled during this process. SBP formulations may also be formatted as a film that is applied to the exterior of a solid implant. Alternatively, SBP formulations may be included into solid implants by spray-drying, spray-coating, and/or milling.

Solid implants prepared with SBP formulations may be of any size and shape. As a non-limiting example, solid implants may be shaped as forceps, speculum, bands, stoppers, screws, tubes, rods, cones, cylinders, teardrops, etc. Solid implants prepared with SBP formulations may be prepared via aqueous processing. The pH of solid implants prepared with SBP formulations may be controlled during preparation and degradation. Solid implants prepared with SBP formulations may comprise a hydrophobic matrix. Solid implants prepared with SBP formulations prepared with high loading of active pharmaceutical ingredient (e.g. at least 50% (w/w)). Delivery of the active pharmaceutical ingredient may be sustained (e.g. for weeks to months). Solid implants prepared with SBP formulations may be biocompatible and/or biodegradable.

Controlled Release

In some embodiments, SBP formulations and related methods described herein be may be used for controlled release of therapeutic agents. As used herein, the term "controlled release" refers to regulated movement of factors from specific locations to surrounding areas. In some embodiments, the specific location is a depot. Controlled release of factors from depots may be regulated by interactions between therapeutic agents and depot components. Such interactions may, for example, modulate therapeutic agent diffusion rate and/or affect therapeutic agent stability and/or degradation. In some embodiments, the depot is an SBP formulation. In some embodiments, factors subject to controlled release from depots are SBP formulations. In some embodiments, therapeutic agents are subject to controlled release from SBP depots.

In some embodiments, SBP formulations may control payload release by extending payload half-life. As used herein, the term "half-life" refers to the length of time necessary for levels of a factor to be reduced (e.g., through clearance or degradation) by 50%. In some embodiments, SBP depots may be used for therapeutic agents, wherein release is facilitated by diffusion. In some embodiments, SBP formulations may be lyophilized together with therapeutic agents. In some embodiments, combined lyophilization may induce further interactions between therapeutic agents and SBP formulations. These interactions may be maintained through SBP preparation and support extended payload release. Payload release may be dependent on SBP degradation and/or dissolution. In some embodiments, SBP β-sheet content is increased (e.g., via water annealing), thereby increasing SBP insolubility in water. Such SBPs may exhibit increased payload release periods. In some embodiments, these SBP formulations may include therapeutic agent stabilizing properties to extend administration periods and/or therapeutic agent half-life.

In some embodiments, methods of increasing payload half-life using SBPs may include any of those described in United States Patent Publication US20100028451, the contents of which are herein incorporated by reference in their entirety. Methods of improving payload half-life may be carried out in vitro or in vivo. In some embodiments, SBP-based methods of improving payload half-life may enable therapeutic indication treatment with fewer doses and/or treatments. Such methods may include any of those described in International Patent Application Publication No. WO2017139684, the contents of which are herein incorporated by reference in their entirety. In some embodiments, payload half-life may be extended by from about 0.01% to about 1%, from about 0.05% to about 2%, from about 1% to about 5%, from about 2% to about 10%, from about 3% to about 15%, from about 4% to about 20%, from about 5% to about 25%, from about 6% to about 30%, from about 7% to about 35%, from about 8% to about 40%, from about 9% to about 45%, from about 10% to about 50%, from about 12% to about 55%, from about 14% to about 60%, from about 16% to about 65%, from about 18% to about 70%, from about 20% to about 75%, from about 22% to about 80%, from about 24% to about 85%, from about 26% to about 90%, from about 28% to about 95%, from about 30% to about 100%, from about 32% to about 105%, from about 34% to about 110%, from about 36% to about 115%, from about 38% to about 120%, from about 40% to about 125%, from about 42% to about 130%, from about 44% to about 135%, from about 46% to about 140%, from about 48% to about 145%, from about 50% to about 150%, from about 60% to about 175%, from about 70% to about 200%, from about 80% to about 225%, from about 90% to about 250%, from about 100% to about 275%, from about 110% to about 300%, from about 120% to about 325%, from about 130% to about 350%, from about 140% to about 375%, from about 150% to about 400%, from about 170% to about 450%, from about 190% to about 500%, from about 210% to about 550%, from about 230% to about 600%, from about 250% to about 650%, from about 270% to about 700%, from about 290% to about 750%, from about 310% to about 800%, from about 330% to about 850%, from about 350% to about 900%, from about 370% to about 950%, from about 390% to about 1000%, from about 410% to about 1050%, from about 430% to about 1100%, from about 450% to about 1500%, from about 480% to about 2000%, from about 510% to about 2500%, from about 540% to about 3000%, from about 570% to about 3500%, from about 600% to about 4000%, from about 630% to about 4500%, from about 660% to about 5000%, from about 690% to about 5500%, from about 720% to about 6000%, from about 750% to about 6500%, from about 780% to about 7000%, from about 810% to about 7500%, from about 840% to about 8000%, from about 870% to about 8500%, from about 900% to about 9000%, from about 930% to about 9500%, from about 960% to about 10000%, or more than 10000%.

Ocular SBPS

SBPs described herein may include ocular SBPs. As used herein, the term "ocular SBP" refers to an SBP used in any application related to the eye. Ocular SBPs may be used in therapeutic applications. Such therapeutic applications may include treating or otherwise addressing one or more ocular indications.

In further embodiments, ocular SBPs may be prepared as eye drops for the treatment of dry eye disease, as described in U.S. Pat. No. 9,394,355, the contents of which are hereby incorporated by reference in their entirety, or formulated for the treatment of corneal injury, as described in International Patent Application Publication Nos. WO2017200659 and WO2018031973; Abdel-Naby et al. (2017) Invest Ophthalmol Vis Sci; 58(3):1425-1433; and Abdel-Naby et al. (2017) PLoS One; 12(11):e0188154, the contents of each of which are hereby incorporated by reference in their entirety.

Ocular SBPs may be prepared in a variety of formats. Some ocular SBPs are in the form of a hydrogel. These hydrogels may vary in viscosity and appear runny. Other ocular SBPs may be in the form of a solution. Some ocular SBPs may be devices. These devices may include medical devices. In some embodiments, solutions may include silk fibroin micelles, as described in Wongpanit et al. (2007) Macromolecular Bioscience 7: 1258-1271, the contents of which are herein incorporated by reference in their entirety. Silk fibroin micelles may be of any size (e.g. between 100 to 200 nm). In some embodiments, ocular SBPs may act as demulcents. The ocular SBPs described herein may relieve irritation or inflammation of the mucous membranes by forming a protective film. In some embodiments, the ocular SBPs of the present disclosure may contain one or more demulcents (e.g. propylene glycol, gelatin, glycerin, carboxymethylcellulose, Dextran 70, methylcellulose, PEG 300, PEG 400, hydroxyethyl cellulose, hydroxypropyl methylcellulose, povidone, polyvinyl alcohol, and polysorbate 80). This film may mimic a mucous membrane. In some embodiments, ocular SBPs may act as a surfactant.

Ocular SBPs may include ocular therapeutic agents. Ocular therapeutic agents may be delivered to subject eyes by release from SBPs while SBPs are in contact with the eyes. Release of ocular therapeutic agents from SBPs may be modulated by one or more of silk fibroin concentration, silk fibroin molecular weight, SBP volume, method used to dry SBPs, ocular therapeutic agent molecular weight, and inclusion of at least one excipient. The ocular therapeutic agents may include any of those described herein. In some embodiments, ocular therapeutic agents include one or more of processed silk, biological agents, small molecules, analgesics, proteins, anti-inflammatory agents, steroids, opiates, sodium channel blockers (e.g. lidocaine and proparacaine), ciclosporin, corticosteroids, tetracyclines, fatty acids, NSAIDs, and VEGF-related agents. Ocular therapeutic agent proteins may include, but are not limited to, lysozyme, bovine serum albumin (BSA), bevacizumab, or VEGF-related agents. NSAIDs may include, but are not limited to, aspirin, carprofen, celecoxib, deracoxib, diclofenac, diflunisal, etodolac, fenoprofen, firocoxib, flurbirofen, ibuprofen, indomethacin, ketoprofen, ketorolac, mefenamic acid, meloxicam, nabumetone, naproxen, oxaprozin, piroxicam, robenacoxib, salsalate, sulindac, and tolmetin. In some embodiments, the SBPs stabilize ocular therapeutic agents included. Ocular SBPs may include ocular therapeutic agent concentrations [expressed as percentage of ocular therapeutic agent weight contributing to total SBP volume] of from about 0.0001% (w/v) to about 98% (w/v). For example, SBPs may include ocular therapeutic agents at a concentration of from about 0.0001% (w/v) to about 0.001% (w/v), from about 0.001% (w/v) to about 0.01% (w/v), from about 0.01% (w/v) to about 1% (w/v), from about 0.05% (w/v) to about 2% (w/v), from about 1% (w/v) to about 5% (w/v), from about 2% (w/v) to about 10% (w/v), from about 4% (w/v) to about 16% (w/v), from about 5% (w/v) to about 20% (w/v), from about 5% (w/v) to about 85% (w/v), from about 8% (w/v) to about 24% (w/v), from about 10% (w/v) to about 30% (w/v), from about 12% (w/v) to about 32% (w/v), from about 14% (w/v) to about 34% (w/v), from about 15% (w/v) to about 95% (w/v), from about 16% (w/v) to about 36% (w/v), from about 18% (w/v) to about 38% (w/v), from about 20% (w/v) to about 40% (w/v), from about 22% (w/v) to about 42% (w/v), from about 24% (w/v) to about 44% (w/v), from about 26% (w/v) to about 46% (w/v), from about 28% (w/v) to about 48% (w/v), from about 30% (w/v) to about 50% (w/v), from about 35% (w/v) to about 55% (w/v), from about 40% (w/v) to about 60% (w/v), from about 45% (w/v) to about 65% (w/v), from about 50% (w/v) to about 70% (w/v), from about 55% (w/v) to about 75% (w/v), from about 60% (w/v) to about 80% (w/v), from about 65% (w/v) to about 85% (w/v), from about 70% (w/v) to about 90% (w/v), from about 75% (w/v) to about 95% (w/v), from about 80% (w/v) to about 96% (w/v), from about 85% (w/v) to about 97% (w/v), from about 90% (w/v) to about 98% (w/v), from about 95% (w/v) to about 99% (w/v), from about 96% (w/v) to about 99.2% (w/v), or from about 97% (w/v) to about 98% (w/v).

Ocular SBPs may have a pH from about 3 to about 10. In some embodiments, the pH is from about 3 to about 6, from about 6 to about 8, or from about 8 to about 10. In some embodiments, the pH of the SBP is about 7.4.

In some embodiments, the ocular SBP is a solution. In some embodiments the ocular SBP is a hydrogel. In some embodiments, the SBP comprises from about 0.0001% to about 35% (w/v) of silk fibroin. In some embodiments the silk fibroin may be included at a concentration (w/w or w/v) of from about 0.0001% to about 0.001%, from about 0.001% to about 0.01%, from about 0.01% to about 1%, from about 0.05% to about 2%, from about 0.1% to about 30%, from about 1% to about 5%, from about 2% to about 10%, from about 3% to about 15%, from about 4% to about 20%, from about 5% to about 25%, from about 6% to about 30%, from about 7% to about 35%, from about 8% to about 40%, from about 9% to about 45%, from about 10% to about 50%, from about 12% to about 55%, from about 14% to about 60%, from about 16% to about 65%, from about 18% to about 70%, from about 20% to about 75%, from about 22% to about 80%, from about 24% to about 85%, from about 26% to about 90%, from about 28% to about 95%, from about 30% to about 96%, from about 32% to about 97%, from about 34% to about 98%, from about 36% to about 98.5%, from about 38% to about 99%, from about 40% to about 99.5%, from about 42% to about 99.6%, from about 44% to about 99.7%, from about 46% to about 99.8%, or from about 50% to about 99.9%.

Ocular SBPs may include one or more excipients. The excipients may include any of those described herein. In some embodiments, the excipients include one or more of sucrose, lactose, phosphate salts, sodium chloride, potassium phosphate monobasic, potassium phosphate dibasic, sodium phosphate dibasic, sodium phosphate monobasic, polysorbate 80, phosphate buffer, phosphate buffered saline, sodium hydroxide, sorbitol, mannitol, lactose USP, Starch 1500, microcrystalline cellulose, potassium chloride, sodium borate, boric acid, sodium borate decahydrate, magnesium chloride hexahydrate, calcium chloride dihydrate, sodium hydroxide, Avicel, dibasic calcium phosphate dehydrate, tartaric acid, citric acid, fumaric acid, succinic acid, malic acid, hydrochloric acid, polyvinylpyrrolidone, copolymers of vinylpyrrolidone and vinylacetate, hydroxypropylcellulose, hydroxyethylcellulose, hydroxypropylmethylcellulose, polyvinyl alcohol, polyethylene glycol, acacia, and sodium carboxymethylcellulose. In some embodiments, excipients may include phosphate buffered saline. In some embodiments, excipients may include phosphate buffer. In some embodiments, excipients may include sucrose. In some embodiments, excipients may include boric acid, sodium borate decahydrate, sodium chloride, potassium chloride, magnesium chloride hexahydrate, calcium chloride dihydrate, sodium hydroxide, and hydrochloric acid. SBPs may include at least one excipient at a concentration of from about 0.0001% to about 50% (w/w or w/v). In some embodiments, SBPs include at least one excipient at a concentration of from about 0.01% to about 1%, from about 0.05% to about 2%, from about 1% to about 5%, from about 2% to about 10%, from about 3% to about 15%, from about 4% to about 20%, from about 5% to about 25%, from about 6% to about 30%, from about 7% to about 35%, from about 8% to about 40%, from about 9% to about 45%, from about 10% to about 50%, from about 12% to about 55%, from about 14% to about 60%, from about 16% to about 65%, from about 18% to about 70%, from about 20% to about 75%, from about 22% to about 80%, from about 24% to about 85%, from about 26% to about 90%, from about 28% to about 95%, from about 30% to about 96%, from about 32% to about 97%, from about 34% to about 98%, from about 36% to about 98.5%, from about 38% to about 99%, from about 40% to about 99.5%, from about 42% to about 99.6%, from about 44% to about 99.7%, from about 46% to about 99.8%, or from about 50% to about 99.9%.

In some embodiments, less than 1% of silk fibroin in an ocular SBP aggregates In some embodiments, less than 0.1% of silk fibroin in an ocular SBP aggregates.

Ocular SBPs may be hydrogels. Such SBPs may include at least one excipient selected from one or more of sorbitol, triethylamine, 2-pyrrolidone, alpha-cyclodextrin, benzyl alcohol, beta-cyclodextrin, dimethyl sulfoxide, dimethylacetamide (DMA), dimethylformamide, ethanol, gamma-cyclodextrin, glycerol, glycerol formal, hydroxypropyl beta-cyclodextrin, kolliphor 124, kolliphor 181, kolliphor 188, kolliphor 407, kolliphor EL (cremophor EL), cremophor RH 40, cremophor RH 60, dalpha-tocopherol, PEG 1000 succinate, polysorbate 20, polysorbate 80, solutol HS 15, sorbitan monooleate, poloxamer-407, poloxamer-188, Labrafil M-1944CS, Labrafil M-2125CS, Labrasol, Gellucire 44/14, Softigen 767, mono- and di-fatty acid esters of PEG 300, PEG 400, or PEG 1750, kolliphor RH60, N-methyl-2-pyrrolidone, castor oil, corn oil, cottonseed oil, olive oil, peanut oil, peppermint oil, safflower oil, sesame oil, soybean oil, hydrogenated vegetable oils, hydrogenated soybean oil, medium chain triglycerides of coconut oil, medium chain triglycerides of palm seed oil, beeswax, d-alpha-tocopherol, oleic acid, medium-chain mono-glycerides, medium-chain di-glycerides, alpha-cyclodextrin, betacyclodextrin, hydroxypropyl-beta-cyclodextrin, sulfo-butylether-beta-cyclodextrin, hydrogenated soy phosphatidylcholine, distearoylphosphatidylglycerol, L-alphadimyristoylphosphatidylcholine, L-alpha-dimyristoylphosphatidylglycerol, PEG 300, PEG 300 caprylic/capric glycerides (Softigen 767), PEG 300 linoleic glycerides (Labrafil M-2125CS), PEG 300 oleic glycerides (Labrafil M-1944CS), PEG 400, PEG 400 caprylic/capric glycerides (Labrasol), polyoxyl 40 stearate (PEG 1750 monosterate), polyoxyl 8 stearate (PEG 400 monosterate), polysorbate 20, polysorbate 80, polyvinyl pyrrolidone, propylene carbonate, propylene glycol, solutol HS15, sorbitan monooleate (Span 20), sulfobutylether-beta-cyclodextrin, transcutol, triacetin, 1-dodecylazacyclo-heptan-2-one, caprolactam, castor oil, cottonseed oil, ethyl acetate, medium chain triglycerides, methyl acetate, oleic acid, safflower oil, sesame oil, soybean oil, tetrahydrofuran, glycerin, and PEG 4 kDa. The SBPs may have an osmolarity of from about 1 mOsm to about 10 mOsm, from about 2 mOsm to about 20 mOsm, from about 3 mOsm to about 30 mOsm, from about 4 mOsm to about 40 mOsm, from about 5 mOsm to about 50 mOsm, from about 6 mOsm to about

60 mOsm, from about 7 mOsm to about 70 mOsm, from about 8 mOsm to about 80 mOsm, from about 9 mOsm to about 90 mOsm, from about 10 mOsm to about 100 mOsm, from about 15 mOsm to about 150 mOsm, from about 25 mOsm to about 200 mOsm, from about 35 mOsm to about 250 mOsm, from about 45 mOsm to about 300 mOsm, from about 55 mOsm to about 350 mOsm, from about 65 mOsm to about 400 mOsm, from about 75 mOsm to about 450 mOsm, from about 85 mOsm to about 500 mOsm, from about 125 mOsm to about 600 mOsm, from about 175 mOsm to about 700 mOsm, from about 225 mOsm to about 800 mOsm, from about 275 mOsm to about 285 mOsm, from about 280 mOsm to about 900 mOsm, or from about 325 mOsm to about 1000 mOsm.

In some embodiments, the viscosity and/or complex viscosity of SBPs is tunable between 1-1000 centipoise (cP). In some embodiments, the viscosity of an SBP is tunable from about 0.0001 to about 1000 Pascal seconds (Pa*s). In some embodiments, the viscosity of an SBP is from about 1 cP to about 10 cP, from about 2 cP to about 20 cP, from about 3 cP to about 30 cP, from about 4 cP to about 40 cP, from about 5 cP to about 50 cP, from about 6 cP to about 60 cP, from about 7 cP to about 70 cP, from about 8 cP to about 80 cP, from about 9 cP to about 90 cP, from about 10 cP to about 100 cP, from about 100 cP to about 150 cP, from about 150 cP to about 200 cP, from about 200 cP to about 250 cP, from about 250 cP to about 300 cP, from about 300 cP to about 350 cP, from about 350 cP to about 400 cP, from about 400 cP to about 450 cP, from about 450 cP to about 500 cP, from about 500 cP to about 600 cP, from about 550 cP to about 700 cP, from about 600 cP to about 800 cP, from about 650 cP to about 900 cP, from about 700 cP to about 1000 cP, from about 1000 cP to about 5000 cP, from about 5000 cP to about 10000 cP, from about 10000 cP to about 20000 cP, from about 20000 cP to about 30000 cP, from about 30000 cP to about 40000 cP, from about 40000 cP to about 50000 cP, from about 50000 cP to about 60000 cP, from about 60000 cP to about 80000 cP, from about 80000 cP to about 90000 cP, or from about 90000 cP to about 100000 cP. In some embodiments, the viscosity of an SBP is from about 0.0001 Pa*s to about 0.001 Pa*s, from about 0.001 Pa*s to about 0.01 Pa*s, from about 0.01 Pa*s to about 0.1 Pa*s, from about 0.1 Pa*s to about 1 Pa*s, from about 1 Pa*s to about 10 Pa*s, from about 2 Pa*s to about 20 Pa*s, from about 3 Pa*s to about 30 Pa*s, from about 4 Pa*s to about 40 Pa*s, from about 5 Pa*s to about 50 Pa*s, from about 6 Pa*s to about 60 Pa*s, from about 7 Pa*s to about 70 Pa*s, from about 8 Pa*s to about 80 Pa*s, from about 9 Pa*s to about 90 Pa*s, from about 10 Pa*s to about 100 Pa*s, from about 100 Pa*s to about 150 Pa*s, from about 150 Pa*s to about 200 Pa*s, from about 200 Pa*s to about 250 Pa*s, from about 250 Pa*s to about 300 Pa*s, from about 300 Pa*s to about 350 Pa*s, from about 350 Pa*s to about 400 Pa*s, from about 400 Pa*s to about 450 Pa*s, from about 450 Pa*s to about 500 Pa*s, from about 500 Pa*s to about 600 Pa*s, from about 550 Pa*s to about 700 Pa*s, from about 600 Pa*s to about 800 Pa*s, from about 650 Pa*s to about 900 Pa*s, from about 700 Pa*s to about 1000 Pa*s, from about 1000 Pa*s to about 2500 Pa*s, from about 2500 Pa*s to about 5000 Pa*s, from about 5000 Pa*s to about 7500 Pa*s, from about 7500 Pa*s to about 10000 Pa*s, from about 10000 Pa*s to about 20000 Pa*s, from about 20000 Pa*s to about 30000 Pa*s, from about 30000 Pa*s to about 40000 Pa*s, or from about 40000 Pa*s to about 50000 Pa*s. In some embodiments, the SBP formulations may shear thin or display shear thinning properties. As used herein, the term "shear thinning" refers to a decrease in viscosity at increasing shear rates. As used herein, the term "shear rate" refers to the rate of change in the ratio of displacement of material upon the application of a shear force to the height of the material. This ratio is also known as strain.

In some embodiments, the storage modulus and/or the loss modulus (G' and G" respectively) of SBPs is tunable between 0.0001-20000 Pascals (Pa). In some embodiments, the storage modulus and/or the loss modulus of SBPs is from about 0.0001 Pa to about 0.001 Pa, from about 0.001 Pa to about 0.01 Pa, from about 0.01 Pa to about 0.1 Pa, from about 0.1 Pa to about 1 Pa, from about 1 Pa to about 10 Pa, from about 2 Pa to about 20 Pa, from about 3 Pa to about 30 Pa, from about 4 Pa to about 40 Pa, from about 5 Pa to about 50 Pa, from about 6 Pa to about 60 Pa, from about 7 Pa to about 70 Pa, from about 8 Pa to about 80 Pa, from about 9 Pa to about 90 Pa, from about 10 Pa to about 100 Pa, from about 100 Pa to about 150 Pa, from about 150 Pa to about 200 Pa, from about 200 Pa to about 250 Pa, from about 250 Pa to about 300 Pa, from about 300 Pa to about 350 Pa, from about 350 Pa to about 400 Pa, from about 400 Pa to about 450 Pa, from about 450 Pa to about 500 Pa, from about 500 Pa to about 600 Pa, from about 550 Pa to about 700 Pa, from about 600 Pa to about 800 Pa, from about 650 Pa to about 900 Pa, from about 700 Pa to about 1000 Pa, from about 1000 Pa to about 1500 Pa, from about 1500 Pa to about 2000 Pa, from about 2000 Pa to about 2500 Pa, from about 2500 Pa to about 3000 Pa, from about 3000 Pa to about 3500 Pa, from about 3500 Pa to about 4000 Pa, from about 4000 Pa to about 4500 Pa, from about 4500 Pa to about 5000 Pa, from about 5000 Pa to about 5500 Pa, from about 5500 Pa to about 6000 Pa, from about 6000 Pa to about 6500 Pa, from about 6500 Pa to about 7000 Pa, from about 7000 Pa to about 7500 Pa, from about 7500 Pa to about 8000 Pa, from about 8000 Pa to about 8500 Pa, from about 8500 Pa to about 9000 Pa, from about 9000 Pa to about 9500 Pa, from about 9500 Pa to about 10000 Pa, from about 10000 Pa to about 11000 Pa, from about 11000 Pa to about 12000 Pa, from about 12000 Pa to about 13000 Pa, from about 13000 Pa to about 14000 Pa, from about 14000 Pa to about 15000 Pa, from about 15000 Pa to about 16000 Pa, from about 16000 Pa to about 17000 Pa, from about 17000 Pa to about 18000 Pa, from about 18000 Pa to about 19000 Pa, or from about 19000 Pa to about 20000 Pa.

In some embodiments, the phase angle of SBPs is tunable between 1°-90°). In some embodiments, the phase angle of SBPs is from about 1° to about 2°, from about 2° to about 3°, from about 3° to about 4°, from about 4° to about 5°, from about 5° to about 6°, from about 6° to about 7°, from about 7° to about 8°, from about 8° to about 9°, from about 9° to about 10°, from about 10° to about 15°, from about 15° to about 20°, from about 20° to about 25°, from about 25° to about 30°, from about 30° to about 35°, from about 35° to about 40°, from about 40° to about 45°, from about 45° to about 50°, from about 50° to about 55°, from about 55° to about 60°, from about 60° to about 65°, from about 65° to about 70°, from about 70° to about 75°, from about 75° to about 80°, from about 80° to about 85°, or from about 85° to about 90°.

In some embodiments, ocular SBPs may demonstrate the effects of interfacial viscosity. In some embodiments, the processed silk of an ocular SBP may migrate to the air-water boundary. In some embodiments this migration may result in an increase in the local concentration at this interface and ultimately generate the effects of interfacial viscosity. The effects of interfacial viscosity may be independent of the concentration of processed silk. In some embodiments, the effects of interfacial viscosity may be mitigated through the incorporation of a surfactant. The surfactant may be any surfactant described herein. In some embodiments, the surfactant is polysorbate 80. In some embodiments, the SBPs are topical ocular drops. In some embodiments these topical ocular drops show the effects of interfacial viscosity, as the thin tear film allows for a large air-water interface. In some embodiments, the processed silk in the ocular SBP and/or topical ocular drops may migrate to the air-water interface. In some embodiments, the ocular SBP and/or topical ocular drops create a viscous film-like layer at the tear film surface. In some embodiments, the effects interfacial viscosity in an ocular SBP and/or topical ocular drops may prevent evaporation of tears. In some embodiments, the effects interfacial viscosity in an ocular SBP and/or topical ocular drops may extend silk residence dwell time on the surface of the eye. In some embodiments, the effects of interfacial viscosity of an ocular SBP and/or topical ocular drops may lead to a reduction in the frequency of application. In some embodiments, ocular SBPs and/or topical ocular drops may have viscosities similar to that of saline (1 cP). In some embodiments, ocular SBPs and/or topical ocular drops may have viscosities similar to that of saline at high shear rates (e.g. greater than or equal to 500 1/s). These shear rates may be similar or identical to those produced during blinking. In some embodiments, the ocular SBPs and/or topical ocular drops may not elicit any unwanted effects with comfort due to the effects of interfacial viscosity.

In some embodiments, ocular SBPs may comprise a solution with a demulcent. In some embodiments, the demulcent is propylene glycol. In some embodiments, ocular SBPs may comprise a solution with 0.001%-10% propylene glycol. In some embodiments, ocular SBPs may comprise a solution with 1% propylene glycol.

In some embodiments, ocular SBPs may comprise a solution in borate (DED) buffer.

In some embodiments, ocular SBPs may comprise a solution in borate (DED) buffer with 1% 480 mb silk fibroin at pH 7.5 with an osmolarity of 150 mOsm/L. These ocular SBPs may be prepared with silk fibroin lyophilized in 50 mM sucrose.

In some embodiments, ocular SBPs may comprise a solution in borate (DED) buffer with 1% 120 mb silk fibroin at pH 7.5 with an osmolarity of 150 mOsm/L. These ocular SBPs may be prepared with silk fibroin lyophilized in 50 mM sucrose.

In some embodiments, ocular SBPs may comprise a solution in borate (DED) buffer with 1% 480 mb silk fibroin at pH 7.5 with an osmolarity of 150 mOsm/L and 1% propylene glycol. These ocular SBPs may be prepared with silk fibroin lyophilized in 50 mM sucrose.

In some embodiments, the ocular SBP may be formulated for topical administration. In some embodiments, ocular SBPs may be formulated for ocular administration. In some embodiments, ocular SBPs are formulated for intraocular administration. In some embodiments, ocular SBPs are formulated for one or more of intravitreal administration, intraretinal administration, intracorneal administration, intrascleral administration, lacrimal administration, punctal administration, administration to the anterior sub-Tenon's, suprachoroidal administration, administration to the posterior sub-Tenon's, subretinal administration, administration to the fornix, administration to the lens, administration to the anterior segment, administration to the posterior segment, macular administration, and intra-aqueous humor administration. Ocular SBPs may be biocompatible, well tolerated, and/or non-immunogenic. Ocular SBPs may be administered as eye drops. Ocular SBPs may be administered as sprays. Ocular SBPs may be biocompatible. Ocular SBPs may be tolerable.

In some embodiments, ocular SBPs may be used as artificial tears. In some embodiments, ocular SBPs may be used in the management of glaucoma. In some aspects, the ocular SBPs useful for the management of glaucoma may be in the format of drops. Eye drops used in managing glaucoma help the eye's fluid to drain better and decrease the amount of fluid made by the eye which decreases eye pressure. Ocular SBPs formatted as eye drops may include prostaglandin analogs, beta blockers, alpha agonists, and carbonic anhydrase inhibitors.

In some embodiments, SBPs formatted as drops may be used to treat ocular allergies. Drops may contain histamine antagonists or nonsteroidal anti-inflammatory drug (NSAIDs), which suppress the optical mast cell responses to allergens including (but not limited to) aerosolized dust particles.

In some embodiments, SBPs formatted as drops may be used to treat conjunctivitis or pink eye. In some embodiments ocular SBPs comprising antibiotics as therapeutic agents may be prescribed when the conjunctivitis is caused by bacteria. In some embodiments, pharmaceutical compositions comprising ocular SBPs may be prepared as artificial tears to help dilute irritating allergens present in the tear film.

In some embodiments, ocular SBPs formatted as drops may include mydriatics, an agent that causes pupil dilation. Mydriatics include but are not limited to phenylephrine, cyclopentolate, tropicamide, hydroxyamphetamine/tropicamide, atropine, cyclopentolate/phenylephrine ophthalmic, homatropine ophthalmic, and scopolamine. Such SBPs may be used in the treatment of ocular indications or in preparation for the diagnosis of ocular conditions.

In some embodiments, ocular SBPs formatted as solutions may be used as contact lens solution. Contact lens solutions are solutions used for the storage of contact lenses in between use of said contact lenses. Contact lenses may be used for vision correction and/or for cosmetic purposes. In some embodiments, the anti-microbial and/or bacteriostatic properties of an SBP may enable the storage of contact lenses while prohibiting the growth of microbes and/or bacteria.

In some embodiments, the ocular SBPs of the present disclosure are biocompatible in the ocular space. In some embodiments, administration of the ocular SBP does not cause local inflammation in the ocular space. In some embodiments, ocular SBP is tolerable in the ocular space. In some embodiments, the retinal tissue remains normal after the administration of the ocular SBP. In some embodiments, the SBPs are biocompatible and tolerable in the ocular space for at least 1 day, at least 3 days, at least 1 week, at least 2 weeks, at least 1 month, at least 3 months, at least 4 months, at least 6 months, or at least 1 year.

In some embodiments, the present disclosure provides methods of treating subjects by contacting them with ocular SBPs. The subjects may have, may be suspected of having, and/or may be at risk for developing one or more ocular indications. Such ocular indications may include any of those described herein. In some embodiments, ocular indications include dry eye disease. In some embodiments, ocular indications include one or more of an infection, refractive errors, age related macular degeneration, cystoid macular edema, cataracts, diabetic retinopathy (proliferative and non-proliferative), glaucoma, amblyopia, strabismus, color blindness, cytomegalovirus retinitis, keratoconus, diabetic macular edema (proliferative and non-proliferative), low vision, ocular hypertension, retinal detachment, eyelid twitching, inflammation, uveitis, bulging eyes, dry eye disease, floaters, xerophthalmia, diplopia, Graves' disease, night blindness, eye strain, red eyes, nystagmus, presbyopia, excess tearing, retinal disorder, conjunctivitis, cancer, corneal ulcer, corneal abrasion, snow blindness, scleritis, keratitis, Thygeson's superficial punctate keratopathy, corneal neovascularization, Fuch's dystrophy, keratoconjunctivitis sicca, iritis, cyclitis, cycloplegia, chorioretinal inflammation (e.g. chorioretinitis, choroiditis, retinitis, retinochoroiditis, pars planitis, Harada's disease, aniridia, macular scars, solar retinopathy, choroidal degeneration, choroidal dystrophy, choroideremia, gyrate atrophy, choroidal hemorrhage, choroidal detachment, retinoschisis, hypertensive retinopathy, Bull's eye maculopathy, epiretinal membrane, peripheral retinal degeneration, hereditary retinal dystrophy, retinitis pigmentosa, retinal hemorrhage, retinal vein occlusion, and separation of retinal layers.

In some embodiments, the present disclosure provides methods of delivering ocular therapeutic agents to subjects by contacting subject eyes with ocular SBPs. Such ocular SBPs may be prepared by combining processed silk with ocular therapeutic agents. The SBPs may be prepared as solutions. The SBPs may be prepared by dissolving processed silk in water or buffer. In some embodiments, the processed silk is silk fibroin. In some embodiments, the processed silk is prepared by degummed degumming for a boiling time selected from a 30-minute boil, a 60-minute boil, a 90-minute boil, a 120-minute boil, and a 480-minute boil. In some embodiments the processed silk is freeze dried in phosphate buffer. In some embodiments, the processed silk is freeze dried with sucrose. In some embodiments, the SBP is stressed. In some embodiments, the SBP is stressed by one or more methods including heating to 60° C. and autoclaving. In some embodiments, the SBP contains phosphate salts. In some embodiments, the SBP is formulated as a hydrogel. In some embodiments the SBP is formulated as a solution. In some embodiments, the SBP solution comprises phosphate buffered saline and trace amounts of phosphate buffer. In some embodiments, the SBP solution comprises sucrose, phosphate buffered saline and/or trace amounts of phosphate buffer.

In some embodiments, the viscosity of an ocular SBP may be tuned by preparation with silk fibroin of different boiling times. In some embodiments, a preparation of an ocular SBP from silk fibroin with a longer boiling time (e.g. 480 mb) may increase the viscosity of the SBP. In some embodiments, a preparation of an ocular SBP from silk fibroin with a shorter boiling time (e.g. 30 mb) may increase the viscosity of the SBP. In some embodiments, the viscosity of an ocular SBP may be tuned by preparation with different concentrations of silk fibroin. In some embodiments, a preparation of an ocular SBP with a lower concentration of silk fibroin may increase the viscosity of the SBP. In some embodiments, a preparation of an ocular SBP with a silk fibroin concentration of below 1% (w/v) may increase viscosity. In some embodiments, a preparation of an ocular SBP with a concentration between 0.005% and 0.5% (w/v) may increase viscosity. In some embodiments, both the boiling time of silk fibroin and the concentration of silk fibroin may be used to tune the viscosity of the SBP. In some embodiments, the viscosity of an ocular SBP may be tuned by preparation with stressed silk fibroin. In some embodiments, both the preparation of an SBP with stressed silk and the concentration of silk fibroin may be used to tune the viscosity of the SBP.

In some embodiments, the shear storage modulus and/or the shear loss modulus of an ocular SBP are tuned by the concentration of silk fibroin. In some embodiments, the shear storage modulus and/or the shear loss modulus of an ocular SBP are tuned by preparation with stressed silk fibroin. In some embodiments, both the preparation of an SBP with stressed silk and the concentration of processed silk may be used to tune the shear storage modulus and the shear loss modulus of the SBP. In some embodiments, the phase angle of the ocular SBP is tuned by preparation with stressed silk fibroin. In some embodiments, the phase angle of the ocular SBP is tuned by the concentration of processed silk.

In some embodiments, the viscosity of an ocular SBP may remain consistent across a range of silk fibroin concentrations. In some embodiments the viscosity of an ocular SBP may remain within 50%, 40%, 30% 20%, 10%, 5%, or 1% upon dilution. An ocular SBP may be diluted between 0 and 20-fold while maintaining a consistent viscosity.

In some embodiments, the ocular SBPs shear thin, or demonstrate shear thinning properties. In some embodiments, the ocular SBPs demonstrate greater shear thinning than commercially available treatments for dry eye disease. In some embodiments, the ocular SBPs of the present disclosure have a higher viscosity at a lower shear rate. In some embodiments, the ocular SBPs of the present disclosure have the viscosity of a gel at a lower shear rate. In some embodiments, the higher viscosity at a lower shear rate tunes the residence time of the SBP. In some embodiments, the residence time is increased. In some embodiments, the ocular SBPs of the present disclosure have a lower viscosity at a higher shear rate. In some embodiments, the ocular SBPs of the present disclosure have the viscosity of a fluid (e.g. liquid) at a higher shear rate. In some embodiments, the lower viscosity at a higher shear rate increases the comfort in the eye. In some embodiments, the shear thinning of tolerable ocular SBPs promotes differentiation.

In some embodiments, the ocular SBPs of the present disclosure are used to treat a subject. In some embodiments, ocular SBPs are used to treat a subject by contacting the subject with an ocular SBP. In some embodiments, the subject has an ocular indication. In some embodiments, the ocular condition is dry eye disease (DED). In some embodiments, the ocular SBP is administered for ocular lubrication. In some embodiments, the ocular SBP acts as artificial tears and/or a tear replacement. In some embodiments, the ocular SBP alleviates the symptoms of DED after administration. These symptoms may include, but are not limited to, ocular discomfort, dryness, grittiness, and pain. These symptoms may be associated with any grade of DED, including mild, moderate, and/or severe DED. In some embodiments, administration of an ocular SBP of the present disclosure may reduce signs of DED. These signs include, but are not limited to, ocular surface staining and/or tear film break-up time. In some embodiments, the ocular SBP improves comfort in the eye. In some embodiments, the ocular SBP is a hydrogel. In some embodiments, the ocular SBP is a solution. In some embodiments, the ocular SBP is hydrophobic. In some embodiments, the ocular SBP is administered to the eye. In some embodiments, the ocular SBP is administered via topical administration. In some embodiments, the ocular SBP is administered as drops. In some embodiments, the ocular SBP is administered as a spray. In some embodiments, the ocular SBP adheres to the ocular surface. In some embodiments, the ocular SBP adheres to the ocular surface in a manner similar to a mucin layer. In some embodiments, the hydrophobicity of the ocular SBP may improve tear film formation and enhance tear production. In some embodiments, the hydrophobicity of the ocular SBP may prevent evaporation. In some embodiments, the residence time of an ocular SBP will be analyzed after ocular SBP administration, using any method known to one skilled in the art. In some embodiments, the efficacy of an ocular SBP will be analyzed after ocular SBP administration, using any method known to one skilled in the art. In some embodiments, the pharmacokinetics of an ocular SBP will be analyzed after ocular SBP administration, using any method known to one skilled in the art. In some embodiments, the irritability of an ocular SBP will be analyzed after ocular SBP administration, using any method known to one skilled in the art. In some embodiments, the use of an ocular SBP to treat irritation will be analyzed after ocular SBP administration, using any method known to one skilled in the art. In some embodiments, the toxicity of an ocular SBP will be analyzed after ocular SBP administration, using any method known to one skilled in the art.

In some embodiments, the ocular SBP has a surface tension similar to that of water. In some embodiments, the ocular SBP has a surface tension similar to that of human tears. In some embodiments, the surfactant properties of ocular SBPs reduce surface tension to magnitude similar to human tears. Human tears have been reported to have a surface tension of 43.6 mN/m, as described in Sweeney et al. (2013) Experimental Eye Research 117: 28-38. In some embodiments, ocular SBPs have a surface tension of about 40-50 mN/m.

Ocular SBPs with a surface tension similar to human tears may allow optimal spreading and tear reformation after blinking. In some embodiments, the surface tension of an ocular SBP may be optimizing to allow for improved spreading and/or tear reformation following blinking. In some embodiments, the surface tension of the ocular SBP may be controlled by the concentration of processed silk. In some embodiments, the surfactant and/or demulcent properties of an ocular SBP may reduce surface tension and provide optimal spreading and/or coating of the ocular surface. In some embodiments, the spreading and/or wetting capabilities of an ocular SBP may be modulated via the surface tension of the ocular SBP. In some embodiments, spreading and/or wetting is improved in ocular SBPs with lower surface tension. In some embodiments, the coating of the ocular surface may be improved in ocular SBPs with lower surface tension.

In some embodiments, ocular SBPs may display a coefficient of friction lower than that of water, as measured by the experimental sliding speeds. In some embodiments, the coefficient of friction of an ocular SBP is slightly lower than that of water. In some embodiments, the ocular SBP is more lubricating that water. In some embodiments, the ocular SBP is slightly more lubricating that water.

In some embodiments, silk fibroin solutions described herein may be prepared by dissolving lyophilized processed silk. That processed silk may be silk fibroin. Silk fibroin may be lyophilized in water, phosphate buffer, sucrose, and any other cryoprotectant and/or buffer described herein. In some embodiments, the formulations with lower molecular weight silk fibroin (480 mb) may be more viscous than the formulations with higher molecular weight silk fibroin (120 mb). Preparation from silk fibroin lyophilized in PB may not affect the viscosity of the formulations.

In some embodiments, silk fibroin may be conjugated with a fluorescent label (e.g. fluorescein isothiocyanate (FITC)). Conjugation may be performed by any method known to one of skill in the art. In some embodiments, the silk fibroin may be conjugated to FITC by mixing the two components together in a sodium bicarbonate solution at a basic pH (e.g. 9.0). The reaction may be performed under light protection at room temperature for at least 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, or overnight. The resulting labeled silk fibroin (FITC-SF) may be dialyzed to remove any impurities. The FITC-SF may be stored at 4° C. until use.

In some embodiments, FITC-SF may be added to solutions of silk fibroin. In the resulting solution, FITC-SF may comprise at 10%, 20%, 30%, or 40% of the total silk fibroin in solution. The total concentration of silk fibroin, with and without FITC-SF, may be any concentration described herein. In some embodiments, formulations of silk fibroin solutions may comprise 1% (w/v) silk fibroin of which 10%, 20%, 30%, or 40% of the total silk fibroin is FITC-SF. In some embodiments, formulations with higher percentages of FITC-SF may have lower complex viscosities. In some embodiments, formulations comprising silk fibroin in which 30% of the silk fibroin is FITC-SF have the most similar rheological properties to solutions of unlabeled silk fibroin.

In some embodiments, silk fibroin solutions may be prepared in any buffer described herein. In some embodiments, silk fibroin solutions may be prepared in a borate (DED) buffer. This buffer may comprise 6 mg/mL boric acid, 0.45 mg/mL sodium borate, 3.4 mg/mL sodium chloride, 1.4 mg/mL potassium chloride, 0.06 mg/mL magnesium chloride, and 0.06 mg/mL calcium chloride, pH 7.3. Silk fibroin solutions may further comprise polyethylene glycol (PG). PG may be present at any concentration described herein. In some embodiments, silk solutions comprise 1% PG. In some embodiments, G' and G" may be measured to be lower for formulations with higher molecular weight silk fibroin. G' and G" may also be increased by the preparation from silk fibroin lyophilized in PB or formulations with PG. The phase angle may be higher for formulations with higher molecular weight silk fibroin.

The irritability and/or tolerability of any silk fibroin solution described herein may be analyzed after administration to the eye of a subject. In some embodiments, topical administration of the solution of silk fibroin may be well tolerated for at least 1 hour, 2 hours, 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 2 weeks, 3 weeks, or 1 month. In some embodiments, no deleterious effects (e.g. to the corneal surface) are detected after administration of an ocular SBP.

In some embodiments, the silk fibroin solutions described herein may be stable under storage. Storage conditions may include, but are not limited to, 4° C., room temperature, and 40° C. Silk fibroin solutions may be stable under the storage conditions for at least 1 hour, 2 hours, 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 2 weeks, 3 weeks, 1 month, 6 months, 1 year, 2 years, 3 years, 4 years, 5 years, or 10 years. In some embodiments, an ocular SBP may have a shelf life of least 1 day, at least 2 days, at least 3 days, at least 4 days, at least 5 days, at least 6 days, at least 7 days, at least 8 days, at least 9 days, at least 10 days, at least 11 days, at least 12 days, at least 13 days, at least 2 weeks, at least 3 weeks, at least 1 month, at least 6 weeks, at least 2 months, at least 10 weeks, at least 3 months, at least 14 weeks, at least 4 months, at least 18 weeks, at least 5 months, at least 22 weeks, at least 6 months, at least 7 months, at least 8 months, at least 9 months, at least 10 months, at least 11 months, at least a year, at least 2 years, at least 3 years, at least 4 years, at least 5 years, or more than 5 years. In some embodiments, ocular SBPs may have a shelf life of about 1 year at room temperature. In some embodiments, ocular SBPs may have a shelf life of about 2 years at room temperature. In some embodiments, ocular SBPs may have a shelf life of about 3 years at room temperature. In some embodiments, ocular SBPs may have a shelf life of about 4 years at room temperature. In some embodiments, ocular SBPs may have a shelf life of about 5 years at room temperature.

In some embodiments, ocular SBPs may be prepared to have desired residence time for their intended use. As used herein, the term "residence time" refers to the average length of time during which a substance (e.g., SBPs) is in a given location or condition. In some embodiments, enhanced residence time may enable convenient dosing for patients, as described in Zhu et al. (2008) Optometry and Vision Science 85(8): E715-E725, the contents of which are herein incorporated by reference in their entirety. The residence time may be modulated via the viscosity of the SBP. In some embodiments the viscosity is about 10-100 cP. In some embodiments, the viscosity is about 20-80 cP. In some embodiments, the viscosity is about 30-60 cP. In some embodiments, higher viscosity of an ocular SBP may lead to a longer residence time of said ocular SBP, as described in Zhu et al. (2007) Current Eye Research 32(3): 177-179, the contents of which are herein incorporated by reference in their entirety. These longer residence times may improve the symptoms benefit of an ocular indication (e.g. DED). In some embodiments, residence time of ocular SBP formulations described herein may vary from a few hours to several weeks. For example, residence time of SBP formulations may be about 1 hour, about 2 hours, about 3 hours, about 4 hours, about 5 hours, about 6 hours, about 7 hours, about 8 hours, about 9 hours, about 10 hours, about 11 hours, about 12 hours, about 13 hours, about 14 hours, about 15 hours, about 16 hours, about 17 hours, about 18 hours, about 19 hours, about 20 hours, about 21 hours, about 22 hours, about 23 hours, about 24 day, about 2 days, about 3 days, about 4 days, about 5 days, about 6 days, about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, one month, 2 months, 6 months, 1 year, or longer than two years. In some embodiments, residence time of ocular SBP may be from about 4 hours to about 10 hours, from about 8 hours to about 12 hours, from about 10 hours to about 14 hours, from about 12 hours to about 16 hours, from about 15 hours to about 20 hours, from about 18 hours to about 22 hours, from about 20 to about 25 hours, from about 22 hours to about 26 hours, from about 24 hour to about 30 hours, from about 20 hour to about 28 hours, or from about 30 hours to about 40 hours. In one embodiment, residence time of an ocular SBP is about 24 hours. In some embodiments, sustained viscosity of an ocular SBP upon dilution may enable longer residence times and enhanced retention.

In some embodiments, ocular SBPs may be prepared to have desired degradation time for their intended use. As used herein, the term "degradation time" refers to the amount of time required for a substance to break down. In some embodiments, degradation times of ocular SBP formulations described herein may vary from a few hours to several weeks. For example, degradation time of ocular SBP formulations may be about 1 hour, about 2 hours, about 3 hours, about 4 hours, about 5 hours, about 6 hours, about 7 hours, about 8 hours, about 9 hours, about 10 hours, about 11 hours, about 12 hours, about 13 hours, about 14 hours, about 15 hours, about 16 hours, about 17 hours, about 18 hours, about 19 hours, about 20 hours, about 21 hours, about 22 hours, about 23 hours, about 24 day, about 2 days, about 3 days, about 4 days, about 5 days, about 6 days, about 1 week, about 2 weeks, about 3 weeks, about 4 weeks, one month, 2 months, 6 months, 1 year, or longer than two years. In some embodiments, degradation time of ocular SBPs may be from about 4 hours to about 10 hours, from about 8 hours to about 12 hours, from about 10 hours to about 14 hours, from about 12 hours to about 16 hours, from about 15 hours to about 20 hours, from about 18 hours to about 22 hours, from about 20 to about 25 hours, from about 22 hours to about 26 hours, from about 24 hour to about 30 hours, from about 20 hour to about 28 hours, or from about 30 hours to about 40 hours.

Delivery

SBPs may be delivered to cells, tissues, organs and/or organisms in naked form. As used herein in, "naked" delivery refers to delivery of an active agent with minimal or with no additional formulation or modification. Naked SBPs may be delivered to cells, tissues, organs and/or organisms using routes of administration known in the art and described herein. In some embodiments, naked delivery may include formulation in a simple buffer such as saline or PBS.

In some embodiments, SBPs may be prepared with one or more cell penetration agents, pharmaceutically acceptable carriers, delivery agents, bioerodible or biocompatible polymers, solvents, and/or sustained-release delivery depots. SBPs may be delivered to cells using routes of administration known in the art and described herein. In some embodiments, SBPs may be formulated for direct delivery to organs or tissues in any of several ways in the art including, but not limited to, direct soaking or bathing, via a catheter, by gels, powder, ointments, creams, gels, lotions, and/or drops, or by using substrates (e.g., fabric or biodegradable materials) coated or impregnated with SBPs.

Routes of Delivery

In some embodiments, SBP formulations may be administered by any route to achieve a therapeutically effective outcome.

These include, but are not limited to enteral (into the intestine), gastroenteral, epidural (into the dura matter), oral (by way of the mouth), transdermal, peridural, intracerebral (into the cerebrum), intracerebroventricular (into the cerebral ventricles), epicutaneous (application onto the skin), intradermal, (into the skin itself), subcutaneous (under the skin), nasal administration (through the nose), intravenous (into a vein), intravenous bolus, intravenous drip, intra-arterial (into an artery), intramuscular (into a muscle), intracranial (into the heart), intraosseous infusion (into the bone marrow), intrathecal (into the spinal canal), intraperitoneal, (infusion or injection into the peritoneum), intrasinal infusion, intravitreal, (through the eye), intravenous injection (into a pathologic cavity) intracavitary (into the base of the penis), intravaginal administration, intrauterine, extra-amniotic administration, transdermal (diffusion through the intact skin for systemic distribution), transmucosal (diffusion through a mucous membrane), transvaginal, insufflation (snorting), sublingual, sublabial, enema, eye drops (onto the conjunctiva), in ear drops, auricular (in or by way of the ear), buccal (directed toward the cheek), conjunctival, cutaneous, dental (to a tooth or teeth), electro-osmosis, endocervical, endosinusial, endotracheal, extracorporeal, hemodialysis, infiltration, interstitial, intra-abdominal, intra-amniotic, intra-articular, intrabiliary, intrabronchial, intrabursal, intracartilaginous (within a cartilage), intracaudal (within the cauda equine), intracisternal (within the cisterna magna cerebellomedularis), intracorneal (within the cornea), dental intracornal, intracoronary (within the coronary arteries), intracorporus cavernosum (within the dilatable spaces of the corporus cavernosa of the penis), intradiscal (within a disc), intraductal (within a duct of a gland), intraduodenal (within the duodenum), intradural (within or beneath the dura), intraepidermal (to the epidermis), intraesophageal (to the esophagus), intragastric (within the stomach), intragingival (within the gingivae), intraileal (within the distal portion of the small intestine), intralesional (within or introduced directly to a localized lesion), intraluminal (within a lumen of a tube), intralymphatic (within the lymph), intramedullary (within the marrow cavity of a bone), intrameningeal (within the meninges), intramyocardial (within the myocardium), intraocular (within the eye), intraovarian (within the ovary), intrapericardial (within the pericardium), intrapleural (within the pleura), intraprostatic (within the prostate gland), intrapulmonary (within the lungs or its bronchi), intrasinal (within the nasal or periorbital sinuses), intraspinal (within the vertebral column), intrasynovial (within the synovial cavity of a joint), intratendinous (within a tendon), intratesticular (within the testicle), intrathecal (within the cerebrospinal fluid at any level of the cerebrospinal axis), intrathoracic (within the thorax), intratubular (within the tubules of an organ), intratumor (within a tumor), intratympanic (within the aurus media), intravascular (within a vessel or vessels), intraventricular (within a ventricle), iontophoresis (by means of electric current where ions of soluble salts migrate into the tissues of the body), irrigation (to bathe or flush open wounds or body cavities), laryngeal (directly upon the larynx), nasogastric (through the nose and into the stomach), occlusive dressing technique (topical route administration which is then covered by a dressing which occludes the area), ophthalmic (to the external eye), oropharyngeal (directly to the mouth and pharynx), parenteral, percutaneous, periarticular, peridural, perineural, periodontal, rectal, respiratory (within the respiratory tract by inhaling orally or nasally for local or systemic effect), retrobulbar (behind the pons or behind the eyeball), intramyocardial (entering the myocardium), soft tissue, subarachnoid, subconjunctival, submucosal, topical, transplacental (through or across the placenta), transtracheal (through the wall of the trachea), transtympanic (across or through the tympanic cavity), ureteral (to the ureter), urethral (to the urethra), vaginal, caudal block, diagnostic, nerve block, biliary perfusion, cardiac perfusion, photopheresis or spinal.

As a non-limiting example, the SBP is in the form of a hydrogel and the route of delivery is topical.

In one embodiment, the amount of the SBP in the formulation can be optimized for a particular Cmax value. When used for delivery (e.g., oral delivery) for small molecules or biologics, the Cmax may be decreased.

Detectable Agents and Labels

In some embodiments, SBP formulations may include detectable labels. As used herein, the term "detectable label" refers to any incorporated compound or entity that facilitates some form of identification. Detectable labels may include, but are not limited to various organic small molecules, inorganic compounds, nanoparticles, enzymes or enzyme substrates, fluorescent materials, luminescent materials (e.g., luminol), bioluminescent materials (e.g., luciferase, luciferin, and aequorin), chemiluminescent materials, radioactive materials (e.g., ^{18}F, ^{67}Ga, ^{81}mKr, ^{82}Rb, ^{111}In ^{123}I, ^{133}Xe, ^{201}Tl, ^{125}I, ^{35}S, ^{14}C, ^{3}H, or ^{999} mTc (e.g., as pertechnetate (technetate(VII), TcO^{4-})), contrast agents (e.g., gold, gold nanoparticles, gadolinium, chelated Gd, iron oxides, superparamagnetic iron oxide (SPIO), monocrystalline iron oxide nanoparticles (MIONs), and ultrasmall superparamagnetic iron oxide (USPIO)), manganese chelates (e.g., Mn-DPDP), barium sulfate, iodinated contrast media (iohexol), microbubbles, or perfluorocarbons). Such optically-detectable labels include for example, without limitation, 4-acetamido-4'-isothiocyanatostilbene-2,2'disulfonic acid; acridine and derivatives (e.g., acridine and acridine isothiocyanate); 5-(2'-aminoethyl)aminonaphthalene-1-sulfonic acid (EDANS); 4-amino-N-[3-vinylsulfonyl)phenyl]naphthalimide-3,5 disulfonate; N-(4-anilino-1-naphthyl)maleimide; anthranilamide; BODIPY; Brilliant Yellow; coumarin and derivatives (e.g., coumarin, 7-amino-4-methylcoumarin (AMC, Coumarin 120), and 7-amino-4-trifluoromethylcoumarin (Coumarin 151)); cyanine dyes; cyanosine; 4',6-diaminidino-2-phenylindole (DAPI); 5' 5"-dibromopyrogallol-sulfonaphthalein (Bromopyrogallol Red); 7-diethylamino-3-(4'-isothiocyanatophenyl)-4-methylcoumarin; diethylenetriamine pentaacetate; 4,4'-diisothiocyanatodihydro-stilbene-2,2'-di sulfonic acid; 4,4'-diisothiocyanatostilbene-2,2'-disulfonic acid; 5-[dimethylamino]-naphthalene-1-sulfonyl chloride (DNS, dansylchloride); 4-dimethylaminophenylazophenyl-4'-isothiocyanate (DABITC); eosin and derivatives (e.g., eosin and eosin isothiocyanate); erythrosin and derivatives (e.g., erythrosin B and erythrosin isothiocyanate); ethidium; fluorescein and derivatives (e.g., 5-carboxyfluorescein (FAM), 5-(4,6-dichlorotriazin-2-yl)aminofluorescein (DTAF), 2',7'-dimethoxy-4'5'-dichloro-6-carboxyfluorescein, fluorescein, fluorescein isothiocyanate, X-rhodamine-5-(and-6)-isothiocyanate (QFITC or XRITC), and fluorescamine); 2-[2-[3-[[1,3-dihydro-1,1-dimethyl-3-(3-sulfopropyl)-2H-benz[e]indol-2-ylidene]ethylidene]-2-[4-(ethoxycarbonyl)-1-piperazinyl]-1-cyclopenten-1-yl]ethenyl]-1,1-dimethyl-3-(3-sulforpropyl)-1H-benz[e]indolium hydroxide, inner salt, compound with n,ndiethylethanamine(1:1) (IR144); 5-chloro-2-[2-[3-[(5-chloro-3-ethyl-2(3H)-benzothiazolylidene) ethylidene]-2-(diphenylamino)-1-cyclopenten-1-yl]ethenyl]-3-ethyl benzothiazolium perchlorate (IR140); Malachite Green isothiocyanate; 4-methylumbelliferone orthocresolphthalein; nitrotyrosine; pararosaniline; Phenol Red; B-phycoerythrin; o-phthaldialdehyde; pyrene and derivatives(e.g., pyrene, pyrene butyrate, and succinimidyl 1-pyrene); butyrate quantum dots; Reactive Red 4 (CIBACRON™ Brilliant Red 3B-A); rhodamine and derivatives (e.g., 6-carboxy-Xrhodamine (ROX), 6-carboxyrhodamine (R6G), lissamine rhodamine B sulfonyl chloride rhodarnine (Rhod), rhodamine B, rhodamine 123, rhodamine X isothiocyanate, sulforhodamine B, sulforhodamine 101, sulfonyl chloride derivative of sulforhodamine 101 (Texas Red), N,N,N',N'tetramethyl-6-carboxyrhodamine (TAMRA) tetramethyl rhodamine, and tetramethyl rhodamine isothiocyanate (TRITC)); riboflavin; rosolic acid; terbium chelate derivatives; Cyanine-3 (Cy3); Cyanine-5 (Cy5); cyanine-5.5 (Cy5.5), Cyanine-7 (Cy7); IRD 700; IRD 800; Alexa 647; La Jolta Blue; phthalo cyanine; and naphthalo cyanine.

In some embodiments, the detectable labels may include non-detectable precursors that becomes detectable upon activation (e.g., fluorogenic tetrazine-fluorophore constructs, tetrazine-BODIPY FL, tetrazine-Oregon Green 488, or tetrazine-BODIPY TMR-X) or enzyme activatable fluorogenic agents (e.g., PROSENSE® (VisEn Medical)). In vitro assays in which enzyme labeled compositions can be used include, but are not limited to, enzyme linked immunosorbent assays (ELISAs), immunoprecipitation assays, immunofluorescence, enzyme immunoassays (EIA), radioimmunoassays (RIA), and Western blot analysis.

In some embodiments, SBP formulations may include may fluorescein isothiocyanate (FITC) as a detectable label. In some embodiments, FITC is conjugated to processed silk. In some embodiments, the processed silk conjugated to FITC is silk fibroin. Conjugation of FITC to silk fibroin may be performed using the standard isothiocyanate coupling protocol. FITC can be attached to silk fibroin via the amine group. The labeled silk fibroin may be purified from the unconjugated fluorescein by gel filtration. The final ratio of labeled and unlabeled silk fibroin can be determined by measuring the absorbance at 280 nm and at 495 nm.

SBP formulations may contain both labeled SBP and free (unlabeled) SBP. In some embodiments, the ratio of labeled SBP to free (unlabeled) SBP may be about 50:1, about 20:1, about 10:1, about 9.5:1, about 9:1, about 8.5:1, about 8:1, about 7.5:1, about 7:1, about 6.5:1, about 6:1, about 5.5:1, about 5:1, about 4.5:1, about 4:1, about 3.5:1, about 3:1, about 7:3, about 2.5:1, about 2:1, about 1.5:1, about 1:1, about 1:1.5, about 1:2, about 1:2.5, about 3:7, about 1:3, about 1:3.5, about 1:4, about 1:4, about 1:4.5, about 1:5, about 1:5.5, about 1:6, about 1:7, about 1:7.5, about 1:8, about 1:8.5, about 1:9, about 1:9.5, about 1:10, about 1:20, or about 1:50. In some embodiments, the ratio of labeled SBP to free (unlabeled) SBP may be from about 10:1 to about 7:1, from about 8:1 to about 5:1, from about 6:1 to about 4:1, from about 5:1 to about 3:1, from about 4:1 to about 2:1, from about 3:1 to about 1.5:1, from about 2:1 to about 1:1, from about 1:1 to about 1:2, from about 1:1.5 to about 1:3, about 1:2 to about 1:4, from about 1:3 to about 1:5, from about 1:4 to about 1:6, from about 1:5 to about 1:8, or from about 1:7 to about 1:10.

In one embodiment, the SBP formulation contains 1% silk fibroin, wherein the ratio of FITC labeled silk fibroin and unlabeled silk fibroin is 4:6.

In one embodiment, the SBP formulation contains 1% silk fibroin, wherein the ratio of FITC labeled silk fibroin and unlabeled silk fibroin is 3:7.

In one embodiment, the SBP formulation contains 1% silk fibroin, wherein the ratio of FITC labeled silk fibroin and unlabeled silk fibroin is 2:8.

In one embodiment, the SBP formulation contains 1% silk fibroin, wherein the ratio of FITC labeled silk fibroin and unlabeled silk fibroin is 1:9.

Depot Administration

In some embodiments, SBPs may be administered by or be used to administer therapeutic agents by depot administration. As used herein, the term "depot" refers to a concentration of one or more agents in a particular region or in association with a composition or device. With depot administration, the one or more agents exit or diffuse from the concentration into surrounding areas. Agents administered by depot administration may be SBPs. In some embodiments, SBPs are depots for therapeutic agents, wherein the therapeutic agents exit or diffuse from the SBPs. In some embodiments, depots are solutions. In some embodiments, depots are gels or hydrogels. In some embodiments, depots are eye drops. In some embodiments, depot administration of an SBP may reduce the number of times a therapeutic agent needs to be administered. In some embodiments, depot administration of an SBP may replace oral administration of a therapeutic agent.

In some embodiments, SBP depots may be used for controlled release of therapeutic agents, wherein release is facilitated by diffusion. Such methods may include any of those described in United States Patent Publication Number US20170333351, the contents of which are herein incorporated by reference in their entirety. Therapeutic agent diffusion may be slowed (i.e., controlled) by SBP depots leading to extended release periods. Extended therapeutic agent release periods may enable longer administration periods. In some embodiments, administration periods are extended by from about 0.01% to about 1%, from about 0.05% to about 2%, from about 1% to about 5%, from about 2% to about 10%, from about 3% to about 15%, from about 4% to about 20%, from about 5% to about 25%, from about 6% to about 30%, from about 7% to about 35%, from about 8% to about 40%, from about 9% to about 45%, from about 10% to about 50%, from about 12% to about 55%, from about 14% to about 60%, from about 16% to about 65%, from about 18% to about 70%, from about 20% to about 75%, from about 22% to about 80%, from about 24% to about 85%, from about 26% to about 90%, from about 28% to about 95%, from about 30% to about 100%, from about 32% to about 105%, from about 34% to about 110%, from about 36% to about 115%, from about 38% to about 120%, from about 40% to about 125%, from about 42% to about 130%, from about 44% to about 135%, from about 46% to about 140%, from about 48% to about 145%, from about 50% to about 150%, from about 60% to about 175%, from about 70% to about 200%, from about 80% to about 225%, from about 90% to about 250%, from about 100% to about 275%, from about 110% to about 300%, from about 120% to about 325%, from about 130% to about 350%, from about 140% to about 375%, from about 150% to about 400%, from about 170% to about 450%, from about 190% to about 500%, from about 210% to about 550%, from about 230% to about 600%, from about 250% to about 650%, from about 270% to about 700%, from about 290% to about 750%, from about 310% to about 800%, from about 330% to about 850%, from about 350% to about 900%, from about 370% to about 950%, from about 390% to about 1000%, from about 410% to about 1050%, from about 430% to about 1100%, from about 450% to about 1500%, from about 480% to about 2000%, from about 510% to about 2500%, from about 540% to about 3000%, from about 570% to about 3500%, from about 600% to about 4000%, from about 630% to about 4500%, from about 660% to about 5000%, from about 690% to about 5500%, from about 720% to about 6000%, from about 750% to about 6500%, from about 780% to about 7000%, from about 810% to about 7500%, from about 840% to about 8000%, from about 870% to about 8500%, from about 900% to about 9000%, from about 930% to about 9500%, from about 960% to about 10000%,

In some embodiments, the controlled release of a therapeutic agent for the treatment of a condition, disease, or indication may be facilitated by the degradation and/or dissolution of SBPs. Such methods may be carried according to those described in International Patent Application Publication Nos. WO2013126799, WO2017165922, and U.S. Pat. No. 8,530,625, the contents of each of which are herein incorporated by reference in their entirety. SBP degradation and/or dissolution may expose increasing amounts of therapeutic agents over time for treatment of therapeutic indications.

In some embodiments, therapeutic agent release from SBPs may be monitored via high performance liquid chromatography (HPLC), ultra-performance liquid chromatography (UPLC), and/or other methods known to those skilled in the art.

SBP hydrogels may be used to extend payload release periods (e.g., as shown for extended release of small molecule in International Patent Application Publication No. WO2017139684, the contents of which are herein incorporated by reference in their entirety. In some embodiments, SBP hydrogels are used to provide extended release of therapeutic agents (e.g., biological agents). Hydrogel networks may stabilize such agents and support their release as the hydrogel degrades. This effect serves to extend agent release and may be modulated by varying factors including processed silk molecular weight, concentration, excipient type, pH, and temperature. In some embodiments, processed silk molecular weight, concentration, excipient type, pH, and processing temperature used to prepare SBPs may be modulated to achieve desired payload release periods for specific therapeutic agents.

In some embodiments, SBPs may be lyophilized together with therapeutic agents. In some embodiments, combined lyophilization may induce further interactions between therapeutic agents and SBPs. These interactions may be maintained through SBP preparation and support extended payload release. Payload release may be dependent on SBP degradation and/or dissolution. In some embodiments, SBP β-sheet content is increased (e.g., via water annealing), thereby increasing SBP insolubility in water. Such SBPs may exhibit increased payload release periods. In some embodiments, these SBPs may include therapeutic agent stabilizing properties to extend administration periods and/or therapeutic agent half-life.

In some embodiments, SBPs described herein maintain and/or improve the controlled delivery of a therapeutic agent. In some embodiments, SBPs lengthen payload release period and/or administration period by at least 1 hour, at least 2 hours, at least 3 hours, at least 4 hours, at least 5 hours, at least 6 hours, at least 7 hours, at least 8 hours, at least 9 hours, at least 10 hours, at least 11 hours, at least 12 hours, at least 13 hours, at least 14 hours, at least 15 hours, at least 16 hours, at least 17 hours, at least 18 hours, at least 19 hours, at least 20 hours, at least 21 hours, at least 22 hours, at least 23 hours, or at least 24 hours. In some embodiments, SBPs lengthen payload release period and/or administration period by at least 1 day, at least 2 days, at least 3 days, at least 4 days, at least 5 days, at least 6 days, at least 7 days, at least 8 days, at least 9 days, at least 10 days, at least 11 days, at least 12 days, at least 13 days, at least 2 weeks, at least 3 weeks, at least 1 month, at least 6 weeks, at least 2 months, at least 10 weeks, or at least 3 months.

In some embodiments, SBPs may be used to modulate depot release of therapeutic agents. Some SBPs may release therapeutic agents according to near zero-order kinetics. In some embodiments, SBPs may release therapeutic agents according to first-order kinetics. In some embodiments, therapeutic agent release rate may be modulated by preparing SBP depots with modification of one or more of density, loading, drying method, silk fibroin molecular weight, and silk fibroin concentration.

In some embodiments, SBPs are prepared to release from about 0.01% to about 1%, from about 0.05% to about 2%, from about 1% to about 5%, from about 2% to about 10%, from about 3% to about 15%, from about 4% to about 20%, from about 5% to about 25%, from about 6% to about 30%, from about 7% to about 35%, from about 8% to about 40%, from about 9% to about 45%, from about 10% to about 50%, from about 12% to about 55%, from about 14% to about 60%, from about 16% to about 65%, from about 18% to about 70%, from about 20% to about 75%, from about 22% to about 80%, from about 24% to about 85%, from about 26% to about 90%, from about 28% to about 95%, from about 30% to about 100% of the total amount of therapeutic or macromolecular therapeutic agent to be delivered.

Therapeutic Applications

In some embodiments, SBP formulations may be used in a variety of therapeutic applications. As used herein, the term "therapeutic application" refers to any method related to restoring or promoting the health, nutrition, and/or well-being of a subject; supporting or promoting reproduction in a subject; or treating, preventing, mitigating, alleviating, curing, or diagnosing a disease, disorder, or condition. As used herein, the term "condition" refers to a physical state of wellbeing. Therapeutic applications may include, but are not limited to, medical applications, surgical applications, and veterinary applications. As used herein, the term "medical application" refers to any method or use that involves treating, diagnosing, and/or preventing disease according to the science of medicine. "Surgical applications" refer to methods of treatment and/or diagnosis that involve operation on a subject, typically requiring incision and the use of instruments. "Veterinary applications" refer to therapeutic applications where the subject is a non-human animal. In some embodiments, therapeutic applications may include, but are not limited to, experimental, diagnostic, or prophylactic applications. In some embodiments, therapeutic applications include preparation and/or use of therapeutic devices. As used herein, the term "therapeutic device" refers to any article prepared or modified for therapeutic use.

SBP formulations used for therapeutic applications may include or may be combined with one or more pharmaceutical compositions, implants, therapeutic agents, coatings, excipients, or devices. In some embodiments, SBP formulations facilitate the delivery and/or controlled release of therapeutic agent payloads. In some embodiments, SBP formulations described herein may be used to stabilize therapeutic agents. Some SBP formulations may be used as tools, materials, or devices in therapeutic applications. Such SBP formulations may include, but are not limited to, delivery vehicles, and scaffolds. In some embodiments, therapeutic applications utilizing SBP formulations may include gene therapy. As used herein, the term "gene therapy" refers to the use of genetic transplantation to address disease and/or genetic disorders. In some embodiments, therapeutic applications utilizing SBP formulations may include gene editing. As used herein, the term "gene editing" refers to any process used to alter a DNA gene sequence at the level of individual nucleotides. In some embodiments, therapeutic applications utilizing SBP formulations may include immunotherapy. As used herein, the term "immunotherapy" refers to treatment of a disease, condition, or indication by modulating the immune system. In some embodiments, therapeutic applications utilizing SBP formulations may include diagnostic applications. In some embodiments, SBP formulations are used as diagnostic tools. In some embodiments, therapeutic applications utilizing SBP formulations may include tissue engineering. In some embodiments, therapeutic applications utilizing SBP formulations may include cell culture. In some embodiments, therapeutic applications utilizing SBP formulations may include surgical applications (e.g. incorporation into surgical tools, devices, and fabrics). In some embodiments, SBP formulations may be or may be included in therapeutic devices. In some embodiments, therapeutic devices may be coated with SBP formulations described herein.

Therapeutic applications of the present disclosure may be applied to a variety of subjects. As used herein, the term "subject" refers to any entity to which a particular process or activity relates to or is applied. Subjects of therapeutic applications described herein may be human or non-human. Human subjects may include humans of different ages, genders, races, nationalities, or health status. Non-human subjects may include non-human animal subjects (also simply referred to herein as "animal subjects"). Animal subjects may be non-human vertebrates or invertebrates. Some animal subjects may be wild type or genetically modified organisms (e.g., transgenic). In some embodiments, subjects include patients. As used herein, the term "patient" refers to a subject seeking treatment, in need of treatment, requiring treatment, receiving treatment, expecting treatment, or who is under the care of a trained (e.g., licensed) professional for a particular disease, disorder, and/or condition.

In some embodiments, SBP formulations may be used to address one or more therapeutic indications. As used herein, the term "therapeutic indication" refers to a disease, disorder, condition, or symptom that may be cured, reversed, alleviated, stabilized, improved, or otherwise addressed through some form of therapeutic intervention (e.g., administration of a therapeutic agent or method of treatment). As a non-limiting example, the therapeutic indication is ophthalmology, ophthalmology-related disease and/or disorder, otology, or an otology-related disease and/or disorder. As a non-limiting example, the ophthalmology-related disease and/or disorder is dry eye disease.

SBP formulation treatment of therapeutic indications may include contacting subjects with SBPs. SBP formulations may include therapeutic agents (e.g., any of those described herein) as cargo or payloads for treatment. In some embodiments, payload release may occur over a period of time (the "payload release period"). The payload release rate and/or length of the payload release period may be modulated by SBP components or methods of preparation.

In some embodiments, ocular SBPs may be used as an anti-inflammatory treatment for dry eye disease, as described in Kim et al. (2017) Scientific Reports 7: 44364, the contents of which are herein incorporated by reference in their entirety. It has been demonstrated that the administration of 0.1 to 0.5% silk fibroin solutions in a mouse model of dry eye disease enhances corneal smoothness and tear production, while reducing the amount of inflammatory markers detected.

Therapeutic Agents

In some embodiments, therapeutic applications involve the use of SBP formulations that are therapeutic agents or are combined with one or more therapeutic agents. As used herein, the term "therapeutic agent" refers to any substance used to restore or promote the health and/or wellbeing of a subject and/or to treat, prevent, alleviate, cure, or diagnose a disease, disorder, or condition. Examples of therapeutic agents include, but are not limited to, adjuvants, analgesic agents, antiallergic agents, antiangiogenic agents, antiarrhythmic agents, antibacterial agents, antibiotics, antibodies, anticancer agents, anticoagulants, antidementia agents, antidepressants, antidiabetic agents, antigens, antihypertensive agents, anti-infective agents, anti-inflammatory agents, antioxidants, antipyretic agents, anti-rejection agents, antiseptic agents, antitumor agents, antiulcer agents, antiviral agents, biological agents, birth control medication, carbohydrates, cardiotonics, cells, chemotherapeutic agents, cholesterol lowering agents, cytokines, endostatins, enzymes, fats, fatty acids, genetically engineered proteins, glycoproteins, growth factors, health supplements, hematopoietics, herbal preparations, hormones, hypotensive diuretics, immunological agents, inorganic synthetic pharmaceutical drugs, ions, lipoproteins, metals, minerals, nanoparticles, naturally derived proteins, NSAIDs, nucleic acids, nucleotides, organic synthetic pharmaceutical drugs, oxidants, peptides, pills, polysaccharides, proteins, protein-small molecule conjugates or complexes, psychotropic agents, small molecules, sodium channel blockers, statins, steroids, stimulants, therapeutic agents for osteoporosis, therapeutic combinations, thrombopoietics, tranquilizers, vaccines, vasodilators, VEGF-related agents, veterinary agents, viruses, virus particles, and vitamins. In some embodiments, SBP therapeutics and methods of delivery may include any of those taught in International Publication Numbers WO2017139684, WO2010123945, WO2017123383, or United States Publication Numbers US20170340575, US20170368236, and US20110171239 the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, SBPs may be used to encapsulate, store, stabilize, preserve, and/or release, in a controlled manner, therapeutic agents. For example, using silk fibroin micrococoons as delivery vehicles for small molecules has been described in Shimanovich et al. (Shimanovich et al. (2015) Nature Communications 8:15902, the contents of which are herein incorporated by reference in their entirety). In some embodiments, SBP formulations may be prepared with therapeutic agents selected from any of those listed in Table 4. In the Table, example categories are indicated for each therapeutic agent. These categories are not limiting and each therapeutic agent may fall under multiple categories (e.g., any of the categories of therapeutic agents described herein).

TABLE 4

THERAPEUTIC AGENTS

Agent	Category
opium	analgesic agent
opiate	analgesic agent
doxycycline monohydrate	antibacterial agent
tigecycline	antibacterial agent
doxycycline hyclate	antibacterial agent
vibramycin	antibacterial agent
doxycycline hydrochloride hemiethanolate hemihydrate	antibacterial agent
doxycycline calcium	antibacterial agent
abciximab	antibody
adalimumab	antibody
adalimumab-atto	antibody
alefacept	antibody
alemtuzumab	antibody
antibody fragment	antibody
antibody-drug conjugate	antibody
atezolizumab	antibody
basiliximab	antibody
belimumab	antibody
bezlotoxumab	antibody
bivalent antibody	antibody
canakinumab	antibody
certolizumab pegol	antibody
cetuximab	antibody
daclizumab	antibody
denosumab	antibody
efalizumab	antibody
golimumab	antibody
inflectra	antibody
ipilimumab	antibody
ixekizumab	antibody
monoclonal antibody	antibody
monovalent antibody	antibody
multivalent antibody	antibody
natalizumab	antibody
nivolumab	antibody
obiltoxaxamab	antibody
olaratumab	antibody
omalizumab	antibody
palivizumab	antibody
panitumumab	antibody
pembrolizumab	antibody
polyclonal antibody	antibody
reslizumab	antibody
rituximab	antibody
secukimmab	antibody
tocilizumab	antibody
trastuzumab	antibody
ustekinumab	antibody
autoantigen	antigen
endogenous antigen	antigen
exogenous antigen	antigen
neoantigen	antigen
tumor antigen	antigen
viral antigen	antigen

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
exogenous antigen	antigen
endogenous antigen	antigen
autoantigen	antigen
neoantigen	antigen
viral antigen	antigen
tumor antigen	antigen
xenogenus (heterologous) antigen	antigen
autologous antigen	antigen
idiotypic antigen	antigen
allogenic (homologous) antigen	antigen
epilope	antigen
tumor-specific antigen	antigen
tumor-associated antigen	antigen
neo-epitope	antigen
allergen	antigen
superantigen	antigen
tolerogen	antigen
immanoglobulin -binding protein	antigen
t-dependent antigen	antigen
t-independent antigen	antigen
immunodominant antigen	antigen
COX-1 inhibitor	anti-inflammatory agent
COX-2 inhibitor	anti-inflammatory agent
bisbiguanides polymeric quaternary ammonium compound	antiseptic agent
chlorhexidine	antiseptic agent
chlorinated phenol	antiseptic agent
ethanol	antiseptic agent
hydrogen peroxide	antiseptic agent
lower alcohol	antiseptic agent
peroxides	antiseptic agent
propanol	antiseptic agent
quaternary amine surfactant	antiseptic agent
silver complex	antiseptic agent
small molecule quaternary ammonium compound	antiseptic agent
5-FU Enhancer	anticancer agent
9-AC	anticancer agent
abraxane	anticancer agent
actinomycin	anticancer agent
AG2037	anticancer agent
AG3340	anticancer agent
Aggrecanase Inhibitor	anticancer agent
alitretinoin	anticancer agent
alkylating agent	anticancer agent
Aminoglutethimide	anticancer agent
Amsacrine (m-AMSA)	anticancer agent
anthracycline	anticancer agent
antimicrobial peptide	anticancer agent
Asparaginase	anticancer agent
Azacitidine	anticancer agent
azathioprine	anticancer agent
Batimastat (BB94)	anticancer agent
BAY 12-9566	anticancer agent
BCH-4556	anticancer agent
bexarotene	anticancer agent
Bis-Naphtalimide	anticancer agent
bleomycin	anticancer agent
Busulfan	anticancer agent
Capecitabine	anticancer agent
Carboplatin	anticancer agent
Carmustaine + Polifepr Osan	anticancer agent
cdk4/cdk2 inhibitor	anticancer agent
chlorambucil	anticancer agent
CI-994	anticancer agent
Cisplatin	anticancer agent
Cladribine	anticancer agent
CS-682	anticancer agent
cyclophosphamide	anticancer agent
cytarabine	anticancer agent
Cytarabine HCl	anticancer agent
cytoskeletal disruptor	anticancer agent
D2163	anticancer agent
dacarbazine	anticancer agent
Dactinomycin	anticancer agent
daunorubicin	anticancer agent

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
Daunorubicin HCl	anticancer agent
DepoCyt	anticancer agent
Dexifosamide	anticancer agent
Docetaxel	anticancer agent
Dolastain	anticancer agent
Doxifluridine	anticancer agent
Doxorubicin	anticancer agent
DX8951f	anticancer agent
E 7070	anticancer agent
EGFR	anticancer agent
Epirubicin	anticancer agent
epothilone	anticancer agent
erlotinib	anticancer agent
Estramustine phosphate sodium	anticancer agent
Etoposide (VP16-213)	anticancer agent
Farnesyl Transferase Inhibitor	anticancer agent
FK 317	anticancer agent
Flavopiridol	anticancer agent
Floxuridine	anticancer agent
Fludarabine	anticancer agent
Fluorouracil (5-FU)	anticancer agent
Flutamide	anticancer agent
Fragyline	anticancer agent
gefitinib	anticancer agent
Gemcitabine	anticancer agent
Hexamethylmelamine (HMM)	anticancer agent
histone deacetylase inhibitor	anticancer agent
hydroxyurea	anticancer agent
Hydroxyurea (hydroxycarbanride)	anticancer agent
idarubicin	anticancer agent
Ifosfamide	anticancer agent
imatinib	anticancer agent
Interferon Alfa-2a	anticancer agent
Interferon Alfa-2b	anticancer agent
Irinotecan	anticancer agent
ISI 641	anticancer agent
kinase inhibitor	anticancer agent
Krestin	anticancer agent
Lemonal DP 2202	anticancer agent
Leuptolide acetate (LHRH-releasing factor analogue)	anticancer agent
Levamisole	anticancer agent
LiGLA (lithium-gamma linolenate)	anticancer agent
Lodine Seed	anticancer agent
Lometexol	anticancer agent
Lomustine (CCNU)	anticancer agent
Marimistat	anticancer agent
mechlorethamine	anticancer agent
Mechlorethamine HCl (nitrogen mustard)	anticancer agent
Megestrol acetate	anticancer agent
Meglamine GLA	anticancer agent
melphalan	anticancer agent
Mercaptopurine	anticancer agent
Mesna	anticancer agent
methotrexate	anticancer agent
methyl glyoxal bis-guanylhydrazone (MGBG)	anticancer agent
Mitoguazone (methyl-GAG	anticancer agent
Mitotane (o.p'-DDD)	anticancer agent
Mitoxantrone HCl	anticancer agent
mitozantrone	anticancer agent
MMI 270	anticancer agent
MMP	anticancer agent
MTA/LY 231514	anticancer agent
MTX	anticancer agent
nitrosourea	anticancer agent
nucleotide analogue	anticancer agent
nucleotide precursor analogue	anticancer agent
ODN 698	anticancer agent
OK-432	anticancer agent
Oral Platinum	anticancer agent
Oral Taxoid	anticancer agent
oxaliplatin	anticancer agent
paclitaxel	anticancer agent
PARP Inhibitor	anticancer agent

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
PD 183805	anticancer agent
Pentostatin (2 deoxycoformycin)	anticancer agent
PKC 412	anticancer agent
platinum based chemotherapeutic	anticancer agent
Plicamycin	anticancer agent
Procarbazine HCl	anticancer agent
PSC 833	anticancer agent
Ralitrexed	anticancer agent
RAS Farnesyl Transferase Inhibitor	anticancer agent
RAS Oncogene Inhibitor	anticancer agent
retinoids	anticancer agent
romidepsin	anticancer agent
Semustine (methyl-CCNU)	anticancer agent
Streptozocin	anticancer agent
Suramin	anticancer agent
tafluposide	anticancer agent
Tamoxifen citrate	anticancer agent
taxane	anticancer agent
Taxane Analog	anticancer agent
taxotere	anticancer agent
Temozolomide	anticancer agent
Teniposide (VM-26)	anticancer agent
Thioguanine	anticancer agent
Thiotepa	anticancer agent
tioguanine	anticancer agent
topoisomerase I inhibitor	anticancer agent
topoisomerase II inhibitor	anticancer agent
Topotecan	anticancer agent
tretinoin	anticancer agent
Tyrosine Kinase	anticancer agent
UFT (Tegafur/Uracil)	anticancer agent
Valtubicin	anticancer agent
vemurafenib	anticancer agent
vinblastine	anticancer agent
Vinblastine sulfate	anticancer agent
vinca alkaloid	anticancer agent
vinca alkaloid derivative	anticancer agent
vincristine	anticancer agent
vindesine	anticancer agent
Vindesine sulfate	anticancer agent
vinorelbine	anticancer agent
vismodegib	anticancer agent
vorinostat	anticancer agent
VX-710	anticancer agent
VX-853	anticancer agent
YM 116	anticancer agent
ZD 0101	anticancer agent
ZD 0473/Anormed	anticancer agent
ZD 1839	anticancer agent
ZD 9331	anticancer agent
2-dimensional tissue	biological
3-dimensional tissue	biological
adenovirus	biological
adipose tissue-derived mesenchymal stem cell	biological
bacteria	biological
bone mesenchymal stem cell	biological
cardiac mesenchymal stem cell	biological
cells	biological
chicken dorsal root ganglion	biological
complex carbohydrate	biological
deoxyribonucleic acid	biological
embryonic stem cell	biological
fibroblast	biological
fungi	biological
gene	biological
hematopoietic stem cell	biological
human corneal epithelial cell	biological
human corneal stromal stem cell	biological
human small intestinal enteroids	biological
lentivirs	biological
limbal epithelial stem cell	biological
lipids	biological
macromolecule	biological
mesenchymal stem cell	biological
microbe	biological

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
microbiome	biological
microorganism	biological
neural stem cells	biological
oligonucleotide	biological
oral keratinocvte	biological
organ	biological
organism	biological
periodontal ligament stem cells	biological
polymer	biological
probiotic	biological
proteins	biological
ribonucleic acid	biological
spore	biological
stem cell	biological
symbiote	biological
T cell	biological
tissue	biological
transfected fibroblast	biological
vesicle	biological
viral particle	biological
virus	biological
abequose	carbohydrate
arabinose	carbohydrate
cellobiose	carbohydrate
derivative of a monosaccharide	carbohydrate
di saccharide	carbohydrate
fructose	carbohydrate
fucose	carbohydrate
galactosamine	carbohydrate
galactose	carbohydrate
glucosamine	carbohydrate
glucose	carbohydrate
glucuronic acid	carbohydrate
iduronic acid	carbohydrate
lactose	carbohydrate
maltose	carbohydrate
mannose	carbohydrate
monosaccharide	carbohydrate
muramic acid	carbohydrate
N-acetylgalactosamine	carbohydrate
N-acetylglucosamine	carbohydrate
N-acetylmuramic acid	carbohydrate
N-acetylneuraminic acid	carbohydrate
oligosaccharide	carbohydrate
rhamnose	carbohydrate
ribose	carbohydrate
sialic acid	carbohydrate
sucrose	carbohydrate
trehalose	carbohydrate
xylose	carbohydrate
adipose tissue-derived mesenchymal stem cell	cell
periodontal ligament stem cell	cell
human small intestinal enteroid	cell
oral keratinocyte	cell
fibroblast	cell
transfected fibroblast	cell
2-dimensional tissue	cell
3-dimensional tissue	cell
T cell	cell
embryonic stem cell	cell
neural stem cell	cell
mesenchymal stem cell	cell
hematopoietic stem ceil	cell
osteoblast	cell
osteoclast	cell
osteocyte	cell
neuron	cell
glial cell	cell
chondrocyte	cell
photoreceptor cell	cell
cone cell	cell
rod cell	cell
corneal cell	cell
keratocyte	cell
comeal endothelial cell	cell

TABLE 4-continued

THERAPEUTIC AGENTS

Agent	Category
brain-derived neurotrophic factor (BDNF)	cytokine
cardiotrophin 1 (CTF1)	cytokine
cardiotrophin-like cytokine factor 1 (CLCF1)	cytokine
cell signal molecule	cytokine
chemokine	cytokine
ciliary neurotrophic factor (CNTF)	cytokine
erythropoietin (EPO)	cytokine
fibroblast growth factor acidic (FGFa)	cytokine
fibroblast growth factor basic (FGFb)	cytokine
IL-18	cytokine
IL-10	cytokine
IL-11	cytokine
IL-12	cytokine
IL-13	cytokine
IL-14	cytokine
IL-15	cytokine
IL-16	cytokine
IL-17	cytokine
IL-19	cytokine
IL-1α	cytokine
IL-1β	cytokine
IL-2	cytokine
IL-20	cytokine
IL-21	cytokine
IL-22	cytokine
IL-23	cytokine
IL-27	cytokine
IL-3	cytokine
IL-4	cytokine
IL-5	cytokine
IL-6	cytokine
IL-7	cytokine
IL-8	cytokine
IL-9	cytokine
interferon	cytokine
interferon-α1	cytokine
interleukin	cytokine
interleukin-1 receptor antagonist (IL-IRA)	cytokine
keratinocyte growth factor 1	cytokine
keratinocyte growth factor 2 (KGF)	cytokine
kit ligand/stem cell factor (KITLG)	cytokine
leptin (LEP)	cytokine
leukemia inhibitory factor (LIF)	cytokine
lymphokine	cytokine
matrix metalloproteinase (MMP)	cytokine
monokine	cytokine
nerve growth factor (NGF)	cytokine
oncostatin M (OSM)	cytokine
prolactin (PRL)	cytokine
TGFβ	cytokine
tissue inhibitor of metalloproteinase (TIMP)	cytokine
transforming growth factor (TGF) α (TGFα)	cytokine
tumor necrosis factor α (TNFα)	cytokine
diacylglycerol	fat
diglycerides	fat
ergosterol	fat
fat-soluble vitamin	fat
glycerol monostearate	fat
glycerophospholipid	fat
glyceryl hydroxystearate	fat
hopanoid	fat
hydroxy-steroid	fat
monoglyceride	fat
monolaurin	fat
oil	fat
palmitin	fat
phosphatidic acid	fat
phosphatidylcholine	fat
phosphatidylethanolamine	fat
phosphatidylserine	fat
phosphoinositides	fat
phospholipids	fat
phosphosphingolipids	fat
phytosterol	fat
sphingolipid	fat
sphingomyelin	fat
stearin	fat
sterol	fat
triglyceride	fat
triolein	fat
wax	fat
fatty acid	fatty acid
essential fatty acid	fatty acid
omega-3 fatty acid	fatty acid
lincoleic acid	fatty acid
omega-6 fatty acid	fatty acid
docosahezaenoic acid	fatty acid
arachidonic acid	fatty acid
omega-9 fatty acid	fatty acid
Hexadecatrienoic acid (HTA)	fatty acid
α-Linolenic acid (ALA)	fatty acid
Stearidonic acid (SDA)	fatty acid
Eicosatrienoic acid (ETE)	fatty acid
Eicosatetraenoic acid (ETA)	fatty acid
Eicosapentaenoic acid (EPA)	fatty acid
Heneicosapentaenoic acid (HPA)	fatty acid
Docosapentaenoic acid (DPA),	fatty acid
Clupanodonic acid	fatty acid
Docosahexaenoic acid (DHA)	fatty acid
Tetracosapentaencic acid	fatty acid
Tetracosahexaenoic acid (Nisinic acid)	fatty acid
5-Dodecenoic acid	fatty acid
7-Tetradecenoic acid	fatty acid
9-Hexadecenoic acid	fatty acid
11-Octadecenoic acid	fatty acid
13-Eicosenoic acid	fatty acid
15-Docosenoic acid	fatty acid
17-Tetracosenoic acid	fatty acid
Linoleic acid (LA)	fatty acid
Gamma-linolenic acid (GLA)	fatty acid
Calendic acid	fatty acid
Eicosadienoic acid	fatty acid
Dihomo-gamma-linolenic acid (DGLA)	fatty acid
Arachidonic acid (AA, ARA)	fatty acid
Docosadienoic acid	fatty acid
Adrenic acid	fatty acid
Osbond acid	fatty acid
Tetracosatetraenoic acid	fatty acid
Tetracosapentaenoic acid	fatty acid
oleic acid	fatty acid
elaidic acid	fatty acid
gondoic acid	fatty acid
mead acid	fatty acid
erucic acid	fatty acid
nervonic acid	fatty acid
ximenic acid	fatty acid
bone morphogenic protein	protein
bone morphogenic-like protein	protein
epidermal growth factor	protein
fibroblast growth factor	protein
insulin like growth factor I	protein
insulin like growth factor II	protein
transforming growth factor	protein
biotin (vitamin B7)	health supplement
iodine	health supplement
niacin (vitamin B3)	health supplement
pantothenic acid (vitamin B5)	health supplement
phosphorus	health supplement
riboflavin (vitamin B2)	health supplement
selenium	health supplement
thiamine (vitamin B1)	health supplement
vanadium	health supplement
vitamin A	health supplement
vitamin B	health supplement

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
vitamin B12	health supplement
vitamin B6	health supplement
vitamin B9	health supplement
vitamin C	health supplement
vitamin D	health supplement
vitamin E	health supplement
vitamin K	health supplement
allspice berry essential oil	herbal preparation
angelica seed essential oil	herbal preparation
anise seed essential oil	herbal preparation
basil	herbal preparation
basil essential oil	herbal preparation
bay essential oil	herbal preparation
bay laurel	herbal preparation
bay laurel essential oil	herbal preparation
bergamot essential oil	herbal preparation
blood orange essential oil	herbal preparation
borage	herbal preparation
camphor essential oil	herbal preparation
caraway	herbal preparation
caraway seed essential oil	herbal preparation
cardamom seed essential oil	herbal preparation
carrot seed essential oil	herbal preparation
cassia essential oil	herbal preparation
catnip	herbal preparation
catnip essential oil	herbal preparation
cedarwood essential oil	herbal preparation
celery seed essential oil	herbal preparation
chamomile german essential oil	herbal preparation
chamomile roman essential oil	herbal preparation
chervil	herbal preparation
chives	herbal preparation
cilantro	herbal preparation
cinnamon bark essential oil	herbal preparation
cinnamon leaf essential oil	herbal preparation
citronella essential oil	herbal preparation
clary sage essential oil	herbal preparation
clove bud essential oil	herbal preparation
cold infusion	herbal preparation
compresses	herbal preparation
cordial	herbal preparation
coriander seed essential oil	herbal preparation
cumin	herbal preparation
cypress essential oil	herbal preparation
decoctions	herbal preparation
dill	herbal preparation
elemi essential oil	herbal preparation
epazote	herbal preparation
essential oils	herbal preparation
eucalyptus essential oil	herbal preparation
fennel	herbal preparation
fennel essential oii	herbal preparation
fir needle essential oil	herbal preparation
flower essence	herbal preparation
frankincense essential oil	herbal preparation
garlic	herbal preparation
geranium essential oil	herbal preparation
ginger essential oil	herbal preparation
granule	herbal preparation
grapefruit pink essential oil	herbal preparation
helichrysum essential oil	herbal preparation
herbal wine	herbal preparation
hop essential oil	herbal preparation
hyssop essential oil	herbal preparation
jasmine absolute oil	herbal preparation
juniper berry essential oil	herbal preparation
labdanum essential oil	herbal preparation
lavender	herbal preparation
lavender absolute oil	herbal preparation
lemon balm	herbal preparation
lemon essential oil	herbal preparation
lemon verbena	herbal preparation
lemongrass	herbal preparation
lemongrass essential oil	herbal preparation
time essential oii	herbal preparation
lovage	herbal preparation
magnolia essential oil	herbal preparation
mandarin essential oil	herbal preparation
margoram essential oil	herbal preparation
marjoram	herbal preparation
Melissa essential oil	herbal preparation
mints	herbal preparation
mugward essential oil	herbal preparation
myrrh essential oil	herbal preparation
myrtle essential oil	herbal preparation
nasturtium	herbal preparation
neroli essential oil	herbal preparation
niaouli essential oil	herbal preparation
nutmeg essential oil	herbal preparation
ointment	herbal preparation
orange sweet essential oil	herbal preparation
oregano	herbal preparation
oregano essential oil	herbal preparation
palmarosa essential oil	herbal preparation
parsley	herbal preparation
patchouli essential oil	herbal preparation
pennyroyal essential oil	herbal preparation
pepper black essential oil	herbal preparation
peppermint essential oil	herbal preparation
petitgram essential oil	herbal preparation
pine needle essential oil	herbal preparation
pink lotus absolute oil	herbal preparation
poultice	herbal preparation
radiata essential oil	herbal preparation
ravensara essential oil	herbal preparation
rose absolute oil	herbal preparation
rose essential oil	herbal preparation
rosemary	herbal preparation
rosemary essential oil	herbal preparation
rosewood essential oil	herbal preparation
sage	herbal preparation
sage essential oil	herbal preparation
salad burnet	herbal preparation
salve	herbal preparation
sambac absolute oil	herbal preparation
sandalwood essential oil	herbal preparation
savory	herbal preparation
scented geranium	herbal preparation
sitz bath	herbal preparation
soak	herbal preparation
sorrel	herbal preparation
spearmint essential oil	herbal preparation
spikenard essential oil	herbal preparation
spruce essential oil	herbal preparation
star anise essential oil	herbal preparation
suppository	herbal preparation
sweet annie essential oil	herbal preparation
syrup	herbal preparation
tangerine essential oil	herbal preparation
tarragon	herbal preparation
tea	herbal preparation
tea tree essential oil	herbal preparation
thyme	herbal preparation
thyme red essential oil	herbal preparation
tincture	herbal preparation
verbena essential oil	herbal preparation
vetiver essential oil	herbal preparation
white lotus absolute oil	herbal preparation
wintergreen essential oil	herbal preparation
wormwood essential oil	herbal preparation
varrow essential oil	herbal preparation
ylang essential oil	herbal preparation
3-ketodesogestrel	hormone
allopregnanolone	hormone
androgen	hormone
androstenediol	hormone
androstenedione	hormone
chlormadinone acetate	hormone
cholesterol	hormone
conjugated estrogen	hormone
dehydroepiandrosterone	hormone
dexamethasone	hormone

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
dihydrotestosterone	hormone
drospirorenone	hormone
estradiol ester	hormone
estradiols	hormone
estriol	hormone
estriol succinate	hormone
estrogen	homo ne
estrone	hormone
estrone sulfate	hormone
ethinyl estradiol	hormone
gestodene	hormone
glucocoriticoid	hormone
levonorgestrel	hormone
mestranol	hormone
mineralocorticoid	hormone
norethisterone acetate	hormone
norgestrel	hormone
polyestriol phosphate	hormone
progesterone	hormone
progestogen	hormone
testosterone	hormone
calcium oxide	ion, metal, or mineral
iron oxide	ion, metal, or mineral
phosphorus oxide	ion, metal, or mineral
iodine oxide	ion, metal, or mineral
magnesium oxide	ion, metal, or mineral
zinc oxide	ion, metal, or mineral
selenium oxide	ion, metal, or mineral
copper oxide	ion, metal, or mineral
manganese oxide	ion, metal, or mineral
chromium oxide	ion, metal, or mineral
molybdenum oxide	ion, metal, or mineral
gold oxide	ion, metal, or mineral
potassium oxide	ion, metal, or mineral
nickel oxide	ion, metal, or mineral
silicon oxide	ion, metal, or mineral
vanadium oxide	ion, metal, or mineral
tin oxide	ion, metal, or mineral
calcium	ion, metal, or mineral
chloride	ion, metal, or mineral
chromium	ion, metal, or mineral
copper	ion, metal, or mineral
gold	ion, metal, or mineral
iron	ion, metal, or mineral
magnesium	ion, metal, or mineral
manganese	ion, metal, or mineral
metal oxide	ion, metal, or mineral
molybdenum	ion, metal, or mineral
nickel	ion, metal, or mineral
potassium	ion, metal, or mineral
silicon	ion, metal, or mineral
silver	ion, metal, or mineral
silver oxide	ion, metal, or mineral
tin	ion, metal, or mineral
zinc	ion, metal, or mineral
nano-hydroxyapatite	nanoparticle
acetominophen	NSAID
carprofen	NSAID
deracoxib	NSAID
fenoprofen	NSAID
firocoxib.	NSAID
flurbirofen	NSAID
mefenamic acid	NSAID
meloxicam	NSAID
robenacoxib	NSAID
aptamer	nucleic acid
antisense nucleic acid	nucleic acid
small interfering RNA (siRNA)	nucleic acid
exon skipping nucleic acid	nucleic acid
RNA editing nucleic acid	nucleic acid
microRNA therapeutic inhibitor (antimiR)	nucleic acid
microRNA therapeutic inhibitor mimic (promiR)	nucleic acid

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
long non-coding RNA modulator	nucleic acid
mRNA	nucleic acid
plasmid	nucleic acid
DNA aptamer	nucleic acid
deoxyribonucleic acidzyme	nucleic acid
RNA aptamer	nucleic acid
RNA decoy	nucleic acid
antisense DNA	nucleic acid
antisense RNA	nucleic acid
ribozyme	nucleic acid
micro ribonucleic acid	nucleic acid
fomivirsen	nucleic acid
viral nucleic acid	nucleic acid
Gendicine (Shenzhen SiBiono GeneTech)	nucleic acid
Advexin (Intro gen)	nucleic acid
nucleic acid vaccines	nucleic acid
fomivirsen sodium (Isis Pharmaceuticals)	nucleic acid
MG98	nucleic acid
ISIS 5132	nucleic acid
DNAzyme	nucleic acid
2,5-diketopiperazine	oxidant or antioxidant
antioxidant	oxidant or antioxidant
melanin	oxidant or antioxidant
oxidants	oxidant or antioxidant
quarternary ammonium chitosan	oxidant or antioxidant
ion	oxidant or antioxidant
mineral	oxidant or antioxidant
vitamin	oxidant or antioxidant
protein	oxidant or antioxidant
hydrogen peroxide	oxidant or antioxidant
ozone	oxidant or antioxidant
nitric acid	oxidant or antioxidant
sulfuric acid	oxidant or antioxidant
oxygen	oxidant or antioxidant
sodium perborate	oxidant or antioxidant
nitrous oxide	oxidant or antioxidant
potassium nitrate	oxidant or antioxidant
sodium bismuthate	oxidant or antioxidant
hypochlorite	oxidant or antioxidant
bleach	oxidant or antioxidant
halogen	oxidant or antioxidant
Cl2	oxidant or antioxidant
F2	oxidant or antioxidant
endogenous oxidant	oxidant or antioxidant
exogenous oxidant	oxidant or antioxidant
hydroxide	oxidant or antioxidant
singlet oxygen	oxidant or antioxidant
superoxide anion	oxidant or antioxidant
hydroxy one radical	oxidant or antioxidant
reactive oxygen species	oxidant or antioxidant
vitamin A	oxidant or antioxidant
beta-carotene	oxidant or antioxidant
carotenoid	oxidant or antioxidant
vitamin C	oxidant or antioxidant
ascorbic acid	oxidant or antioxidant
vitamin E	oxidant or antioxidant
tocopherol	oxidant or antioxidant
tocotrienol	oxidant or antioxidant
selenium	oxidant or antioxidant
glutatione peroxidase	oxidant or antioxidant
zinc	oxidant or antioxidant
catalase	oxidant or antioxidant
superoxide dismutase	oxidant or antioxidant
copper	oxidant or antioxidant
manganese	oxidant or antioxidant
glutathione	oxidant or antioxidant
polyphenol	oxidant or antioxidant
tirilazad	oxidant or antioxidant
allupurinol	oxidant or antioxidant
acetylcysteine	oxidant or antioxidant
lipoic acid	oxidant or antioxidant
carotene	oxidant or antioxidant
ubiquinol	oxidant or antioxidant
BHA	oxidant or antioxidant

TABLE 4-continued

THERAPEUTIC AGENTS

Agent	Category
BHT	oxidant or antioxidant
Anidulafungin	peptide
Atosiban acetate	peptide
Bacitracin	peptide
Bivalirudin	peptide
Bortezomib	peptide
Buserelin	peptide
Calcitonin	peptide
Captopril	peptide
Carbetocin acetate	peptide
Caspofungin	peptide
Cetrorelix acetate	peptide
Colistin	peptide
cyclic dipeptide	peptide
cyclic peptide	peptide
Cyclosporine	peptide
Daptomycin	peptide
Degarelix acetate	peptide
Enalapril maleate	peptide
Enfuvittide	peptide
Eptifibatide	peptide
Exenatide	peptide
Glutathione	peptide
Goserelin	peptide
Human calcitonin	peptide
Ianreotide acetate	peptide
Icatibant acetate	peptide
Lepirdin	peptide
Liraglutide	peptide
Lisinopril	peptide
Lypressin	peptide
Nafarelin acetate	peptide
Nesiritide	peptide
Oxytocin	peptide
RGD peptide	peptide
r-hirudin	peptide
Salmon calcitonin	peptide
Saralasin acetate	peptide
Somatostatin acetate	peptide
Spaglumat magnesium	peptide
Thymalfasin	peptide
Tirofib an	peptide
Vapreotide acetate	peptide
Ziconotide	peptide
adrenocorticotropic hormone (ACTH)	protein
Alpha interferon	protein
antibody	protein
Anakinra	protein
antigen	protein
Beta interferon	protein
Bone morphogenetic protein (BMP)	protein
bone-motphogenic protein 2	protein
chimeric protein	protein
Coagulation Factor IX	protein
Colony stimulating growth factor (CSF)	protein
Desmopressin	protein
Etanercept	protein
Factor IX	protein
Factor VII	protein
Factor VIII	protein
Follicle stimulating hormone (FSH)	protein
Gamma interferon	protein
gastrin prolactin	protein
Granulocyte colony-stimulating factor (G-CSF)	protein
Granulocyte macrophage colony stimulating factor (GM-CSF)	protein
growth hormone (GH)	protein
Human chorionic gonadotropin (HCG)	protein
infliximab	protein
insulin	protein
Insulin glargine	protein
Insulin-like growth factor (IGF)	protein

TABLE 4-continued

THERAPEUTIC AGENTS

Agent	Category
kallikrein	protein
kerantinocyte growth factor (KGF)	protein
Luteinizing hormone (LH)	protein
Macrophage colony stimulating factor (M-CSF)	protein
Neurotrophic growth factor (NGF)	protein
obesity protein (leptin)	protein
Octreotide	protein
Osteoprotegerin (OPG)	protein
pancreatic RNAase	protein
Pegfilgrastim	protein
peptides	protein
platelet activating factor acetyl hydrolase	protein
Platelet-derived growth factor (PDGF)	protein
processed silk	protein
sene in	protein
silk	protein
silk fibroin	protein
Stem cell factor (SCF)	protein
streptokinase	protein
Superoxide dismutase (SOD)	protein
synthetic protein	protein
thrombopoietin	protein
Thyroid stimulating hormone (TSH)	protein
tissue plasminogen activator (TPA)	protein
Tumor necrosis factor (TNF)	protein
tumor necrosis factor binding protein (TNFbp)	protein
urokinase	protein
Vascular endothelial growth factor (VEGF)	protein
antibacterial agent	small molecule
antifungal agent	small molecule
antimalarial agent	small molecule
antiseptic	small molecule
nonsteroidal anti-inflammatory drugs (NSAIDs)	small molecule
stimulant	small molecule
tranquilizers	small molecule
acetaminophen	small molecule-analgesic agent
alcohols	small molecule-analgesic agent
alcuronium	small molecule-analgesic agent
alfentanyl	small molecule-analgesic agent
amethocaine	small molecule-analgesic agent
amobarbital	small molecule-analgesic agent
anileridine	small molecule-analgesic agent
atracurium	small molecule-analgesic agent
bupivacaine	small molecule-analgesic agent
buprenorphine	small molecule-analgesic agent
butorphanol	small molecule-analgesic agent
cannabinoid	small molecule-analgesic agent
cannabis	small molecule-analgesic agent
cisatracurium	small molecule-analgesic agent
cocaine	small molecule-analgesic agent
codiene	small molecule-analgesic agent
COX-2 inhibitor	small molecule-analgesic agent
decamethonium	small molecule-analgesic agent
diacetyl morphine	small molecule-analgesic agent
diamorphine	small molecule-analgesic agent
diazepam	small molecule-analgesic agent
dibucaine	small molecule-analgesic agent
doxacurium	small molecule-analgesic agent
etomidate	small molecule-analgesic agent
fentanyl	small molecule-analgesic agent
gallamine	small molecule-analgesic agent
hydrocodone	small molecule-analgesic agent
hydromorphone	small molecule-analgesic agent
ketamine	small molecule-analgesic agent
levobupivacaine	small molecule-analgesic agent
levorphanol	small molecule-analgesic agent
lidocaine	small molecule-analgesic agent
lorazepam	small molecule-analgesic agent
meperidine (pethidine)	small molecule-analgesic agent
mepivacaine	small molecule-analgesic agent

TABLE 4-continued

THERAPEUTIC AGENTS

Agent	Category
methadone	small molecule-analgesic agent
methohexital	small molecule-analgesic agent
metocurine	small molecule-analgesic agent
midazolam	small molecule-analgesic agent
morphine	small molecule-analgesic agent
morphine glucuronide	small molecule-analgesic agent
morphine sulfate	small molecule-analgesic agent
nalbuphine	small molecule-analgesic agent
NSAID	small molecule-analgesic agent
opioid agonist	small molecule-analgesic agent
opioid antagonist	small molecule-analgesic agent
opioids	small molecule-analgesic agent
oxycodone	small molecule-analgesic agent
oxymorphone	small molecule-analgesic agent
pancuronium	small molecule-analgesic agent
pentazocine	small molecule-analgesic agent
pipecuronium	small molecule-analgesic agent
prilocaine	small molecule-analgesic agent
procaine	small molecule-analgesic agent
propoxyphene	small molecule-analgesic agent
rapacuronium	small molecule-analgesic agent
remifentanil	small molecule-analgesic agent
rocuronium	small molecule-analgesic agent
ropivacaine	small molecule-analgesic agent
succinylcholine	small molecule-analgesic agent
sufentanil	small molecule-analgesic agent
thiamylal	small molecule-analgesic agent
thiopental	small molecule-analgesic agent
tubocurarine	small molecule-analgesic agent
vecuronium	small molecule-analgesic agent
amikacin	small molecule-antibacterial agent
amoxicillin	small molecule-antibacterial agent
ampicillin	small molecule-antibacterial agent
azithromycin	small molecule-antibacterial agent
azlocillin	small molecule-antibacterial agent
aztreonam	small molecule-antibacterial agent
capreomycin	small molecule-antibacterial agent
carbenicillin	small molecule-antibacterial agent
cefaclor	small molecule-antibacterial agent
cefadroxil	small molecule-antibacterial agent
cefalexin	small molecule-antibacterial agent
cefalotin	small molecule-antibacterial agent
cefamandole	small molecule-antibacterial agent
cefazolin	small molecule-antibacterial agent
cefdinir	small molecule-antibacterial agent
cefditoren	small molecule-antibacterial agent
cefepime	small molecule-antibacterial agent
cefixime	small molecule-antibacterial agent
cefoperazone	small molecule-antibacterial agent
cefotaxime	small molecule-antibacterial agent
cefoxitin	small molecule-antibacterial agent
cefpodoxime	small molecule-antibacterial agent
cefprozil	small molecule-antibacterial agent
ceftaroline fosamil	small molecule-antibacterial agent
ceftazidime	small molecule-antibacterial agent
ceftibuten	small molecule-antibacterial agent
ceftizoxime	small molecule-antibacterial agent
ceftobiprole	small molecule-antibacterial agent
ceftriaxone	small molecule-antibacterial agent
cefuroxime	small molecule-antibacterial agent
cilastatin	small molecule-antibacterial agent
ciprofolaxin	small molecule-antibacterial agent
clarithromycin	small molecule-antibacterial agent
clindamycin	small molecule-antibacterial agent
clofazimine	small molecule-antibacterial agent
cloxacillin	small molecule-antibacterial agent
cycloserine	small molecule-antibacterial agent
dalbavancin	small molecule-antibacterial agent
dapsone	small molecule-antibacterial agent
demeclocycline	small molecule-antibacterial agent
dicloxacillin	small molecule-antibacterial agent
dirithromycin	small molecule-antibacterial agent
doripenem	small molecule-antibacterial agent
doxycycline	small molecule-antibacterial agent
enoxacin	small molecule-antibacterial agent
ertapenem	small molecule-antibacterial agent

TABLE 4-continued

THERAPEUTIC AGENTS

Agent	Category
erythromycin	small molecule-antibacterial agent
ethambutol	small molecule-antibacterial agent
ethionamide	small molecule-antibacterial agent
flucloxacillin	small molecule-antibacterial agent
furazolidone	small molecule-antibacterial agent
gatifloxacin	small molecule-antibacterial agent
geldanamycin	small molecule-antibacterial agent
gemifloxacin	small molecule-antibacterial agent
gentamicin	small molecule-antibacterial agent
grepafloxacin	small molecule-antibacterial agent
herbimycin	small molecule-antibacterial agent
imipeneum	small molecule-antibacterial agent
isoniazid	small molecule-antibacterial agent
kanamycin	small molecule-antibacterial agent
levofloxacin	small molecule-antibacterial agent
linezolid	small molecule-antibacterial agent
linomycin	small molecule-antibacterial agent
lomefloxacin	small molecule-antibacterial agent
loracarbef	small molecule-antibacterial agent
mafenide	small molecule-antibacterial agent
meropenem	small molecule-antibacterial agent
methicillin	small molecule-antibacterial agent
mezlocillin	small molecule-antibacterial agent
minocycline	small molecule-antibacterial agent
moxifloxacin	small molecule-antibacterial agent
nafcillin	small molecule-antibacterial agent
nalidixic acid	small molecule-antibacterial agent
neomycin	small molecule-antibacterial agent
netilmicin	small molecule-antibacterial agent
nitrofurantoin	small molecule-antibacterial agent
norfloxacin	small molecule-antibacterial agent
ofloxacin	small molecule-antibacterial agent
oritavancin	small molecule-antibacterial agent
oxacillin	small molecule-antibacterial agent
oxytetracycline	small molecule-antibacterial agent
paromomycin	small molecule-antibacterial agent
penicillin	small molecule-antibacterial agent
penicillin G	small molecule-antibacterial agent
penicillin V	small molecule-antibacterial agent
piperacillin	small molecule-antibacterial agent
posizolid	small molecule-antibacterial agent
pyrazinamide	small molecule-antibacterial agent
radezolid	small molecule-antibacterial agent
rifampicin	small molecule-antibacterial agent
rifaximin	small molecule-antibacterial agent
roxithromycin	small molecule-antibacterial agent
sparfloxacin	small molecule-antibacterial agent
spectinomycin	small molecule-antibacterial agent
spiramycin	small molecule-antibacterial agent
streptomycin	small molecule-antibacterial agent
sulfacetamide	small molecule-antibacterial agent
sulfadiazine	small molecule-antibacterial agent
sulfadimethoxine	small molecule-antibacterial agent
sulfamethizole	small molecule-antibacterial agent
sulfamethoxazole	small molecule-antibacterial agent
sulfanilimide	small molecule-antibacterial agent
sulfasalazine	small molecule-antibacterial agent
sulfisoxazole	small molecule-antibacterial agent
teicoplanin	small molecule-antibacterial agent
telavancin	small molecule-antibacterial agent
telithromycin	small molecule-antibacterial agent
temafloxacin	small molecule-antibacterial agent
temocillin	small molecule-antibacterial agent
tetracycline	small molecule-antibacterial agent
ticarcillin	small molecule-antibacterial agent
tobramycin	small molecule-antibacterial agent
torezolid	small molecule-antibacterial agent
troleandomycin	small molecule-antibacterial agent
trovafloxacin	small molecule-antibacterial agent
vancomycin	small molecule-antibacterial agent
5-flnorocytosine	small molecule-antifungal
abafungin	small molecule-antifungal
albaconazole	small molecule-antifungal
amorolfin	small molecule-antifungal
amphotericin B	small molecule-antifungal
benzoic acid	small molecule-antifungal

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
bifonazole	small molecule-antifungal
butenafine	small molecule-antifungal
butoconazole	small molecule-antifungal
candicidin	small molecule-antifungal
caspofungin	small molecule-antifungal
ciclopirox	small molecule-antifungal
clotrimazole	small molecule-antifungal
crystal violet	small molecule-antifungal
econazole	small molecule-antifungal
efinaconazole	small molecule-antifungal
epexiconazole	small molecule-antifungal
fenticonazole	small molecule-antifungal
filipin	small molecule-antifungal
fluconazole	small molecule-antifungal
flucytosine	small molecule-antifungal
griseofulvin	small molecule-antifungal
haloprogin	small molecule-antifungal
hamycin	small molecule-antifungal
isavuconazole	small molecule-antifungal
isoconazole	small molecule-antifungal
itraconazole	small molecule-antifungal
ketoconazole	small molecule-antifungal
luliconazole	small molecule-antifungal
micafungin	small molecule-antifungal
miconazole	small molecule-antifungal
naftifine	small molecule-antifungal
natamycin	small molecule-antifungal
nystatin	small molecule-antifungal
omoconazole	small molecule-antifungal
oxiconazole	small molecule-antifungal
posaconazole	small molecule-antifungal
propiconazole	small molecule-antifungal
ravuconazole	small molecule-antifungal
rimocidin	small molecule-antifungal
sertaconazole	small molecule-antifungal
sulconazole	small molecule-antifungal
terbinafine	small molecule-antifungal
terconazole	small molecule-antifungal
tioconazole	small molecule-antifungal
tolnaftate	small molecule-antifungal
undecylenic acid	small molecule-antifungal
voriconazole	small molecule-antifungal
aminoquinoline	small molecule-antimalarial
amodiaquine	small molecule-antimalarial
antifolate	small molecule-antimalarial
artemether	small molecule-antimalarial
artemisinin derivative	small molecule-antimalarial
artemotil	small molecule-antimalarial
artesunate	small molecule-antimalarial
atovaquone	small molecule-antimalarial
biguanide	small molecule-antimalarial
bisphosphonate	small molecule-antimalarial
chloproguanil	small molecule-antimalarial
chloroquine	small molecule-antimalarial
cinchona alkaloid	small molecule-antimalarial
dermaseptin	small molecule-antimalarial
DHA-piperaquine	small molecule-antimalarial
diaminopyrimidine	small molecule-antimalarial
dihydroartemisinin	small molecule-antimalarial
doxycillin	small molecule-antimalarial
halofantrine	small molecule-antimalarial
lumefantrine	small molecule-antimalarial
melfoquine	small molecule-antimalarial
N-acetyl cysteine	small molecule-antimalarial
piperaquine	small molecule-antimalarial
primaquine	small molecule-antimalarial
proguanil	small molecule-antimalarial
pyremethamine	small molecule-antimalarial
pyronaridine	small molecule-antimalarial
quercitin	small molecule-antimalarial
quinidine	small molecule-antimalarial
quinine	small molecule-antimalarial
sulfadoxine-pyrimethamine	small molecule-antimalarial
sulfonamide	small molecule-antimalarial
tafenoquine	small molecule-antimalarial
trimethoprim	small molecule-antimalarial

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
choline salicylate	small molecule-antipyretic
magnesium salicylate	small molecule-antipyretic
metamizole	small molecule-antipyretic
nimesulide	small molecule-antipyretic
phenazone	small molecule-antipyretic
salicylate	small molecule-antipyretic
sodium salicylate	small molecule-antipyretic
aspirin	small molecule-NSAID
celecoxib	small molecule-NSAID
diclofenac	small molecule-NSAID
diflunisal	small molecule-NSAID
etodolac	small molecule-NSAID
fenoprofen	small molecule-NSAID
flurbiprofen	small molecule-NSAID
ibuprofen	small molecule-NSAID
indomethacin	small molecule-NSAID
ketoprofen	small molecule-NSAID
ketorolac	small molecule-NSAID
mechlofenamic acid	small molecule-NSAID
nabumetone	small molecule-NSAID
naproxen	small molecule-NSAID
oxaprozin	small molecule-NSAID
piroxicam	small molecule-NSAID
roficoxib	small molecule-NSAID
salsalate	small molecule-NSAID
sulindac	small molecule-NSAID
tolfenamic acid	small molecule-NSAID
tolmetin	small molecule-NSAID
sodium channel blocker	sodium channel blocker
alkaloid	sodium channel blocker
saxitoxin	sodium channel blocker
neosaxitoxin	sodium channel blocker
tetrodoxin	sodium channel blocker
class I antiarrhythmic agent	sodium channel blocker
quinidine	sodium channel blocker
procainamide	sodium channel blocker
disopryamide	sodium channel blocker
tocainide	sodium channel blocker
mexiletine	sodium channel blocker
proparacaine	sodium channel blocker
flecainide	sodium channel blocker
propafenone	sodium channel blocker
moricizine	sodium channel blocker
atorvastatin	statin
cerivastatin	statin
fluvastatin	statin
lovastatin	statin
mevastatin	statin
pitavastatin	statin
pravastatin	statin
rosuvastatin	statin
simvastatin	statin
statin	statin
steroid	steroid
corticosteroid	steroid
triamcinolone	steroid
cortisone	steroid
prednisone	steroid
methylprenisolone	steroid
prednisolone	steroid
betamethasone	steroid
dexamethasone	steroid
hydrocortisone	steroid
deflazacort	steroid
fludrocortisone	steroid
Anadrol	steroid
Anavar	steroid
Clenbuterol	steroid
Clomid	steroid
Cytomel	steroid
Deca Durabolin	steroid
Dianabol	steroid
Equipoise	steroid
Halotestin	steroid
Human Growth Hormone	steroid
Insulin	steroid

TABLE 4-continued

THERAPEUTIC AGENTS

Agent	Category
Lasix	steroid
Methyltestosterone	steroid
Nolvadex	steroid
Omnadren	steroid
Primobolan	steroid
Sustanon	steroid
Cypionate	steroid
Enanthate	steroid
Propionate	steroid
Testosterone	steroid
Trenbolone	steroid
Winstrol	steroid
Flunisolide	steroid
Budesonide	steroid
Mometasone	steroid
Ciclesonide	steroid
Fluticasone	steroid
Beclomethasone	steroid
glutocorticoid	steroid
minerolocorticoid	steroid
corticosterone	steroid
aldosterone	steroid
Hydrocortisone	steroid
methylprednisolone	steroid
prednisolone	steroid
prednisone	steroid
triamcinolone	steroid
Amcinonide	steroid
budesonide	steroid
desonide	steroid
fluocinolone acetonide	steroid
fluocinonide	steroid
halcinonide	steroid
triamcinolone acetonide	steroid
Beclometasone	steroid
betamethasone	steroid
dexamethasone	steroid
fluocortolone	steroid
halometasone	steroid
mometasone	steroid
Alclometasone dipropionate	steroid
betamethaso ne dipropionate	steroid
betamethasone valerate	steroid
clobetasol propionate	steroid
clobetasone butvrate	steroid
fluprednidene acetate	steroid
mometasone furoate	steroid
Ciclesonide	steroid
cortisone acetate	steroid
hydrocortisone aceponate	steroid
hydrocortisone acetate	steroid
hydrocortisone buteprate	steroid
hydrocortisone butyrate	steroid
hydrocortisone valerate	steroid
prednicarbate	steroid
tixocottol pivalate	steroid
3,4-methylenedioxymethamphetamine	stimulant
amphetamines	stimulant
caffeine	stimulant
ephedrine	stimulant
mephedrone	stimulant
methamphetamine	stimulant
methylenedioxy pyrovalerone	stimulant
methylphenidate	stimulant
nicotine	stimulant
phenylpropanolamine	stimulant
propylhexedrine	stimulant
pseudoephedrine	stimulant
imatinib (Gleevac)	therapeutic combination
all-trans-retinoic acid	therapeutic combination
monoclonal antibody treatment	therapeutic combination
gemtuzumab	therapeutic combination
ozogamicin	therapeutic combination
chemotherapy	therapeutic combination
chlorambucil	therapeutic combination
prednisone	therapeutic combination

TABLE 4-continued

THERAPEUTIC AGENTS

Agent	Category
prednisolone	therapeutic combination
vincristine	therapeutic combination
cytarabine	therapeutic combination
clofarabine	therapeutic combination
farnesyl transferase inhibitor	therapeutic combination
decitabine	therapeutic combination
inhibitor of MDR1	therapeutic combination
rituximab	therapeutic combination
interferon-α	therapeutic combination
anthracy cline drug	therapeutic combination
daunorubicin	therapeutic combination
idambicin	therapeutic combination
L-asparaginase	therapeutic combination
doxorubicin	therapeutic combination
cyclophosphamide	therapeutic combination
bleomycin	therapeutic combination
fludarabine	therapeutic combination
etoposide	therapeutic combination
pentostatin	therapeutic combination
cladribine	therapeutic combination
bone marrow transplant	therapeutic combination
stem cell transplant	therapeutic combination
radiation therapy	therapeutic combination
anti-metabolite drug	therapeutic combination
methotrexate	therapeutic combination
6-mercapto purine	therapeutic combination
acepromazine	tranquilizer
alpha blockers	tranquilizer
alpha-adrenergic agonist	tranquilizer
antihistamine	tranquilizer
azapirone	tranquilizer
barbiturate	tranquilizer
benperidol	tranquilizer
benzamidine	tranquilizer
benzodiazepine	tranquilizer
beta-blocker	tranquilizer
bromantane	tranquilizer
bromperidol	tranquilizer
butyrophenone	tranquilizer
carbamates	tranquilizer
carpipramine	tranquilizer
chiorpromazine	tranquilizer
chiorprothixene	tranquilizer
clocapramine	tranquilizer
clopenthixol	tranquilizer
clorotepine	tranquilizer
cyamemazine	tranquilizer
diphenylbutylpiperidine	tranquilizer
dixyrazine	tranquilizer
droperidol	tranquilizer
emoxypine	tranquilizer
fabomotizole	tranquilizer
flupentixol	tranquilizer
fluphenazine	tranquilizer
fluspirilene	tranquilizer
gamma aminobutyric acid	tranquilizer
haloperidol	tranquilizer
inhalants	tranquilizer
levomepromazine	tranquilizer
loxapine	tranquilizer
mebicar	tranquilizer
mentyl isovalerate	tranquilizer
meso ridazine	tranquilizer
molindone	tranquilizer
monoamine oxidase inhibitors	tranquilizer
moperone	tranquilizer
mosapramine	tranquilizer
penfluridol	tranquilizer
perazine	tranquilizer
periciazine	tranquilizer
perphenazine	tranquilizer
phenothiazine	tranquilizer
pimozide	tranquilizer
pipamperone	tranquilizer
pipotiazine	tranquilizer
pregabalin	tranquilizer

TABLE 4-continued

THERAPEUTIC AGENTS	
Agent	Category
prochlorperazine	tranquilizer
promazine	tranquilizer
promethazine	tranquilizer
propofol	tranquilizer
prothipendyl	tranquilizer
racetam	tranquilizer
selank	tranquilizer
selective serotonin reuptake inhibitors	tranquilizer
serotonin-norepinephrine reuptake inhibitor	tranquilizer
sulpiride	tranquilizer
sultopride	tranquilizer
sympatholytic	tranquilizer
thioproperazine	tranquilizer
thioridazine	tranquilizer
thiothixene	tranquilizer
thioxanthene	tranquilizer
timiperone	tranquilizer
tricyclic	tranquilizer
trifluoperazine	tranquilizer
triflupromazine	tranquilizer
veralipride	tranquilizer
zuclopenthixol	tranquilizer
3-(4-Bromo-2,6-difluoro-benzyloxy)-5-[3-(4-pyrrolidin 1-yl-butyl)-ureido]-isothiazole-4-carboxylic acid amide hydrochloride	VEGF-related agent
5-((7- Benzyloxyquinazolin-4-yl) amino)-4-fluoro-2 -methyl phenol	VEGF-related agent
hydrochloride aflibercept	VEGF-related agent
AG-013958 (Pfizer Inc.)	VEGF-related agent
Angiogenesis inhibitor	VEGF-related agent
angiostatin	VEGF-related agent
angiozyme	VEGF-related agent
anti-VEGF antibody	VEGF-related agent
arresten	VEGF-related agent
AVASTIN ®	VEGF-related agent
axitinib	VEGF-related agent
bevacizumab	VEGF-related agent
canstatin	VEGF-related agent
cediranib	VEGF-related agent
combretastatin	VEGF-related agent
Combretastatin A4 Prodrug (CA4P)	VEGF-related agent
combstatin	VEGF-related agent
endogenous peptide	VEGF-related agent
EVIZON ™ (squalamine lactate)	VEGF-related agent
Fumagillin	VEGF-related agent
Fumagillin analogtie	VEGF-related agent
glufanide disodium	VEGF-related agent
JSM6427 (Jerini AG)	VEGF-related agent
LUCENTIS ®	VEGF-related agent
MACUGEN ®	VEGF-related agent
multitargeted human epidermal receptor (HER) 1/2 and vascular endothelial growth factor receptor (VEGFR) 1/2 receptor family tyrosine kinases inhibitor	VEGF-related agent
N-(4-bromo-2- fluorophenyl)-6-methoxy-7-[(1-methylpiperidin-4-yl)methoxy]quinazolin-4-amine	VEGF-related agent
N-(4-bromo-2-fluorophenyl)-6-methoxy-7-[(1-methylpiperidin-4-yl) methoxy]quinazol in-4-amine	VEGF-related agent
N,2-dimethyl-6-(2-(1-methyl-1H-imidazol-2- yl)thieno[3,2-b]pyridin-7-yloxy)benzo[b]thiophene-3-carboxamide	VEGF-related agent
pan-VEGF-R-kinase inhibitor	VEGF-related agent
pegaptanib	VEGF-related agent
protein kinase inhibitor	VEGF-related agent
ranibizumab	VEGF-related agent
shark cartilage	VEGF-related agent
shark cartilage derivative	VEGF-related agent
short interfering RNA (siRNA)	VEGF-related agent
siRNA-based VEGFR 1 inhibitor	VEGF-related agent
soluble ectodomain of the VEGP receptor	VEGF-related agent
sorafenib	VEGF-related agent
synthetic peptide	VEGF-related agent
thalidomide	VEGF-related agent
thalidomide derivative	VEGF-related agent
thrombospondin	VEGF-related agent
tivozanib	VEGF-related agent
toll-like receptor agonist	VEGF-related agent
tumstatin	VEGF-related agent
tyrosine kinase inhibitor of the RET/PTC oncogenic kinase	VEGF-related agent
vatalanib	VEGF-related agent
VEGF agonist	VEGF-related agent
VEGF antagonist	VEGF-related agent
VEGF nucleic acid ligand	VEGF-related agent
VEGF therapeutic agent	VEGF-related agent
VEGF-R1 inhibitor	VEGF-related agent
VEGF-R2 inhibitor	VEGF-related agent
VEGFR2-selective monoclonal antibody	VEGF-related agent
VEGF-Trap	VEGF-related agent
β2-glycoprotein 1	VEGF-related agent

Processed Silk

In some embodiments, SBP formulations may include processed silk as an active therapeutic component. The processed silk may include, but is not limited to one or more of silk fibroin, fragments of silk fibroin, chemically altered silk fibroin, and mutant silk fibroin. Therapeutic applications including such SBPs may include any of those taught in International Publication Number WO2017200659; Aykac et al. (2017) Gene s0378-1119(17)30865-8; and Abdel-Naby (2017) PLoS One 12(11):e0188154, the contents of each of which are herein incorporated by reference in their entirety. Processed silk may be administered as a therapeutic agent for treatment of a localized indication or for treatment of an indication further from the SBP application site. In some embodiments, therapeutic agents are combinations of processed silk and some other active component. In some embodiments, therapeutic agent activity requires cleavage or dissociation from silk. Therapeutic agents may include silk fibroin and/or chemically modified silk fibroin. In some embodiments, such therapeutic agents may be used to treat burn injury, inflammation, wound healing, or corneal injury. These and other treatments may be carried out according to any of the methods described in International Publication Number WO2017200659; United States Publication Number US20140235554; Aykac et al. (2017) Gene s0378-1119 (17)30868-30865; or Abdel-Naby (2017) PLoS One 12(11): e0188154, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, SBPs are silk fibroin solutions used to facilitate wound healing, as described in Park et al. (2017) Acta Biomater 67:183-195, the contents of which are herein incorporated by reference in their entirety. These SBPs may enhance wound healing via a nuclear factor kappa enhancer binding protein (NF-κB) signaling pathway. In some embodiments, SBPs are therapeutic agents used to facilitate delivery and/or release of therapeutic agent payloads. Such therapeutic agents and/or methods of use may include, but are not limited to, any of those described in International Publication Number WO2017139684, the contents of which are herein incorporated by reference in their entirety.

Lubricants

In some embodiments, processed silk and/or SBPs may be used as a lubricant. In some embodiments, processed silk may be selected base on or prepared to maximize its use as a lubricant. As used herein, the term "lubricant" refers to a substance that reduces the friction between two or more surfaces. In some embodiments, the surfaces in need of lubrication may be part of a subject. In some embodiments, surfaces in need of lubrication include, but are not limited to, the body, eyes, ears, skin, scalp, mouth, vagina, nose, hands, feet, and lips. In some embodiments, SBPs are used for ocular lubrication. As used herein, the term "ocular lubrication" refers to a method of the reduction of friction and/or irritation in the eye. In some embodiments, processed silk and/or SBPs may be used to reduce friction caused by dryness, as taught in U.S. Pat. No. 9,907,836 (the content of which is herein incorporated by reference in its entirety). This dryness may be dryness in the eye.

In some embodiments, the coefficient of friction of an SBP is approximately that of naturally occurring, biological, and/or protein lubricants (e.g. lubricin). In some embodiments, SBPs may display a coefficient of friction lower than that of water, as measured by the experimental sliding speeds. In some embodiments, the coefficient of friction of an SBP is slightly lower than that of water. In some embodiments, the SBP is more lubricating that water. In some embodiments, the SBP is slightly more lubricating that water. In some embodiments, SBPs may be incorporated into a lubricant. Such methods may include any of those presented in International Patent Application Publication No. WO2013163407, the contents of which are herein incorporated by reference in their entirety. In some embodiments, processed silk and/or SBPs may be used as an excipient. In some embodiments, processed silk and/or SBPs may be used as an excipient to prepare a lubricant.

Lubricants comprising SBPs may be prepared in any format described herein. Non-limiting examples include solutions, gels, hydrogels, creams, drops, and sprays.

In some embodiments, an SBP is a lubricant prepared in the format of a nasal spray.

In some embodiments, an SBP is a lubricant prepared in the format of eye drops.

In some embodiments, an SBP is a lubricant prepared in the format of ear drops.

Biological Agents

In some embodiments, therapeutic agents include biological agents (also referred to as "biologics" or "biologicals"). As used herein, a "biological agent" refers to a therapeutic substance that is or is derived from an organism or virus. Examples of biological agents include, but are not limited to, proteins, organic polymers and macromolecules, carbohydrates, complex carbohydrates, nucleic acids, cells, tissues, organs, organisms, DNA, RNA, oligonucleotides, genes, and lipids. Biological agents may include processed silk. In some embodiments, biological agents may include any of the biologicals and compounds associated with specific categories of biological agents listed in Table 4, above. In some embodiments, biological agents may include any of those taught in International Publication Numbers WO2010123945 or WO2017123383, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, SBP formulations may be used to deliver or administer biological agents. In some embodiments, delivery may include controlled release of one or more biological agents. Delivery may be carried out in vivo. In some embodiments, delivery is in vitro. Processed silk may be used to facilitate delivery and/or maintain stability of biological agents.

Antibodies

In some embodiments, SBP formulations may include one or more antibodies. As used herein, the term "antibody" refers to a class of immune proteins that bind to specific target antigens or epitopes. Herein, "antibody" is used in the broadest sense and embraces various natural and derivative formats that include, but are not limited to monoclonal antibodies, polyclonal antibodies, multispecific antibodies (e.g., bispecific antibodies that bind to two different epitopes), antibody conjugates (e.g., antibodies conjugates with therapeutic agents, cytotoxic agents, or detectable labels), antibody variants [e.g., antibody mimetics, chimeric antibodies (e.g., having components from two or more antibody types or species), and synthetic variants], and antibody fragments. Antibodies are typically amino acid-based but may include post-translational or synthetic modifications. In some embodiments, SBPs may be used to facilitate antibody delivery, as taught in International Publication Number WO2017139684 and Guziewicz et al. (2011) Biomaterials 32(10):2642-2650, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, SBPs may be used to improve antibody stability.

In some embodiments, antibodies are VEGF antagonist or agonists. Non-limiting examples of monoclonal antibody therapeutic agents include canakinumab, palivizumab, panitumumab, inflectra, adalimumab-atto, alemtuzumab, nivolumab, ustekinumab, alefacept, ixekizumab, obiltoxaxamab, golimumab, pembrolizumab, atezolizumab, tocilizumab, basiliximab, abciximab, denosumab, omalizumab, belimumab, efalizumab, natalizumab, ustekinumab, trastuzumab, bezlotoxumab, adalimumab, rituximab, daclizumab, secukinumab, cetuximab, reslizumab, olaratumab, ipilimumab, ixekizumab, certolizumab pegol, and daclizumab. In some embodiments, antibodies may include, but are not limited to, any of those listed in Table 4, above.

Antigens

In some embodiments, SBP formulations include antigens. As used herein, the term "antigen" refers to any substance capable of provoking an immune response. In some embodiments, antigens include processed silk. In some embodiments, antigens include any of those presented in Table 4, above. In some embodiments, SBPs may be used to facilitate antigen delivery. In some embodiments, SBPs may stabilize included antigens. In some embodiments, SBPs are or are included in vaccines. Vaccines that include processed silk and methods of using such vaccines may include any of those taught in United States Publication Number US20170258889 or in Zhang et al. (2012) PNAS 109(30):11981-6 (retracted), the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, formulation of an antigen with processed silk may be used to facilitate the delivery of said antigen in a vaccine, as taught in Zhang et al. (2012) PNAS 109(30):11981-6 (retracted).

Carbohydrates

In some embodiments, SBP formulations include carbohydrates. As used herein, the term "carbohydrate" refers to any members of a class of organic compounds that typically have carbon, oxygen, and hydrogen atoms and include, but are not limited to, simple and complex sugars. In some embodiments, carbohydrates may be monosaccharides or derivatives of a monosaccharides (e.g., ribose, glucose, fructose, galactose, mannose, abequose, arabinose, fucose, rhamnose, xylose, glucuronic acid, galactosamine, glucosamine, N-acetylgalactosamine, N-acetylglucosamine, iduronic acid, muramic acid, sialic acid, N-acetylmuramic acid, and N-acetylneuraminic acid). In some embodiments, carbohydrates may include disaccharides (e.g., sucrose, lactose, maltose, trehalose, and cellobiose). In some embodiments, carbohydrates are oligosaccharides or polysaccharides. In some embodiments, incorporation of carbohydrates may be used to stabilize SBPs. Such methods of use may include any of those taught in Li et al. (2017) Biomacromolecules 18(9):2900-5, the contents of which are herein incorporated by reference in their entirety. In some embodiments, carbohydrates may include, but are not limited to, any of those listed in Table 4, above.

Cells, Tissues, Microorganisms, and Microbiomes

In some embodiments, SBP formulations include cells, tissues, organs, and/or organisms. In some embodiments, such agents are used for direct treatment. In other embodiments, cell- or tissue-based therapeutic agents are incorporated into SBPs to prepare model systems. Such methods may include any of those taught in International Publication Number WO2017189832; Chen et al. (2017) PLoS One, 12(11):e0187880; or Chen et al. (2017) Stem Cell Research and Therapy 8:260, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, incorporated cells are stem cells (e.g., see International Publication Number WO2017189832; Chendang et al. (2017) J Biomaterials and Tissue Engineering 7:858-862; Xiao et al. (2017) Oncotarget 8(49):86471-89487; Ciocci et al. (2017) Int J Biol Macromol 50141-8130(17):32839-8; Li et al. (2017) Stem Cell Res Ther 8(1):256; or Ruan et al. (2017) Biomed Pharmacother 97:600-6, the contents of each of which are herein incorporated by reference in their entirety). Examples of cell- or tissue-based therapeutic agents include, but are not limited to, human corneal stromal stem cells, human corneal epithelial cells, chicken dorsal root ganglions, bone mesenchymal stem cells, limbal epithelial stem cells, cardiac mesenchymal stem cells, adipose tissue-derived mesenchymal stem cells, periodontal ligament stem cells, human small intestinal enteroids, oral keratinocytes, fibroblasts, transfected fibroblasts, any 2-dimensional tissue, and any 3-dimensional tissue, T cells, embryonic stem cells, neural stem cells, mesenchymal stem cells, and hematopoietic stem cells. In some embodiments, cells used as therapeutic agents may include, but are not limited to, any of those listed in Table 4, above.

In some embodiments, therapeutic agents include bacteria or other microorganisms. Such therapeutic agents may be used to alter a microbiome. Examples of bacteria or other microorganisms that may be used as therapeutic agents in SBPs include any of those described in US Patent Numbers U.S. Pat. Nos. 9,688,967 and 9,688,967; US Publication Numbers US20170136073, US20170128499, US20160206666, US20170067065, and US20170014457; and International Publication Numbers WO2017123676, WO2017123675, WO2017123610, WO2017123592, WO2017123418, WO2016210384, WO2017075485, WO2017023818, WO2016210373, WO2017040719, WO2016210378, and WO2016106343, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, SBP formulations include cellular therapeutics, such as bacteria and/or other microorganisms. As used herein, the term "microorganism" refers to a microscopic living thing (e.g. bacteria and/or fungi). In some embodiments, SBPs may be used to deliver cellular therapeutics (e.g., bacteria and/or other microorganisms) to alter or improve the microbiome of a subject or patient. In some embodiments, bacteria and/or other microorganisms used as therapeutic agents may include, but are not limited to, any of those described in US Patent Numbers 9,688,967, or 9,688,967; in US Patent Publication Numbers US20170136073, US20170128499, US20160206666, US20170067065, or US20170014457; or in International Publication Numbers WO2017123676, WO2017123675, WO2017123610, WO2017123592, WO2017123418, WO2016210384, WO2017075485, WO2017023818, WO2016210373, WO2017040719, WO2016210378, WO2016200614, WO2017087580, or WO2016106343, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, said bacteria and/or microorganisms are formulated as a part of SBPs. In some embodiments, the bacteria and/or microorganisms may be supported during delivery using SBPs. In some embodiments, bacteria and/or other microorganisms used as therapeutic agents may be engineered, e.g., by any method described in the US Patent Numbers 9,688,967 or 9,487,764; or in International Publication Numbers WO2016200614 and WO2017087580, the contents of each of which are herein incorporated by reference in their entirety.

In some embodiments, SBPs described herein maintain and/or improve the stability of bacteria and/or other microorganisms. The maintenance and/or improvement of stability may be determined by comparing stability with SBP compositions to stability with compositions lacking SBPs or to standard compositions in the art. Maintenance and/or improvement of stability may be found or appreciated where superior or durational benefits are observed with SBPs. In some embodiments, SBPs maintain and/or improve the stability of bacteria and/or other microorganisms that can be used in bacterial or microbial therapy.

In some embodiments, bacteria and/or other microorganisms may be used as biopesticides. As used herein, the term "biopesticide" refers to a composition with a bacteria, microorganisms, or biological cargo used to harm, kill, or prevent the spread of pests. Biopesticides have been used in agricultural development, as described in U.S. Pat. No. 6,417,163, the contents of which are herein incorporated by reference in their entirety. In some embodiments, SBPs that include bacteria, microorganisms, and/or microbiomes, may be used as biopesticides to support agricultural applications.

In some embodiments, bacteria and/or other microorganisms formulated as a part of SBPs may include one or more of the following microbes: *Abiotrophia, Abiotrophia defectiva, Acetanaerobacterium, Acetanaerobacterium elongatum, Acetivibrio, Acetivibrio bacterium, Acetobacterium, Acetobacterium woodii, Acholeplasma, Acidaminococcus, Acidaminococcus fermentans, Acidianus, Acidianus brierleyi, Acidovorax, Acinetobacter, Acinetobacter guillouiae, Acinetobacter junii, Actinobacillus, Actinobacillus M*1933/96/1, *Actinomyces, Actinomyces* ICM34, *Actinomyces* ICM41, *Actinomyces* ICM54, *Actinomyces lingnae, Actinomyces odontolyticus, Actinomyces oral, Actinomyces phi, Adlercreutzia, Adlercreutzia equolifaciens, Adlercreutzia intestinal, Aerococcus, Aeromonas, Aeromonas* 165C, *Aeromonas hydrophila, Aeromonas* RC50, *Aeropyrum, Aeropyrum pernix, agglomerans, Aggregatibacter, Agreia, Agreia bicolorata, Agromonas, Agromonas* CS30, *Akkermansia, Akkermansia muciniphila, Alistipes, Alistipes* ANH, *Alistipes* AP11, *Alistipes bacterium, Alistipes* CCUG, *Alistipes* DJF B185, *Alistipes* DSM, *Alistipes* EBA6-25cl2, *Alistipes finegoldii, Alistipes indistinctus, Alistipes* JC136, *Alistipes* NML05A004, *Alistipes onderdonkii, Alistipes putredinis, Alistipes* RMA, *Alistipes senegalensis, Alistipes shahii, Alis-*

tipes Smarlab, Alkalibaculum, Alkaliflexus, Allisonella, Allisonella histaminiformans, Alloscardovia, Alloscardovia omnicolens, Anaerofilum, Anaerofustis, Anaerofustis stercorihominis, Anaeroplasma, Anaerostipes, Anaerostipes 08964, *Anaerostipes* 494a, *Anaerostipes* 5_1_63FAA, *Anaerostipes* AIP, *Anaerostipes bacterium, Anaerostipes butyraticus, Anaerostipes caccae, Anaerostipes hadrum, Anaerostipes* IE4, *Anaerostipes indolis, Anaerostipes* ly-2, *Anaerotruncus, Anaerotruncus colihominis, Anaerotruncus* NML, *Aquincola, Arcobacter, Arthrobacter, Arthrobacter* FVI-1, *Asaccharobacter, Asaccharobacter celatus, Asteroleplasma, Atopobacter, Atopobacter phocae, Atopobium, Atopobium parvulum, Atopobium rimae, Bacteriovorax, Bacteroides, Bacteroides* 31SF18, *Bacteroides* 326-8, *Bacteroides* 35AE31, *Bacteroides* 35AE37, *Bacteroides* 35BE34, *Bacteroides* 4072, *Bacteroides* 7853, *Bacteroides acidifaciens, Bacteroides* API, *Bacteroides* AR20, *Bacteroides* AR29, *Bacteroides* B2, *Bacteroides bacterium, Bacteroides barnesiae, Bacteroides* BLBE-6, *Bacteroides* BV-1, *Bacteroides caccae, Bacteroides* CannelCatfish9, *Bacteroides cellulosilyticus, Bacteroides chinchillae, Bacteroides* CIP 103040, *Bacteroides clarus, Bacteroides coprocola, Bacteroides coprophilus, Bacteroides* D8, *Bacteroides* DJF_B097, *Bacteroides* dnLKV2, *Bacteroides* dnLKV7, *Bacteroides* dnLKV9, *Bacteroides dorei, Bacteroides* EBA5-17, *Bacteroides eggerthii, Bacteroides enrichment, Bacteroides* F-4, *Bacteroides faecichinchillae, Bacteroides faecis, Bacteroides fecal, Bacteroides finegoldii, Bacteroides fragilis, Bacteroides gallinarum, Bacteroides helcogenes, Bacteroides* icl292, *Bacteroides intestinalis, Bacteroides massiliensis, Bacteroides mpnisolate, Bacteroides* NB-8, *Bacteroides* new, *Bacteroides* NLAE-zl-c204, *Bacteroides* NLAE-zl-c205, *Bacteroides* NLAE-zl-c206, *Bacteroides* NLAE-zl-c207, *Bacteroides* NLAE-zl-c211, *Bacteroides* NLAE-zl-c218, *Bacteroides* NLAE-zl-c257, *Bacteroides* NLAE-zl-c260, *Bacteroides* NLAE-zl-c261, *Bacteroides* NLAE-zl-c263, *Bacteroides* NLAE-zl-c308, *Bacteroides* NLAE-zl-c315, *Bacteroides* NLAE-zl-c322, *Bacteroides* NLAE-zl-c324, *Bacteroides* NLAE-zl-c331, *Bacteroides* NLAE-zl-c339, *Bacteroides* NLAE-zl-c36, *Bacteroides* NLAE-zl-c367, *Bacteroides* NLAE-zl-c375, *Bacteroides* NLAE-zl-c376, *Bacteroides* NLAE-zl-c380, *Bacteroides* NLAE-zl-c391, *Bacteroides* NLAE-zl-c459, *Bacteroides* NLAE-zl-c484, *Bacteroides* NLAE-zl-c501, *Bacteroides* NLAE-zl-c504, *Bacteroides* NLAE-zl-c515, *Bacteroides* NLAE-zl-c519, *Bacteroides* NLAE-zl-c532, *Bacteroides* NLAE-zl-c557, *Bacteroides* NLAE-zl-c57, *Bacteroides* NLAE-zl-c574, *Bacteroides* NLAE-zl-c592, *Bacteroides* NLAE-zl-cl3, *Bacteroides* NLAE-zl-cl58, *Bacteroides* NLAE-zl-cl59, *Bacteroides* NLAE-zl-cl61, *Bacteroides* NLAE-zl-cl63, *Bacteroides* NLAE-zl-cl67, *Bacteroides* NLAE-zl-cl72, *Bacteroides* NLAE-zl-cl8, *Bacteroides* NLAE-zl-cl82, *Bacteroides* NLAE-zl-cl90, *Bacteroides* NLAE-zl-cl98, *Bacteroides* NLAE-zl-g209, *Bacteroides* NLAE-zl-g212, *Bacteroides* NLAE-zl-g213, *Bacteroides* NLAE-zl-g218, *Bacteroides* NLAE-zl-g221, *Bacteroides* NLAE-zl-g228, *Bacteroides* NLAE-zl-g234, *Bacteroides* NLAE-zl-g237, *Bacteroides* NLAE-zl-g24, *Bacteroides* NLAE-zl-g245, *Bacteroides* NLAE-zl-g257, *Bacteroides* NLAE-zl-g27, *Bacteroides* NLAE-zl-g285, *Bacteroides* NLAE-zl-g288, *Bacteroides* NLAE-zl-g295, *Bacteroides* NLAE-zl-g296, *Bacteroides* NLAE-zl-g303, *Bacteroides* NLAE-zl-g310, *Bacteroides* NLAE-zl-g312, *Bacteroides* NLAE-zl-g327, *Bacteroides* NLAE-zl-g329, *Bacteroides* NLAE-zl-g336, *Bacteroides* NLAE-zl-g338, *Bacteroides* NLAE-zl-g347, *Bacteroides* NLAE-zl-g356, *Bacteroides* NLAE-zl-g373, *Bacteroides* NLAE-zl-g376, *Bacteroides* NLAE-zl-g380, *Bacteroides* NLAE-zl-g382, *Bacteroides* NLAE-zl-g385, *Bacteroides* NLAE-zl-g4, *Bacteroides* NLAE-zl-g422, *Bacteroides* NLAE-zl-g437, *Bacteroides* NLAE-zl-g454, *Bacteroides* NLAE-zl-g455, *Bacteroides* NLAE-zl-g456, *Bacteroides* NLAE-zl-g458, *Bacteroides* NLAE-zl-g459, *Bacteroides* NLAE-zl-g46, *Bacteroides* NLAE-zl-g461, *Bacteroides* NLAE-zl-g475, *Bacteroides* NLAE-zl-g481, *Bacteroides* NLAE-zl-g484, *Bacteroides* NLAE-zl-g5, *Bacteroides* NLAE-zl-g502, *Bacteroides* NLAE-zl-g515, *Bacteroides* NLAE-zl-g518, *Bacteroides* NLAE-zl-g521, *Bacteroides* NLAE-zl-g54, *Bacteroides* NLAE-zl-g6, *Bacteroides* NLAE-zl-g8, *Bacteroides* NLAE-zl-g80, *Bacteroides* NLAE-zl-g98, *Bacteroides* NLAE-zl-gl17, *Bacteroides* NLAE-zl-gl05, *Bacteroides* NLAE-zl-gl27, *Bacteroides* NLAE-zl-gl36, *Bacteroides* NLAE-zl-gl43, *Bacteroides* NLAE-zl-gl57, *Bacteroides* NLAE-zl-g167, *Bacteroides* NLAE-zl-gl71, *Bacteroides* NLAE-zl-gl87, *Bacteroides* NLAE-zl-gl94, *Bacteroides* NLAE-zl-gl95, *Bacteroides* NLAE-zl-gl99, *Bacteroides* NLAE-zl-h207, *Bacteroides* NLAE-zl-h22, *Bacteroides* NLAE-zl-h250, *Bacteroides* NLAE-zl-h251, *Bacteroides* NLAE-zl-h28, *Bacteroides* NLAE-zl-h313, *Bacteroides* NLAE-zl-h319, *Bacteroides* NLAE-zl-h321, *Bacteroides* NLAE-zl-h328, *Bacteroides* NLAE-zl-h334, *Bacteroides* NLAE-zl-h390, *Bacteroides* NLAE-zl-h391, *Bacteroides* NLAE-zl-h414, *Bacteroides* NLAE-zl-h416, *Bacteroides* NLAE-zl-h419, *Bacteroides* NLAE-zl-h429, *Bacteroides* NLAE-zl-h439, *Bacteroides* NLAE-zl-h444, *Bacteroides* NLAE-zl-h45, *Bacteroides* NLAE-zl-h46, *Bacteroides* NLAE-zl-h462, *Bacteroides* NLAE-zl-h463, *Bacteroides* NLAE-zl-h465, *Bacteroides* NLAE-zl-h468, *Bacteroides* NLAE-zl-h471, *Bacteroides* NLAE-zl-h472, *Bacteroides* NLAE-zl-h474, *Bacteroides* NLAE-zl-h479, *Bacteroides* NLAE-zl-h482, *Bacteroides* NLAE-zl-h49, *Bacteroides* NLAE-zl-h493, *Bacteroides* NLAE-zl-h496, *Bacteroides* NLAE-zl-h497, *Bacteroides* NLAE-zl-h499, *Bacteroides* NLAE-zl-h50, *Bacteroides* NLAE-zl-h531, *Bacteroides* NLAE-zl-h535, *Bacteroides* NLAE-zl-h8, *Bacteroides* NLAE-zl-h120, *Bacteroides* NLAE-zl-h15, *Bacteroides* NLAE-zl-h162, *Bacteroides* NLAE-zl-h17, *Bacteroides* NLAE-zl-h174, *Bacteroides* NLAE-zl-h18, *Bacteroides* NLAE-zl-h188, *Bacteroides* NLAE-zl-h192, *Bacteroides* NLAE-zl-h194, *Bacteroides* NLAE-zl-h195, *Bacteroides* NLAE-zl-p208, *Bacteroides* NLAE-zl-p213, *Bacteroides* NLAE-zl-p228, *Bacteroides* NLAE-zl-p233, *Bacteroides* NLAE-zl-p267, *Bacteroides* NLAE-zl-p278, *Bacteroides* NLAE-zl-p282, *Bacteroides* NLAE-zl-p286, *Bacteroides* NLAE-zl-p295, *Bacteroides* NLAE-zl-p299, *Bacteroides* NLAE-zl-p301, *Bacteroides* NLAE-zl-p302, *Bacteroides* NLAE-zl-p304, *Bacteroides* NLAE-zl-p317, *Bacteroides* NLAE-zl-p319, *Bacteroides* NLAE-zl-p32, *Bacteroides* NLAE-zl-p332, *Bacteroides* NLAE-zl-p349, *Bacteroides* NLAE-zl-p35, *Bacteroides* NLAE-zl-p356, *Bacteroides* NLAE-zl-p370, *Bacteroides* NLAE-zl-p371, *Bacteroides* NLAE-zl-p376, *Bacteroides* NLAE-zl-p395, *Bacteroides* NLAE-zl-p402, *Bacteroides* NLAE-zl-p403, *Bacteroides* NLAE-zl-p409, *Bacteroides* NLAE-zl-p412, *Bacteroides* NLAE-zl-p436, *Bacteroides* NLAE-zl-p438, *Bacteroides* NLAE-zl-p440, *Bacteroides* NLAE-zl-p447, *Bacteroides* NLAE-zl-p448, *Bacteroides* NLAE-zl-p451, *Bacteroides* NLAE-zl-p476, *Bacteroides* NLAE-zl-p478, *Bacteroides* NLAE-zl-p483, *Bacteroides* NLAE-zl-p489, *Bacteroides* NLAE-zl-p493, *Bacteroides* NLAE-zl-p557, *Bacteroides* NLAE-zl-p559, *Bacteroides* NLAE-zl-p564, *Bacteroides* NLAE-zl-p565, *Bacteroides* NLAE-zl-p572, *Bacteroides* NLAE-zl-p573, *Bacteroides* NLAE-zl-p576, *Bacteroides* NLAE-zl-p591, *Bacteroides* NLAE-zl-p592, *Bacteroides*

NLAE-zl-p631, *Bacteroides* NLAE-zl-p633, *Bacteroides* NLAE-zl-p696, *Bacteroides* NLAE-zl-p7, *Bacteroides* NLAE-zl-p720, *Bacteroides* NLAE-zl-p730, *Bacteroides* NLAE-zl-p736, *Bacteroides* NLAE-zl-p737, *Bacteroides* NLAE-zl-p754, *Bacteroides* NLAE-zl-p759, *Bacteroides* NLAE-zl-p774, *Bacteroides* NLAE-zl-p828, *Bacteroides* NLAE-zl-p854, *Bacteroides* NLAE-zl-p860, *Bacteroides* NLAE-zl-p886, *Bacteroides* NLAE-zl-p887, *Bacteroides* NLAE-zl-p900, *Bacteroides* NLAE-zl-p909, *Bacteroides* NLAE-zl-p913, *Bacteroides* NLAE-zl-p916, *Bacteroides* NLAE-zl-p920, *Bacteroides* NLAE-zl-p96, *Bacteroides* NLAE-zl-pl04, *Bacteroides* NLAE-zl-pl05, *Bacteroides* NLAE-zl-pl08, *Bacteroides* NLAE-zl-pl32, *Bacteroides* NLAE-zl-pl33, *Bacteroides* NLAE-zl-pl51, *Bacteroides* NLAE-zl-pl57, *Bacteroides* NLAE-zl-pl66, *Bacteroides* NLAE-zl-p167, *Bacteroides* NLAE-zl-pl71, *Bacteroides* NLAE-zl-pl78, *Bacteroides* NLAE-zl-pl87, *Bacteroides* NLAE-zl-pl91, *Bacteroides* NLAE-zl-pl96, *Bacteroides nordii, Bacteroides oleiciplenus, Bacteroides ovatus, Bacteroides paurosaccharolyticus, Bacteroides plebeius, Bacteroides* R6, *Bacteroides rodentium, Bacteroides* S-17, *Bacteroides* S-18, *Bacteroides salyersiae, Bacteroides* SLC1-38, *Bacteroides* Smarlab, *Bacteroides* 'Smarlab, *Bacteroides stercorirosoris, Bacteroides stercoris, Bacteroides* str, *Bacteroides thetaiotaomicron, Bacteroides* TP-5, *Bacteroides uniformis, Bacteroides vulgatus, Bacteroides* WA1, *Bacteroides* WH2, *Bacteroides* WH302, *Bacteroides* WH305, *Bacteroides* X077B42, *Bacteroides* XB12B, *Bacteroides* XB44A, *Bacteroides xylanisolvens, Barnesiella, Barnesiella intestinihominis, Barnesiella* NSB1, *Barnesiella viscericola, Bavariicoccus, Bdellovibrio, Bdellovibrio oral, Bergeriella, Bifidobacterium, Bifidobacterium* 103, *Bifidobacterium* 108, *Bifidobacterium* 113, *Bifidobacterium* 120, *Bifidobacterium* 138, *Bifidobacterium* 33, *Bifidobacterium* Acbbto5, *Bifidobacterium adolescentis, Bifidobacterium* Amsbbtl2, *Bifidobacterium angulatum, Bifidobacterium animalis, Bifidobacterium bacterium, Bifidobacterium bifidum, Bifidobacterium* Bisn6, *Bifidobacterium* Bma6, *Bifidobacterium breve, Bifidobacterium catenulatum, Bifidobacterium choerinum, Bifidobacterium coryneforme, Bifidobacterium dentium, Bifidobacterium* DJF_WC44, *Bifidobacterium* F-10, *Bifidobacterium* F-11, *Bifidobacterium* group, *Bifidobacterium* hl2, *Bifidobacterium* HMLN1, *Bifidobacterium* HMLN12, *Bifidobacterium* HMLNS, *Bifidobacterium* iarfr2341d, *Bifidobacterium* iarfr642d48, *Bifidobacterium* icl332, *Bifidobacterium indicum, Bifidobacterium kashiwanohense, Bifidobacterium* LISLUCIII-2, *Bifidobacterium longum, Bifidobacterium* M45, *Bifidobacterium merycicum, Bifidobacterium minimum, Bifidobacterium* MSX5B, *Bifidobacterium oral, Bifidobacterium* PG12A, *Bifidobacterium* PL1, *Bifidobacterium pseudocatenulatum, Bifidobacterium pseudolongum, Bifidobacterium pullorum, Bifidobacterium ruminantium, Bifidobacterium* S-10, *Bifidobacterium saeculare, Bifidobacterium saguini, Bifidobacterium scardovii, Bifidobacterium simiae, Bifidobacterium* SLPYG-1, *Bifidobacterium stercoris, Bifidobacterium* TM-7, *Bifidobacterium* Trm9, *Bilophila, Bilophila* NLAE-zl-h528, *Bilophila wadsworthia, Blautia, Blautia bacterium, Blautia* CE2, *Blautia* CE6, *Blautia coccoides, Blautia* DJF_VR52, *Blautia* DJF_VR67, *Blautia* DJF_VR70kl, *Blautia formate, Blautia glucerasea, Blautia hansenii, Blautia* icl272, *Blautia* 1E5, *Blautia* K-1, *Blautia luti, Blautia* M-1, *Blautia mpnisolate, Blautia* NLAE-zl-c25, *Blautia* NLAE-zl-c259, *Blautia* NLAE-zl-c51, *Blautia* NLAE-zl-c520, *Blautia* NLAE-zl-c542, *Blautia* NLAE-zl-c544, *Blautia* NLAE-zl-h27, *Blautia* NLAE-zl-h316, *Blautia* NLAE-zl-h317, *Blautia obeum, Blautia producta, Blautia productus, Blautia schinkii, Blautia* Ser5, *Blautia* Ser8, *Blautia* WAL, *Blautia wexlerae, Blautia* YHC-4, *Brenneria, Brevibacterium, Brochothrix, Brochothrix thermosphacta, Buttiauxella, Buttiauxella* 57916, *Buttiauxella gaviniae, Butyricicoccus, Butyricicoccus bacterium, Butyricimonas, Butyricimonas* 180-3, *Butyricimonas* 214-4, *Butyricimonas bacterium, Butyricimonas* GD2, *Butyricimonas synergistica, Butyricimonas virosa, Butyrivibrio, Butyrivibrio fibrisolvens, Butyrivibrio hungatei, Caldimicrobium, Caldisericum, Campylobacter, Campylobacter coli, Campylobacter hominis, Capnocytophaga, Carnobacterium, Carnobacterium alterfunditum, Caryophanon, Catenibacterium, Catenibacterium mitsuokai, Catonella, Caulobacter, Cellulophaga, Cellulosilyticum, Cetobacterium, Chelatococcus, Chlorobium, Chryseobacterium, Chryseobacterium* A1005, *Chryseobacterium* KJ9C8, *Citrobacter, Citrobacter* 1, *Citrobacter* 191-3, *Citrobacter agglomerans, Citrobacter amalonaticus, Citrobacter ascorbata, Citrobacter bacterium, Citrobacter* BinzhouCLT, *Citrobacter braakii, Citrobacter enrichment, Citrobacter* F24, *Citrobacter* F96, *Citrobacter farmeri, Citrobacter freundii, Citrobacter gillenii, Citrobacter* HBKC_SR1, *Citrobacter* HD4.9, *Citrobacter hormaechei, Citrobacter* ka55, *Citrobacter lapagei, Citrobacter* LAR-1, *Citrobacter ludwigii, Citrobacter* MEBS, *Citrobacter* MS36, *Citrobacter murliniae, Citrobacter* NLAE-zl-c269, *Citrobacter* P014, *Citrobacter* P042bN, *Citrobacter* P046a, *Citrobacter* P073, *Citrobacter* SR3, *Citrobacter* Tl, *Citrobacter* tnt4, *Citrobacter* tnt5, *Citrobacter* trout, *Citrobacter* TSA-1, *Citrobacter werkmanii, Cloacibacillus, Cloacibacillus* adv66, *Cloacibacillus* NLAE-zl-p702, *Cloacibacillus* NMLO5A017, *Cloacibacterium, Collinsella, Collinsella aerofaciens, Collinsella* A-1, *Collinsella* AUH-Julong21, *Collinsella bacterium, Collinsella* CCUG, *Comamonas, Comamonas straminea, Comamonas testosteroni, Conexibacter, Coprobacillus, Coprobacillus bacterium, Coprobacillus cateniformis, Coprobacillus* TM-40, *Coprococcus, Coprococcus* 14505, *Coprococcus bacterium, Coprococcus catus, Coprococcus comes, Coprococcus eutactus, Coprococcus nexile, Coraliomargarita, Coraliomargarita fucoidanolyticus, Coraliomargarita marisfiavi, Corynebacterium, Corynebacterium amycolatum, Corynebacterium durum, Coxiella, Cronobacter, Cronobacter dublinensis, Cronobacter sakazakii, Cronobacter turicensis, Cryptobacterium, Cryptobacterium curtum, Cupriavidus, Cupriavidus eutropha, Dechloromonas, Dechloromonas HZ, Desulfobacterium, Desulfobulbus, Desulfopila, Desulfopila* La4.1, *Desulfovibrio, Desulfovibrio* D4, *Desulfovibrio desulfuricans, Desulfovibrio* DSM 12803, *Desulfovibrio enrichment, Desulfovibrio fairfieldensis, Desulfovibrio* LNB1, *Desulfovibrio piger, Dialister, Dialister* E2_20, *Dialister* GBA27, *Dialister invisus, Dialister oral, Dialister succinatiphilus, Dorea, Dorea* auhjulong64, *Dorea bacterium, Dorea formicigenerans, Dorea longicatena, Dorea mpnisolate, Dysgonomonas, Dysgonomonas gadei, Edwardsiella, Edwardsiella tarda, Eggerthella, Eggerthella* El, *Eggerthella lenta, Eggerthella* MLG043, *Eggerthella* MVA1, *Eggerthella* S6-C1, *Eggerthella* SDG-2, *Eggerthella sinensis, Eggerthella* str, *Enhydrobacter, Enterobacter, Enterobacter* 1050, *Enterobacter* 112, *Enterobacter* 1122, *Enterobacter* 77000, *Enterobacter* 82353, *Enterobacter* 9C, *Enterobacter* ASC, *Enterobacter adecarboxylata, Enterobacter aerogenes, Enterobacter agglomerans, Enterobacter* AJAR-A2, *Enterobacter amnigenus, Enterobacter asburiae, Enterobacter* B 1(2012), *Enterobacter* B363, *Enterobacter* B509, *Enterobacter bacterium, Enterobacter* Badong3, *Enterobacter* BEC441, *Enterobacter* C8, *Enterobacter canceroge-*

nus, Enterobacter cloacae, Enterobacter CO, *Enterobacter* core2, *Enterobacter cowanii, Enterobacter* dc6, *Enterobacter* DRSBII, *Enterobacter* enrichment, *Enterobacter* FL13-2-1, *Enterobacter* GIST-NKst9, *Enterobacter* GIST-NKst10, *Enterobacter* GJ1-11, *Enterobacter* gx-148, *Enterobacter hormaechei, Enterobacter* I-Bh20-21, *Enterobacter* ICB113, *Enterobacter kobei, Enterobacter* KW14, *Enterobacter ludwigii, Enterobacter* M10_1B, *Enterobacter* M1R3, *Enterobacter marine, Enterobacter* NCCP-167, *Enterobacter* of *Enterobacter oryzae, Enterobacter oxytoca, Enterobacter* P101, *Enterobacter* SEL2, *Enterobacter* SI 1, *Enterobacter* SPh, *Enterobacter* SSASP5, *Enterobacter terrigena, Enterobacter* TNT3, *Enterobacter* TP2MC, *Enterobacter* TS4, *Enterobacter* TSSAS2-48, *Enterobacter* ZYXCA1, *Enterococcus, Enterococcus* 020824/02-A, *Enterococcus* 1275b, *Enterococcus* 16C, *Enterococcus* 48, *Enterococcus* 6114, *Enterococcus* ABRIINW-H61, *Enterococcus asini, Enterococcus avium, Enterococcus azikeevi, Enterococcus bacterium, Enterococcus* BBDP57, *Enterococcus* BPH34, *Enterococcus* Bt, *Enterococcus canis, Enterococcus casselif avus, Enterococcus* CmNA2, *Enterococcus* Da-20, *Enterococcus devriesei, Enterococcus dispar, Enterococcus* DJF_030, *Enterococcus* DMB4, *Enterococcus durans, Enterococcus* enrichment, *Enterococcus* F81, *Enterococcus faecalis, Enterococcus faecium, Enterococcus* fcc9, *Enterococcus fecal, Enterococcus flavescens, Enterococcus fluvialis, Enterococcus* FR-3, *Enterococcus* FUA3374, *Enterococcus gallinarum, Enterococcus* GSC-2, *Enterococcus* GYPB01, *Enterococcus hermanniensis, Enterococcus hirae, Enterococcus lactis, Enterococcus malodoratus, Enterococcus manure, Enterococcus marine, Enterococcus* MNC1, *Enterococcus moraviensis, Enterococcus* MS2, *Enterococcus mundtii, Enterococcus* NAB 15, *Enterococcus* NBRC, *Enterococcus* NLAE-zl-c434, *Enterococcus* NLAE-zl-g87, *Enterococcus* NLAE-zl-gl06, *Enterococcus* NLAE-zl-h339, *Enterococcus* NLAE-zl-h375, *Enterococcus* NLAE-zl-h381, *Enterococcus* NLAE-zl-h383, *Enterococcus* NLAE-zl-h405, *Enterococcus* NLAE-zl-p401, *Enterococcus* NLAE-zl-p650, *Enterococcus* NLAE-zl-pl16, *Enterococcus* NLAE-zl-pl48, *Enterococcus pseudoavium, Enterococcus* R-25205, *Enterococcus raffinosus, Enterococcus rottae, Enterococcus* RU07, *Enterococcus saccharolyticus, Enterococcus saccharominimus, Enterococcus sanguinicola, Enterococcus* SCA16, *Enterococcus* SCA2, *Enterococcus* SE138, *Enterococcus* SF-1, *Enterococcus sulfureus, Enterococcus* SV6, *Enterococcus* te32a, *Enterococcus* te42a, *Enterococcus* te45r, *Enterococcus* te49a, *Enterococcus* te51a, *Enterococcus* te58r, *Enterococcus* te59r, *Enterococcus* te61r, *Enterococcus* te93r, *Enterococcus* te95a, *Enterococcus tela, Enterorhabdus, Enterorhabdus caecimuris, entomophaga, Erwinia, Erwinia agglomerans, Erwinia enterica, Erwinia rhapontici, Erwinia tasmaniensis, Erysipelotrichaceae incertae sedis, Erysipelotrichaceae incertae sedis aff Erysipelotrichaceae incertae sedis bacterium, Erysipelotrichaceae incertae sedis biforme, Erysipelotrichaceae incertae sedis* C-1, *Erysipelotrichaceae incertae sedis cylindroides, Erysipelotrichaceae incertae sedis* GK12, *Erysipelotrichaceae incertae sedis innocuum, Erysipelotrichaceae incertae sedis* NLAE-zl-c332, *Erysipelotrichaceae incertae sedis* NLAE-zl-c340, *Erysipelotrichaceae incertae sedis* NLAE-zl-g420, *Erysipelotrichaceae incertae sedis* NLAE-zl-g425, *Erysipelotrichaceae incertae sedis* NLAE-zl-g440, *Erysipelotrichaceae incertae sedis* NLAE-zl-g463, *Erysipelotrichaceae incertae sedis* NLAE-zl-h340, *Erysipelotrichaceae incertae sedis* NLAE-zl-h354, *Erysipelotrichaceae incertae sedis* NLAE-zl-h379, *Erysipelotrichaceae incertae sedis* NLAE-zl-h380, *Erysipelotrichaceae incertae sedis* NLAE-zl-h385, *Erysipelotrichaceae incertae sedis* NLAE-zl-h410, *Erysipelotrichaceae incertae sedis tortuosum, Escherichia/Shigella, Escherichia/Shigella* 29(2010), *Escherichia/Shigella* 4091, *Escherichia/Shigella* 4104, *Escherichia/Shigella* 8gw18, *Escherichia/Shigella* A94, *Escherichia/Shigella albertii, Escherichia/Shigella* B-1012, *Escherichia/Shigella* B4, *Escherichia/Shigella bacterium, Escherichia/Shigella* BBDP15, *Escherichia/Shigella* BBDP80, *Escherichia/Shigella boydii, Escherichia/Shigella carotovorum, Escherichia/Shigella* CERAR, *Escherichia/Shigella coli, Escherichia/Shigella* DBC-1, *Escherichia/Shigella* dc262011, *Escherichia/Shigella dysenteriae, Escherichia/Shigella* enrichment, *Escherichia/Shigella escherichia, Escherichia/Shigella* fecal, *Escherichia/Shigella fergusonii, Escherichia/Shigella flexneri, Escherichia/Shigella* GDR05, *Escherichia/Shigella* GDR07, *Escherichia/Shigella* H7, *Escherichia/Shigella* marine, *Escherichia/Shigella* ML2-46, *Escherichia/Shigella* mpnisolate, *Escherichia/Shigella* NA, *Escherichia/Shigella* NLAE-zl-g330, *Escherichia/Shigella* NLAE-zl-g400, *Escherichia/Shigella* NLAE-zl-g441, *Escherichia/Shigella* NLAE-zl-g506, *Escherichia/Shigella* NLAE-zl-h204, *Escherichia/Shigella* NLAE-zl-h208, *Escherichia/Shigella* NLAE-zl-h209, *Escherichia/Shigella* NLAE-zl-h213, *Escherichia/Shigella* NLAE-zl-h214, *Escherichia/Shigella* NLAE-zl-h4, *Escherichia/Shigella* NLAE-zl-h435, *Escherichia/Shigella* NLAE-zl-h81, *Escherichia/Shigella* NLAE-zl-p21, *Escherichia/Shigella* NLAE-zl-p235, *Escherichia/Shigella* NLAE-zl-p237, *Escherichia/Shigella* NLAE-zl-p239, *Escherichia/Shigella* NLAE-zl-p25, *Escherichia/Shigella* NLAE-zl-p252, *Escherichia/Shigella* NLAE-zl-p275, *Escherichia/Shigella* NLAE-zl-p280, *Escherichia/Shigella* NLAE-zl-p51, *Escherichia/Shigella* NLAE-zl-p53, *Escherichia/Shigella* NLAE-zl-p669, *Escherichia/Shigella* NLAE-zl-p676, *Escherichia/Shigella* NLAE-zl-p717, *Escherichia/Shigella* NLAE-zl-p731, *Escherichia/Shigella* NLAE-zl-p826, *Escherichia/Shigella* NLAE-zl-p877, *Escherichia/Shigella* NLAE-zl-p884, *Escherichia/Shigella* NLAE-zl-pl26, *Escherichia/Shigella* NLAE-zl-pl98, *Escherichia/Shigella* NMU-ST2, *Escherichia/Shigella* ocl 82011, *Escherichia/Shigella* of *Escherichia/Shigella proteobacterium, Escherichia/Shigella* Ql, *Escherichia/Shigella sakazakii, Escherichia/Shigella* SF6, *Escherichia/Shigella* sm1719, *Escherichia/Shigella* SOD-7317, *Escherichia/Shigella sonnei, Escherichia/Shigella* SW86, *Escherichia/Shigella vulneris, Ethanoligenens, Ethanoligenens harbinense, Eubacterium, Eubacterium* ARC-2, *Eubacterium callanderi, Eubacterium* E-1, *Eubacterium* G3(2011), *Eubacterium infirmum, Eubacterium limosum, Eubacterium methylotrophicum, Eubacterium* NLAE-zl-p439, *Eubacterium* NLAE-zl-p457, *Eubacterium* NLAE-zl-p458, *Eubacterium* NLAE-zl-p469, *Eubacterium* NLAE-zl-p474, *Eubacterium* oral, *Eubacterium saphenum, Eubacterium sulci, Eubacterium* WAL, *Euglenida, Euglenida longa, Faecalibacterium, Faecalibacterium bacterium, Faecalibacterium canine, Faecalibacterium* DJF VR20, *Faecalibacterium* icl379, *Faecalibacterium prausnitzii, Filibacter, Filibacter globispora, Flavobacterium, Flavobacterium* SSL03, *Flavonifractor, Flavonifractor* AUH-JLC235, *Flavonifractor* enrichment, *Flavonifractor* NLAE-zl-c354, *Flavonifractor orbiscindens, Flavonifractor plautii, Francisella, Francisella piscicida, Fusobacterium, Fusobacterium nucleatum, Gardnerella, Gardnerella vaginalis, Gemmiger, Gemmiger* DJF_VR33k2, *Gemmiger formicilis, Geobacter*, GHAPRB1, *Gordonibacter, Gordonibacter bacterium, Gor-*

donibacter intestinal, Gordonibacter pamelaeae, Gp2, Gp21, Gp4, Gp6, *Granulicatella, Granulicatella adiacens, Granulicatella* enrichment, *Granulicatella* oral, *Granulicatella paraadiacens, Haemophilus, Hafnia, Hafnia* 3-12 (2010), *Hafnia alvei, Hafnia* CC16, *Hafnia proteus, Haliea, Hallella, Hallella seregens, Herbaspirillum, Herbaspirillum* 022S4-11, *Herbaspirillum seropedicae, Hespellia, Hespellia porcina, Hespellia stercorisuis, Holdemania, Holdemania* AP2, *Holdemania filiformis, Howardella, Howardella ureilytica, Hydrogenoanaerobacterium, Hydrogenoanaerobacterium saccharovorans, Hydrogenophaga, Hydrogenophaga bacterium, Ilumatobacter, inulinivorans, Janthinobacterium, Janthinobacterium* C30An7, *Jeotgalicoccus, Klebsiella, Klebsiella aerogenes, Klebsiella bacterium, Klebsiella* E1L1, *Klebsiella* EB2-THQ, *Klebsiella* enrichment, *Klebsiella* F83, *Klebsiella* ggl60e, *Klebsiella* Gl-6, *Klebsiella granulomatis, Klebsiella* HaNA20, *Klebsiella* HF2, *Klebsiella* ii_3_chl_1, *Klebsiella* KALA-ICIBA17, *Klebsiella* kpu, *Klebsiella* M3, *Klebsiella* MB45, *Klebsiella milletis, Klebsiella* NCCP-138, *Klebsiella* okl_1_9_S16, *Klebsiella okl_1_9*_S54, *Klebsiella planticola, Klebsiella pneumoniae, Klebsiella poinarii, Klebsiella* PSB26, *Klebsiella* RS, *Klebsiella* Sel4, *Klebsiella* SRC_DSD12, *Klebsiella* tdl53s, *Klebsiella* TG-1, *Klebsiella* TPS 5, *Klebsiella variicola, Klebsiella* WB-2, *Klebsiella* Y9, *Klebsiella* zlmy, *Kluyvera, Kluyvera* An5-1, *Kluyvera cryocrescens, Kocuria, Kocuria* 2216.35.31, *Kurthia, Lachnobacterium, Lachnobacterium* C12b, *Lachnospiracea incertae sedis, Lachnospiracea incertae sedis bacterium, Lachnospiracea incertae sedis contortum, Lachnospiracea incertae sedis* Eg2, *Lachnospiracea incertae sedis eligens, Lachnospiracea incertae sedis ethanolgignens, Lachnospiracea incertae sedis galacturonicus, Lachnospiracea incertae sedis gnavus, Lachnospiracea incertae sedis hallii, Lachnospiracea incertae sedis hydrogenotrophica, Lachnospiracea incertae sedis* ID5, *Lachnospiracea incertae sedis intestinal, Lachnospiracea incertae sedis mpnisolate, Lachnospiracea incertae sedis pectinoschiza, Lachnospiracea incertae sedis ramulus, Lachnospiracea incertae sedis rectale, Lachnospiracea incertae sedis* RLB1, *Lachnospiracea incertae sedis rumen, Lachnospiracea incertae sedis* SY8519, *Lachnospiracea incertae sedis torques, Lachnospiracea incertae sedis uniforme, Lachnospiracea incertae sedis ventriosum, Lachnospiracea incertae sedis xylanophilum, Lachnospiracea incertae sedis* ye62, *Lactobacillus, Lactobacillus* 5-1-2, *Lactobacillus* 66c, *Lactobacillus acidophilus, Lactobacillus arizonensis, Lactobacillus* B5406, *Lactobacillus brevis, Lactobacillus casei, Lactobacillus crispatus, Lactobacillus curvatus, Lactobacillus delbrueckii, Lactobacillus fermentum, Lactobacillus gasseri, Lactobacillus helveticus, Lactobacillus hominis, Lactobacillus* ID9203, *Lactobacillus* IDSAc, *Lactobacillus intestinal, Lactobacillus johnsonii, Lactobacillus lactis, Lactobacillus manihotivorans, Lactobacillus mucosae, Lactobacillus* NA, *Lactobacillus oris, Lactobacillus* P23, *Lactobacillus* P8, *Lactobacillus paracasei, Lactobacillus paraplantarum, Lactobacillus pentosus, Lactobacillus plantarum, Lactobacillus pontis, Lactobacillus* rennanqilfyl4, *Lactobacillus* rennanqilyf9, *Lactobacillus reuteri, Lactobacillus rhamnosus, Lactobacillus salivarius, Lactobacillus sanfranciscensis, Lactobacillus suntoryeus, Lactobacillus* T3R1C1, *Lactobacillus vaginalis, Lactobacillus zeae, Lactococcus, Lactococcus* 56, *Lactococcus* CR-3175, *Lactococcus* CW-1, *Lactococcus* D8, *Lactococcus* Da-18, *Lactococcus* DAP39, *Lactococcus delbrueckii, Lactococcus* F116, *Lactococcus fujiensis, Lactococcus* G22, *Lactococcus garvieae, Lactococcus lactis, Lactococcus manure, Lactococcus* RTS, *Lactococcus* SXVIII1(2011), *Lactococcus* TP2MJ, *Lactococcus* TP2ML, *Lactococcus* TP2MN, *Lactococcus* U5-1, *Lactonifactor, Lactonifactor bacterium, Lactonifactor longoviformis, Lactonifactor* NLAE-zl-c533, *Leclercia, Lentisphaera, Leuconostoc, Leuconostoc carnosum, Leuconostoc citreum, Leuconostoc garlicum, Leuconostoc gasicomitatum, Leuconostoc gelidum, Leuconostoc inhae, Leuconostoc lactis, Leuconostoc* MEBE2, *Leuconostoc mesenteroides, Leuconostoc pseudomesenteroides, Limnobacter, Limnobacter* spf3, *Luteolibacter, Luteolibacter bacterium, Lutispora, Marinifilum, Marinobacter, Marinobacter arcticus, Mariprofundus, Marvinbryantia, Megamonas, Megasphaera, Melissococcus, Melissococcus faecalis, Methanobacterium, Methanobacterium subterraneum, Methanobrevibacter, Methanobrevibacter arboriphilus, Methanobrevibacter millerae, Methanobrevibacter olleyae, Methanobrevibacter oralis, Methanobrevibacter* SM9, *Methanobrevibacter smithii, Methanosphaera, Methanosphaera stadtmanae, Methylobacterium, Methylobacterium adhaesivum, Methylobacterium bacterium, Methylobacterium* iEID, *Methylobacterium* MP3, *Methylobacterium oryzae, Methylobacterium* PB132, *Methylobacterium* PB20, *Methylobacterium* PB280, *Methylobacterium* PDD-23b-14, *Methylobacterium radiotolerans, Methylobacterium* SKJH-1, *Mitsuokella, Mitsuokella jalaludinii, Morganella, Morganella morganii, Moritella, Moritella* 2D2, *Moryella, Moryella indoligenes, Moryella naviforme, Mycobacterium, Mycobacterium tuberculosis, Negativicoccus, Nitrosomonas, Nitrosomonas eutropha, Novosphingobium, Odoribacter, Odoribacter laneus, Odoribacter splanchnicus, Olsenella, Olsenella* 1832, *Olsenella* F0206, *Orbus, Orbus gilliamella, Oribacterium, Oscillibacter, Oscillibacter bacterium, Oscillibacter enrichment, Owenweeksia, Oxalobacter, Oxalobacter formigenes, Paludibacter, Pantoea, Pantoea eucalypti, Papillibacter, Papillibacter cinnamivorans, Parabacteroides, Parabacteroides* ASF519, *Parabacteroides* CR-34, *Parabacteroides distasonis, Parabacteroides* DJF B084, *Parabacteroides* DJF B086, *Parabacteroides* dnLKV8, *Parabacteroides* enrichment, *Parabacteroides* fecal, *Parabacteroides goldsteinii, Parabacteroides gordonii, Parabacteroides johnsonii, Parabacteroides merdae, Parabacteroides mpnisolate, Parabacteroides* NLAE-zl-p340, *Paraeggerthella, Paraeggerthella hongkongensis, Paraeggerthella* NLAE-zl-p797, *Paraeggerthella* NLAE-zl-p896, *Paraprevotella, Paraprevotella clara, Paraprevotella xylaniphila, Parasutterella, Parasutterella excrementihominis, Pectobacterium, Pectobacterium carotovorum, Pectobacterium wasabiae, Pediococcus, Pediococcus* te2r, *Pedobacter, Pedobacter* b3Nlb-b5, *Pedobacter daechungensis, Peptostreptococcus, Peptostreptococcus anaerobius, Peptostreptococcus stomatis, Phascolarctobacterium, Phascolarctobacterium faecium, Photobacterium, Photobacterium* MIE, *Pilibacter, Planctomyces, Planococcaceae incertae sedis, Planomicrobium, Plesiomonas, Porphyrobacter, Porphyrobacter* KK348, *Porphyromonas, Porphyromonas asaccharolytica, Porphyromonas bennonis, Porphyromonas canine, Porphyromonas somerae, Prevotella, Prevotella bacterium, Prevotella* BI-42, *Prevotella bivia, Prevotella buccalis, Prevotella copri, Prevotella* DJF_B112, *Prevotella mpnisolate, Prevotella oral, Propionibacterium, Propionibacterium acnes, Propionibacterium freudenreichii, Propionibacterium* LG, *Proteiniborus, Proteiniphilum, Proteus, Proteus* HS7514, *Providencia, Pseudobutyrivibrio, Pseudobutyrivibrio bacterium, Pseudobutyrivibrio fibrisolvens, Pseudobutyrivibrio ruminis, Pseudochrobactrum, Pseudoflavonifractor, Pseudoflavonifractor* asf500, *Pseudoflavonifractor* *bacterium, Pseudoflavonifractor capillosus, Pseudoflavonifractor* NML, *Pseudomonas, Pseudomonas* 1043, *Pseudomonas* 10569, *Pseudomonas* 11-44, *Pseudomonas* 127(39-ix), *Pseudomonas* 12A_19, *Pseudomonas* 145 (38zx), *Pseudomonas* 22010, *Pseudomonas* 32010, *Pseudomonas* 34t20, *Pseudomonas* 3C_10, *Pseudomonas* 4-5(2010), *Pseudomonas* 4-9(2010), *Pseudomonas* 6-13.J, *Pseudomonas* 63596, *Pseudomonas* 82010, *Pseudomonas* a001-142L, *Pseudomonas aeruginosa, Pseudomonas agarici, Pseudomonas* all 1-5, *Pseudomonas* a101-18-2, *Pseudomonas* amspl, *Pseudomonas* AU2390, *Pseudomonas* AZ18R1, *Pseudomonas azotoformans, Pseudomonas* B122, *Pseudomonas* B65(2012), *Pseudomonas bacterium, Pseudomonas* BJSX, *Pseudomonas* BLH-8D5, *Pseudomonas* BWDY-29, *Pseudomonas* CA18, *Pseudomonas* Cantasl2, *Pseudomonas* CB11, *Pseudomonas* CBZ-4, *Pseudomonas cedrina, Pseudomonas* CGMCC, *Pseudomonas* CL16, *Pseudomonas* CNE, *Pseudomonas corrugata, Pseudomonas cuatrocienegasensis, Pseudomonas* CYEB-7, *Pseudomonas* D5, *Pseudomonas* DAP37, *Pseudomonas* DB48, *Pseudomonas deceptionensis, Pseudomonas* Den-05, *Pseudomonas* DF7EH1, *Pseudomonas* DhA-91, *Pseudomonas* DVS14a, *Pseudomonas* DYJK4-9, *Pseudomonas* DZQS, *Pseudomonas* E11_ICE19B, *Pseudomonas* E2.2, *Pseudomonas* e2-CDC-TB4D2, *Pseudomonas* EM189, *Pseudomonas* enrichment, *Pseudomonas extremorientalis, Pseudomonas* FAIR/BE/F/GH37, *Pseudomonas* FAIR/BE/F/GH39, *Pseudomonas* FAIR/BE/F/GH94, *Pseudomonas* FLM05-3, *Pseudomonas fluorescens, Pseudomonas fragi, Pseudomonas* 'FSL, *Pseudomonas* G1013, *Pseudomonas gingeri, Pseudomonas* HC2-2, *Pseudomonas* HC2-4, *Pseudomonas* HC2-5, *Pseudomonas* HC4-8, *Pseudomonas* HC6-6, *Pseudomonas* Hg4-06, *Pseudomonas* HLB8-2, *Pseudomonas* HLS12-1, *Pseudomonas* HSF20-13, *Pseudomonas* HW08, *Pseudomonas* IpA-92, *Pseudomonas* IV, *Pseudomonas* JCM, *Pseudomonas jessenii, Pseudomonas* JSPB5, *Pseudomonas* K3R3.1A, *Pseudomonas* KB40, *Pseudomonas* KB42, *Pseudomonas* KB44, *Pseudomonas* KB63, *Pseudomonas* KB73, *Pseudomonas* KK-21-4, *Pseudomonas* KOPRI, *Pseudomonas* L1R3.5, *Pseudomonas* LAB-27, *Pseudomonas* LAB-44, *Pseudomonas* LclO-2, *Pseudomonas libanensis, Pseudomonas* Ln5C.7, *Pseudomonas* LS197, *Pseudomonas lundensis, Pseudomonas marginalis, Pseudomonas* MFY143, *Pseudomonas* MFY146, *Pseudomonas* MY 1412, *Pseudomonas* MY1404, *Pseudomonas* MY1416, *Pseudomonas* MY 1420, *Pseudomonas* N14zhy, *Pseudomonas* NBRC, *Pseudomonas* NCCP-506, *Pseudomonas* NFU20-14, *Pseudomonas* NJ-22, *Pseudomonas* NJ-24, *Pseudomonas* Nj-3, *Pseudomonas* Nj-55, *Pseudomonas* Nj-56, *Pseudomonas* Nj-59, *Pseudomonas* Nj-60, *Pseudomonas* Nj-62, *Pseudomonas* Nj-70, *Pseudomonas* NP41, *Pseudomonas* OCW4, *Pseudomonas* OW3-15-3-2, *Pseudomonas* P2(2010), *Pseudomonas* P3(2010), *Pseudomonas* P4(2010), *Pseudomonas* PD, *Pseudomonas* PF1B4, *Pseudomonas* PF2M10, *Pseudomonas* PILH1, *Pseudomonas* P1(2010), *Pseudomonas poae, Pseudomonas proteobacterium, Pseudomonas* ps4-12, *Pseudomonas* ps4-2, *Pseudomonas* ps4-28, *Pseudomonas* ps4-34, *Pseudomonas* ps4-4, *Pseudomonas psychrophila, Pseudomonas putida, Pseudomonas* R-35721, *Pseudomonas* R-37257, *Pseudomonas* R-37265, *Pseudomonas* R-37908, *Pseudomonas* RBE2CD-42, *Pseudomonas* regd9, *Pseudomonas* RKS7-3, *Pseudomonas* S2, *Pseudomonas* seawater, *Pseudomonas* SGb08, *Pseudomonas* SGb396, *Pseudomonas* SGbl20, *Pseudomonas* sgn, *Pseudomonas* 'Shk, *Pseudomonas stutzeri, Pseudomonas syringae, Pseudomonas taetrolens, Pseudomonas tolaasii, Pseudomonas trivialis, Pseudomonas* TUT1023, *Pseudomonas* W15Feb26, *Pseudomonas* W15Feb4, *Pseudomonas* W15Feb6, *Pseudomonas* WD-3, *Pseudomonas* WR4-13, *Pseudomonas* WR7 #2, *Pseudomonas* Y1000, *Pseudomonas* ZS29-8, *Psychrobacter, Psychrobacter* umbl3d, *Pyramidobacter, Pyramidobacter piscolens, Rahnella, Rahnella aquatilis, Rahnella carotovorum, Rahnella* GIST-WP4wl, *Rahnella* LR113, *Rahnella* Z2-S1, *Ralstonia, Ralstonia bacterium, Raoultella, Raoultella* B 19, *Raoultella* enrichment, *Raoultella planticola, Raoultella* sv6xvii, *Raoultella* SZ015, RBE1CD-48, *Renibacterium, Renibacterium* G20, rennanqilfyl0, *Rhizobium, Rhizobium leguminosarum, Rhodococcus, Rhodococcus erythropolis, Rhodopirellula, Riemerella, Riemerella anatipestifer, Rikenella, Robinsoniella, Robinsoniella peoriensis, Roseburia, Roseburia* 11SE37, *Roseburia bacterium, Roseburia cecicola, Roseburia* DJF_VR77, *Roseburia faecis, Roseburia fibrisolvens, Roseburia hominis, Roseburia intestinalis, RoseibaciUus, Rothia, Rubritalea, Ruminococcus, Ruminococcus* 25F6, *Ruminococcus albus, Ruminococcus bacterium, Ruminococcus bromii, Ruminococcus callidus, Ruminococcus champanellensis, Ruminococcus* DJF_VR87, *Ruminococcus flavefaciens, Ruminococcus gauvreauii, Ruminococcus lactaris, Ruminococcus* NK3A76, *Ruminococcus* YE71, *Saccharofermentans, Salinicoccus, Salinimicrobium, Salmonella, Salmonella agglomerans, Salmonella bacterium, Salmonella enterica, Salmonella freundii, Salmonella hermannii, Salmonella paratyphi, Salmonella* SL0604, *Salmonella subterranea, Scardovia, Scardovia* oral, *Schwartzia, Sedimenticola, Sediminibacter, Selenomonas, Selenomonas* fecal, *Serpens, Serratia, Serratia* 1135, *Serratia* 136-2, *Serratia* 5.1R, *Serratia* AC-CS-1B, *Serratia* AC-CS-B2, *Serratia aquatilis, Serratia bacterium, Serratia* BS26, *Serratia carotovorum, Serratia* DAP6, *Serratia* enrichment, *Serratia* F2, *Serratia ficaria, Serratia fonticola, Serratia grimesii, Serratia* J 145, *Serratia* JM983, *Serratia liquefaciens, Serratia marcescens, Serratia plymuthica, Serratia proteamaculans, Serratia proteolyticus, Serratia* ptz-16s, *Serratia quinivorans, Serratia* SBS, *Serratia* SS22, *Serratia* trout, *Serratia* UA-G004, *Serratia* White, *Serratia* yellow, *Shewanella, Shewanella baltica, Slackia, Slackia intestinal, Slackia isoflavoniconvertens, Slackia* NATTS, *Solibacillus, Solobacterium, Solobacterium moorei, Spartobacteria genera incertae sedis, Sphingobium, Sphingomonas, Sporacetigenium, Sporobacter, Sporobacterium, Sporobacterium olearium, Staphylococcus, Staphylococcus epidermidis, Staphylococcus* PCA17, stellenboschense, *Stenotrophomonas, Streptococcus, Streptococcus* 15, *Streptococcus* 1606-02B, *Streptococcus agalactiae, Streptococcus alactolyticus, Streptococcus anginosus, Streptococcus bacterium, Streptococcus bovis, Streptococcus* ChDC, *Streptococcus constellatus, Streptococcus* CR-314S, *Streptococcus criceti, Streptococcus cristatus, Streptococcus downei, Streptococcus dysgalactiae, Streptococcus* enrichment, *Streptococcus equi, Streptococcus equinus, Streptococcus* ES11, *Streptococcus eubacterium, Streptococcus fecal, Streptococcus gallinaceus, Streptococcus gallolyticus, Streptococcus gastrococcus, Streptococcus genomosp, Streptococcus gordonii, Streptococcus infantarius, Streptococcus intermedius, Streptococcus* Je2, *Streptococcus* JS-CD2, *Streptococcus* LRC, *Streptococcus luteciae, Streptococcus lutetiensis, Streptococcus* M09-11185, *Streptococcus mitis, Streptococcus mutans, Streptococcus* NA, *Streptococcus* NLAE-zl-c353, *Streptococcus* NLAE-zl-p68, *Streptococcus* NLAE-zl-p758, *Streptococcus* NLAE-zlp807, *Streptococcus* oral, *Streptococcus oralis, Streptococcus parasanguinis, Streptococcus phocae, Streptococcus pneu-

moniae, Streptococcus porcinus, Streptococcus pyogenes, Streptococcus S 16-08, *Streptococcus salivarius, Streptococcus sanguinis, Streptococcus sobrinus, Streptococcus suis, Streptococcus symbiont, Streptococcus thermophilus, Streptococcus* TW1, *Streptococcus vestibularis, Streptococcus warneri, Streptococcus* XJ-RY-3, *Streptomyces, Streptomyces malaysiensis, Streptomyces* MVCS6, *Streptophyta, Streptophyta cordifolium, Streptophyta ginseng, Streptophyta hirsutum, Streptophyta oleracea, Streptophyta sativa, Streptophyta sativum, Streptophyta sativus, Streptophyta tabacum*, Subdivision3 genera *incertae sedis, Subdoligranulum, Subdoligranulum bacterium, Subdoligranulum* icl393, *Subdoligranulum icl*395, *Subdoligranulum variabile, Succiniclasticum, Sulfuricella, Sulfuro spirillum, Sutterella, Syntrophococcus, Syntrophomonas, Syntrophomonas bryantii, Syntrophus, Tannerella, Tatumella, Thermo gymnomonas, Thermofilum, Thermogymnomonas, Thermovirga, Thiomonas, Thiomonas* ML1-46, *Thorsellia, Thorsellia carsonella*, TM7 genera *incertae sedis, Trichococcus, Turicibacter, Turicibacter sanguinis, Vagococcus, Vagococcus* bfsl 1-15, *Vampiro vibrio, Vampirovibrio, Varibaculum, Variovorax, Variovorax* KS2D-23, *Veillonella, Veillonella dispar, Veillonella* MSA 12, *Veillonella* OK8, *Veillonella* oral, *Veillonella parvula, Veillonella tobetsuensis, Vibrio, Vibrio* 3C1, *Victivallis, Victivallis vadensis, Vitellibacter, wadsworthensis, Wandonia, Wandonia haliotis, Weissella, Weissella cibaria, Weissella confusa, Weissella oryzae, Yersinia, Yersinia* 9gw38, *Yersinia* A125, *Yersinia aldovae, Yersinia aleksiciae, Yersinia* b702011, *Yersinia bacterium, Yersinia bercovieri, Yersinia enterocolitica, Yersinia frederiksenii, Yersinia intermedia, Yersinia kristensenii, Yersinia* MAC, *Yersinia massiliensis, Yersinia mollaretii, Yersinia nurmii, Yersinia pekkanenii, Yersinia pestis, Yersinia pseudotuberculosis, Yersinia rohdei, Yersinia ruckeri, Yersinia* s4fe31, *Yersinia* sl0fe31, *Yersinia* sl7fe31, and *Yersinia* YEM17B.

The names of the microbes provided herein, may optionally include the strain name.

Cytokines

In some embodiments, SBP formulations include cytokines. As used herein, the term "cytokine" refers to a class of biological signaling molecules produced by cells that regulate cellular activity in surrounding or distant cells. In some embodiments, the cytokine is a lymphokine, monokine, growth factor, colony-stimulating factor (CSF), transforming growth factor (TGF), tumor necrosis factor (TNF), chemokine, and/or interleukin. Examples of cytokines include, but are not limited to, brain-derived neurotrophic factor (BDNF), cardiotrophin-like cytokine factor 1 (CLCF1), ciliary neurotrophic factor (CNTF), cardiotrophin 1 (CTF1), epidermal growth factor (EGF), erythropoietin (EPO), fibroblast growth factor acidic (FGFa), fibroblast growth factor basic (FGFb), granulocyte colony stimulating factor (G-CSF), growth hormone, granulocyte-macrophage colony stimulating factor 2 (GM-CSF), interferon-al, interleukin(IL)-1 (IL-1), IL-la, IL-1(3, IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-11, IL-12, IL-13, IL-14, IL-15, IL-16, IL-17, IL-18, IL-19, IL-20, IL-21, IL-22, IL-23, IL-27, interleukin-1 receptor antagonist (IL-1RA), keratinocyte growth factor 1 and 2 (KGF), kit ligand/stem cell factor (KITLG), leptin (LEP), leukemia inhibitory factor (LIF), nerve growth factor (NGF), oncostatin M (OSM), platelet derived growth factors, prolactin (PRL), thrombopoietin (THPO), transforming growth factor (TGF) α (TGFα), TGFβ, tumor necrosis factor α (TNFα), vascular endothelial growth factor (VEGF), tissue inhibitor of metalloproteinase (TIMP), matrix metalloproteinase (MMP), any of the interferons, any of the interleukins, any of the lymphokines, any of the cell signal molecules, and any structural or functional molecule thereof. In some embodiments, cytokines may include, but are not limited to, any of those listed in Table 4, above.

Lipids

In some embodiments, therapeutic agents include lipids. As used herein, the term "lipid" refers to members of a class of organic compounds that include fatty acids and various derivatives of fatty acids that are soluble in organic solvents, but not in water. Examples of lipids include, but are not limited to, fats, triglycerides, oils, waxes, sterols (e.g. cholesterol, ergosterol, hopanoids, hydroxysteroids, phytosterol, and steroids), stearin, palmitin, triolein, fat-soluble vitamins (e.g., vitamins A, D, E, and K), monoglycerides (e.g. monolaurin, glycerol monostearate, and glyceryl hydroxystearate), diglycerides (e.g. diacylglycerol), phospholipids, glycerophospholipids (e.g., phosphatidic acid, phosphatidylethanolamine, phosphatidylcholine, phosphatidylserine, phosphoinositides), sphingolipids (e.g., sphingomyelin), and phosphosphingolipids. In some embodiments, lipids may include, but are not limited to, any of those listed (e.g., fats and fatty acids) in Table 4, above.

Macromolecules

In some embodiments, therapeutic agents include macromolecules, cells, tissues, organs, and/or organisms. Examples of macromolecules include, but are not limited to, proteins, polymers, carbohydrates, complex carbohydrates, lipids, nucleic acids, oligonucleotides, and genes. Macromolecules may be expressed (e.g. expression in *Escherichia coli*) or they may be chemically synthesized (e.g. solid phase synthesis, and/or polymer forming chain reactions).

Nucleic Acids

In some embodiments, SBP formulations include nucleic acids. As used herein, the term "nucleic acid" refers to any polymer of nucleotides (natural or non-natural) or derivatives or variants thereof. Nucleic acids may include deoxyribonucleic acid (DNA) or ribonucleic acid (RNA). In some embodiments, SBP formulations may enhance the stability of the nucleic acid (e.g. RNA), as taught in He et al. (2018) ACS Biomaterials Science and Engineering/4(5):1708-1715, the contents of which are herein incorporated by reference in their entirety. In some embodiments, nucleic acids may be polynucleotides or oligonucleotides. Some nucleic acids may include aptamers, plasmids, small interfering RNA (siRNA), microRNAs, or viral nucleic acids. In some embodiments, nucleic acids may encode proteins. In some embodiments, SBPs including therapeutic agent nucleic acids may include any of those described in International Publication Number WO2017123383, the contents of which are herein incorporated by reference in their entirety. In some embodiments, nucleic acids may include, but are not limited to, any of those listed in Table 4, above.

In some embodiments, nucleic acids may include a "CELiD" DNA as described in Li et al. (2013) PLoS One. 8(8):e69879, the contents of which are herein incorporated by reference in their entirety. CELiD DNA is a eukaryotic vector DNA that includes an expression cassette flanked by adeno-associated virus (AAV) inverted terminal repeats.

Proteins

In some embodiments, SBP formulations may include biological agents that are or include proteins. As used herein, the term "protein" generally refers to polymers of amino acids linked by peptide bonds and embraces "peptides" and "polypeptides." In some SBPs, the biological agent protein included is processed silk. Classes of proteins used as biological agent may include, but are not limited to, antigens, antibodies, antibody fragments, cytokines, peptides, hormones, enzymes, oxidants, antioxidants, synthetic proteins, and chimeric proteins. In some embodiments, proteins include any of those presented in Table 4, above. In some embodiments, proteins are combined with processed silk to improve protein stability.

In some embodiments, SBP formulations include peptides. The term "peptide" generally refers to shorter proteins of about 50 amino acids or less. Peptides with only two amino acids may be referred to as "dipeptides." Peptides with only three amino acids may be referred to as "tripeptides." Polypeptides generally refer to proteins with from about 4 to about 50 amino acids. SBPs that include peptides may include any of those described in International Publication Numbers WO2017123383 and WO2010123945, the contents of each of which are herein incorporated by reference in their entirety. Peptides may be obtained via any method known to those skilled in the art. In some embodiments, peptides may be expressed in culture. In some embodiments, peptides may be obtained via chemical synthesis (e.g. solid phase peptide synthesis). In some embodiments, peptides are used to functionalize SBPs, for example, as taught in International Publication Number WO2010123945.

In some embodiments, SBP formulations are used to facilitate peptide delivery, for example, according to the methods presented in International Publication Number WO2017123383. In some embodiments, peptides include RGD peptides, for example, as taught in Kambe et al. (2017) Materials 10(10):1153, the contents of which are herein incorporated by reference in their entirety. Non-limiting examples of peptide therapeutic agents include, but are not limited to, Degarelix acetate, Liraglutide, Cyclosporine, Eptifibatide, Dactinomycin, Spaglumat magnesium, Colistin, Nafarelin acetate, Somatostatin acetate, Buserelin, Enfuvirtide, Octreotide, lanreotide acetate, Caspofungin, Nesiritide, Goserelin, Salmon calcitonin, Lepirudin or r-hirudin, Daptomycin, Exenatide, Carbetocin acetate, Tirofiban, Glutathione, Cetrorelix acetate, Enalapril maleate, Bivalirudin, Vapreotide acetate, Icatibant acetate, Human calcitonin, Oxytocin, Atosiban acetate, Bacitracin, Lypressin, Vancomycin, Captopril, Anidulafungin, Bortezomib, Saralasin acetate, Calcitonin, Thymalfasin, Ziconotide, and Lisinopril. In some embodiments, peptides may include any of those presented in Table 4, above.

Synthetic/Chimeric Proteins

In some embodiments, SBP formulations include synthetic proteins. As used herein, the term "synthetic" refers to any article produced through at least some human manipulation. Synthetic proteins may be identical to proteins found in nature or may have one or more distinguishing features. Distinguishing features may include, but are not limited to, differences in amino acid sequences, incorporation of non-natural amino acids, post-translational modifications, and conjugation to non-protein moieties (e.g., some antibody drug conjugates). Synthetic proteins may be expressed in vitro or in vivo. Synthetic proteins may also be chemically synthesized (e.g. by solid phase peptide synthesis). In some embodiments, synthetic proteins are made from a combination of expression and chemical synthesis (e.g. native chemical ligation or enzyme catalyzed protein ligation).

In some embodiments, synthetic proteins include chimeric or fusion proteins. As used herein, the term "fusion protein" refers to a substance that includes two or more protein components that are conjugated through at least one chemical bond. As used herein, the term "chimeric protein" refers to a protein that includes segments from at least two different sources (e.g., from two different species or two different isotypes or variants from a common species). Chimeric proteins may be produced via the expression of two or more ligated genes encoding different proteins. Chimeric proteins may be produced via chemical synthesis. In some embodiments, chimeric proteins are made from a combination of expression and chemical synthesis (e.g. native chemical ligation or enzyme catalyzed protein ligation). In some embodiments, synthetic proteins or chimeric proteins may include, but are not limited to, any of those listed in Table 4, above.

Viruses and Viral Particles

In some embodiments, SBP formulations include viruses or viral particles. Viruses and viral particles may be used to transfer nucleic acid into cells for genetic manipulation, gene therapy, gene editing, protein expression, or to inhibit protein expression. In some embodiments, SBPs be prepared with viral or viral particle payloads. In some embodiments, payload release may occur over a period of time (the payload release period). The payload release rate and/or length of the payload release period may be modulated by SBP components or methods of preparation. Examples of viruses and viral particles may include, but are not limited to, any of those presented in Table 4, above. In some embodiments, SBP formulations with a virus or viral particle may enhance the stability of said virus or viral particle, as taught in WO2018053524A1, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, the virus or viral particle payloads prepared with SBP formulations may include, but are not limited to, adeno-associated virus, lentivirus, alphavirus, enterovirus, pestivirus, baculovirus, herpesvirus, Epstein Barr virus, papovavirus, poxvirus, vaccinia virus, herpes simplex virus, and/or a viral particle thereof.

Oxidants/Antioxidants

In some embodiments, therapeutic agents include oxidants or antioxidants. As used herein, the term "oxidant" refers to a substance that oxidizes (i.e., strips electrons from) another substance. Inhibitors of oxidation are referred to herein as "antioxidants." The use of oxidants and/or antioxidants as therapeutic agents may include any of the methods taught, for example, in International Publication Number WO2017137937; Min et al. (2017) Int J Biol Macromol s0141-8130(17):32855-32856; or Manchineella et al. (2017) European Journal of Organic Chemistry 30:4363-4369, the contents of each of which are herein incorporated by reference in their entirety. Oxidant or antioxidant therapeutic agents may be included in SBPs for treatment of indications requiring localized treatment or for indications requiring activity more distant from an administration site. In some embodiments, incorporation of oxidants or antioxidants may be used to modulate SBPs stability or degradation. In some embodiments, oxidants or antioxidants may be polymers. Such polymers may include quaternary ammonium chitosan and melanin. Examples of such therapeutic agents include those taught in International Publication Number WO2017137937 and Min et al. (2017) Int J Biol Macromol s0141-8130(17):32855-32856, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, oxidants or antioxidants include small molecules, metals, ions, minerals, vitamins, peptides, and/or proteins. In some embodiments, antioxidants include cyclic dipeptides or 2,5-diketopiperazines. Such antioxidants may include any of those taught in Manchineella et al. (2017) European Journal of Organic Chemistry 30:4363-4369, the contents of which are herein incorporated by reference in their entirety. In some embodiments, oxidants or antioxidants may include, but are not limited to, any of those listed in Table 4, above.

Small Molecules

In some embodiments, SBP formulations include small molecule therapeutic agents. As used herein, the term "small molecule" refers to a low molecular weight compound, typically less than 900 Daltons. Many small molecules are able to pass through cell membranes, making them attractive candidates for therapeutic applications. SBPs may be combined with any small molecules to carry out a variety of therapeutic applications. Such small molecules may include small molecule drugs approved for human treatment. Some small molecules may be hydrophobic or hydrophilic. Small molecules may include, but are not limited to, antibacterial agents, antifungal agents, anti-inflammatory agents, non-steroidal anti-inflammatory drugs, antipyretics, analgesics, antimalarial agents, antiseptics, hormones, stimulants, tranquilizers, and statins. In some embodiments, small molecules may include any of those listed in Table 4, above.

In some embodiments, SBP formulations may be used to encapsulate, store and/or release, in a controlled manner, small molecules. For example, using silk fibroin micrococoons as delivery vehicles for small molecules has been described in Shimanovich et al. (Shimanovich et al. (2015) Nature Communications 8:15902, the contents of which are herein incorporated by reference in their entirety).

Angiogenesis Modulators

In some embodiments, therapeutic agents include modulators of angiogenesis. Such therapeutic agents may include vascular endothelial growth factor (VEGF)-related agents. As used herein, the term "VEGF-related agent" refers to any substance that affects VEGF expression, synthesis, stability, biological activity, degradation, receptor binding, cellular signaling, transport, secretion, internalization, concentration, or deposition (e.g., in extracellular matrix).

In some embodiments, VEGF-related agents are angiogenesis inhibitors. In some embodiments, the angiogenesis inhibitor includes any of those taught in International Publication Number WO2013126799, the contents of which are herein incorporated by reference in their entirety. In some embodiments, VEGF-related agents may include antibodies. VEGF-related agents may include VEGF agonists, including, but not limited to, toll-like receptor agonists. In some embodiments, the therapeutic agent is a VEGF antagonist. VEGF agonists or antagonists may be small molecules. In some embodiments, VEGF agonists or antagonists may be macromolecules or proteins. Angiogenesis inhibitors may include, but are not limited to, MACUGEN® or another VEGF nucleic acid ligand; LUCENTIS®, AVASTIN®, or another anti-VEGF antibody; combretastatin or a derivative or prodrug thereof such as Combretastatin A4 Prodrug (CA4P); VEGF-Trap (Regeneron); EVIZON™ (squalamine lactate); AG-013958 (Pfizer, Inc.); JSM6427 (Jerini AG); a short interfering RNA (siRNA) that inhibits expression of one or more VEGF isoforms (e.g., $VEGF_{165}$); an siRNA that inhibits expression of a VEGF receptor (e.g., VEGFR1), endogenous or synthetic peptides, angiostatin, combstatin, arresten, tumstatin, thalidomide, thalidomide derivatives, canstatin, endostatin, thrombospondin, and β2-glycoprotein 1. In some embodiments, VEGF-related agents may include, but are not limited to any of those listed in Table 4, above.

Antibacterial Agents

In some embodiments, therapeutic agents include antibacterial agents. As used herein, the term "antibacterial agent" refers to any substance that harms, kills, or otherwise inhibits the growth and/or reproduction of bacteria. Antibacterial agents may include, but are not limited to, any of those listed in Table 4, above.

Antifungal Agents

In some embodiments, therapeutic agents include antifungal agents. As used herein, the term "antifungal agent" refers to any substance that harms, kills, or otherwise inhibits the growth and/or reproduction of fungi. Antifungal agents may include, but are not limited to, any of those listed in Table 4, above.

Analgesic Agents

In some embodiments, therapeutic agents include analgesic agents. As used herein, the term "analgesic agent" refers to any substance used to reduce or alleviate pain. Analgesic agents may include, but are not limited to, any of those listed in Table 4, above.

Antipyretics

In some embodiments, therapeutic agents include antipyretics. As used herein, the term "antipyretic" refers to any substance used to reduce or alleviate fever. Examples of antipyretics include, but are not limited to, any NSAID, acetaminophen, aspirin and related salicylates (e.g. choline salicylate, magnesium salicylate, and sodium salicylate), ibuprofen, naproxen, ketoprofen, nimesulide, phenazone, metamizole, and nabumetone. In some embodiments, antipyretics may include, but are not limited to, any of those listed in Table 4, above.

Antimalarial Agents

In some embodiments, therapeutic agents include antimalarial agents. As used herein, the term "antimalarial agent" refers to any substance that harms, kills, or otherwise inhibits the growth and/or reproduction of plasmodium parasites. Examples of antimalarial agents may include, but are not limited to, any of those listed in Table 4, above.

Antiseptic Agents

In some embodiments, therapeutic agents include antiseptic agents. As used herein, the term "antiseptic agent" refers to any substance that harms, kills, or otherwise inhibits the growth and/or reproduction of microorganisms. Examples of antiseptics include, but are not limited to, iodine, lower alcohols (ethanol, propanol, etc.), chlorhexidine, quaternary amine surfactants, chlorinated phenols, biguanides, bisbiguanides polymeric quaternary ammonium compounds, silver and its complexes, small molecule quaternary ammonium compounds, peroxides, and hydrogen peroxide. In some embodiments, antiseptic agents may include any of those listed in Table 4, above.

Hormones

In some embodiments, therapeutic agents include hormones. As used herein, the term "hormone" refers to a cellular signaling molecule that promotes a response in cells or tissues. Hormones may be produced naturally by cells. In some embodiments, hormones are synthetic. Examples of hormones include, but are not limited to, any steroid, dexamethasone, allopregnanolone, any estrogen (e.g. ethinyl estradiol, mestranol, estradiols and their esters, estriol, estriol succinate, polyestriol phosphate, estrone, estrone sulfate and conjugated estrogens), any progestogen (e.g. progesterone, norethisterone acetate, norgestrel, levonorgestrel, gestodene, chlormadinone acetate, drospirorenone, and 3-ketodesogestrel), any androgen (e.g. testosterone, androstenediol, androstenedione, dehydroepiandrosterone, and dihydrotestosterone), any mineralocorticoid, any glucocorticoid, cholesterols, and any hormone known to those skilled in the art. In some embodiments, hormones may include, but are not limited to, any of those listed in Table 4, above.

Non-Steroidal Anti-Inflammatory Drugs

In some embodiments, therapeutic agents include non-steroidal anti-inflammatory drugs. A nonsteroidal anti-inflammatory drug (NSAID) is a class of non-opioid analgesics used to reduce inflammation and associated pain. NSAIDs may include, but are not limited to, any of those listed in Table 4, above. In some embodiments, the NSAID is celecoxib. Some SBPs include gels or hydrogels that are combined with NSAIDs (e.g., celecoxib). Such SBPs may be used as carriers for NSAID payload delivery. NSAID delivery may include controlled release of the NSAID.

Stimulants

In some embodiments, therapeutic agents include stimulants. As used herein, the term "stimulant" refers to any substance that increases subject physiological or nervous activity. Examples of stimulants include, but are not limited to, amphetamines, caffeine, ephedrine, 3,4-methylenedioxymethamphetamine, methylenedioxypyrovalerone, mephedrone, methamphetamine, methylphenidate, nicotine, phenylpropanolamine, propylhexedrine, pseudoephedrine, and cocaine. In some embodiments, stimulants may include, but are not limited to, any of those listed in Table 4, above.

Tranquilizers

In some embodiments, therapeutic agents include tranquilizers. As used herein, the term "tranquilizer" refers to any substance used to lower subject anxiety or tension. Examples of tranquilizers include, but are not limited to, barbiturates, benzodiazepines, carbamates, antihistamines, opioids, antidepressants (e.g. monoamine oxidase inhibitors, tetracyclic antidepressants, tricyclic antidepressants, selective serotonin reuptake inhibitors, and serotonin-norepinephrine reuptake inhibitors), sympatholytics (e.g. alpha blockers, beta-blockers, and alpha-adrenergic agonists), mebicar, fabomotizole, selank, bromantane, emoxypine, azapirones, pregabalin, mentyl isovalerate, propofol, racetams, alcohols, inhalants, any butyrophenone (e.g. benperidol, bromperidol, droperidol, haloperidol, moperone, pipamperone, and timiperone), any diphenylbutylpiperidine (e.g. fluspirilene, penfluridol, and pimozide), any phenothiazine (e.g. acepromazine, chlorpromazine, cyamemazine, dixyrazine, fluphenazine, levomepromazine, levomepromazine, mesoridazine, perazine, periciazine, perphenazine, pipotiazine, prochlorperazine, promazine, promethazine, prothipendyl, thioproperazine, thioridazine, trifluoperazine, and triflupromazine), any thioxanthene (e.g. chlorprothixene, clopenthixol, flupentixol, thiothixene, and zuclopenthixol), any benzamidine (e.g. sulpiride, sultopride, and veralipride), any tricyclic (e.g. carpipramine, clocapramine, clorotepine, loxapine, and mosapramine), gamma aminobutyric acid, and molindone. In some embodiments, tranquilizers may include, but are not limited to, any of those listed in Table 4, above.

Statins

In some embodiments, therapeutic agents include statins. As used herein, the term "statin" refers to a class of compounds that inhibit hydroxy-methylglutaryl-coenzyme A reductase (HMG-CoA reductase), a key enzyme in cholesterol biosynthesis. Statins are referred to herein in the broadest sense and include statin derivatives such as ester derivatives or protected ester derivatives. Examples of statins include, but are not limited to, rosuvastatin, pitavastatin, pravastatin, fluvastatin, cerivastatin, atorvastatin, simvastatin, mevastatin, and lovastatin. In some embodiments, statins may include, but are not limited to, any of those listed in Table 4, above.

Anti-Cancer Agents

In some embodiments, therapeutic agents include anti-cancer agents. As used herein, the term "anticancer agent" refers to any substance used to kill cancer cells or inhibit cancer cell growth and/or cell division. Anticancer agents that target tumor cells are referred to herein as "antitumor agents." Such anticancer agents may reduce tumor mass and/or volume. Anticancer agents that are chemical substances are referred to herein as "chemotherapeutic agents." Examples of antitumor agents include, but are not limited to, busulphan, cisplatin, cyclophosphamide, MTX, daunorubicin, doxorubicin, melphalan, vincristine, vinblastine, chlorabucil, any alkylating agent (e.g. cyclophosphamide, mechlorethamine, chlorambucil, melphalan, dacarbazine, nitrosoureas, and temozolomide), any anthracycline (e.g. daunorubicin, doxorubicin, epirubicin, idarubicin, mitozantrone, and valrubicin), any cytoskeletal disruptors or taxanes (e.g. paclitaxel, docetaxel, abraxane, and taxotere), any epothilones, any histone deacetylase inhibitors (e.g. vorinostat and romidepsin), any topoisomerase I inhibitors (e.g. irinotecan and topotecan), any topoisomerase II inhibitors (e.g. etoposide, teniposide, and tafluposide), kinase inhibitors (e.g. bortezomib, erlotinib, gefitinib, imatinib, vemurafenib, and vismodegib), nucleotide and precursor analogues (e.g. azacitidine, azathioprine, capecitabine, cytarabine, doxifluridine, fluorouracil, gemcitabine, hydroxyurea, mercaptopurine, methotrexate, and tioguanine), antimicrobial peptides (e.g. bleomycin and actinomycin), platinum based chemotherapeutics (e.g. carboplatin, cisplatin, oxaliplatin), retinoids (e.g. tretinoin, alitretinoin, and bexarotene), and vinca alkaloids and derivatives (e.g. vinblastine, vincristine, vindesine, and vinorelbine). In some embodiments, anticancer agents may include, but are not limited to, any of those listed in Table 4, above.

Herbal Preparations

In some embodiments, therapeutic agents include herbal preparations. As used herein, the term "herbal preparation" refers to any substance derived or extracted from vegetation. These preparations may include, but are not limited to, tea, decoctions, cold infusions, tinctures, cordials, herbal wines, granules, syrups, essential oils (e.g. allspice berry essential oil, angelica seed essential oil, anise seed essential oil, basil essential oil, bay laurel essential oil, bay essential oil, bergamot essential oil, blood orange essential oil, camphor essential oil, caraway seed essential oil, cardamom seed essential oil, carrot seed essential oil, cassia essential oil, catnip essential oil, cedarwood essential oil, celery seed essential oil, chamomile german essential oil, chamomile roman essential oil, cinnamon bark essential oil, cinnamon leaf essential oil, citronella essential oil, clary sage essential oil, clove bud essential oil, coriander seed essential oil, cypress essential oil, elemi essential oil, eucalyptus essential oil, fennel essential oil, fir needle essential oil, frankincense essential oil, geranium essential oil, ginger essential oil, grapefruit pink essential oil, helichrysum essential oil, hop essential oil, hyssop essential oil, juniper berry essential oil, labdanum essential oil, lemon essential oil, lemongrass essential oil, lime essential oil, magnolia essential oil, mandarin essential oil, margoram essential oil, Melissa essential oil, mugward essential oil, myrrh essential oil, myrtle essential oil, neroli essential oil, niaouli essential oil, nutmeg essential oil, orange sweet essential oil, oregano essential oil, palmarosa essential oil, patchouli essential oil, pennyroyal essential oil, pepper black essential oil, peppermint essential oil, petitgram essential oil, pine needle essential oil, radiata essential oil, ravensara essential oil, rose essential oil, rosemary essential oil, rosewood essential oil, sage essential oil, sandalwood essential oil, spearmint essential oil, spikenard essential oil, spruce essential oil, star anise essential oil, sweet annie essential oil, tangerine essential oil, tea tree essential oil, thyme red essential oil, verbena essential oil, vetiver essential oil, wintergreen essential oil, wormwood essential oil, yarrow essential oil, ylang essential oil, jasmine absolute oil, lavender absolute oil, pink lotus absolute oil, rose absolute oil, sambac absolute oil, and white lotus absolute oil), flower essences, sitz baths, soaks, pills, suppositories, poultices, compresses, salves, and ointments. Examples of herbs to be incorporated include, but are not limited to, sage, thyme, cumin, basil, bay laurel, borage, caraway, catnip, chervil, chives, cilantro, dill, epazote, fennel, garlic, lavender, lemongrass, lemon balm, lemon verbena, lovage, marjoram, mints, nasturtium, parsley, oregano, rosemary, salad burnet, savory, scented geranium, sorrel, and tarragon. In some embodiments, herbal preparations may include, but are not limited to, any of those listed in Table 4, above.

Health Supplements

In some embodiments, therapeutic agents include health supplements. As used herein, the term "health supplement" refers to any substance used to provide a nutrient, vitamin, or other beneficial compound that is typically lacking from a normal diet or is complimentary to such substances present in a normal diet. Examples of health supplements include, but are not limited to, vitamin A, vitamin B, vitamin C, vitamin D, vitamin E, vitamin K, thiamin, riboflavin, niacin, vitamin B6, vitamin B12, biotin, pantothenic acid, calcium, iron, phosphorus, iodine, magnesium, zinc, selenium, selenium, copper, manganese, chromium, molybdenum, chloride, potassium, nickel, silicon, vanadium, and tin. In some embodiments, health supplements may include, but are not limited to, any of those listed in Table 4, above.

Ions, Metals, Minerals

In some embodiments, therapeutic agents include ions, metals, and/or minerals. Examples include, but are not limited to, calcium, iron, phosphorus, iodine, magnesium, zinc, selenium, selenium, copper, manganese, chromium, molybdenum, gold, silver, chloride, potassium, nickel, silicon, vanadium, and tin. In some embodiments, therapeutic agents include oxides (e.g. silver oxide). In some embodiments, ions, metals, and/or minerals may be present in nanoparticles. Such nanoparticles may include any of those taught in Mane et al. (2017) Scientific Reports 7:15531; and Babu et al. (2017) J Colloid Interface Sci 513:62-72, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, ions, metals, and/or minerals may include, but are not limited to, any of those listed in Table 4, above.

Vitamins

In some embodiments, therapeutic agents include vitamins or vitamin analogues. As used herein, the term "vitamin" refers to a nutrient that must be obtained through diet (i.e., is not synthesized endogenously or is synthesized endogenously, but in insufficient amounts). Examples of vitamins include, but are not limited to, vitamin A, vitamin B-1, vitamin B-2, vitamin B-3, vitamin B-5, vitamin B-6, vitamin B-7, vitamin B-9, vitamin B-12, vitamin C, vitamin D, vitamin E, and vitamin K. In some embodiments, vitamins may include, but are not limited to, any of those listed in Table 4, above.

Other Therapeutic Agents

Other therapeutic agents may include, but are not limited to, anthocyanidin, anthoxanthin, apigenin, dihydrokaempferol, eriodictyol, fisetin, flavan, flavan-3,4-diol, flavan-3-ol, flavan-4-ol, flavanone, flavanonol, flavonoid, furanoflavonols, galangin, hesperetin, homoeriodictyol, isoflavonoid, isorhamnetin, kaempferol, luteolin, myricetin, naringenin, neoflavonoid, pachypodol, proanthocyanidins, pyranoflavonols, quercetin, rhamnazin, tangeritin, taxifolin, theaflavin, thearubigin, chondrocyte-derived extracellular matrix, macrolide, erythromycin, roxithromycin, azithromycin and clarithromycin. In some embodiments, other therapeutic agents may include, but are not limited to, any of those listed in Table 4, above.

Controlled Degradation

In some embodiments, SBP formulations may be used for controlled degradation, and to modulate the stability and storage of therapeutic agents or other materials (e.g., agricultural compositions, agricultural products, materials, devices, and excipients). As used herein, the term "controlled degradation" refers to regulated loss of structure, function, and/or other physical and chemical properties. Such SBP formulations may be used to stabilize therapeutic agents used in therapeutic applications. In some embodiments, SBP formulations are used to maintain and/or improve the stability of therapeutic agents during storage. In some embodiments SBP formulations may be used to enhance the degradation rate of therapeutic agents. In some embodiments, an SBP formulation may increase the rate of degradation of a therapeutic agent (e.g. a protein) during storage. In some embodiments, preparation of an SBP formulation with a therapeutic agent may increase the rate of degradation of said therapeutic agent (e.g. a protein) during storage.

Use as a Preservation/Stabilizing Agent

In some embodiments, SBP formulations may be used to preserve or stabilize therapeutic agents or other materials (e.g., agricultural compositions, agricultural products, materials, devices, and excipients). Such SBPs may be used to stabilize therapeutic agents used in therapeutic applications. In some embodiments, SBP formulations are used to maintain and/or improve the stability of therapeutic agents during lyophilization. The maintenance and/or improvement of stability during lyophilization may be determined by comparing products lyophilized with SBP formulations to products lyophilized with non-SBP formulation. Maintenance and/or improvement of stability during lyophilization will be found or appreciated by those of skill in the art when products lyophilized with SBP formulations are determined to impart superior or durational benefits over non-SBP formulations or those standard in the art.

In some embodiments, lyophilization may be utilized for long term storage of processed silk formulations. In some embodiments processed silk formulations are stored frozen. In some embodiments, processed silk formulations may be frozen without altering the protein quality. In some embodiments, processed silk formulations may be frozen without altering rheological properties, protein size, and aggregation of said formulations. In some embodiments, processed silk formulations may be prepared as solutions, and then frozen. In some embodiments these solutions may then be thawed. The thawed solutions may exhibit less than 10%, less than 5%, less than 3%, less than 1%, less than 0.1%, less than 0.01%, or less than 0.001% aggregation of protein.

In some embodiments, silk fibroin processing will be more efficient and cheaper if there is no need for lyophilization. In some embodiments, removing a drying condition will require that silk fibroin solutions are stable through a freeze/thaw process. This will allow for aseptic preparation and shipment of drug substance from the manufacturing site to the fill/finish facility. In some embodiments, these silk fibroin solutions may comprise silk fibroin at various concentrations, as well as cryoprotectants (sucrose and trehalose), to improve the freeze-thaw stability of dialyzed drug substance. In some embodiments, these solutions may comprise silk fibroin at a concentration of from about 0.01% to about 0.1%, about 0.1% to about 1%, about 1% to about 2%, about 2% to about 3%, about 3% to about 4%, about 4% to about 5%, about 5% to about 10%, about 10% to about 15%, about 15% to about 20%, or about 20% to about 30% (w/v) silk fibroin. In some embodiments, the cryoprotectant is sucrose or trehalose. Cryoprotectents may be included at a concentration of 0 to about 150 mM. In some embodiments, the rheological properties of the silk fibroin solutions may remain the same following freeze/thaw. In some embodiments, the molecular weight by SEC may not change following freeze/thaw. In some embodiments, the lowest increase in aggregation may be seen in formulations with sucrose as an excipient.

In some embodiments, the SBP formulations maintain and/or improve therapeutic agent stability by at least 1 day, at least 2 days, at least 3 days, at least 4 days, at least 5 days, at least 6 days, at least 7 days, at least 8 days, at least 9 days, at least 10 days, at least 11 days, at least 12 days, at least 13 days, at least 2 weeks, at least 3 weeks, at least 1 month, at least 6 weeks, at least 2 months, at least 10 weeks, at least 3 months, at least 14 weeks, at least 4 months, at least 18 weeks, at least 5 months, at least 22 weeks, at least 6 months, at least 7 months, at least 8 months, at least 9 months, at least 10 months, at least 11 months, at least a year, at least 2 years, at least 3 years, at least 4 years, at least 5 years, or more than 5 years.

Ocular Therapeutic Agents

In some embodiments, therapeutic agents include ocular therapeutic agents. As used herein, the term "ocular therapeutic agent" refers to any compound that has a healing, corrective, diagnostic, and/or prophylactic effect and/or elicits a desired biological and/or pharmacological effect on the eye. In some embodiments, ocular therapeutic agents include one or more of processed silk, biological agents, small molecules, proteins, anti-inflammatory agents, steroids, opiates, analgesics, ciclosporin, corticosteroids, tetracyclines, essential fatty acids, sodium channel blockers, nonsteroidal anti-inflammatory drugs, cyclosporine, lifitegrast, and vascular endothelial growth factor-related agents. Ocular therapeutic agent proteins may include, but are not limited to, lysozyme, bovine serum albumin (BSA), bevacizumab, or VEGF-related agents. In some embodiments, ocular therapeutic agents may be used to treat one or more of the ocular therapeutic indications described herein.

Bacteriostatic Agents

In some embodiments, therapeutic agents include bacteriostatic agents. As used herein, the term "bacteriostatic agent" or "bacteriostat" refers to a substance that prevents bacterial reproduction and may or may not kill said bacteria. Bacteriostatic agents prevent the growth of bacteria. Non-limiting examples of bacteriostatic agents include antibiotics, antiseptics, disinfectants, and preservatives. Other non-limiting examples of bacteriostatic agents include tigecycline, trimethoprim, oxazolidinone, tetracyclines, novobiocin, clindamycin, nitrofurantoin, ethambutol, sulfonamides, macrolides, lincosamides, spectinomycin, and chloramphenicol.

Non-Steroidal Anti-Inflammatory Drugs

Therapeutic agents may include nonsteroidal anti-inflammatory drugs. A nonsteroidal anti-inflammatory drug (NSAID) is a class of non-opioid analgesics used to reduce inflammation and associated pain. NSAIDs may include small molecules. NSAIDs may include, but are not limited to, aspirin, carprofen, celecoxib, deracoxib, diclofenac, diflunisal, etodolac, fenoprofen, firocoxib, flurbirofen, ibuprofen, indomethacin, ketoprofen, ketorolac, mefenamic acid, meloxicam, nabumetone, naproxen, oxaprozin, piroxicam, robenacoxib, salsalate, sulindac, and tolmetin. In some embodiments, NSAIDs may be used to treat one or more of the ocular therapeutic indications described herein. In some embodiments, the NSAID is celecoxib. Some SBPs include gels or hydrogels that are combined with NSAIDs (e.g., celecoxib). Such SBPs may be used as carriers for NSAID payload delivery. NSAID delivery may include controlled release of the NSAID.

Therapeutic Indications

In some embodiments, SBPs are used to address one or more therapeutic indications. As used herein, the term "therapeutic indication" refers to a disease, disorder, condition, or symptom that may be cured, reversed, alleviated, stabilized, improved, or otherwise addressed through some form of therapeutic intervention (e.g., administration of a therapeutic agent or method of treatment).

SBP treatment of therapeutic indications may include contacting subjects with SBPs. SBPs may include therapeutic agents (e.g., any of those described herein) as cargo or payloads for treatment. In some embodiments, payload release may occur over a period of time (the "payload release period"). The payload release rate and/or length of the payload release period may be modulated by SBP components or methods of preparation.

Ocular Indications

In some embodiments, therapeutic indications include ocular indications. As used herein, the term "ocular indication" refers to any therapeutic indication related to the eye. Treatment of such indications in subjects may include contacting subjects with SBPs. SBPs and SBP formulations may include therapeutic agents (e.g., any of those described herein) as cargo or payloads for treatment. In some embodiments, payload release may occur over a period of time (the payload release period). The payload release rate and/or length of the payload release period may be modulated by SBP components or methods of preparation. In some embodiments, SBPs may be provided in the form of a solution or may be incorporated into a solution for ocular administration. Such solutions may be administered topically (e.g., in the form of drops, creams, or sprays) or by injection. In some embodiments, SBPs may be provided in the format of a lens or may be incorporated into lenses that are placed on eye. In some embodiments, SBPs are provided in the form of implants or are incorporated into implants that may be placed around the eye, on a surface of the eye, in a periocular space or compartment, or intraocularly. Implants may be solid or gelatinous (e.g., a gel or slurry) and may be in the form of a bleb, rod, or plug. Some gelatinous implants may harden after application. In some embodiments, implants include punctal plugs. Such plugs may be inserted into tear ducts. In some embodiments, SBPs may be used to repair ocular damage. In some embodiments, the SBP adheres to the ocular surface. In some embodiments, the SBP adheres to the ocular surface in a manner similar to a mucin layer.

Non-limiting examples of ocular indications include infection, refractive errors, age related macular degeneration, cataracts, diabetic retinopathy (proliferative and non-proliferative), cystoid macular edema, glaucoma, amblyopia, strabismus, color blindness, cytomegalovirus retinitis, keratoconus, diabetic macular edema (proliferative and non-proliferative), low vision, ocular hypertension, retinal detachment, eyelid twitching, inflammation, uveitis, bulging eyes, dry eye disease, floaters, xerophthalmia, diplopia, Graves' disease, night blindness, eye strain, red eyes, nystagmus, presbyopia, excess tearing, retinal disorders (e.g. age related macular degeneration), conjunctivitis, cancer, corneal ulcer, corneal abrasion, snow blindness, scleritis, keratitis, Thygeson's superficial punctate keratopathy, corneal neovascularization, Fuch's dystrophy, keratoconjuctitivis sicca, iritis, chorioretinal inflammation (e.g. chorioretinitis, choroiditis, retinitis, retinochoroiditis, pars planitis, and Harada's disease), aniridia, macular scars, solar retinopathy, choroidal degeneration, choroidal dystrophy, choroideremia, gyrate atrophy, choroidal hemorrhage, choroidal detachment, retinoschisis, hypertensive retinopathy, Bull's eye maculopathy, epiretinal membrane, peripheral retinal degeneration, hereditary retinal dystrophy, retinitis pigmentosa, retinal hemorrhage, separation of retinal layers, retinal vein occlusion, and other visual impairments. In some embodiments, ocular indications include inflammation of the eye.

Dry Eye

In one embodiment, the ocular indications which may be treated with the SBPs described herein may be dry eye. "Dry eye", "dry eye syndrome," "dry eye disease", or "DED" is a condition involving a lack of hydration on the eye surface that may be caused by one or more of a variety of factors (e.g., cellular/tissue dysfunction or environmental irritants). After the development of symptoms in a subject or patient, an optometrist and/or ophthalmologist may conduct and ocular exam and test for additional signs in the cornea and tears (e.g. ocular surface staining, corneal fluorescein and conjunctival staining, or tear film break-up time). Symptoms of DED may result from lack of tear production, improper tear or film production, alterations in tear or film composition and/or alteration in tear or film clearance. General symptoms of DED include, but are not limited to, ocular discomfort, dryness, conjunctival redness, grittiness, pain, burning, stinging, and any other symptom described in Moshifar et al. (2014) Clinical Ophthalmology 8:1419-1433, the contents of which are herein incorporated by reference in their entirety.

Symptoms may vary with the severity level of DED, which is graded on a scale of 1-4, as described in Behrens et al. (2006) Cornea 25: 900-907, the contents of which are herein incorporated by reference in their entirety. Mild and/or episodic DED (grade 1) has signs and symptoms which may include, but are not limited to, no or episodic mild fatigue, a variable Schirmer score, no to mild conjunctival injection, variable tear film break-up time (TFBUT), no to mild conjunctival staining, variable meibomian gland dysfunction (MGD), no to mild corneal staining, and no to mild corneal/and or tear signs. Mild DED may be caused by environmental stress. Moderate episodic and/or chronic DED (grade 2) has signs and symptoms which may include, but are not limited to, episodic visual symptoms that may annoy and/or limit activity, a Schirmer score of less than 10 mm/5 min, no to mild conjunctival injection, variable conjunctival staining, variable MGD, TFBUT of less than 10 minutes, variable corneal staining, and mild debris with a variable meniscus. Moderate DED may be brought on due to stress. Severe DED (grades 3-4) may be frequent, annoying, chronic, activity limiting, disabling, constant, and brought on without stress. Signs and symptoms of severe DED may include, but are not limited to, conjunctival injection (+/− and +/++), a Schirmer score of less than 5 mm/5 min (in some cases less than 2 mm/5 min), moderate to marked conjunctival staining, frequent MGD (optionally including trichiasis, keratinization, and symblepharon), moderate to marked corneal staining (optionally with severe punctate erosions), a TFBUT of less than 5 minutes (in some cases immediate), filamentary keratitis, mucus clumping, increased tear debris, and ulceration.

After diagnosis of DED, a treatment for DED (e.g. artificial tears) may be administered to the subject. Non-limiting examples of current DED treatments include education, counseling, environmental management, tear supplementation (with or without preservatives), prescription drugs (e.g. cyclosporine and lifitegrast), punctal plugs, and surgery. Side effects of current DED treatments include, but are not limited to, burning, redness, discomfort, discharge, pain, blurring, eye irritation, and changes in taste. Current DED treatments may also require frequent administration, as described in Moshifar et al. (2014) Clinical Ophthalmology 8:1419-1433.

In some embodiments, SBPs used to treat dry eye are provided as or included in solutions or devices. Solutions may include silk fibroin micelles, as described in Wongpanit et al. (2007) Macromolecular Bioscience 7: 1258-1271, the contents of which are herein incorporated by reference in their entirety. Solutions and SBPs to treat dry eye may be administered topically (e.g., by cream, spray, microspheres, or drops) or by injection to periocular or intraocular areas. Solutions may include viscous solutions, such as gels or slurries. The viscosity of such a solution may modulate properties of a resulting artificial tear replacement. Low viscosity artificial tear replacements may have shorter residence times and less efficacy. High viscosity artificial tear replacements may result in side effects including, but not limited to, blurred vision and discomfort. Devices for the treatment of DED may include, but are not limited to, solutions, implants, microspheres, hydrogels, lenses, artificial tear replacements, contact lens solution, and plugs. Devices may be hardened structures or gelatinous. In some embodiments, devices are gelatinous or prepared as a slurry, but harden after placement. Devices may include lacrimal or punctal plugs that treat dry eye via tear duct insertion. Punctal plugs may be prepared with or without therapeutic agent payloads.

SBPs used to treat dry eye may include therapeutic agent payloads. The therapeutic agents may include any of those described herein. In some embodiments, therapeutic agents include one or more of cyclosporine, corticosteroids, tetracyclines, lifitegast, NSAIDs, anti-inflammatory agents, opiates, analgesics, and essential fatty acids. In some embodiments, processed silk is the therapeutic agent. Therapeutic agent release from SBPs may occur over an extended payload release period. The payload release period may be from about 1 hour to about 48 hours, from about 1 day to about 14 days, or from about 1 week to about 52 weeks, or more than 52 weeks.

In some embodiments, ocular SBPs may be used as an anti-inflammatory treatment for dry eye disease, as described in Kim et al. (2017) Scientific Reports 7: 44364, the contents of which are herein incorporated by reference in their entirety. It has been demonstrated that the administration of 0.1 to 0.5% silk fibroin solutions in a mouse model of dry eye disease enhances corneal smoothness and tear production, while reducing the amount of inflammatory markers detected. In some embodiments, the SBPs described herein may be used to treat dry eye disease in humans. In some embodiments, the SBPs described herein may be used to treat dry eye disease in non-human primates. In some embodiments, the SBPs described herein may be used to treat dry eye disease in canines (e.g. dogs). In some embodiments, the SBPs described herein may be used to treat dry eye disease in felines (e.g. cats).

Combinations

In some embodiments, SBPs may be administered in combination with other therapeutic agent and/or methods of treatment, e.g., with known pharmaceuticals and/or known therapeutic methods, such as, for example, those which are currently employed for treating these disorders. For example, SBPs used to treat ocular indications may be administered in combination with other therapeutic agents used to treat ocular indications.

Pharmaceutical Compositions

In some embodiments, SBPs are or are included in pharmaceutical compositions. As used herein, the term "pharmaceutical composition" refers to a composition designed and/or used for medicinal purposes (e.g., the treatment of a disease).

In some embodiments, pharmaceutical compositions include one or more excipients and/or one or more therapeutic agents. Excipients and/or therapeutic agents included in pharmaceutical compositions may include, but are not limited to, any of those described herein. Relative amounts of therapeutic agents, excipient, and/or any additional ingredients in pharmaceutical compositions may vary, depending upon the identity, size, and/or condition of subjects being treated and further depending upon routes by which compositions are administered. For example, the compositions may include from about 0.0001% to about 99% (w/v) of a therapeutic agent.

Some excipients may include pharmaceutically acceptable excipients. The phrase "pharmaceutically acceptable" as used herein, refers to suitability within the scope of sound medical judgment for contacting subject (e.g., human or animal) tissues and/or bodily fluids with toxicity, irritation, allergic response, or other complication levels yielding reasonable benefit/risk ratios. As used herein, the term "pharmaceutically acceptable excipient" refers to any ingredient, other than active agents, that is substantially nontoxic and non-inflammatory in a subject. Pharmaceutically acceptable excipients may include, but are not limited to, solvents, dispersion media, diluents, inert diluents, buffering agents, lubricating agents, oils, liquid vehicles, dispersion or suspension aids, surface active agents, isotonic agents, thickening or emulsifying agents, preservatives, and the like, as suited to the particular dosage form desired. Various excipients for formulating pharmaceutical compositions and techniques for preparing the composition are known in the art (see Remington: The Science and Practice of Pharmacy, 21^{st} Edition, A. R. Gennaro, Lippincott, Williams & Wilkins, Baltimore, MD, 2006; incorporated herein by reference in its entirety). The use of a conventional excipient medium may be contemplated within the scope of the present disclosure, except insofar as any conventional excipient medium may be incompatible with a substance or its derivatives, such as by producing any undesirable biological effect or otherwise interacting in a deleterious manner with any other component(s) of pharmaceutical compositions.

A pharmaceutical composition in accordance with the present disclosure may be prepared, packaged, and/or sold in bulk, as a single unit dose, and/or as a plurality of single unit doses. As used herein, a "unit dose" refers to a discrete amount of the pharmaceutical composition comprising a predetermined amount of therapeutic agent or other compound. The amount of therapeutic agent may generally be equal to the dosage of therapeutic agent administered to a subject and/or a convenient fraction of such dosage including, but not limited to, one-half or one-third of such a dosage.

In some embodiments, pharmaceutical compositions may include between 0.0001 to 35% (w/v) silk fibroin. In some embodiments, the pharmaceutical compositions may include silk fibroin in concentrations from about 0.0001% (w/v) to about 0.001% (w/v), from about 0.001% (w/v) to about 0.01% (w/v), from about 0.01% (w/v) to about 0.1% (w/v), from about 0.1% (w/v) to about 1% (w/v), from about 1% (w/v) to about 5% (w/v), from about 5% (w/v) to about 10% (w/v), from about 10% (w/v) to about 20% (w/v), or from about 20% (w/v) to about 35% (w/v).

Dosing

In some embodiments, the present disclosure provides methods of administering pharmaceutical compositions that are or include SBPs to subjects in need thereof. Such methods may include providing pharmaceutical compositions at one or more doses and/or according to a specific schedule. In some embodiments, doses may be determined based on desired amounts of therapeutic agent or SBP to be delivered. Doses may be adjusted to accommodate any route of administration effective for a particular therapeutic application. The exact amount required will vary from subject to subject, depending on the species, age, and general condition of the subject, the severity of the disease, the particular composition, its mode of administration, its mode of activity, and the like. The frequency of dosing required will also vary from subject to subject, depending on the species, age, and general condition of the subject, the severity of the disease, the particular composition, its mode of administration, its mode of activity, and the like.

SBPs may be formulated in dosage unit form. Such forms may allow for ease of administration and uniformity of dosage. Total daily SBP usage may be decided by an attending physician within the scope of sound medical judgment. The specific therapeutically effective, prophylactically effective, or appropriate imaging dose level for any particular patient will depend upon a variety of factors including the disorder being treated and the severity of the disorder; the activity of the specific compound employed; the specific composition employed; the age, body weight, general health, sex and diet of the patient; the time of administration, route of administration, and rate of excretion of the specific compound employed; the duration of the treatment; drugs used in combination or coincidental with the specific compound employed; and like factors well known in the medical arts.

In some embodiments, pharmaceutical compositions that are or include SBPs may include a therapeutic agent or SBP at a concentration of from about 10 ng/mL to about 30 ng/mL, from about 12 ng/mL to about 32 ng/mL, from about 14 ng/mL to about 34 ng/mL, from about 16 ng/mL to about 36 ng/mL, from about 18 ng/mL to about 38 ng/mL, from about 20 ng/mL to about 40 ng/mL, from about 22 ng/mL to about 42 ng/mL, from about 24 ng/mL to about 44 ng/mL, from about 26 ng/mL to about 46 ng/mL, from about 28 ng/mL to about 48 ng/mL, from about 30 ng/mL to about 50 ng/mL, from about 35 ng/mL to about 55 ng/mL, from about 40 ng/mL to about 60 ng/mL, from about 45 ng/mL to about 65 ng/mL, from about 50 ng/mL to about 75 ng/mL, from about 60 ng/mL to about 240 ng/mL, from about 70 ng/mL to about 350 ng/mL, from about 80 ng/mL to about 400 ng/mL, from about 90 ng/mL to about 450 ng/mL, from about 100 ng/mL to about 500 ng/mL, from about 0.01 μg/mL to about 1 μg/mL, from about 0.05 μg/mL to about 2 μg/mL, from about 1 μg/mL to about 5 μg/mL, from about

2 μg/mL to about 10 μg/mL, from about 4 μg/mL to about 16 μg/mL, from about 5 μg/mL to about 20 μg/mL, from about 8 μg/mL to about 24 μg/mL, from about 10 μg/mL to about 30 μg/mL, from about 12 μg/mL to about 32 μg/mL, from about 14 μg/mL to about 34 μg/mL, from about 16 μg/mL to about 36 μg/mL, from about 18 μg/mL to about 38 μg/mL, from about 20 μg/mL to about 40 μg/mL, from about 22 μg/mL to about 42 μg/mL, from about 24 μg/mL to about 44 μg/mL, from about 26 μg/mL to about 46 μg/mL, from about 28 μg/mL to about 48 μg/mL, from about 30 μg/mL to about 50 μg/mL, from about 35 μg/mL to about 55 μg/mL, from about 40 μg/mL to about 60 μg/mL, from about 45 μg/mL to about 65 μg/mL, from about 50 μg/mL to about 75 μg/mL, from about 60 μg/mL to about 240 μg/mL, from about 70 μg/mL to about 350 μg/mL, from about 80 μg/mL to about 400 μg/mL, from about 90 μg/mL to about 450 μg/mL, from about 100 μg/mL to about 500 μg/mL, from about 0.01 mg/mL to about 1 mg/mL, from about 0.05 mg/mL to about 2 mg/mL, from about 1 mg/mL to about 5 mg/mL, from about 2 mg/mL to about 10 mg/mL, from about 4 mg/mL to about 16 mg/mL, from about 5 mg/mL to about 20 mg/mL, from about 8 mg/mL to about 24 mg/mL, from about 10 mg/mL to about 30 mg/mL, from about 12 mg/mL to about 32 mg/mL, from about 14 mg/mL to about 34 mg/mL, from about 16 mg/mL to about 36 mg/mL, from about 18 mg/mL to about 38 mg/mL, from about 20 mg/mL to about 40 mg/mL, from about 22 mg/mL to about 42 mg/mL, from about 24 mg/mL to about 44 mg/mL, from about 26 mg/mL to about 46 mg/mL, from about 28 mg/mL to about 48 mg/mL, from about 30 mg/mL to about 50 mg/mL, from about 40 mg/mL to about 100 mg/mL, from about 100 mg/mL to about 200 mg/mL, from about 200 mg/mL to about 300 mg/mL, from about 300 mg/mL to about 400 mg/mL, or more than 400 mg/mL.

In some embodiments, pharmaceutical compositions that are or include SBPs may be administered at a dose that provides subjects with a mass of therapeutic agent or SBP per unit mass of the subject (e.g., mg therapeutic agent or SBP per kg of subject [mg/kg]). In some embodiments, therapeutic agents or SBPs are administered at a dose of from about 1 ng/kg to about 5 ng/kg, from about 2 ng/kg to about 10 ng/kg, from about 4 ng/kg to about 16 ng/kg, from about 5 ng/kg to about 20 ng/kg, from about 8 ng/kg to about 24 ng/kg, from about 10 ng/kg to about 30 ng/kg, from about 12 ng/kg to about 32 ng/kg, from about 14 ng/kg to about 34 ng/kg, from about 16 ng/kg to about 36 ng/kg, from about 18 ng/kg to about 38 ng/kg, from about 20 ng/kg to about 40 ng/kg, from about 22 ng/kg to about 42 ng/kg, from about 24 ng/kg to about 44 ng/kg, from about 26 ng/kg to about 46 ng/kg, from about 28 ng/kg to about 48 ng/kg, from about 30 ng/kg to about 50 ng/kg, from about 35 ng/kg to about 55 ng/kg, from about 40 ng/kg to about 60 ng/kg, from about 45 ng/kg to about 65 ng/kg, from about 50 ng/kg to about 75 ng/kg, from about 60 ng/kg to about 240 ng/kg, from about 70 ng/kg to about 350 ng/kg, from about 80 ng/kg to about 400 ng/kg, from about 90 ng/kg to about 450 ng/kg, from about 100 ng/kg to about 500 ng/kg, from about 0.01 μg/kg to about 1 μg/kg, from about 0.05 μg/kg to about 2 μg/kg, from about 1 μg/kg to about 5 μg/kg, from about 2 μg/kg to about 10 μg/kg, from about 4 μg/kg to about 16 μg/kg, from about 5 μg/kg to about 20 μg/kg, from about 8 μg/kg to about 24 μg/kg, from about 10 μg/kg to about 30 μg/kg, from about 12 μg/kg to about 32 μg/kg, from about 14 μg/kg to about 34 μg/kg, from about 16 μg/kg to about 36 μg/kg, from about 18 μg/kg to about 38 μg/kg, from about 20 μg/kg to about 40 μg/kg, from about 22 μg/kg to about 42 μg/kg, from about 24 μg/kg to about 44 μg/kg, from about 26 μg/kg to about 46 μg/kg, from about 28 μg/kg to about 48 μg/kg, from about 30 μg/kg to about 50 μg/kg, from about 35 μg/kg to about 55 μg/kg, from about 40 μg/kg to about 60 μg/kg, from about 45 μg/kg to about 65 μg/kg, from about 50 μg/kg to about 75 μg/kg, from about 60 μg/kg to about 240 μg/kg, from about 70 μg/kg to about 350 μg/kg, from about 80 μg/kg to about 400 μg/kg, from about 90 μg/kg to about 450 μg/kg, from about 100 μg/kg to about 500 μg/kg, from about 0.01 mg/kg to about 1 mg/kg, from about 0.05 mg/kg to about 2 mg/kg, from about 1 mg/kg to about 5 mg/kg, from about 2 mg/kg to about 10 mg/kg, from about 4 mg/kg to about 16 mg/kg, from about 5 mg/kg to about 20 mg/kg, from about 8 mg/kg to about 24 mg/kg, from about 10 mg/kg to about 30 mg/kg, from about 12 mg/kg to about 32 mg/kg, from about 14 mg/kg to about 34 mg/kg, from about 16 mg/kg to about 36 mg/kg, from about 18 mg/kg to about 38 mg/kg, from about 20 mg/kg to about 40 mg/kg, from about 22 mg/kg to about 42 mg/kg, from about 24 mg/kg to about 44 mg/kg, from about 26 mg/kg to about 46 mg/kg, from about 28 mg/kg to about 48 mg/kg, from about 30 mg/kg to about 50 mg/kg, from about 35 mg/kg to about 55 mg/kg, from about 40 mg/kg to about 60 mg/kg, from about 45 mg/kg to about 65 mg/kg, from about 50 mg/kg to about 75 mg/kg, from about 60 mg/kg to about 240 mg/kg, from about 70 mg/kg to about 350 mg/kg, from about 80 mg/kg to about 400 mg/kg, from about 90 mg/kg to about 450 mg/kg, from about 100 mg/kg to about 500 mg/kg, from about 0.01 g/kg to about 1 g/kg, from about 0.05 g/kg to about 2 g/kg, from about 1 g/kg to about 5 g/kg, or more than 5 g/kg.

In some embodiments, pharmaceutical compositions that are or include SBPs may be administered at a dose sufficient to yield desired therapeutic agent or SBP concentration levels in subject tissue or fluids (e.g., blood, plasma, urine, etc.). In some embodiments, doses are adjusted to achieve subject therapeutic agent or SBP concentration levels in subject tissues or fluids of from about 1 pg/mL to about 5 pg/mL, from about 2 pg/mL to about 10 pg/mL, from about 4 pg/mL to about 16 pg/mL, from about 5 pg/mL to about 20 pg/mL, from about 8 pg/mL to about 24 pg/mL, from about 10 pg/mL to about 30 pg/mL, from about 12 pg/mL to about 32 pg/mL, from about 14 pg/mL to about 34 pg/mL, from about 16 pg/mL to about 36 pg/mL, from about 18 pg/mL to about 38 pg/mL, from about 20 pg/mL to about 40 pg/mL, from about 22 pg/mL to about 42 pg/mL, from about 24 pg/mL to about 44 pg/mL, from about 26 pg/mL to about 46 pg/mL, from about 28 pg/mL to about 48 pg/mL, from about 30 pg/mL to about 50 pg/mL, from about 35 pg/mL to about 55 pg/mL, from about 40 pg/mL to about 60 pg/mL, from about 45 pg/mL to about 65 pg/mL, from about 50 pg/mL to about 75 pg/mL, from about 60 pg/mL to about 240 pg/mL, from about 70 pg/mL to about 350 pg/mL, from about 80 pg/mL to about 400 pg/mL, from about 90 pg/mL to about 450 pg/mL, from about 100 pg/mL to about 500 pg/mL, from about 0.01 ng/mL to about 1 ng/mL, from about 0.05 ng/mL to about 2 ng/mL, from about 1 ng/mL to about 5 ng/mL, from about 2 ng/mL to about 10 ng/mL, from about 4 ng/mL to about 16 ng/mL, from about 5 ng/mL to about 20 ng/mL, from about 8 ng/mL to about 24 ng/mL, from about 10 ng/mL to about 30 ng/mL, from about 12 ng/mL to about 32 ng/mL, from about 14 ng/mL to about 34 ng/mL, from about 16 ng/mL to about 36 ng/mL, from about 18 ng/mL to about 38 ng/mL, from about 20 ng/mL to about 40 ng/mL, from about 22 ng/mL to about 42 ng/mL, from about 24 ng/mL to about 44 ng/mL, from about 26 ng/mL to about 46 ng/mL, from about 28 ng/mL to about 48 ng/mL, from about 30 ng/mL to about 50 ng/mL, from about 35 ng/mL to about 55 ng/mL, from about 40 ng/mL to about 60 ng/mL, from about 45 ng/mL to about 65 ng/mL, from about 50 ng/mL to about 75 ng/mL, from about 60 ng/mL to about 240 ng/mL, from about 70 ng/mL to about 350 ng/mL, from about 80 ng/mL to about 400 ng/mL, from about 90 ng/mL to about 450 ng/mL, from about 100 ng/mL to about 500 ng/mL, from about 0.01 μg/mL to about 1 μg/mL, from about 0.05 μg/mL to about 2 μg/mL, from about 1 μg/mL to about 5 μg/mL, from about 2 μg/mL to about 10 μg/mL, from about 4 μg/mL to about 16 μg/mL, from about 5 μg/mL to about 20 μg/mL, from about 8 μg/mL to about 24 μg/mL, from about 10 μg/mL to about 30 μg/mL, from about 12 μg/mL to about 32 μg/mL, from about 14 μg/mL to about 34 μg/mL, from about 16 μg/mL to about 36 μg/mL, from about 18 μg/mL to about 38 μg/mL, from about 20 μg/mL to about 40 μg/mL, from about 22 μg/mL to about 42 μg/mL, from about 24 μg/mL to about 44 μg/mL, from about 26 μg/mL to about 46 μg/mL, from about 28 μg/mL to about 48 μg/mL, from about 30 μg/mL to about 50 μg/mL, from about 35 μg/mL to about 55 μg/mL, from about 40 μg/mL to about 60 μg/mL, from about 45 μg/mL to about 65 μg/mL, from about 50 μg/mL to about 75 μg/mL, from about 60 μg/mL to about 240 μg/mL, from about 70 μg/mL to about 350 μg/mL, from about 80 μg/mL to about 400 μg/mL, from about 90 μg/mL to about 450 μg/mL, from about 100 μg/mL to about 500 μg/mL, from about 0.01 mg/mL to about 1 mg/mL, from about 0.05 mg/mL to about 2 mg/mL, from about 1 mg/mL to about 5 mg/mL, from about 2 mg/mL to about 10 mg/mL, from about 4 mg/mL to about 16 mg/mL, from about 5 mg/mL to about 20 mg/mL, from about 8 mg/mL to about 24 mg/mL, from about 10 mg/mL to about 30 mg/mL, from about 12 mg/mL to about 32 mg/mL, from about 14 mg/mL to about 34 mg/mL, from about 16 mg/mL to about 36 mg/mL, from about 18 mg/mL to about 38 mg/mL, from about 20 mg/mL to about 40 mg/mL, from about 22 mg/mL to about 42 mg/mL, from about 24 mg/mL to about 44 mg/mL, from about 26 mg/mL to about 46 mg/mL, from about 28 mg/mL to about 48 mg/mL, from about 30 mg/mL to about 50 mg/mL, from about 35 mg/mL to about 55 mg/mL, from about 40 mg/mL to about 60 mg/mL, from about 45 mg/mL to about 65 mg/mL, from about 50 mg/mL to about 75 mg/mL, from about 60 mg/mL to about 240 mg/mL, from about 70 mg/mL to about 350 mg/mL, from about 80 mg/mL to about 400 mg/mL, from about 90 mg/mL to about 450 mg/mL, from about 100 mg/mL to about 500 mg/mL, from about 0.01 g/mL to about 1 g/mL.

In some embodiments, pharmaceutical compositions that are or include SBPs are provided in one or more doses and are administered one or more times to subjects. Some pharmaceutical compositions are provided in only a single administration. Some pharmaceutical compositions are provided according to a dosing schedule that include two or more administrations. Each administration may be at the same dose or may be different from a previous and/or subsequent dose. In some embodiments, subjects are provided an initial dose that is higher than subsequent doses (referred to herein as a "loading dose"). In some embodiments, doses are decreased over the course of administration. In some embodiments, dosing schedules include pharmaceutical composition administration from about every 2 hours to about every 10 hours, from about every 4 hours to about every 20 hours, from about every 6 hours to about every 30 hours, from about every 8 hours to about every 40 hours, from about every 10 hours to about every 50 hours, from about every 12 hours to about every 60 hours, from about every 14 hours to about every 70 hours, from about every 16 hours to about every 80 hours, from about every 18 hours to about every 90 hours, from about every 20 hours to about every 100 hours, from about every 22 hours to about every 120 hours, from about every 24 hours to about every 132 hours, from about every 30 hours to about every 144 hours, from about every 36 hours to about every 156 hours, from about every 48 hours to about every 168 hours, from about every 2 days to about every 10 days, from about every 4 days to about every 15 days, from about every 6 days to about every 20 days, from about every 8 days to about every 25 days, from about every 10 days to about every 30 days, from about every 12 days to about every 35 days, from about every 14 days to about every 40 days, from about every 16 days to about every 45 days, from about every 18 days to about every 50 days, from about every 20 days to about every 55 days, from about every 22 days to about every 60 days, from about every 24 days to about every 65 days, from about every 30 days to about every 70 days, from about every 2 weeks to about every 8 weeks, from about every 3 weeks to about every 12 weeks, from about every 4 weeks to about every 16 weeks, from about every 5 weeks to about every 20 weeks, from about every 6 weeks to about every 24 weeks, from about every 7 weeks to about every 28 weeks, from about every 8 weeks to about every 32 weeks, from about every 9 weeks to about every 36 weeks, from about every 10 weeks to about every 40 weeks, from about every 11 weeks to about every 44 weeks, from about every 12 weeks to about every 48 weeks, from about every 14 weeks to about every 52 weeks, from about every 16 weeks to about every 56 weeks, from about every 20 weeks to about every 60 weeks, from about every 2 months to about every 6 months, from about every 3 months to about every 12 months, from about every 4 months to about every 18 months, from about every 5 months to about every 24 months, from about every 6 months to about every 30 months, from about every 7 months to about every 36 months, from about every 8 months to about every 42 months, from about every 9 months to about every 48 months, from about every 10 months to about every 54 months, from about every 11 months to about every 60 months, from about every 12 months to about every 66 months, from about 2 years to about 5 years, from about 3 years to about 10 years, from about 4 years to about 15 years, from about 5 years to about 20 years, from about 6 years to about 25 years, from about 7 years to about 30 years, from about 8 years to about 35 years, from about 9 years to about 40 years, from about 10 years to about 45 years, from about 15 years to about 50 years, or more than every 50 years.

In some embodiments, pharmaceutical compositions that are or include SBPs may be administered at a dose sufficient to provide a therapeutically effective amount of therapeutic agents or SBPs. As used herein, the term "therapeutically effective amount" refers to an amount of an agent sufficient to achieve a therapeutically effective outcome. As used herein, the term "therapeutically effective outcome" refers to a result of treatment where at least one objective of treatment is met. In some embodiments, a therapeutically effective amount is provided in a single dose. In some embodiments, a therapeutically effective amount is administered according to a dosing schedule that includes a plurality of doses. Those skilled in the art will appreciate that in some embodiments, a unit dosage form may be considered to include a therapeutically effective amount of a particular agent or entity if it includes an amount that is effective when administered as part of such a dosage regimen.

Administration

In some embodiments, pharmaceutical compositions that are or include SBPs may be administered according to one or more administration routes. In some embodiments, administration is enteral (into the intestine), transdermal, intravenous bolus, intralesional (within or introduced directly to a localized lesion), intrapulmonary (within the lungs or its bronchi), diagnostic, intraocular (within the eye), transtympanic (across or through the tympanic cavity), intravesical infusion, sublingual, nasogastric (through the nose and into the stomach), spinal, intracartilaginous (within a cartilage), insufflation (snorting), rectal, intravascular (within a vessel or vessels), buccal (directed toward the cheek), dental (to a tooth or teeth), intratesticular (within the testicle), intratympanic (within the aurus media), percutaneous, intrathoracic (within the thorax), submucosal, cutaneous, epicutaneous (application onto the skin), dental intracornal, intramedullary (within the marrow cavity of a bone), intra-abdominal, epidural (into the dura matter), intramuscular (into a muscle), intralymphatic (within the lymph), iontophoresis (by means of electric current where ions of soluble salts migrate into the tissues of the body), subcutaneous (under the skin), intragastric (within the stomach), nasal administration (through the nose), transvaginal, intravenous drip, endosinusial, intraprostatic (within the prostate gland), soft tissue, intradural (within or beneath the dura), subconjunctival, oral (by way of the mouth), peridural, parenteral, intraduodenal (within the duodenum), intracisternal (within the cisterna magna cerebellomedularis), periodontal, periarticular, biliary perfusion, intracoronary (within the coronary arteries), intrathecal (within the cerebrospinal fluid at any level of the cerebrospinal axis), intrameningeal (within the meninges), intracavernous injection (into a pathologic cavity) intracavitary (into the base of the penis), intrabiliary, subarachnoid, intrabursal, ureteral (to the ureter), intratendinous (within a tendon), auricular (in or by way of the ear), intracardiac (into the heart), enema, intraepidermal (to the epidermis), intraventricular (within a ventricle), intramyocardial (within the myocardium), intratubular (within the tubules of an organ), vaginal, sublabial, intracorporus cavernosum (within the dilatable spaces of the corporus cavernosa of the penis), intradermal (into the skin itself), intravitreal (through the eye), perineural, cardiac perfusion, irrigation (to bathe or flush open wounds or body cavities), in ear drops, endotracheal, intraosseous infusion (into the bone marrow), caudal block, intrauterine, transtracheal (through the wall of the trachea), intra-articular, intracorneal (within the cornea), endocervical, extracorporeal, intraspinal (within the vertebral column), transmucosal (diffusion through a mucous membrane), topical, photopheresis, oropharyngeal (directly to the mouth and pharynx), occlusive dressing technique (topical route administration which is then covered by a dressing which occludes the area), transplacental (through or across the placenta), intrapericardial (within the pericardium), intraarterial (into an artery), interstitial, intracerebral (into the cerebrum), intracerebroventricular (into the cerebral ventricles), peridural, intrapleural (within the pleura), infiltration, intrabronchial, intrasinal (within the nasal or periorbital sinuses), intraductal (within a duct of a gland), transdermal (diffusion through the intact skin for systemic distribution), intracaudal (within the cauda equine), nerve block, retrobulbar (behind the pons or behind the eyeball), intravenous (into a vein), intra-amniotic, conjunctival, intrasynovial (within the synovial cavity of a joint), gastroenteral, intraluminal (within a lumen of a tube), intrathecal (into the spinal canal), electro-osmosis, intraileal (within the distal portion of the small intestine), intraesophageal (to the esophagus), extra-amniotic administration, hemodialysis, intragingival (within the gingivae), intratumor (within a tumor), eye drops (onto the conjunctiva), laryngeal (directly upon the larynx), urethral (to the urethra), intravaginal administration, intramyocardial (entering the myocardium), intraperitoneal (infusion or injection into the peritoneum), respiratory (within the respiratory tract by inhaling orally or nasally for local or systemic effect), intradiscal (within a disc), ophthalmic (to the external eye), and/or intraovarian (within the ovary).

In some embodiments, pharmaceutical compositions that are or include SBPs may be administered by auricular administration, intraarticular administration, intramuscular administration, intrathecal administration, extracorporeal administration, buccal administration, intrabronchial administration, conjunctival administration, cutaneous administration, dental administration, endocervical administration, endosinusial administration, endotracheal administration, enteral administration, epidural administration, intra-abdominal administration, intrabiliary administration, intrabursal administration, oropharyngeal administration, interstitial administration, intracardiac administration, intracartilaginous administration, intracaudal administration, intracavernous administration, intracerebral administration, intracorporous cavernosum, intracavitary administration, intracomeal administration, intracisternal administration, cranial administration, intracranial administration, intradermal administration, intralesional administration, intratympanic administration, intragingival administration, intraovarian administration, intraocular administration, intradiscal administration, intraductal administration, intraduodenal administration, ophthalmic administration, intradural administration, intraepidermal administration, intraesophageal administration, nasogastric administration, nasal administration, laryngeal administration, intraventricular administration, intragastric administration, intrahepatic administration, intraluminal administration, intravitreal administration, intravesicular administration, intralymphatic administration, intramammary administration, intramedullary administration, intrasinal administration, intrameningeal administration, intranodal administration, intraovarian administration, intrapulmonary administration, intrapericardial administration, intraperitoneal administration, intrapleural administration, intrapericardial administration, intraprostatic administration, intrapulmonary administration, intraluminal administration, intraspinal administration, intrasynovial administration, intratendinous administration, intratesticular administration, subconjunctival administration, intracerebroventricular administration, epicutanous administration, intravenous administration, retrobulbar administration, periarticular administration, intrathoracic administration, subarachnoid administration, intratubular administration, periodontal administration, transtympanic administration, transtracheal administration, intratumor administration, vaginal administration, urethral administration, intrauterine administration, oral administration, gastroenteral administration, parenteral administration, sublingual administration, ureteral administration, percutaneous administration, peridural administration, transmucosal administration, perineural administration, transdermal administration, rectal administration, soft tissue administration, intraarterial administration, subcutaneous administration, topical administration, extra-amniotic administration, insufflation, enema, eye drops, ear drops, or intravesical infusion. In some embodiments, the SBPs described herein may be administered via injection. Injection site reactions may be monitored via any method known to one skilled in the art.

In some embodiments, SBPs may be administered for localized treatment (e.g., see United States Publication Numbers US20170368236 and US20110171239, the contents of each of which are herein incorporated by reference in their entirety). In some embodiments, SBPs may be administered for treatment of areas located further away from administration sites (e.g., see Aykac et al. (2017) Gene s0378-1119(17)30868-30865, the contents of which are herein incorporated by reference in their entirety).

In some embodiments, administration includes ocular administration. As used herein, the term "ocular administration" refers to delivery of an agent to an eye. Ocular administration may include, but is not limited to, topical administration (e.g., using eye drops, ointments, sprays, or creams), intraocular administration, intravitreal administration, intraretinal administration, intracorneal administration, intrascleral administration, lacrimal administration, punctal administration, administration to the anterior sub-Tenon's, suprachroidal administration, administration to the posterior sub-Tenon's, subretinal administration, administration to the fornix, administration to the lens, administration to the anterior segment, administration to the posterior segment, macular administration, and intra-aqueous humor administration. Administration may include intravitreal injection.

In some embodiments, the SBPs described herein may be administered as an eye drop. In some embodiments, the SBPs described herein may be administered as a spray. In some embodiments, the SBPs described herein may be administered by injection. In some embodiments, the SBPs described herein may be administered by lens application. In some embodiments, the SBPs described herein may be administered as a plug. In some embodiments, the SBPs are administered as a lacrimal plug. In some embodiments, the SBPs are administered as a punctal plug.

In some embodiments, SBP administration or SBP-based therapeutic agent administration occurs over a period of time, referred to herein as the "administration period." During administration periods, administration may be continuous or may be separated into two or more administrations. In some embodiments, administration periods may be from about 1 min to about 30 min, from about 10 min to about 45 min, from about 20 min to about 60 min, from about 40 min to about 90 min, from about 2 hours to about 10 hours, from about 4 hours to about 20 hours, from about 6 hours to about 30 hours, from about 8 hours to about 40 hours, from about 10 hours to about 50 hours, from about 12 hours to about 60 hours, from about 14 hours to about 70 hours, from about 16 hours to about 80 hours, from about 18 hours to about 90 hours, from about 20 hours to about 100 hours, from about 22 hours to about 120 hours, from about 24 hours to about 132 hours, from about 30 hours to about 144 hours, from about 36 hours to about 156 hours, from about 48 hours to about 168 hours, from about 2 days to about 10 days, from about 4 days to about 15 days, from about 6 days to about 20 days, from about 8 days to about 25 days, from about 10 days to about 30 days, from about 12 days to about 35 days, from about 14 days to about 40 days, from about 16 days to about 45 days, from about 18 days to about 50 days, from about 20 days to about 55 days, from about 22 days to about 60 days, from about 24 days to about 65 days, from about 30 days to about 70 days, from about 2 weeks to about 8 weeks, from about 3 weeks to about 12 weeks, from about 4 weeks to about 16 weeks, from about 5 weeks to about 20 weeks, from about 6 weeks to about 24 weeks, from about 7 weeks to about 28 weeks, from about 8 weeks to about 32 weeks, from about 9 weeks to about 36 weeks, from about 10 weeks to about 40 weeks, from about 11 weeks to about 44 weeks, from about 12 weeks to about 48 weeks, from about 14 weeks to about 52 weeks, from about 16 weeks to about 56 weeks, from about 20 weeks to about 60 weeks, from about 2 months to about 6 months, from about 3 months to about 12 months, from about 4 months to about 18 months, from about 5 months to about 24 months, from about 6 months to about 30 months, from about 7 months to about 36 months, from about 8 months to about 42 months, from about 9 months to about 48 months, from about 10 months to about 54 months, from about 11 months to about 60 months, from about 12 months to about 66 months, from about 2 years to about 5 years, from about 3 years to about 10 years, from about 4 years to about 15 years, from about 5 years to about 20 years, from about 6 years to about 25 years, from about 7 years to about 30 years, from about 8 years to about 35 years, from about 9 years to about 40 years, from about 10 years to about 45 years, from about 15 years to about 50 years, or more than 50 years.

Agricultural Applications and Products

In some embodiments, SBP formulations are prepared for use in agriculture. As used herein, the term "agriculture" refers to the cultivation of plants and animals to produce products useful for individual, communal, industrial, or commercial purposes. SBPs may be agricultural compositions. In some embodiments, SBPs may include an agricultural composition. As used herein, the term "agricultural composition" refers to any substance used in or produced by agriculture. In some embodiments, SBPs may be used to improve the growth, production, the shelf-life and stability of agricultural products. As used herein, the term "agriculture product" refers to any product of agriculture (e.g., food, medicines, materials, biofuels, etc.). In some embodiments, SBP formulations may be used in a variety of agricultural applications (e.g., agricultural SBP formulations). As used herein, the term "agricultural application" refers to any method used to improve, promote or increase the production of products obtained through the cultivation of plants and animals, for the benefit of individuals, communities, or commercial entities.

In some embodiments, agricultural compositions described herein are used for agricultural and environmental development. In some embodiments, SBP formulations may be used to improve the growth and production of agricultural products. These agricultural products may be plants, animals, plant agricultural products, or animal agricultural products. In some embodiments, SBP formulation administration may result in increased weight, biomass, growth, offspring production, product levels, and/or product size of one or more agricultural products. In some embodiments, the agricultural SBP formulations may include soil or locus stabilizers.

Cargo

In some embodiments, agricultural SBP formulations are used to facilitate delivery of cargo that enhance agricultural product health, yield, half-life and/or stability. In some embodiments, SBP formulations may be the cargos. In some embodiments, cargos may include, but are not limited to, therapeutic agents, small molecules, chemicals, nutrients, micronutrients, macronutrients, pest control agents, pesticides, antibiotics, antifungal, fungicide, virus, virus fragment, virus particle, herbicide, insecticide, fertilizers, pH modulators, soil stabilizers, and flowability agents. In some embodiments, the efficacy of the cargo is improved by formulation within an agricultural SBP formulations. In some embodiments, agricultural SBP formulations may encapsulate cargo for extended and/or controlled release.

In some embodiments, cargos for use in SBP formulations may be selected from any of those listed in Table 5.

TABLE 5

CARGO	
Payload	Category
amikacin	antibiotic
amoxicillin	antibiotic
ampicillin	antibiotic
azithromycin	antibiotic
azlocillin	antibiotic
aztreonam	antibiotic
capreomycin	antibiotic
carbenicillin	antibiotic
cefaclor	antibiotic
cefadroxil	antibiotic
cefalexin	antibiotic
cefalothin	antibiotic
cefamandole	antibiotic
cefazolin	antibiotic
cefdinir	antibiotic
cefditoren	antibiotic
cefepime	antibiotic
cefixime	antibiotic
cefoperazone	antibiotic
cefotaxime	antibiotic
cefoxitin	antibiotic
cefpodoxime	antibiotic
cefprozil	antibiotic
ceftaroline fosamil	antibiotic
ceftazidime	antibiotic
ceftibuten	antibiotic
ceftizoxime	antibiotic
ceftobiprole	antibiotic
ceftriaxone	antibiotic
cefuroxime	antibiotic
cilastatin	antibiotic
ciprofolaxin	antibiotic
clarithromycin	antibiotic
clindamycin	antibiotic
clofazimine	antibiotic
cloxacillin	antibiotic
cycloserine	antibiotic
dalbavancin	antibiotic
dapsone	antibiotic
daptomycin	antibiotic
demeclocycline	antibiotic
dicloxacillin	antibiotic
dirithromycin	antibiotic
doripenem	antibiotic
doxycycline	antibiotic
enoxacin	antibiotic
ertapenem	antibiotic
ethambutol	antibiotic
ethionamide	antibiotic
flucloxacillin	antibiotic
furazolidone	antibiotic
gatifloxacin	antibiotic
geldanamycin	antibiotic
gemifloxacin	antibiotic
gentamicin	antibiotic
grepafloxacin	antibiotic
herbimycin	antibiotic
imipeneum	antibiotic
isoniazid	antibiotic
kanamycin	antibiotic
levofloxacin	antibiotic
linezolid	antibiotic
linomycin	antibiotic
lomefloxacin	antibiotic
loracarbef	antibiotic
mafenide	antibiotic
meropenem	antibiotic
methicillin	antibiotic
mezlocillin	antibiotic
minocycline	antibiotic
moxifloxacin	antibiotic
nafcillin	antibiotic
nalidixic acid	antibiotic
neomycin	antibiotic
netilmicin	antibiotic
nitrofurantoin	antibiotic

TABLE 5-continued

CARGO	
Payload	Category
norfloxacin	antibiotic
ofloxacin	antibiotic
oritavancin	antibiotic
oxacillin	antibiotic
oxytetracycline	antibiotic
paromomycin	antibiotic
penicillin G	antibiotic
penicillin V	antibiotic
piperacillin	antibiotic
posizolid	antibiotic
pyrazinamide	antibiotic
radezolid	antibiotic
rifampicin	antibiotic
rifaximin	antibiotic
roxithromycin	antibiotic
sparfloxacin	antibiotic
spectinomycin	antibiotic
spiramycin	antibiotic
sulfacetamide	antibiotic
sulfadiazine	antibiotic
sulfadimethoxine	antibiotic
sulfamethizole	antibiotic
sulfamethoxazole	antibiotic
sulfanilimide	antibiotic
sulfasalazine	antibiotic
sulfisoxazole	antibiotic
teicoplanin	antibiotic
telavancin	antibiotic
telithromycin	antibiotic
temafloxacin	antibiotic
temocillin	antibiotic
ticarcillin	antibiotic
tobramycin	antibiotic
torezolid	antibiotic
troleandomycin	antibiotic
trovafloxacin	antibiotic
vancomycin	antibiotic
erythromycin	antibiotic; endoparasiticide
penicillin	antibiotic; endoparasiticide
streptomycin	antibiotic; endoparasiticide
tetracycline	antibiotic; endoparasiticide
5-fluorocytosine	antifungal
abafungin	antifungal
albaconazole	antifungal
allylamine	antifungal
amorolfin	antifungal
amphotericin B	antifungal
anidulafungin	antifungal
aurone	antifungal
balsam	antifungal
benzoic acid	antifungal
bifonazole	antifungal
butenafine	antifungal
butoconazole	antifungal
candicidin	antifungal
caspofungin	antifungal
ciclopirox	antifungal
clotrimazole	antifungal
crystal violet	antifungal
echinocandin	antifungal
econazole	antifungal
efinaconazole	antifungal
epoxiconazole	antifungal
fenticonazole	antifungal
filipin	antifungal
fluconazole	antifungal
flucytosine	antifungal
griseofulvin	antifungal
haloprogin	antifungal
hamycin	antifungal
imidazole	antifungal
isavuconazole	antifungal
isoconazole	antifungal
itraconazole	antifungal
ketoconazole	antifungal
luliconazole	antifungal

TABLE 5-continued

CARGO	
Payload	Category
micafungin	antifungal
miconazole	antifungal
miltefosine	antifungal
naftitine	antifungal
natamycin	antifungal
nystatin	antifungal
omoconazole	antifungal
orotomide	antifungal
oxiconazole	antifungal
polyene antifungal	antifungal
posaconazole	antifungal
propiconazole	antifungal
ravuconazole	antifungal
rimocidin	antifungal
sertaconazole	antifungal
sulconazole	antifungal
terbinafine	antifungal
terconazole	antifungal
thiazole	antifungal
tioconazole	antifungal
tolnaftate	antifungal
triazole	antifungal
undecylenic acid	antifungal
voriconazole	antifungal
bacterial cell	biologic
microbiome	biologic
microorganism	biologic
rhizobia bacteria	biologic
symbiote	biologic
virus	biologic
virus fragment	biologic
virus particle	biologic
fungicide	biologic; pesticide
Pour-Ons	ectoparasiticide
Sprays	ectoparasiticide
Dips	ectoparasiticide
Ear Tags	ectoparasiticide
Collars	ectoparasiticide
Oral Tablets	ectoparasiticide
Other Ectoparasiticides	ectoparasiticide
Spot-Ons	ectoparasiticide
ectoparaciticide	ectoparasiticide
pyrethroid	ectoparasiticide
carbamate	ectoparasiticide
water-insoluble organo-phospphorus compound	ectoparasiticide
benzoyl urea	ectoparasiticide
formamidine	ectoparasiticide
triazine	ectoparasiticide
avermectin	ectoparasiticide
milbemycin	ectoparasiticide
flumethrin	ectoparasiticide
alphamethrin	ectoparasiticide
pirimphos methyl	ectoparasiticide
pirimphos ethyl	ectoparasiticide
mylbemycin	ectoparasiticide
moxidectin	ectoparasiticide; endoparasiticide
ivermectin	ectoparasiticide; endoparasiticide; insecticide
doramectin	ectoparasiticide; endoparasiticide; insecticide
abamectin	ectoparasiticide; insecticide
pyrethrin	ectoparasiticide; insecticide
cyhalothrin	ectoparasiticide; insecticide
amitraz	ectoparasiticide; insecticide
deltamethrin	ectoparasiticide; insecticide
diazinon	ectoparasiticide; insecticide
macrocyclic lactones	endecticide
benzimidazole	endecticide
pro-benzimidazole	endecticide
imidazothiazole	endecticide
tetrahydropyrimidine	endecticide
organophosphate	endecticide

TABLE 5-continued

CARGO	
Payload	Category
Endoparasiticide	endoparasiticide
Oral Liquids	endoparasiticide
Oral Solids	endoparasiticide
Injectables	endoparasiticide
Feed Additives	endoparasiticide
Endectocides	endoparasiticide
Tetramisole	endoparasiticide
dexamisole	endoparasiticide
Milbemycin oxime	endoparasiticide
Nemadectin	endoparasiticide
Albendazole	endoparasiticide
Clorsulon	endoparasiticide
Cydectin	endoparasiticide
Diethyicarbamazine	endoparasiticide
Febantel	endoparasiticide
Fenbendazole	endoparasiticide
Haloxon	endoparasiticide
Levamisole	endoparasiticide
Mebendazole	endoparasiticide
Morantel	endoparasiticide
Oxyclozanide	endoparasiticide
Oxibendazole	endoparasiticide
Oxfendazole	endoparasiticide
Oxamniquine	endoparasiticide
Pyrantel	endoparasiticide
Praziquantel	endoparasiticide
Thiabendazole	endoparasiticide
cyclosporin	endoparasiticide
sulfonamide	endoparasiticide
cephalosporin	endoparasiticide
cephamycin	endoparasiticide
aminoglucosid	endoparasiticide
trimethoprim	endoparasiticide
dimetridazole	endoparasiticide
framycetin	endoparasiticide
fruazolidone	endoparasiticide
plenromutilin	endoparasiticide
a compound active against protozoa	endoparasiticide
piperazine	endoparasiticide; endecticide
emamectin	endoparasiticide; insecticide
eprinomectin	endoparasiticide; insecticide
milbemectin	endoparasiticide; insecticide
permethrin	endoparasiticide; insecticide
selamectin	endoparasiticide, insecticide
trichlorfon	endoparasiticide; insecticide
ammonium nitrate	fertilizer
binary fertilizer	fertilizer
compound fertilizer	fertilizer
diammonium phosphate	fertilizer
fertilizer	fertilizer
monoammonium phosphate	fertilizer
multinutrient fertilizer	fertilizer
natural fertilizer	fertilizer
nitrogen fertilizer	fertilizer
NK fertilizer	fertilizer
NP fertilizer	fertilizer
NPK fertilizer	fertilizer
organic fertilizer	fertilizer
phosphate fertilizer	fertilizer
PK fertilizer	fertilizer
potassium fertilizer	fertilizer
single-nutrient fertilizer	fertilizer
superphosphate	fertilizer
synthetic fertilizer	fertilizer
urea	fertilizer
binapacryl	fungicide
Bisphenol A	fungicide
copper 8-hydroxyquinoline	fungicide
copper sulfate	fungicide
mercuric chloride	fungicide
phenol	fungicide
phenylmercuric oleate	fungicide
tributyltin chloride	fungicide
tributyltin triacetate	fungicide
pentachlorophenol	fungicide; insecticide
Bupirimate	fungicide; pesticide

TABLE 5-continued

CARGO	
Payload	Category
Captan	fungicide; pesticide
Carbendazim	fungicide; pesticide
Chloranil	fungicide; pesticide
antibiotic	general
fertilizer	general
nutrient	general
pH modulator	general
small molecule	general
soil stabilizer	general
therapeutic agent	general
2,4,5-T	herbicide
2,4-D	herbicide
atrazine	herbicide
chlorophenoxy acid	herbicide
cynazine	herbicide
glyphosate	herbicide
hexazinone	herbicide
MCPA	herbicide
metribuzin	herbicide
organic phosphorus herbicide	herbicide
silvex	herbicide
simazine	herbicide
triazine herbicide	herbicide
bromacil	herbicide, pesticide
Chloramben	herbicide; pesticide
Chlorfenac	herbicide; pesticide
Chlorsulfuron	herbicide; pesticide
Abscisic acid	hormone
Auxins	hormone
Cytokinins	hormone
Ethylene	hormone
Gibberellins	hormone
steroid	hormone
dexamethasone	hormone
allopregnanolone	hormone
estrogen	hormone
ethinylestradiol	hormone
mestranol	hormone
estradiols	hormone
estriol	hormone
estriolsuccinate	hormone
polyestriolphosphate	hormone
estrone	hormone
estronesulfate	hormone
conjugatedestrogens	hormone
progesterone	hormone
norethisteroneacetate	hormone
norgestrel	hormone
levonorgestrel	hormone
gestodene	hormone
chlormadinoneacetate	hormone
drospirorenone	hormone
3-ketodesogestrel	hormone
androgen	hormone
testosterone	hormone
androstenediol	hormone
androstenedione	hormone
dehydroepiandrosterone	hormone
dihydrotestosterone	hormone
anymineralocorticoid	hormone
anyglucocoriticoid	hormone
cholesterols	hormone
1,2-dichloropropane	insecticide
acephate	insecticide
acetamiprid	insecticide
acethion	insecticide
acetoprole	insecticide
acrinathrin	insecticide
acrylonitrile	insecticide
alanycarb	insecticide
aldicarb	insecticide
aldnrn	insecticide
aldoxycarb	insecticide
allethrin	insecticide
allosamidin	insecticide
allyxycarb	insecticide

TABLE 5-continued

CARGO	
Payload	Category
alpha-cypermethrin	insecticide
amidithion	insecticide
aminocarb	insecticide
amiton	insecticide
anabasine	insecticide
athidathion	insecticide
azadirachtin	insecticide
azamethiphos	insecticide
azinphos-ethyl	insecticide
azinphos-methyl	insecticide
azothoate	insecticide
barium hexafluorosilicate	insecticide
barthrin	insecticide
bendiocarb	insecticide
benfuracarb	insecticide
bensultap	insecticide
beta-cyfluthrin	insecticide
beta-cypermethrin	insecticide
bifenthrin	insecticide
bioallethrin	insecticide
bioethanomethrin	insecticide
biopermethrin	insecticide
bioresmethrin	insecticide
bistrifluron	insecticide
borax	insecticide
botanical insecticide	insecticide
bromfenvinfos	insecticide
bromo-DDT	insecticide
bromophos-ethyl	insecticide
bufencarb	insecticide
buprofezin	insecticide
butacarb	insecticide
butathiofos	insecticide
butocarboxim	insecticide
butonate	insecticide
butoxycarboxim	insecticide
cadusafos	insecticide
calcium arsenate	insecticide
calcium polysulfide	insecticide
camphechlor	insecticide
carbanolate	insecticide
carbofuran	insecticide
carbon disulfide	insecticide
carbon tetrachloride	insecticide
carbosulfan	insecticide
cartap	insecticide
chlorbicyclen	insecticide
chlordane	insecticide
chlordecone	insecticide
chlorethoxyfos	insecticide
chlorfenapyr	insecticide
chlorfenvinphos	insecticide
chlorfluazuron	insecticide
chlormephos	insecticide
chloroform	insecticide
chloropicrin	insecticide
chlorphoxim	insecticide
chlorprazophos	insecticide
chlorpyrifos-methyl	insecticide
chlorthiophos	insecticide
chromafenozide	insecticide
cinerin I	insecticide
cinerin II	insecticide
cismethrin	insecticide
cloethocarb	insecticide
closantel	insecticide
clothianidin	insecticide
copper acetoarsenite	insecticide
copper arsenate	insecticide
coumaphos	insecticide
coumithoate	insecticide
crotamiton	insecticide
crotoxyphos	insecticide
crufomate	insecticide
cryolite	insecticide
cyanofenphos	insecticide

TABLE 5-continued

CARGO	
Payload	Category
cyanophos	insecticide
cyanthoate	insecticide
cyclethrin	insecticide
cycloprothrin	insecticide
cyfluthrin	insecticide
cypermethrin	insecticide
cyphenothrin	insecticide
cyromazine	insecticide
cythioate	insecticide
DDT	insecticide
decarbofuran	insecticide
demephion	insecticide
demephion-O	insecticide
demephion-S	insecticide
demeton	insecticide
demeton-methyl	insecticide
demeton-O	insecticide
demeton-O-methyl	insecticide
demeton-S	insecticide
demeton-S-methyl	insecticide
demeton-S-methylsulphon	insecticide
diafenthiuron	insecticide
dialifos	insecticide
dicapthon	insecticide
dichlofenthion	insecticide
dichlorvos	insecticide
dicresyl	insecticide
dicrotophos	insecticide
dicyclanil	insecticide
dieldrin	insecticide
diflubenzuron	insecticide
dilor	insecticide
dimefox	insecticide
dimetan	insecticide
dimethoate	insecticide
dimethrin	insecticide
dimethylvinphos	insecticide
dimetilan	insecticide
dinex	insecticide
dinoprop	insecticide
dinosam	insecticide
dinotefuran	insecticide
diofenolan	insecticide
dioxabenzofos	insecticide
dioxacarb	insecticide
dioxathion	insecticide
disulfoton	insecticide
dithicrofos	insecticide
d-limonene	insecticide
ecdysterone	insecticide
empenthrin	insecticide
endosulfan	insecticide
endothion	insecticide
endrin	insecticide
epofenonane	insecticide
esfenvalerate	insecticide
etaphos	insecticide
ethiofencarb	insecticide
ethion	insecticide
ethiprole	insecticide
ethoate-methyl	insecticide
ethoprophos	insecticide
ethyl formate	insecticide
ethylene dibromide	insecticide
ethylene dichloride	insecticide
ethylene oxide	insecticide
etofenprox	insecticide
etrimfos	insecticide
famphur	insecticide
fenamiphos	insecticide
fenazaflor	insecticide
fenchlorphos	insecticide
fenethacarb	insecticide
fenfluthrin	insecticide
fenitrothion	insecticide
fenobucarb	insecticide

TABLE 5-continued

CARGO	
Payload	Category
fenoxacrim	insecticide
fenoxycarb	insecticide
fenpirithrin	insecticide
fenpropathrin	insecticide
fensulfothion	insecticide
fenthion	insecticide
fenthion-ethyl	insecticide
fenvalerate	insecticide
fipronil	insecticide
flonicamid	insecticide
flucofuron	insecticide
flucycloxuron	insecticide
flucythrinate	insecticide
flufenerim	insecticide
flufenoxuron	insecticide
flufenprox	insecticide
fluvalinate	insecticide
fonofos	insecticide
formetanate	insecticide
formothion	insecticide
formparanate	insecticide
fosmethilan	insecticide
fospirate	insecticide
fosthietan	insecticide
furathiocarb	insecticide
furethrin	insecticide
gamma-cyhalothrin	insecticide
halfenprox	insecticide
halofenozide	insecticide
heptachlor	insecticide
heptenophos	insecticide
heterophos	insecticide
hexaflumuron	insecticide
hydramethylnon	insecticide
hydrogen cyanide	insecticide
hydroprene	insecticide
hyquincarb	insecticide
imidacloprid	insecticide
imiprothrin	insecticide
indoxacarb	insecticide
isazofos	insecticide
isobenzan	insecticide
isodrin	insecticide
isofenphos	insecticide
isoprocarb	insecticide
isoprothiolane	insecticide
isothioate	insecticide
isoxathion	insecticide
isoxazole	insecticide
jasmolin I	insecticide
jasmolin II	insecticide
jodfenphos	insecticide
juvenile hormone I	insecticide
juvenile hormone II	insecticide
juvenile hormone III	insecticide
kelevan	insecticide
kinoprene	insecticide
lambda-cyhalothrin	insecticide
lead arsenate	insecticide
leptophos	insecticide
lirimfos	insecticide
lufenuron	insecticide
lythidathion	insecticide
malathion	insecticide
malonoben	insecticide
mazidox	insecticide
mecarbam	insecticide
mecarphon	insecticide
menazon	insecticide
mephosfolan	insecticide
mercurous chloride	insecticide
mesulfenfos	insecticide
methacrifos	insecticide
methamidophos	insecticide
methidathion	insecticide
methiocarb	insecticide

TABLE 5-continued

CARGO	
Payload	Category
methocrotophos	insecticide
methomyl	insecticide
methoprene	insecticide
methoxychlor	insecticide
methoxyfenozide	insecticide
methyl bromide	insecticide
methylchloroform	insecticide
methylene chloride	insecticide
metofluthrin	insecticide
metolcarb	insecticide
metoxadiazone	insecticide
mevinphos	insecticide
mexacarbate	insecticide
mipafox	insecticide
mirex	insecticide
monocrotophos	insecticide
morhothion	insecticide
naftalofos	insecticide
naled	insecticide
naphthalene	insecticide
nicotine	insecticide
nifluridide	insecticide
nitenpyram	insecticide
nithiazine	insecticide
nitrilacarb	insecticide
novaluron	insecticide
noviflumuron	insecticide
omethoate	insecticide
oxamyl	insecticide
oxydemeton-methyl	insecticide
oxydeprofos	insecticide
oxydisulfoton	insecticide
para-dichlorobenzene	insecticide
parathion	insecticide
parathion-methyl	insecticide
penfluron	insecticide
phenkapton	insecticide
phenothrin	insecticide
phenthoate	insecticide
phorate	insecticide
phosalone	insecticide
phosfolan pirimetaphos	insecticide
phosmet	insecticide
phosnichlor	insecticide
phosphamidon	insecticide
phosphine	insecticide
phoxim	insecticide
phoxim-methyl	insecticide
pirimicarb	insecticide
pirimiphos-ethyl	insecticide
pirimiphos-methyl	insecticide
potassium arsenite	insecticide
potassium thiocyanate	insecticide
pp'-DDT	insecticide
prallethrin	insecticide
precocene I	insecticide
precocene II	insecticide
precocene III	insecticide
primidophos	insecticide
profenofos	insecticide
profluthrin	insecticide
promacyl	insecticide
promecarb	insecticide
propaphos	insecticide
propetamphos	insecticide
propoxur	insecticide
prothidathion	insecticide
prothiofos	insecticide
prothoate	insecticide
protrifenbute	insecticide
pyraclofos	insecticide
pyrazophos	insecticide
pyresmethrin	insecticide
pyrethrin I	insecticide
pyrethrin II	insecticide
pyridaben	insecticide

TABLE 5-continued

CARGO	
Payload	Category
pyridalyl	insecticide
pyridaphenthion	insecticide
pyrimidifen	insecticide
pyrimitate	insecticide
pyriproxyfen	insecticide
quassia	insecticide
quinalphos	insecticide
quinalphos-methyl	insecticide
quinothion	insecticide
rafoxanide	insecticide
resmethrin	insecticide
rotenone	insecticide
ryania	insecticide
sabadilla	insecticide
schradan	insecticide
silafluofen	insecticide
sodium arsenite	insecticide
sodium fluoride	insecticide
sodium hexafluorosilicate	insecticide
sodium thiocyanate	insecticide
sophamide	insecticide
spinosad	insecticide
spiromesifen	insecticide
sulcofuron	insecticide
sulfluramid	insecticide
sulfotep	insecticide
sulfuryl fluoride	insecticide
sulprofos	insecticide
tau-fluvalinate	insecticide
tazimcarb	insecticide
tebufenozide	insecticide
tebufenpyrad	insecticide
tebupirimfos	insecticide
teflubenzuron	insecticide
tefluthrin	insecticide
temephos	insecticide
terallethrin	insecticide
terbufos	insecticide
tetrachlorethane	insecticide
tetrachlorvinphos	insecticide
tetramethrin	insecticide
theta-cypermethrin	insecticide
thiacloprid	insecticide
thiamethoxam	insecticide
thicrofos	insecticide
thiocarboxime	insecticide
thiocyclam	insecticide
thiodicarb	insecticide
thiofanox	insecticide
thiometon	insecticide
thiosultap	insecticide
thuringiensin	insecticide
tolfenpyrad	insecticide
tralomethrin	insecticide
transfluthrin	insecticide
transpermethrin	insecticide
triarathene	insecticide
triazamate	insecticide
triazophos	insecticide
trichlormetaphos-3	insecticide
trichloronat	insecticide
trifenofos	insecticide
triflumuron	insecticide
trimethacarb	insecticide
triprene	insecticide
vamidothion	insecticide
vaniliprole	insecticide
xylylcarb	insecticide
zeta-cypermethrin	insecticide
zolaprofos	insecticide
α-ecdysone	insecticide
bromophos	insecticide; pesticide
chlordimeform	insecticide; pesticide
Carbaryl	insecticide; pesticide
Carbophenothion	insecticide; pesticide
Chlorpyrifos	insecticide; pesticide

TABLE 5-continued

CARGO	
Payload	Category
amino acid	macronutrient
amylopectin	macronutrient
amylose	macronutrient
arachidic acid	macronutrient
behenic acid	macronutrient
butyric acid	macronutrient
capric acid	macronutrient
caprioic acid	macronutrient
caprylic acid	macronutrient
carbohydrate	macronutrient
cerotic acid	macronutrient
cervonic acid	macronutrient
clupanodonic acid	macronutrient
eicosen	macronutrient
erucic acid	macronutrient
essential fatty acid	macronutrient
fat	macronutrient
fructose	macronutrient
galactose	macronutrient
glucose	macronutrient
heptadecanoic acid	macronutrient
lactose	macronutrient
lauric acid	macronutrient
lignoceric acid	macronutrient
linoleic acid	macronutrient
Macronutrient	macronutrient
maltose	macronutrient
margaric acid	macronutrient
monounsaturated fat	macronutrient
myristic acid	macronutrient
myristol	macronutrient
nervonic acid	macronutrient
oleic acid	macronutrient
palmitic acid	macronutrient
palmitoyl	macronutrient
pentadecanoic acid	macronutrient
polyunsaturated fat	macronutrient
protein	macronutrient
ribose	macronutrient
saturated fat	macronutrient
stearic acid	macronutrient
steridonic acid	macronutrient
sucrose	macronutrient
timnodonic acid	macronutrient
α-linoleic acid	macronutrient
calcium	micronutrient
chloride	micronutrient
chromium	micronutrient
copper	micronutrient
iodine	micronutrient
iron	micronutrient
magnesium	micronutrient
manganese	micronutrient
mineral	micronutrient
molybdenum	micronutrient
nickel	micronutrient
phosphorus	micronutrient
potassium	micronutrient
selenium	micronutrient
silicon	micronutrient
tin	micronutrient
vanadium	micronutrient
vitamin	micronutrient
vitamin A	micronutrient
vitamin B-1	micronutrient
vitamin B-12	micronutrient
vitamin B-2	micronutrient
vitamin B-3	micronutrient
vitamin B-5	micronutrient
vitamin B-6	micronutrient
vitamin B-7	micronutrient
vitamin B-9	micronutrient
vitamin C	micronutrient
vitamin D	micronutrient
vitamin E	micronutrient
vitamin K	micronutrient

TABLE 5-continued

CARGO	
Payload	Category
zinc	micronutrient
adhesive	pest control agent
allomone	pest control agent
anti-disease agent	pest control agent
antifeedant	pest control agent
antifungal	pest control agent
behavior-modifying compound	pest control agent
bird repellent	pest control agent
black pepper	pest control agent
caffeine	pest control agent
capsaicin	pest control agent
capsaicin oleoresin	pest control agent
catnip oil	pest control agent
chemosterilant	pest control agent
chili powder	pest control agent
complex sugar	pest control agent
dill	pest control agent
ginger	pest control agent
gum	pest control agent
herbicide	pest control agent
insect attractant	pest control agent
insect repellent	pest control agent
insecticide	pest control agent
kairomone	pest control agent
mammal repellent	pest control agent
mating disrupter	pest control agent
monoterpenoid	pest control agent
paprika	pest control agent
pest control agent	pest control agent
pesticide	pest control agent
phenolic compound	pest control agent
pheromone	pest control agent
red pepper	pest control agent
acaricide	pesticide
algicide	pesticide
avicide	pesticide
Bacillus thuringiensis endotoxin polypeptide	pesticide
bactericide	pesticide
Bis(p-chlorophenoxy)methane	pesticide
Bitertanol	pesticide
Bromadiolone	pesticide
Bromethalinlin	pesticide
Bromopropylate	pesticide
Busulfan	pesticide
Butrylin	pesticide
Cambendazole	pesticide
Candicidin	pesticide
Candidin	pesticide
Chloramphenacol	pesticide
Chlorbetamide	pesticide
Chlorothion	pesticide
Chlorphenesin	pesticide
molluscicide	pesticide
nematicide	pesticide
rodenticide	pesticide
virucide	pesticide
biopolymer	soil stabilizer
chemical	soil stabilizer
co-polymer	soil stabilizer
enzyme	soil stabilizer
fiber reinforcement agent	soil stabilizer
flowability agent	soil stabilizer
hydrophilic agent	soil stabilizer
hydrophobic agent	soil stabilizer
ionic stabilizer	soil stabilizer
polymer	soil stabilizer
resin	soil stabilizer
salt	soil stabilizer
surfactant	soil stabilizer
chelated micronutrient	therapeutic agent
microbe	therapeutic agent
non-chelated micronutrient	therapeutic agent
probiotic	therapeutic agent

In one embodiment, the cargo for use in SBP formulations may be hormone analogue such as, but not limited to, Deslorelin.

Coating

In some embodiments, agricultural SBP formulations may include one or more coatings. As used herein, the term "coating" refers to any substance that is applied to the surface of another substance. In some embodiments, the coating may be functional, decorative or both. Coatings may be applied to completely cover the surface. Coating may also be applied to partially cover the surface. In some embodiments, coatings may include processed silk. In some aspects, the coating may be an SBP formulation. Coatings may also include but are not limited to any of the cargos described in Table 5. In some embodiments, the coating may be a plant coating. In some embodiments, the coating may be a seed coating. In some embodiments, the coating may be a leaf coating. In some embodiments, agricultural compositions described herein, such as coatings, may be able to penetrate plants, leaves, seeds, roots, and/or any other part of the plant described herein. In some embodiments, the SBP coating may be used for one or more applications, including, but not limited to, protection of a seed, plant, planting substrate, agricultural product, or device; fertilizing and/or promoting germination of a coated seed or plant; encasing a payload; delivering a payload; modulating nutrient and/or water uptake; stabilizing a payload; and/or controlling the release of a payload. In some embodiments, SBP coatings may be applied to a fruit or a vegetable to prevent spoilage.

Agricultural Products

In some embodiments, agricultural SBP formulations may include one or more agricultural products. These agricultural products may be plants, animals, plant agricultural products, and animal agricultural products.

In some embodiments, agricultural SBP formulations may include plants. The methods and SBPs of the present disclosure may have applications in plants. In some embodiments, SBPs will serve as agricultural composition to facilitate the production of plants. In some embodiments the plants are agricultural plants i.e., plants for farming purposes. In some embodiments, the plants are silvicultural plants, i.e. plants for the controlling the growth, health, establishment, composition, and quality of forests. In some embodiments, the plants are ornamental plants. In some embodiments, the plants are edible plants. In some embodiments, the plants are horticultural plants. In some embodiments, the plants are natural or wild-type plants. In other embodiments, the plants are genetically modified plants. In some aspects, the plants are medicinal plants. In some embodiments, the agricultural products may be portions of plants.

In some embodiments, agricultural products may include animals and/or animal agricultural products. In some embodiments, the animals used with agricultural compositions of the present disclosure include but are not limited to cows, bulls, sheep, goat, bison, turkey, buffalo, pigs, poultry, horses, alpaca, llama, camels, rabbits, guinea pigs, fish, shrimps, crustaceans, mollusks, insects, silk worms, bees, and crickets.

In some embodiments, the agricultural SBP formulations may be or may include one or more animal agricultural products. Animal agricultural products may include, but are not limited to milk, butter, cheese, yogurt, whey, curds, meat, oil, fat, blood, amino acids, hormones, enzymes, wax, feathers, fur, hide, bones, gelatin, horns, ivory, wool, venom, tallow, silk, sponges, manure, eggs, pearl culture, honey, and food dye. In some embodiments, the animal agricultural

product is a dairy product. Non-limiting examples of dairy products include milk, cream, cheese, clotted cream, sour cream, gelato, ghee, infant formula, powdered milk, butter, crème fraiche, ice cream, yoghurt, curds, whey, custard, dulce de leche, evaporated milk, eggnog, frozen yoghurt, frozen custard, buttermilk, formula, casein, condensed milk, cottage cheese, and cream cheese.

Pest Control Agent

In some embodiments, agricultural SBP formulations may include pest control agents. In some aspects, the SBPs may be a pest control agent. As used herein, the term "pest" refers to any organism that harms, irritates, causes discomfort, or generally annoys another organism. In some embodiments, the pest control agent may optionally include a pesticide. In some embodiments, pesticides used in agricultural compositions may be selected from any of those listed in Table 5. Pesticides may include, but are not limited to parasiticides, insecticides, herbicides, antifungal or fungicide, anti-disease agents, behavior-modifying compounds, adhesives (e.g. gums), acaricide, algicide, avicide, bactericide, molluskicide, biocides, miticides, nematicide, rodenticide, and a virucide.

Biological Systems

In some embodiments, agricultural SBP formulations described herein include biological systems. As used herein, the term "biological system" refers to a network of interrelated substances and/or organisms. These biological systems may include systems of symbiotes, microbiomes and/or probiotics. The compositions provided herein may include a SBPs and an active amount of beneficial microbes/probiotics. In some embodiments, SBP formulations may be used as stabilizers in the microbial compositions. In some embodiments, these microbiomes or symbiotes may incorporate species of fungi or bacteria. In some embodiments, the fungi are from the *Aspergillus* genus. In some embodiments, the bacteria are from the *Streptomyces* genus. In some embodiments, SBP biological systems may be used as biopesticides. As used herein, the term "biopesticide" refers to a composition with a bacteria, microorganism, or biological cargo that displays pesticidal activity. In some embodiments, SBP biological systems may be applied as a coating to a plant. The coating may be applied to the whole plant, or to any part of the plant described in the present disclosure. In some embodiments, the coating may be applied to a seed.

In some embodiments, the biological systems may be used to enable nitrogen fixation. These microbes, microorganisms, and/or microbiomes may incorporate *rhizobia* bacteria. *Rhizobia* bacteria enable nitrogen fixation in plants that do not independently fix nitrogen, such as legumes (Zahran et al. (1999) Microbiology and Molecular Biology Reviews 63(4):968-989, the contents of which are herein incorporated by reference in its entirety). In some embodiments, the biological systems described herein deliver *rhizobia* bacteria for the growth and production of other plants. In some embodiments, the SBP agricultural compositions described herein may be formulated with the nutrients needed to promote the growth of *rhizobia* bacteria. The beneficial microbe and/or probiotic can be any beneficial microbe and/or probiotic known in the art.

In some embodiments, SBP biological systems formulations may include microbes, microorganisms, and/or microbiomes that promote plant growth. SBP biological systems may include one or more microbes, microorganisms and/or microbiomes that promote plant growth. Such microbes, microorganisms, and/or microbiomes may include, but are not limited to, *Algoriphagus ratkowskyi, Altererythrobacter luteolus, Alternaria thalictrigena, Arthrobacter agilis,* *Arthrobacter arilaitensis, Arthrobacter aurescens, Arthrobacter citreus, Arthrobacter crystallopoeietes, Arthrobacter globiformis, Arthrobacter humicola, Arthrobacter oryzae, Arthrobacter oxydans, Arthrobacter pascens, Arthrobacter ramosus, Arthrobacter tumbae, Aspergillus fumigatiaffinis, Bacillus aquimaris, Bacillus benzoevorans, Bacillus cibi, Bacillus herbersteinensis, Bacillus idriensis, Bacillus licheniformis, Bacillus niacin, Bacillus psychordurans, Bacillus simplex, Bacillus simplex* 11, *Bacillus simplex* 237, *Bacillus simplex* 30N-5, *Bacillus subtilis* 30VD-1, *Bartonella elizabethae, Citricoccus alkalitolerans, Citricoccus nitrophenolicus, Cladosporium sphaerospermum, Curtobacterium flaccumfaciens, Exiguobacterium aurantiacum, Fusarium equiseti, Fusarium oxysporum, Georgenia ruanii, Halomonas aquamarina, Kocuria rosea, Massilia timonae, Mesorhizobium loti, Microbacterium aerolatum, Microbacterium oxydans, Microbacterium paludicola, Microbacterium paraoxydans, Microbacterium phyllosphaerae, Microbacterium testaceum, Micrococcus luteus, Mycobacterium sacrum, Nocardiopsis quinghaiensis, Oceanobacillus picturae, Ochroconis sp., Olivibacter soli, Paenibacillus tundrae, Penicillium chrysogenum, Penicillium commune, Phoma betae, Planococcus maritimus, Planococcus psychrotoleratus, Planomicrobium koreense, Planomicrobium okeanokoites, Promicromonospora kroppenstedtii, Pseudomonas brassicacearum, Pseudomonas fluorescens, Pseudomonas frederiksbergensis, Pseudomonas fulva, Pseudomonas geniculata, Pseudomonas gessardii, Pseudomonas libanensis, Pseudomonas mosselii, Pseudomonas plecoglossicida, Pseudomonas putida, Pseudomonas stutzeri, Pseudomonas syringae, Rhodococcus jostii, Sinorhizobium medicae, Sinorhizobium meliloti, Staphylococcus succinus, Stenotrophomonas maltophilia, Stenotrophomonas rhizophila, Streptomyces althioticus, Streptomyces azureus, Streptomyces bottropensis, Streptomyces candidus, Streptomyces chryseus, Streptomyces cirratus, Streptomyces coeruleofuscus, Streptomyces durmitorensis, Streptomyces flaveus, Streptomyces fradeiae, Streptomyces griseoruber, Streptomyces griseus, Streptomyces halstedii, Streptomyces marokkonensis, Streptomyces olivoviridis, Streptomyces peucetius, Streptomyces phaeochromogenes, Streptomyces pseudogriseolus, Terribacillus halophilus, Virgibacillus halodenitrificans*, and *Williamensia muralis*. In further embodiments, such plant growth-promoting microbes, microorganisms, and/or microbiomes may be selected from any of those microbial isolates described in US Publication Number US20140342905, and International Publication Number WO2014201044, the contents of which are hereby incorporated by reference in their entirety.

In some embodiments, the SBP biological system formulations contains a mixture of two or more microbes and/or microorganisms. In some embodiments, the mixture might be a mixture of generic microbes. In some embodiments, formulation with one or more microbes enhances the viability of said microbes. In some embodiments, the SBP biological system may also include one or more excipients (e.g. sugar, mannitol, trehalose, buffer salts, PEGs, Poloxamer-188, Poloxamer-407, glycerol, HPMC, HEC, CMC, glycerol formal, propylene glycol, propylene carbonate, sorbitol, and/or polysorbate-80). The excipients may be included at concentrations between 0.1%-50% (w/w or w/v). The concentration of processed silk may be any of those described in the present disclosure. The SBP biological systems may be made from processed silk prepared with any boiling time described in the present disclosure (e.g. 90mb, 120mb, and 480mb) and one or more microbes. In some embodiments, the pH of the SBP biological system is between about 4 and about 6. In some embodiments, the osmolarity of the SBP biological system is between about 200 mOsm/L to about 400 mOsm/L. The osmolarity may also be from about 290 mOsm/L to about 320 mOsm/L.

In some embodiments, the SBP biological system formulations may be prepared as a solution. These solutions may be made from processed silk prepared with any boiling time described in the present disclosure (e.g. 90mb, 120mb, and 480mb) and the bacteria. The solutions may be prepared with any concentration of processed silk described herein (e.g. 0.5%, 1%, 5%). The solutions may be prepared with any solvent and/or buffer described in the present disclosure.

In some embodiments, the SBP biological system formulations may be prepared as a lyophilized powder. These powders may be prepared from processed silk prepared with any boiling time described in the present disclosure (e.g. 90mb, 120mb, and 480mb) and the bacteria. The processed silk may be mixed with one or more microbes and then lyophilized. The lyophilized powder may be prepared from solutions of any concentration of processed silk described herein (e.g. 0.5%, 1%, 5%). The lyophilized powder may be prepared from processed silk formulated with a sugar (e.g. sugar, mannitol, or trehalose) to aid in the stability of bacteria. The lyophilized silk may be reconstituted in a solvent (e.g. water or buffer). The lyophilized powder may be reconstituted in any solvent and/or buffer described in the present disclosure.

In some embodiments, the SBP biological system formulations may be prepared as insoluble powder, particles, cakes and/or films. To prepare the powder, particles, cakes and/or films, gels and/or other SBP formulations may be formed and then dried. In some embodiments, the gels and/or other SBP formulations may be lyophilized to form cakes and/or films. Powders, particles, cakes and films may also utilize excipients (e.g. gelling agents) as well as sonication, or pH changes to induce beta-sheet formation prior to or during drying or lyophilization. These excipients may be powders under typical environmental conditions (e.g. MW PEG's, Poloxamer-188, etc.). The total solid content of the powders, particles, cakes and/or films may be between 3-40% (w/w or w/v). The solid content included buffer, silk, and any excipients included.

In some embodiments, the SBP biological systems formulations may be insoluble. These SBP biological system formulations may be lyophilized powders. The SBP biological system formulations may also be prepared by milling insoluble solids (e.g. cakes or films) into a powder. In some embodiments, the SBP biological system formulations may be digested. In some embodiments, insoluble SBP biological system formulations are formulated for spray drying. The total solid content of the formulations for spray drying may be between 5-40% (w/w or w/v). The solid content included buffer, silk, and any excipients included.

Agricultural Therapeutic Agent

In some embodiments, agricultural applications involve the use of SBP formulations which can be agricultural therapeutic agents or are combined with one or more agricultural therapeutic agents. Examples of SBP therapeutic agents include, but are not limited to, adjuvants, analgesic agents, antiallergic agents, antiangiogenic agents, antiarrhythmic agents, antibacterial agents, antibiotics, antibodies, anticancer agents, anticoagulants, antidementia agents, antidepressants, antidiabetic agents, antigens, antihypertensive agents, anti-infective agents, anti-inflammatory agents, antioxidants, antipyretic agents, anti-rejection agents, antiseptic agents, antitumor agents, antiulcer agents, antiviral agents, biological agents, birth control medication, carbohydrates, cardiotonics, cells, chemotherapeutic agents, cholesterol lowering agents, cytokines, endostatins, enzymes, fats, fatty acids, genetically engineered proteins, glycoproteins, growth factors, health supplements, hematopoietics, herbal preparations, hormones, hypotensive diuretics, immunological agents, inorganic synthetic pharmaceutical drugs, ions, lipoproteins, metals, minerals, nanoparticles, naturally derived proteins, NSAIDs, nucleic acids, nucleotides, organic synthetic pharmaceutical drugs, oxidants, peptides, pills, polysaccharides, proteins, protein-small molecule conjugates or complexes, psychotropic agents, small molecules, sodium channel blockers, statins, steroids, stimulants, therapeutic agents for osteoporosis, therapeutic combinations, thrombopoietics, tranquilizers, vaccines, vasodilators, VEGF-related agents, veterinary agents, viruses, virus particles, and vitamins. In some embodiments, SBP therapeutics and methods of delivery may include any of those taught in International Patent Publication Numbers WO2017139684, WO2010123945, WO2017123383, or United States Publication Numbers US20170340575, US20170368236, and US20110171239 the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, the agricultural therapeutic agent may be a pest control agent. In some embodiments, examples of pest control agents that may be useful as agricultural therapeutic agent include, but are not limited to parasiticides, insecticides, antifungal or fungicide, anti-disease agents, acaricide, algicide, avicide, bactericide, nematicide, and a virucide.

In some embodiments, the subject in the context of an agricultural therapeutic agent may refer to one or more plants.

In some embodiments, the subject in the context of an agricultural therapeutic agent may refer to one or more non-human animals.

In some embodiments, the agricultural therapeutic agent may be nucleic acids. Nucleic acids may include DNA and/or RNA. In some embodiments, nucleic acids may be polynucleotides or oligonucleotides. Exemplary nucleic acids may include, but are not limited to, aptamers, plasmids, siRNA, microRNAs, or viral nucleic acids. In some embodiments, nucleic acids may encode a therapeutic peptide or protein, such as any one of those described herein. In some embodiments, SBPs may be used to improve the stability of composition comprising the nucleic acids. In some embodiments, SBPs may be used to facilitate the delivery of the nucleic acids to a plant.

Agriculture Devices

In some embodiments, agricultural SBP formulations may be or may include may be used to improve the growth and production of agricultural products by utilizing said composition with an agricultural device. An agricultural device is a device or machine that assists in agricultural production. The agricultural SBP formulations may comprise any format described in the present disclosure (e.g. hydrogel). In some embodiments, SBP formulations may be utilized as an agricultural device, as taught in in United States Patent Publication US20030198659 (the contents of which are herein incorporated by reference in its entirety). In some embodiments, SBP formulations may comprise one or more components of an agricultural device. In some embodiments, SBP formulations may be used in conjunction with another agricultural device. Agricultural devices that may incorporate SBP formulations include, but are not limited to, agricultural equipment, crop storage devices (e.g. bale bags), landscaping fabrics (e.g. polypropylene and burlap blankets), and pest control devices. In one embodiments, the agricultural equipment may comprise a silk-coated microporous pipeline, as taught in Chinese Patent Publication, CN102407193, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, SBP formulations are or are used with agricultural devices used for pest control and are referred to as pest control agents. In some embodiments, SBPs that include one or more pest control agents are used as coatings to coat agricultural pest control devices. Devices may be carriers used to spread pest control agents included in carrier coatings. The carriers may be seeds. SBP seed coatings (e.g., seed coating compositions) provided herein may offer advantages with respect to the variety of cargo that can be formulated (small molecules, proteins, DNA, microbes, viruses), the ability to tailor the release rate of the cargo, stabilization of the cargo, efficient seed coating, break-down into non-toxic peptides, and/or a significantly reduced propensity to produce dust that can contaminate surrounding environments. The latter property, along with the controlled and delayed release of the active ingredient significantly reduces the contamination of surrounding environments by the active ingredient. These properties will likely mitigate the collateral damage to important pollinator populations. In addition, the compositions (e.g., seed coating compositions) provided herein impart advantages vs. seed flow and plantability that are due to the physical properties of silk fibroin such as a very low coefficient of friction.

In some embodiments, SBP formulation agricultural devices described herein may be used in the field of animal husbandry. In some embodiments, SBP formulation agricultural devices described herein may be include a component or the whole of animal housing in the field of animal husbandry. SBP formulations may be used in animal housing applications to provide optimal temperature, humidity, radiation, air flow, precipitation and light required to keep the animal safe, healthy and comfortable.

Animals require healthy environments that permit the production, and quality of the non-human animals, as well as that of the animal agricultural products. Examples of animal housing include, but are not limited to, blankets, bedding, clothing, footwear (e.g. horseshoes), feeding equipment (e.g. bowls and water bottles), brushes, bandages, barns, coops, cages, stalls, liners, enclosures, ropes, ties, pens, flooring, shelters, sheds, stalls, ventilations systems, and wires.

In some embodiments, SBP formulation agricultural devices may be used to aid the health and production of animals. In some embodiments, SBP formulations may be used in the treatment of mastitis. Transition from the dry period prior to lactation to lactation is a high-risk period for agricultural animals such as cows. During the period, the mammary gland (udder) may become infected with bacteria resulting in inflammation. In some embodiments, SBPs may be used in the treatment of mastitis. SBP formulations may be or may include antibiotics effective against one or more mastitis causing bacteria. SBP formulations may also be formatted into plugs and inserted into the teat canal (e.g., a teat sealant). In some embodiments, SBP formulations may be prepared as solutions and injected into the teat canal by an injection apparatus (e.g., a syringe, a needle, etc.). Formation of the plug may occur during injection and/or after injection. In some embodiments, SBP formulations may be formatted into films that is applied to the exterior of the teats. SBPs may be useful, both in treating and preventing mastitis. In some embodiments, SBP formulations may be used as a form of birth control, to regulate the production of animals, such as cows In some embodiments, SBP formulations may be or may include hormones and/or birth control agents. SBP formulations may be prepared as implants or depots and injected into the subject (cattle).

Aquaculture Products

In some embodiments, agricultural SBP formulations may be used in the preparation of aquaculture products. As used herein, the term "aquaculture" generally refers to the farming of aquatic animals (e.g., fish, crustaceans, mollusks) or the cultivation of aquatic plants (e.g., algae). As a non-limiting example, agricultural SBP formulations may be used in the preparation of aquaculture feeds for various aquatic animals including, but not limited to, carp, salmon, catfish, tilapia, cod, trout, milkfish, eel, shrimp, crawfish, crab, oyster, mussel, clam, jellyfish, sea cucumbers and sea urchins.

Delivery

In some embodiments, the delivery of the agricultural SBP formulations described herein may occur through controlled release. In some embodiments, the agricultural SBP formulations may be utilized for the local delivery of cargo. In some embodiments, the agent may be a chemical for use in any one agricultural applications described in the present disclosure. In some embodiments, SBP formulations described herein may enable the controlled delivery of cargos that have a shorter half-life when delivered without SBP formulations, therein enhancing the time for which the therapeutic agent may be effective, as taught in United States Patent Publication US20100028451, the contents of which are herein incorporated by reference in its entirety. In some embodiments, SBP formulations may enhance the residence time of a cargo. In some embodiments the SBP formulation delivery may be targeting to the entire plant, or animal; or it may be targeted to a portion of the plant or animal. In some embodiments, the portion of the plant may be leaf, root, bark, phloem, seed, and/or fruit.

In some embodiments, the controlled release of the SBP formulations for agricultural applications may be facilitated by diffusion of SBP formulations into the surrounding environment. This phenomenon has been observed in pharmaceutical compositions for animal subjects, as taught in United States Patent Publication No. US20170333351, the contents of which are herein incorporated by reference in its entirety. In some embodiments, the controlled release of SBPs for an agricultural application may be facilitated by the degradation and/or dissolution of SBPs. The degradation and/or dissolution has been employed for pharmaceutical compositions for animal subjects, as taught in International Patent Publications WO2013126799, WO2017165922, and U.S. Pat. No. 8,530,625, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, both the diffusion and the degradation and/or dissolution of SBPs may facilitate the controlled release of the agricultural compositions for agricultural applications.

Applications

In some embodiments, the agricultural SBP formulations may be used to increase biomass, increase product yield, and/or enhance offspring production of plants, plant agricultural products, animals, and animal agricultural products. In some embodiments, SBPs may be used in the field of farming. As used herein, "farming" refers to the technique of growing crops or keeping animals for food and materials. Agricultural SBP formulations may be used in arable farming to grow crops, and/or pastoral farming SBP formulations may be utilized to improve one or more aspects of farming such as, but not limited to, plant growth, yield, reproduction, soil properties, weed control, pest control, disease control, product preservation, and/or treatment, environmental factors such as controlling access to water, air, and/or sunlight. In some embodiments, SBPs may be used to mitigate crop damage. In some embodiments, the agricultural SBP formulations of the present disclosure may be used to tune properties of soil. In some embodiments, the agricultural SBP formulations of the present disclosure are used as agents of weed control. In some embodiments, seeds may be treated with agricultural SBP formulations to increase germination, seedling vigor, and seedling size. In some aspects, seeds may be treated with agricultural SBP formulations to increase seed storage, and shelf life of the seed, such that the seedlings produced upon germination of stored seeds are superior to seeds that stored without SBPs. In some embodiments, the agricultural SBP formulations described herein may be used to enhance plant germination. As used herein, the term "germination" refers to growth from a seed or spore. In some embodiments, agricultural products may be treated with agricultural SBP formulations to improve preservation, the shelf life, the physical appearance, and/or freshness of the agricultural products. In some aspects, agricultural products may be treated with SBPs to preserve the products such that they are superior in agricultural SBP formulations and appearance to products untreated agricultural products. In some embodiments, SBPs described herein may be used to control the access of the plant, animal or agricultural product to environmental factors such as water, air and/or sunlight. In some embodiments, agricultural SBP formulations may be used to modulate different aspects of the environment such as, but are not limited to, water, air, humidity, and light.

In some embodiments, the agricultural SBP formulations of the present disclosure may be or may include photodegradable film. Agricultural SBP formulations may be prepared to be photosensitive or may include photosensitive agents that degrade upon exposure to light, (see Chinese Patent Publication CN105199353 and International Patent Publication WO2017123383; the contents of each of which are herein incorporated by reference in their entirety). Photosensitive agents may be chemicals, small molecules, or a drug. Photodegradable SBPs may be prepared in any format (e.g. films, microspheres, nanospheres, and any format described in the present disclosure).

Animals

In some embodiments, agricultural SBP formulations may be used to improve characteristics of animal, and/or increase the yield and quality of animal agricultural products. In some embodiments, the agricultural products include, but are not limited to, milk, butter, cheese, yogurt, whey, curds, meat, oil, fat, blood, amino acids, hormones, enzymes, wax, feathers, fur, hide, bones, gelatin, horns, ivory, wool, venom, tallow, silk, sponges, manure, eggs, pearl culture, honey, and food dye.

In some embodiments, agricultural SBP formulations of the present disclosure may be used in animal agricultural products to facilitate the release of fragrance, flavor, or other compounds responsible for odor and/or flavor, as taught in United States Patent Publication No. US20150164117, the contents of which are herein incorporated by reference in their entirety.

In some embodiments, agricultural SBP formulations may incorporate animal feed or beverage. In some embodiments, agricultural SBP formulations may include health supplements, produce supplements, hormone supplements, and/or agricultural therapeutic agents to improve the health and viability of the animals. In some embodiments, agricultural SBP formulations may include animal feed such as forage, fodder, or a combination of forage and fodder. Examples of forage include, but are not limited to, plant derived material (e.g. leaves and stems), hay, grass, silage, herbaceous legumes, tree legumes, and crop residue. Examples of fodder include, but are not limited to, hay, straw, silage, compressed and pelleted feeds, oils, mixed rations, fish meal, meat and bone meal, molasses, oligosaccharides, seaweed, seeds, grains (e.g. maize, soybeans, wheat, oats, barley, rise, peanuts, corn, and sorghum), crop residues (e.g. stover, copra, straw, chaff, and sugar beet waste), sprouted grains and legumes, brewer's spent grains, yeast extract, compounded feeds (e.g. meal type, pellets, nuts, cakes, and crumbles), cut grass and other forage plants, bran, concentrate mix, oilseed prescake (e.g. cottonseed, safflower, soybean peanut, and groundnut), horse gram, clipping waste, and legumes.

In some embodiments, agricultural SBP formulations described herein may be used to improve the yield of animal agricultural products by improving the health of non-human animals. In some embodiments, SBPs described herein may be used to improve the production capabilities of non-human animals. In some embodiments, SBPs described herein may be used to improve the breeding of non-human animals. In some embodiments, SBPs described herein may be used to improve the health, production, breeding, or a combination thereof in non-human animals.

In some embodiments, agricultural SBP formulations of this disclosure may be used to deliver health supplements to a non-human animal. These health supplements may improve the health of said non-human animals. SBPs may deliver said health supplements as a payload. SBPs may be incorporated into the feed, housing, or any other component or tool of animal husbandry that would enable the delivery of the payload. Examples of health supplements include, but are not limited to, vitamin A, vitamin B, vitamin C, vitamin D, vitamin E, vitamin K, thiamin, riboflavin, niacin, vitamin B6, vitamin B12, biotin, pantothenic acid, calcium, iron, phosphorus, iodine, magnesium, zinc, selenium, selenium, copper, manganese, chromium, molybdenum, chloride, potassium, nickel, silicon, vanadium, and tin.

In some embodiments, agricultural SBP formulations of this disclosure may be used to deliver supplements to a non-human animal that improve the yield and/or quality of the animal agricultural products. These health supplements may improve the production capabilities of said non-human animals. SBPs may include said supplements as a payload. Examples of supplements include, but are not limited to, vitamins, minerals, ions, nutrients, and hormones. In some embodiments, the SBPs may be used to stimulate animal appetite.

In some embodiments, agricultural SBP formulations of this disclosure may be used to deliver hormones to a non-human animal. SBPs may deliver said hormones as a payload. Examples of hormones include, but are not limited to, any steroid, dexamethasone, allopregnanolone, any estrogen (e.g. ethinyl estradiol, mestranol, estradiols and their esters, estriol, estriol succinate, polyestriol phosphate, estrone, estrone sulfate and conjugated estrogens), any progestogen (e.g. progesterone, norethisterone acetate, norgestrel, levonorgestrel, gestodene, chlormadinone acetate, drospirorenone, and 3-ketodesogestrel), any androgen (e.g. testosterone, androstenediol, androstenedione, dehydroepiandrosterone, and dihydrotestosterone), any mineralocorticoid, any glucocoriticoid, cholesterols, and any hormone known to those skilled in the art. In some embodiments, any of the hormones described herein.

In some embodiments, agricultural SBP formulations of this disclosure may be used to deliver birth control agents to a non-human animal. These agents of disease control may improve the health, growth, and/or increase the yield of the agricultural product from said non-human animals. SBPs may be or may include birth control as cargo. SBPs may be incorporated into the feed, housing, or any other component or tool of animal husbandry that would enable the delivery of the payload. In some embodiments, SBPs may be used in conjunction with other forms of birth control, such as surgical procedures (e.g. spaying and neutering). Examples of birth control agents, include, but are not limited to, pills, ointments, implants, surgical procedures, hormones, patches, barriers, and injections.

In one embodiment, agricultural SBP formulations may be used to deliver birth control agents to cattle. Cattle birth control is important for producers to maintain herd genetic traits, reduce disease transmission, as well as eliminating the need for separate breeding pastures. The SBPs may provide controlled release of the birth control agent to the cattle. The birth control agents may include, but are not limited to, gonadorelin, gonadorelin acetate, progesterone, dinoprost tromethamine, and cloprostenol sodium, and any combination thereof.

Pest Control

In some embodiments, agricultural SBP formulations may be used in pest control of plants, animals, plant agricultural products, and/or animal agricultural products. agricultural SBP formulations may be or may include pest control agents described herein. In some embodiments, agricultural SBP formulations pest control devices may be used in pest control. Pest control agents and devices described herein may be applied directly to the pest; a pest susceptible surface such as the locus or planting substrate where the plant is growing e.g. soil; a pest habitat and/or the animal affected by the pest. In some embodiments, SBPs may be used to reduce the drift of a pest control agent to a surrounding environment.

Disease Control

In some embodiments, agricultural SBP formulations may be useful in disease control of plants, and/or animals. In some embodiments, disease may be caused by disease agents. As used herein, the term "disease agent" refers to any biological pathogen that causes a disease. In some embodiments, the disease agent may be a parasite.

In some embodiments, the agricultural SBP formulations of the present disclosure may be used to treat plant diseases. In some embodiments, the present disclosure relates to the use SBPs as a matrix for formulations of disease inhibitory agents. In some embodiments, formulations of silk fibroin containing active ingredients with the ability to prevent the infection of plants, or of controlling disease in plants already infected with disease. More specifically, compositions including a silk fibroin and an inhibitory agent (e.g., 10 antibiotics with the ability to prevent the infection of citrus trees with Las, or of controlling citrus greening in citrus trees already infected with Las). In some embodiments, the agricultural SBP formulations may be or may include therapeutic agents and/or agricultural therapeutic agents to enable disease control. SBPs offer advantages for treating plant disease in their ability to tune the release rate, stabilization, and are biodegradable.

In some embodiments, hydrogels or other formats of agricultural SBP formulations described herein may be utilized to inject and form drug depots in the phloem and provide effective and long-term treatment of affected plants or agricultural products, or protection of susceptible plants and agricultural products.

In some embodiments, agricultural SBP formulations of this disclosure may be used to deliver agents of disease control to a non-human animal. These agents of disease control may improve the health of said non-human animals. Agricultural SBP formulations may deliver said agents of disease control as a payload. SBPs may be incorporated into the feed, housing, or any other component or tool of animal husbandry that would enable the delivery of the payload. In some embodiments, SBPs for disease control may be administered to treat a disease. In some embodiments, SBPs for disease control may be administered as a prophylactic to prevent the onset and/or spread of disease. Examples of agents of disease control include, but are not limited to, biologics, small molecules, vitamins, minerals, herbal preparations, health supplements, ions, metals, carbohydrates, fats, hormones, proteins, peptides, antibiotics and other anti-infective agents, hematopoietics, thrombopoietics, agents, antidementia agents, antiviral agents, antiangiogenic proteins (e.g. endostatin), antitumoral agents (chemotherapeutic agents), antipyretics, analgesics, anti-inflammatory agents, anti-infective, antiulcer agents, antiallergic agents, antidepressants, psychotropic agents, cardiotonics, antiarrhythmic agents, vasodilators, antihypertensive agents such as hypotensive diuretics, antidiabetic agents, anti-rejection agents, anticoagulants, cholesterol lowering agents, therapeutic agents for osteoporosis, bone morphogenic proteins, bone morphogenic-like proteins, enzymes, vaccines, immunological agents and adjuvants, naturally derived proteins, genetically engineered proteins, chemotherapeutic agents, cytokines, growth factors (e.g. epidermal growth factor, fibroblast growth factor, insulin like growth factor I and II, transforming growth factors, and vascular endothelial growth factors), nucleotides and nucleic acids, steroids carbohydrates and polysaccharides, glycoproteins, lipoproteins, viruses and virus particles, conjugates or complexes of small molecules and proteins, or mixtures thereof, and organic or inorganic synthetic pharmaceutical drugs.

Material Science Applications

In some embodiments, SBP formulations may be prepared for use in one or more material science (MS) applications (e.g., MS SBP formulations). As used herein, the term "material science application" refers to any method related to development, production, synthesis, use, degradation, or disposal of materials. As used herein, the term "material" refers to a substance or chemical substance that may be used for the fabrication, production, and/or manufacture of an article. MS SBP formulations may be materials or may be combinations of processed silk with one or more materials. Examples of materials include, but are not limited to, adhesives, aquaculture products, biomaterials, composting agents, conductors, devices, electronics, emulsifiers, fabrics, fibers, fillers, films, filters, food products, heaters, insulators, lubricants, membranes, metal replacements, micelles, microneedles, microneedle arrays, microspheres, nanofibers, nanomaterials, nanoparticles, nanospheres, paper, paper additives, particles, plastics, plastic replacements, polymers, sensors, solar panels, spheres, sun screens, taste-masking agents, textiles, thickening agents, topical creams or ointments, optical devices, vasolines, and composites thereof. In some embodiments, materials comprising SBPs described herein may be used as a plastic, plastic supplement, or a plastic replacement, as taught in Yu et al. and Chantawong et al. (Yu et al. (2017) Biomed Mater Res A doi. 10.1002/jbm.a.36297; Chantawong et al. (2017) Mater Sci Mater Med 28(12):191), the contents of which are herein incorporated by reference in their entirety.

Consumer Products

In some embodiments, MS SBP formulations or materials comprising SBP formulations may be used to produce or may be incorporated into consumer products. As used herein, the term "consumer products" refers to goods or merchandise purchasable by the public. Consumer products may include, but are not limited to, agricultural products, therapeutic products, veterinary products, and products for household use. Non-limiting examples of consumer products include cleaning supplies, sponges, brushes, cloths, protectors, sealant, adhesives, lubricants, protectants, labels, paint, clothing, insulators, devices, bandages, screens, electronics, batteries, and surfactants.

Surfactant Materials

In some embodiments, MS SBP formulations may be used as a surfactant. In some embodiments, SBP materials may reduce the surface tension of liquids. In some embodiments, the SBP materials may be used to tune the surface tension of liquids. In some embodiments, the SBP may be a surfactant. In some embodiments, the surfactant may be prepared from SBPs. In some embodiments, silk is used in the preparation of surfactant using any of the methods described in Chinese patent publication CN105380891, the contents of which are herein incorporated by reference in their entirety. In some embodiments, SBP surfactants may be more environmentally friendly than existing surfactants. In some embodiments, MS SBP formulations have the surface tension of water. In some embodiments, SBPs have the surface tension of tears.

Lubricant

In some embodiments, MS SBP formulations may be used as lubricants, to reduce friction between two or more surfaces. In some embodiments, the SBP is a lubricant. In some embodiments, the SBP is an excipient in a lubricant. In some embodiments, the SBP is prepared from processed silk, oils, water, and other materials as described in Chinese Patent Publication Number CN101725049, the contents of which are herein incorporated by reference in their entirety. Lubricants can be prepared from SBPs in many formats, including, but not limited to, capsules, coatings, emulsions, fibers, films, foams, gels, grafts, hydrogels, membranes, microspheres, nanoparticles, nanospheres, organogels, particles, powders, rods, scaffolds, sheets, solids, solutions, sponges, sprays, suspensions, and vapors. In some embodiments, an SBP lubricant may comprise silk microspheres. In some embodiments, the microspheres may be prepared with a phospholipid coating as described in United States Patent Application Publication Number US20150150993A1, the contents of which are herein incorporated by reference in their entirety. In some embodiments, the lubricants may be used on a material surface, non-limiting examples of which include gears, machinery, vacuums, plastics, threads, wood, furniture, and other items. In some embodiments, the lubricants may be used on a biological surface, non-limiting examples of which include bones, joints, eyes, and mucosal membranes. In some embodiments, the coefficient of friction of an SBP is approximately that of naturally occurring, biological and/or protein lubricants (e.g. lubricin). In some embodiments, SBPs may be incorporated into a lubricant. Such methods may include any of those presented in International Publication No. WO2013163407, the contents of which are herein incorporated by reference in their entirety. In some embodiments, processed silk and/or SBPs may be used as an excipient to prepare a lubricant.

Device Materials

In some embodiments, MS SBP formulations may be used in the fabrication, production, and/or manufacture of a device, e.g., as taught in European Patent Number EP2904133, U.S. Pat. No. 9,802,374, and United States Patent Application Publication Number US20170312387, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, the device is a medical device (e.g. surgical devices, implants, dental devices, dental implants, diagnostic device, hospital equipment, etc.). In some embodiments, the device is an electronic device (e.g. diagnostic device, hospital equipment, implants, etc.).

The term "medical device" refers to any device, product, equipment or material having surfaces that contact tissue, blood, or other bodily fluids of a subject in the course of their use or operation. Exemplary medical devices include, but are not limited to, absorbable and nonabsorbable sutures, access ports, amniocentesis needles, arterial catheters, arteriovenous shunts, artificial joints, artificial organs, artificial urinary sphincters, bandages, biliary stents, biopsy needles, blood collection tubes, blood filters, blood oxygenators, blood pumps, blood storage bags, bolts, brain and nerve stimulators, calipers, cannulas, cardiac defibrillators, cardioverter defibrillators, castings, catheter introducers, catheter sheaths s, catheters, chemical sensors, clips or fasteners, contraceptive devices, coronary stents, dialysis catheters, dialysis devices, dilators, drain tubes, drainage tubes, drug infusion catheters and guidewires, electrodes, endoscopes, endotracheal tubes, feminine hygiene products, fetal monitors, Foley catheters, forceps, gastroenteric tubes, genitourinary implants, guide wires, halo systems, heart valves, hearing aids, hydrocephalus shunts, implants, infusion needles, inserters, intermittent urinary catheters, intraurethral implants, introducers, introducer needles, irrigators, joint prostheses, knives, long-term central venous catheters, long-term tunneled central venous catheters, long-term urinary devices, monitors, nails, nasogastric tubes, needles, neurological stents, nozzles, nuts, obdurators, orthopedic implants, orthopedic devices, osteoports, pacemaker capsules, pacemaker leads, pacemakers, patches, penile prostheses, peripheral venous catheters, peripherally insertable central venous catheters, peritoneal catheters, peritoneal dialysis catheters, personal hygiene items, pins, plates, probes, prostheses, pulmonary artery Swan-Ganz catheters, pulse generators, retractors, rods, scaffolding, scalpels, screws, sensors, short-term central venous catheters, shunts, small joint replacements, specula, spinal stimulators, stents, stints, stylets, suture needles, suturing materials, syringes, temporary joint replacements, tissue bonding urinary devices, tracheostomy devices, transducers, trocars, tubes, tubing, urethral inserts, urinary catheters, urinary dilators, urinary sphincters, urological stents, valves, vascular catheters, vascular catheter ports, vascular grafts, vascular port catheters, vascular stents, wire guides, wires, wirings, wound drains, wound drain tubes, and wound dressings.

In some embodiments, the medical device may be an ocular device, such as, but not limited to, contact lens (hard or soft), intraocular lens, corneal onlay, ocular inserts, artificial cornea and membranes, eye bandages, and eyeglasses.

In some embodiments, the medical device may be a dental device, such as, but not limited to, dental flossers, dental flossing devices, dental threaders, dental stimulators, dental picks, dental massagers, proxy brushes, dental tapes, dental fillings, dental implants, orthodontic arch wire, and other orthodontic devices or prostodontic devices.

In some embodiments, the device may be any one of the following devices: audio players, bar code scanners, cameras, cell phones, cellular phones, car audio systems, communication devices, computer components, computers, credit cards, depth finders, digital cameras, digital versatile discs (DVDs), electronic books, electronic games and game systems, emergency locator transmitters (ELTs), emergency position-indicating radio beacons (EPIRBs), fish finders, global positioning system (GPS), home security systems, image play back devices, media players, mobile computers, mobile phones, MP3 players, music players, notebook computers, pagers, palm pilots, personal computers, personal digital assistants (PDAs), personal locator beacons, portable books, portable electronic devices, portable game consoles, radar displays, radios, remote control device, satellite phones, smart cards, smartphones, speakers, tablets, telephones (e.g. cellular and standard), televisions, video cameras, video players, automobiles, boats, and aircraft.

In some embodiments, MS SBP formulations are used as, or incorporated into, the coating materials of a device. In some embodiments, the coating may be functional, decorative or both. Coatings may be applied to completely cover the surface. Coating may also be applied to partially cover the surface. Devices coated with SBPs may be more biocompatible and/or less-immunogenic.

Antibiotic Materials

In some embodiments, MS SBP formulations may be used as materials due to their antibiotic properties. Such methods may include any of those described in European Patent Number EP3226835 and Mane et al. (2017) Scientific Reports 7:15531, the contents of each of which are herein incorporated by reference in their entirety. These antibiotic properties may be a general property of SBPs. In some embodiments, SBPs materials with antibiotic properties may include antibiotic cargo. In some embodiments, SBP materials may include antibiotic wound-healing materials (e.g., see Babu et al. (2017) J Colloid Interface Sci 513:62-72, the contents of which are herein incorporated by reference in their entirety).

Synthetic Materials

In some embodiments, MS SBP formulations are combined with synthetic materials. Such SBPs may be used to form scaffolds (e.g., see Lo et al. (2017) J Tissue Eng Regen Med doi.10.1002/term.2616, the contents of which are herein incorporated by reference in their entirety). In some embodiments, SBPs described herein are utilized to coat other materials. Such SBPs may include any of those described in Ai et al. (2017) International Journal of Nanomedicine 12:7737-7750, the contents of which are herein incorporated by reference in their entirety. In some embodiments, SBPs include plastics (e.g. thermoplastics, bioplastics, polyethylene, ultra-high-molecular-weight polyethylene, polypropylene, polystyrene, and polyvinyl chloride). In some embodiments, SBPs include plastic replacements. In some embodiments, SBPs include electronic materials or insulators.

In some embodiments, SBPs include polyolefins, polymers, and/or particles. In some embodiments, SBP materials may be prepared and used according to the methods of preparation and use described in European Patent Numbers EP3226835, EP3242967, and EP2904133, United States Publication Numbers US20170333351 and US20170340575, and Cheng et al. (2017) ACS Appl Mater Interfaces doi.10.1021/acsami.7b13460, the contents of each of which are herein incorporated by reference in their entireties.

In some embodiments, MS SBP formulations may be used as a plastic replacement in various products. Conventional plastic is made from petroleum products, primarily oil. It does not biodegrade and is harmful to the environment. MS

SBP formulations are an attractive alternative to synthetic plastics due to their biocompatibility and biodegradability. As a non-limiting example, SBPs may be used as a plastic replacement in the production of water bottles and food containers. As another non-limiting example, MS SBP formulations may be used as a plastic replacement in the preparation of coating materials on a fabric or a cloth. Coatings used on apparels, such as a waterproof jacket or athletic shirt, are generally made of synthetic polymers and may release micro-plastic particles into water during a wash cycle. Using MS SBP formulations in replacement of synthetic polymers may help eliminate this problem.

Nanomaterials

In some embodiments, MS SBP formulations include nanomaterials (e.g. nanoparticles, nanofibrils, nanostructures, and nanofibers), as taught in International Patent Application Publication No. WO2017192227, Xiong et al., and Babu et al. (Xiong et al. (2017) ACS Nano 11(12): 12008-12019.; Babu et al. (2017) J Colloid Interface Sci 513:62-72), the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, the nanoparticles may include excipients. In some embodiments, the nanoparticles may include, but are not limited to, any of those excipients listed in Table 1, above.

Cosmetics

In some embodiments, MS SBP formulations are or used in the preparation of cosmetics. In some embodiments, SBPs are active substances in said cosmetics, e.g., as taught in U.S. Pat. No. 6,280,747 and United States Publication Number US20040170590, the contents of each of which are herein incorporated by reference in their entirety. In some embodiments, SBPs are added as a thickening agent, e.g., as taught in United States Publication Number US20150079012, the contents of which are herein incorporated by reference in their entirety. In some embodiments, cosmetics may incorporate SBPs for stabilization and/or preservation of cosmetic components (e.g., see Li et al. (2017) Biomacromolecules 19(9):2900-2905, the contents of which are herein incorporated by reference in their entirety). In some embodiments, MS SBP formulations may be incorporated into cosmetics as a lubricant. In some embodiments, MS SBP formulations may be used as a plastic replacement in the preparation of cosmetics. As a non-limiting example, MS SBP formulations may be formatted as microbeads to be used in replacement of plastic microbeads in facial scrubs and toothpastes. Examples of cosmetics include, but are not limited to, shampoos, conditioners, lotions, foundations, concealers, eye shadows, powders, lipsticks, lip glosses, ointments, mascara, gels, sprays, eye liners, liquids, solids, eyebrow mascaras, eyebrow gels, hairspray, moisturizers, dyes, minerals, perfumes, colognes, rouges, natural cosmetics, synthetic cosmetics, soaps, cleansers, deodorants, creams, towelettes, bath oils, bath salts, body butters, nail polish, hand sanitizer, primers, plumpers, balms, contour powders, bronzers, setting sprays, and setting powders.

Thickening Agents

In some embodiments, MS SBP formulations may be or may be combined with thickening agents. As used herein, the term "thickening agent" refers to a substance used to increase viscosity of another material, typically without altering any properties of the other material. In some embodiments, SBP thickening agents may be used in paints, inks, explosives, cosmetics, foods, or beverages.

In some embodiments, SBP thickening agents may be used in products for human consumption (e.g., as taught in United States Publication No. US20150079012, the contents of which are herein incorporated by reference in their entirety). SBP biocompatibility, biodegradability, and low toxicity make SBPs attractive tools for thickening materials designed for human consumption. In some embodiments, SBP thickening agents may be used to increase the viscosity of a food item. Examples of food items include, but are not limited to, puddings, soups, sauces, gravies, yogurts, oatmeals, chilis, gumbos, chocolates, and stews. In some embodiments, SBP thickening agents may be used to increase the viscosity of beverages. Examples of beverages include, but are not limited to, shakes, drinkable yogurts, milks, creams, sports drinks, protein shakes, diet supplement beverages, and coffee creamers.

In some embodiments, SBP thickening agents may be added to cosmetics (e.g., as taught in United States Publication Number US20150079012, the contents of which are herein incorporated by reference in their entirety. Such cosmetic products may include, but are not limited to, shampoos, conditioners, lotions, foundations, concealers, eye shadows, powders, lipsticks, lip glosses, ointments, mascara, gels, sprays, eye liners, liquids, solids, eyebrow mascaras, eyebrow gels, hairspray, moisturizers, dyes, minerals, perfumes, colognes, rouges, natural cosmetics, synthetic cosmetics, soaps, cleansers, deodorants, creams, towelettes, bath oils, bath salts, body butters, nail polish, hand sanitizer, primers, plumpers, balms, contour powders, bronzers, setting sprays, and setting powders.

Military Applications

In some embodiments, SBP formulations may be used in military applications. For example, SBP formulations may be incorporated in military fabrics. Such fabrics may be used in items such as, but not limited to, ponchos, tents, uniforms, vests, backpacks, personal protective equipment (PPE), linings, cords, ropes, and cables, webbings, straps and sheaths, helmet coverings, flags, bedsheets and mattress fabrics, ribbons, hats, gloves, masks, boots, suits and belts. As another example, SBP formulations may be used in the manufacture of a military device or gear. Non-limiting examples of military devices or gears include goggles, sunglasses, telescopes, binoculars, monoculars, flashlight, torches, watches, compasses, whistle, shields, knee caps, water bottles, flasks, and cameo face paint.

The details of one or more embodiments of the disclosure are set forth in the accompanying description above. Although any materials and methods similar or equivalent to those described herein can be used in the practice or testing of the present disclosure, the preferred materials and methods are now described. Other features, objects and advantages of the disclosure will be apparent from the description. In the description, the singular forms also include the plural unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. In the case of conflict, the present description will control.

Definitions

Linear viscoelastic region: As used herein, the term "linear viscoelastic region" or "LVR" refers to the range in amplitude of deformation in which the shear loss modulus and the shear storage modulus do not vary and the structure of the material does not change.

Phase angle: As used herein, the term "phase angle" refers to the difference in the stress and strain applied to a material during the application of oscillating shear stress. The lag between stress and strain represents a measure of a material's viscoelasticity.

Shear: As used herein, the term “shear” or “shear force” or refers to a force applied to a material that is parallel to the surface of said material.

Shear loss modulus: As used herein, the term “shear loss modulus” or “G''” refers to the measure of a material’s ability to dissipate energy, usually in the form of heat.

Shear rate: As used herein, the term “shear rate” refers to the rate of change in a material’s strain over time.

Shear storage modulus: As used herein, the term “shear storage modulus” or “G'” refers to the measure of a material’s elasticity or reversible deformation as determined by the material’s stored energy.

Shear stress: As used herein, the term “shear stress” refers to the force applied parallel to a material divided by the area along the force.

Strain: As used herein, the term “strain” refers to the ratio of displacement of material upon the application of a shear force to the height of the material.

Strain sweep: As used herein, the term “strain sweep” refers to a range of strain measurements over which an experimental parameter may be determined.

Tune: As used herein, the term “tune” refers to the control or modulation of a physical or chemical property. Physical or chemical properties of SBPs may be tuned through the control of silk fibroin molecular weight, silk concentration, the incorporation of excipients, and any other method of control described in the present disclosure.

Viscoelasticity: As used herein, the term “viscoelasticity” refers to the range of viscous and elastic properties of materials under stress.

Viscosity: As used herein, the term “viscosity” refers to a material’s ability to resist deformation due to shear forces, and the ability of a fluid to resist flow. It is reflected by the shear stress over the shear rate.

The details of one or more embodiments of the disclosure are set forth in the accompanying description below. Although any materials and methods similar or equivalent to those described herein can be used in the practice or testing of the present disclosure, the preferred materials and methods are now described. Other features, objects and advantages of the disclosure will be apparent from the description. In the description, the singular forms also include the plural unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. In the case of conflict, the present description will control.

The present disclosure is further illustrated by the following non-limiting examples.

EXAMPLES

Example 1: Silk Fibroin Isolation

Silk yarn, purchased from Jiangsu SOHO International Group, was degummed to remove sericin. 30 grams of cut silk yarn were boiled or heated or degummed at 100° C. in 3 L of deionized (DI) water with 0.02M sodium carbonate with stirring. The yarn was then transferred to a new boiling 0.02M sodium carbonate aqueous solution and boiled at 100° C. with stirring. The total degumming time was discussed in terms of minute boil, or “mb.” The total degumming time was 480 minutes, or 480 mb.

Alternatively, Silk yarn, purchased from Jiangsu SOHO International Group, was degummed to remove sericin. 30 grams of cut silk yarn were boiled or heated or degummed at 85° C. in 3 L of deionized (DI) water with 0.5M sodium carbonate with stirring. The total degumming time was 240 minutes or 240 mb. Alternatively, Silk yarn, purchased from Jiangsu SOHO International Group, was degummed to remove sericin. 30 grams of cut silk yarn were boiled or heated or degummed at 90° C. in 3 L of deionized (DI) water with 0.5M sodium carbonate with stirring. The total boiling or degumming time was 120 minutes or 120 mb.

The fibroin was washed in 3×3 L exchanges of 70° C. DI water for 20 minutes each followed by 3×3 L exchanges of room temperature (RT) DI water for 20 minutes each. The silk fibroin was dried overnight, weighed, and dissolved at 20% (w/v) in a 9.3M aqueous solution of lithium bromide (from Sigma-Aldrich, St. Louis, MO) for 5 hours at 60° C. The resulting fibroin solution was dialyzed against water (150 mL of fibroin solution against 5 L of water) at room temperature in 3.5 kDa or 50 kDa regenerated cellulose dialysis tubing for 48 hours, with 6 water changes to remove the excess salt. The conductivity was tested and recorded after each water change with a digital quality tester. When the conductivity was under 5 ppm, the fibroin solution was determined to be ready.

The resulting solution was centrifuged for 20 minutes at 3,900 RPM and 4° C. to remove insoluble particles. The supernatant was collected, and samples of the supernatant were diluted at 1:20 and 1:40 in water or 10 mM phosphate buffer. Samples for a standard curve were prepared for an A280 assay by diluting pre-measured fibroin solutions to 5, 2.5, 1.25, 0.625, 0.3125, and 0 mg/mL in water or 10 mM phosphate buffer. The silk concentration of the 1:20 and 1:40 diluted silk fibroin samples was measured against the standard curve by the absorbance at 280 nm.

The silk fibroin solutions were diluted to a final concentration of 3% (w/v) in 10 mM phosphate buffer (from Sigma Aldrich Fine Chemicals, St. Louis, MO), pH 7.4, and/or in 5% sucrose, and/or in water, and they were filtered through a 0.2 μm filter using a vacuum filter unit. 10 mL of each solution was aliquoted into 50 mL conical tubes, snap frozen, or frozen to −80° C., in liquid nitrogen for 10 minutes, transferred for 20 minutes in −80° C., and lyophilized for 72 hours.

Silk yarn, purchased from Jiangsu SOHO International Group, was degummed to remove sericin and modify silk fibroin molecular weight. 20 grams of cut silk yarn were split into four cotton drawstring bags with 5 grams of silk per bag. The four bags of yarn were then immersed in 2 L of degumming solution at either 0.05 M, 0.1 M, or 0.5 M sodium carbonate. The temperature of the solution was controlled with a ChefSteps Joule Sous Vide (ChefSteps, Seattle, WA, USA) at either 80, 85, 90° C. A control degumming was performed with four bags of yarn in 0.02 M carbonate at 100° C. Every two hours for each degumming, a bag would be taken from the solution and the pH of the solution would be measured. The silk fibroin in the bag was thoroughly rinsed under DI water and allowed to dry overnight in a fume hood. Table 6 lists the degumming conditions, pH values of the degumming solution at the time the silk fibroin was removed, and the dry weight of the silk fibroin after degumming.

TABLE 6

SILK FIBROIN DEGUMMING CONDITIONS, PH VALUES, AND DRY WEIGHTS

Lot #	Silk Degumming Time (hr)	Silk Degumming Sodium Carbonate Concentration (M)	Silk Degumming Temperature (° C.)	Final Degumming Solution pH	Final Silk Fibroin Dry Weight (g)
111.2	2	0.02	100	10.53	3.726
111.4	4	0.02	100	10.52	3.515
111.6	6	0.02	100	10.76	3.293
111.8	8	0.02	100	11.18	3.131
108A.2	2	0.05	80	11.11	3.776
108A.4	4	0.05	80	10.77	3.737
108A.6	6	0.05	80	10.84	3.717
108A.8	8	0.05	80	10.64	3.657
108B.2	2	0.05	85	10.77	3.870
108B.4	4	0.05	85	10.68	3.730
108B.6	6	0.05	85	10.93	3.670
108B.8	8	0.05	85	10.75	3.500
108C.2	2	0.05	90	10.74	3.730
108C.4	4	0.05	90	10.64	3.560
108C.6	6	0.05	90	10.91	3.390
108C.8	8	0.05	90	10.68	3.160
109A.2	2	0.1	80	10.67	3.804
109A.4	4	0.1	80	10.58	3.726
109A.6	6	0.1	80	11.13	3.528
109A.8	8	0.1	80	11.07	3.386
109B.2	2	0.1	85	10.76	3.684
109B.4	4	0.1	85	10.38	3.558
109B.6	6	0.1	85	11.10	3.257
109B.8	8	0.1	85	11.04	3.073
109C.2	2	0.1	90	10.95	3.690
109C.4	4	0.1	90	10.42	3.333
109C.6	6	0.1	90	11.02	2.755
109C.8	8	0.1	90	10.83	2.477
110A.2	2	0.5	80	10.65	3.717
110A.4	4	0.5	80	10.98	3.459
110A.6	6	0.5	80	11.09	2.958
110A.8	8	0.5	80	11.07	2.654
110B.2	2	0.5	85	11.05	3.443
110B.4	4	0.5	85	10.97	2.967
110B.6	6	0.5	85	11.37	2.153
110B.8	8	0.5	85	11.30	1.891
110C.2	2	0.5	90	10.99	3.054
110C.4	4	0.5	90	10.92	2.502
110C.6	6	0.5	90	11.34	1.785
110C.8	8	0.5	90	11.34	0.522

The silk fibroin samples were weighed and dissolved at 20% (w/v) in a 9.3M aqueous solution of lithium bromide (from Sigma-Aldrich, St. Louis, MO) for 5 hours at 60° C. The resulting fibroin solution was dialyzed against water at 4° C. in 50 kDa regenerated cellulose dialysis tubing for 48 hours, with 6 water changes to remove the excess salt. The conductivity was recorded after each water change with a digital quality tester. When the conductivity was under 5 ppm, the fibroin solution was determined to be ready.

The resulting solution was centrifuged for 20 minutes at 3,900 RPM and 4° C. to remove insoluble particles. The supernatant was collected, and samples of the supernatant were assessed for silk fibroin concentration via gravimetric analysis. Weigh boats were tared and 500 of solution was added onto it. The solution was dried overnight at 60° C. before a final weight was measured. The resulting weight of silk fibroin remaining was used to determine the silk fibroin percentage (w/v).

Silk fibroin solutions were diluted to a final concentration of 5% (w/v) and 50 mM sucrose (from Sigma Aldrich Fine Chemicals, St. Louis, MO). If the solution was below 5% (w/v) after dialysis, the solution was diluted as little as possible in order to reach a final concentration of 50 mM sucrose. Ten milliliters of each solution was aliquoted into 50 mL conical tubes, frozen overnight at −80° C., and then stored at −20° C. until use.

Example 2: Silk Fibroin Isolation and Preparation by Gravimetry

Silk yarn, purchased from Jiangsu SOHO International Group, was degummed to remove sericin. 30 grams of cut silk yarn were boiled at 100° C. in 3 L of deionized (DI) water with 0.02M sodium carbonate with stirring. The yarn was then transferred to a new boiling 0.02M sodium carbonate aqueous solution and boiled at 100° C. with stirring. The total degumming time was discussed in terms of minute boil, or "mb." The total degumming time was 480 minutes, or 480 mb.

Alternatively, Silk yarn, purchased from Jiangsu SOHO International Group, was degummed to remove sericin. 30 grams of cut silk yarn were boiled at 85° C. in 3 L of deionized (DI) water with 0.5M sodium carbonate with stirring. The total degumming time was 240 minutes or 240 mb. Alternatively, Silk yarn, purchased from Jiangsu SOHO International Group, was degummed to remove sericin. 30 grams of cut silk yarn were boiled at 90° C. in 3 L of deionized (DI) water with 0.5M sodium carbonate with stirring. The total boiling time was 120 minutes or 120 mb.

The fibroin was washed in 3×3 L exchanges of 70° C. DI water for 20 minutes each followed by 3×3 L exchanges of room temperature (RT) DI water for 20 minutes each. The silk fibroin was dried overnight, weighed, and dissolved at 20% (w/v) in a 9.3M aqueous solution of lithium bromide (from Sigma-Aldrich, St. Louis, MO) for 5 hours at 60° C. The resulting fibroin solution was dialyzed against water (150 mL of fibroin solution against 5L of water) at room temperature in a 3.5 kDa or 50 kDa regenerated cellulose dialysis tubing for 48 hours, with 6 water changes to remove the excess salt. The conductivity was tested and recorded after each water change with a digital quality tester. When the conductivity was under 5 ppm, the fibroin solution was determined to be ready.

The resulting solution was centrifuged for 20 minutes at 3,900 RPM and 4° C. to remove insoluble particles. The supernatant was collected, and concentration of silk fibroin was determined via gravimetry. The weight of dried silk fibroin remaining was divided by the sample volume to determine the concentration of silk fibroin in solution. Samples of the supernatant were added to pre-tared weigh boats and dried at 60° C. overnight.

The silk fibroin solutions were diluted to a final concentration of 3% (w/v) in 10 mM phosphate buffer (from Sigma Aldrich Fine Chemicals, St. Louis, MO), pH 7.4, or to 5% (w/v) silk fibroin in 50 mM sucrose and they were filtered through a 0.2 μm PES filter using a vacuum filter unit. For those solutions diluted in phosphate buffer, 10 mL of each solution was aliquoted into 50 mL conical tubes, snap frozen in liquid nitrogen for 10 minutes, transferred for 20 minutes in −80° C., and lyophilized for 72 hours. For those solutions diluted in sucrose, 10 mL of each solution was aliquoted into 50 mL conical tubes, frozen overnight at −80° C., and then stored at −20° C. Some aliquots were lyophilized. Some aliquots were stored at 2-8° C. prior to freezing at −80° C.

Example 3: Silk Fibroin Isolation and Molecular Weight Determination

Silk yarn, purchased from Jiangsu SOHO International Group, was degummed to remove sericin. 30 grams of cut silk yarn were boiled at 100° C. in 3 L of deionized (DI) water with 0.02M sodium carbonate with stirring. The yarn was then transferred to a new boiling 0.02 M sodium carbonate aqueous solution and boiled at 100° C. with stirring. The total boiling time, or degumming time, was discussed in terms of minute boil, or "mb." The total boiling times ranged from 30-480 mb. The fibroin was dried overnight, weighed, and dissolved at 20% (w/v) in a 9.3 M aqueous solution of lithium bromide (from Sigma-Aldrich, St. Louis, MO) for 5 hours at 60° C. The resulting fibroin solution was dialyzed against water at room temperature in 50 kDa regenerated cellulose dialysis tubing for 48 hours, with 6 water changes to remove the excess salt. The conductivity was tested and recorded after each water change with a digital quality tester. When the conductivity was under 5 ppm, the fibroin solution was determined to be ready.

The resulting solution was centrifuged for 20 minutes at 3,900 RPM and 4° C. to remove insoluble particles. The supernatant was collected, and the concentration of silk fibroin was determined via gravimetry. Samples of the supernatant were added to pre-tared weigh boats and dried at 60° C. overnight. The dried weight of silk fibroin was used to determine the concentration in solution. The silk fibroin solutions were diluted to a final concentration of 5% (w/v) in 50 mM sucrose, separated into 10 mL was aliquots in 15 mL conical tubes, and frozen at ≤−20° C.

Silk fibroin molecular weight was determined using a UPLC SEC method. A Waters Acquity H-Class UPLC (Waters, Milford, MA) equipped with a Waters BEH 200 Å 1.7 μm, 4.8×150 cm UPLC SEC column was used. Sample temperature was maintained at 4° C. An isocratic flow at 0.3 mL/min of 100 mM Tris-HCl with 400 mM sodium perchlorate, pH 8.0 was used to elute SF from the column. Elution was monitored at 280 nm. Molecular weights were calculated using Waters BEH200 Protein Standard Mix (Waters, Milford, MA). Some preparations of 480mb had an average molecular weight of 30 kDa. Other preparations resulted in molecular weights between 20 kDa and 40 kDa.

The molecular weight of each batch collected was determined by HPLC-SEC. A Waters Acquity H-Class UPLC (Waters, Milford, MA) equipped with a Waters BEH 200 Å 1.7 mm, 4.8×300 cm UPLC SEC column (from Waters, Milford, MA) was used. Sample temperature was maintained at 4° C. An isocratic flow at 0.3 mL/min of mobile phase (100 mM Tris-HCl with 400 mM sodium perchlorate (Alfa Aesar, Ward Hill, MA), pH 8.0 was used to elute SF from the column. Mobile phase was prepared by dissolving 12.1 g of Tris base (VWR, Pittsburgh, PA) and 48.1 g of sodium perchlorate (Alfa Aesar, Ward Hill, MA) in 800 mL of DI water. The pH was adjusted to 8.0 with 6N HCl. DI water was added to a final volume of 1 L. The solution was sterile filtered through a 0.2 μm polyethersulfone membrane filter. Silk fibroin was diluted to 1% (w/v) in mobile phase after thawing before being run on SEC. Silk fibroin elution was monitored at 280 nm. Molecular weights were calculated using a standard curve prepared from Waters BEH 200 Å Protein SEC Standard Mix (Waters, Milford, MA). Table 7 lists these molecular weights for the select batches analyzed.

TABLE 7

SEC MEASUREMENTS OF SILK FIBROIN SOLUTION FORMULATIONS

Lot #	Average MW (kDa)	Average Viscosity at 1 1/s (cP)
111.2	117.56	30.89
111.4	60.74	39.00
111.6	42.41	76.21
111.8	34.03	97.66
108A.2	—	—
108A.4	—	—
108A.6	—	—
108A.8	93.51	15.22
108B.2	—	—
108B.4	—	—
108B.6	—	—
108B.8	45.49	53.41
108C.2	122.52	12.54
108C.4	65.01	23.90
108C.6	49.87	54.18
108C.8	31.45	74.70
109A.2	—	—
109A.4	—	—
109A.6	—	—
109A.8	43.95	43.38
109B.2	118.93	27.46
109B.4	64.36	25.68
109B.6	38.21	82.10
109B.8	31.75	104.60
109C.2	82.76	17.12
109C.4	36.73	49.41
109C.6	25.04	106.95
109C.8	22.04	142.48
110A.2	68.74	18.16
110A.4	35.47	42.74
110A.6	22.59	78.72
110A.8	21.32	128.03
110B.2	37.88	41.37
110B.4	22.13	89.69
110B.6	17.60	178.79
110B.8	—	—
110C.2	24.25	99.74
110C.4	17.60	140.56
110C.6	—	—
110C.8	—	—

A decrease in silk fibroin molecular weight was observed over time under each degumming condition. This is a result of the hydrolysis occurring during the degumming process in the basic sodium carbonate buffer. Longer degumming times lead to decreased average silk fibroin MW. Silk fibroin degradation was also observed with increasing sodium carbonate concentration. For example, increasing the sodium carbonate concentration from 0.05 M to 0.5 M while at the same temperature (90° C.) and time (2 hrs.) decreased the average molecular weight of silk fibroin from 122.52 kDa to 24.25 kDa. Temperature also impacted the silk fibroin hydrolysis with increasing temperature leading to lower molecular weights. An example of this was shown with silk that was degummed with 0.5 M sodium carbonate for 2 hrs. The sample at 80° C. showed a molecular weight of 68.74 kDa, while increasing the temperature to 90° C. decreased the molecular weight average of the silk fibroin to 24.35 kDa. These results indicated that by controlling the time of degumming, the sodium carbonate concentration, and the temperature, it is possible to control the hydrolysis of silk fibroin.

Example 4: Formulation of Silk Fibroin Solutions

To prepare the solutions, dry silk fibroin of a designated mb, which had been lyophilized in phosphate buffer, was reconstituted in phosphate buffered saline to afford 30% and

10% (w/v) silk fibroin solutions. Formulations from silk of a 30mb were prepared from silk fibroin lyophilized with sucrose. Additional dilutions with phosphate buffered saline were performed on the 10% (w/v) silk fibroin solutions to obtain solutions with 5%, 2.5%, 1%, 0.5% and 0.1% (w/v) silk fibroin. The formulations were prepared as described in Table 8. The samples in Table 8 are named by the process used to prepare and formulate each solution. For example, in the sample named "30mb; sln; 10% SFf; 18% Suc" "30mb" refers to silk degummed with a 30 minute boil, "sln" refers to the formulation of the sample as a solution, "10% SFf" refers to a formulation with 10% (w/v) silk fibroin, and "18% Suc" refers to a formulation with 18% sucrose (w/v).

TABLE 8

SILK FIBROIN SOLUTION FORMULATIONS

Sample	Silk Fibroin Boil Time (mb)	Formulation Silk Concentration (w/v %)	Sample Name
1A	30	30	30 mb; sln; 30% SFf
1B	30	10	30 mb; sln; 10% SFf; 18% Suc
1C	30	5	30 mb; sln; 5% SFf; 9% Suc
1D	30	2.5	30 mb; sln; 2.5% SFf; 4.5% Suc
1E	30	1	30 mb; sln; 1% SFf; 1.8% Suc
1F	30	0.5	30 mb; sln; 0.5% SFf; 0.9% Suc
1G	30	0.1	30 mb; sln; 0.1% SFf 0.18% Suc
2A	90	30	90 mb; sln; 30% SFf
2B	90	10	90 mb; sln; 10% SFf
2C	90	5	90 mb; sln; 5% SFf
2D	90	2.5	90 mb; sln; 2.5% SFf
2E	90	1	90 mb; sln; 1% SFf
2F	90	0.5	90 mb; sln; 0.5% SFf
2G	90	0.1	90 mb; sln; 0.1% SFf
3A	120	30	120 mb; sln; 30% SFf
3B	120	10	120 mb; sln; 10% SFf
3C	120	5	120 mb; sln; 5% SFf
3D	120	2.5	120 mb; sln; 2.5% SFf
3E	120	1	120 mb; sln; 1% SFf
3F	120	0.5	120 mb; sln; 0.5% SFf
3G	120	0.1	120 mb; sln; 0.1% SFf
4A	480	30	480 mb; sln; 30% SFf
4B	480	10	480 mb; sln; 10% SFf
4C	480	5	480 mb; sln; 5% SFf
4D	480	2.5	480 mb; sln; 2.5% SFf
4E	480	1	480 mb; sln; 1% SFf
4F	480	0.5	480 mb; sln; 0.5% SFf
4G	480	0.1	480 mb; sln; 0.1% SFf

To prepare the following solutions in Table 9, silk fibroin of a designated mb, which had either been frozen in 50 mM sucrose or stored at 4° C. in phosphate buffer, was used. A phosphate buffered saline stock solution containing sodium phosphate dibasic, potassium phosphate monobasic, and sodium chloride was prepared to reach final concentrations of 10 mM phosphate and 125 mM sodium chloride and buffered to pH 7.5. The formulations were prepared as described in Table 6. The samples in Table 9 are named by the process used to prepare and formulate each solution with each degumming process. For example, in the sample named "2% SFf; sln; 1.71% Suc; phosphate buffer", "1% SFf" refers to a formulation with 1% (w/v) silk fibroin, "sln" refers to the formulation of the sample as a solution, and "1.71% Suc" refers to a formulation with 1.71% (w/v), or 50 mM, sucrose. All formulations are in a "phosphate buffer" which refers to a vehicle containing final concentrations of 10 mM phosphate and 125 mM sodium chloride at pH 7.5. It is noted that the three variables that differ in the samples are carbonate concentration, temperature, and time duration. By altering these three variables, similar molecular weights can be achieved without having to boil the solution or keep the solution at a high or boiling temperature for a full eight hours.

TABLE 9

SILK FIBROIN FORMULATIONS (PREPARED WITH EACH SF FROM EXAMPLE 1)

Sample	SF Conc. (%)	Description
1	2	2% SFf; sln; phosphate buffer
2	2.5	2.5% SFf; sln; phosphate buffer
3	3	3% SFf; sln; phosphate buffer
4	2	2% SFf; sln; 1.71% Suc; phosphate buffer
5	2.5	2.5% SFf; sln; 1.71% Suc; phosphate buffer
6	3	3% SFf; sln; 1.71% Suc; phosphate buffer

Example 5: Formulation of Silk Fibroin Solutions with Lyophilized or Freeze-Thaw Silk Fibroin

To prepare the solutions, silk fibroin was either reconstituted from a dry state or thawed from a frozen state. Dry silk fibroin of a designated mb, which had been lyophilized in phosphate buffer or sucrose, was reconstituted in phosphate buffered saline to afford 10% (w/v) silk fibroin solutions. Frozen silk fibroin solutions were left at room temperature until completely thawed and then gently mixed before use. Additional dilutions with phosphate buffered saline were performed on the 10% and 5% (w/v) silk fibroin solutions to obtain solutions with 5%, 1%, 0.5%, 0.1%, and 0.05% (w/v) silk fibroin. The formulations were prepared as described in Table 10. The samples in Table 10 are named by the process used to prepare and formulate each solution. For example, in the sample named "90mb; sln; 5% SFf; 1.7% Suc; frozen" "90mb" refers to silk degummed with a 90-minute boil, "sln" refers to the formulation of the sample as a solution, "5% SFf" refers to a formulation with 5% (w/v) silk fibroin, "1.7% Suc" refers to a formulation with 1.7% sucrose (w/v), "frozen" refers to the starting silk fibroin stock being frozen, and "1yo" refers to the starting silk fibroin stock being lyophilized.

TABLE 10

SILK FIBROIN SOLUTION FORMULATIONS PREPARED WITH LYOPHILIZED OR FREEZE-THAW SILK FIBROIN

Sample	Silk Fibroin Boil Time (mb)	Silk Concentration (w/v %)	Sample Name
1	90	5	90 mb; sln; 5% SFf; 1.7% Suc; frozen
2	90	1	90 mb; sln; 1% SFf; 0.34% Suc; frozen
3	90	0.5	90 mb; sln; 0.5% SFf; 0.17% Suc; frozen
4	90	0.1	90 mb; sln; 0.1% SFf; 0.034% Suc; frozen
5	90	0.05	90 mb; sln; 0.05% SFf; 0.017% Suc; frozen
6	120	5	120 mb; sln; 5% SFf; 1.7% Suc; frozen
7	120	1	120 mb; sln; 1% SFf; 0.34% Suc; frozen
8	120	0.5	120 mb; sln; 0.5% SFf; 0.17% Suc; frozen
9	120	0.1	120 mb; sln; 0.1% SFf; 0.034% Suc:
10	120	0.05	120 mb; sln; 0.05% SFf; 0.017% Suc; frozen
11	480	5	480 mb; sln; 5% SFf; 1.7% Suc; frozen

TABLE 10-continued

SILK FIBROIN SOLUTION FORMULATIONS PREPARED WITH LYOPHILIZED OR FREEZE-THAW SILK FIBROIN

Sample	Silk Fibroin Boil Time (mb)	Silk Concentration (w/v %)	Sample Name
12	480	1	480 mb; sln; 1% SFf; 0.34% Suc; frozen
13	480	0.5	480 mb; sln; 0.5% SFf; 0.17% Suc; frozen
14	480	0.1	480 mb; sln; 0.1% SFf; 0.034% Suc; frozen
15	480	0.05	480 mb; sln; 0.05% SFf; 0.017% Suc; frozen
16	480	5	480 mb; sln; 5% SFf; 1.7% Suc; lyo
17	480	1	480 mb; sln; 1% SFf; 0.34% Suc; lyo
18	480	0.5	480 mb; sln; 0.5% SFf; 0.17% Suc; lyo
19	480	0.1	480 mb; sln; 0.1% SFf; 0.034% Suc; lyo
20	480	0.05	480 mb; sln; 0.05% SFf; 0.017% Suc; lyo

Example 6: Formulation of Silk Fibroin Solutions with Sucrose, Surfactant, and Borate Buffer

To prepare the solutions, silk fibroin of a designated 480 mb (prepared as in Examples 1 or 2), which had been frozen in 50 mM sucrose, was thawed to room temperature. A borate buffer stock solution containing borate, potassium chloride, magnesium chloride, and calcium chloride was prepared and buffered to pH 7.3. A 10% (v/v) polysorbate 80 stock solution was prepared. The formulations and controls (no silk fibroin) were prepared as described in Table 11. The samples in Table 11 are named by the process used to prepare and formulate each solution. For example, in the sample named "480mb; sln; 1% SFf; 0.34% Suc; borate buffer" "480mb" refers to silk degummed with a 480-minute boil, "sln" refers to the formulation of the sample as a solution, "1% SFf" refers to a formulation with 1% (w/v) silk fibroin, "0.34% Suc" refers to a formulation with 0.34% (w/v) sucrose, and "0.2% T80" refers to a formulation containing 0.2% (v/v) polysorbate 80. All formulations are in a "borate buffer" which refers to a vehicle containing final concentrations of 99.3 mM borate, 18.8 mM potassium chloride, 0.6 mM magnesium chloride, and 0.5 mM calcium chloride

TABLE 11

SILK FIBROIN FORMULATIONS WITH SUCROSE AND BORATE BUFFER

Sample	SF Conc. (%)	Polysorbate 80 Conc. (%)	Description
1	0	0	0% SFf; sln; borate buffer
2	0	0.2	0% SFf; sln; borate buffer; 0.2% T80
3	1	0	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer
4	1	0.2	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer; 0.2% T80

Example 7: Additional Formulation of Silk Fibroin Solutions with Sucrose, Surfactant, and Borate Buffer

To prepare the solutions, silk fibroin of a designated 480 mb (prepared as in Examples 1 or 2), which had been frozen

in 50 mM sucrose, was thawed to room temperature. A borate buffer stock solution containing borate, potassium chloride, magnesium chloride, and calcium chloride was prepared and buffered to pH 7.3. A 10% (v/v) polysorbate 80 stock solution was prepared. The formulations and controls (no silk fibroin) were prepared as described in Table 12. The samples in Table 12 are named by the process used to prepare and formulate each solution. For example, in the sample named "480mb; sln; 1% SFf; 0.34% Suc; borate buffer" "480mb" refers to silk degummed with a 480 minute boil, "sln" refers to the formulation of the sample as a solution, "1% SFf" refers to a formulation with 1% (w/v) silk fibroin, and "0.34% Suc" refers to a formulation with 0.34% (w/v) sucrose. All formulations are in a "borate buffer" which refers to a vehicle containing final concentrations of 99.3 mM borate, 18.8 mM potassium chloride, 0.6 mM magnesium chloride, and 0.5 mM calcium chloride.

TABLE 12

SILK FIBROIN FORMULATIONS WITH 480 MB AND SUCROSE AND BORATE BUFFER

Sample	SF Conc. (%)	Sucrose Conc. (%)	Description
1	5	1.71	480 mb; 5% SFf; sln; 1.71% Suc; borate buffer
2	1	0.34	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer
3	0.5	0.17	480 mb; 0.5% SFf; sln; 0.17% Suc; borate buffer
4	0.1	0.03	480 mb; 0.1% SFf; sln; 0.03% Suc; borate buffer
5	0.05	0.01	480 mb; 0.1% SFf; sln; 0.02% Suc; borate buffer

Example 8. Rheological Measurements of Silk Solutions

Rheological measurements were performed on a Bohlin C-VOR 150 with a 4°/40 mm cone and plate geometry with a 0.5 mm gap. 1.4 mL of each sample were pipetted onto a Peltier plate system that maintained a steady temperature of 25° C. throughout the test. Four tests were then run. The first test comprised a strain sweep from 0.1% to 10% strain at 1 Hz over the course of 160 seconds (s), with 50 readings taken. This was to determine the linear viscoelastic region (LVR) of each sample. The second test comprised a strain hold at 5% (or 1% if the LVR was smaller) and 1 Hz for 145 s, with 15 readings taken. The third test comprised a shear rate sweep from 0.1 1/s to 10 1/s over the course of 100 s, followed by a hold at 10 1/s for 20 s, with 70 readings taken. Lastly, a shear rate hold was conducted at 1 1/s for 135 s, with 15 readings taken. The data was then analyzed to determine the average shear storage modulus (the elastic modulus (G')), shear loss modulus (the viscous modulus (G")), phase angle, and viscosity at shear rates of 1 1/s and 10 1/s. The results of the rheology measurements are listed in Table 13 and Table 18.

TABLE 13

RHEOLOGY MEASUREMENTS OF SILK FIBROIN SOLUTION FORMULATIONS AND CONTROLS

Sample	Description	Avg. Viscosity at 1 1/s (cP)	Avg. Viscosity at 10 1/s (cP)	Avg. G' (Pa)	Avg. G" (Pa)	Avg. Phase Angle (°)
—	Refresh Liquigel	79.90	46.29	1.69E−01	3.46E−01	64.05
—	SYSTANE ® lubricant eye drop (Alcort)	20.23	12.06	2.06E−02	7.83E−02	75.44
—	SYSTANE ® lubricant eye gel (Alcon)	1073.57	549.81	2.20E+00	3.03E+00	53.98
1B	10% 30 mb	71.58	47.41	7.60E−02	4.12E−01	79.56
1C	5% 30 mb	63.99	16.37	3.96E−01	5.67E−01	55.19
1D	2.5% 30 mb	69.77	13.22	3.24E−01	3.70E−01	48.99
1E	1% 30 mb	42.44	17.32	1.33E+01	3.57E+00	15.15
1F	0.5% 30 mb	31.17	10.18	3.12E−01	2.65E−01	40.42
1G	0.1% 30 mb	233.61	36.57	1.47E+01	3.40E+00	13.13
2B	10% 90 mb	12.89	6.50	1.11E−02	5.77E−02	79.50
2C	5% 90 mb	11.51	5.36	4.28E−02	4.36E−02	45.40
2D	2.5% 90 mb	13.00	4.22	4.09E−02	4.35E−02	46.23
2E	1% 90 mb	20.36	5.92	4.79E−02	7.50E−02	57.45
2F	0.5% 90 mb	51.25	10.44	1.37E+00	1.77E+00	52.64
2G	0.1% 90 mb	242.98	37.12	7.15E+00	2.47E+00	19.06
3A	30% 120 mb	148.68	125.72	2.90E−02	8.00E−01	87.93
3B	10% 120 mb	11.83	4.69	6.78E−03	4.15E−02	80.08
3C	5% 120 mb	8.74	4.15	2.77E−02	3.71E−02	53.00
3D	2.5% 120 mb	8.00	1.70	3.25E−02	2.79E−02	40.78
3E	1% 120 mb	13.23	4.58	1.10E−02	5.23E−02	78.14
3F	0.5% 120 mb	46.98	12.72	4.30E−01	3.45E−01	38.85
3G	0.1% 120 mb	469.61	74.17	6.69E+00	3.30E+00	26.22
4A	30% 480 mb	110.03	43.84	5.76E−01	5.30E−01	42.89
4B	10% 480 mb	55.18	12.76	3.40E−01	2.37E−01	35.14
4C	5% 480 mb	67.55	13.98	4.03E−01	2.91E−01	35.99
4D	2.5% 480 mb	77.45	15.06	4.86E−01	3.34E−01	34.61
4E	1% 480 mb	113.09	20.44	6.51E−01	4.55E−01	35.16
4F	0.5% 480 mb	255.44	36.18	2.46E+00	1.52E+00	31.70
4G	0.1% 480 mb	432.52	47.05	1.37E+01	3.30E+00	13.52
5B	180 mg/mL Sucrose	10.54	2.41	6.08E−03	3.97E−02	81.30
5E	18 mg/mL Sucrose	10.86	1.92	1.01E−02	3.43E−02	73.85
5G	1.8 mg/mL Sucrose	14.96	2.75	9.53E−03	3.56E−02	74.70
6	1× PBS	13.41	3.03	3.26E−02	3.55E−02	46.66

Table 14 provides the standard deviation of the rheology measurements in Table 13.

TABLE 14

STANDARD DEVIATIONS OF THE RHEOLOGY MEASUREMENTS OF SILK FIBROIN SOLUTION FORMULATIONS AND CONTROLS

Sample	Description	Std Dev. Avg. Viscosity at 1 1/s (cP)	Std. Dev. Avg. Viscosity at 10 1/s (cP)	Std. Dev. Avg. G' (Pa)	Std. Dev. Avg. G" (Pa)	Std. Dev. Avg. Phase Angle (°)
—	Refresh Liquigel	6.07	6.84	1.35E−02	9.89E−03	1.81
—	SYSTANE ® lubricant eye drop (Alcon)	7.60	6.33	8.25E−03	6.22E−03	5.24
—	SYSTANE ® lubricant eye gel (Alcon)	7.08	5.45	1.41E−02	1.54E−02	0.11
1B	10% 30 mb	7.17	5.61	1.26E−02	6.37E−03	1.68
1C	5% 30 mb	7.94	6.65	5.69E−02	5.06E−02	1.72
1D	2.5% 30 mb	4.40	6.59	5.67E−02	4.85E−02	2.17
1E	1% 30 mb	18.27	6.03	1.38E+00	1.97E−01	0.68
1F	0.5% 30 mb	7.75	5.78	1.41E−02	8.38E−03	1.24

TABLE 14-continued

STANDARD DEVIATIONS OF THE RHEOLOGY MEASUREMENTS OF SILK FIBROIN SOLUTION FORMULATIONS AND CONTROLS

Sample	Description	Std Dev. Avg. Viscosity at 1 1/s (cP)	Std. Dev. Avg. Viscosity at 10 1/s (cP)	Std. Dev. Avg. G' (Pa)	Std. Dev. Avg. G'' (Pa)	Std. Dev. Avg. Phase Angle (°)
1G	0.1% 30 mb	13.66	6.30	1.32E+00	1.34E−01	0.75
2B	10% 90 mb	5.17	1.82	9.02E−03	1.16E−02	6.91
2C	5% 90 mb	6.03	1.33	7.01E−03	7.83E−03	6.04
2D	2.5% 90 mb	7.96	2.10	7.76E−03	1.14E−02	7.90
2E	1% 90 mb	7.18	1.63	6.50E−03	5.90E−03	4.02
2F	0.5% 90 mb	10.85	1.59	2.39E−01	3.21E−02	4.55
2G	0.1% 90 mb	39.66	1.86	4.02E−01	5.14E−02	0.69
3A	30% 120 mb	4.99	1.35	1.02E−02	1.06E−02	0.71
3B	10% 120 mb	4.28	1.06	4.27E−03	8.25E−03	6.47
3C	5% 120 mb	3.51	1.54	8.24E−03	8.37E−03	12.40
3D	2.5% 120 mb	5.11	1.25	6.70E−03	6.08E−03	9.61
3E	1% 120 mb	3.06	1.56	7.58E−03	8.57E−03	7.62
3F	0.5% 120 mb	9.42	1.60	3.59E−02	1.11E−02	2.53
3G	0.1% 120 mb	32.15	4.17	9.19E−02	4.70E−02	0.59
4A	30% 480 mb	7.24	1.89	9.07E−02	3.09E−02	3.21
4B	10% 480 mb	4.88	1.29	4.11E−02	1.17E−02	2.33
4C	5% 480 mb	5.80	1.70	4.65E−02	1.99E−02	1.79
4D	2.5% 480 mb	5.61	1.48	4.89E−02	1.63E−02	1.68
4E	1% 480 mb	5.25	1.65	8.22E−02	2.58E−02	2.18
4F	0.5% 480 mb	8.00	1.31	2.16E−01	6.53E−02	1.29
4G	0.1% 480 mb	30.88	2.73	5.66E−01	5.21E−02	0.44
5B	180 mg/mL Sucrose	5.59	1.71	4.41E−03	1.03E−02	5.51
5E	18 mg/mL Sucrose	7.62	1.06	7.04E−03	8.92E−03	10.50
5G	1.8 mg/mL Sucrose	7.38	1.61	5.75E−03	7.31E−03	10.01
6	1× PBS	4.56	1.14	5.57E−03	1.01E−02	8.93

All formulations were observed to demonstrate increased viscosity relative to the controls with sucrose or PBS (samples 5B-6). The viscosities of commercially available dry eye treatments (Refresh Liquigel, SYSTANE® lubricant eye drops (Alcon), and SYSTANE® lubricant eye gel (Alcon)) were also measured as a point of comparison. In general, the solutions prepared from silk fibroin with the longest boiling time (480 mb) and the shortest boil time (30 mb) were more viscous than the other solutions. At higher concentrations, the silk fibroin with a 480 mb and with a 30 mb demonstrated higher viscosities than the other formulations. When the concentration was below 1% (w/v) silk fibroin, the formulations prepared from silk fibroin with longer boiling times demonstrated higher viscosities. A relationship between the concentration of silk fibroin and the viscosity of the solutions was also observed regardless of the boiling time of the silk fibroin. When the silk fibroin concentration was varied between 1-30% (w/v), the viscosity remained consistent among samples prepared from silk fibroin with the same boiling times. When the silk fibroin concentration was between 0.1-0.5%, the solutions demonstrated an increase in viscosity as compared to their more concentrated counterparts. In addition, the storage modulus was higher for formulations with lower concentrations of silk fibroin, and the phase angle was measured to be lower for formulations with lower concentrations of silk fibroin. All of the formulations shear thinned, demonstrating lower viscosities at higher shear rates.

To prepare the solutions for rheological analysis, frozen silk fibroin solutions were thawed at room temperature and gently mixed before use. A borate buffer was prepared at 5× the final desired concentration by dissolving 1500 mg of boric acid, 213 mg of sodium borate decahydrate, 350 mg of potassium chloride, 850 mg of sodium chloride, 32 mg of magnesium chloride hexahydrate, and 20 mg of calcium chloride in DI water to reach 50 mL final. Each silk fibroin solution, after thawing, was diluted with DI water and this 5× borate buffer to reach 1% silk fibroin (w/v) final in 1× borate buffer (which indicates 6 mg/mL boric acid, 0.852 mg/mL sodium borate decahydrate, 1.4 mg/mL potassium chloride, 3.4 mg/mL sodium chloride, 0.128 mg/mL magnesium chloride hexahydrate, and 0.08 mg/mL calcium chloride dihydrate.) Rheological measurements were performed on a Bohlin C-VOR 150 with a 4° C./40 mm cone and plate geometry with a 0.15 mm gap. 1.2 mL of each sample were pipetted onto a Peltier plate system that maintained a steady temperature of 25° C. throughout the test. One test was then run. The test comprised of a pre-shear of 1 1/s for 20s followed by a shear rate hold at 1 1/s for 90 s, with 30 readings taken. The data was then analyzed to determine the average viscosity. The results of the rheology measurements are listed in Table 15.

TABLE 15

RHEOLOGY MEASUREMENTS OF SILK FIBROIN SOLUTION FORMULATIONS

Lot #	Average Viscosity at 1 1/s (cP)	Std. Dev. Of Viscosity at 1 1/s (cP)
111.2	30.89	5.03
111.4	39.00	6.11
111.6	76.21	6.31
111.8	97.66	14.15
108A.2	—	—
108A.4	—	—
108A.6	—	—

TABLE 15-continued

RHEOLOGY MEASUREMENTS OF SILK FIBROIN SOLUTION FORMULATIONS		
Lot #	Average Viscosity at 1 1/s (cP)	Std. Dev. Of Viscosity at 1 1/s (cP)
108A.8	15.22	4.01
108B.2	—	—
108B.4	—	—
108B.6	—	—
108B.8	53.41	4.41
108C.2	12.54	3.12
108C.4	23.90	3.54
108C.6	54.18	5.58
108C.8	74.70	5.99
109A.2	—	—
109A.4	—	—
109A.6	—	—
109A.8	43.38	6.58
109B.2	27.46	8.29
109B.4	25.68	6.41
109B.6	82.10	4.84
109B.8	104.60	11.71
109C.2	17.12	3.91
109C.4	49.41	4.55
109C.6	106.95	10.12
109C.8	142.48	10.65
110A.2	18.16	3.97
110A.4	42.74	5.62
110A.6	78.72	5.98
110A.8	128.03	6.75
110B.2	41.37	5.20
110B.4	89.69	5.75
110B.6	178.79	4.62
110B.8	—	—
110C.2	99.74	4.36
110C.4	140.56	10.33
110C.6	—	—
110C.8	—	—

The viscosity of silk fibroin solutions increased with decreasing molecular weight. This was observed within the same conditions (sodium carbonate concentration and temperature) over the degumming time course. As time increased and molecular weight decreased, viscosity increased. Overall, silk fibroin solutions fell into three ranges. Solutions with average molecular weight ≤34.04 kDa displayed the highest viscosity that ranged from 78.72 to 178.79 centipoise (cPs). Silk fibroin solutions with an average molecular weights that ranged from 35.47 to 60.74 kDa showed slightly lower viscosities which ranged from 39.00 to 82.10 cPs. The lowest viscosity was observed in the solutions with the highest average molecular weights of 64.36 to 122.52 kDa. These solutions displayed viscosities of 12.54 to 30.89 cPs. This effect of lower viscosity silk fibroin solutions which displayed higher viscosity is most likely due to the interfacial partitioning of silk fibroin to the air water interface. The lower molecular weight silk fibroin proteins most likely organize at the air-water interface more efficiently allowing for higher local concentrations and increased local viscosity.

Example 9: Rheological Measurements of Silk Solutions with Lyophilized or Freeze-Thaw Silk Fibroin

Rheological measurements were performed on a Bohlin C-VOR 150 with a 4°/40 mm cone and plate geometry with a 0.15 mm gap. 1.2 mL of each sample were pipetted onto a Peltier plate system that maintained a steady temperature of 25° C. throughout the test. Three tests were then run. The first test comprised a strain sweep from 0.1% to 10% strain at 1 Hz over the course of 130 seconds (s), with 40 readings taken. This was to determine the linear viscoelastic region (LVR) of each sample. The second test comprised a strain hold at 5% (or 1% if the LVR was smaller) and 1 Hz for 160 s, with 12 readings taken. The third test comprised a shear rate hold at 1 1/s for 30 s, with 30 readings taken, followed by a hold at 10 1/s for 30 s, with 30 readings taken. The data was then analyzed to determine the average shear storage modulus (the elastic modulus (G')), shear loss modulus (the viscous modulus (G")), phase angle, and viscosity at shear rates of 1 1/s and 10 1/s. The results of the rheology measurements are listed in Table 16 and Table 17.

TABLE 16

RHEOLOGY MEASUREMENTS OF SILK FIBROIN SOLUTION FORMULATIONS WITH LYOPHILIZED OR FREEZE-THAW SILK FIBROIN, AND CONTROLS						
Sample	Description	Avg. Viscosity at 1 1/s (cP)	Avg. Viscosity at 10 1/s (cP)	Avg. G' (Pa)	Avg. G" (Pa)	Avg. Phase Angle (°)
1	5% 90 mb	24.21	12.23	7.89E−02	1.79E−01	66.03
2	1% 90 mb	36.63	14.35	9.42E−02	1.52E−01	57.61
3	0.5% 90 mb	65.94	19.57	1.86E−01	2.62E−01	54.71
4	0.1% 90 mb	565.71	136.56	2.75E+00	1.02E+00	21.18
5	0.05% 90 mb	492.38	79.65	9.81E+00	2.62E+00	15.21
6	5% 120 mb	36.63	10.35	1.69E−01	3.04E−01	61.01
7	1% 120 mb	23.09	10.90	4.52E−02	9.52E−02	64.35
8	0.5% 120 mb	50.90	16.62	1.08E−01	1.72E−01	57.92
9	0.1% 120 mb	302.36	43.97	1.98E+00	1.21E+00	31.59
10	0.05% 120 mb	145.83	29.29	3.79E+00	1.25E+00	18.32
11	5% 480 mb	54.85	16.64	1.95E−01	2.44E−01	51.50
12	1% 480 mb	129.58	26.62	9.48E−01	9.51E−01	45.37
13	0.5% 480 mb	117.32	21.45	8.24E−01	7.93E−01	44.19
14	0.1% 480 mb	111.52	23.66	9.07E−01	6.55E−01	36.39
15	0.05% 480 mb	120.76	35.64	1.17E+00	5.39E−01	26.95
16	5% 480 mb Lyo	74.02	20.63	3.51E−01	3.66E−01	46.41
17	1% 480 mb Lyo	155.43	29.82	9.23E−01	8.88E−01	44.04
18	0.5% 480 mb Lyo	180.84	31.41	1.25E+00	1.09E+00	41.36
19	0.1% 480 mb Lyo	127.93	36.58	1.08E+00	1.05E+00	47.47
20	0.05% 480 mb Lyo	80.88	22.45	1.11E+00	1.08E+00	50.41

Table 17 provides the standard deviation of the rheology measurements in Table 16.

TABLE 17

STANDARD DEVIATIONS OF THE RHEOLOGY MEASUREMENTS OF SILK FIBROIN SOLUTION FORMULATIONS AND CONTROLS

Sample	Description	Std. Dev. Avg. Viscosity at 1 1/s (cP)	Std. Dev. Avg. Viscosity at 10 1/s (cP)	Std. Dev. Avg. G' (Pa)	Std. Dev. Avg. G'' (Pa)	Std. Dev. Avg. Phase Angle (°)
1	5% 90 mb	5.62	0.37	3.18E−02	8.32E−02	1.36
2	1% 90 mb	12.48	2.48	5.13E−03	4.53E−02	6.48
3	0.5% 90 mb	4.41	0.28	1.92E−03	2.36E−02	2.74
4	0.1% 90 mb	102.17	5.17	1.15E+00	2.43E−01	3.80
5	0.05% 90 mb	390.47	49.59	4.17E+00	9.28E−01	1.13
6	5% 120 mb	8.51	0.53	2.84E−02	4.00E−02	0.85
7	1% 120 mb	6.26	0.38	1.79E−02	6.64E−03	6.59
8	0.5% 120 mb	4.88	0.47	2.51E−02	3.45E−02	0.94
9	0.1% 120 mb	210.48	18.49	2.20E−01	2.42E−02	2.17
10	0.05% 120 mb	113.49	18.36	1.07E+00	3.10E−01	0.47
11	5% 480 mb	4.75	0.19	1.61E−02	1.96E−02	0.01
12	1% 480 mb	21.17	3.70	1.84E−01	1.47E−01	1.21
13	0.5% 480 mb	15.19	0.71	1.02E−01	8.08E−02	0.52
14	0.1% 480 mb	15.68	1.88	2.03E−01	7.53E−02	9.20
15	0.05% 480 mb	93.34	29.99	6.85E−01	1.76E−01	7.04
16	5% 480 mb Lyo	0.44	0.12	1.97E−02	1.98E−03	1.73
17	1% 480 mb Lyo	3.23	3.43	4.82E−02	3.72E−03	1.64
18	0.5% 480 mb Lyo	13.39	0.77	2.37E−01	1.61E−01	1.23
19	0.1% 480 mb Lyo	73.44	29.01	8.81E−01	5.87E−01	9.63
20	0.05% 480 mb Lyo	49.81	8.37	1.19E+00	8.62E−01	13.37

The solutions prepared from silk fibroin with the longest boiling time (480 mb) were more viscous than solutions prepared with shorter boil times. At higher concentrations (5%, 1%, and 0.5% (w/v)), the 480mb silk fibroin solutions demonstrated higher viscosities than the shorter boil time formulations. However, when the concentration of silk fibroin was below 0.5% (w/v), the formulations prepared from shorter boil times demonstrated higher viscosities. A relationship between the concentration of silk fibroin and the viscosity of the solutions was also observed. The general trend showed that the viscosity of the solutions with 5% (w/v) silk fibroin was lower than solutions with 1% (w/v) silk fibroin. At concentrations of 0.1% and 0.05% (w/v) silk fibroin, the viscosities of the 90mb and 120mb samples increased, whereas that of the 480mb samples remained constant. These trends hold for oscillatory properties as well, with the G' and G'' values following the viscosity trends. The phase angle displayed lower values with increased viscosity, indicating a more viscous material. The 480mb samples prepared with lyophilized silk fibroin showed slightly lower viscosities at 5%, 1%, and 0.5% (w/v) silk fibroin than the solutions prepared with frozen silk fibroin stock. The lyophilized solution formulations matched the frozen silk fibroin samples at 0.1% and 0.05% (w/v). All of the formulations shear thinned, demonstrating lower viscosities at higher shear rates.

Example 10. Rheological Measurements Regarding the Interfacial Viscosity of Silk Fibroin Solutions

The effect of adding a surfactant on the viscosity properties of silk fibroin solution formulations was assessed. Similar to a surfactant, hydrophobic proteins have been shown to migrate to the air-water boundary, resulting in an increase in the local concentration at this interface. This increase in the local concentration of a protein, such as silk fibroin, can lead to an increase in apparent viscosity. This effect is termed "interfacial viscosity." Incorporation of a surfactant, a molecule that can better and more efficiently associate to the air-water interface, will block the hydrophobic protein association and will negate any increase in viscosity due to the protein buildup at the boundary. We assessed the viscosity properties of silk fibroin formulations with and without the presence of a surfactant using a rotational rheometer.

Rheological measurements were performed on a Bohlin C-VOR 150 with a 4°/40 mm cone and plate geometry with a 0.15 mm gap. 1.2 mL of each sample were pipetted onto a Peltier plate system that maintained a steady temperature of 25° C. throughout the test. The test comprised a logarithmic shear rate sweep from 0.01 1/s to 1000 1/s over the course of 606 s with 100 viscosity readings. The test was performed on silk fibroin formulations after their preparation. The formulations tested were prepared using methods described in the previous examples. A standard oil (100 cP) was also evaluated to ensure accuracy of the measurements across this large shear range. The viscosity measurements are listed in Table 18 and Table 19.

TABLE 18

SHEAR VISCOSITY OF SILK FIBROIN FORMULATIONS

Shear Rate (1/s)	100 cP Oil Std. Viscosity (cP)	0% SF; sln; borate buffer Average Viscosity (cP)	0% SF; sln; borate buffer; 0.2% T80 Average Viscosity (cP)	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer Average Viscosity (cP)	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer; 0.2% T80 Average Viscosity (cP)
0.10	137.67	26.15	12.85	470.03	27.73
0.11	136.36	50.98	15.89	432.36	43.58
0.13	125.49	29.37	6.47	404.69	44.84
0.15	70.37	26.54	7.76	356.65	36.53
0.16	106.89	27.47	31.77	337.79	28.39

TABLE 18-continued

SHEAR VISCOSITY OF SILK FIBROIN FORMULATIONS

Shear Rate (1/s)	100 cP Oil Std. Viscosity (cP)	0% SF; sln; borate buffer Average Viscosity (cP)	0% SF; sln; borate buffer; 0.2% T80 Average Viscosity (cP)	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer Average Viscosity (cP)	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer; 0.2% T80 Average Viscosity (cP)
0.18	106.75	18.73	9.84	316.25	23.51
0.21	112.05	22.77	24.58	285.04	28.04
0.23	92.07	22.73	28.37	263.86	26.75
0.26	105.37	21.92	18.64	239.22	25.60
0.29	100.39	24.97	12.64	230.36	30.24
0.33	84.72	9.69	22.31	186.51	24.47
0.37	83.06	20.69	23.15	180.08	14.57
0.41	89.68	17.10	17.00	158.27	15.79
0.46	100.45	15.32	20.27	151.15	13.29
0.52	87.69	38.20	24.42	128.86	16.82
0.59	98.09	11.24	14.67	127.66	11.09
0.66	96.52	8.89	18.03	114.52	5.56
0.74	97.34	11.06	12.89	106.66	11.27
0.83	98.63	4.54	6.75	92.54	5.40
0.93	95.38	6.67	7.43	83.53	8.81
1.05	101.01	6.33	4.18	78.63	7.58
1.18	92.76	7.82	4.52	67.05	7.37
1.32	94.63	6.73	5.97	62.58	6.30
1.48	99.90	6.72	5.21	59.37	8.20
1.67	100.40	4.79	3.91	51.99	4.71
1.87	102.84	3.86	5.53	47.74	3.83
2.10	91.83	4.23	6.45	39.50	3.43
2.36	96.82	4.01	5.81	41.94	4.20
2.66	99.03	6.72	3.93	35.41	4.89
2.98	94.40	5.33	3.68	35.36	3.86
3.35	96.11	0.50	4.35	34.85	4.08
3.76	96.71	3.55	5.41	29.08	4.23
4.23	97.02	2.21	4.33	27.18	4.87
4.75	99.08	0.95	2.55	22.68	1.61
5.33	104.28	2.95	2.14	23.08	3.29
5.99	107.99	25.60	26.36	41.52	25.51
6.73	111.10	5.88	9.62	21.58	7.56
7.56	97.59	3.23	3.80	16.63	3.21
8.49	99.17	2.17	1.81	18.22	5.88
9.54	99.03	5.64	5.74	13.38	1.57
10.72	102.48	2.78	3.99	15.76	3.10
12.04	61.92	142.42	7.63	3.31	22.65
13.53	98.09	4.35	2.71	241.74	2.13
15.19	98.74	1.85	4.38	11.42	3.33
17.07	99.86	2.81	4.11	10.51	3.26
19.17	98.95	3.18	4.35	9.39	6.58
21.54	97.41	3.18	2.96	9.97	3.74
24.19	104.21	6.94	6.47	11.80	1.11
27.18	98.30	4.27	2.99	8.54	3.52
30.53	98.69	3.65	3.59	8.52	4.35
34.29	98.21	3.38	4.13	8.39	3.23
38.52	98.29	3.50	3.80	7.28	3.88
43.27	98.38	3.22	3.45	7.08	3.25
48.61	97.86	3.85	3.16	6.27	3.25
54.60	99.51	3.36	3.42	6.12	3.54
61.34	327.32	5.99	7.34	9.85	7.05
68.90	123.58	29.33	8.68	10.15	7.91
77.40	117.30	8.22	6.40	9.54	6.55
86.94	117.13	8.21	6.74	11.35	8.58
97.66	112.80	7.91	8.28	10.49	6.77
109.71	110.36	7.44	7.57	10.42	8.22
123.22	106.51	7.15	8.42	10.83	8.44
138.43	105.91	7.45	7.74	9.62	8.33
155.50	103.33	7.24	6.80	6.40	7.41
174.68	102.19	7.60	8.91	9.05	8.01
196.24	101.06	8.03	8.25	9.54	8.32
220.42	100.93	7.96	8.04	9.17	8.11
247.61	99.67	7.64	7.83	9.02	8.09
278.14	92.73	10.03	11.21	12.20	13.06
312.44	22.66	20.41	20.35	12.57	45.20
350.98	105.62	17.09	18.27	8.17	9.58
394.29	99.61	7.23	12.32	17.93	11.21
443.08	92.48	28.72	21.22	10.89	5.05
497.49	92.60	9.49	23.01	16.97	21.32
558.85	99.96	17.15	24.88	4.26	29.54
627.73	95.56	21.40	18.93	32.16	11.94
705.17	85.14	1.67	8.02	6.44	5.21
792.18	95.23	3.81	3.77	6.85	1.22
889.90	88.54	4.51	5.69	11.00	9.08
999.53	91.39	11.65	6.41	12.41	11.06

TABLE 19

STANDARD DEVIATION OF SHEAR VISCOSITY OF SILK FIBROIN FORMULATIONS

Shear Rate (1/s)	0% SFf; sln; borate buffer Standard Deviation (cP)	0% SF; sln; borate buffer; 0.2% T80 Standard Deviation (cP)	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer Standard Deviation (cP)	480 mb; 1% SF; sln; 0.34% Suc; borate buffer; 0.2% T80 Standard Deviation (cP)
0.10	183.21	259.48	370.04	121.43
0.11	113.76	28.46	397.78	379.43
0.13	48.59	117.59	345.17	153.38
0.15	172.35	239.23	250.15	130.21
0.16	274.44	79.64	356.87	269.20
0.18	108.99	79.30	211.10	131.32
0.21	62.00	102.73	137.40	150.49
0.23	174.88	59.71	246.07	212.97
0.26	41.72	88.83	155.11	82.60
0.29	33.42	66.11	137.49	264.30
0.33	44.47	14.79	164.96	30.89
0.37	28.32	39.87	208.57	39.79
0.41	84.89	61.00	162.86	87.22
0.46	61.60	33.17	119.86	12.43
0.52	74.92	20.12	117.38	23.07
0.59	46.31	6.54	92.01	95.79
0.66	12.20	18.90	101.13	49.55
0.74	23.71	15.03	77.60	11.40
0.83	22.47	40.76	70.79	40.35
0.93	14.94	23.77	40.81	24.14
1.05	15.01	7.93	75.35	20.04
1.18	6.46	7.53	37.63	40.49
1.32	24.69	3.23	59.65	36.10
1.48	5.95	5.47	39.75	22.63
1.67	10.29	17.84	39.72	14.55
1.87	17.73	9.92	32.55	14.91
2.10	10.80	18.38	26.43	7.00
2.36	10.97	5.47	27.16	19.05
2.66	8.06	7.19	15.91	8.66
2.98	4.69	11.54	8.37	6.22
3.35	10.58	12.60	17.45	0.82
3.76	10.44	5.84	7.57	8.83
4.23	12.37	8.87	5.75	7.85
4.75	5.90	9.31	7.37	4.44
5.33	1.61	11.87	17.69	5.67
5.99	2.31	7.17	6.03	4.25
6.73	5.34	8.04	3.08	1.72
7.56	7.60	1.76	9.48	2.28
8.49	1.49	5.16	6.15	6.15
9.54	4.67	2.76	5.88	1.95
10.72	5.77	3.23	1.84	2.55
12.04	4.66	2.09	2.04	2.11

TABLE 19-continued

STANDARD DEVIATION OF SHEAR VISCOSITY OF SILK FIBROIN FORMULATIONS

Shear Rate (1/s)	0% SFf; sln; borate buffer Standard Deviation (cP)	0% SF; sln; borate buffer; 0.2% T80 Standard Deviation (cP)	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer Standard Deviation (cP)	480 mb; 1% SF; sln; 0.34% Suc; borate buffer; 0.2% T80 Standard Deviation (cP)
13.53	1.37	1.29	2.55	1.63
15.19	2.98	3.42	5.36	2.47
17.07	1.72	2.94	4.59	2.52
19.17	2.58	1.77	5.37	4.18
21.54	1.80	1.58	6.85	3.32
24.19	0.80	1.48	4.69	1.72
27.18	2.55	1.89	5.21	1.48
30.53	1.90	0.58	4.83	2.71
34.29	0.43	2.07	2.32	1.85
38.52	0.70	0.67	0.98	0.82
43.27	1.10	2.68	5.94	1.81
48.61	1.07	0.63	2.43	1.25
54.60	2.98	2.07	1.49	3.06
61.34	4.46	8.70	6.58	5.84
68.90	5.94	3.88	3.98	2.44
77.40	2.75	2.41	4.45	0.96
86.94	1.25	0.49	2.28	0.79
97.66	2.37	4.26	2.66	0.57
109.71	3.43	3.01	3.25	1.72
123.22	232.03	2.38	3.76	17.99
138.43	3.46	1.59	391.52	1.72
155.50	2.03	0.43	0.47	0.45
174.68	0.15	0.55	0.38	0.98
196.24	1.13	1.23	0.63	5.03
220.42	1.17	0.38	1.10	0.31
247.61	3.81	3.80	4.33	1.79
278.14	1.05	0.43	1.08	0.59
312.44	0.32	0.08	0.55	0.75
350.98	0.39	0.35	0.68	0.60
394.29	0.14	0.83	1.29	0.21
443.08	0.23	0.41	0.92	0.71
497.49	1.07	0.18	1.42	0.87
558.85	0.31	0.26	0.74	0.28
627.73	2.30	0.40	0.93	1.17
705.17	40.59	0.89	1.09	1.04
792.18	1.24	1.04	0.91	1.30
889.90	1.61	0.20	1.74	0.69
999.53	1.22	1.43	0.53	0.49

The results displayed that silk fibroin formulations without surfactant display shear thinning, reaching viscosities of 470 cP at 0.1 1/s which is reduced to 79 cP at 1 1/s and 13 cP at 10 1/s. When shear rate was increased above 50 cP, the silk fibroin formulation with no surfactant showed viscosity similar to buffer controls (no silk fibroin). When surfactant was added to the silk fibroin formulation, it displayed the same shear viscosity profile as the buffer controls across the entire range of shear rates. Viscosity reached a maximum of 45 cP at a shear rate of 0.13 1/s, which fell below 10 cP at shear rates of 0.6 1/s and above. The controls (+/− polysorbate 80) showed the same trends in viscosity over the shear range tested. This increase in viscosity of silk fibroin formulations that was reduced with the addition of a surfactant (polysorbate 80) indicated that silk fibroin in solution possesses interfacial properties that lead to an increase in apparent viscosity.

Example 11: Rheological Measurements of Silk Fibroin Solutions Using Capillary Rheology

Capillary rheology was utilized to further understand the interfacial viscosity rheology effects of silk fibroin in solution. Unlike the rotational rheometer, a capillary rheometer has a very low air-water interface. This will greatly reduce if not eliminate interfacial viscosity effects that are observed using other rheologic methods (cone and plate rotational rheology). Samples were evaluated using two tests. First, samples were evaluated using a VROC® microVISC™ capillary viscometer (Rheosense) at 25° C. at 500 1/s shear rate. Samples were then measured on the VROC® Initium automated viscometer (Rheosense) using the B05 chip. Aliquots of 40 μL for each sample were loaded into inserts and then, using a benchtop centrifuge, they were centrifuged at 8000 rpm for 20 seconds to remove any bubbles. The inserts with the sample volume were then placed in capped vials and loaded onto the Initium sample tray. The software was then programmed to run a shear rate sweep with 5 segments for each sample. The shear rates selected for each sample were determined by logic in the Initium software (3808, 4760, 6427, 16150, and 19040 1/s). All measurements were taken at 25° C. Each viscosity value reported is the average of 10 repeat measurements. The formulations were prepared as described in the previous examples, such as Example 6.

Results showed that silk fibroin solution formulations and buffer controls, in the presence or absence of a surfactant, displayed Newtonian viscosity at shear rates from 500 1/s to 19040 1/s. Borate buffer displayed viscosity that ranged from 0.910 cP to 1.00 cP over this range. The addition of polysorbate 80 to this buffer slightly increased the viscosity (0.931-1.07 cP). The silk fibroin formulations displayed higher viscosity than the buffer controls with the same trends. Silk fibroin without surfactant had a slightly lower viscosity over this shear range than the formulation containing polysorbate 80. Silk fibroin formulations displayed viscosities of 1.05 cP-1.19 cP while the addition of surfactant very slightly increased the viscosity to 1.06 cP-1.31 cP over the same shear range (Table 20 and Table 21).

TABLE 20

CAPILLARY VISCOSITY OF SILK FIBROIN FORMULATIONS

Temperature (° C.)	Shear Rate (1/s)	0% SFf; sln; borate buffer Average Viscosity (cP)	0% SFf; sln; borate buffer; 0.2% T80 Average Viscosity (cP)	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer Average Viscosity (cP)	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer; 0.2% T80 Average Viscosity (cP)
25	500	1.00	1.07	1.18	1.31
25	3808	0.92	0.93	1.05	1.07
25	4759	0.92	0.94	1.06	1.07
25	6466	0.91	0.94	1.05	1.06
25	10310	0.91	0.94	1.05	1.07
25	19040	0.92	0.93	1.05	1.06

TABLE 21

STANDARD DEVIATION OF CAPILLARY VISCOSITY OF SILK FIBROIN FORMULATIONS					
Temperature (° C.)	Shear Rate (1/s)	0% SFf; sln; borate buffer Standard Deviation	0% SFf; sln; borate buffer; 0.2% T80 Standard Deviation	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer Standard Deviation	480 mb; 1% SFf; sln; 0.34% Suc; borate buffer; 0.2% T80 Standard Deviation
25	500	0.006	0.003	0.004	0.006
25	3808	0.005	0.004	0.004	0.003
25	4759	0.006	0.003	0.002	0.004
25	6466	0.007	0.006	0.004	0.007
25	10310	0.000	0.000	0.000	0.000
25	19040	0.001	0.001	0.004	0.003

The capillary rheology data is similar to the rotational data at these shear rates greater than or equal to 500 1/s for all of the formulations.

Example 12: Rheological Measurements of Stressed Silk Solutions

Formulations were prepared by dissolving lyophilized silk fibroin, which had been lyophilized in phosphate buffer, into phosphate buffered saline. Rheological measurements were performed on a Bohlin C-VOR 150 with a 4°/40 mm cone and plate geometry with a 0.5 mm gap. 1.4 mL of each sample were pipetted onto a Peltier plate system that maintained a steady temperature of 25° C. throughout the test. Five tests were then run. The first test comprised a strain sweep from 0.1% to 10% strain at 1 Hz over the course of 160 s, with 50 readings taken. This was to determine the linear viscoelastic region (LVR) of each sample. The second test comprised a strain hold at 5% (or 1% if the LVR was smaller) and 1 Hz for 145 s, with 15 readings taken. The third test comprised a shear rate sweep from 0.1 1/s to 10 1/s over the course of 100 s, followed by a hold at 10 1/s for 20 s. 70 readings were taken. Fourth, a shear rate hold was conducted at 1 1/s for 135 s, with 15 readings taken. Lastly, a shear ramp was conducted from 0.01 1/s to 1 1/s over 140s with 20 readings. The data was then analyzed to determine the average shear storage modulus (G'), shear loss modulus (G''), phase angle, and viscosity at shear rates of 0.1 1/s, 1 1/s, and 10 1/s. The experiments were performed on silk fibroin formulations after their preparation. The formulations were then stressed by incubation at 60° C. overnight, and the samples that did not gel were tested again. The formulations were then stressed by autoclave and the samples that did not gel were tested for a third time. The formulations were stressed to simulate a long shelf-life and terminal sterilization, and to see if the protein maintained rheological integrity. The formulations tested were described in Table 22, and the rheological measurements were listed in Table 23 and Table 24. The samples in the tables are named by the process used to prepare and formulate each solution. For example, in the sample named "480mb; sln; 1% SFf; 60° C.; Auto" the "480mb" refers to the silk degummed with a 480-minute boil, "sln" refers to the formulation of the sample as a solution, "1% SFf" refers to a formulation with 1% (w/v) silk fibroin, "60° C." refers to a preparation stressed by incubation at 60° C. overnight, and "Auto" refers to a preparation stressed by incubation at 60° C. overnight followed by autoclave. Table 22 provides characteristics of the formulations containing stressed silk fibroin.

TABLE 22

SILK SOLUTION FORMULATIONS WITH STRESSED SILK FIBROIN				
Sample	Description	Stress Condition	480 mb Silk % (w/v)	Sample Name
—	REFRESH LIQUIGEL ® lubricant eye gel (Allergan)	None	0	REFRESH LIQUIGEL ® lubricant eye gel (Allergan)
—	SYSTANE ® lubricant eye drop (Alcon)	None	0	SYSTANE ® lubricant eye drop (Alcon)
	SYSTANE ® lubricant eye gel (Alcon)	None	0	SYSTANE ® lubricant eye gel (Alcon)
1	10% Unstressed	None	10	480 mb; sln; 10% SFf
2	5% Unstressed	None	5	480 mb; sln; 5% SFf
3	1% Unstressed	None	1	480 mb; sln; 1% SFf
4	0.5% Unstressed	None	0.5	480 mb; sln; 0.5% SFf
5	0.1% Unstressed	None	0.1	480 mb; sln; 0.1% SFf
6	0.05% Unstressed	None	0.05	480 mb; sln; 0.05% SFf
7	0.01% Unstressed	None	0.01	480 mb; sln; 0.01% SFf
8	0.005% Unstressed	None	0.005	480 mb; sln; 0.005% SFf
9	1x PBS	None	0	PBS Control
—	REFRESH LIQUIGEL ® lubricant eye gel (Allergan) 60° C.	60° C. Overnight	0	REFRESH LIQUIGEL ® lubricant eye gel (Allergan); 60° C.
—	SYSTANE ® lubricant eve drop (Alcon)60° C.	60° C. Overnight	0	SYSTANE ® lubricant eye drop (Alcon); 60° C.
—	SYSTANE ® lubricant eve gel (Alcon)60° C.	60° C. Overnight	0	SYSTANE ® lubricant eye gel (Alcon) 60° C.
10	10% 60° C.	60° C. Overnight	10	480 mb; sln; 10% SFf; 60° C.
11	5% 60° C.	60° C. Overnight	5	480 mb; sln; 5% SFf; 60° C.
12	1% 60° C.	60° C. Overnight	1	480 mb; sln; 1% SFf; 60° C.
13	0.5% 60° C.	60° C. Overnight	0.5	480 mb; sln; 0.5% SFf; 60° C.
14	0.1% 60° C.	60° C. Overnight	0.1	480 mb; sln; 0.1% SFf; 60° C.
15	0.05% 60° C.	60° C. Overnight	0.05	480 mb; sln; 0.05% SFf; 60° C.
16	0.01% 60° C.	60° C. Overnight	0.01	480 mb; sln; 0.01% SFf; 60° C.
17	0.005% 60° C.	60° C. Overnight	0.005	480 mb; sln; 0.005% SFf; 60° C.

TABLE 22-continued

SILK SOLUTION FORMULATIONS WITH STRESSED SILK FIBROIN				
Sample	Description	Stress Condition	480 mb Silk % (w/v)	Sample Name
18	1x PBS 60° C.	60° C. Overnight	0	PBS control; 60° C.
19	1% Autoclave	60° C. and Autoclave	1	480 mb; sln; 1% SFE 60° C.; Auto
20	0.5% Autoclave	60° C. and Autoclave	0.5	480 mb; sln; 0.5% SFf; 60° C.; Auto
21	0.1% Autoctave	60° C. and Autoclave	0.1	480 mb, sln; 0.1% SFf; 60° C.; Auto
22	0.05% Autoclave	60° C. and Autoclave	0.05	480 mb; sln; 0.05% SFf; 60° C.; Auto
23	0.01% Autoclave	60° C. and Autoclave	0.01	480 mb; sln; 0.01% SFf; 60° C.; Auto
24	0.005% Autoclave	60° C. and Autoclave	0.005	480 mb; sln; 0.005% SFf; 60° C.; Auto
25	1x PBS Autoclave	60° C. and Autoclave	0	PBS control; 60° C.; Auto

Table 23 provides rheological data for the formulations described in Table 22.

TABLE 23

RHEOLOGICAL MEASUREMENTS OF SILK SOLUTIONS WITH STRESSED SILK FIBROIN							
Sample	Description	Avg. Viscosity at 0.01 1/s (cP)	Avg. Viscosity at 1 1/s (cP)	Avg. Viscosity at 10 1/s (cP)	Avg. G' (Pa)	Avg. G'' (Pa)	Avg. Phase Angle (°)
—	REFRESH LIQUIGEL ® lubricant eye gel (Allergan)	—	79.90	46.29	0.17	0.35	64.05
—	SYSTANE ® lubricant eye drop (Alcon)	—	20.23	12.06	0.02	0.08	75.44
	SYSTANE ® lubricant eye gel (Alcon)	—	1073.57	549.81	2.20	3.03	53.98
1	10% Unstressed	483.28	61.29	13.90	0.18	0.19	46.54
2	5% Unstressed	1241.98	77.42	15.54	0.33	0.25	37.93
3	1% Unstressed	1211.40	152.93	24.45	0.82	0.59	36.14
4	0.5% Unstressed	3168.00	164.13	23.96	0.87	0.58	34.04
5	0.1% Unstressed	62996.50	848.57	71.80	6.80	2.17	17.65
6	0.05% Unstressed	78834.00	1012.95	73.62	9.55	2.21	13.07
7	0.01% Unstressed	42258.50	279.10	25.40	9.80	2.00	11.63
8	0.005% Unstressed	13412.20	141.06	21.93	8.61	1.58	10.48
9	1× PBS	448.98	10.33	2.90	0.01	0.04	68.70
—	REFRESH LIQUIGEL ® lubricant eye gel (Allergan) 60° C.	752.92	40.80	32.99	0.03	0.21	80.72
—	SYSTANE ® lubricant eye drop (Alcon) 60° C.	1581.90	16.88	10.60	0.03	0.07	70.70
—	SYSTANE ® lubricant eye gel (Alcon) 60° C.	3035.00	926.24	501.00	1.89	2.80	56.00
10	10% 60° C.	1586.30	62.18	13.58	0.22	0.22	45.50
11	5% 60° C.	1783.60	62.33	12.19	0.25	0.21	41.18
12	1% 60° C.	2922.10	139.74	22.57	0.74	0.54	36.53
13	0.5% 60° C.	3902.25	143.14	22.19	0.73	0.53	35.88
14	0.1% 60° C.	16141.50	250.78	33.08	3.32	1.17	19.61
15	0.05% 60° C.	30653.00	206.13	23.65	6.22	1.47	13.47
16	0.01% 60° C.	27437.00	120.19	13.38	6.49	1.33	11.62
17	0.005% 60° C.	27917.00	149.65	19.14	8.77	1.49	9.70
18	1× PBS 60° C.	109.72	9.85	2.82	0.03	0.03	42.80
19	1% Autoclave	4274.55	141.91	21.93	0.91	0.57	32.09
20	0.5% Autoclave	5146.85	150.73	23.07	1.12	0.62	29.17
21	0.1% Autoclave	6566.45	110.14	14.90	1.59	0.56	20.14
22	0.05% Autoclave	9674.40	182.30	24.15	4.83	1.33	15.09
23	0.01% Autoclave	19294.50	113.32	11.52	4.48	0.75	9.50
24	0.005% Autoclave	707.12	7.10	3.51	0.05	0.03	37.33
25	1× PBS Autoclave	2833.30	83.75	14.97	0.44	0.33	36.70

Table 24 provides the standard deviations for the rheological measurements described in Table 23.

TABLE 24

STANDARD DEVIATIONS OF THE RHEOLOGICAL MEASUREMENTS OF SILK SOLUTIONS WITH STRESSED SILK FIBROIN

Sample	Description	Std. Dev. Avg. Viscosity at 0.01 1/s (cP)	Std. Dev. Avg. Viscosity at 1 1/s (cP)	Std. Dev. Avg. Viscosity at 10 1/s (cP)	Std. Dev. Avg. G' (Pa)	Std. Dev. Avg. G" (Pa)	Std. Dev. Avg. Phase Angle (°)
—	REFRESH LIQUIGEL ® lubricant eye gel (Allergan)	—	6.07	6.84	0.01	0.01	1.81
—	SYSTANE ® lubricant eye drop (Alcon)	—	7.60	6.33	0.01	0.01	5.24
	SYSTANE ® lubricant eye gel (Alcon)	—	7.08	5.45	0.01	0.02	0.11
1	10% Unstressed	0.00	5.11	0.64	0.02	0.02	5.94
2	5% Unstressed	1494.72	8.11	2.23	0.09	0.04	3.45
3	1% Unstressed	122.90	4.47	1.07	0.11	0.00	3.52
4	0.5% Unstressed	561.44	9.15	0.43	0.07	0.05	0.08
5	0.1% Unstressed	19771.41	7.32	14.43	0.22	0.29	1.71
6	0.05% Unstressed	5313.20	174.63	19.72	0.67	0.20	0.26
7	0.01% Unstressed	16087.39	130.05	17.16	0.35	0.03	0.30
8	0.005% Unstressed	14438.84	20.29	7.83	1.15	0.18	0.24
9	1× PBS	0.00	3.78	1.99	0.01	0.01	12.65
—	REFRESH LIQUIGEL ® lubricant eye gel (Allergan) 60° C.	0.00	5.41	1.50	0.01	0.01	2.03
—	SYSTANE ® lubricant eye drop (Alcon) 60° C.	0.00	4.63	1.96	0.01	0.01	6.10
—	SYSTANE ® lubricant eye gel (Alcon) 60° C.	0.00	6.00	1.30	0.01	0.01	0.14
10	10% 60° C.	155.14	2.05	0.34	0.04	0.02	2.51
11	5% 60° C.	192.33	17.86	3.46	0.09	0.07	0.77
12	1% 60° C.	491.86	7.55	0.02	0.04	0.01	2.09
13	0.5% 60° C.	535.77	1.70	0.52	0.03	0.03	0.41
14	0.1% 60° C.	2926.71	16.47	2.71	0.50	0.16	0.31
15	0.05% 60° C.	15813.74	47.39	8.45	1.12	0.11	1.34
16	0.01% 60° C.	20776.21	18.64	2.47	2.82	0.59	0.03
17	0.005% 60° C.	7448.66	63.16	6.20	0.75	0.06	0.42
18	1× PBS 60° C.	0.00	5.20	1.19	0.01	0.01	10.68
19	1% Autoclave	729.80	29.98	3.22	0.24	0.19	1.97
20	0.5% Autoclave	525.73	6.93	1.23	0.02	0.10	3.36
21	0.1% Autoclave	4329.26	115.43	14.66	0.98	0.28	2.37
22	0.05% Autoclave	2750.08	75.77	9.32	2.15	0.74	1.79
23	0.01% Autoclave	4848.63	23.18	0.65	0.92	0.17	0.17
24	0.005% Autoclave	509.23	0.08	0.15	0.00	0.01	10.04
25	1× PBS Autoclave	0.00	3.28	1.03	0.06	0.03	1.44

All of the formulations measured in Table 23 shear thinned, demonstrating lower viscosities at higher shear rates. The observed shear thinning was more pronounced in the silk fibroin formulations than in the corresponding experimental controls, which included commercially available treatments for dry eye (REFRESH LIQUIGEL® lubricant eye gel (Allergan), SYSTANE® lubricant eye drop (Alcon), SYSTANE® lubricant eye gel (Alcon)). At lower shear rates (1 1/s), the silk fibroin formulations exhibited viscosities similar to that of a gel drop. At higher shear rates (10 1/s), the silk fibroin formulations exhibited viscosities similar to that of a liquid drop. The viscosities of the formulations were measured to be higher at lower concentrations of silk fibroin; however, the viscosity peaked when the formulations were between 0.1-0.5% silk fibroin. When the silk fibroin concentration was 0.1% (w/v) or below, the viscosity decreased with the use of stressed silk.

As seen in Table 23, the average shear storage modulus (G') was measured to be higher for formulations with a lower concentration of silk fibroin. When the silk fibroin concentration was 0.1% (w/v) or below, the shear storage modulus was lower for formulations prepared from stressed silk fibroin than from unstressed. The average phase angle was also measured to be lower for formulations with a lower concentration of silk fibroin. For the unstressed formulations and heat stressed formulations (60° C. Overnight), the average shear loss modulus (G") was measured to be higher for formulations with a lower concentration of silk fibroin. When the silk fibroin concentration was below 0.1% (w/v), the average shear loss modulus was lower for formulations prepared from stressed silk fibroin than for formulations prepared from unstressed silk fibroin.

Example 13: Preparation of Fluorescein Isothiocyanate (FITC)-Labeled Silk Fibroin (FITC-SF) Solution

Silk fibroin (SF) was labeled with fluorescein isothiocyanate (FITC) for the residence time studies described in the Examples herein. 420 mg of sodium bicarbonate was dissolved in 9 mL of deionized (DI) water. The pH was adjusted to 9.0 using 1 N sodium hydroxide and 1 N hydrochloric acid. A quantity of DI water sufficient to raise the volume to 10 mL was added to prepare a 0.5 M sodium bicarbonate solution.

1.5 M hydroxylamine was prepared by dissolving 262 mg hydroxylamine in 2.0 mL of water. The pH was adjusted to 8.5 using 10 N sodium hydroxide, and a quantity of DI water sufficient to raise the volume to 2.5 mL was added.

Immediately before performing the labeling reaction, FITC (three 10 mg vials, ThermoFisher) was dissolved in 0.5 mL of dimethyl sulfoxide (DMSO; Sigma) resulting in a 20 mg/mL solution of FITC in DMSO.

All buffers and solutions were filtered through 0.2 μm filters under aseptic conditions with the exception of the silk fibroin solution and the FITC in DMSO solution.

A 5% (w/v) silk fibroin solution (480mb; Batch 88) containing 50 mM sucrose was thawed. 1 mL 0.5 M sodium bicarbonate buffer was added to a vial containing 4 mL of the thawed 5% (w/v) silk fibroin solution. If needed, the pH was adjusted to between 8.5-9.0 using 1N sodium hydroxide. A sample of the silk fibroin solution was retained as a control.

The labeling reaction was performed by adding 1.44 mL FITC in DMSO to 4.5 mL of the silk fibroin solution. The vial was kept protected from light at room temperature (RT) for 2 hours on a rocker resulting in FITC-labeled silk fibroin (FITC-SF).

The control sample of silk fibroin solution was prepared by adding 1.4 mL of DMSO to 4.5 mL of the thawed 5% (w/v) silk fibroin solution. The mixture was incubated at RT for 2 hours on a rocker protected from light.

After the two-hour incubation, 0.6 mL hydroxylamine solution was added to each reaction, and the mixture was placed on a rocker at RT for one hour. The pH was then adjusted to 7.0 using 1 N hydrochloric acid. Each solution was transferred to separate 20 kDa dialysis cassettes. Each solution was protected from light while dialyzed against 4.5 L of water with four complete exchanges at 4° C. over 72 hours. Dialyzed solutions were filtered under aseptic conditions through a 0.2 μm polyethersulfone (PES) filter unit. Final solutions were stored in sterile containers at 4° C. until use.

Example 14: Confirmation of the Conjugation of FITC to Silk Fibroin

High performance liquid chromatography (HPLC) was used to confirm conjugation is successful in the FITC-labeled silk fibroin (FITC-SF) solution. An Agilent 1260 BioInert HPLC system equipped with a Waters X-Bridge Protein BEH SEC, 200 Å, 3.5 μm column was used. An isocratic flow of mobile phase (100 mM Tris-HCl with 400 mM sodium perchlorate, pH 8.5) at 0.86 mL/min was used to elute analytes. Successful FITC labelling of SF was determined by monitoring protein absorbance at 280 nm and FITC emission at 525 nm following excitation at 490 nm.

Samples were diluted to 1% (w/v) prior to injection. The data showed overlapping UV and fluorescence profiles for the silk fibroin and FITC-SF, which would represent successful conjugation since the molecular weight of unconjugated FITC is smaller than the silk fibroin (400 Da vs. >6 kDa, respectively) and would elute much later during this analysis.

Example 15: Preparation of Formulations of FITC-SF

Silk fibroin solutions containing varying percentages of FITC-SF were prepared. FITC-SF was prepared as described in Example 13. A frozen stock of 5% (w/v) silk fibroin solution (5% SF) in water was thawed for preparation of these formulations.

Stock solutions, including a 5× Borate Buffer, a 1× Borate Buffer, and a 0.86% (w/v) solution of FITC-SF in water, were prepared. The 5× Borate Buffer was prepared by combining the following: 1.5 g boric acid, 213.1 mg sodium borate decahydrate, 350 mg potassium chloride, 32 mg magnesium chloride hexahydrate, 20 mg calcium chloride dihydrate, and 40 mL DI water. The pH was adjusted to 7.3 using 1 N hydrochloric acid and 1 N sodium hydroxide. A quantity of deionized (DI) water sufficient to raise the volume to 50 mL was added. The 1× Borate Buffer (150 mOsm/L) was prepared by combining 4 mL of the 5× Borate Buffer with 12 mL DI water. The pH was adjusted to 7.3 using 1 N hydrochloric acid and 1 N sodium hydroxide. A quantity of DI water sufficient to raise the volume to 20 mL was added. The FITC-SF solution described in Example 13 was diluted with water to a concentration of 0.86% (w/v) after dialysis.

1% (w/v) silk fibroin solutions were prepared, in which varying percentages of the silk fibroin in solution was FITC-SF. These formulations were prepared by combining volumes of the solutions described below. Table 25 shows the volume of components in each formulation of the resulting 1% (w/v) silk fibroin solution (150 mOsm/L).

TABLE 25

FORMULATION VOLUMES

Sample	% FITC-SF of the 1% SF formulation	5% Borate buffer [mL]	DI water [mL]	5% SF [mL]	FITC-SF [mL]
289-1	SF Control	0.8	2.0	0.8	None
289-2	40%	0.8	0.5	0.48	1.850
289-3	30%	0.8	1.0	0.56	1.387
289-4	20%	0.8	1.5	0.64	0.924
289-5	10%	0.8	1.75	0.72	0.462

For all the formulations, the pH was adjusted to 7.3 using sodium hydroxide and hydrochloric acid, and DI water sufficient to raise the volume to 4 mL was added. The final product was obtained through filtration with a 0.2 μm syringe filter.

Example 16: Effects of FITC Conjugation on the Viscosity of a Silk Fibroin Formulation

A Bholin C-VOR 150 rheometer (Malvern) with a 4°/40 mm cone and plate was used to measure rheology characteristics at 25° C. and a gap of 0.15 mm.

To analyze the complex viscosity of each formulation, thirty samples were put in a strain hold at 5%, each with a delay time of 5 s, an integration time of 1s, and a wait time of 4 s. Table 26 provides the oscillation results. As used herein, the term "complex viscosity" refers to viscosity, measured under oscillatory conditions, that is dependent on the percent strain and frequency. "G'" ("G prime") represents the storage modulus, and "G"" ("G double prime") represents the loss modulus.

TABLE 26

OSCILLATION DATA

Sample	% FITC-SF of the 1% SF formulation	Complex Viscosity [cP]	Standard Deviation	G' [Pa]	Standard Deviation	G" [Pa]	Standard Deviation
289-1a	0%	130.6	7.6	0.686	0.053	0.415	0.023
289-1b	0%	58.9	8.9	0.311	0.059	0.198	0.026
Average	0%	91	35.8	0.486	0.198	0.299	0.113
289-2a	40%	45.1	7.3	0.209	0.040	0.190	0.032
289-2b	40%	29.9	6.9	0.128	0.044	0.135	0.029
Average	40%	37.5	10.4	0.168	0.058	0.162	0.041
289-3a	30%	83.4	9.3	0.430	0.060	0.299	0.027
289-3b	30%	97.8	12.2	0.505	0.076	0.348	0.033
Average	30%	90.6	12.9	0.467	0.077	0.323	0.039
289-4a	20%	184.6	15.3	0.930	0.104	0.691	0.032
289-4b	20%	34.2	6.5	0.137	0.039	0.162	0.038
Average	20%	143.6	69.8	0.714	0.372	0.547	0.243
289-5a	10%	122.9	81.9	0.602	0.432	0.479	0.291
289-5b	10%	247.2	38.6	1.235	0.188	0.941	0.158
Average	10%	185.1	89.1	0.919	0.459	0.710	0.328

Effects of conjugation of FITC to silk fibroin on shear viscosity were also analyzed. Fifteen samples were held at a shear rate of 1 1/s. As used herein, the term "shear viscosity" or "viscosity" refers to the ratio between shear stress and shear rate. Shear viscosity is measured under shear conditions, which means that steady, simple shear in the same direction is applied to the sample. Results are shown in Table 27.

TABLE 27

SHEAR VISCOSITY DATA

Sample	% FITC-SF of the 1% SF formalation	Shear Viscosity (cP)	Stand. Dev.
289-1a	0%	76.2	11.4
289-1b	0%	50.7	4.1
Average	0%	63.4	15.5
289-2a	40%	39.3	5.0
289-2b	40%	33.5	4.7
Average	40%	36.4	5.6
289-3	30%	61.9	5.2
289-3b	30%	85.3	6.9
Average	30%	73.6	13.3
289-4a	20%	91.7	11.9
289-4b	20%	23.3	6.6
Average	20%	46.7	33.9
289-5a	10%	35.2	3.3
289-5b	10%	98.2	4.9
Average	10%	66.7	12.9

Higher percentages of FITC-SF in the formulations was observed to decrease complex viscosity. The complex viscosity, G', G", and shear viscosity of the samples with 30% FITC-SF were determined to be the most similar to those of the control silk fibroin solution. As a result, silk fibroin solutions in future studies will comprise 30% FITC-SF. 30% FITC-SF was chosen in order to balance maintaining viscosity with effective labeling, because a 10% FITC-SF ration may result in weak labeling.

Example 17: Preparation of Silk Fibroin Formulations for Irritability Analysis in Rabbits

The irritability in the eyes of rabbits was compared for silk fibroin formulations with different concentrations and molecular weights of silk fibroin. The materials used in the formulations are shown in Table 28.

TABLE 28

MATERIALS FOR SILK FIBROIN FORMULATIONS

Product	Supplier	Catalog Identifier	Lot Number
Lyophilized LMW Silk-Fibroin (480 mb)	Cocoon Biotech	—	Batch #080-3C
Lyophilized LMW Silk-Fibroin (120 mb)	Cocoon Biotech	—	Batch #068-C
Boric Acid	Fisher	A73-500	163991
Sodium Chloride	Sigma	S7653-1KG	SLBS2340V
Sodium Borate Decahydrate	Sigma	31457-100G	BCBS7522V
Potassium Chloride	Sigma	12636-250G	BCBV6226
Magnesium Chloride Hexahydrate	Sigma	M7304-100G	SLBS4877
Calcium Chloride Dihydrate	Fisher	C70-500	160760

Table 29 described the formulations analyzed in this Example. They were selected based on viscosity and surface tension.

TABLE 29

SILK FIBROIN FORMULATIONS

Sample	Description	Silk-Fibroin Boil Time (minutes)	Silk-Fibroin Conc. (%)
1	DED Buffer (Vehicle)	N/A	N/A
2	1% 480 mb in DED Buffer (Silk Fibroin-LMW)	480	1
3	1% 120 mb in DED Buffer (Silk Fibroin-HMW)	120	1

TABLE 29-continued

SILK FIBROIN FORMULATIONS			
Sample	Description	Silk-Fibroin Boil Time (minutes)	Silk-Fibroin Conc. (%)
4	1% 480 mb in DED Buffer with 1% Polyethylene Glycol (Silk Fibroin-LMW with Excipient)	480	1

The DED buffer (Vehicle) was prepared by dissolving: 1.2 g boric acid, 170.5 mg sodium borate decahydrate, 680 mg sodium chloride, 280 mg potassium chloride, 25.6 mg magnesium chloride hexahydrate, and 15.9 mg calcium chloride dihydrate in a 250 mL beaker containing 190 mL DI water. The pH was adjusted to 7.3 final using 1 N sodium hydroxide and 1 N hydrochloric acid. A quantity of DI water sufficient to raise the volume to 200 mL was added.

The 1% 480 mb in DED buffer (Silk Fibroin-LMW) formulation was prepared by reconstituting 300 mg of 480 mb silk fibroin to 5% (w/v) by adding 5.7 mL DED buffer. The silk fibroin was allowed to dissolve for 30 minutes or until clear. The solution was diluted to 1% (w/v) silk fibroin by adding 24 mL of DED buffer.

The 1% 120mb in DED buffer (Silk Fibroin-HMW) formulation was prepared by reconstituting 300 mg of 120 mb silk fibroin to 5% (w/v) by adding 5.7 mL of DED buffer. The silk fibroin was allowed to dissolve for 30 minutes or until clear. The solution was diluted to 1% (w/v) silk fibroin by adding 24 mL of DED buffer.

The 1% 480mb in DED Buffer with 1% Propylene Glycol (Silk Fibroin-LMW with Excipient) formulation was prepared as follows. Propylene glycol may act as a demulcent. 1.25% propylene glycol in DED buffer was prepared by mixing 0.375 mL propylene glycol with 29.625 mL of DED buffer. One tube (300 mg) of 480mb silk fibroin was reconstituted to 5% (w/v) by adding 5.7 mL DED buffer. The silk fibroin was allowed to dissolve for 30 minutes or until clear. The solution was diluted to 1% (w/v) SF and 1% propylene glycol by adding 24 mL of 1.25% propylene glycol in DED buffer.

Final formulations were filtered through a 0.2 μm filtration unit with polyethersulfone membranes and stored at 4° C. For each formulation, ten 1.75 mL aliquots were placed in separate sterile 2 mL polypropylene Eppendorf® tubes (1 tube/day+3 extra). All samples were stored at 4° C.

Example 18: Rheological Studies of Silk Fibroin Formulations for Irritability Studies

Silk fibroin formulations were prepared from the protocol as described in Example 17. Silk fibroin formulations were prepared from silk fibroin lyophilized either in water or in phosphate buffer (PB). The formulations tested were presented in Table 30. Some formulations were prepared with polypropylene glycol (PG). All formulations were prepared in Dry Eye Disease (DED) buffer, a borate buffer comprising 6 mg/mL boric acid, 0.45 mg/mL sodium borate, 3.4 mg/mL sodium chloride, 1.4 mg/mL potassium chloride, 0.06 mg/mL magnesium chloride, and 0.06 mg/mL calcium chloride, pH 7.3.

TABLE 30

FORMULATIONS FOR IRRITABILITY AND RHEOLOGY STUDIES				
Sample	Description	Silk Lyophilization Buffer	Silk Conc. (%) (w/v)	PG Conc. (%)
254-1	DED Buffet	—	0	0
254-2	1% 480 mb in DED	Water	1	0
254-3	1% 120 mb in DED	Water	1	0
254-4	1% 480, 1% PG in DED	Water	1	1
254-4V	DED Buffer, 1% PG	—	0	1
254-2PB	1% 480 mb (PB) in DED	PB	1	0
254-3PB	1% 120 mb (PB) in DED	PB	1	0
254-4PB	1% 480 (PB), 1% PG in DED	PB	1	1

The rheological properties of the silk fibroin formulations were then analyzed. The experiments were conducted on a Bohlin C-VOR 150 rotational rheometer with a 4°/40 mm cone and plate geometry, a 0.15 mm gap size, and a temperature of 25° C. The first test comprised a strain sweep from 0.1% to 10% strain at 1 Hz over the course of 160 s, with 50 readings taken. This was to determine the linear viscoelastic region (LVR) of each sample. The second test comprised a strain hold at 5% (or 1% if the LVR was smaller) and 1 Hz for 145 s, with 15 readings taken. The third test comprised a shear ramp from 0.1 1/s to 10 1/s over 100 s, followed by a shear hold for 20 s, for a total time of 120 s, with 70 samples. A fourth test was also conducted with a shear rate of 1 1/s, a time of 135 s, and 15 samples. The data from the rheological experiments are presented in Table 31.

TABLE 31

RHEOLOGICAL STUDIES OF SILK FIBROIN FORMULATIONS FOR IRRITABILITY STUDIES							
Sample	254-1	254-2	254-3	254-4	254-2PB	254-3PB	254-4PB
Viscosity at 1 1/s (cP)	7.07	62.85	38.95	81.96	57.78	43.14	56.30
Viscosity Standard deviation (1 1/s)	4.25	3.23	2.64	4.59	3.45	4.29	6.03
Viscosity at 10 1/s (cP)	4.76	12.86	11.64	16.95	11.69	10.42	10.92
Viscosity Standard Deviation (10 1/s)	5.67	5.68	6.79	6.22	1.69	1.04	1.52

TABLE 31-continued

RHEOLOGICAL STUDIES OF SILK FIBROIN FORMULATIONS FOR IRRITABILITY STUDIES							
Sample	254-1	254-2	254-3	254-4	254-2PB	254-3PB	254-4PB
G' (Pa)	2.24E−02	2.41E−01	8.76E−02	4.39E−01	3.07E−01	1.48E−01	4.24E−01
G' Standard Deviation	9.10E−03	5.26E−02	1.87E−02	4.72E−02	3.72E−02	1.66E−02	5.22E−02
G" (Pa)	4.15E−02	2.43E−01	1.35E−01	3.74E−01	2.45E−01	2.09E−01	3.55E−01
G" Standard Deviation	1.03E−02	2.76E−02	1.23E−02	2.31E−02	1.87E−02	1.09E−02	2.91E−02
Phase Angle (°)	61.11	45.73	57.27	40.56	38.71	54.76	40.05
Phase Angle Standard Deviation	13.39	3.48	5.16	1.61	1.94	2.66	1.63

All of the formulations showed evidence of shear thinning; the viscosities were measured to be lower at higher shear rates. The formulations with lower molecular weight silk fibroin (480 mb) were more viscous than the formulations with higher molecular weight silk fibroin (120 mb). Preparation from silk fibroin lyophilized in PB did not affect the viscosity of the formulations. Formulations with PG were more viscous than formulations without this excipient. The G' and G" were measured to be lower for formulations with higher molecular weight silk fibroin. G' and G" were also increased by the preparation from silk fibroin lyophilized in PB or formulations with PG. The phase angle was higher for formulations with higher molecular weight silk fibroin.

Example 19: Analysis of Irritability of Silk Fibroin Formulations in Rabbits

Non-GLP ocular irritability/tolerability was evaluated by measuring the effects of topical administration of the silk fibroin formulations described herein in New Zealand White (NZW) Rabbits. Animals received seven days of four times daily (QID) topical administrations of test article or vehicle (bilateral) and underwent full ophthalmic exams at baseline, post the first dose on Day 1, post the last dose on Day 1, post the last dose on Day 4, and post the last dose on Day 8.

Rabbits were individually housed in suspended wire bottom caging in a procedure room. Rabbit Diet 5P25 (lab diet specially formulated per lab animal diet) was provided according to U.S. Department of Agriculture (USDA) guidelines. Filtered tap water or spring water was provided to the animals ad libitum. Environmental controls were set to maintain temperatures 16-22° C. (61-72° F.) with relative humidity between 50±20%. A 12-hour light/12-hour dark cycle was maintained. A staff veterinarian was available as needed throughout the study. Prior to study initiation, the veterinarian released the animals from quarantine.

Animals were assigned to dose groups after a baseline ocular examination. Table 32 provides the group assignments for the 16 animals selected for the study.

TABLE 32

GROUP ASSIGNMENTS FOR ANIMALS							
Group	Number of Animals	Silk Fibroin minute boil (mb)	Route	No. of Doses per Day	SF Dose Concentration % (w/v)	Propylene Glycol Dose Concentration (mg/mL)	Dose Volume (μL/eye)
1/Vehicle	4	N/A	Topical	QID	0	0	40
2/Silk Fibroin-LMW Concentration 1	4	480	Topical	QID	10	0	40
3/Silk Fibroin-HMW Concentration 2	4	120	Topical	QID	10	0	40
4/Silk Fibroin-LMW + 1% PG Formulation	4	480	Topical	QID	10	10	40

40 μL of test article or vehicle (bilateral) was administered to the ocular surface of the rabbit using a calibrated micropipette. Care was taken to ensure no dose fell out of the rabbit's eye. QID dosing occurred at approximately 8 am, 11 am, 2 pm, and 4 pm. All times are ±60 minutes.

The weight of each animal was measured on Day 1 and Day 8. Values were rounded to the nearest 0.1 kg. A full ophthalmic exam was performed at baseline, twice on Day 1 (after the first dose and after the fourth dose), Day 4, and Day 8. After all animals were dosed, cage side observations were made for each animal specifically focusing on the eyes. If any indication of ocular pain was observed (squinting, pawing at the eye, or other signs of distress), the technician took the animal out and performed a gross examination of the eye. Gross exams observed the general appearance of the eye (whether or not there is hyperemia, discharge, swelling, squinting, etc.). All exams were 15 minutes post dose.

A hand-held slit-lamp was used to assess anterior abnormalities. If tolerability issues arose, the fundus of the eye was observed using an indirect ophthalmoscope. Prior to posterior examination, animals were dilated with a mydriatic (tropicamide).

Animals underwent full front of the eye exams and a quick vitreal scan via slitlamp. Endpoints included hyperemia, chemosis, discharge, aqueous flare, aqueous cells, presentation of keratic precipitates, and vitritis (as noted via slitlamp). Throughout the study, animals showed no signs of ocular discomfort or pain and the majority of endpoints were scored at zero.

Results indicate the silk fibroin formulations and vehicle are well tolerated in the eyes of New Zealand White rabbits. Results indicate that topical administration of the solution of silk fibroin was tolerated well throughout the duration of the study. Observations and clinical scoring of endpoints maintained baseline levels throughout the entire study. The concentrations of this test article show that it is safe to be administered topically. There were no significant endpoint changes in the hyperemia and discharge evaluations, and the remaining endpoints maintained scores of zero throughout all evaluations. Animals showed no signs of ocular pain or discomfort throughout all dosing time points as well.

Example 20: Ocular Silk Fibroin Residence

Formulation Preparation

To track the residence of the silk fibroin (SF) formulations in the eye, SF formulations were labelled with fluorescent dye, FITC. 5% (w/v) SF, (480mb) degummed as described above, was prepared in water and 50 mM sucrose. This solution was separated into 5 mL aliquots and frozen to −80° C. to be thawed as needed. FITC labelled silk fibroin (FITC-SF) was prepared as described in Example 13. A 5× stock of the borate (DED) buffer was prepared by combining the following components: 1.5 g boric acid, 213.1 mg sodium borate decahydrate, 350 mg potassium chloride, 32 mg magnesium chloride hexahydrate, 20 mg calcium chloride dihydrate, and 40 mL deionized (DI) water. The pH was adjusted 7.3 using 1 N hydrochloric acid and 1 N sodium hydroxide and brought up to 50 mL with DI water. The 5× stock was prepared so that the final solution of FITC-SF and unlabeled silk fibroin would be in a 1×DED buffer, the contents of which were described in earlier examples. 0.86% (w/v) FITC-SF in water was used to prepare the 30% FITC-SF formulation. A sample of 1% SF in DED with an osmolarity of 150 mOsm/L was prepared by combining 3 mL 5% SF in water, 3 mL 5× borate buffer, and 9 mL DI water in a 20 mL conical tube. The pH was adjusted to 7.3 using 1 N hydrochloric acid and 1 N sodium hydroxide and brought up to 15 mL with DI water. A second sample, a 30% FITC-SF formulation (containing 1% SF in DED with 150 mOsm/L), was also prepared. 5.201 mL 0.86% (w/v) FITC-SF was combined with 2.1 mL 5% (w/v) SF in water, 3.0 mL 5× borate buffer and 4.699 mL DI water. The pH of the solution was adjusted to 7.3 using 1 N hydrochloric acid and 1 N sodium hydroxide and brought up to 15 mL with DI water. The resulting solution has FITC labelled SF and unlabeled SD in the ratio of 3:7. Both formulations were filtered with a 0.2 μm syringe filter.

Rheology of Formulations

The complex viscosity, elastic modulus (G'), viscous modulus (G"), phase angle, and shear viscosity of the formulations to be tested for residence time were measured. The rheological properties were measured on a Bohlin C-VOR 150 rotational rheometer with a 4°/40 mm cone and plate geometry, a gap size of 0.15 mm, a temperature of 25° C., a frequency of 1 Hz, and pre-shear conditions of 1 1/s for 10s with a 10 second equilibration. The experiment consisted of two tests. The first test involved a strain hold at 5% with 30 samples, each with a delay time of 5 s, an integration time of 1s, and a wait time of 4 s. The second test used a hold at a shear rate of 1 1/s for 15 samples. The results of the experiments are presented in Table 33.

TABLE 33

RHEOLOGY DATA OF SILK FIBROIN FORMULATIONS FOR RESIDENCE STUDIES

	Sample							
	1% SF in borate buffer				30% FITC-SF Formulation			
Sample	291-1a	291-1b	291-1c	Average	291-2a	291-2b	291-2c	Average
Complex viscosity (cP)	89.64	217.01	315.31	207.32	252.18	238.09	325.11	271.79
Complex viscosity standard deviation	18.48	46.29	50.70	101.52	35.54	40.71	48.58	56.39
G' (Pa)	0.42	1.10	1.58	—	1.13	1.08	1.50	—
G' standard deviation	0.11	0.28	0.31	—	0.21	0.25	0.29	—
G" (Pa)	0.38	0.81	1.19	—	1.11	1.03	1.38	—
G" standard deviation	0.06	0.12	0.12	—	0.11	0.12	0.14	—
Phase Angle (°)	42.95	37.14	37.44	39.18	44.94	44.44	43.05	44.14
Phase angle standard deviation	4.90	3.78	2.94	4.73	2.97	4.03	3.20	3.47
Shear Viscosity (cP)	71.32	82.03	154.21	102.52	100.58	90.42	140.34	110.45
Shear viscosity standard deviation	8.86	19.86	28.30	42.32	21.79	37.13	41.39	40.12

These results suggested that the incorporation of FITC-SF into the silk fibroin solution formulation produced formulations with similar properties.

Ocular Residence Time

The fluorescence intensity of the formulations was established in vitro. The FITC-SF formulation was diluted 10×, 50×, and 100× in saline solution. 15 μL of the formulations were pipetted onto the end of a wicking strip. The formulations utilized include undiluted FITC-SF and 10×, 50×, and 100× dilutions of the FITC-SF solution in saline. Images of each of the wicking strips were taken through an orange filter with illumination using a cobalt blue light and the fluorescence brightness was analyzed and quantified using Fiji Image software. A segment of the image just adjacent to the end of the wicking strip was used to normalize the background brightness of the images. After brightness correction, a circle was drawn to encompass the end of the wicking strip. The brightness intensity of the wicking strip was then recorded. Two images of each sample were analyzed and averaged to generate the value for each timepoint from each animal. There were 6 animals per timepoint. The results are shown in Table 34.

TABLE 34

FLUORESCENCE INTENSITY			
% FITC-SF Formulation	Dilution	Corrected Intensity Avg.	Standard Deviation
100	0	86.1	0.8
10	10X	40.2	7.5
2	50X	17.5	3.4
1	100X	4.2	0.1
0	100 (Saline)	0.0	0.1

As the concentration of FITC-SF in the formulation decreased, the fluorescence intensity also decreased. The R squared value obtained was 0.935 suggesting a strong correlation between concentration of FITC and fluorescence brightness measured.

To measure the ocular residence time of the FITC-SF formulation, 40 μL of FITC-SF formulation was administered into the lower left lid of each rabbit. Eyes were manually blinked 2-3 times prior to the first sampling (T0). Sampling was performed by placing a wicking strip into the bottom eyelid of the rabbit for 5 sec. to collect ≤5 μL of tears. Images of the wicking strips were taken and the fluorescence was quantified as described above. Sampling was performed at T0, 15 minutes, 30 minutes, 1 hour, 2 hours, and 24 hours. The eyes were also scored clinically for the presence of fluorescence using the Fluorescence Intensity (FI) scale just prior to sampling. Images of the eyes were taken through an orange filter with illumination using a cobalt blue light.

Table 35 shows the Fluorescence Intensity (FI) scale for the tears collected from the fornix of the tear lake.

TABLE 35

FLUORESCENCE INTENSITY (FI) SCALE	
Score	Description
0	No fluorescence
1	≤25% fluorescence
2	≤50% fluorescence
3	≤75% fluorescence
4	≥75% fluorescence

Based on the FI scale in Table 35, the clinical fluorescence score for each time point was calculated as shown in Table 36.

TABLE 36

CLINICAL FLUORESCENCE SCORE			
Time (min)	Average	St Dev	SEM
0	4.0	0.0	0.0
0.25	2.0	0.6	0.2
0.5	1.3	0.8	0.2
1	1.0	0.6	0.2
2	0.2	0.4	0.1
24	0.0	0.0	0.0

Clinical Fluorescence Score decreased over time indicating that the FITC-SF is cleared from the tear lake over time. The Clinical Fluorescence Score decreased to less than one by 2 hours, indicating that the residence time of the FITC-SF formulations in the eye is ≤2 hours. The reduction in Clinical Fluorescence Score over time was strongly correlated with an R squared value of 0.97.

Images of the wicking strips used to collect the tears from the rabbits were quantified as shown in Table 37. As described above, images were analyzed using Fiji Image software. A segment of the image just adjacent to the end of the wicking strip was used to normalize the brightness of the images. After brightness correction, a circle was drawn to encompass the end of the wicking strip. The brightness intensity of the wicking strip was then recorded. Two images of each sample were analyzed and averaged to generate the value for the timepoint from each animal. There were a total of 6 animals per timepoint.

TABLE 37

QUANTIFIED FLUORESCENCE			
Time (hrs)	Intensity Average	Standard Deviation	SEM
0	139.8	20.7	1.7
0.25	10.1	25.1	2.1
0.5	97 5	14.7	1.2
1	93.8	20.5	1.7
2	86.3	6.3	0.5
24	89.3	4.5	0.4

Similar to the Clinical Fluorescence Score, the quantified fluorescence decreased over time. A sharp reduction in the quantified fluorescence was observed by 0.5 hours which slowly plateaued over the next hour to reach the values close to the lower limit of detection (86.3) by 2 hours. The lower limit of detection was determined from the dilution study of the fluorescence intensity and was approximately 2% of the initial formulation (a 50× dilution).

Example 21: Tribological Analysis of Silk Fibroin

In this test silk fibroin solutions were analyzed for their lubricating properties by applying a layer of the test sample to a substrate and forcing an upper geometry to slide against it at a number of speeds while under a defined load. Testing was performed using a research rheometer (DHR2, TA Instruments) fitted with a custom 3-balls-on-plate setup with a pliant lower substrate. A tribology assembly was employed that comprised a geometry of 3 glass spheres that slide against a pliant lower substrate, under a defined load of 1 N, onto which the sample has been spread. The rotational angular velocity is ramped from 0.05 rad/s to 20 rad/s, 8 points per decade, with each point maintained for 20 s with the coefficient of friction averaged over the final 15 s. The lower surface was made to hold 25° C. throughout the analysis. This test was performed in triplicate for each sample. The test was performed on silk fibroin formulation 480mb; 1% SFf; sln; 0.34% Suc; borate buffer. The formulation tested was prepared as described in previous examples. The silk fibroin solution exhibited a clear "stick-slip" behavior across all sliding speeds. Stick-slip behavior is the spontaneous jerking motion that may occur while two objects are sliding over each other.

Example 22: Surface Tension Measurements of Silk Fibroin Solutions Using Dynamic Analysis

Surface tension was employed to determine the spreading properties of silk fibroin solutions. The test analyzed the relationship between the volume, density, and shape of the liquid drop suspended from the end of a needle to calculate the surface tension of that liquid. It was performed on a drop shape analyzer (DSA30R, Krüss Scientific) fitted with the pendant drop module. An aliquot of each sample was equilibrated to 25° C. immediately prior to testing. A pendant drop was formed in a cuvette with a saturated atmosphere to minimize evaporation. This drop was imaged, and the surface tension measured every second over a period of 600s. This test was performed in triplicate for each sample. The tests were performed on silk fibroin formulations after their preparation. Deionized water served as a control. The formulations tested were prepared as described in previous examples. The results of the experiments are displayed in Table 38.

TABLE 38

SURFACE TENSION OF SILK FIBROIN SOLUTIONS

	Sample 1 Surface Tension (mN/m)	Sample 2 Surface Tension (mN/m)	Sample 3 Surface Tension (mN/m)	Average Surface Tension (mN/m)	RSD (%)	Standard Dev.
DI Water	71.32	71.46	71.46	71.41	0.092	0.066
480 mb; 5% SFf; sln; 1.71% Suc; borate buffer	42.48	42.15	42.08	42.24	0.412	0.174
480 mb; 1% SFf; sln; 0.34% Suc; borate buffer	44.27	44.57	44.82	44.55	0.505	0.225
480 mb; 0.5% SPf; sln; 0.17% Suc; borate buffer	45.80	45.99	46.03	45.94	0.218	0.100
480 mb; 0.1% SFf; sln; 0.03% Suc; borate buffer	46.37	46.72	46.89	46.66	0.464	0.217
480 mb; 0.05% SFf; sln; 0.01% Suc; borate buffer	49.06	49.12	49.10	49.09	0.051	0.025

Increasing silk fibroin concentration lead to a decrease in calculated surface tension. The presence of 0.05% (w/v) silk fibroin decreased the surface tension to 49.99 mN/m. This is a large decrease from water which displayed a surface tension of 71.41 mN/m. As silk concentration in the samples increased, the surface tension was further decreased reaching a low of 42.24 mN/m with the 5% (w/v) silk fibroin sample (Table 38).

Example 23: Treatment of Dry Eye Disease in Humans

To demonstrate safety and tolerability, optimize a formulation or dose, and show efficacy compared to a placebo, the following study will be performed in humans and animals. Both a single site, open label, placebo controlled safety study and a multi-site, randomized, double-masked, parallel group, placebo or vehicle-controlled, efficacy study will be performed with about 30 subjects and up to 150 patients, respectively. Standardized inclusion and exclusion criteria will be used for enrollment. The subjects will be randomized to a treatment arm after a wash-out or two-week vehicle run-in period.

Treatment arms will be the silk formulation administered four times a day (QID), twice a day (BID), or once a day (QD) compared to a placebo group (patients will serve as their own placebo) and will include between 8-10 subjects per arm. Treatment duration will be one month. The primary endpoint will be safety as measured by visual acuity, slit lamp biomicroscopy of intraocular pressure (IOP), fundoscopy, and adverse event query. The secondary endpoint will be efficacy as measured by corneal fluorescein staining, staining of individual regions of cornea (e.g. inferior, central, superior, nasal, and temporal), conjunctival staining, tear film break-up time, Schirmer's Test, and conjunctival redness. All the results are expected to change from the baseline. Ocular discomfort, such as dryness, grittiness, burning, stinging, etc., and the ocular surface disease index (OSDI) may also be used to characterize the results.

Example 24: Stability of Silk Fibroin Formulations

The stability of multiple silk fibroin formulations may be compared after storage at 4° C. for three weeks, room temperature for three weeks, or 40° C. for one week. A silk fibroin solution was formulated with a concentration of 0.5%, 1%, or 3% (w/v) silk fibroin with either 10 mM phosphate buffer (pH 7.5) or 10 mM borate buffer (pH 7.5) to result in an osmolarity of 150 mOsm/L or 290 mOsm/L. Aggregation was observed to be below 0.1% for all formulations and under all conditions. Silk fibroin formulations maintain their silk fibroin concentration, physical properties, and rheological properties after storage.

Example 25: Preparation of SBP Hydrogel Formulations

Silk was degummed using either a 90 or a 480-minute boil (90mb and 480mb), processed as described in Example 1 and lyophilized in 10 mM phosphate buffer (PB). Lyophilized silk fibroin was dissolved in ultrapure water containing Tween-80 to obtain 10, 20, and 30% (w/v) silk fibroin with 0.4% Tween-80. The hydrogels were formed by mixing with a gelling agent (e.g. PEG 4 kDa). 2.5 mL of the silk solution was mixed with 2.5 mL one of the following excipients: (i) 20% P188, (ii) 80% PEG 4 kDa, or (iii) 80% glycerol. 1 N hydrochloric acid was added to the formulations containing 80% PEG 4 kDa, to afford a final concentration of 15 mM hydrochloric acid. The final pH of all formulations was estimated to be approximately 7.4+/−0.1. The SBP hydrogel formulations are summarized in Table 39 below. The samples in Table 39 are named by the process used to prepare and formulate each hydrogel. For example, the sample named "90mb; hyd; 10% st; 5% SFf; 10% P188f" refers to a formulation where the silk was degummed with a 90-minute boil (90mb), "hyd" refers to the formulation of the sample as a hydrogel, "10% st" refers to the % (w/v) concentration of the stock solution of silk fibroin, "5% SFf" refers to a formulation with 5% (w/v) final silk fibroin concentration, and "10% P188f" refers to a formulation with 10% P188.

TABLE 39

SBP HYDROGEL FORMULATIONS

No.	Sample name	Boil Time (min)	[Silk] (%)	Excipient name	[Excipient] (%)	[Phosphate Buffer] (mM)
1	90 mb; hyd; 10% st; 5% SFf; 10% P188f	90	5	P188	10	16.7
2	90 mb; hyd; 10% st; 5% SFf; 40% PEG4kf	90	5	PEG 4k	40	16.7
3	90 mb; hyd; 10% st; 5% SFf; 40% Glycf	90	5	Glycerol	40	16.7
4	90 mb; hyd; 20% st; 10% SFf; 10% P188f	90	10	P188	10	33.3
5	90 mb; hyd; 20% st 10% SFf; 40% PEG4kf	90	10	PEG 4k	40	33.3
6	90 mb; hyd; 20% st; 10% SFf; 40% Glycf	90	10	Glycerol	40	33.3
7	480 mb; hyd; 10% st; 5% SFf; 10% P188f	480	5	P188	10	16.7
8	480 mb; hyd; 10% st; 5% SFf; 40% PEG4kf	480	5	PEG 4k	40	16.7
9	480 mb; hyd; 10% st; 5% SFf; 40% Glycf	480	5	Glycerol	40	16.7
10	480 mb; hyd; 20% st; 10% SFf; 10% P188f	480	10	P188	10	33.3
11	480 mb; hyd; 20% st; 10% SFf; 40% PEG4kf	480	10	PEG 4k	40	33.3
12	480 mb; hyd; 20% st; 10% SFf; 40% Glycf	480	10	Glycerol	40	33.3
13	480 mb; hyd; 30% st; 15% SFf; 10% P188f	480	15	P188	10	50
14	480 mb; hyd; 30% st; 15% SFf; 40% PEG4kf	480	15	PEG 4k	40	50
15	480 mb; hyd; 30% st; 15% SFf; 40% Glycf	480	15	Glycerol	40	50

Example 26: Freeze Thaw Studies with Varying Silk Fibroin Concentrations and Buffers

To date, lyophilization has been utilized for long term storage of silk fibroin. However, this can be costly and time-consuming in product manufacturing. In this study, various silk fibroin concentrations and buffer conditions were examined to test the feasibility of freezing silk fibroin without altering the protein quality. The samples were evaluated using rheology for viscosity and modulus, and size exclusion chromatography (SEC) for molecular weight and aggregation.

This study evaluated the properties of formulations with silk fibroin concentration varied at 1% and 3% (w/v) in phosphate buffer, borate buffer, or phosphate buffered saline, with or without propylene glycol. The formulation in each sample is described in Table 40. In the Table, "PB" is potassium phosphate buffer (from Sigma Aldrich Fine Chemicals, St. Louis, MO), pH 7.4; "PBS" is 1× phosphate buffered saline; "DED" is 1× borate buffer as described in Example 15; and "PG" is propylene glycol. The "baseline" measurements were made prior to freezing at −80° C.

The borate buffer stock solution (5×) was prepared by dissolving 3000 mg boric acid, 426.25 mg sodium borate decahydrate, 1700 mg of sodium chloride, 700 mg potassium chloride, 64 mg magnesium chloride hexahydrate, and 40 mg calcium chloride dihydrate in DI water, adjusting pH to 7.3 final using 1 N sodium hydroxide and 1 N hydrochloric acid, and filling up to 100 mL with DI water. The borate buffer was filtered through 0.2 μm polyethersulfone membrane filtration units prior to use. The final formulations contained 1× of this buffer (6 mg/mL boric acid, 0.45 mg/mL sodium borate, 3.4 mg/mL sodium chloride, 1.4 mg/mL potassium chloride, 0.06 mg/mL magnesium chloride, and 0.06 mg/mL calcium chloride, pH 7.3).

TABLE 40

FORMULATIONS WITH VARYING SILK FIBROIN CONCENTRATIONS AND BUFFERS

Sample No.	Description	Silk %	Buffer	Frozen
82-1	3% Water, Baseline	3	Water	—
82-1A	3% Water, Not Frozen	3	Water	N
82-1B	3% Water, Frozen	3	Water	Y
82-2	3% PB, Baseline	3	10 mM PB	—
82-2A	3% PB, Not Frozen	3	10 mM PB	N
82-2B	3% PB, Frozen	3	10 mM PB	Y
82-3	3% DED, Baseline	3	DED	—
82-3A	3% DED, Not Frozen	3	DED	N
82-3B	3% DED, Frozen	3	DED	Y
82-4	1% PBS, Baseline	1	PBS	—
82-4A	1% PBS, Not Frozen	1	PBS	N
82-4 B	1% PBS, Frozen	1	PBS	Y
82-5	1% DED, Baseline	1	DED	—
82-5A	1% DED, Not Frozen	1	DED	N
82-5B	1% DED, Frozen	1	DED	Y
82-6	1% PBS + PG, Baseline	1	PBS + 1% PG	—
82-6A	1% PBS + PG, Not Frozen	1	PBS + 1% PG	N
82-6B	1% PBS + PG, Frozen	1	PBS + 1% PG	Y
82-7	1% DED + PG, Baseline	1	DED + 1% PG	—

TABLE 40-continued

FORMULATIONS WITH VARYING SILK FIBROIN CONCENTRATIONS AND BUFFERS

Sample No.	Description	Silk %	Buffer	Frozen
82-7A	1% DED + PG, Not Frozen	1	DED + 1% PG	N
82-7B	1% DED + PG, Frozen	1	DED + 1% PG	Y

The rheology results and the size exclusion chromatography (SEC) results for the solutions are presented in Table 41 and Table 42, respectively, where viscosity, elastic modulus (G'), viscous modulus (G"), and phase angle were determined. G', G" and phase angle were measured using a Bohlin C-VOR 150 rotational rheometer. The temperature was held constant at 25° C. via a Peltier Plate system, and the gap was held at 0.15 mm. First a strain ramp was applied from 0.0014 to 5% strain with a constant frequency of 1 Hz, with 30 samples taken, each with a delay time of 9 s and an integration time of 1 s. Next a logarithmic shear ramp was applied from 0.00014 to 1 1/s, with holds at 0.00014, 0.001, 0.01, 0.1, and 1 1/s. Overall, the rheological properties did not change significantly following freezing with the exception of the 1% (w/v) silk fibroin in borate buffer with propylene glycol. The molecular weight as measured by SEC did not change following freeze/thaw, nor did the total peak area. Increases in aggregation were observed for all solutions containing borate buffer with the greatest increase in aggregation occurring with the addition of propylene glycol. Given these results, phosphate buffer performed better upon freeze/thaw than borate buffer.

TABLE 41

RHEOLOGICAL PROPERTIES AFTER FREEZE THAW

		Viscosity (Pa*s)		G' (Pa)		G" (Pa)		Phase Angle (°)	
Sample No.	Frozen	Avg.	Standard Dev.	Avg.	Standard Dev.	Avg.	Standard Dev.	Avg.	Standard Dev.
82-1	—	65.19	2.25	4.03E−01	0	4.47E−01	0	48.01	0
82-1A	N	50.91	5.69	1.45E−01	2.66E−02	2.53E−01	4.07E−02	59.73	7.91
82-1B	Y	49.77	5.1	1.45E−01	9.34E−03	2.24E−01	1.52E−02	57.02	2.78
82-2	—	38.18	3.17	3.28E−01	0	3.06E−01	0	43.05	0
82-2A	N	60.42	7.66	3.33E−01	5.78E−02	3.43E−01	5.08E−03	46.08	5.55
82-2B	Y	51.63	4.71	3.19E−01	6.16E−02	3.07E−01	1.44E−02	44.22	6.25
82-3	—	81.34	3.01	2.80E−01	0	3.67E−01	0	52.73	0
82-3A	N	79.23	1.75	3.16E−01	4.08E−02	3.97E−01	1.91E−02	51.65	2.23
82-3B	Y	76.64	23.4	4.28E−01	3.36E−02	4.20E−01	4.56E−02	44.45	2.47
82-4	—	93.78	4.89	5.84E−01	0	5.13E−01	0	41.28	0
82-4A	N	75.67	4.05	2.93E−01	4.79E−02	3.25E−01	3.87E−02	48.08	5.01
82-4B	Y	62.5	9.67	3.12E−01	5.82E−02	3.16E−01	1.02E−01	44.66	4.43
82-5	—	115.1	7.78	5.66E−01	0	7.49E−01	0	52.89	0
82-5A	N	82.94	19.47	4.47E−01	1.25E−01	4.76E−01	1.15E−01	47.04	1.69
82-5B	Y	72.05	15.33	3.17E−01	4.23E−02	3.53E−01	6.76E−02	47.83	3.86
82-6	—	91.32	3.57	3.17E−01	0	4.34E−01	0	53.85	0
82-6A	N	64.97	5.43	2.62E−01	1.31E−02	3.25E−01	3.00E−02	51.08	3.82
82-6B	Y	52.32	10.46	2.03E−01	5.77E−02	2.70E−01	8.27E−02	52.86	0.99
82-7	—	113.82	5.15	5.33E−01	0	6.14E−01	0	49.04	0
82-7A	N	87.53	24.46	4.40E−01	1.17E−01	4.67E−01	9.24E−02	46.96	2.26
82-7B	Y	48.81	12.4	2.59E−01	3.78E−02	2.86E−01	2.78E−02	47.92	6.46

TABLE 42

MOLECULAR WEIGHT, PEAK AREA AND AGGREGATION AFTER FREEZE THAW							
		MW (Da)		Peak Area		Aggregation (%)	
Sample No.	Frozen	Avg.	Standard Dev.	Avg.	Standard Dev.	Avg.	Standard Dev.
82-1	—	18218	0	25183515	0	0	0
82-1A	N	18043	1190	24592650	210277	0	0
82-1B	Y	18223	1008	24541530	207449	0	0
82-2	—	18075	0	26649930	0	0	0
82-2A	N	17943	1006	26433232	96728	0	0
82-2B	Y	18023	987	26385803	9567	0	0
82-3	—	18123	0	26161084	0	0	0
82-3A	N	18008	1069	25858311	65572	0	0
82-3B	Y	17953	1110	25937330	1196173	0.32	0.09
82-4	—	18434	0	8206505	0	0	0
82-4A	N	18381	1237	8102340	43772	0	0
82-4B	Y	18375	980	8008933	38407	0.04	0.01
82-5	—	18627	0	8590083	0	0	0
82-5A	N	18301	884	8529287	16053	0	0
82-5B	Y	18182	964	8130534	115760	1.31	0.36
82-6	—	18603	0	8191389	0	0	0
82-6A	N	18283	1139	8137815	13188	0	0
82-6B	Y	18345	1085	8130532	18074	0	0
82-7	—	18675	0	8613387	0	0	0
82-7A	N	18262	925	8527920	57162	0	0
82-7B	Y	17305	991	7919038	87323	7.27	1.7

Example 27: Freeze Thaw Studies with Varying Cryoprotectants

Silk fibroin processing will be more efficient and cheaper if there is no need for lyophilization. Removing a drying condition will require that silk fibroin solutions are stable through a freeze-thaw process. This will allow for aseptic preparation and shipment of silk fibroin from the manufacturing site to the fill/finish facility. This study investigated the effect of silk fibroin concentration as well as cryoprotectants (sucrose and trehalose) on the freeze-thaw stability silk fibroin.

The silk fibroin concentration was varied from 0.5% to 5% (w/v) in 10 mM phosphate buffer and the concentration of each cryoprotect was varied from 10 to 150 mM. The formulation in each sample is described in Table 43. All formulations contained 10 mM phosphate buffer at pH 7.4 and the baseline measurements were made prior to freezing at −80° C. The five 5 mL replicates were prepared for each group in separate 15 cc. conical tubes. In the Table, "SF" refers to silk fibroin.

TABLE 43

FORMULATIONS WITH VARYING SILK FIBROIN CONCENTRATIONS AND CRYOPROTECTANTS				
Sample No.	Description	Silk Fibroin Conc. (mg/mL)	Cryo-protectant	[Cryo-protectant] mM
83-1	0.5% SF in 10 mM phosphate, pH 7.4	5	—	—
83-2	1% SF in 10 mM phosphate, pH 7.4	10	—	—
83-3	2.5% SF in 10 mM phosphate, pH 7.4	25	—	—
83-4	5% SF in 10 mM phosphate, pH 7.4	50	—	—
83-5	5% SF, 10 mM sucrose in 10 mM phosphate, pH 7.4	50	Sucrose	10
83-6	5% SF, 50 mM sucrose in 10 mM phosphate, pH 7.4	50	Sucrose	50
83-7	5% SF, 100 mM sucrose in 10 mM phosphate, pH 7.4	50	Sucrose	100
83-8	5% SF, 150 mM sucrose in 10 mM phosphate, pH 7.4	50	Sucrose	150
83-9	5% SF, 10 mM trehalose in 10 mM phosphate, pH 7.4	50	Trehalose	10
83-10	5% SF, 50 mM trehalose in 10 mM phosphate, pH 7.4	50	Trehalose	50
83-11	5% SF, 100 mM trehalose in 10 mM phosphate, pH 7.4	50	Trehalose	100
83-12	5% SF, 150 mM trehalose in 10 mM phosphate, pH 7.4	50	Trehalose	150

The silk batch number 83 had a concentration of dialyzed silk fibroin of 5.9% as determined using UV-Vis assessment. The phosphate stock buffer (100 mM) was prepared by combining 20 mL of 100 mM monobasic potassium phosphate and 80 mL of 100 mM dibasic potassium phosphate and adjusting the pH to 7.4 using the 100 mM mono or dibasic phosphate solutions. The 1 M sucrose, 67 mM phosphate stock was prepared by dissolving 3.42 g sucrose in 6.7 mL of 100 mM phosphate buffer and filling up to 10 mL with DI water. The 0.67 M sucrose, 67 mM phosphate stock was prepared by dissolving 2.29 g of sucrose in 6.7 mL of 100 mM phosphate buffer and filling up to 10 mL with DI water. The 1 M sucrose stock was prepared by dissolving

3.42 g of sucrose in 6.7 mL of DI water and filling up to 10 mL with DI water. The 1 M trehalose, 67 mM phosphate stock was prepared by dissolving 3.42 g of trehalose in 6.7 mL of 100 mM phosphate buffer and filing up to 10 mL with DI water. The 0.67 M trehalose, 67 mM phosphate stock was prepared by dissolving 3.42 g of trehalose in 6.7 mL of 100 mM phosphate buffer and filing up to 10 mL with DI water. The 1 M trehalose stock was prepared by dissolving 3.42 g of trehalose in 6 mL DI water and filling up to 10 mL with DI water.

Each of the formulations was prepared as follows. Sample 83-1 (0.5% SF in 10 mM phosphate, pH 7.4) was prepared by adding 2.54 mL 5.9% silk fibroin stock, 3 mL 100 mM phosphate, pH 7.4, and 24.46 mL DI water in a 50 cc conical tube. Sample 83-2 (1% SF in 10 mM phosphate, pH 7.4) was prepared by adding 5.08 mL 5.9% silk fibroin stock, 3 mL 100 mM phosphate, pH 7.4, and 21.92 mL DI water in a 50 cc conical tube. Sample 83-3 (2.5% SF in 10 mM phosphate, pH 7.4) was prepared by adding 12.7 mL 5.9% silk fibroin stock, 3 mL 100 mM phosphate, pH 7.4, and 14.3 mL DI water in a 50 cc conical tube. Sample 83-4 (5% SF in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 3 mL 100 mM phosphate, pH 7.4, and 1.6 mL DI water in a 50 cc conical tube. Sample 83-5 (5% SF, 10 mM sucrose in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 3 mL 100 mM phosphate, pH 7.4, 300 μL 1M sucrose, and 1.3 mL DI water in a 50 cc conical tube. Sample 83-6 (5% SF, 50 mM sucrose in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 3 mL 100 mM phosphate, pH 7.4, 1.5 mL 1M sucrose, and 100 μL DI water in a 50 cc conical tube. Sample 83-7 (5% SF, 100 mM sucrose in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 4.5 mL 0.67M sucrose, 67 mM phosphate buffer, and 100 μL DI water in a 50 cc conical tube. Sample 83-8 (5% SF, 150 mM sucrose in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 4.5 mL 1M sucrose, 67 mM phosphate buffer, and 100 μL DI water in a 50 cc conical tube. Sample 83-9 (5% SF, 10 mM trehalose in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 3 mL 100 mM phosphate, pH 7.4, 300 μL 1M trehalose, and 1.3 mL DI water in a 50 cc conical tube. Sample 83-10 (5% SF, 50 mM trehalose in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 3 mL 100 mM phosphate, pH 7.4, 1.5 mL 1M trehalose, and 100 μL DI water in a 50 cc conical tube. Sample 83-11 (5% SF, 100 mM trehalose in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 4.5 mL 0.67M trehalose, 67 mM phosphate buffer, and 100 μL DI water in a 50 cc conical tube. Sample 83-12 (5% SF, 150 mM trehalose in 10 mM phosphate, pH 7.4) was prepared by adding 25.4 mL 5.9% silk fibroin stock, 4.5 mL 1M trehalose, 67 mM phosphate buffer, and 100 μL DI water in a 50 cc conical tube.

The rheology results and the size exclusion chromatography (SEC) results are presented in Table 44 and Table 45, respectively. In all conditions, the rheological properties of the silk fibroin solutions remained the same or showed a slight increase following freeze/thaw. The molecular weight by SEC did not change following freeze/thaw, however, a very slight increase in aggregation was observed for all samples. When sucrose or trehalose was used as an excipient, aggregation was shown to remain the same or decrease compared the control (no cryoprotectant at 5% (w/v).

TABLE 44

RHEOLOGICAL PROPERTIES AFTER FREEZE THAW

		Viscosity (Pa*s)		G' (Pa)		G" (Pa)		Phase Angle (°)	
Sample No.	Frozen	Avg.	Standard Dev.	Avg.	Standard Dev.	Avg.	Standard Dev.	Avg.	Standard Dev.
83-1	—	—	—	—	—	—	—	—	—
83-1	Y	—	—	—	—	—	—	—	—
83-2	—	208.24	10.52	1.07E+00	7.56E−02	7.49E−01	2.55E−02	34.99	1.99
83-2	Y	265.7	39.98	1.32E+00	1.21E−01	1.01E+00	2.54E−01	37.1	4.24
83-3	—	—	—	—	—	—	—	—	—
83-3	Y	—	—	—	—	—	—	—	—
83-4	—	86.71	7.93	4.11E−01	4.73E−02	3.57E−01	2.74E−02	41.11	2.41
83-4	Y	120.28	15.33	5.48E−01	6.75E−02	5.19E−01	7.26E−02	43.69	1.75
83-5		118.53	11.54	5.74E−01	6.46E−02	4.73E−01	4.48E−02	39.57	2.35
83-5	Y	122.43	34.8	5.60E−01	1.47E−01	5.26E−01	1.64E−01	43.11	2.12
83-6	—	—	—	—	—	—	—	—	—
83-6	Y	134.4	1.76	6.40E−01	5.47E−02	5.47E−01	5.07E−02	40.67	5.03
83-7	—	82.57	10.4	3.92E−01	6.08E−02	3.39E−01	3.53E−02	41.01	2.92
83-7	Y	108.28	7.24	5.04E−01	4.65E−02	4.55E−01	4.39E−02	42.28	3.87
83-8	—	94.85	11.97	4.52E−01	7.23E−02	3.87E−01	4.19E−02	40.82	3.53
83-8	Y	112.06	32.05	5.22E−01	1.57E−01	4.71E−01	1.28E−01	42.43	1.1
83-9	—	84.11	10.1	3.95E−01	5.80E−02	3.50E−01	3.68E−02	41.66	2.93
83-9	Y	130.05	15.31	6.09E−01	7.05E−02	5.43E−01	6.83E−02	41.88	1.4
83-10	—	79.88	9.05	3.49E−01	6.10E−02	3.57E−01	4.44E−02	45.84	5.67
83-10	Y	105.12	30.49	4.74E−01	1.36E−01	4.58E−01	1.40E−01	44.22	3.07
83-11	—	70.31	8.22	3.26E−01	4.58E−02	2.97E−01	3.46E−02	42.5	3.28
83-11	Y	92.24	44.09	4.36E−01	1.95E−01	3.79E−01	2.00E−01	40.63	3.51
83-12	—	80.52	9.49	3.75E−01	6.29E−02	3.37E−01	3.00E−02	42.25	4.2
83-12	Y	113.08	17.18	5.32E−01	7.895−02	4.70E−01	7.47E−02	41.64	0.93

TABLE 45

MOLECULAR WEIGHT AND AGGREGATION AFTER FREEZE THAW

Sample No.	Frozen	MW (Da) Avg.	MW (Da) Standard Dev.	Aggregation (%) Avg.	Aggregation (%) Standard Dev.
83-1	—	—	—	—	—
83-1	Y	22730	166	0.17	0.01
83-2	—	—	—	—	—
83-2	Y	22768	162	0.25	0.09
83-3	—	—	—	—	—
83-3	Y	22659	229	0.11	0.04
83-4	—	23175	—	0.00	—
83-4	Y	22688	77	0.08	0.03
83-5	—	23151	—	0.00	—
83-5	Y	22491	595	0.09	0.03
83-6	—	—	—	—	—
83-6	Y	22377	408	0.08	0.05
83-7	—	23004	—	0.00	—
83-7	Y	22815	503	0.07	0.01
83-8	—	22955	—	0.00	—
83-8	Y	22818	224	0.07	0.05
83-9	—	23175	—	0.02	—
83-9	Y	22296	648	0.06	0.04
83-10	—	23102	—	0.00	—
83-10	Y	2.2506	555	0.18	0.09
83-11	—	23028	—	0.00	—
83-11	Y	22583	402	0.10	0.02
83-12	—	23004	—	0.00	—
83-12	Y	22486	325	0.06	0.02

EQUIVALENTS AND SCOPE

Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments in accordance with the disclosure described herein. The scope of the present disclosure is not intended to be limited to the above Description, but rather is as set forth in the appended claims.

In the claims, articles such as "a," "an," and "the" may mean one or more than one unless indicated to the contrary or otherwise evident from the context. Claims or descriptions that include "or" between one or more members of a group are considered satisfied if one, more than one, or all of the group members are present in, employed in, or otherwise relevant to a given product or process unless indicated to the contrary or otherwise evident from the context. The disclosure includes embodiments in which exactly one member of the group is present in, employed in, or otherwise relevant to a given product or process. The disclosure includes embodiments in which more than one, or the entire group members are present in, employed in, or otherwise relevant to a given product or process.

It is also noted that the term "comprising" is intended to be open and permits but does not require the inclusion of additional elements or steps. When the term "comprising" is used herein, the term "consisting of" is thus also encompassed and disclosed.

Where ranges are given, endpoints are included. Furthermore, it is to be understood that unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or subrange within the stated ranges in different embodiments of the disclosure, to the tenth of the unit of the lower limit of the range, unless the context clearly dictates otherwise.

In addition, it is to be understood that any particular embodiment of the present disclosure that falls within the prior art may be explicitly excluded from any one or more of the claims. Since such embodiments are deemed to be known to one of ordinary skill in the art, they may be excluded even if the exclusion is not set forth explicitly herein. Any particular embodiment of the compositions of the disclosure (e.g., any antibiotic, therapeutic or active ingredient; any method of production; any method of use; etc.) can be excluded from any one or more claims, for any reason, whether or not related to the existence of prior art.

It is to be understood that the words which have been used are words of description rather than limitation, and that changes may be made within the purview of the appended claims without departing from the true scope and spirit of the disclosure in its broader aspects.

While the present disclosure has been described at some length and with some particularity with respect to the several described embodiments, it is not intended that it should be limited to any such particulars or embodiments or any particular embodiment, but it is to be construed with references to the appended claims so as to provide the broadest possible interpretation of such claims in view of the prior art and, therefore, to effectively encompass the intended scope of the disclosure. The present disclosure is further illustrated by the following nonlimiting examples.

```
                         SEQUENCE LISTING

Sequence total quantity: 1
SEQ ID NO: 1            moltype = AA  length = 6
FEATURE                 Location/Qualifiers
source                  1..6
                        mol_type = protein
                        note = Silk fibroin heavy chain repeat unit
                        organism = Synthetic construct
SEQUENCE: 1
GAGAGS                                                            6
```

The invention claimed is:

1. A method of preparing processed silk fibroin and controlling hydrolysis of silk fibroin, comprising

providing raw silk,

degumming the raw silk in 0.5 M sodium carbonate at a temperature of 80 to 90° C., and for a time of 120 to 480 minutes, to provide degummed silk fibers,

dissolving the degummed silk fibers in 5 to 13 M lithium bromide at a concentration of 10% (w/v) to 30% (w/v), and

removing the lithium bromide by Tangential Flow Filtration to provide the processed silk fibroin,

wherein the combination of sodium carbonate concentration, time and temperature of the degumming control the hydrolysis of the silk fibroin, and wherein the processed silk fibroin has an average molecular weight of 15 kDa to 55 kDa by HPLC SEC using a Waters BEH200 Protein Standard Mix.

2. The method of claim **1**, wherein the raw silk is silk fibers comprising silk fibroin and sericin.

3. The method of claim **1**, wherein dissolving comprises dissolving the degummed raw silk at 20% w/v in 9.3 M lithium bromide for 5 hours at 60° C.

4. The method of claim **1**, wherein the processed silk fibroin has an average molecular weight of 20 kDa to 40 kDa by HPLC SEC using a Waters BEH200 Protein Standard Mix.

5. The method of claim **1**, wherein the processed silk fibroin can be dissolved to provide a 10% to 30% (w/v) silk fibroin solution.

6. The method of claim **1**, wherein a 0.5%, 1%, or 3% (w/v) solution of the processed silk fibroin in 10 mM phosphate buffer (pH 7.5) or 10 mM borate buffer (pH 7.5) to result in an osmolarity of 150 mOsm/L or 290 mOsm/L is stable after storage at 4° C. for three weeks, room temperature for three weeks, or 40° C. for one week determined as aggregation below 0.1%.

* * * * *